

Engineering Materials

Carlton A. Taft
Sergio Ricardo de Lazaro *Editors*

Progress in Hydrogen Energy, Fuel Cells, Nano-Biotechnology and Advanced, Bioactive Compounds

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Engineering Materials

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Carlton A. Taft · Sergio Ricardo de Lazaro
Editors

Progress in Hydrogen Energy, Fuel Cells, Nano-Biotechnology and Advanced, Bioactive Compounds

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Preface

We overview state-of-the-art topics for quantum dots synthesis, hydrogen fuel cells, perovskite photovoltaic solar cells, hydrogen generation and storage, spintronics, quantum computing, and artificial intelligence. We also do computational research and review chapters involving bioactive/medicinal/phytotechnological compounds. Advanced biotechnological strategies are presented.

Chapter “[Synthesis of Colloidal Semiconductor Nanocrystals for Enhanced Optoelectronic Technologies](#)” explores the synthesis and importance of quantum dot technology.

Chapter “[Overview: Fuel Cells, Hydrogen Energy, Science, Materials, Nanotechnology, Artificial Intelligence and State of the Art](#)” discusses the fuel cells technology underlining, quantum dots/artificial intelligence potential increasing participation, as one of the avenues for future perspectives.

Chapter “[Two-Dimensional Materials Applied to Hydrogen Storage](#)”, discusses the 2D materials as promising candidates due to their unique properties such as tunable electronic properties and high surface area.

Chapter “[Biohydrogen Production in a Biorefinery Using Industrial Effluents: Challenges, Opportunities and Economic Analysis](#)” discusses the technological/economical viewpoint strategies to improve biohydrogen production.

In Chapter “[In Silico Quantum Simulation on the \$\text{MnTiO}_3\$, \$\text{FeTiO}_3\$, and \$\text{NiTiO}_3\$ Materials in Ilmenite Structure as Candidates to Spintronic Devices](#)”, the authors investigate using computational density functional theory magnetic ordering as a route to analyze potential candidates for spintronic devices such as quantum computers.

Chapter “[Theoretical and Experimental Insight for the Design and Characterization of Perovskite-Based Solar Cell](#)” reports the theoretical/computational/simulations (including DFT) of perovskite photovoltaic solar cells in order to study the performance of different architectures of solar cells.

Chapter “[Biohydrogen Generation from Agroindustrial Waste: Challenges and Opportunities](#)” explores the utilization of agroindustrial/urban biomasses for biohydrogen generation and identifies bottlenecks, cost-effectiveness, H_2 -producing microbes, and future prospects for microbial hydrogen.

Chapter “Antiallergic, Antioxidant, Anti-Inflammatory and Immunostimulant Potential of Kefiran Postbiotic: Molecular Docking, Prediction of Pharmacokinetic Properties and Biological Activity” reports the *in silico* studies of kefiran in order to identify and characterize biological targets of medicinal interest.

Chapter “Computational Simulation of Molecular Docking and Adme/Tox Analysis of Triazole-Based Kojic Acid Analogues” utilizes the theoretical chemistry methods for validation of new inhibitors for TYR, a constituent of the biosynthetic steps of melanogenesis.

Chapter “Moonlighting Proteins: Importance and Current Scenarios” elucidates the molecular mechanisms and structural adaptations of moonlight proteins highlighting their impact on different pathogenesis, related databases, and main forms of prediction applying bioinformatic tools for therapeutic purposes.

Chapter “Use of Phytotherapeutics in the Amazon with a Focus on the Metropolitan Region of Belém: Traditional Applications and Computational Studies of Bioactive Compounds” reviews the traditional usage of phytotherapeutics with advanced computational strategies to promote health and well-being while preserving biodiversity/cultural knowledge of the Amazon.

Chapter “Drug-Like Properties and Therapeutical Potential of *Calendula officinalis* L. Active Ingredients” presents the anti-cancer, antimicrobial, antiparasitic, antioxidant, anti-inflammatory, hypoglycemic, hypolipidemic, and wound-healing properties of the *Calendula* medicinal herb.

Chapter “Herbal Medicines: From History to Current Research—A Comprehensive Survey” offers a review of herbal medicines including a historical perspective including advances and processes involved in their extraction and production.

Chapter “The Role of Artificial Intelligence in the Discovery of New Drugs” analyzes the process of accelerating multiple phases by analyzing vast amounts of biological and chemical data to identify patterns that would be difficult to detect manually.

Chapter “Laccase: A Catalyst for Change in Eco-Friendly Industrial Processes” underlines the challenges to enhance the position of laccase as a pivotal player in sustainable biotechnological processes such as bioenergy.

Chapter “Photostimulation Applied to the Production of Biopolymers (Exopolysaccharides)” underlines the photobiomodulation for biosynthesis of microbial biopolymers as promising tools in health, environment, and industry.

Chapter “Sustainable Horizons: Navigating Challenges in Butanol Production from Lignocellulosic By-Products” underlines the approaches such as synthetic biology, nanotechnology, enzymology, and boosting techniques for butanol microbial enhancement.

Chapter “The Role of Composting in Sustainable Agriculture and Waste Management” discusses the materials and innovative research to improve sustainability, carbon capture, and bioenergy.

Chapter “Exploring the Landscape of Fermentation: Products, Processes, and Outlook” discusses the fermentation bioprocesses providing forecasts for the biotechnological future.

Chapter “[A Narrative Assessment of Free Webservers for Pharmacokinetic and Toxicological Predictions Using In Silico Tools](#)” discusses the features, benefits, and drawbacks of open-access online servers that perform toxicological and pharmacokinetic predictions.

Rio de Janeiro, Brazil
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Synthesis of Colloidal Semiconductor Nanocrystals for Enhanced Optoelectronic Technologies



Felipe M. Pinto, Swayandipta Dey, Carlton A. Taft, and Felipe La Porta

Abstract Colloidal semiconductor nanocrystals, likewise known as quantum dots, have captured significant attraction owing to their outstanding properties (arising from quantum confinement effects) and potential for groundbreaking applications in optoelectronic technologies. This chapter explores the diverse methods for synthesizing quantum dots, delving into the intricacies of both top-down and bottom-up approaches. We emphasize the ongoing efforts to refine these methods, addressing challenges like size distribution control, surface passivation, and defect minimization. Understanding the interplay between quantum dot structure, defects, and their resulting properties is crucial for tailoring quantum dots for specific applications. Herein, we will delve into the critical role of surface ligands play in stabilizing quantum dots and influencing their behavior, highlighting the importance of post-synthesis modifications for optimizing their functionality. The chapter also explores the remarkable impact of quantum dots on optoelectronic technologies, paving the way for a brighter, more technological future. Their tunable bandgaps and bright, narrow-linewidth emission make them ideal for high-performance light-emitting diodes, lasers, photodetectors, and also solar cells. We delve into the recent progress in quantum dot-based optoelectronics, showcasing advancements in efficiency, color purity, and stability. While the field of quantum dot research has seen significant

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improvement, we acknowledge the remaining challenges, including device stability and integration issues. Looking towards the future, we discuss the ongoing exploration of novel quantum dot materials and synthesis techniques, promising to unlock even more significant potential for these remarkable nanocrystals in shaping the future of optoelectronics.

1 Introduction

The 1980s marked a turning point in materials science with the independent discoveries of Aleksey Ekimov and Louis Brus. Ekimov observed nanocrystals embedded in glasses [1–3], while Brus demonstrated the remarkable size-dependent quantum effects in semiconductor nanoparticles, now known as quantum dots (QDs) [4, 5]. These QDs, specifically made from cadmium sulfide (CdS) and zinc sulfide (ZnS) colloidal nanocrystals, exhibited captivating optical properties that could be tuned by manipulating their size. However, initial synthesis methods often resulted in QDs with inconsistent quality and numerous defects. This challenge was addressed by Moungi Bawendi in 1993, who introduced a revolutionary synthesis technique that yielded high-quality, uniform QDs [6]. Bawendi’s method enabled precise control and uniformity in the growth of these colloidal nanocrystals. The profound impact of these discoveries was acknowledged with the 2023 Nobel Prize in Chemistry, awarded to Ekimov, Brus, and Bawendi for their pioneering work [7].

So, what exactly are QDs? These remarkable QDs, whose properties are dictated by their tiny small size (typically ranging from 1 to 20 nm in diameter), are colloidal nanocrystals now considered the building blocks of nanotechnology with the potential to transform various scientific and technological fields [8, 9]. Their unique physicochemical properties stem from the ability to confine electrons in all three spatial dimensions, leading to discrete energy levels (molecule-like behavior) [10–12]. This phenomenon, termed quantum confinement, is responsible for the remarkable tunability of their physical properties. Moreover, QDs can readily be integrated with other materials, forming heterostructures with entirely new physical properties, further expanding their versatility and potential applications [13–17]. Consequently, QDs are actively explored for a huge range of groundbreaking applications, spanning optoelectronics to biomedicine [7, 18–24]. However, challenges such as the toxicity of certain materials and the need for more scalable production methods still need to be overcome for QDs to reach their full potential for shaping the future of diverse fields [7].

This review delves into the captivating world of QDs, exploring and unveiling their fascinating scientific nature and immense potential for transforming the future. By understanding these synthesis methods and the fundamental properties of QDs, we can unlock their vast application potential. This review will explore how a comprehensive spectroscopic toolkit helps elucidate order–disorder effects at various length scales and their impact on size-dependent properties. We will begin by examining

and discussing synthetic strategies in Sect. 2. Section 3 will briefly overview relevant QD properties and their applications in optoelectronics devices. Finally, the main summary and outlook according to the key insights gained from the synthesis and understanding of their fundamental properties are detailed in Sect. 4.

2 An Overview of the QD Synthesis Methods

It is well known that the synthesis method used generally depends on the desired physical properties of the target QDs, including their size, shape, composition, surface chemistry, as well as, even the presence of imperfections within the nanocrystals [25–28]. Generally, the QD synthesis strategies are broadly classified into two well-established types: top-down and bottom-up approaches (Fig. 1). The top-down approach involves carving out QDs from a larger piece of semiconductor. As we know, the inherent precision of manipulating pre-existing materials is a key advantage, enabling fine control over QD size and placement within larger structures [29, 30]. This control is vital for applications requiring specific spatial arrangements of QDs, such as in the fabrication of optoelectronic devices. At the same time, the bottom-up approach involves the meticulous assembling of QDs atom by atom or molecule by molecule. This approach offers greater flexibility in terms of composition and the ability to create QDs with diverse morphologies [25–28]. Both methods have their advantages and challenges which will be detailed below.

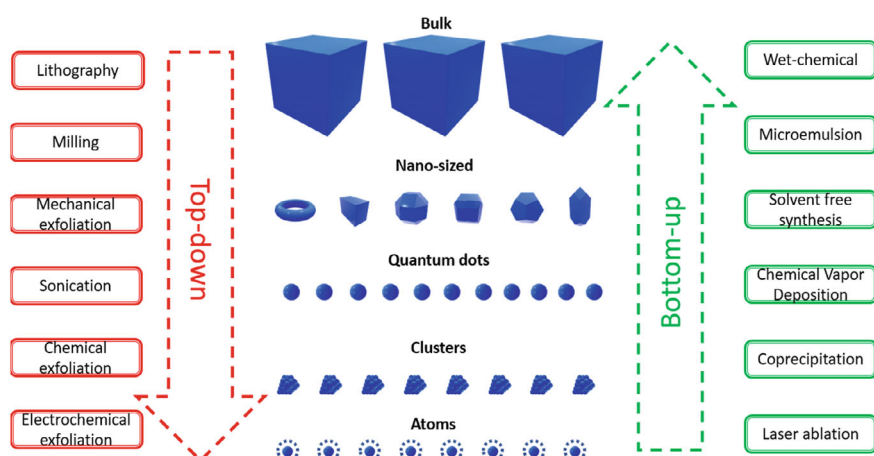


Fig. 1 Schematic overview of both top-down and bottom-up approaches for the QDs synthesis

2.1 Top-Down Fabrication

2.1.1 Lithography and Focused Ion Beam Milling: Precision Through Physical Vacuum-Based Methods

The top-down approach to QD synthesis utilizes established techniques like lithography and milling to carve nanometer-sized structures from bulk semiconductors. This approach offers unparalleled control over the spatial arrangement of QDs, enabling the design and fabrication of intricate semiconductor architectures with tailored functionalities [31].

Lithography, in particular, has been widely adopted in QD fabrication [29, 32]. Originally developed for the semiconductor industry's production of integrated circuits, its ability to create nanoscale patterns with exceptional precision has proven invaluable for QD synthesis [33]. The fundamental principle of lithography involves the use of light or electrons to pattern a photosensitive polymer layer, known as a photoresist. Upon exposure, the photoresist undergoes a change in solubility, allowing for the removal of either the exposed or unexposed regions during development. This creates a patterned photoresist mask that can be used to etch or deposit materials onto the underlying substrate with nanoscale accuracy [32]. Several lithography techniques find application in QD synthesis, each offering distinct advantages and limitations.

- I. Photolithography, the most common variant, utilizes ultraviolet light for exposure, providing a cost-effective solution for large-scale production [34]. Yet, its resolution is, of course, limited by the wavelength of light used [35, 36].
- II. Electron beam lithography, on the other hand, uses a focused beam of electrons to write patterns directly onto the photoresist, offering unmatched precision [37]. But at the expense of slower processing speeds and higher costs due to the need for vacuum environments [37, 38].
- III. Nanoimprint lithography, a relatively new technique, utilizes a mold for high-resolution patterning [39]. This technique can achieve very high resolution and is potentially cheaper and faster than electron beam lithography. However, it requires the fabrication of specialized molds, which can be challenging (Fig. 2a–e).

Additionally, the choice of lithography technique allows for the creation of QDs with diverse shapes, including spheres, rods, wires, and more complex structures. Photolithography remains the go-to for mass production, while electron beam lithography excels in research settings where precision is paramount [8–12]. Nanoimprint lithography provides a cost-effective solution for replicating established complex patterns [39]. By leveraging a prefabricated mold, nanoimprint lithography enables the precise transfer of intricate designs onto a substrate, significantly reducing production time and cost compared to direct-write techniques. This makes it particularly appealing for the fabrication of large-area QD arrays or devices with complex geometries [40–42]. The versatility of lithographic techniques empowers scientists

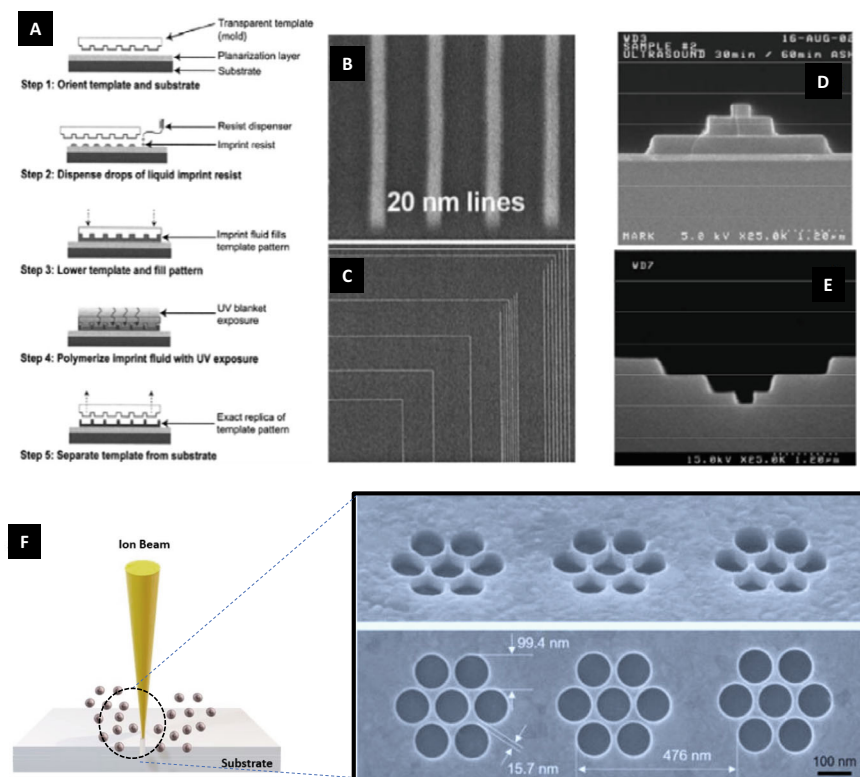


Fig. 2 Nanopattern fabrication using the step and flash imprint lithography (SFIL) process and FIB milling. **a** Schematic illustration of the SFIL workflow. **b** High-resolution nanopattern (20 nm linewidth) fabricated via SFIL. **c** Demonstrated reusability of an SFIL template (40 nm lines) for over 1500 imprints. **d, e** Multilayered SFIL template and its corresponding imprint pattern replicated in a resist material. Adapted with permission from [39]. Copyright 2007 © Wiley. **f** FIB milling process for creating hollow nanopillar heaters (HNHs) for potential nanotechnology applications. Adapted with permission from [44]. Copyright 2021 © MDPI. Adapted with permission from [45]. Copyright 2019 © Walter de Gruyter GmbH

to tailor the size, shape, and, ultimately, the functionality of QDs [29, 32]. By meticulously controlling the exposure and development processes, researchers can define the dimensions of individual QDs and the spacing between them. This fine-tuning enables the optimization of QD properties such as bandgap energy, emission wavelength, and quantum yield, which are directly linked to their performance in optoelectronic applications [43].

In QD synthesis, the patterned photoresist serves as a mask during the following etching or deposition steps, enabling proper control over the particle size as well as morphology [31]. This control hinges on carefully calibrated exposure and development parameters. As electronics continue to shrink, advancements in high-precision lithography are essential. Lin et al. [46] introduced a novel immersion detector

for lithography light sources, demonstrating remarkable sensitivity even in low-light conditions with high external quantum efficiency (PLQY), and showcasing its real-time monitoring capability.

Electron beam lithography, in particular, enables direct patterning of intricate QD nanostructures [31]. However, recent research by Petkov et al. [47] revealed the unexpected influence of substrate material on electron beam lithography outcomes. They found that achieving clean, precise patterns with hydrogen silsesquioxane (HSQ) resist was notably easier on silicon than germanium substrates. Using the Monte-Carlo method to simulate e-beam interactions with pure Si, pure Ge, and Ge/Si epitaxial layers on Si substrates revealed that the observed limitations are directly related to the generation of back-scattered electrons (BSEs). Also, the researchers also noted variations in brightness within the patterns when viewed under a scanning electron microscope (SEM). The study highlights that brightness variation in SEM images across neighboring lines is crucial for understanding resist clearance, impacting the subsequent etching step for pattern transfer [47]. A deeper understanding of how substrate material influences electron beam lithography processes will enable scientists to refine nanofabrication techniques, leading to smaller and more precise features, particularly in the development of cutting-edge laterally stacked group IV multi-channel field effect transistors (FETs).

The ongoing drive for miniaturization has spurred the development of scanning probe lithography (SPL), particularly mechanical SPL (m-SPL) [48]. This versatile and cost-effective technique offers nanoscale resolution and has recently been enhanced by integration with heat and electrical energy sources [49]. These advancements position m-SPL as a leading contender for achieving sub-nanometer precision in nanofabrication.

Focused ion beam (FIB) milling, a mechanical material removal technique, has distinct advantages in QD synthesis [50]. By employing a tightly focused ion beam, FIB milling allows for the precise removal of material, enabling the creation of intricate nanostructures and well-defined QDs with tailored properties [50]. Its potential integration with existing semiconductor fabrication processes makes FIB milling promising for scalable QD production [32, 50]. Moreover, FIB techniques offer exceptional lateral precision in QD etching.

2.1.2 Ball Milling

High-energy ball milling has emerged as a versatile and compelling technique in the relentless pursuit of novel nanomaterials [51–55]. In this method, the zirconium balls collide with the material under high energy, effectively grinding it into nanoscale particles [56, 57]. Overall, this mechanical approach offers several advantages, including scalability and the ability to synthesize a huge range of nanomaterials.

For instance, Youh et al. [51] successfully produced graphene QDs with narrow size distribution (1.0–4.0 nm) through ball milling. The resulting GQDs exhibited excellent stability, with their dispersions remaining well-mixed for at least 6 months, showing no signs of settling or clumping. The narrow size distribution, blue

light emission, and minimal functional groups make these graphene QDs promising candidates for various emerging optoelectronic applications.

Xu et al. [32] report a novel, high-yield (>15 wt%) approach for mass-producing various QDs from readily available bulk materials, including lead sulfide (PbS), CdS, copper sulfide (CuS), ferrous sulfide (FeS), and also the ZnS. Figure 3a illustrates the schematic overview of the fabrication QDs process employed in the study. Their two-step approach, involving silica-assisted ball milling and sonication-assisted solvent treatment, yielded QDs with intrinsic properties, excellent water solubility, and exceptional photoluminescence, broadening the scope for their applications. Figure 3b complements this information by showcasing photographs of the QD dispersions in various solvents and powders. Additionally, Fig. 3c presents zeta potential plots of the aqueous (re)dispersions for these QDs, confirming their satisfactory stability and water solubility—fundamental properties for solution-based processing in diverse applications of interest [32]. Protesescu et al. [58] demonstrated the power of ball milling in synthesizing highly emissive lead halide perovskite nanocrystals (Fig. 3d). Despite being an unconventional method for these materials, ball milling produced nanocrystals with optoelectronic quality comparable to those from the traditional hot-injection method, offering a simpler and potentially more scalable alternative.

A particularly attractive advancement in ball milling is the development of solvent-free synthesis methods, which eliminate the need for harmful organic solvents—a major source of environmental pollution [60]. Solvent-free ball milling not only minimizes hazardous waste but also enhances energy efficiency by removing the need for solvent recovery systems [61]. Therefore, this approach aligns with the principles of green chemistry, promoting sustainable and environmentally responsible manufacturing practices.

The solvent-free ball milling process relies on the direct reaction of solid precursors, resulting in the formation of QDs (Fig. 3e). To elucidate the reactivity of molecular solids in ball milling using a solvent-free approach, Kaupp proposed a three-stage mechanism [62]. This mechanism suggests that:

- I. Reactants initially permeate through a phase resembling an amorphous solid, triggering a chemical reaction.
- II. The product phase undergoes nucleation and subsequent expansion.
- III. Finally, the product detaches, exposing a fresh reactant surface for further reaction cycles [59].

Hence, solvent-free ball milling synthesis offers several advantages over traditional solvent-based methods, including simplified procedures, reduced contamination risks, and precise control over QD properties through manipulation of reaction parameters like temperature and time [63]. Recent studies have further highlighted the potential of solvent-free ball milling. Hu et al. [64] developed a scalable method for synthesizing and functionalizing atomically thin MoS₂ nanosheets using GQDs as exfoliation agents. This mechanochemical approach produced high-quality, solvent-stable nanosheets with exceptional catalytic activity for hydrogen evolution. Zhang et al. [65] proposes a solvent-free ball milling to synthesize ultrafine Mn₂O₃

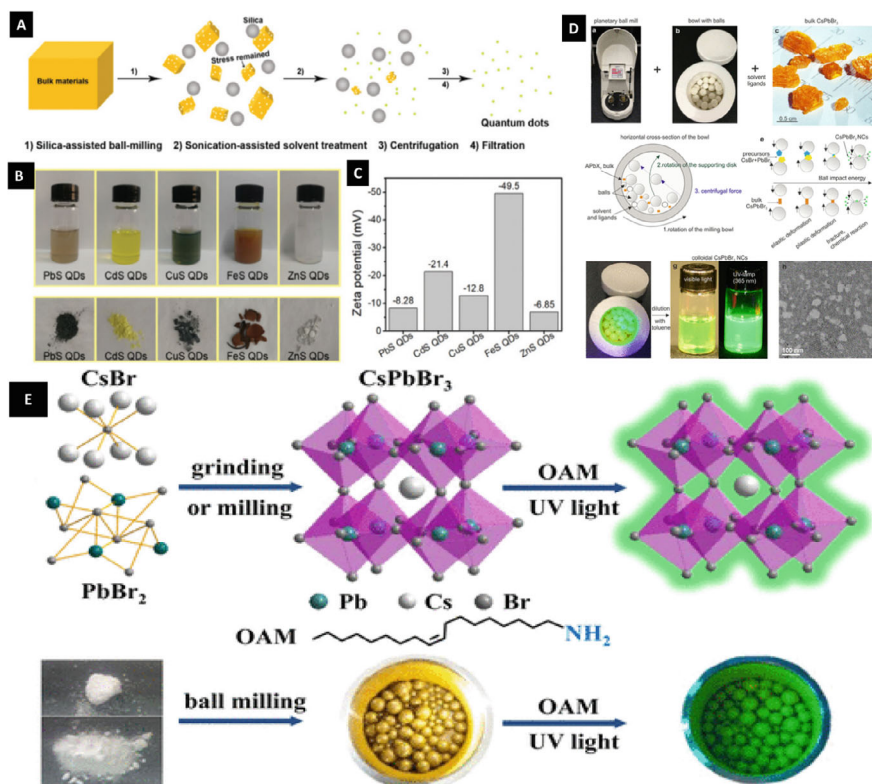


Fig. 3 Top-down manufacturing process of the QDs materials. **a** Schematic representation of the key steps involved in QDs synthesis. **b** Photographs showcase both the dispersions and powders of the QDs obtained through this process. The top panel displays the aqueous dispersions, while the bottom panel exhibits the powdered form. **c** Zeta potential measurements of the QDs in aqueous dispersions (0.1 mg mL^{-1}). Adapted with permission from [32]. Copyright 2021 © Royal Society of Chemistry. **d** Schematic representation of the ball milling process for the cesium lead bromide synthesis. Adapted with permission from [58]. Copyright 2018 © American Chemical Society. **e** Diagram illustrating the alterations in CsPbX₃ QDs sample color and fluorescence under UV light exposure during the ball milling process. Adapted with permission from [59] © 2017 American Chemical Society

QDs embedded into N-doped carbon matrix (2D composites). These composites exhibited robust interfacial interactions and demonstrated promising performance in sodium-ion hybrid capacitors. Ren et al. [66] employed a solvent-free route to create highly crystalline Cu/In dual-single-atom-doped mesoporous TiO₂ photocatalysts, showcasing enhanced photocatalytic activity in the reduction of Cr⁶⁺, a harmful environmental pollutant.

2.1.3 Sonication Method

The sonication method has been widely utilized for synthesizing semiconductor QDs. This method harnesses the power of high-energy cavitation induced by ultrasonic waves in liquid media to fragment bulk materials into nanoscale particles. The resulting QDs exhibit uniform size distribution, high purity, and can be produced in large quantities, making sonication a valuable tool in nanomaterials research and applications [67]. Particularly, the sonication process involves the propagation of ultrasonic waves through a liquid medium, inducing rapid pressure fluctuations. These fluctuations lead to the formation and subsequent collapse of microscopic bubbles (cavitation) [68]. The implosion of these bubbles generates intense localized shear forces and high temperatures, effectively breaking down bulk materials into smaller nanoscale particles [68]. Notably, this sonication process offers several advantages, including simplicity, scalability, and its capability to yield high-quality QDs with minimal contamination [69].

Sonication has been successfully employed to synthesize a diverse range of QDs. For instance, Chen et al. [70] demonstrated the synthesis of high-quality $\text{CH}_3\text{NH}_3\text{PbX}_3$ ($\text{X} = \text{Cl}, \text{Br}, \text{and I}$) hybrid halide perovskite QDs with varying compositions and uniform size using sonication treatment. Coleman et al. [71] demonstrated the successful synthesis of a variety of QDs based on transition metal dichalcogenide (TMDC), including MoS_2 , WS_2 , MoSe_2 , MoTe_2 , TaSe_2 , NbSe_2 , NiTe_2 , and Bi_2Te_3 , through a combination of sonication and solvent treatment. These studies highlighted the efficacy of sonication in producing QDs with high yields and excellent photoluminescent properties, suitable for optoelectronic devices and beyond [72]. Also, it is well-known that the sonication can be integrated with other techniques, i.e., enhancing its capabilities in controlled synthesis of QDs, significantly [73].

A recent study by Ali et al. [73] presents a novel synthesis method for MoS_2 QDs utilizing a sonication-assisted ion-intercalation process with sodium hydroxide to help separate the layers. Atomic force microscopy (AFM) measurements confirmed the successful production of single or few-layered QDs, with a thickness of less than 1.6 nm. Additionally, the X-ray diffraction (XRD) analysis provided evidence that the characteristic signature of bulk 2H- MoS_2 disappeared, indicating successful exfoliation. The researchers also used high-resolution transmission electron microscopy (HRTEM) to examine the size of the QDs. Their analysis revealed an average lateral size of approximately 3.7 nm, with some variation.

Matyszczyk et al. [74] utilized sonication to synthesize SnS and SnS₂ QDs for environmental remediation applications, demonstrating their efficacy in photo- and sonocatalytic degradation of azo-dye Metanil Yellow. Both QDs were prepared using the sonication method using tin chloride and thioacetamide solutions, with varied sonication times. SnS₂ QDs synthesized for 120 min exhibited the highest photocatalytic efficiency (85.2% color removal), while SnS QDs under the same conditions exhibited the best sonocatalytic performance (66.9% color removal). These findings highlight the potential of sonication synthesis for creating colloidal nanoparticles with specific catalytic properties. By fine-tuning the process, scientists can develop even more efficient catalysts for environmental remediation and other applications.

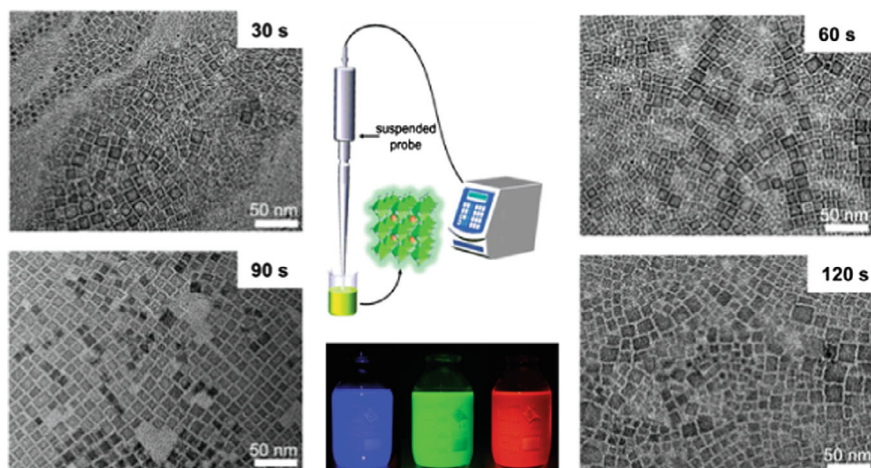


Fig. 4 Schematic representation of the sonication synthesis of FAPbX₃ perovskite QDs. Adapted with permission from [75]. Copyright 2020 © Wiley

Dou et al. [75] reported a new, simpler method to create large quantities of high-quality FAPbX₃ perovskite QDs, where X can be bromine (Br), a mix of chlorine (Cl) and bromine (Br), or a mix of bromine (Br) and iodine (I), with tunable emission colors using a high-powered ultrasound technique at room temperature (Fig. 4). As a result, the FAPbX₃ QDs are all similar in size and emit light very efficiently, with a photoluminescence quantum yield of up to 93%. Figure 4 shows TEM images of cubic FAPbBr₃ QDs obtained by sonication synthesis. From this TEM data, we can clearly observe how the particle size of these QDs changes with growth duration. Their average side length increases from 6.7 nm for 30 s growth to 12.0 nm for 120 s growth. Furthermore, the researchers were able to create a high-performance green light-emitting diode (LED) using these QDs, specifically FAPbBr₃. This LED achieved an impressive maximum current efficiency of 61.3 cd A⁻¹, as well as, the PLQY of about 14.1% [75].

Hence, the sonication has proven to be a versatile and powerful tool for synthesizing a wide array of QDs. Its simplicity, scalability, and ability to produce high-quality nanocrystals with minimal contamination make it a valuable asset in nanomaterials research and applications. Ongoing research exploring novel combinations of sonication with other techniques promises to further expand the capabilities of this method in the controlled synthesis of QDs for diverse applications, including optoelectronics, catalysis, and environmental remediation [58–67].

2.1.4 Mechanical, Chemical and/or Electrochemical Exfoliations

Mechanical, chemical, and electrochemical exfoliation techniques offer a valuable top-down approach for synthesizing high-quality semiconductor QDs, providing

distinct advantages in scalability and control over size and morphology [76]. By leverage physical, chemical and electrochemical forces, these methods can produce QDs from bulk materials, enabling the creation of QDs with tailored properties for diverse applications [68–76].

Mechanical exfoliation is particularly effective for producing QDs from layered materials, e.g., the MoS_2 , a layered TMDC, is a prime example of a material well-suited for exfoliation-based QD synthesis [77]. Owing to the weak interlayer van der Waals forces between layers facilitate exfoliation into monolayer or few-layer nanosheets, which can be further fragmented into QDs of the desired size [76].

Wu et al. [78] demonstrated a groundbreaking method for constructing high-quality moiré superlattices. Their technique utilizes a gold-template-assisted mechanical exfoliation process. This approach overcomes the difficulties of precisely aligning small, exfoliated 2D crystals and eliminates the need for wet-transfer procedures. It allows for the creation of moiré superlattices from various types of 2D materials, including semiconductors, superconductors, and magnets. The quality of the resulting layers is confirmed using photoluminescence, Raman spectroscopy, and electrical transport measurements. Notably, this method enables the construction of diverse moiré superlattices, including twisted homobilayers with exceptional twisting accuracy (around 0.3 degrees), twisted heterobilayers with sub-degree alignment precision, and even multilayer superlattices exhibiting moiré patterns, interlayer excitons, and second harmonic generation. This advancement offers a powerful tool for scientists to explore and potentially discover novel phenomena within moiré superlattices.

Kancharla and Tadi [79] developed a lead sensor using MoS_2 QDs prepared by a mechanical shear exfoliation method. The simple and cheap method involves creating MoS_2 QDs and placing them on a pencil graphite electrode. The resulting sensor exhibits a linear detection range from 5.66×10^{-9} M to 491.93×10^{-9} M with a low detection limit of 1.96×10^{-9} M. Furthermore, the applicability of the sensor is validated by tests with real-world soil and water samples demonstrate high accuracy, making it promising for environmental monitoring.

Chemical etching, another top-down technique, utilizes chemical reactants to selectively remove material from a semiconductor substrate, allowing for precise control over QD size and shape [80]. Alwehaibi et al. [81] demonstrated the synthesis of MoS_2 QDs with strong UV absorption and a well-defined bandgap through liquid exfoliation, leading to the fabrication of a novel photodetector with broad spectral response and high sensitivity. Gopalakrishnan et al. [77] utilized a hybrid sonication-assisted liquid exfoliation method, which combines bath and probe sonication, to scalable produce of MoS_2 nanostructures with QDs embedded within few-layered MoS_2 sheets. These nanostructures can be efficiently extracted as precipitates using less polar volatile solvents like chloroform and show great promise for hydrogen evolution reaction owing to their excellent electrocatalytic activity, as illustrated in transmission electron microscopy (TEM) images shown in Fig. 5a, b. In this direction, the integration of top-down synthesis with precise chemical treatments offers a powerful toolkit for fabricating QDs with tailored properties. As research in this field

continues to advance, we can anticipate further innovations in exfoliation techniques, unlocking the full potential of QDs for a wide range of emerging applications.

Electrochemical exfoliation, unlike mechanical or chemical methods, employs an electric field within an electrolytic solution to induce the detachment of atomic layers

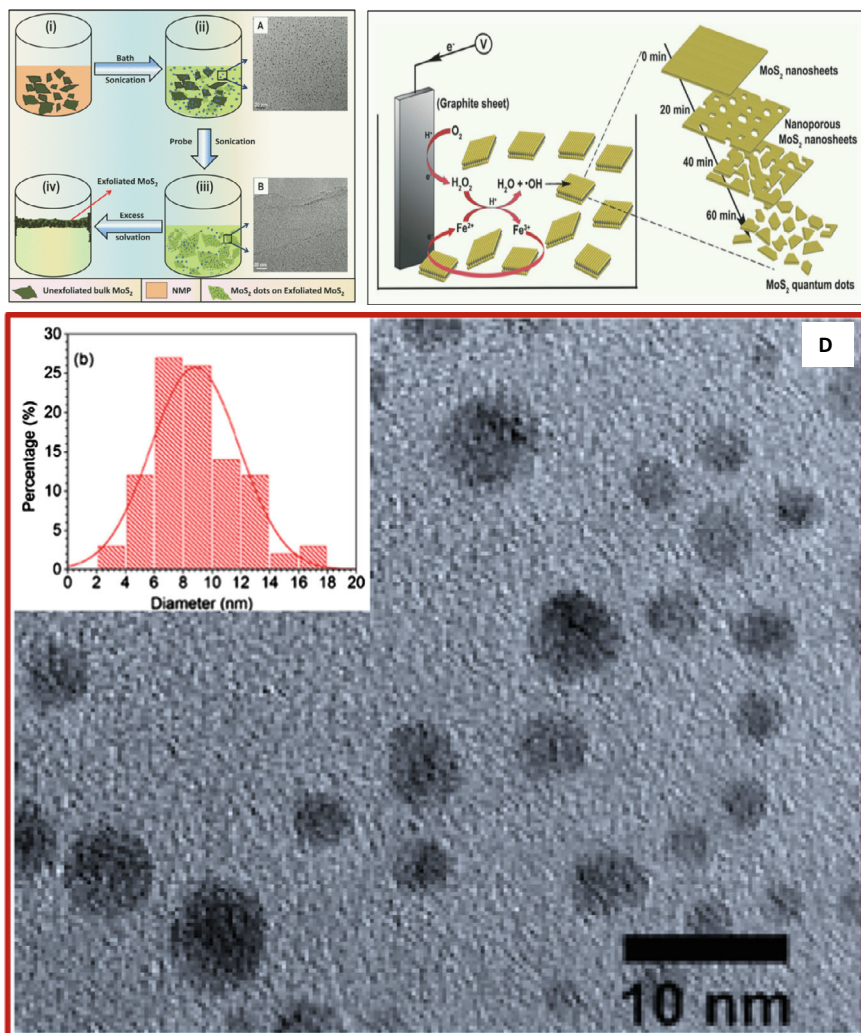


Fig. 5 Synthesis of MoS₂ QDs and carbon dots via exfoliation. **a, b** Schematic and TEM images illustrating the liquid exfoliation process for synthesizing MoS₂ QDs (approximately 2 nm in size) embedded in MoS₂ nanosheets. Adapted with permission from [77]. Copyright 2014 © American Chemical Society. **c** Schematic illustration of the MoS₂ QDs synthesis assisted by the electro-Fenton process. Adapted with permission from [82]. Copyright 2014 © Royal Society of Chemistry. **d** TEM image of carbon dots synthesized via electrochemical exfoliation of graphite, with inset showing particle size distribution. Adapted with permission from [83]. Copyright 2016 © Wiley

from a bulk material [76]. Under an applied potential, intercalation (the insertion of ions between layers) occurs within the material, leading to the weakening of interlayer bonds and subsequent exfoliation. This method is particularly suitable for synthesizing QDs from materials like graphite and TMDCs [84]. For example, Joseph and Anappara [83] successfully produced carbon dots from graphite rods using this strategy (Fig. 4d). In general, the electrochemical exfoliation offers the advantage of precise control over the exfoliation process, enabling the production of QDs with uniform size distribution and tailored properties. Additionally, its compatibility with existing fabrication systems makes it scalable for large-scale production.

Li et al. [82] present a straightforward, cost-effective, and efficient method for synthesizing MoS₂ QDs from MoS₂ nanosheets via a controllable electro-Fenton reaction involving hydroxyl radicals. This approach addresses a gap in MoS₂ research, where QDs have received less attention compared to nanosheets. Notably, the resulting zero-dimensional MoS₂-QDs boast smaller sizes and show strong quantum confinement and edge effects, that is, leading to unique properties not observed in larger nanosheets. The study investigates the interaction between MoS₂ nanosheets and hydroxyl radicals, demonstrating the formation of MoS₂-QDs through controlled etching of the nanosheets (Fig. 5c). By adjusting reaction time to control etching degree and resultant porosity, as evidenced in TEM analyzes. Compared to MoS₂ nanosheets, both MoS₂-QDs and nanoporous MoS₂ nanosheets offer significantly more active sites at the edges, enhancing their potential for various MoS₂-based applications. In short, the combination of top-down synthesis with precise chemical treatments offers a powerful approach for fabricating QDs. As our understanding of QD synthesis and design continues to advance, we can expect further breakthroughs in this field, unlocking the full potential of QDs for a huge range of cutting-edge applications.

2.2 *Bottom-Up Approaches to QD Synthesis*

Bottom-up approaches to QD synthesis offer a fascinating glimpse into the world of material construction on an atomic level. Imagine meticulously assembling builds complex structures from fundamental building blocks, atom/molecule by atom/molecule. Hence, this bottom-up approach offers greater control over the size, shape, and composition of the as-prepared QD materials, i.e., leading to structures with specific desired nanoscale properties [85, 86]. Bottom-up QD synthesis techniques can be adapted to various advanced materials systems, i.e., allowing for the fabrication of QDs from a wide range of semiconductors. However, this synthetic strategy can also be complex and require careful control over various reaction parameters. Additionally, some bottom-up techniques, such as solution-based synthesis, can be scaled up to produce large quantities of QDs [87–89]. As research continues, scientists are developing new and improved bottom-up techniques that are increasingly attractive for QD engineering, making them even more valuable for various QD-based technologies.

2.2.1 Hydro (Solvo-) Thermal Method as Strategy for QD Synthesis

An extremely popular solution-based synthesis strategy for fabrication of semiconductor QDs with specific functionalities is hydrothermal method. This method involves sealing precursor solutions containing metal cations and coordinating ligands in a high-pressure vessel and subjecting them to controlled heating (Fig. 6a) [90]. These conditions can accelerate reaction rates, promote the formation of complex structures, and improve material properties. When using different non-aqueous solvents, the process is often referred to as solvothermal synthesis [91]. Through this hydro (solvo-) thermal process enables precise control over particle size and shape of the QDs by facilitating their nucleation and growth [92].

One key advantage of the hydro (solvo-) thermal method is its ability to produce high-quality semiconductor QDs at comparatively lower temperatures than other techniques [91]. For example, CdS QDs synthesized via hydrothermal methods have exhibited remarkable size uniformity and optical properties suitable for various optoelectronic applications [92]. Researchers developed a facile hydrothermal method to create carbon dots that emit a rainbow of colors (blue to red) under a single source of ultraviolet light (Fig. 6b) [94]. These carbon dots boast tunable photoluminescence and high efficiencies (up to 35%). Further purification yielded samples with exceptional uniformity, each emitting a single color consistently. Interestingly, despite similar size and core structure, variations in surface chemistry, particularly oxidation levels, led to the observed color shift (440–625 nm). Unlike traditional quantum dots where size dictates color, this research suggests a unique mechanism based on surface properties. With excellent luminescence and low toxicity, these carbon dots

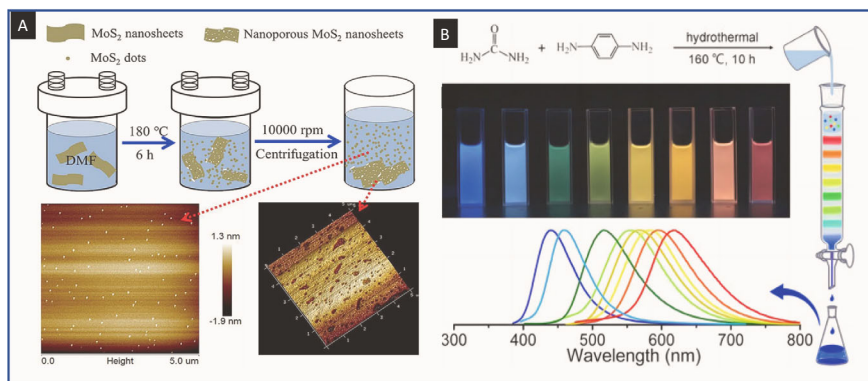


Fig. 6 Hydro (solvo) thermal synthesis of QDs. **a** Schematic illustration and atomic force microscopy (AFM) images of MoS₂ QDs synthesized using a hydro (solvo) thermal cutting integrated method. Adapted with permission from [93]. Copyright 2018 © Elsevier. **b** Hydrothermal synthesis of different types of carbon dots that glow under ultraviolet. These QDs were prepared from a 1:1 molar ratio of urea and p-phenylenediamine and subsequently separated into fractions with distinct optical properties using silica column chromatography through changing the solvent polarity. Adapted with permission from [94]. Copyright 2016 © American Chemical Society

hold promise for multicolor cellular imaging using a single light source. Notably, the red-emitting carbon dots, due to their deeper tissue penetration, could be particularly valuable for live-animal imaging in mice [94].

Also, solvothermal synthesis offers a distinct advantage in achieving tunable optical properties and high quantum yields for indium phosphide (InP) QDs, i.e., making them promising candidates for photovoltaic and light-emitting diode (LED) applications [95].

Exploring the use of alternative solvents in the hydro(solve-)thermal approaches, e.g., such as ionic liquids (ILs) and supercritical fluids, represents the current strategy used to refine this bottom-up QD synthesis [87, 96–99]. Hence, these alternative solvents offer advantages over traditional aqueous media, allowing for the production of high-performance QDs with enhanced physicochemical properties. For instance, Chen et al. [96] reported synthesizing water-soluble Ag-In-Zn-S (AIZS) QDs by means of the ILs-assisted hydrothermal approach and their application in white light-emitting diodes (LEDs). This white LEDs combined AIZS with green $\text{Lu}_3\text{Al}_5\text{O}_{12}:\text{Ce}^{3+}$ phosphors and a blue InGaN chip. The resulting device exhibited an excellent luminous efficacy of about 85.2 lm/W, highlighting the promising performance of AIZS QDs for this application.

Sun et al. [100] reported a novel solvothermal method using soft templates to synthesize ultrathin $\gamma\text{-Ga}_2\text{O}_3$ for far-UVC photodetectors. This synthesized $\gamma\text{-Ga}_2\text{O}_3$ exhibit strong absorption peak in the far-UVC range (190–230 nm) with a sharp absorption edge below 250 nm. The absorption coefficients of the $\gamma\text{-Ga}_2\text{O}_3$ increase significantly with decreasing diameters, both experimentally and theoretically confirmed. Self-powered photoelectrochemical-type photodetectors based on 1.7 nm diameter Ga_2O_3 demonstrated exceptional far-UVC detection performance.

Beyond conventional heating, hydro (solvo) thermal reactions can be further enhanced by incorporating other technologies like electrochemistry, microwave irradiation, and sonochemistry [91, 101–112]. These combined approaches leverage the synergistic effects of each technique, potentially unlocking new reaction pathways with improved efficiency and faster reaction times [91].

2.2.2 Microemulsion Method as Strategy for QD Synthesis

Between the wet-chemical methods are generally used, microemulsion, in particular, has been a popular method due to its controllability and effectiveness [113]. However, the impressive capabilities of the microemulsion method have a surprisingly long history. The pioneering work of Schulman and Hoar in 1943 laid the foundation for this technique, demonstrating its potential for various applications [114]. From this perspective, the microemulsion offers versatility in the bottom-up synthesis of semiconductor QDs with precise control over size and morphology [115]. In this technique, nanoscale droplets of water or oil dispersed in a surfactant solution act as confined reaction vessels for QDs formation [116] (Fig. 7). By adjusting the composition of the microemulsion system and reaction conditions, researchers can achieve precise control over QDs size and shape [116]. The microemulsion environment

provides a confined space that controls the growth and prevents aggregation of the QD nuclei, resulting in monodisperse nanoparticles [117, 118]. ZnSe QDs synthesized via microemulsion methods, for instance, exhibit narrow size distribution and excellent colloidal stability, essential for applications in infrared photodetectors and solar cells [119]. He et al. [120] recently reported a one-pot microemulsion synthesis of FeOOH QDs for the effective removal of copper cations from aqueous solutions. This research highlights the potential of microemulsion-synthesized QDs for environmental remediation technologies.

Microemulsion, particularly the water-in-oil (W/O) type, provides an ideal environment for nanoparticle preparation [121]. This process creates reverse micelles. These nano-sized structures consist of water droplets are encapsulated by a layer of surfactant and cosurfactant molecules, remaining stably dispersed within the oil phase [121]. It is well-known that the reverse micelle method offers the potential to synthesize QDs with shapes that mirror their initial nuclei. Hence, this approach is particularly advantageous for creating QDs with specific functionalities that are dependent on their shape. Additionally, it is well-known that the vigorous stirring of the micelle suspension can enhance the rate of the reaction by promoting collisions

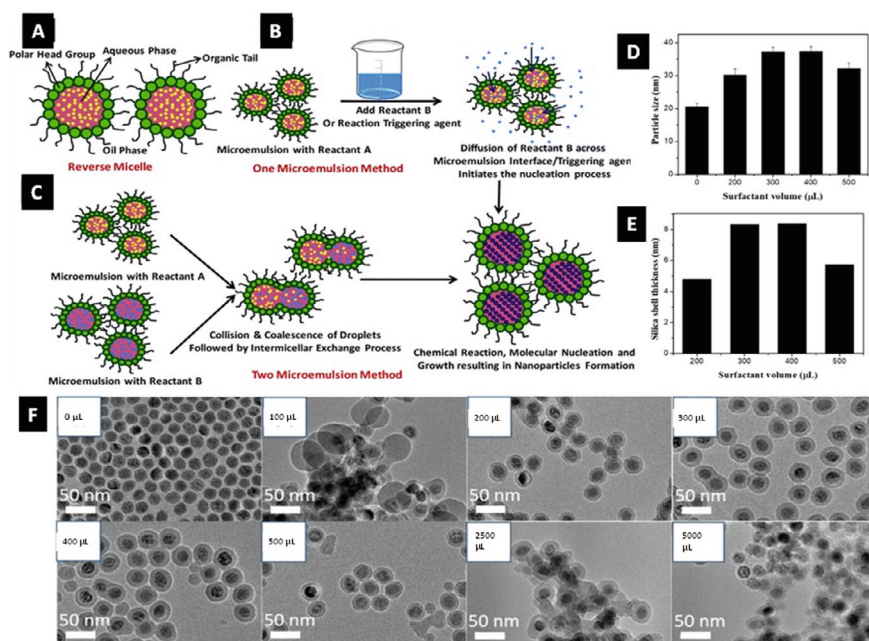


Fig. 7 Reverse microemulsion method for QD synthesis. **a** Formation of a reverse micellar system. **b** Single microemulsion method. **c** Dual microemulsion method. Adapted with permission from [127]. Copyright 2015 © Royal Society of Chemistry. Effect of surfactant volume on **d** particle size and **e** silica shell thickness of Fe₃O₄@SiO₂, as determined by TEM analysis (shown in panel **f**). Adapted with permission from [128] Copyright 2020 © MDPI

between micelles, facilitating the exchange of reactants within the confined environment [122–124]. The technique is also recognized for its soft chemistry approach, requiring no extreme pressure or temperature control, and being easy to handle without the need for special or expensive equipment [125]. Hence, microemulsion processes are a popular method for synthesizing QDs at room temperature [126].

One challenge associated with the reverse micelle method is achieving high fluorescent yields in the resulting QDs. This can be attributed to factors like surface defects and incomplete passivation of the QD surface. Additionally, the presence of impurities and defects within the QDs themselves can arise during synthesis [124, 129]. Despite these limitations, researchers are continuously exploring strategies to overcome them and improve the performance of reverse micelle-synthesized QDs. This includes developing novel purification techniques to minimize impurities and exploring new surface passivation methods to enhance fluorescent yields. As research progresses, we can expect the reverse micelle approach to become even more effective for producing high-quality QDs with tailored properties.

For example, Pham et al. [128] studied how changes in the reverse microemulsion process affect the size and thickness of the silica shell when coating Fe_3O_4 nanoparticles (Fig. 7d–f). Their goal was to create $\text{Fe}_3\text{O}_4@\text{SiO}_2$ particles with strong magnetic properties. While Mehra et al. [130] report the creation of microemulsions using a new biobased IL surfactant, lutidinium dioctyl sulfosuccinate ([Lut] [AOT]), with [BMIM][PF6] as the nonpolar phase and ethanolammonium formate (HO-EAF) or ethylammonium formate (EAF) as polar components. Hence, the high emulsifying capacity of [Lut] [AOT] eliminated the need for a cosurfactant. HO-EAF induced an electrolytic effect at the micelle interface, while EAF resulted in swollen micelles that penetrated the microemulsion droplets' core. These droplets were used as nanoreactors to synthesize spherical core–shell $\text{PbSAgS}@\text{COOH}$ nanoparticles (NPs) at room temperature, achieving precise size and morphology control. EAF provided a larger monophasic region and produced higher quality NPs than HO-EAF. The synthesized NPs had excellent optical properties, high quantum yield, and effectively degraded polluting dyes in aqueous solutions, paving the way for further nanomaterial synthesis research in IL-confined environments.

Tarkas et al. [131] employed a surfactant-free microemulsion approach to synthesize PbS quantum dots, ranging in size from 13.17 to 26.91 nm. They observed a decrease in the band gap with increasing microemulsion concentration and particle size, attributable to quantum confinement effects. The optimal concentration for producing uniformly distributed, small PbS nanoparticles was found to be 0.0125 M. Stabilization and growth of spherical PbS in the microemulsion were attributed to van der Waals forces. Utilizing this optimal concentration, they constructed a highly stable photodetector (FTO/ TiO_2 /PbS/PANI/NiS/C) with a rise time of approximately 0.39 s, a decay time of approximately 0.22 s, a photoresponsivity of approximately $5.466 \mu\text{A W}^{-1}$, PLQY of approximately $0.116 \times 10^{-4}\%$, and a detectivity of 6.83×10^7 Jones. This method offers a simple, cost-effective approach for creating efficient and stable photodetectors.

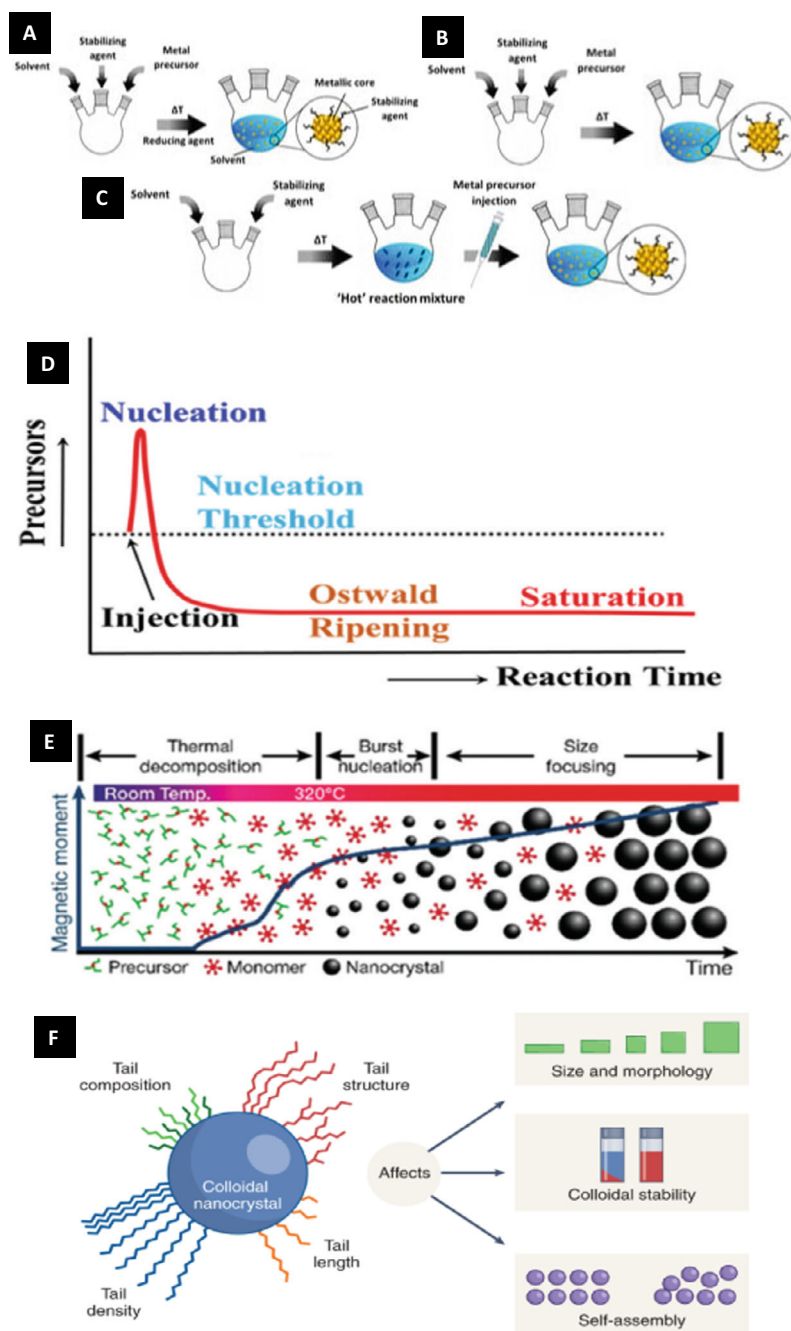
2.2.3 Colloidal Synthesis as Strategy for QD Synthesis

The pioneering work of Murray, Norris, and Bawendi in 1993 [6] revolutionized the field of QD synthesis. They demonstrated the crucial role of organic solvents in controlling the nucleation and growth process of particles, leading to the now widely recognized strategy of colloidal synthesis. Colloidal synthesis encompasses a range of methods, each with distinct advantages for tailoring QD properties. Chemical reduction, a prevalent technique, involves dissolving metal precursors in a solvent with a protective agent, followed by the introduction of a reducing agent at elevated temperatures to trigger nanoparticle formation (Fig. 8a) [132]. Thermal decomposition offers two alternatives: one that mirrors chemical reduction but relies solely on high temperatures to drive the process (Fig. 8b), and another, known as hot injection, where a preheated solvent-protective agent mixture is rapidly injected with a metal precursor solution (Fig. 8c) [132]. This hot injection method, based on the LaMer model, often results in highly monodispersed QDs [132]. Hence, it is well-known that the choice of precursors, solvents, and stabilizing ligands during nucleation significantly impacts the final size and uniformity of the QDs [109–111, 124]. Precise control over reaction parameters, such as temperature, precursor concentration, and time, allows researchers to fine-tune QD size distribution and properties [108, 112].

Colloidal synthesis has not only revolutionized the production of monodispersed QDs but has also paved the way for large-scale production, enabling advancements in optoelectronics, photoelectrocatalysis, and biomedicine [136]. It enables the synthesis of a diverse range of QDs with controlled size and shape, encompassing materials from semiconductors of group II–VI [138–141], III–V [138, 142–145], IV–VI [146, 147], and even group IV elements [148, 149]. The fine-tuned control over growth kinetics and the compatibility of solution-based deposition techniques with various substrates make it ideal for manufacturing large-area devices and integrating QDs into diverse platforms [150]. Additionally, solution-based deposition techniques employed in colloidal synthesis are compatible with high-throughput manufacturing and a wide variety of substrates [9, 151]. This compatibility allows for seamless integration of QDs with various platforms, including silicon electronics, plastic circuits, fiber optics, and even fabrics, highlights the vast potential applications of QDs across diverse fields [9, 152].

Colloidal synthesis offers a versatile approach for precisely controlling the growth of QDs, which depending on reaction conditions following different mechanisms [153] such as Ostwald ripening [154] or LaMer burst nucleation (rapid, simultaneous growth of all QDs) [155]. Surface ligands, typically organic molecules, play a critical role in both stabilizing the QDs (ligands prevent QD agglomeration by introducing steric repulsion between them in solution) and influencing their properties (including solubility, stability, and compatibility with diverse environments) [156, 157].

The surface ligands found in QDs, typically represented by voluminous organic molecules like oleic acid and oleylamine, play a crucial role in establishing repulsive forces among dispersed QDs within a solvent. This phenomenon contributes to their colloidal stability [156, 157]. When QDs are deposited onto solid substrates,



◀**Fig. 8** Synthesis protocols for colloidal QDs. **a** Chemical reduction, **b** thermal decomposition (precursor pre-added), and **c** thermal decomposition (hot injection). Adapted with permission from [132] Copyright 2020 © MDPI. **d** Diagram of the hot injection method based on the La Mer model, illustrating the growth of monodispersed colloidal QDs starting from nucleation. **e** Non-injection heating method where all precursors are combined at room temperature and then heated to initiate nucleation and growth. Adapted with permission from [160]. Copyright 2019 © Royal Society of Chemistry. **f** Schematic illustration demonstrating the impact of surface ligands on the synthesis of quantum dots. Adapted with permission from [161]. Copyright 2022 © Springer Nature

the resulting quantum dot films can exhibit either a glassy or partially ordered structure [158]. This outcome is influenced by factors such as the monodispersity of nanocrystals, the kinetics of solvent drying, and the interplay among surface ligands [159].

Following initial synthesis, various post-synthesis modifications can further enhance QD functionality for specific applications [87]. These modifications include:

- **Ligand Exchange:** Replacing the original ligands with new ones can introduce desired functionalities like improved solubility, stability, or compatibility with different materials [162] This is crucial for adapting QDs to specific applications, such as bioimaging or device integration.
- **Surface Functionalization:** Attaching specific molecules to the QD surface allows for precise tailoring of properties like charge, hydrophilicity, or bioconjugation potential [163].
- **Shell Growth:** Adding an inorganic/organic shell around the QD core can enhance stability, increase quantum yield, and modify the absorption/emission properties [164].

Colloidal synthesis has evolved from a proof-of-concept technique to an essential tool in the fabrication of a wide range of nanomaterials [87]. Its continuous refinement has led to innovative applications, particularly in the realms of photovoltaics, LEDs, and biomedicine [165].

For instance, Ag-In-S/ZnS QDs synthesized via thermal decomposition demonstrate tunable bandgaps and high photovoltaic efficiencies, showcasing the scalability of this method for QD-based solar cells [137]. Kayal and De [166] successfully prepared MoS₂ QDs using thermal decomposition in octadecene, with ammonium tetrathiomolybdate as the precursor. These MoS₂ QDs demonstrated exceptional efficiency as catalysts for the synthesis of α -amino phosphonates. This research highlights the potential of thermally decomposed QDs as catalysts for sustainable chemical transformations.

Despite its successes, colloidal synthesis, especially thermal decomposition, faces challenges such as high energy consumption and poor water dispersibility of the resulting QDs due to hydrophobic ligands [167]. However, ongoing research into novel precursors, precise reaction control, and integration with other techniques holds promise for further advancements in colloidal synthesis, leading to the development of next-generation QDs with enhanced properties and a wider range of applications.

Hence, the evolution of colloidal synthesis, coupled with post-synthesis modifications, has not only solidified its role as a cornerstone of nanomaterial fabrication but also opened up a myriad of possibilities for the future of QD technology. As research progresses, we can anticipate even more sophisticated and tailored QDs, driving innovation in fields as diverse as energy, healthcare, and electronics [128–159].

2.2.4 Molecular Beam Epitaxy and Metalorganic Chemical Vapor Deposition Methods as a Strategy for QD Synthesis

Advanced thin-film deposition techniques such as molecular beam epitaxy (MBE) and metalorganic chemical vapor deposition (MOCVD) enable the realization of high-quality epitaxial QDs [168]. These methods offer unparalleled control over the deposition of QDs onto crystalline substrates, as documented in various studies [169–171]. Notably, the epitaxial growth facilitated by MBE and MOCVD ensures the precise alignment of crystal lattices between the deposited QDs and the substrate. This lattice matching is a critical factor contributing to the enhanced optical properties, uniformity, and overall quality of the QDs [171]. The primary distinction between MBE and MOCVD is in their film thickness/composition precision. While MOCVD boasts significantly faster growth rates, making it ideal for large-scale QD production, its control over film thickness/composition is less precise compared to MBE [172–174]. Hence, this can potentially lead to compositional variations within the QDs. Researchers are continuously exploring ways to improve both MBE and MOCVD for QD growth, enabling the creation of even more sophisticated and functional QDs for next-generation technologies.

Holmes et al. [175] devised a method for fabricating precisely positioned QDs on nanowires using MOCVD (Fig. 9). In this fabrication process, sapphire substrates were initially coated with a 25 nm low-temperature aluminum nitride (AlN) layer followed by a subsequent deposition of a 25 nm silicon dioxide (SiO₂) layer. To define the growth locations for the gallium nitride (GaN) nanowires, precisely positioned arrays of apertures (each with a diameter of 25 nm) were created using electron beam lithography and reactive ion etching techniques. Notably, the GaN nanowires were grown under specific conditions, including a pressure of 76 Torr, and significantly reduced V/III ratios compared to standard GaN thin-film growth. Subsequently, a layer of aluminum gallium nitride (Al_xGa_{1-x}N) with a nominal aluminum mole fraction (x) of 0.8 was deposited around the GaN nanowires [175]. This deposition resulted in the formation of a core-shell structure. Finally, GaN QDs were selectively grown on the nanowire tips during a brief 8-s GaN growth step. This meticulously controlled fabrication process grants precise three-dimensional control over the placement of each QDs. The precise lateral (XY) positioning of the materials is dictated by the underlying GaN nanowire, which is itself determined by the patterned apertures in the SiO₂ layer. The vertical (Z) position is governed by the height of the GaN nanowires, which were approximately 700 nm in this study [175].

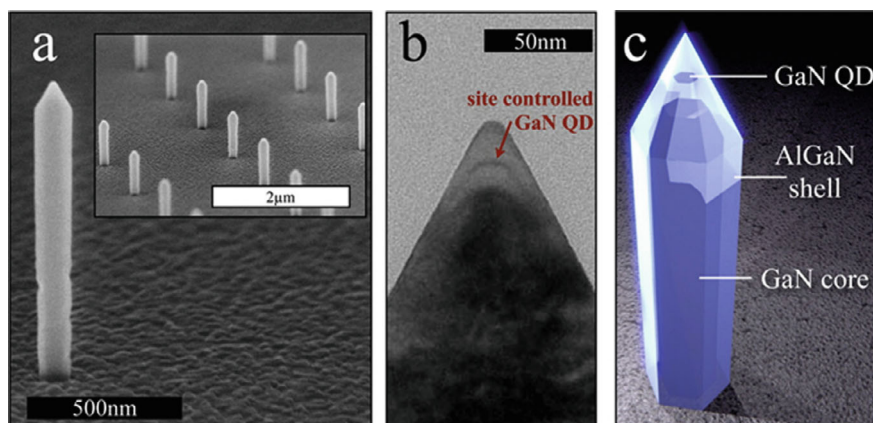


Fig. 9 Site-Controlled Nanowire QDs using selective area MOCVD. **a** SEM image showcasing a single nanowire precisely grown on a patterned SiO₂ substrate. The inset displays an array of nanowires with a center-to-center spacing of 2 μm. **b** TEM image reveals the formation of a single QDs located near the tip of a single nanowire. **c** Schematic representation of a nanowire containing a single QDs. (Adapted from Ref. [175] Copyright 2014 American Chemical Society)

2.2.5 Chemical Vapor Deposition

Chemical Vapor Deposition (CVD) is a well-established technique known for its precise control over the size and morphology of synthesized QDs [9]. In this process, volatile precursors are transported to a heated substrate, where they react or decompose on a substrate, leading to the formation of the desired QDs. The CVD process typically involves the vaporization of precursor materials, often in form of powders, within a high-temperature zone [176, 177]. These vaporized precursors are then transported downstream and deposited onto a substrate at a specific temperature zone, leading to the formation of QDs with desired characteristics. A typical CVD setup (Fig. 10) consists of a horizontal tube furnace with an alumina or quartz tube positioned within [178]. The source material, usually in powder form, is placed in the center of this tube. Key parameters like temperature, pressure, and evaporation time are meticulously controlled. Before initiating the process, the reaction chamber is evacuated to a low pressure using a mechanical rotary pump. Once the desired reaction temperature is reached, the source material is heated, causing it to evaporate. An inert carrier gas, such as argon, then transports the vaporized precursors downstream towards the cooler region of the tube. Finally, these precursors deposit onto the growth substrate or the inner wall of the alumina/quartz tube located at the downstream end [178, 179].

Vapor–Liquid–Solid (VLS) growth mechanism is a well-established method for the synthesis of various nanostructures, particularly nanowires [180]. Typically, a metal nanoparticle serves as the catalyst that guides anisotropic growth. During VLS growth, the gas-phase reactants interact with the nanoparticles at high temperatures, forming a supersaturated melt. Subsequently, the precipitation of reactants leads

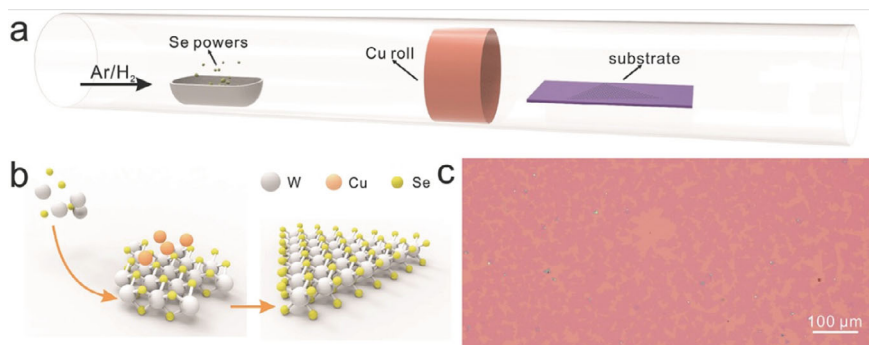


Fig. 10 **a** Schematic illustration of the chemical vapor deposition process used to obtain WSe₂ single crystals. **b** Diagram showing the self-limited process used to produce single crystals of WSe₂. **c** Low magnification optical image of the strictly monolayer WSe₂ crystals prepared using this method. Adapted with permission from [182]. Copyright 2016 © Wiley

to the nucleation and axial elongation (growth along a specific axis) of crystalline nanowires [181]. Notably, the diameter of the resulting nanowires closely resembles the size of the initial catalyst droplet, as it dictates the growth direction.

The CVD method offers several advantages over other synthesis methods. It is capable of producing high-quality QDs with uniform size and shape [179, 183]. Despite these advantages, there are also challenges associated with the CVD method. The process requires high temperatures and precise control over the reaction conditions. Moreover, the precursors used in the process can be toxic or hazardous, posing safety and environmental concerns.

Zhao et al. [184] report a novel method for synthesizing Bi₁₃S₁₈I₂ thin film using mist CVD. This bismuth-based ternary compound, known for its low toxicity and tunable electrical and optical properties, was used to create a photoelectrochemical-type photodetector. The detector was constructed based on the Bi₁₃S₁₈I₂ thin film synthesized under optimal temperature. The photodetection performance was evaluated with considerations on electrolyte composition, working temperature, and bias voltage. This research provides insights into the controllable synthesis and potential applications of Bi₁₃S₁₈I₂ materials.

2.2.6 Coprecipitation

Among various QD synthesis methods, coprecipitation has emerged as a popular technique owing to its simplicity and accessibility [185]. This method involves the controlled precipitation of a solid from a solution under conditions that promote the formation of QDs with desired properties [186]. The process is typically carried out at room temperature and does not require any specialized equipment, making it a budget-friendly approach for QD production [187]. By adjusting factors such as temperature, pH, and the selection of an appropriate solvent and precipitating

agent, researchers can tailor the properties of the synthesized QDs, making it suitable for targeted material development [188]. Despite its advantages, coprecipitation also presents certain challenges, such as controlling the size distribution of the resulting QDs. This can lead to a broad size distribution, which can affect the physical properties of the QDs [185].

For instance, Chang et al. [189] developed a rapid and efficient co-precipitation method to synthesize composites of aqueous CdTe quantum dots (QDs) embedded in inorganic salts like NaCl for white LED applications. The tight salt matrix offers protection, resulting in composites with enhanced stability, processability, and high photoluminescence properties (Fig. 11a–d). Zhang et al. [190] developed a novel method for synthesizing stable ternary $\text{Pb}_{1-x}\text{Cd}_x\text{S}$ QDs through in situ coprecipitation of $\text{Pb}_{4-n}\text{Cd}_n\text{O}_4$ bimetallic clusters in an aqueous solution. These QDs exhibited a uniform size of approximately 4.0 nm and demonstrated second near-infrared (NIR-II) emission at 1100 nm with a remarkable PLQY of about 17.72%. The QDs demonstrated excellent stability, retaining 90.9% of their initial PLQY after 15 days when stored at 4 °C in colloidal form, and showing no spectral changes after 5 months when stored as powder. Due to their high PLQY and good water compatibility, these $\text{Pb}_{1-x}\text{Cd}_x\text{S}$ QDs offer promising potential for in vivo vasculature and lymphatic system imaging in the NIR-II window, even at low power densities (10 mW cm^{-2}).

Guan et al. [191] developed a ZnO/Bi-QDs/BiOBr photocatalyst with a lamellar porous structure designed to enhance dye adsorption and control reactive groups for photodegradation and H_2 production. Notably, the photocatalyst demonstrated rapid degradation of rhodamine B (RhB) dye within 8 min under visible light, further reduced to 2 min at pH 5. Additionally, complete degradation was observed within 120 min under natural sunlight. This study offers a promising strategy and design concept for photocatalysts capable of simultaneous photocatalytic H_2 production and organic dye photodegradation, utilizing a coprecipitation synthesis method.

Mohanapriya et al. [192] explored the synthesis of cobalt-doped CdS QDs through a straightforward co-precipitation process. Their research revealed that increasing the cobalt content from 0 to 10% resulted in a reduction of crystallite size from 5.9 to 2.9 nm, as determined by X-ray diffraction analysis. Furthermore, Tauc plot analysis demonstrated that Co-doping led to an increase in the bandgap of the CdS quantum dots from 3.12 to 3.45 eV. Additionally, they observed that the photoluminescence peak intensity reached a maximum at 10% cobalt doping. Fourier transform infrared and Raman spectroscopy confirmed Co substitution and CdS formation. TEM confirmed nearly spherical particles with sizes of 7.2–3.9 nm, suitable for various applications.

2.2.7 Laser Ablation

Nanoparticle fabrication encompasses a diverse toolbox of techniques, including bottom-up approaches in both liquid and vapor phases. Liquid-phase bottom-up methods, such as sol–gel and chemical reduction, offer cost-effective synthesis

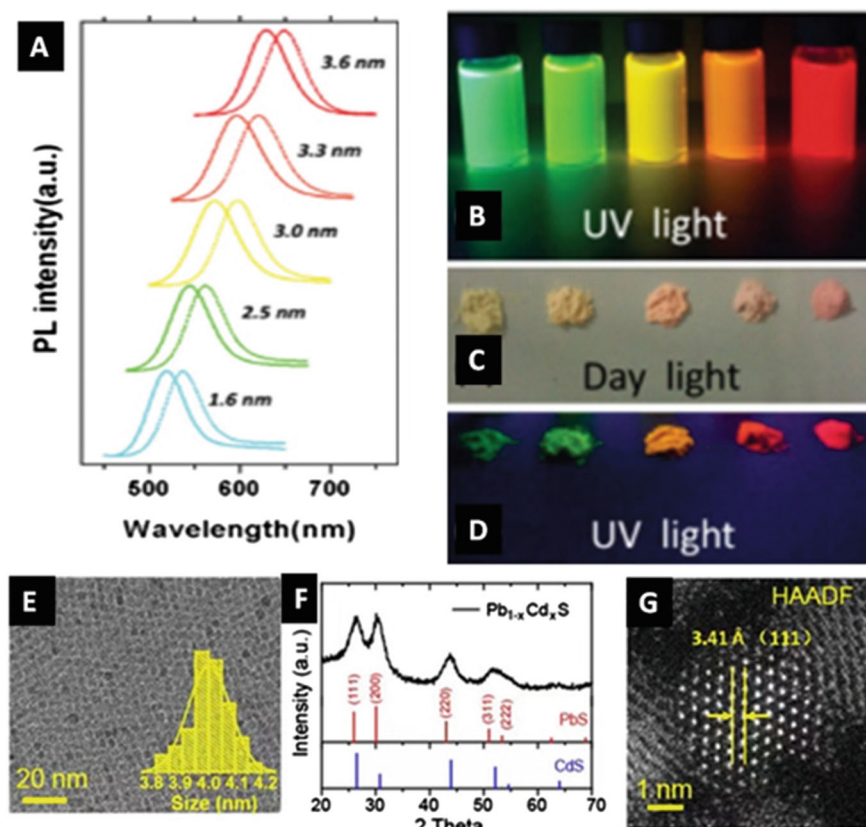


Fig. 11 **a** PL spectra at room temperature of CdTe QDs prepared by coprecipitation method. **b** Photographs of CdTe QDs dispersed in water under UV light, **c** CdTe@NaCl powders under daylight, and CdTe@NaCl powders under UV light (365 nm), revealing their varying luminescence properties. Adapted with permission from [190]. Copyright 2022 © Wiley **e** TEM image with inset of particle size distribution histograms, **f** XRD patterns and **g** HAADF-STEM image of ternary Pb_{1-x}Cd_xS QDs. Adapted with permission from [189]. Copyright 2015 © Royal Society of Chemistry

with precise control over nanoparticle structures at the laboratory scale. Meanwhile, vapor-phase processes, such as physical/chemical vapor deposition and flame synthesis, excel in producing high-purity nanoparticles through continuous flow reactors. In both bottom-up approaches, solid nanoparticles form via the nucleation of supersaturated species from precursor reactions or solid evaporation.

Laser ablation, employing laser energy for solid target material ablation, yields high-purity nanoparticles unaffected by reactor contamination (Fig. 12a) [193]. However, conventional laser ablation poses challenges in size distribution, agglomeration, and crystal structure control due to random molecular motion. Advanced

laser ablation techniques have emerged to address these limitations, enabling the fabrication of morphology-controlled nanoparticles.

Laser Ablation Synthesis in Solution (LASIS) stands out as a promising method for producing ultrapure and less toxic nanoparticles, offering an environmentally friendly alternative to conventional chemical synthesis approaches [196]. Unlike traditional methods that utilize potentially hazardous chemicals, LASIS generates high-purity nanoparticle colloids without the need for precursors or reducing agents,

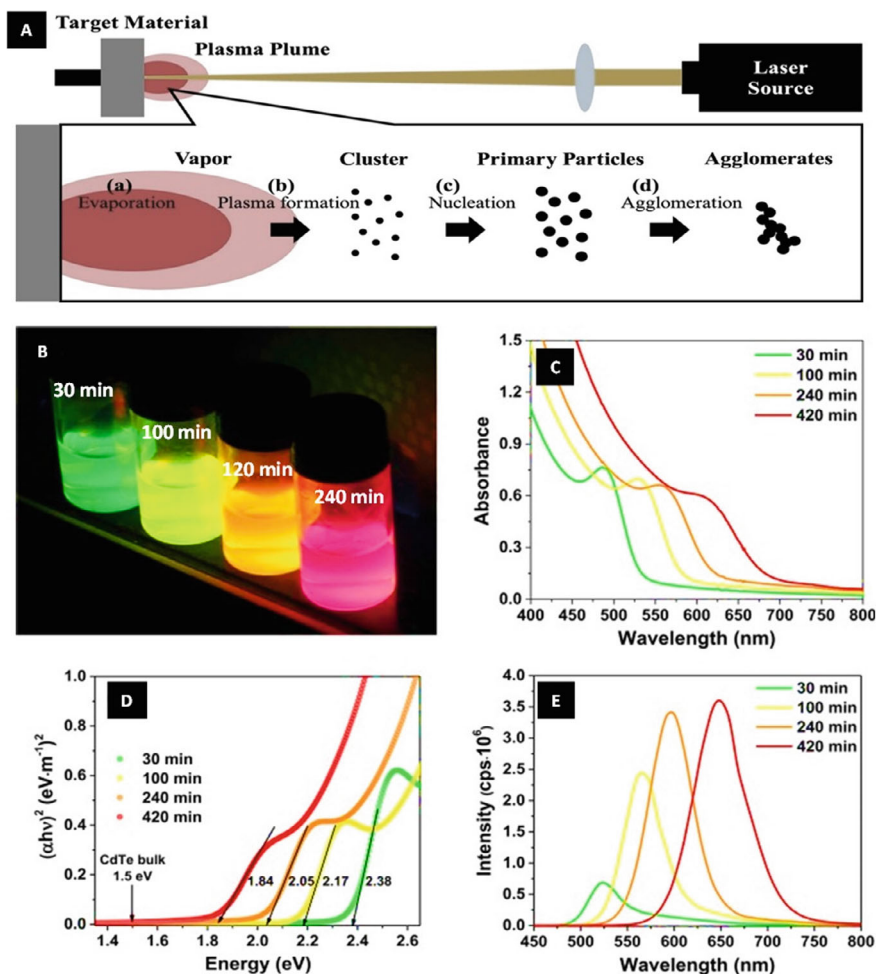


Fig. 12 a Diagram illustrating the particle formation process during laser ablation. Adapted with permission from [194]. Copyright 2017 © Hosokawa Powder Technology Foundation. **b–e** Optical properties of CdTe-MPA prepared by laser ablation method. Adapted with permission from [195]. Copyright 2022 © Elsevier

minimizing environmental impact and waste generation. Additionally, LASIS facilitates the synthesis of various nanoparticle types in different solvents, utilizing established laser-based systems for setup. The applicability of LASIS extends beyond conventional nanoparticle production. This versatile technique can also be employed for the synthesis of semiconductor QDs. By utilizing LASIS, researchers can achieve precise control over the size, composition, and morphology of QDs. This enables the tailoring of their properties for specific applications such as LEDs, solar cells, and biological imaging. LASIS offers a versatile and environmentally friendly route for QD fabrication, contributing significantly to the advancement of semiconductor nanotechnology and its diverse applications [196].

A novel, environmentally friendly approach for synthesizing CdTe QDs was demonstrated by Silva et al. [195], utilizing a bottom-up method assisted by reactive laser ablation in aqueous media, achieving a maximum PLQY of about 28% and tunable luminescence ranging from green to red (Fig. 12). Sahoo et al. [197] synthesized MoS₂ QDs utilizing the femtosecond pulsed laser ablation in liquid (fs-PLAL) technique, allowing control over optical, structural, and electronic properties. By adjusting parameters such as pulse width, ablation power, and ablation time, this approach offers the potential to tailor MoS₂ QDs for diverse optoelectronic applications. Hassan et al. [198] prepared PbS QD for the first time using a Nd-Yag laser with a λ of 1064 nm and varying energy levels (5, 6.5, 8 J cm⁻²) with a fixed number of pulses (600 pulses). Benzene was employed as the liquid medium for ablation to prevent QDs from clumping. XRD patterns revealed a cubic PbS nanocrystalline phase, while the direct optical energy gap was found to range from 2 to 2.5 eV. PL emission spectra showed a peak around 400–470 nm, indicating the crystallite sizes ranged from 4.5 to 7 nm. In a related study, Buerkle et al. [199] investigates the impact of quantum confinement and surface orientations on the electronic properties of NiO QDs. Their research compared NiO nanocrystals synthesized using atmospheric-pressure microplasma and Their research compared NiO nanocrystals synthesized using atmospheric-pressure microplasma and fs-PLAL. Both methods yielded quantum-confined nanocrystals with a face-centered cubic lattice. While the electronic properties of microplasma-synthesized NiO nanocrystals aligned with established literature, those produced by fs-laser ablation displayed unique electronic characteristics, notably a high-lying lowest unoccupied molecular orbital (LUMO) level. Density functional theory calculations corroborated that these level positions correlate with different polar and nonpolar facets of the nanocrystal surface.

3 Physical Properties of the QDs for Applications in Optoelectronic Technologies

QDs exhibit outstanding physicochemical properties owing to known quantum confinement effects [200]. Particularly, this quantum confinement allows for precise tuning of their bandgap energies by adjusting their size, enabling controlled light

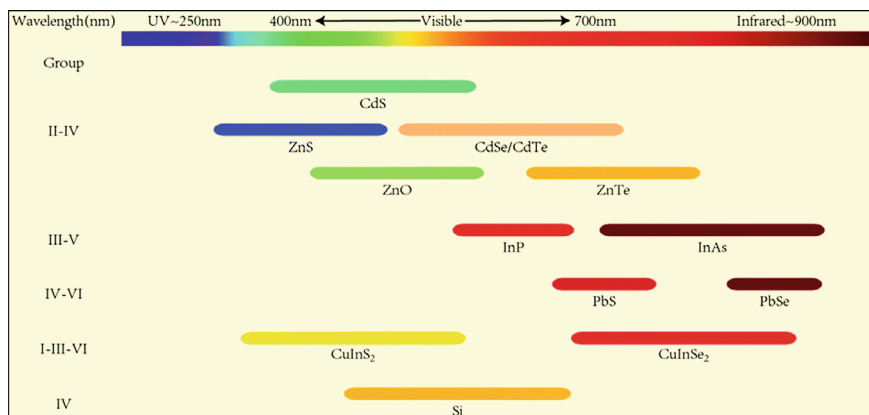


Fig. 13 Emission spectral range of extensively studied semiconductor nanocrystals, demonstrating the tunability of emission wavelengths across the visible and near-infrared spectrum by varying material composition. (Adapted from Ref. [115] © Royal Society of Chemistry)

emission across the entire visible spectrum and into the infrared and ultraviolet regions [201]. This tunability, along with other unique characteristics like high quantum yield and photostability, makes QDs highly versatile for a wide range of applications in optoelectronics, photonics, biotechnology, and quantum computing [202]. Semiconductor QDs, typically composed of metal and nonmetal elements, are categorized into groups such as II–VI, III–V, IV–VI, I–III–VI and IV [191–231]. As illustrated in Fig. 13, different types of QDs display adjustable photoluminescence and emission spectra. Owing to their quantum confinement effects, QDs exhibit size-dependent properties, and hence, the color of emitted light by QDs can be controlled simply by adjusting their size [148].

Beyond their optical properties, the surface chemistry of QDs plays a crucial role in their behavior and functionality. By modifying the QD surface with various ligands, polymers, or inorganic coatings, researchers can tailor their properties to suit specific applications in fields like bioimaging, drug delivery, and environmental sensing [203]. QDs have also shown promise in quantum computing as potential qubits, the fundamental units of quantum information processing [204]. Researchers are actively exploring the manipulation of QD spin and charge states to develop scalable and reliable qubit architectures for future quantum computers [205].

Advancements in colloidal synthesis, epitaxial growth, and nanostructuring techniques have enabled the fabrication of QDs with complex morphologies, such as core–shell, alloyed, and hierarchical structures [206, 207]. These developments open up new avenues for exploring fundamental physical properties and engineering novel device functionalities [204–209].

In addition to their individual properties, the assembly of QDs into semiconducting solids offers further opportunities for creating materials with emergent electronic properties distinct from those of individual QDs [208]. These QD solids have been successfully integrated into next-generation optoelectronic technologies, including

transistors and highly sensitive photodetectors capable of detecting light across a wide spectral range [208].

Understanding size quantization in colloidal QDs is crucial for tailoring their optoelectronic properties. Theoretical studies have yielded analytical relationships and sizing functions to predict bandgap based on QD size [209]. However, these methods have limitations. They often rely on simplified models with redundant parameters that can lead to inaccurate predictions, especially when extrapolating beyond the range of data used to develop the model [210]. Aubert et al. [211] proposed new sizing function that incorporates a correction for non-parabolic energy bands (Fig. 14). This approach offers more accurate predictions of size quantization for various semiconductor materials, including groups IV, III–V, II–VI, and IV–VI, as well as metal-halide perovskite semiconductors. It can be readily adapted for negative-gap semiconductors and non-spherical QDs. By refining the model with experimental data and introducing the Bohr diameter as a fitting parameter, the approach reveals significant variations in the band gap-to-size relationship for different crystal structures of CdSe, highlighting its ability to capture these material-specific nuances.

Thermal effects can also influence the bandgap of QDs. As temperature increases, in particular, the material expands slightly owing to thermal expansion, resulting in decrease of bandgap [212, 213]. Furthermore, changes in the QD volume due to strain can affect the bandgap in a similar manner to applying hydrostatic pressure [214].

One defining characteristics that make QDs highly attractive for many technological applications is, of course, their ability to emit bright and narrow-linewidth light [215–217]. Achieving this desirable combination of bright emission with a narrow linewidth in QDs necessitates meticulous control over the synthesis process. Techniques such as precise manipulation of the internal shell thickness and adjustments to the surface chemical composition have been employed to achieve narrower emission linewidths [218]. Hence, the combination of bright emission and narrow linewidth in QDs has significant implications for various applications. These QDs are ideal candidates for optoelectronic devices such as LEDs and display technologies due to their superior color purity and tunability [219, 220]. Furthermore, their exceptional emission properties make them promising candidates for quantum light sources, which are essential components in quantum communication, sensing, and computation technologies [9].

The crafting of the desired functionalization of QDs is controlled by ligands that passivate the surfaces and have a critical aspect in their physicochemical properties [221, 222]. For instance, the use of different ligands has been shown to affect the emission properties of PbS QDs [221, 223]. In addition to ligand control, the surface chemistry of QDs can also be tuned through the use of various synthesis and post-synthesis methods [222]. Researchers have developed a method using reactive secondary phosphine sulfide precursors, applied in a layer-by-layer approach, to precisely control the chemical composition of the outer layer (surface) of CdS QDs [222]. By carefully crafting the surface chemistry, researchers can tailor the properties of QDs for specific high-performance applications, such as photovoltaics and LEDs [200]

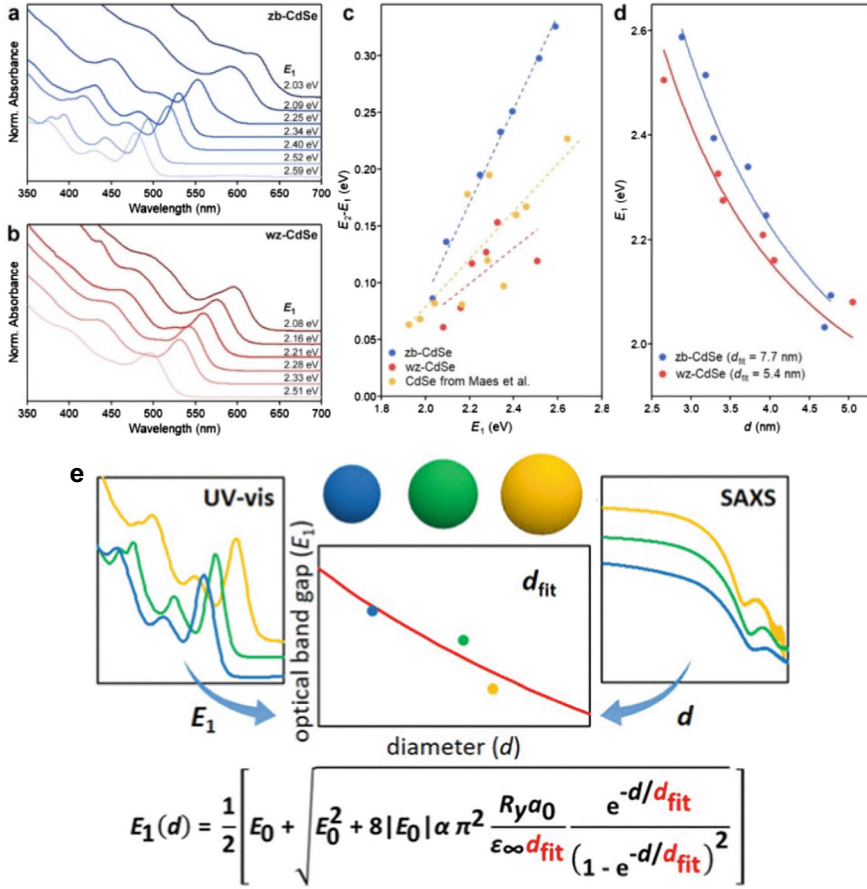


Fig. 14 Comparative analysis of optical properties and size quantization in CdSe QDs. **a** Absorption spectra of zinc blende CdSe QDs. **b** Absorption spectra of wurtzite CdSe QDs. **c** Relationship between the energy difference of first and second excitonic transitions (E_2-E_1) and the first transition energy (E_1) for both zinc blende and wurtzite CdSe QD samples. **d** Optical band gap energy as a function of QD diameter, fitted with a mathematical model. **e** Schematic representation of the model equation and its adaptation to experimental data. (Adapted from Ref. [211] © American Chemical Society)

QD-based LEDs are attracting significant interest in next-generation display and lighting owing to their high color purity, wide color gamut, low power consumption, and wide viewing angles compared to traditional technologies [213, 224]. The remarkable PL emission properties of QDs are harnessed in display and lighting applications through the use of color-converting QD films [224].

As LED chip size shrinks below 20 μm to meet the demands of high-resolution displays, conventional phosphor color conversion methods struggle to deliver sufficient brightness and color yield [225, 226]. This limitation stems from the low light

absorption efficiency of these phosphors. QDs emerge as a promising solution due to their unique properties, remarkable PL emission, narrow bandwidth emission, color tuneability, high quantum yield, and nanoscale size [225]. These characteristics enable QDs to convert light from LEDs efficiently and produce a wider range of colors with exceptional detail, making them a powerful solution for next-generation micro-LED (μ -LED) displays [136].

Despite these advancements, there are still challenges to be addressed. The primary hurdle lies in ensuring device stability, encompassing operational stability, shelf stability, and maintaining efficiency over time (efficiency roll-off). Additionally, unresolved issues in device physics require further investigation. Researchers are actively exploring solutions to these challenges, paving the way for future advancements in QD-LED technology [213, 224].

QDs lasers represent a significant leap forward in optoelectronics, offering unique properties that open doors to exciting applications. These lasers leverage the exceptional light-emitting characteristics of quantum-confined semiconductors by utilizing QDs as the active light-emitting region [227]. The need for high-performance, energy-efficient light sources drives the development of QD lasers. QDs confine charge carriers within all three dimensions, resulting in an electronic structure similar to atoms. This unique property leads to discrete energy levels and remarkably narrow emission linewidths [227].

Recent advancements have achieved superior device reliability in fabricating III-V compound semiconductor QD lasers on Si substrates using a blanket heteroepitaxy technique at elevated temperatures [228]. However, thick layers introduced for defect management hinder light from effectively coupling from the QD region to the waveguides embedded in the silicon-on-insulator platform [228]. Previously, two major obstacles limited the realization of QD lasers: rapid optical gain decay owing to a non-radiative process called Auger recombination, and the instability of QD solids under the high current densities necessary for lasing operation [228]. Fortunately, researchers have successfully addressed these challenges, paving the way for developing electrically pumped QD lasers capable of functioning at ultra-high current densities exceeding $1,000 \text{ A cm}^{-2}$ [13].

QDs have become a powerful tool in optical sensing and imaging due to their unique properties. These nanoscale semiconductors, owing to their size-dependent optical properties, have found significant applications in the development of high-sensitivity sensors and high-resolution imaging systems as well as high-efficiency solar cells [229]. One exciting advancement in this field is the creation of multispectral colloidal QD imagers. These imagers, fabricated using a direct optical lithography technique on colloidal QDs, can detect light across a broad spectrum, ranging from ultraviolet to short-wave infrared [230]. They have demonstrated the ability to capture high-resolution single-color images and even merge these images to create multispectral data [138].

In addition to multispectral imaging, QDs have also been used for dual-band infrared imaging [231]. Researchers have developed a two-terminal QD detector with a switchable spectral response [231]. By adjusting a bias voltage, this detector can rapidly switch between capturing short-wave infrared and mid-wave infrared

light, with modulation frequencies as high as 100 kHz [213]. The tunable optical properties, high photostability, and versatility of QDs make them highly suitable for various medical imaging applications as well [231].

The development of infrared (IR) QDs poses ongoing challenges due to complexities in synthesis and characterization. However, significant progress has been made in the past decade, expanding the range of available IR QDs and their potential applications in photovoltaics, infrared LEDs, telecommunications, and in vivo imaging [232]. QD-sensitized solar cells (QDSSCs) and QD light-emitting diodes (QLEDs) are two notable examples of how QDs are revolutionizing the field of optoelectronics, offering enhanced efficiency, tunability, and stability compared to traditional technologies [233].

In addition to QDSSCs, researchers are exploring the exciting potential of QD light-emitting diodes (QLEDs) as promising artificial light sources. QLEDs, along with QDSSCs, offer unique advantages such as controllability, flexibility, portability, and lightweight design, making them attractive options for integration into a wide range of applications. The unique PL characteristics of QDs make them ideal for use in color-conversion films, which are increasingly being applied in display and lighting technologies [234]. Quantum light generation, a fascinating area of research at the quantum frontier, has been revolutionized by the advent of QD materials [126]. These emitters exhibit exceptional promise due to their high brightness, ability to function under everyday lighting conditions, and compatibility with various electronic and photonic platforms [235]. This breakthrough has sparked significant research efforts to discover novel emitters in diverse materials and develop innovative manipulation techniques for precise control over their properties. Despite these advancements, there are still challenges to be addressed. Device stability, encompassing operational stability, shelf stability, and maintaining efficiency over time (efficiency roll-off), is a key hurdle to be overcome [213, 232, 235].

4 Conclusions and Outlook

QD-based technologies have firmly established themselves as a promising frontier in nanotechnology, offering a plethora of opportunities across various scientific domains [9]. Their exceptional light emission properties and tunable energy states render them ideal candidates for next-generation displays, high-efficiency solar cells, and advanced bio-imaging tools, among other applications [9]. However, to fully realize the transformative potential of QDs, several key challenges must be addressed. Precise control over size, composition, and surface chemistry remains crucial for consistent and reliable device performance [211]. Additionally, the seamless integration of QDs with existing technologies like fiberoptic networks needs further exploration [236].

Despite these challenges, the field of QD research is witnessing remarkable progress. Recent breakthroughs in synthesis and surface functionalization techniques have enabled the fine-tuning of QD properties, paving the way for commercialization

in display and lighting technologies [232, 237]. Continued research and innovation, particularly in addressing device stability and integration issues, will be instrumental in unlocking the vast potential of QDs and ushering in a new era of technological advancements across a multitude of fields.

The future of QDs is undeniably bright. As researchers delve deeper into their fundamental properties and explore novel synthesis and integration strategies, we can anticipate a surge of groundbreaking applications that will reshape industries and enhance our daily lives. QDs are poised to revolutionize not only optoelectronics but also fields like medicine, energy, and environmental science. With their unique capabilities and untapped potential, QDs hold the key to a future where technology seamlessly integrates with our world, offering unprecedented solutions to global challenges.

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Overview: Fuel Cells, Hydrogen Energy, Science, Materials, Nanotechnology, Artificial Intelligence and State of the Art



Carlton Anthony Taft and Jose Gabriel Solano Canchaya

Abstract Fuel Cells (FCs), typically hydrogen energy based, obtained from various sources, are important electricity producing technologies that could help provide the renewable, sustainable energy we need, with mostly water as by-product, whereas electrical power output could be sustained indefinitely. The FCs anode/electrolyte/cathode technology converts the fuel's chemical energy into electricity with near-zero emissions, high efficiency, low maintenance, high energy density, no need to recharge, reusability of heat and long usage time. Pure hydrogen can be used as fuel whereas for oxidizing agent the ambient oxygen containing air can be used. Hydrogen (green, blue, grey ...), versatile energy carrier, can be produced from thermochemical, electrochemical, biochemical technologies (electrolysis, steam reforming, photolysis, fermentation) using various types of feedstocks, stored (liquid, compressed, cryogenic, solid state) and transported (tanks, pipelines, trucks) in the **challenging hydrogen technology**. We overview the field of hydrogen energy and FCs indicating actual state of the art as well as future trends and perspectives. We summarize the fundamental science/technology of FCs, materials, operation temperature ranges, hydrogen oxidation reactions (HOR), oxygen reduction reactions (ORR), devices typically named after electrolytes used, i.e. solid oxide fuel cells (SOFC), alkaline fuel cell (AFC), phosphoric acid fuel cell (PAFC), proton exchange membrane fuel cells (PEMFC) and molten carbonate fuel cell (MCFC). We also comment on other FCs including direct ethanol fuel cells (DEFC), direct formic acid fuel cells (DFACs), direct methanol fuel cells (DMFC), direct borohydride fuel cell (DBFC), direct carbon fuel cells (DCFC), direct glycerol fuel cells (DGFCs), biofuel cells (BFCs), glucose fuel cells (GFCs), regenerative fuel cells (RFCs), indicating reactions, underlying their advantages/disadvantages. We also underline the

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materials used for anode/cathode/electrolytes including *perovskites* (*tunable*), Au, Pt (*costly*), Pd, Ru, Ir, Ni, Ag, graphene, graphene oxide, carbon based, transition metal based, nanotechnological materials (NMs) including multiwalled nanotubes, nanorods, nanowires, nanosheets, core shell nanoparticles, nanofibers, carbon dots, quantum dots. The usage of NMs can address restrictions such as expensive materials that hinders commercialization and increase overall efficiency via effects including high surface area/size. ***Quantum dot NMs should play important roles in the fuel cell technology (addressing some of the challenges in the hydrogen production, storage, transportation, anode/cathode/electrolyte technology)***. The FCs have wide application range (stationary, mobile, portable) including applications in diversified areas such as electronic, power plants, passenger vehicles, housing, trucks, trains, unmanned aerial vehicles, submarines, aircrafts, three wheeled vehicles, industry, space shuttles, stationary energy generation, uninterrupted power supply, hybrid power, residential cogeneration, personal, wearable, portable power generators, mobile phones, laptops. Examples are given for simulation and artificial intelligence (AI) usage as important tools (artificial neural networks, particle swarm optimization) for design of improved cost, efficiency/performance, commercially accessible fuel cells, providing accurate diagnoses/quick data collection.

1 Introduction

Fuels cells (FCs), which are mostly efficient hydrogen energy based technologies, known in principle as early as 1893, have advanced significantly over the decades, gradually starting to replace some traditional technologies in energy production and industry becoming in recent years hot topics of interest with considerable potential for enhanced sustainability. Since they can be used as stationary, portable or mobile applications they have an exceptional wide range functionality. Using fuels such as pure hydrogen together with an associated oxidizing agent, they can produce electrical energy generated through an electrochemical reaction between fuel and oxidant [1–67].

The hydrogen energy, anode/electrolyte/cathode fuel cell technology, is sustainable, with near-zero emissions, high efficiency, few moving parts, low noise, low maintenance, potential for indefinitely sustained electrical power, wide range applicability. However, hydrogen production, storage, transportation, cost are challenges of this technology.

Hydrogen can be produced from thermochemical, electrochemical, biochemical technologies such as (steam reforming, gasification, photolysis, water electrolysis), using various feedstocks, stored and transported in different forms for multi-functional usage.

We present an overview of FCs, i.e. State of the Art, Science, Materials, Advantages, Disadvantages, Hydrogen Production, Storage, Transportation, Artificial Intelligence, Nanotechnology, Perspectives, Future Trends.

We summarize the fundamental science, hydrogen energy, technological anode, electrolyte, cathode structure, materials, temperature range, ORR, HOR reactions, devices typically named after electrolyte used are discussed, i.e. solid oxide fuel cells (**SOFC**), alkaline fuel cell (**AFC**), phosphoric acid fuel cell (**PAFC**), proton exchange membrane fuel Cells (**PEMFC**) and molten carbonate fuel cell (**MCFC**).

We also comment on other **FCs**, i.e., direct ethanol fuel cells (**DEFC**), direct formic acid fuel cell (**DFAFC**), direct methanol fuel cells (**DMFC**), direct borohydride fuel cell (**DBFC**), direct carbon fuel cells (**DCFC**), direct glycerol fuel cell (**DGFC**), biofuel cells (**BFC**), glucose fuel cell (**GFC**), microbial fuel cells (**MFC**), regenerative Fuel Cell (**RFC**).

We discuss anode/cathode/electrolyte material technology underlining *perovskites* and summarize usage of **nanotechnological materials** including multiwalled nanotubes, nanowires, nanosheets, core shell nanoparticles, carbon nanofibers, dots, quantum dots. *Perovskites, Quantum dot NMs should play important roles in the hydrogen based fuel cell technology.*

We discuss and give examples of usage of artificial intelligence (**AI**) methodologies that can help provide quick data collection, accurate diagnosis, improved efficiency and performance serving as tools to aid design of upcoming generation commercially accessible fuel cells.

2 Solid Oxide Fuel Cells

The solid oxide fuel cell (**SOFC**), which can be traced back to 1899, has become a development direction, research hotspot of fuel cell technology, and a good choice among sustainable technologies to efficiently convert fuels to electricity with low pollutant emissions, with however challenges for marketing. Based on the operating temperatures of the cells, they can be classified into three types, i.e. HT-SOFCs, or high temperature; IT-SOFCs, or intermediate temperature and LT-SOFCs, low temperature SOFCs [12–20].

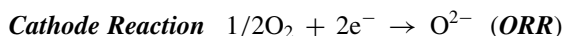
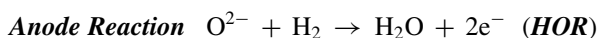
Notably, the HT-SOFCs work at high temperatures (900–1000 °C) and were developed in the initial stage of this technology whereas such temperatures are required to obtain adequate ionic conductivity. Nonetheless although these high temperatures improve charge conduction/reaction rates they also present difficulties in applications (material degradation, cost, incompatibility).

IT-SOFCs were introduced in order to reduce operating temperatures affording temperatures of 650–800 °C, whereas nanostructured materials are preferred at lower temperatures up to 650 °C and are easier to operate, using less costly materials and cell fabrication with long-term durability.

LT-SOFCs (up to 650 °C) have high conversion efficiencies and are among the growing research fields of the present era requiring development of critical anodes, cathodes, electrolytes and in-depth understanding of the material science that yields necessary conductivity, electrocatalytic activity, chemical compatibility and thermal stability.

In general, development of SOFC devices for practical/commercial deployment is associated with technological challenges including research of energy materials (cost, fabrication procedures, material selection) whereas electrolyte materials in particular have received significant boost pursuing the goals of increased SOFC performance.

A single solid oxide fuel cell (SOFC) is composed of an anode, a cathode and an electrolyte. Via the oxygen reduction reaction (ORR), oxygen (O_2) is reduced to oxygen ions (O^{2-}) on the cathode surface. Via the electrolyte, the oxygen ions are transferred to the anode. Hydrogen (H_2) is then oxidized at the anode surface (HOR) by O^{2-} forming H_2O and releasing electrons that pass through the external circuit. A closed circuit is formed as the electrons flow back to the cathode.



Since its inception, this technology has made significant progress but still faces challenges (instability, high cost) due considerable to high operating temperatures.

SOFCs offer potential to directly employ existing fossil energy resources or renewable biofuels, large-scale energy/heat power generation/distribution, multifuel compatibility, low pollutant emissions, good efficiency to convert fuels to electricity, energy supply in aerospace vehicles, ships and others, with however, high production cost with the use of expensive materials/high temperatures imposing barriers for widespread commercialization. In order to expedite commercialization of SOFCs a sustained journey involving novel system designs/innovative functional materials is required. It is necessary that the SOFC operates in the 600–800 °C IT range for widespread applications which is hindered by the cathode's often unsatisfactory performance due to polarization loss at the reduced operational temperatures.

However, various factors are essential for effective low temperature work. It is essential to develop appropriate cell components (electrolytes, cathodes and anodes) whereas sufficient material science background is necessary in order to obtain materials with required electrocatalytic activity, electrical conductivity, chemical compatibility and thermal stability. It is necessary to study diverse materials and combinations thereof to obtain desirable operating cost, fabrication and temperatures for SOFC requirements.

2.1 Sofc Electrolytes

In a SOFC, the primary role of a solid electrolyte is to conduct ions, O^{2-} or H^+ (nature of the specific material), between the electrodes driving the electrons to move through the outside external circuit to complete the overall electrochemical reaction whereas

the core electrolytes influences the essential temperature of the fuel cell. The electrolytes used in SOFCs could be an oxide ion conductor, proton ion conductor or even a dual ion conductor which decides performance and reactions of each cell configuration. The electrolytes must in addition to high ionic conductivity have thermal, chemical, mechanical companionability with adjoining cell components, low electronic conductivity (avoid short-circuits).

SOFC electrolyte materials have been of significant interest during last decades whereas material scientists have aimed to improve performance with the challenge of limited ionic conduction. Among the ionic conductors explored are doped $\text{CeO}_2/\text{Bi}_2\text{O}_3/\text{ZrO}_2$, LaGaO_3 -based perovskite oxide, $\text{Ca}_{12}\text{Al}_{14}\text{O}_{33}$, $\text{Ln}_{10}(\text{SiO}_4)_6\text{O}_3$ [$\text{Ln} = \text{La}, \text{Nd}, \text{Gd}, \text{Sm}, \text{Dy}$], $\text{Na}_{0.5}\text{Bi}_{0.5}\text{TiO}_3$ and $\text{La}_2\text{Mo}_2\text{O}_9$ -based fast oxide ion conductors.

Selected for SOFCs academic/industrial usage are the ionic conductors of (O^{2-}) , i.e., zirconia and the proton conducting materials (barium cerate, barium zirconate). Indicating at high operating temperatures (above 700°C) high ionic conduction, YSZ (Yttrium-stabilized zirconia) is the common choice indicating as well ability to work at wide of range oxygen partial pressures with insufficient ionic conductivity however at low temperatures. At present, implemented doped LaGaO_3 -based ionic conductors and acceptor doped ceria undergo short circuits due to reductions in Ce^{4+} as well as chemical reactivity with other electrode materials. Other tantalates and niobates doped with rare-earths and used for electrolytes still require more study.

2.2 SOFC Cathodes

The first SOFC cathode materials used precious metals including Pt, Au, Pd loaded on oxides such as CeO_2 which was however soon modified due to difficulties of large scale commercialization from limited resources and high cost. Subsequently, perovskite oxides ($\text{L}_{1-x}\text{Sr}_x\text{MnO}_3$) with good oxygen catalytic properties in the range ($800\text{--}1000^\circ\text{C}$) were considered although their ORR occurred only at the electrolyte and cathode interface resulting in limited cathode catalytic efficiency.

2.3 Mixed Ionic and Electronic Conductors

Mixed ionic and electronic conductors (MIEC) were then developed including $\text{La}_{0.6}\text{Sr}_{0.4}\text{Co}_{0.2}\text{Fe}_{0.8}\text{O}_3$ and $\text{SmBaFe}_2\text{O}_5$ (SBF). The MIEC used as cathodes in SOFC increased the ORR efficiency. Some conductors also displayed excellent electrochemical performance in the IT range.

2.4 Protonic Ceramic Fuel Cells

In the protonic ceramic fuel cells (**PCFC**) instead of oxide ions, protons pass through the electrolytes at reduced temperatures where the proton conducting activation energy is less. The PCFC also indicate below 600 °C high working performance and efficiency whereas in addition the by-product water generated at cathodes prevents anode fuel dilution.

2.5 Single, Electrolyte-Free and Double Component Fuel Cells

The SOFC devices have recently gone through a major revolution with introduction of double layer fuel cells, electrolyte-free and single layer component fuels cells (**DLFC**, **EFFC** and **SLFC** respectively). The DLFC indicates considerable advancement potential. In the SOFC (conventional three-layer configuration) there is no electron passage with electrolytes for ion conduction whereas electrons are released at anode as H₂ fuel is oxidized and oxygen is reduced to ions at cathode producing electricity from combination with electrons from external circuit. The single component fuel cells (**SCFCs**) does not use cathodes or anodes (electrolyte membrane or electrodes) offering a simpler cost-effective “three in one” system (co-existence of semiconductors and ion conductors) that could contribute to fuel cell commercialization performing processes of fuel cell, nano-redox at both particle/nanoscale levels avoiding the internal short-circuiting.

2.6 Semiconductors in Fuel Cells

Amphoteric oxide semiconductor cathodes/anodes which are separated by ionic conducting electrolyte (n and p-type regimes) can also be used in the fuel cell technology. The electrolyte (ion conducting materials) have same properties as intrinsic semiconductors (with larger bandgap values). The cathode/anode placed in oxidizing/reducing environments are responsible for ORR/HOR indicating high hole/electronic conductivity. Some oxides, including perovskites have mixed ionic and electronic conductivity.

2.7 Semiconductor Membrane Fuel Cells

Some materials appear to contradict traditional SOFC knowledge whereas semiconductor membrane materials exhibit typically much higher electrical conductivity in

comparison to ionic conductivity. Semiconductor-ionic membrane or semiconductor membrane fuel cells (**SMFC**) is another important area of research involving interfaces developed among semiconductor-semiconductor, semiconductor-insulator, oxides, semiconductor-ionic with influence on charge conduction and bandgap properties. These fuel cells can yield changes in ionic transport properties related to interfacial/surface state/electronic band modifications since changes in semiconductors and their heterostructure composites can indicate similar ionic conduction and power of the fuel cells. Across interfaces modified by energy bands the ionic charge transport process could have a similar origin.

Electric fields along interfaces and local electron charge density of composite heterostructure oxide materials determines ion transportation. The literature indicates that the ionic conductivity is significantly improved by usage of semiconductor-ionic composite heterostructure materials. As an example, compared to YSZ, the SrTiO_3 composite mixture has indicated superior conductivity.

By using a n-type anode, an intrinsic semiconductor-type and a p-type cathode, the SOFCs can be related to p-i-n semiconductor device. By removal of i-type electrolyte a p-n heterojunction is obtained or double layer fuel cell (**DLFC**). New semiconductor heterojunction-based fuel cells, using built-in electric field energy and band/alignment for the electron transport, supports superionic charge transport along the interfaces that can improve the device performance.

For SOFCs, the materials and bipolar plates represents most of total costs of fuel cell stacks which is much higher than that of SMFCs. The future lies in developing semiconductor-ionic-based materials as basis for upcoming generation advanced functional materials from combination of semiconductor energy band and built in electric field theory (BIEF) to develop superionic materials with self-driven ionic transport at low temperatures.

2.8 SOFC Electrode, Electrolyte Semiconductor Materials

The currently used materials for SOFC electrodes/electrolytes can be considered as semiconductors from materials science viewpoint, i.e., wide bandgap semiconductors for ionic conducting electrolytes or narrow bands (triple/mixed) for charge conducting electrodes. Notable, different types of electrolyte materials have been used to address issues such as ionic conductivity at low temperatures.

2.9 SOFC Electrolytes and Nanotechnology

Reducing the thickness of electrolytes from micrometer to nanometer scale can increase the ionic conductivity since the ohmic resistance of the electrolytes is directly related to thickness yielding reduction/improvement of charge conduction at low temperatures. One alternative is using nanograin electrolytes with superionic charge

conduction in a composite interface along the grain boundary (GB) and interface. There is growing international interest in developing nanocomposite materials for advanced fuel cells including fast ion conducting nanocomposite electrolytes (core-shell and nanowire structural ceria-based nanocomposites), yielding multi-ion charge conduction, excellent LT-SOFCs performance, i.e. outstanding electrolytes for SOFC applications.

2.10 Perovskite SOFC

In recent years MIEC cathode materials such as Fe based double perovskite oxides indicated potential to increase oxygen vacancies and ion mobility making easier oxygen dissociation/adsorption in the oxide [12–14]. These perovskites also indicate sites for property modification, good structural stability and cost-effectiveness. The general formula of a simple perovskite oxide is ABO_3 . In this formula an alkali or lanthanide rare earth metal occupies the A site and a transition metal occupies the B site. It can be described by an idealized P_{m-3m} cubic crystal structure where corners are shared by BO_6 octahedrons forming a lattice skeleton. The A sites are localized in the skeleton void centers.

On the other hand, the double perovskite oxide consists of different (2) perovskite structures (ABO_3 and $A'B'O_3$). In one structure, B-site double perovskite (A, A' occupied by same element, B, B' are different). In the A-site double perovskite A, A' are different, B, B' are same).

Alkali Earth metals (Ba^{2+} , Sr^{2+}), large rare earth ions (La^{3+} , Sm^{3+} , Nd^{3+} , Gd^{3+} , Pr^{3+}), mostly occupy A sites. Lower valence transition metals (Fe^{3+} , Co^{3+} , Ni^{2+}) mostly and higher valence states (MO^{6+} , N^{B5+} , W^{6+}) typically occupy the B, B' sites. In order to minimize the lattice distortion and balance charge distribution the B and B' site elements are typically orderly arranged within structure. B–O–B is formed via oxygen atoms connecting B and B' atoms.

Notably the SFM and AAB_2O_5 double perovskite families have been identified for SOFC technology with high potential as cathode materials. The SFM compounds are of actual research interest indicating a highly symmetrical structure and is a promising SOFC cathode material as well as a MIEC whose structural symmetry leads to electron conduction, indicating also modest thermal coefficient (compatible with typical electrolytes) as well as excellent structural stability.

The $A_2BB'O_6$ family including Sr_2FeNbO_6 , Sr_2NiMoO_6 , Sr_2CoMoO_6 and Sr_2FeMoO_6 (SFM) have been identified as cathode materials, research hotspot of ceramic materials promising for SOFC. The $AA'B_2O_5$ family which includes $LnBaX_2O_5$ ($X = Mn, Fe, Co$) have been identified as cathode materials which can indicate layered structures with excellent oxygen diffusion within structure and dynamic oxygen exchange on surfaces as well as appropriate thermal expansion coefficient. Nonetheless, in order to be a cathode material which is effective in IT-SOFC technology they still lack sufficient catalytic and conductive activities. Consequently, there has been considerable focus on $LnBaFe_2O_5$ ($Ln = Gd, Sm, Nd, Pr, La$) as well

as SFM in order to improve their ORR catalytic activity and DC conductivity for usage as SOFC cathode materials. To achieve this, one approach is via surface modifications, another method is forming composite cathodes, and a third method is via doping modifications replacing O^{2-} (anion doping).

Originating from $SrFeO_3$, $SrFeO_3$, double perovskite has high degree of oxygen ordering but limited mobility and concentration of oxygen vacancies, motivation Mo-doping in Fe-sites which increases electron conduction/oxygen vacancies leading to actual important cathode candidates.

Synthesis of $Sr_2Fe_{1.5}Mo_{0.5}O_6$ (SFM_{0.5}), pure phase double perovskite (excellent catalytic performance, conductivity) was soon recognized as ideal candidate as cathode for SOFC technology initiating extensive effort by research community to further improve the performance as a SOFC cathode material using in the synthesis a Fe-rich chemical precursor. Notably, the doping is classified into groups using doping elements with different ionic radius (same Sr^{2+} valence states) and groups involving elements with different valence states. A-site absence modification has also been used to improve ORR activity.

SFM_{0.5} B-site doping can be divided into a first group whereas dopants (varied valence states including transition metals) can influence oxygen vacancies, equilibrium, ORR cathode activity and DC conductivity. Another group of dopants (Ga, Nb, Sc) have indicated influence on ORR activity and conductivity of SFM_{0.5}.

Taking into consideration valence electron densities and electronegativities the anion doping substitution of O^{2-} has also been used for improving SFM_{0.5} electrochemical performance via weakening of M–O bonding and increasing lattice oxygen activity without affecting structure. A second composite cathode can also be formed by combining SFM_{0.5} with a second phase (ionic conductor with low thermal expansion and high ionic conductivity via mechanical mixing) in order to extend cathode boundaries and support ORR activity and better match electrolytes used in SOFC technology. With time, SFM_{0.5} DC conductivity has improved via modifications summarized above maintaining reasonable ORR cathode activity.

Relevant research on $LnBaFe_2O_5$ ($Ln = La, Pr, Nd, Sm, Gd$) involves various doping elements at different sites in order to improve cathode electrochemical performance and has attracted attention because its conductivity and ORR catalytic activity dictates good electrochemical cathode performance depending on the doping Ln element used which can indirectly influence for example the cubic linear O–Fe–O bonds. Both the A and B site doping with numerous transition metals and stable elements has been investigated to improve ORR and conductivity. In-situ precipitation, surface impregnation were also investigated to enhance the oxygen catalytic activity. Notable, elemental doping can influence the conductivity and catalytic activity in Fe-based double perovskites (not always in same direction) with potential to further enhance advanced SOFC technology.

Summarizing, modifications of both B^- , A^- sites as well as doping elements on the perovskite structure affects the metal/oxygen defects and stability which impacts the cathode performance (conductivity, catalytic activity, structure, chemical stability, thermal expansion). Notably, for advanced SOFC technology there is still

potential to further enhance the Fe-based double perovskite cathodes electrochemical performance.

2.11 SOFC Perspectives

Cathodes, anodes and electrolytes of SOFCs have been modified during last decades with usage of new materials in order to improve prevention of electronic conduction, oxygen ionic conductivity, mechanical properties, operation temperatures, electrode compatibility as well as chemical, thermodynamic stability. The SOFC (electrolyte-based YSZ) is commercially available with high cost manufacturing. The technology of SMFC could use wide range low cost manufacturing materials whereas issues such as scalability, stability needs to be addressed.

2.12 SOFC Barriers

Barriers to commercial usage of SOFCs include sulfur/coke/surface poisoning, electrochemical reaction in electrolytes, thermal mismatch of cell components, cathode rate determining factors, chemical instability, incompatibility, poor thermal resistance, complex hydrocarbon reforming, non efficient stacks, incomplete fuel utilization, degradation of components at high temperatures, lower power at lower temperatures, high temperature instability and corrosion.

3 Alkaline Fuel Cells

Historically, the transport area was one of first benefitted from development of alkaline fuel cells (AFCs), which was also the first working cell to deliver power and used, inclusively for Moon Apollo missions, Gemini Spacecraft and Space Shuttle Orbiter. Nonetheless, it was recognized in the 1970s that the AFCs had technological/economical disadvantages including intolerance to CO and liquid electrolyte immobilization and attention was temporarily changed to other cells including phosphoric acid fuel cells (PAFCs) and proton exchange membrane fuel cells (PEMFCs). These nonetheless, costly cells stimulated renewed interest in AFCs within framework of alkaline anion exchange membrane fuel cells (AAEMFC) where some technical problems of conventional AFCs are solved [21–28].

3.1 Working Principles of AFC

The general principles that govern fuel cell operation are used for AFCs whereas at the anode the input fuel reacts and at cathode the oxidant reacts resulting in water and electrical power with the usage of alkaline electrolytes determining the main difference which are most often hydroxide ion conductors. The chemical reactions are also the hydrogen oxidation reaction (HOR) Herewith, also the cathodes, opposite to anodes, receive oxygen/air yielding solvation, water production and formation of OH^- , i.e. ORR reactions whereas the overall reaction produce heat/water/electrons and consequently a electromotive force. The durability/power of the AFC is mostly dependent on state/behaviour of cathode (polarization losses).

Hydrogen oxidation reaction (HOR) $\text{H}_2 + 2\text{OH}^- \rightarrow 2\text{H}_2\text{O} + 2\text{e}^-$
(reaction of OH^- in the electrolyte with incoming hydrogen which yields water and electrons)

Oxygen reduction reaction (ORR) $\text{O}_2 + 2\text{H}_2\text{O} + 4\text{e}^- \rightarrow 4\text{OH}^-$
(Cathodes supplied with air/oxygen is solvated with water produced, forming OH^-)

3.2 Catalyst ORR Materials

3.2.1 Precious Metals

Platinum

Polycrystalline metal Pt is a good ORR catalysis, with scarcity, high cost and insufficient durability. Pt nanoparticles supported on a carbon black matrix (Pt/C) can enhance ORR/reducing high costs of Pt. Core-shell structures can be used where Pt serve as the shell and a relatively cheaper metal used as the core via nanostructures supported on multi-walled carbon nanotubes (MWCNT).

Various PtM (including Fe, Co, Cu) nanowires perform well for HOR activity.

Gold

Gold (Au) (100) is active crystalline electrocatalyst for alkaline ORRs exhibiting excellent activities at the nanoscale using carbon/graphene/graphene oxide supports.

Palladium

Palladium (Pd), alloyed/pure, ORR electrocatalyst, comparable catalytic activity to Pt with surface nano particle areas much higher than that of Pt which can be further increased by alloying. Pd core shell nanoparticles structures supported on graphene oxide nanosheets have also been constructed to enhance performance/lower costs.

Silver

Silver (Ag) is cheaper than Pd and Pt with comparable ORR and can be used as nanocatalysts deposited on mesoporous carbon support materials and Fe–N–C.

3.2.2 Carbon-Based

Graphene, carbon black, carbon nanofibers, carbon nanotubes are important catalysts with large surface area/electronic conductivity/ORR which can be improved by doping with other elements including phosphorus, boron, fluorine, sulfur, nitrogen-doped carbon dots, carbon nanotubes/nanofibers and carbon transition metal hybrids.

3.2.3 Transition Metal Oxides

The transition spinel oxides e.g. Co, Mn, Cu, Zn, Fe, Ni, and spinel/nanocarbon hybrids often indicate high ORR whereas electrochemical properties often depends on structure, composition.

3.2.4 Perovskite Oxides

The adaptable perovskite structures can incorporate different dopants at its lattice sites which can yield different crystal phases, crystallite sizes and impact catalytic reactions [2, 11, 12].

3.2.5 Recent ORR Catalysts

Recent investigated good ORR catalysts includes $\text{NiFe}_{1.75}\text{Co}_{0.25}\text{O}_4/\text{C}$, NiCo_2O_4 , $\text{Mn}_{1.5}\text{Co}_{1.5}\text{O}_4$, $\text{NiFe}_2\text{O}_4/\text{C}$, $\text{ZnCoMnO}_4/\text{N-rGO}$, $\text{Co}_3\text{O}_4\text{-N-Gr}$, $\text{NiO-Co}_3\text{O}_4$, CoCuMnO_x , B- and N-doped carbon dots on carbon nanotubes, P-, N-doped carbon nanosheets [18–22].

3.3 *Anode HOR Catalysts*

3.3.1 Precious Metals

Platinum

Pt, within different combinations, similar to ORRs, also has high electrocatalytic performances in hydrogen reduction (HOR) as well as oxidation reactions. Pt, PtM(Fe, Co, Cu) nanoparticles on nanowires and core–shell nanoparticles acquired high HOR electrocatalytic activity including single-dimensional materials. Pt represents the pinnacle of electrocatalytic performance lying two orders of magnitude above its alkaline counterpart and since there is no apparent breakthrough to make Pt absolute, the effects of lowering particle size by using nanomaterials have been prioritized.

Ruthenium

Ru nanoparticles, metal nanocrystallites, core–shell electrocatalyst materials, alloyed carbon nanoparticle materials, mesoporous carbon supported and ternary metallic catalysts have also indicated enhanced catalytic activity.

Iridium

Ir is excellent alternative due to high stability and cheaper price whereas catalytic performance can be improved by nanoparticle alloying with metals.

Transition Metals

The abundant Ni catalysts have high surface area, moderate costs, corrosion immunity in highly concentrated alkaline solutions with however insignificant/unstable HOR activity more indicated as support material. However, the activity can be improved by using carbon-supported Ni nanoparticles as well as alloying with other metals and loading on different supports such as carbon and its nanotubes.

3.4 *HOR Exchange Current Densities*

The literature indicates a wide range for HOR electrocatalyst specific exchange current densities for various materials, including (Pt, Ni, Pd, Co, Ir) [18–22].

3.5 *Liquid Electrolytes*

AFCs use metal electrodes and liquid alkaline electrolytes that are placed back-to-back with a thin electrically insulating diaphragm which prevents air and hydrogen from mixing, enabling necessary chemical reactions. The permeable gas diffusion electrodes (GDE) allows flow of liquid electrolyte into gas channels simultaneously with supply of air to active catalyst sites.

3.6 *Anionic Exchange Membrane Fuel Cells (AEMFCs)*

AFCs takes a great step towards further improvement with usage of AEMFCs in which there is replacement of liquid electrolytes in traditional AFCs although long-term stability and usage of platinum group metals (PGMs) are a challenge emphasizing necessary continued search for new non-PGM materials. The AEMFC uses for electrolytes (quaternary ammonia/piperidinium based polymers), for anode catalysts include Pt, Pd, NiMo, CoNiMo, Ni₃N and for cathode catalysts Pt, PtRu, Pd, Ag, Co₃O₄, MnO₂. Challenges of anion exchange membranes (AEMs) include stability, performance, durability, low-cost production, mechanical strength, effects of humidity, offering advantages such as decreased gas crossover, better CO₂ tolerance, purer hydrogen, higher open circuit voltage (OCV) and increased efficiency avoiding problems of flooding and weeping.

The experimental AEMs are solid polymer membranes made of a polymer backbone (poly(aryl piperidinium)), polyphenylene polynorbornene, poly(ether-ether ketone), polyethylene, poly(phenylene oxide), poly(ether sulphone), polysulphone onto which functional cationic end-groups are attached assuring ionic conductivity (mobile OH⁻ anions). A challenge of AEMs is keeping simultaneously high mechanical strength and anionic conductivity and avoid long term degradation.

3.7 *Anionic Ionomer Fuel Cells*

Another important field of research is the production of anionic ionomers (which can be produced by using a mixture of water and solvent) whereas high quality analogues have not yet been created whereas due to cathode/anode characteristics should be created separately from AEM. The ionomer-HOR catalyst interaction indicate that the adsorption of phenyl groups (ionomer components) impacts strongly AEMFC anode performance.

DFT studies indicated that AEMFC performance increases as the adsorption of phenyl groups weakens whereas PtRu catalysts are less affected by phenyl groups making it preferred to Pt on the anode [22].

3.8 Goals

Goals include development of peak-efficient, direct hydrogen, long term durable fuel cell power with mass-production, cost effective economical accessibility. Making accessible medium scale distributed generation fuel cells power systems operating on renewable fuels as well as development of low cost reversible fuel cells for energy storage applications.

4 Phosphoric Acid Fuel Cells (PAFC)

Phosphoric acid fuel cells, using phosphoric acid (H_3PO_4), (PA), as an electrolyte, and hydrogen (diverse sources) as fuel (oxidation at anode, reduction at cathode, electrons via electrolyte and external circuit), has good chemical, electrical and thermal stability, quick start, eco-friendly nature and is a first-generation fuel with successful commercialization, installed worldwide in hospitals, buses, trains, submarines, centers, hotels, schools, airports, heating for public baths, and residential systems for operations at temperatures between 150 and 200 °C, scaling up of stacks from 20 to 500 kW, output with good overall efficiency, electricity yield higher than 40%, good CO tolerance rate, whereas 85% of the vapor could be used in cogeneration and can be employed using available distribution/refueling systems. Future progress include usage of other fuel types, progress to cheaper/substitutes for Pt and progress to large-scale commercial applications [29, 30].

In a completely dissociated ionic state phosphoric acid is a good ionic conductor, whereas at lower temperatures it is a poor ionic conductor and the percentage of CO in the platinum electrocatalyst can become too high. However, in a low CO regime, impure hydrogen can be used increasing potential of fuels, although platinum cost could be high and the output low compared to other fuel cell technologies.

Typically, PAFC involves PA trapped in an electrolyte matrix, with electrodes (anodes, cathodes) having diffusion layers whereas the fuels source and oxidant circulate via bipolar plates (BPPs) on opposite electrolyte sites with hydrogen oxidizing to electrons and protons at anode. In this process the electrons yields current passing through the external circuit while the protons pass through the electrolyte and the water byproduct (can be used for heating) is formed at the cathode as electrons and protons combine.

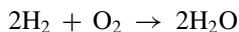
At the anode the oxidation reaction (HOR) splits the expelled hydrogen into 4 protons and 4 electrons



and at the cathode via the reduction reaction (ORR) four electrons and four protons combine with oxygen to form water.



whereas the overall reaction is



HOR starts with transportation of H_2 gas molecules towards electrode surface followed by adsorption of H_2 on electrode surface (chemisorption of hydrogen atoms) allowing electron transfer and releasing H^+ ions into the electrolyte and finally transport of H^+ ions in the sea of protons in the electrolyte. The slowest step determining the overall HOR ratio is the release of a proton in the electrolyte.

The choice of catalyst is extremely important for reaction rates as thermodynamics represents quantity of energy possible to harness from the reaction, as well as electrons and currents which depends on the choice of material and design of electrodes which catalyzes the process in search for minimum activation energies whereas catalysts should be used having high exchange current density, electrical conductivity, mechanical stability, porosity, ease of manufacturing and low corrosion. The adsorption bonds should not be too strong or too weak and their should be a fine balance for conversion of the chemisorbed atom to electron and proton transfer.

Since HOR is the easiest process it can be used to better understand importance of catalyst/support materials whereas for example the PAFC can be supplied with reformat fuel containing CO requiring the HOR catalyst to be tolerant to CO. The more complex ORR often requires higher quantity of catalyst.

The materials used for PAFCs, include for catalysts (Pt, Pt-Co, Pt-V, Pt-Cr, Pt-Ni, Pt-Sn, Pt-Ru) whereas alloying of Pt with other materials enhance durability, reduce poisoning. Non precious catalysts including Fe-Nx, Fe-N-C lowers catalytic activity.

For catalyst support includes CNTs, activated carbon black, CNFs, graphene, NCNTs which yields improved stability, higher surface area, higher electronic conductivity as well as TiO_2 , SnO_2 , SiC, SiC/C, IrO_2 , WO_3 with lower electronic stability and conductivity.

For membranes includes nafion/ZrSPP composites, nafion/(sulfonated and silica), higher water uptake, need to maintain membrane hydrated at elevated temperature, preserve proton conductivity. Matrix for holding PA includes SiC, PBI, polyacetalimide, polyolefin and polytetrafluoroethylene (in order to maintain chemical, thermal, mechanical stability in the acid).

For gas diffusion layers uses carbon paper, cloth with microporous layer, hydrophobic agents, with μm thickness offers mechanical strength to catalyst layer and can hold PA.

The bipolar plates are made of Al, Ti, Cu, Steel with coating materials made of NiP, Au, graphene, ZrN, TiN which enhances corrosion resistance and durability whereas steel offers machinability, high strength and low cost.

Based on the electrolyte used the PAFCs can be classified into PA doped in polybenzimidazole membrane and PA impregnated in silicon carbide. There are

various challenges faced for scale-up of stack including design issues of electrodes, stack component/quality and operational time issues.

Various considerations are necessary for scaling up of stacks for PAFC in order to provide to all cells (micro and macro) similar eco-systems. First, design must consider planarity and crushing of electrodes and the clamp pressure non-uniformity as well as stack component quality. Operational issues such as failure attributed to thermal expansion and fault in stacks. PAFCs can also be coupled with heat driven refrigerators. Evaluation of steady state/dynamic PAFC one/two/three dimensional models have been reported in the literature.

4.1 Conclusions

PAFCs are environmental friendly, high efficient electrical utility applications using wide range of fuels including natural gas, nafta and biogas and are thus good candidates for industrial energy. They are known for their high efficiency and environmentally friendly nature for electric utility applications. Also, the wide range of fuels that can be used for PAFC applications ranging from natural gas, naphtha, LPG to biogas makes PAFCs desirable candidate for industrial energy. Although initially designed for stationary applications they are now used in trains, buses, submarines and scaling up power plant stacks.

5 Proton Exchange Membrane Fuel Cells (PEMFC)

The **PEMFC**, also known as the “Polymer” Exchange Membrane Fuel Cell serves as both electrolyte and divider of the anode and cathode whereas the solid polymer membrane, which can be made from acidified Teflon is the PEMFC basic structural component. Due to their energy densities, high power and low pollution emissions they are promising electrochemical power source that convert fuel cells chemical energy directly to electrical energy via electrochemical reactions

$\text{H}_2 \rightarrow 2\text{H}^+ + 2\text{e}^-$ at the anode and $2\text{O}_2 + 2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2\text{O}$ at the cathode.

There is an increasing need for low-power/small-scale storage/source of electricity. The PEMFC has gained attention as one of the clean energy technologies that use hydrogen as fuel converting into electricity with water and heat as by-products. A single PEMFC is clear/simple whereas individual cells can be combined forming stacks with other advantages including short start-up times, low operating conditions and absence of noise allowing industrial applications (vehicles, portable energy, aircraft, power generation, stationary generation) [31–39].

In these cells the proton exchange membrane (PEM) serves as the fuel-blocking membrane and also as the main conductor of protons and expected to simultaneously yield excellent stability and high proton conductivity by controlling, organic–inorganic hybrid composite membranes, chemical structure as well as nanofiber composite membranes which offers stability, long-range order and strong proton conductivity. High performance can be obtained via acid-rich layers, proton transport channel and acid–base interactions.

After splitting at the anode into protons the hydrogen fuel travels across the PEMFC membrane using a system pathway. Electrons gather as electrical current via connection of the two electrodes with an external circuit. In addition, electrons from external electrical circuit mix with airborne oxygen and protons passing through electrolyte using network of channels (cathode).

This technology functions well in mobile and fixed applications and is promising for new fuel cell vehicles. For commercialization concerns related to cost, infrastructure, durability, usage of pure hydrogen are obstacles for wide deployment. Deficiency in the supply of high purity hydrogen could contaminate the anode electro-catalyst.

The performance can be enhanced by raising the temperature allowing increased reaction rates, heat usage, decreased voltage loss and enhanced resistance to carbon monoxide. Notable, water builds up at cathode which must be removed continuously in order to promote more reactions. The PEMFCs run typically at 70–90 °C.

The comparison of specific energies for selected energy storage/electrochemical devices including lead-acids, supercapacitors, solid-state batteries, nickel-metal hydride, lithium-sulfur, zinc-air, lithium-ion batteries have indicated for PEMFCs advantages of low operating temperatures, low pollution emissions and high specific energy (1000 Wh kg⁻¹). The usage of stored hydrogen offers as fuel a high specific energy/conversion of ~60% for PEMFC. The environmental friendliness and low operating temperatures make PEMFCs very attractive for small-scale applications.

During the last decades, the PEMFCs have been widely used successfully as portable, stationary generation as well as transportation and can thus be divided into first, small scale (less than 10 kW) used for unmanned aerial vehicles, portable power generation and residential power supply. The second range involve the medium-scale applications (10–100 kW) which includes distributed generation, material handling trucks and automobile applications. The third scale is large-scale (>100 kW) appropriate airplanes and heavy-duty vehicles. The typical specific power of the PEMFC is thus known to have a wide range encouraging the success of commercialization of both large and medium-scale PEMFCs.

The small-scale PEMFCs are alternative sources for energy/electric propulsion of low-power vehicles. Let us look at some of these alternatives. We can start with the remotely operated unmanned aerial vehicles (UAV) used as air transport for both civilian and military for which energy/power density, weight determines endurance, power density, acceleration, capability, volume, range of system whereas cost affects the availability. Notable, small UAVs with Li-ion batteries yields high battery capacity, power and increased safety although there could be a conflict between endurance and battery weight.

Small-scale PEMFCs, in contrast to internal combustion engines and batteries, can provide UAVs with advanced features (reliability, energy density, long endurance, low or zero emissions, lower thermal infrared characteristics, lower noise, different power ranges, easy upgrade of fuel cell systems, air-cooled open-cathode, in series stacking). PEMFC and Li-ion battery hybrid UAVs power systems often compose UAVs increasing dynamic response, peak power, improved efficiency, lighter systems, flight power, increased flight endurance and reasonable fuel consumption.

Notable, the crucial endurance capacity of UAVs is mostly determined by hydrogen storage (metal hydride hydrogen, compressed hydrogen, liquid hydrogen, cryogenic compressed hydrogen). The material-based hydrogen storage technology, due to minimal supporting infrastructure requirements, has the advantage of an improved energy density for hydrogen storage whereas, the storage technology does not affect the UAV technology.

Underwater vehicles UVs (conventional/submarines, autonomous underwater vehicles (AUVs), miniature submarines) increasingly use small-scale PEMFCs providing alternative power source technology due to their improve stealth capabilities, stable operation, higher efficiency/energy density, endurance and longer ranges. Their wide power range meets requirements for varying sizes of UVs.

For light traction vehicles, small-scale PEMFCs are also proposed and classified in (a) less than 400 W PEMFC-powered bicycles, (b) less than 2 kW PEMFC-powered golf carts, (c) less than 4 kW large consumer market PEMFC-powered scooters, i.e., two wheeled transport; (d) few kW to tens of KW for PEMFC-powered trucks for material handling, i.e., forklifts and trucks with 100,000 h runtime.

For small scale stationary energy generation, small-scale PEMFCs have advantages (higher power generation efficiency, fast startup, zero emissions, compactness, low noise level) offering uninterrupted power supply and residential cogeneration in commercial/remote areas.

Reliable and 'uninterrupted power supply' (UPS), i.e. (supercapacitors, Pb-acid batteries, engine-generator sets, flywheels, millisecond start-up) for users in commercial/remote areas promoted by small-scale PEMFCs offer advanced features such as low cost daily maintenance, adaptability for complex operating environments as well as zero emissions. Standalone wind-PEMFC hybrid power systems are able to supply for 1 year constant power for homes.

Residential cogeneration systems can also be provided by small-scale PEMFCs with low pollution and high overall efficiency providing as well domestic/central water/household cooling/heating yielding energy cascade of efficient utilization using natural gas or methanol reforming yielding an energy utilization efficiency) which can be improved to ~60–90%. Nonetheless, in remote areas, hydrogen transportation via usage of other technologies including pipelines or other means of transporting hydrogen or hydrocarbons are expensive.

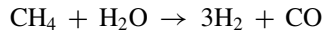
For portable fields, the small-scale PEMFCs indicate some good advantages (high energy density, longer duration, off-grid power generation, adaptability extreme environments) with portable supply potential for offgrid, light outdoor, emergency, personal usage including disasters such as floods, earthquakes as well as wearable/portable power generators with thermal signatures and low acoustics. Hydrogen

supply systems reduces the PEMFC compactness whereas systems with high hydrogen storage/density, quick charge rates, hydrogen release, improved safety, strong reversibility is crucial for increasing overall device energy density. Notable, the proton exchange membrane water photoelectrochemical cell is used for water splitting under vapor feeding [36].

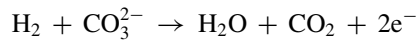
6 Molten Carbonate Fuel Cell (MCFC)

The **MCFC**, uses molten carbonate for the electrolyte and traditionally runs at intermediate temperatures of 650 °C which yields adequate ionic conductivity and fuel flexibility. MCFC handles various fuel types (versatile) including, natural gas, light alcohols/hydrocarbons and hydrogen without usage of noble metal catalysts, and can reduce amount of carbon dioxide emissions compared to other cells [40].

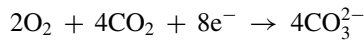
In the presence of water methane goes through steam reforming where it is added at the anode as hydrogen, carbon dioxide and carbon monoxide followed by reactions that consume hydrogen, carbon, generating at the anode electrons



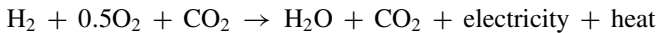
Using carbonate ions available at the electrolyte we can have



At the cathode



The overall reaction



To continue improving the MCFCs more research into fundamental of electrode processes, cathodic oxygen reduction, mechanisms underlying cell component corrosion, impact of carbonate melt wettability on MCFC materials, material stability and lowering cost for commercialization.

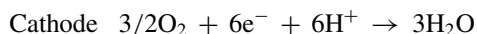
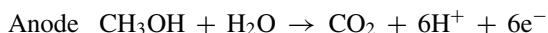
7 Direct Fuel Cells (DLFCs, DMFC, DEFC, DBFC, DCFC, DFAFC, DGFC)

7.1 Direct Liquid Fuel Cells (DLFCs)

DLFCs using liquid fuels, small organic molecules including ethanol, formic acid, methanol have low hazardous emissions, high energy-conversion efficiencies, convenient storage and transportation, can be easily mass produced with actual industrial system, for the direct conversion into electrical energy of the fuel's chemical energy with potential applications in portable electronic devices, aerospace components and electric vehicles [41, 42].

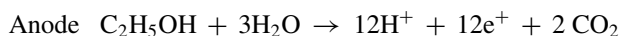
7.1.1 Direct Methanol Fuel Cell (DMFC)

The **DMFC**, using a polymer electrolyte membrane, have long lifetime, low temperature operation, clean energy source, portability whereas methanol participate in FC reactions. At the anode, methanol is reformed into carbon dioxide whereas water/steam is formed from oxygen in air at the cathode. Advantages includes high energy density of liquid methanol compared with hydrogen (gas). Disadvantages include methanol membrane cross-over, catalyst poisoning, water/methanol concentration maintenance, fuel toxicity, CO₂ removal [41]. At the anode/cathode the reactions are



7.1.2 Direct Ethanol Fuel Cell (DEFC)

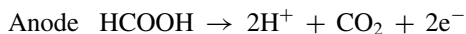
The **DEFC** is similar to the **DMFC**, using instead directly ethanol (C₂H₅OH) instead of the more toxic methanol in the FC, is easier for consumer usage, with a higher specific energy compared to methanol, can be obtained in large quantities from biomass renewable resources, absorbing as well carbon dioxide emitted into atmosphere and overcoming some of storage/infrastructure challenges. The reactions at anode/cathode are [41]



Water consumed/produced at anode/cathode. Protons transported across membrane to cathode to react with oxygen producing water whereas electrons transported using external circuit from anode to cathode.

7.1.3 Direct Formic Acid Fuel Cell (DFAFC)

In **DFAFC** oxidation occurs at the anode with formation of carbon dioxide, protons which react with oxygen at the cathode. This technology overcomes problems such as storage, low flammable chemical, safe handling and transportation of H_2 and is considered a potential automobile application [26].



7.2 Direct Borohydride Fuel Cell (DBFC)

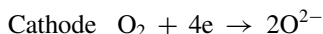
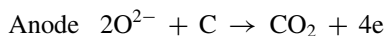
The **DBFC** use metal borohydrides (sodium and potassium borohydride) for fuels, air/oxygen/hydrogen peroxide as oxidants, with comparatively higher specific energy density compared to **DMFC**, for example. The electrochemicals can involve transport of hydroxyls and cations (H^+ , Na^+). They have potential for mobile phones/laptops and other applications. In addition, they do not require expensive platinum catalysts and have relative high power densities, reducing number/cost of stacks [31]. However the electrocatalyst influences considerable the borohydride oxidation reactions.

The main reaction for the electro-oxidation of BH_4 is as follows



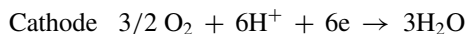
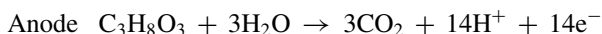
7.3 Direct Carbon Fuel Cell (DCFC)

DCFC are made of anodes and cathodes separated by electrolyte (electronic insulator, ionic conductor). Solid carbon fuel supplies the anode chamber which reacts at the electrode to form gaseous product. They can be fueled with biochar from biomass carbonization as green fuel. They yield high fuel utilization, with a wide range of fuels whereas pure CO_2 is by-product which can be sequestered [53]. The reactions at anode/cathode are



7.4 Direct Glycerol Fuel Cells (DGFC)

Glycerol is a co-product of biodiesel manufacture whose oxidation in fuel cells is operationally feasible. In **DGFCs** the fuel is fed directly to the anode whereas however, the glycerol oxidation is significantly impacted by the slow chemical reactions within the process [60].

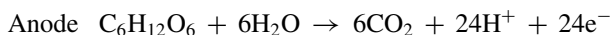


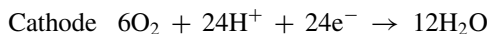
7.5 Biofuel Cells (BFCs), MFCs, EFCs

Biofuel cells (**BFCs**) utilize microorganisms or enzymes to generate electricity decomposing and catalyzing biological organic matter which can be classified as enzymatic biofuels (**EBFCs**) and microbial biofuel cells (**MBFCs**). In **EBFCs**, fuel can be oxidized using enzymes as a catalyst. The **MBFCs** generate electricity via bacteria mimicking their interactions in nature. These **MBFCs** have been used in energy recovery, wireless sensor networks, and toxic metal elimination during wastewater treatment [44, 45].

7.6 Glucose Fuel Cells (GFCs)

Glucose (renewable, easily available, cheap, abundant, non-toxic, easy to store) is an important energetic source for numerous living things which can be used at anode glucose oxidation reactions as well as ORRs at a cathode in **GFCs** with good power density and small efficiency loss with potential for implantable or wearable devices [43].





In general, based on catalysts used, GFCs can be classified as enzymatic/nonenzymatic fuel cells whereas nonenzymatic GFCs have been mostly used due to disadvantages in enzymatic GFCs including difficult long time storage of activity, unstable output power, poisoning, complex immobilization procedures despite excellent reaction rates and selectivity [45].

7.7 Regenerative Fuel Cells (RFC)

The **RFC** generates electricity and heat as conventional fuel cells and can also do water electrolysis whereas with supply of electricity to electrodes water can be split generating for storage of pure hydrogen and oxygen [5, 46]. One example is the solid oxide regenerative fuel cell (SORFC) using at high temperatures less electricity for the electrolysis process.

7.8 Hydrogen Production, Storage, Transportation

Hydrogen is world's lightest, amplest, simplest element but is not a main energy source, whereas it must be created from another source and transported for usage, functioning as a secondary energy carrier of electricity. Hydrogen has a higher calorific value compared to most fuels. Liquid hydrogen has a higher gravimetric energy density but lower volumetric energy density requiring larger tanks. It also poses some risk when used since it has a relatively low ignition temperature and is a flammable gas. It may also pass through materials inducing degradation, leaking, embrittlement in some materials due to their small molecular size. Environmental friendly, it gives up with oxidation two electrons to form water [1–67].

Notably, hydrogen, an invisible gas, has a wide color spectrum, i.e., blue hydrogen, green hydrogen yellow hydrogen, brown hydrogen, pink hydrogen. **Green** hydrogen uses clean electricity (wind, solar) for water electrolysis whereas water is split into oxygen and hydrogen without carbon dioxide emission. This technology yields, however, only a small worldwide percentage due to expensive production. Via steam reforming, **blue** hydrogen can be produced from natural gas and heated water (steam) yielding hydrogen and carbon dioxide as a by-product whereas CCS (carbon capture and storage) can be used to store this carbon. If creation of greenhouse gases are not avoided a 'low carbon blue hydrogen' is produced. Using steam methane reformation, **grey** hydrogen is produced from methane or natural gas, without capture in the process, of greenhouse gases. In contrast to green energy, **brown** and **black** hydrogen use lignite (brown) or (black) coal in the process. Nuclear energy powered electrolysis produces **pink** hydrogen. **Red** or **purple** hydrogen are nuclear-produced. **Turquoise** hydrogen uses methane pyrolysis to produce solid carbon and hydrogen. **Yellow**

hydrogen is obtained from electrolysis via solar power. **White** hydrogen occurs naturally in underground deposits from fracking. Due to high cost, the green hydrogen (water electrolysis) involves less than 0.1% of overall hydrogen generation which is expected to increase with growing demand.

Technologies for hydrogen production include AER (Absorption Enhanced Reforming), gasification (biomass coal, coke, oil), supercritical water gasification, SMR (Steam Methane Reforming), ATR (AutoThermal Reforming), water electrolysis (Alkaline, PEM, SOEC (Solid Oxide Electrolyser Cell), microbial), chemical looping gasification, pyrolysis, chlor-alkali electrolysis, molten metal methane splitting, fermentation (dark, photochemical), direct photobiological splitting, thermochemical cycle, methanation, plasma methane splitting, solar thermal dissociation, methanol synthesis, ammonia synthesis and FT (Fischer–Tropsch) synthesis [1–67].

Despite numerous technologies, actually, most hydrogen is created by catalytical interaction of methane with heated steam creating hydrogen, carbon monoxide and some carbon dioxide and elimination of impurities. The reaction of carbon monoxide, steam with catalyst next creates hydrogen and carbon dioxide. Hydrogen can be produced as well from fossil fuels (gasoline, propane, coal) using the steam reforming process. Hydrogen is also produced by water electrolysis whereas it is divided into hydrogen and oxygen (only by product) inside an electrolytic cell. Photoelectrochemical water splitting, specialized semiconductors and sunlight can be used to produce hydrogen.

Notable, nano materials (NP) including (Ag, Au, TiO₂, Fe NP) play important roles in nano-photo catalysis production of hydrogen. Biohydrogen production includes dark and photofermentation (organic substrates converted to hydrogen and CO₂ via photosynthetic bacteria that captures light) as well as indirect and direct biophotolysis (dissociation of water under solar radiation forming oxygen and hydrogen) using algae and bacteria. Ammonia can be decomposed to nitrogen and hydrogen by heating to gaseous state whereas it is split in a reactor into nitrogen and hydrogen.

Notable, semiconductor materials such as carbon nitride and cadmium sulfide are excellent candidates for photocatalytic hydrogen production due to their narrow band gap (absorb visible light) once quantum dots are used for boosting charge separation efficiency which is important to enhance the photocatalytic hydrogen production [66].

Hydrogen could be produced and transferred to vehicles at a central station. Usage otherwise of a refuelling network and a central production plant requires hydrogen transport facilities to deliver hydrogen to the stations. Other important factors, however, influencing decision-making, are the hydrogen production cost and reliability of on-site productions. Notably, concerns during production, transmission, storing, utilisation and distribution as well as cost, safety and reliability offers limitations of hydrogen energy technology usage in transport.

It is also of interest to analyze advantages/disadvantages of maintaining hydrogen in the liquid, compressed or solid state. We note that compressed H₂ is a highly efficient, mature technology that uses however, costly materials, has high pressure leakage, issues in fast refuelling and small quantity storage. Liquid hydrogen affords

high quantity and storage efficiency, low storage pressure, low temperatures, evaporation losses. The solid state is a simple and safe process offering high volume and weight capacity but is a maturing technology with increasing quantity storage and still high materials requirements. Solid-state nanoparticles have received considerable attention due to their adaptable characteristics, large surface area and potential for reversible uptake/release of hydrogen.

Technologies for hydrogen storage include liquefied hydrogen, compressed hydrogen, cryogenic liquid, organometallic skeletons, porous materials, metal hydrides, carbon based materials, liquid organic hydrogen carbons, liquid organic hydro-carbons, high entropy alloys, underground caverns, and pressurized tanks. A recent publication indicated that underground storage shows lowest storage cost, followed by compressed hydrogen and liquid hydrogen storage [20]. There is however ongoing growing progress regarding material storage. In the solid form hydrogen can be stored using materials at near ambient pressure/temperatures addressing safety concerns for on-board transportation underlying solid state storing as a technically advantageous/secure option.

In the materials hydrogen is stored via (a) chemical chemisorption (adsorption) with energies between -4.0 and -2.0 eV (high energy dehydrogenation requirements) and dissociates into individual atoms, (b) preferred reversible physisorption, via van-der Waals interactions, with hydrogen in gas form whereas it is easier to recover hydrogen but at high operating temperatures. In a third suggested quasi-molecular bonding mode (between chemisorption and physisorption) $\sim (-0.2$ to -0.7 eV) there should be charge transfer between hydrogen molecules and storage material yielding hydrogen molecules with good potential for storage at ambient temperatures and pressures.

Notwithstanding, storage capacity/dynamics of solid-state hydrogen is strongly influenced by the materials surface interactions whereas solid-state nanoparticles, including carbon based materials and metal hydrides offer great promise for hydrogen storage due to their adaptable characteristics including large surface area and reversible hydrogen release/uptake with however slow kinetics, stability and high operating temperatures. Carbon based materials including graphene and nanotubes indicate comparatively lower storage capability but excellent surface area and reduced cost with however propensity for time deterioration and synthesis difficulties. Heteroatoms doped on quantum dots can form with hydrogen strong interactions with an increased adsorption capacity which is promising for hydrogen storage. Notable, quantum confined graphene based materials called, i.e., poly aromatic hydrocarbons (PAHs) or graphene quantum dots demonstrate increased hydrogen storage capacity. It has been indicated that heteroatoms doped on QDs form with hydrogen strong interactions which increases capacity for adsorption offering increased hydrogen storage. Two-dimensional quantum dot materials are also notable efficient for hydrogen storage [66].

Technologies for transportation include LOHC tanks, H_2 injection into natural gas grid, H_2 pipeline, hydrogen refuelling stations, cryogenic tanks, trucks, high-pressure tanks. Hydrogen refuelling stations will be important for commercialisation of hydrogen powered automobiles. Carbon-free ammonia (NH_3), is a good alternative

carrier for H_2 with advantages such as high energy density, enhanced safety, ease of detection, higher liquefaction temperature, simplified storage, transportation and green/sustainable/clean energy source.

Notable for the transport technology, the pipelines are highly efficient, offering large quantity transport with however high infrastructure cost. The tube (trailer/rail) is suitable for small scale, small distance delivery, but is however an energy intensive process. The highly efficient tanker (truck/rail) is an energy intensive process with boil-off losses for liquid. Usage of ships offers high capacity transport with boil-off losses for liquid and is expected to be commercially available this decade.

Technologies for hydrogen utilization include Scooters, Ships, Ferrys, Trucks, Trams, Vehicles, Turbines, Combined Heat and Power Systems, Aircrafts, Submarines.

8 Nanotechnology (NT)

NT is the science of synthesis/development/characterization of materials at nanometer scales with at least one dimension between 1 and 100 nm whereas these nanomaterials (NMs) indicate enhanced properties compared to bulk including rich/surface/interface effects, high surface-to-volume ratios, distinct chemical/physical properties, higher electronic/ionic conductivities, improved electrocatalytic activity, less expensive materials, possible quantum limits and based on spatial confinement degree can be placed in four major classification, i.e. zero, one, two, three and four dimensional materials [1, 46–52].

Effectively, the usual physical science rules may not apply if the material only has one or more nanoscale dimensions whereas their reactivity rate, strength and electrical conductivity could surprisingly increase. The electrocatalytic activity of nanofiber electrodes is for example greatly enhanced. Usage of nanotechnology could also contribute to making FCs more environmental sustainable, friendly, via green manufacturing processes [1].

In general, the usage of nanoengineered materials in fuel cells technology has showed increasing interest due to the influence of these nanostructured materials on the fuel cells performance introducing innovations able to overcome challenges for commercialization.

The usage of nanomaterials as FCs components (catalysts, electrodes, membrane/electrolytes) can avoid restrictions such as expensive materials, fuel crossover and increase performance, overall efficiency via the nanomaterials unique size effect and high surface area, enhancing effectively the active sites for electrode reactions.

In general a larger number of atoms become available as active sites on the surfaces with decreasing particle size yielding highly active nanocatalysts including metal nanoparticles/metal oxides, exfoliated clays, carbon-based nanomaterials (fullerenes, graphenes, carbon nanotubes) used directly as nanocatalysts or as support for metals. Noble metal decorated porous carbon nanoarchitectures with large accessible surface areas, reliable structural stability, three-dimensional cross-linked porous networks,

homogeneous dispersion (of noble metals), and outstanding electrical conductivity exhibit exceptional electrocatalytic performances [51].

Although graphene/graphene oxide-based catalysts show promise in fuel cells, there are concerns regarding affordability leading to more environmental friendly, cost effective preparation of carbon nanomaterials from plastic waste. Nitrogen-doped graphene oxide decorated core-shell nano-structured catalysts were used for PEMFCs enhancing efficiency, stability, electrochemical surface, and durability which is supported by DFT-based first principles simulations [55]. Carbon dots (zero dimension less than 10 nm in size) modified MFCs indicated higher maximum power density and open-circuit voltage [47].

Quantum dots (QDs), are three dimensional semiconductor nanomaterials, few nanometers in size, differing from those of larger particles, with quantized energy levels, quantum confinement, increasing bandgaps with decreasing size, indicating advantageous properties such as tunable band gaps. Doped quantum dots graphene carriers indicated promising improved electrocatalytic ORR for PEMFC. Graphene QDs have also been used as metal-free highly active electrocatalyst in alkaline electrolyte for ORR. MoS₃ based QDs and Ag₂S QDs cocatalysts with graphitic nitride, for hydrogen evolution, indicated a very high enhanced hydrogen yield. Enhanced electricity was generated from green algae using quantum dots decorated photoanodes in bioelectrochemical fuel cells [46–52].

9 Artificial Intelligence for Fuel Cell Devices

The usage of simulation tools/artificial intelligence (AI) to reduce time/cost of FCs testing/optimization/efficiency is increasingly important. Energy management in fuel cell/hybrid vehicles is challenging whereas prediction, data-driven control, optimization and vehicle to everything (V2X) strategies are appropriate [53–63].

The neural networks (NN), random forest, genetic algorithm, particle swarm optimization, support vector machine and extreme learning machine are among the AI tools that can be used for optimization of FCs operational parameters. Effectively, the AI algorithms are providing fast data collecting, accurate diagnosis becoming beneficial, popular tools for simulating FCs systems nature and appropriate for optimizing their essential operational parameters [53–63].

FCs normally work within a limited range of humidity and confined temperature resulting in defects, difficult regulations and deterioration. Approaches for diagnosis include restricted input information (rapid, unobtrusive) whereas advanced characterization can be more accurate with invasive/delayed tests. AI shows promise for both quick data collecting and accurate diagnoses [57].

Among the AI tools, neural networks (NN) is inspired by the human brain, which shows computers how to process data, i.e. machine learning, using connected neurons/nodes in layered structures. Neuron activity can be expressed in a binary way with obvious connection to on/off behaviour of bits in computer/machines. AI NNs (ANN) can handle performance faults prediction, variations of catalysis and

electrochemistry, non-linear complex processes and are suitable to foster integration of electrochemical devices in fuel cell systems, coping with variable operating systems and facilitating management.

The particle swarm optimization (PSO) tool optimizes a problem by computation, iteratively improving population of candidate solutions (particles) which are moved in search space using mathematical formula (particle velocity, position). The Hopfield optimization method uses one layer of fully connected recurrent neurons via converging interactive processes. The back propagation (BP) tool uses a feed-forward, multi-layer network trained according to error BP propagation algorithms [53–63].

Among FCs, variability of SOFC applications and non-linearity of its electrochemical systems supports the implementation of AI techniques for optimization of usage of this technology. Among these techniques, based on the PSO algorithm, the NN method has been used to study current/voltage characteristics of SOFC for marine applications (ships) and compared with Hopfield and BP algorithms with significant potential in SOFC optimization whereas the voltage/current characteristic curves under different fuel flow-air volume ratios as well as stack models electrical behaviour could be accurately predicted. ANNs have also been used on experimental data sets to forecast the SOFC polarization behaviour [53–63].

Increasingly important for mapping performance as well as fault diagnosis related to fuel cells, ANNs can handle prediction, performance variation, integration of electrochemical devices in energy systems, facilitating management and coping with the operating conditions and variability of systems such as gas-turbines, engines, heat exchangers, cogeneration units with excellent relevance to performance mapping [53–63].

Supported by AI for green urban transport, a hydrogen fuel cell prototype vehicle was developed focusing on a three-wheeled prototype, integrating energy storage systems via battery, solar and hydrogen-powered drives. For fuel cell vehicle control and detecting the environment, AI, via reinforcement learning, uses energy management to select energy source (optimal) for operations using layers of maps (high-fidelity). The environment-sensing technology is designed to distinguish, detect, select environmental elements using image segmentation camera images providing feedback to user via displays [54].

Carbon rich biochar/biomass (produced from carbonization) have potential for usage in DCFCs to produce sustainable electricity. There are, however, gaps in knowledge regarding electrochemical mechanisms of their transfer phenomena in the complex microstructure of the different DCFC zones. This challenge can be addressed using AI algorithms allowing prediction/optimization of key operating parameters of DCFCs using the available experimental data [53].

AI algorithms were used for boosting power density of PEMFCs. Considering factors such as membrane compression, pressure, relative humidity, the output power density of the PEM-FC was simulated using an ANFIS (Adaptive Neuro-Fuzzy Inference System) model which establishes a nonlinear mapping of inputs using membership functions within the layers. The Salp swarm algorithm (SSA) could be

used to obtain input control optimal parameters which are treated during optimization process as decision variables in order to maximize the power density [60].

ANN has been applied to fault diagnosis of hybrid batteries/fuel cells/electric vehicles, integration of electrochemical devices, variable/wide range of operating conditions/systems/applications, fault diagnosis, energy management, integration in complex systems, grid-connected power plants, subcomponent-level modelling, power density, voltage behavior of low/high temperature FCs, heat exchange, fuel performance/variation, wide range of SOFC applications, effects of pressure, relative humidity, membrane compression, fault diagnosis, defects, hastened deterioration, maximizing output power density considering factors such as pressure, membrane compression, relative humidity, temperature impacts on the electrolyte conductivity, chemical kinetics of internal reforming and gas shift reactions and fast evaluation of numerous vital/dynamic possibilities of interest.

Conclusions Artificial intelligence, tunable perovskites and advanced nanotechnological quantum dot materials are underlined to play important roles in various phases of the hydrogen production, storage, transportation and anode/cathode/electrolyte hydrogen fuel cell technology.

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Two-Dimensional Materials Applied to Hydrogen Storage



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Abstract Hydrogen as a clean and efficient energy has received much attention due to its potential to address global energy challenges. However, efficient storage methods continue to be a critical bottleneck and, therefore, a challenge for the scientific community. In this sense, several material classes have been proposed as possible candidates for this finding. Among them, two-dimensional (2D) materials have emerged as promising candidates for hydrogen storage due to their unique properties, including high surface area, tunable electronic properties, and chemical stability. In this chapter, the recent advancements in 2D materials for hydrogen storage were highlighted. Hydrogen adsorption mechanisms on 2D materials and their performance, such as gravimetric and volumetric hydrogen storage capacities, kinetics, and reversibility, were emphasized. The analysis and design of efficient devices to store H₂ are validated through computational simulations and available experimental data, which enables a comprehensive analysis of hydrogen storage on 2D materials.

Keywords 2D materials · H₂ · Mxene · Graphene · TMD

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1 Introduction

Recently, a notable advancement in renewable energy technologies has been noticed [1–7]. Solar photovoltaic cells and wind turbines are illustrative examples. The perovskite solar cells [8] have increased energy conversion efficiencies, making solar power more accessible and affordable. Wind turbine design enhancements have led to more efficient turbines, unlocking an increased potential for electric energy generation [9]. A fact that illustrates wind-based energy advancement is the development of offshore wind farms to take advantage of wind patterns over the ocean [10].

Energy storage technologies also play a key role in the energetic transition. Lithium-ion and solid-state batteries are reported as more efficient and cost-effective, demonstrating better storage of intermittent renewable energy [11, 12]. These batteries have been shown to be critical for equilibrating the supply and demand relation, especially in grid applications and electric vehicles. Alternative solutions, such as flow batteries and supercapacitors, have been studied and can offer new possibilities and mechanisms for energy storage [13].

Carbon capture, utilization, and storage (CCUS) technologies actually represent important devices for decarbonization, particularly for hard-to-abate industrial sectors [14]. The main function of this mechanism is based on CO₂ emissions captured from power plants and industrial processes and later storage for obtaining valuable products like chemicals, fuels, and building materials [15, 16]. Improved capture methods are reported as recent advancements in CCUS. Between them are amine-based [17] and solid [18] sorbents. Enhanced geological storage techniques and monitoring systems are also responsible for the safety and reliability of CO₂ storage [19].

In the movement towards sustainable energy, the use of hydrogen (H₂) as a fuel is considered a promising approach. Hydrogen is recognized as a versatile energy source and can be used in different areas. In transportation, hydrogen powers fuel cells in electric transport, thus replacing gasoline and diesel [20]. Hydrogen has been reported as an alternative to the use of natural gas in industrial processes such as steel and cement production, glass manufacturing and refineries [21, 22]. Moreover, hydrogen can be mixed with natural gas for heating and power generation [23], gradually decarbonizing these sectors.

Large-scale processes are necessary in a hydrogen-based economy to supply the energetic demand and reduce the environmental impacts. Hydrogen production can be classified as grey, blue, turquoise, and green, given criteria such as development stage, required feedstock and resources (such as natural gas, oil, coal, biomass, and water), and GHG emissions [24, 25], as illustrated in Fig. 1.

Grey hydrogen is mainly produced via steam methane reforming (SMR), a process that demands fossil resources such as natural gas and, consequently, is associated with CO₂ emissions [27]. In the SMR process, the methane with steam is submitted at high pressure and temperature to obtain hydrogen and CO₂. An efficiency of up to 90.7% on a higher heating value basis can be obtained [28]. However, this methodology

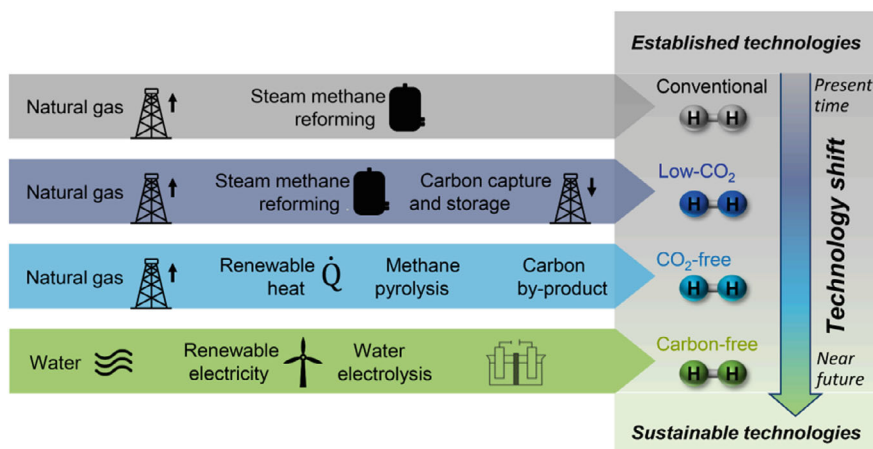


Fig. 1 Common differentiation of hydrogen production pathways into conventional, low- CO_2 , CO_2 -free, and carbon-free production routes and frequently associated colors. Adapted with permission from [26]. Copyright 2024 Elsevier

results in significant CHG emissions. A simplified scheme of a modern SMR plant is shown in Fig. 2.

Blue hydrogen follows the same principle as grey hydrogen production, as represented in Fig. 3. The difference refers to the implementation of carbon capture and storage (CCS) technologies [30]. During the process, the CO_2 can be stored in geological formations or utilized in other industrial applications. The CCS mechanism mitigates CO_2 emissions significantly. At the same time, it is responsible for reducing SMR efficiency.

Turquoise hydrogen is generated via methane pyrolysis (MP), as denoted by Fig. 4. In this process, the methane is decomposed into hydrogen and solid carbon without

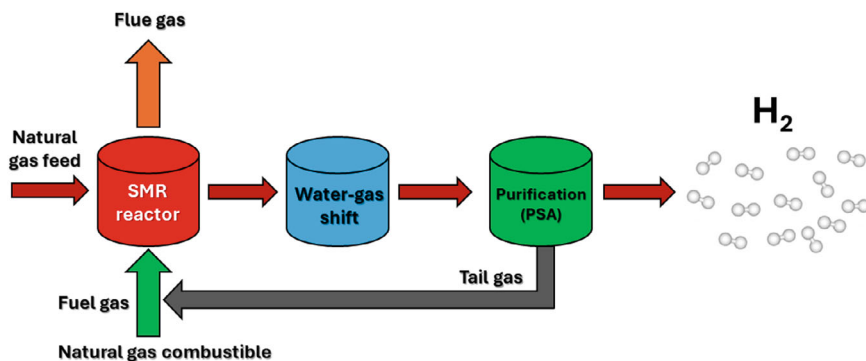


Fig. 2 Diagram of a typical modern SMR plant. Adapted with permission from [29]. Copyright 2024 Elsevier

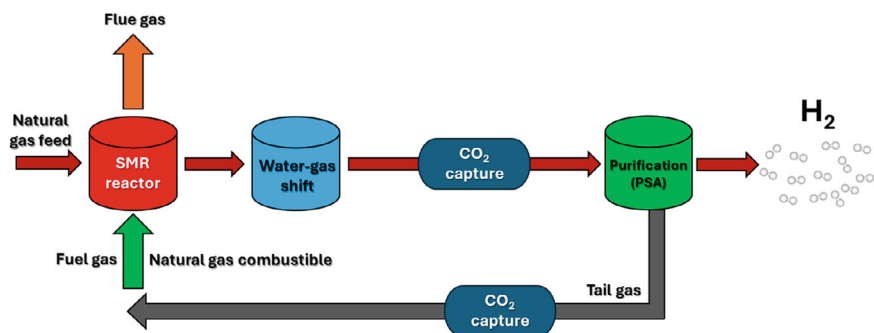


Fig. 3 CO₂ capture during H₂ production by steam methane reforming. Adapted with permission from [26]. Copyright 2024 Elsevier

directly producing CO₂ emissions [31]. To apply this technique, high temperatures and various catalysts are required to be employed to improve efficiency [32, 33]. The solid carbon by-product can be used in industrial applications, thereby increasing economic value. Although handling the solid carbon by-product and expanding the process to commercial levels remain problems, turquoise hydrogen is viewed as a viable bridging technology for generating hydrogen while lowering carbon emissions.

Green hydrogen production uses renewable energy sources such as wind, solar, or hydroelectric power [34, 35]. The fundamental element is the splitting of water into hydrogen and oxygen, which results in a completely clean and sustainable process without emitting CO₂ [36]. A representation of a polymer electrolyte membrane for water electrolysis is obtained from Fig. 5. A challenge for large-scale production of green hydrogen is related to the high cost of electrolysis technology [37]. Currently, the cost of manufacturing green hydrogen by electrolysis is still higher than SMR technique with carbon capture. Numerous efforts have been made to overcome this problem, as can be noticed in the diverse scientific publications that explore different strategies and materials for efficient and cost-effective water splitting [38, 39].

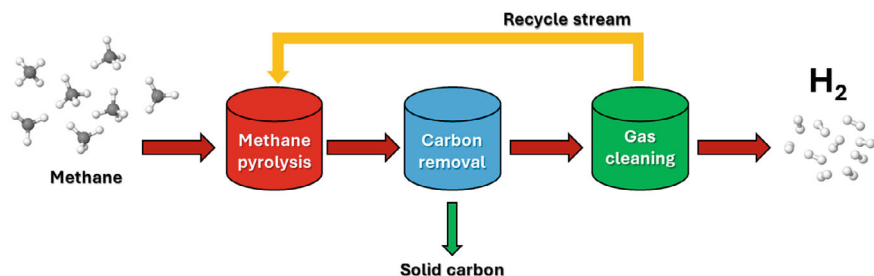


Fig. 4 Diagram of H₂ production by methane pyrolysis. Adapted with permission from [26]. Copyright 2024 Elsevier

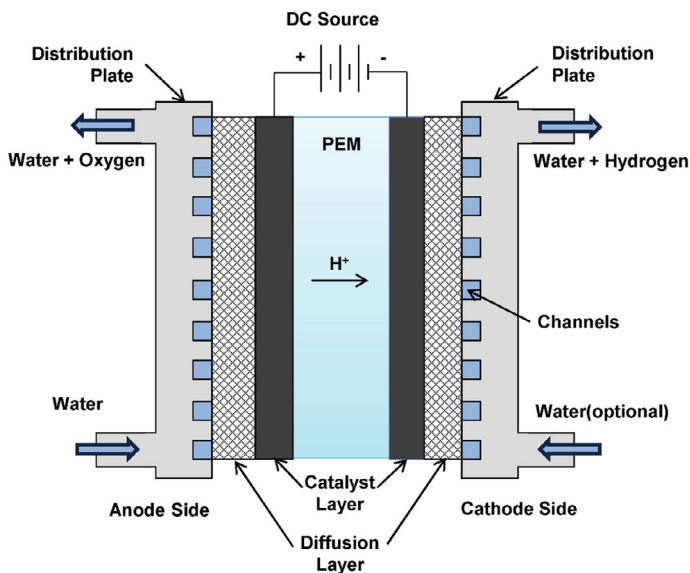


Fig. 5 Schematic representation of a polymer electrolyte membrane for electrolysis cell. Reproduced from [40] with CC-BY licensing permission

Another factor that can be considered is related to hydrogen storage, which is a critical point and represents an obstacle to its large-scale adoption. Hydrogen has a low energy density by volume, requiring efficient storage solutions to be practical. Furthermore, hydrogen storage technologies contribute to the integration of renewable energy sources, such as wind and photovoltaic power generation, further highlighting the importance of efficient hydrogen storage systems in the energy landscape [6]. The traditional methods of H₂ storage include compression [9, 10] and cryogenic hydrogen [11, 12] or both simultaneous strategies (cryo-compression) [13].

Recently, adsorption in solids has emerged as a promising alternative, enhancing the storage rate under adsorption and desorption processes [14, 15]. In this context, the US Department of Energy (DOE) has defined a target for hydrogen storage systems to achieve a gravimetric density of 5.5 wt% hydrogen [41]. Achieving this benchmark is crucial to the efficiency of hydrogen storage solutions, especially for applications like fuel cell vehicles. An illustration of the methodology is shown in Fig. 6. In this mechanism, the H₂ molecules are adsorbed on the surface of a solid material through weak van der Waals forces. This process requires low temperatures and moderate pressures. Physical adsorption is highlighted as the main strategy of hydrogen storage for two-dimensional (2D) materials. In particular, 2D materials have been targeted due to their high surface area and tunable properties.

From a technological perspective, 2D materials provide high surface-to-volume ratios [43], which maximize the interaction sites for hydrogen adsorption. Scientifically, the ability to manipulate the electronic properties and surface chemistry of

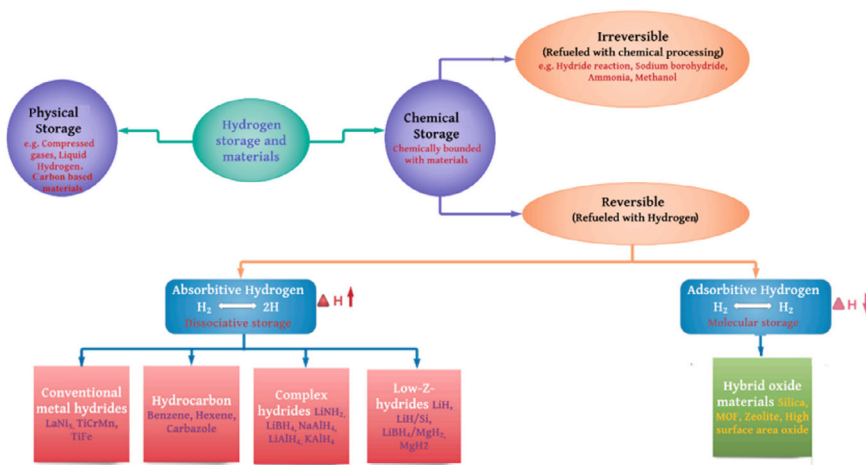


Fig. 6 Different methodologies have been reported for hydrogen storage. Reprinted with permission from [42]. Copyright 2024 Elsevier

2D materials enables precise control over hydrogen binding energies. This control is crucial for achieving the right balance between hydrogen adsorption and desorption, which is essential for practical storage systems. Moreover, the lightweight nature of 2D materials contributes to higher gravimetric storage capacities, making them more efficient compared to traditional storage methods. By exploring these materials, sustainable and economically viable hydrogen storage solutions can be achieved. In this way, this book chapter discusses the role of hydrogen storage on 2D materials in the energetic transition. The properties and potential applications of these materials in hydrogen storage are not only scientifically intriguing but also hold the key to addressing pressing energy and environmental challenges.

2 Hydrogen Storage on Two-Dimensional (2D) Materials

It is known that physisorption is, in general, faster and more effective than chemisorption [44, 45]. Physisorption occurs via weak van der Waals interactions, which makes the adsorption/desorption processes more likely to occur. These weaker interactions maintain the substrate integrity and offer the possibility of reuse. This process is exothermic, and its adsorption enthalpy can be close to -20 to -40 kJ/mol in several cases. Differently, the chemisorption causes the chemical degradation of the adsorbent and reduces its lifespan for hydrogen storage. Therefore, for hydrogen storage purposes, the adsorption energies between -0.10 and -0.60 eV are desired for reversible H_2 storage at room temperature and ambient pressure, a process in the range of physisorption and chemisorption mechanisms.

Furthermore, physisorption is more selective, enabling it to adsorb certain substances over others preferentially [46], which can avoid the introduction of undesired substances. Here, the concern is the amount of hydrogen packaged, i.e., the highest volumetric density is suitable, and the advantage of 2D materials is the high superficial area as compared to other materials.

In this sense, we cannot fail to begin our explanation without talking about one of the most cited materials in the recent scientific literature. In our discussion, we spotlight graphene [48] due to its high surface area, exceptional mechanical strength, and electrical conductivity [47]. It can adsorb hydrogen molecules on its surface and within its layered structure. Another important class of materials in the scientific spotlight is the transition metal dichalcogenides (TMDs), such as MoS_2 and WS_2 [48]. These are considered promising due to their tunable electronic properties and layered structures, which may lead to hydrogen intercalation. In the past decade, the advent of Mxenes has piqued researchers interest, particularly in the field of energy [49]. Mxenes stand out due to their layered structure and the remarkable ability to fine-tune interlayer spacing, offering huge flexibility. The extensive experimental investigations into Mxenes, coupled with numerous theoretical reports, especially in the context of hydrogen storage, underscore the immense promise this 2D material holds for hydrogen production.

Therefore, this topic is devoted to presenting the main properties and applications of graphene, TMDs, MXenes and other particular 2D materials from the physical and chemical point of view, incorporating both theoretical and experimental studies, where available. Also, it was discussed how their features become physisorption, one of the most recently reported mechanisms for hydrogen storage. Diverse strategies such as doping, decoration, heterostructure design, and number of layers control have been employed to engineer these materials at the atomic level and optimize the hydrogen adsorption and release kinetics.

2.1 Graphene for Hydrogen Storage

Graphene and its derivatives are considered a promising alternative for the next generation of material used for hydrogen storage. Graphene is characterized by its planar monolayer structure derived from graphite, featuring a perfect 2D crystal lattice of sp^2 -hybridized carbon atoms arranged in a honeycomb pattern [50]. Several fields, including chemistry, physics, and materials science, have extensively explored graphene honeycomb lattice for a broad range of applications [51–56]. Key attributes such as a high surface-to-volume ratio [57], specific surface area [58], exceptional electron mobility [59–61], mechanical strength [62], and excellent electrical [62–64] and thermal [65] conductivities make graphene one of the most promising materials of the twenty-first century.

Its remarkable properties open up possibilities for electrochemical energy storage in supercapacitors [66, 67], batteries [68–70], catalysts [71, 72], and, particularly, based hydrogen systems [73–75]. Its two-dimensional nature allows for efficient

access to adsorption sites, enhancing hydrogen storage capacity. Functionalizing graphene with dopant atoms or functional groups can enhance its hydrogen adsorption properties, improving both storage capacity and kinetics [76–81]. However, challenges remain, such as achieving high hydrogen storage capacities at ambient conditions, overcoming diffusion limitations, and ensuring reversible hydrogen adsorption and desorption.

A recent approach involves decorating graphene with alkali elements by Ataca et al. [82] has reported that Li decoration can achieve a gravimetric density above 10 wt%, reported a gravimetric density of 12.8 wt%, with each Li adsorbing at four H_2 molecules. On the other hand, Du et al. [83], employing the Li-decorated porous graphene, reached a gravimetric capacity of 12 wt%. The decoration of holey graphene (C_2N) nanosheets with magnesium (Mg), sodium (Na), and potassium (K) has also been explored [84]. The metal atoms are strongly bonded to the monolayer, preventing cluster formation. The findings indicate that metal decoration enhances hydrogen adsorption, with storage capacities of 5.5 wt% for Na and 6.9 wt% for Mg.

Another innovative approach involves the use of super alkali cluster decoration. This methodology aims to prevent metal clusterization and provide more adsorption sites for H_2 molecules. Qi et al. [85] investigate the influence of NLi_4 cluster decoration on graphene for hydrogen storage, as can be seen in Fig. 7. Their findings demonstrate that NLi_4 clusters can be stably anchored on graphene. Due to their larger size compared to single Li atoms, each NLi_4 cluster can adsorb up to 10 H_2 molecules, achieving an impressive storage capacity of 10.75 wt% with an adsorption energy of -0.21 eV/ H_2 , indicative of physisorption. Thermodynamic analysis suggests that this system enables reversible hydrogen storage at ambient temperatures.

Another promising solution for hydrogen storage in graphene requires the combination of metal decoration and doping. For instance, N-doped defective graphene

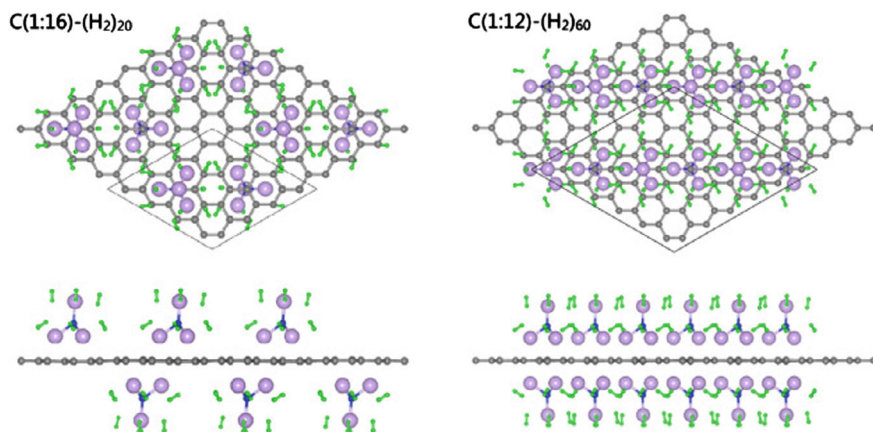


Fig. 7 Optimized configurations of NLi_4 decorated graphene for the adsorption of H_2 molecules in the configurations C(1:16) and C(1:12). Reproduced with permission from [85]. Copyright 2024 Elsevier

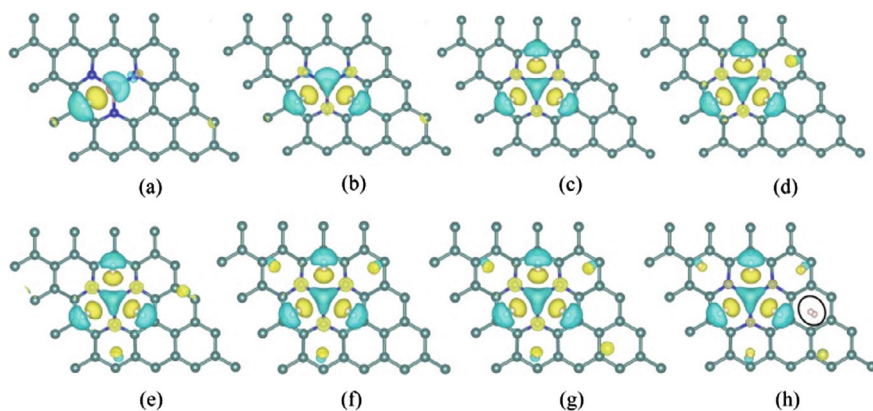


Fig. 8 The charge density difference of several H_2 molecules absorbed on the N-doped defective graphene. The plot uses the yellow color to indicate charge accumulation, while the cyan color indicates charge loss. Reproduced with permission from [86]. Copyright 2024 Elsevier

decorated with Mg atoms has recently been used as a platform for hydrogen storage [86], as denoted in Fig. 8. The authors highlight that functionalization with Mg, N atoms, and vacancy defects transforms graphene from a semi-metal into a metal. The findings demonstrate that this system possesses excellent thermodynamic stability, with adsorption of seven H_2 molecules per site and adsorption energies ranging from -0.21 to -0.15 eV. The polarization of H_2 and orbital hybridization between H atoms and the substrate explains the adsorption mechanism. Desorption temperatures above 206 K suggest this system is a potential alternative for reversible hydrogen storage materials.

The design of heterostructures based on graphene has been largely explored to achieve improved hydrogen storage capacities [87, 88]. In recent research, Kumar et al. [89] investigated hydrogen capacity on the β 12-borophene monolayer and β 12-borophene/graphene bilayer heterostructure, as shown in Fig. 9, to enhance substrate properties for hydrogen physisorption. The outcomes show that H_2 adsorption energies vary between the monolayer and bilayer heterostructure. Synergetic effects related to the interlayer arrangement in the heterostructure are responsible for the enhanced H_2 adsorption. With Li decoration, an optimal gravimetric density of 13.64 wt% can be achieved at ambient conditions.

Recently, a novel in-plane hybrid structure of hexagonal boron nitride (h-BN) and graphene was reported as a promising platform for hydrogen storage [90]. In this structure, represented in Fig. 10, carbon-boron and carbon-nitrogen interfaces occur, and different BN and graphene concentrations were studied to determine the hydrogen storage potential. As expected, the band gap energy reduces with the increase in graphene concentration, while the increase in the BN concentration is responsible for increasing the band gap energy. It is worth noting that the BN and graphene structural properties, such as bond length and bond angles, remained

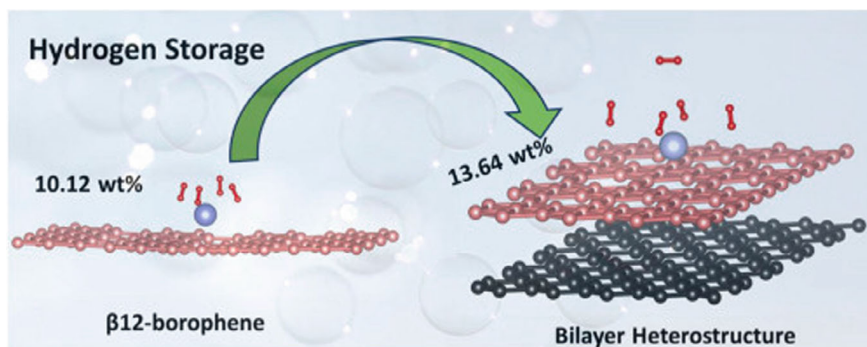


Fig. 9 Illustrative scheme comparing the H_2 storage on Li-decorated $\beta 12$ -borophene monolayer and $\beta 12$ -borophene/graphene heterostructure. Reprinted with permission from [89]. Copyright 2024 Elsevier

the same for both graphene and boron nitride hybrid systems. The pristine C-B-terminated structure showed an average adsorption energy of -0.046 to -0.076 eV/ H_2 without field influence and upon dual-site hydrogen molecules addition. Theoretical hydrogen capacities ranging from 10.18 to 10.38 wt% were achieved. When an external electric field is applied, the adsorption energies increase linearly due to the polarization of the adsorbed H_2 molecules.

An alternative proposed for a graphene-based system for hydrogen storage consists of the intercalation of Pd atoms between two graphene monolayers, culminating in the sandwich graphene-Pd(T)-graphene (G-Pd(T)-G, as named by the authors) [91]. A binding energy of 4.16 eV for the interaction of Pd with the graphene bilayer was reported, which is higher than the cohesion energy of Pd atoms, namely

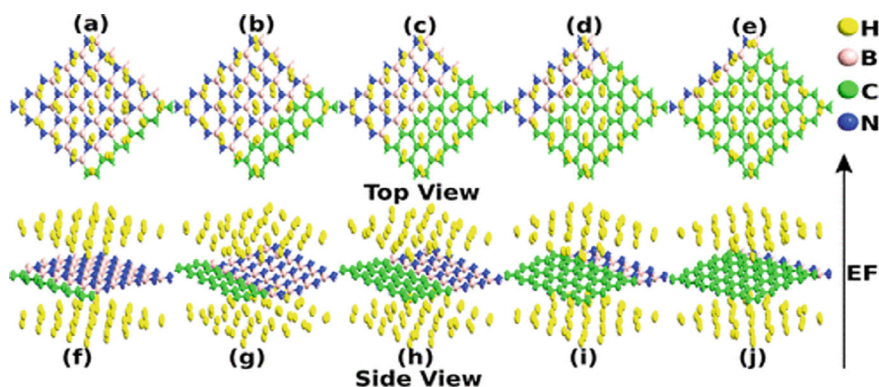


Fig. 10 Top and side view of the minimum energy configurations of H_2 molecules adsorbed for several compositions of in-plane h-BN/graphene heterostructure. Atoms are labeled with respective colors and are presented on the right side. Reprinted with permission from [90]. Copyright 2024 American Chemical Society

3.89 eV. This effect directly impacts the adsorption properties by preventing the clustering of Pd. Finally, it was found that three H_2 molecules can be successfully absorbed on the G-Pd(T)-G, with an average adsorption energy of 0.22 eV. Figure 11 illustrates the systems analyzed, where it can be noticed that the addition of H_2 molecules increases the interlayer distance. The adsorption of four H_2 molecules was also evaluated, obtaining an adsorption energy of -0.28 eV, which denotes the instability of the system with more than three H_2 molecules. It is worth noting that in this work, the signs of cohesive and adsorption energies are inverted in relation to the usual convention. This difference is due to the specific definition utilized by the author for these energy calculations, which does not affect the overall discussion.

At the present moment, only the physisorption process has been discussed. However, chemisorption mechanisms also can be successfully employed with higher gravimetric densities. In 2021, Morse et al. [64] obtained a hydrogen storage capacity of 7.7 wt% for the chemically hydrogenated graphene, which is illustrated in Fig. 12. This system confines hydrogen in the form of a chemically bound solid, resulting in high volumetric density. Via thermogravimetric mass loss analysis (TGA-MS), the authors demonstrated that the thermal decomposition of hydrogen begins at temperatures above 400°C , releasing H_2 molecules as hydrogen gas. A peak release is observed at 480°C , with the baseline returning to normal at 625°C . An interesting analysis was performed to understand the correlation between the temperature and

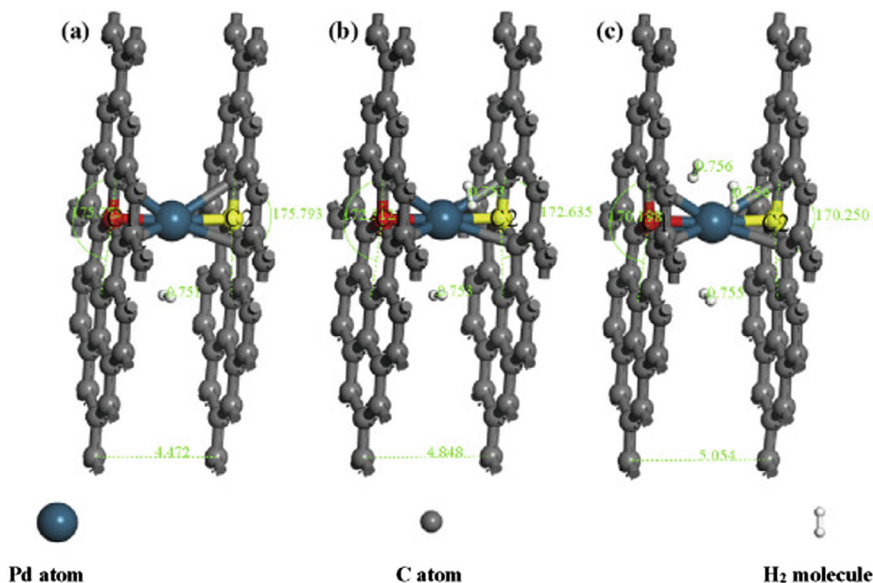


Fig. 11 Panels a, b, and c display the geometric structures of the system following the adsorption and relaxation of 1, 2, and 3 H_2 molecules, respectively. Reproduced from [91] with CC-BY licensing permission

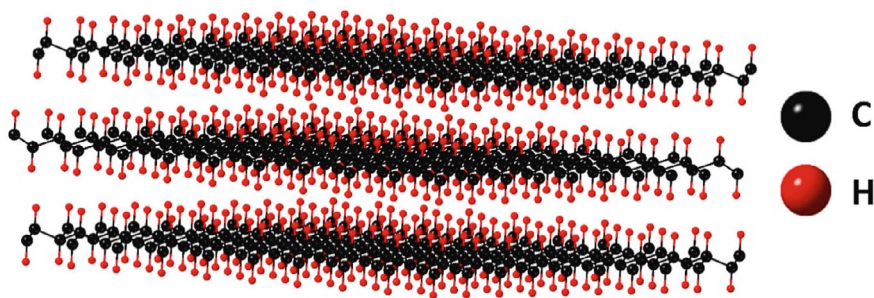


Fig. 12 Idealized bulk structure for pristine, multilayer graphene. Black spheres represent carbon atoms, and red spheres represent hydrogen atoms. Reprinted with permission from [92]. Copyright 2024 Elsevier

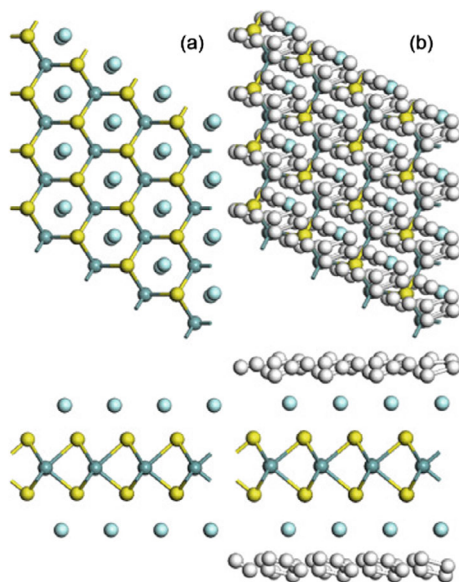
partial hydrogen gas release. Samples were individually heated in a sealed and evacuated pressure reactor at several temperatures and time scales, then removed and characterized by TGA-MS. The data demonstrate that when the samples were heated at 450 °C or below, only a fraction of the stored hydrogen was lost. When subjected to an isothermal exposure at 450 °C, a significant amount of hydrogen was lost. These results show that both temperature and heating time can be adjusted to release the suitable hydrogen amount and control the flux.

2.2 Transition Metal Dichalcogenides (TMDs) for Hydrogen Storage

Transition metal dichalcogenides (TMDs) represent another class of interesting 2D materials that have been extensively employed in hydrogen storage [93, 94]. This material family comprises 40 distinct compounds, which are characterized by the general formula MX_2 , where **M** stands for a transition metal, and **X** denotes a chalcogen element, as can be seen in Fig. 13a. TMDs are characterized as lamellar compounds with strong in-plane bonding and weak Van der Waals interlayer interactions type, which allows the isolation of individual layers [95]. These layers typically consist of a transition metal atom layer sandwiched between two chalcogenide layers [96].

TMDs exist in various structural phases, with three primary polytypic forms distinguished by the spatial arrangement of the transition metal atoms, as shown in Fig. 13b. These include the 2H phase, characterized by trigonal prismatic coordination; the 1T phase, which features octahedral coordination; and the 1T' phase, a distorted variant of the octahedral structure [97, 98]. The arrangement of the three atomic layers (X–M–X) varies among TMDs. The 2H-TMDs display a Bernal stacking pattern, known as ABA, while the 1T-TMDs are arranged in a rhombohedral, ABC stacking sequence [48].

Fig. 14 **a** Optimized structure of the MoS_2 with full Y coverage on both sides and **b** its hydrogen adsorption system. Reproduced from [102] with CC-BY licensing permission



MoSe_2 hetero-structure bilayer employing DFT simulations. The authors highlight the $\text{MoSe}_2\text{:Cu@Se}$ and $\text{MoSe}_2\text{:Ni@Se}$ systems due to the promising adsorption energies of -0.362 and -0.307 eV, respectively, for the $\text{MoSe}_2\text{:2Ni@2Se}$ structure, which is represented in Fig. 15, a gravimetric capacity of 3.6 wt% was reached, with an average adsorption energy per hydrogen molecule of -0.15 eV/ H_2 .

In the current year, Altuntepe et al. [104] analyzed the hydrogen storage on MoS_2 layers obtained by the liquid phase exfoliation method. The AFM (Fig. 16) was used to demonstrate the formation of few-layer nanosheets characterized by thicknesses from 2 up to 6 nm, denoting 2–5 layers. The hydrogen storage capacity was evaluated under 1, 3, 5, 7 and 9 bar hydrogen pressure. It was verified that the hydrogen leads to a red shift for A_{1g} and E_{2g} peaks of pristine MoS_2 . The exfoliation considerably modifies the specific surface area, with reported values of 12.31 m^2/g for the bulk and 23.16 m^2/g for exfoliated MoS_2 . The weight percent storage for H- MoS_2 and H-exfoliated MoS_2 were 1.2 and 2.4%, respectively. This study demonstrates the influence of specific surfaces for hydrogen storage.

Specifically, TMDs and their heterostructures are considered promising for hydrogen evolution as an inexpensive and sustainable alternative to metal-based catalysts [100, 101]. However, there are challenges associated with TMDs in hydrogen storage. The photocatalytic activity of TMDs can be limited, but it can be improved through strategies such as elemental doping, heterojunction formation, and morphological tuning [100]. Additionally, while TMD quantum dots (QDs) show promise due to their large absorption coefficient and abundant active sites, their development is still in the nascent stages, and comprehensive summaries on their performance in solar-to-hydrogen conversion are scarce [105].

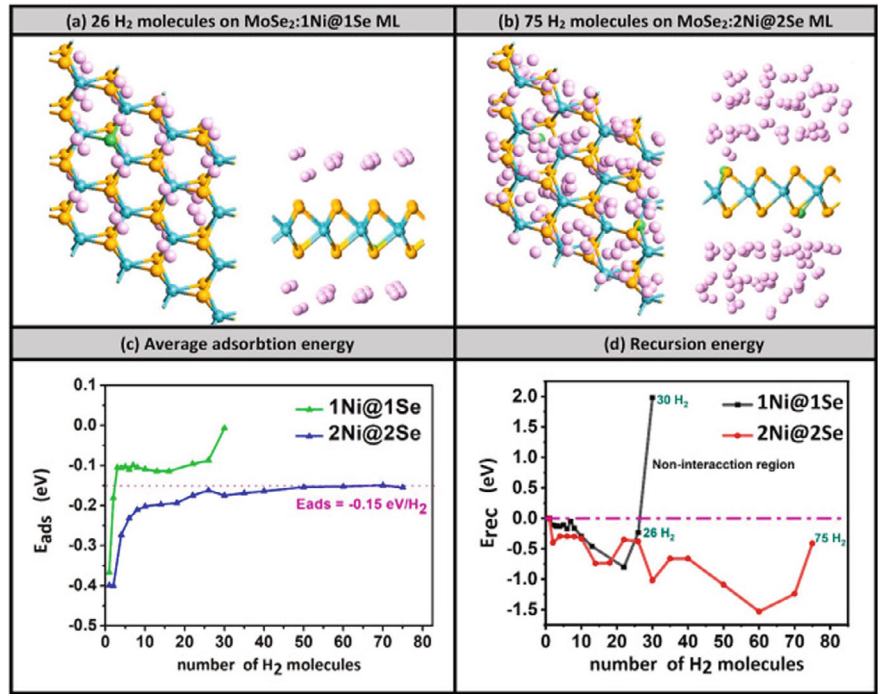
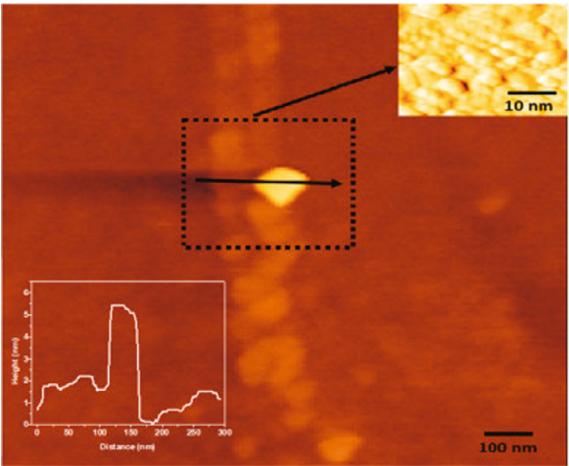


Fig. 15 **a** MoSe₂:1Ni@1Se system with the highest storage capacity of 26H₂ molecules, **b** MoSe₂:2Ni@2Se structure with the highest storage capacity of 75H₂ molecules, **c** average adsorption energy per H₂ molecule, and **d** recursive energy per 4H₂ molecules versus the number of molecules deposited on 1 Ni and 2 Ni doped MoSe₂, respectively. Reprinted from [103] with CC-BY licensing permission

Fig. 16 Exfoliated MoS₂ spin-coated onto Si/SiO₂ substrates. AFM images and corresponding line scans of exfoliated MoS₂ flakes on Si/SiO₂ substrates. Reprinted with permission from Ref. [104]. Copyright 2024 Elsevier



In their DFT study, Zhang et al. [106] explored Li-decorated 2D germanium selenide (2D GeSe) as a promising material for hydrogen storage. Their comprehensive analysis focused on the maximum hydrogen storage capacity and gravimetric density calculations. The authors observed a positive correlation between the gravimetric density of hydrogen adsorption and the incremental addition of H_2 molecules to the Li-decorated 2D GeSe substrate. They calculated the average adsorption energy in relation to the number of H_2 molecules adsorbed, finding an optimal adsorption energy of approximately 0.1 eV per H_2 molecule. When H_2 molecules were adsorbed solely on one side of the substrate, a gravimetric density of 2.85 wt% was achieved, with an adsorption energy of -0.133 eV, ensuring the adherence of H_2 molecules to the substrate. Conversely, dual-sided adsorption resulted in an adsorption energy of -0.12 eV and a significant gravimetric density of 5.53 wt%. The range of adsorption energy for H_2 molecules on Li-decorated 2D GeSe, spanning from -0.2 to -0.1 eV per H_2 molecule, indicates a favorable environment for reversible hydrogen storage.

When examining WS_2 nanostructures, numerous studies highlight their potential for hydrogen storage. Zhou et al. [107] used the local density approximation (LDA) to demonstrate that the $1T'$ - WS_2 phase exhibits a higher capacity for hydrogen molecule adsorption compared to the $2H$ - WS_2 phase. The $1T'$ - WS_2 phase has an average adsorption energy ranging from -0.2 to -0.6 eV, which is considered optimal for hydrogen storage. This phase can achieve a hydrogen adsorption ratio of 2.7 wt%, exceeding the 2.4 wt% capacity of the $2H$ - WS_2 phase. Additionally, Bin Xu et al. [108] investigated hydrogen storage in an Sc-decorated WS_2 nanoribbons complex using DFT. They found that while H_2 storage in Sc/ WS_2 monolayers is not feasible due to weak interactions between Sc and WS_2 , each adsorbed Sc atom can capture four H_2 molecules with an average adsorption energy of approximately 0.2 eV.

2.3 MXenes for Hydrogen Storage

MXenes are a class of 2D transition metal carbides, nitrides, and carbonitrides that have recently gained attention for their potential applications in energy storage [109, 110]. MXenes are derived from layered MAX phases, where the A-layer (usually an element from groups 13 or 14 of the periodic table) is selectively etched away, leaving behind a 2D structure with a high surface area and abundant surface terminations [49].

Additionally, MXenes often possess metallic conductivity, enhancing the dissociation and diffusion of hydrogen molecules [111]. MXenes also exhibit good thermal and chemical stability [42]. One of the great advantages of using MXenes in energetic devices is due to their composition by transition metal, which promotes a stronger interaction between the molecule and the substrate compared to carbon-based materials.

The MXene terminations also play a significant role in the H_2 storage. Depending on the MAX phase treatment, different functional groups can be found, which provide new adsorption sites and distinct approaches to the inclusion of hydrogen between

the layers. Figure 17 depicts the experimental route to obtain the Mxene sheets for hydrogen storage. Following the functionalization mechanisms, Chu and Lau [112] appoint the significant effect of $-O$, $-OH$, $-H$ and $-F$ functional groups originated by the etching process using the DFT approach. Without surface termination, the H_2 storage on the Ti_3C_2 Mxene occurs guided by a chemisorption led by H_2 dissociation and $Ti-H$ formation. Meanwhile, the uniform or mixed surface termination with $-O$, $-OH$, $-H$ and $-F$ decreases the strong interaction between the hydrogen molecules and the Mxene, and a physisorption is found (-0.08 to -0.14 eV).

In 2013, Hu et al. [113] studied Ti_2C Mxenes via DFT simulations as an H_2 capture platform. The author found a gravimetric density of 8.6 wt%. Subsequently, it was concluded that the H_2 diffusion through the adsorption sites of Ti_2C is suitable due to its low energy barrier (0.12 eV), which facilitates the molecule release processes. By another theoretical study, a gravimetric capacity of 6.4 wt% is reported for Cr_2C Mxene by a reversible process (Kubas-type interaction), being guided with adsorption energies between 0.1 and 0.4 eV [114].

Besides carbides Mxenes, Ti_2N structures were also computationally investigated. In the absence of any functionalization, Li et al. [115] report a maximum capacity of 8.55 wt% for Ti_2N Mxene. That magnitude of H_2 is severally mitigated to 5.18, 5.38 and 5.31 wt% when functionalized with $-F$, $-O$ and $-OH$, respectively, denoting the huge effect of the surface terminations. Experimentally, Mxenes have been a key factor in increasing H_2 storage in solid solutions or composites. It is possible to mention the compound $(Ti_{0.5}V_{0.5})_3C_2$ as a catalytic agent for Mg-based hydrides,

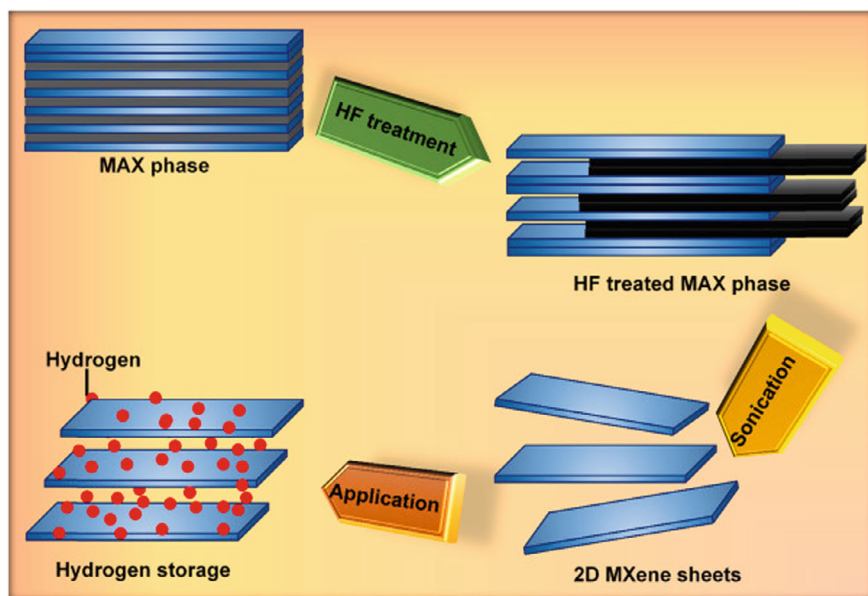


Fig. 17 The experimental process to store H_2 in the 2D Mxene sheets. Reprinted with permission from [42]. Copyright 2024 Elsevier

which reduces the dehydrogenation process by a temperature difference of 70 °C, where 5 wt% of H₂ is released at 250 °C in a time of 20 min, therefore, a superior performance of traditional catalytic supports such as Nb₂O₅ [116]. Kinetic properties involving the release of H₂ in Mg were also investigated in the laboratory with the Nb₄C₃T_x Mxene [117]. As a reinforcement of sodium allanate (NaH/Al) and lithium borohydride (LiBH₄) composites, the Ti₃C₂ Mxene also presents very promising hydrogen retention performances [118].

In recent years, great efforts have been made to understand the energetic properties of Mxenes on a large scale and under different conditions. Due to this, the study of Mxene multilayer structures attracts both theoretical and experimental approaches, aiming for a global assessment of H₂ retention and release. In 2021, Liu et al. [119] proposed that the Ti₂CT_x Mxene multilayer incompletely etched with hydrofluoric acid (HF) has a gravimetric capacity of 8.8 wt% under 60 bar of pressure and a temperature of 25 °C. At ambient conditions (25 °C and 1 bar), the retention capacity reaches ~4 wt%. Figure 18 illustrates the H₂ storage mechanism on the Mxene multilayer.

Also, in this work, the authors apply the DFT method to evaluate the impact of the interlayer spacing between Ti₂C on the H₂ adsorption mechanisms. Weak chemical adsorption is unveiled when the distance between the layers is 6.8 Å, where the adsorption energy varies from −0.134 to −0.128 eV until the end of hydrogen loading on the surface, which indicates a good ability to retain H₂. By increasing the distance between Ti₂C layers to 8 Å, the authors observed a weak physisorption (−0.068 eV), making it impossible to store H₂ even in low temperature and

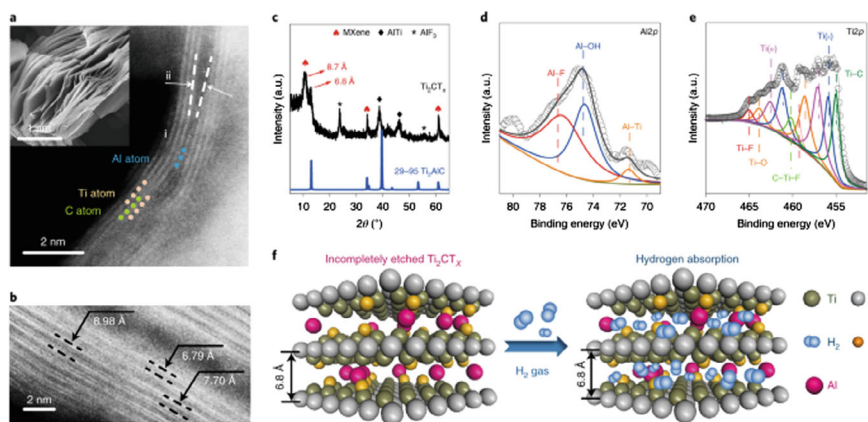


Fig. 18 Structural characterization of Ti₂CT_x. **a**, HAADF-STEM image of two adjacent Ti₂CT_x nanosheets, showing the residual Al atoms due to incomplete etching (region i) and a bell-mouth morphology (region ii). Inset: scanning electron microscopy image of the Ti₂CT_x stack. **b**, HAADF-STEM image of multilayered Ti₂CT_x. **c**, XRD patterns of Ti₂CT_x with the reference of Ti₂AlC. 29–05 is the powder diffraction file number of Ti₂AlC. **d**, **e**, XPS spectra of elemental Al (**d**) and Ti (**e**) in Ti₂CT_x. **f**, Schematic of the hydrogen storage process in the narrow interlayer space of Ti₂CT_x. Reprinted with permission from [119]. Copyright 2024 Elsevier

high-pressure conditions, highlighting the influence of the interlayer spacing as a crucial factor in the H_2 adsorption strength. Using characterization techniques such as excited photoelectron spectroscopy (XPS) and Fourier transform infrared spectroscopy (FTIR), results indicate chemical adsorption of H_2 atoms above Ti sites and high absorption peaks originated by the interaction of H with the functional agents, $-O$ and $-F$, which act synergistically in the retention and release of the molecule.

The close relationship between the interlayer spacing and the H_2 adsorption was also explored theoretically in many other multilayer carbide Mxenes. As studied by Zheng et al. [120], the expansion of interlayer spacing for Ti_2C leads to a smaller H_2 adsorption magnitude, which trend is followed for the M_2C ($M = Sc, Ti, V, Cr, Zr, Nb, Mo, Hf, \text{ and } Ta$) Mxenes, except for Cr_2C and V_2C that present more negative values as a function of interlayer expansion. At 10% of expansion, the authors found a strong physisorption for Hf_2C ($E_{ads} = -0.24$ eV) and Zr_2C ($E_{ads} = -0.18$ eV) Mxenes, superior to those registered for Ti_2C ($E_{ads} = -0.135$ eV).

2.4 Other 2D Materials for Hydrogen Storage

Recently, different 2D structures have been investigated as promising platforms for energy storage. Focusing on the adsorption mechanism, which is the key factor of hydrogen storage capability in solid-state systems, the replacement of the carbon atom for another element or compound to create binary and ternary 2D materials has been used as a strategy to retain more H_2 molecules. An interesting way to boost the hydrogen capacity of some 2D materials is their functionalization, resulting in better interaction between the H_2 molecules and the substrate, as visualized in Fig. 19.

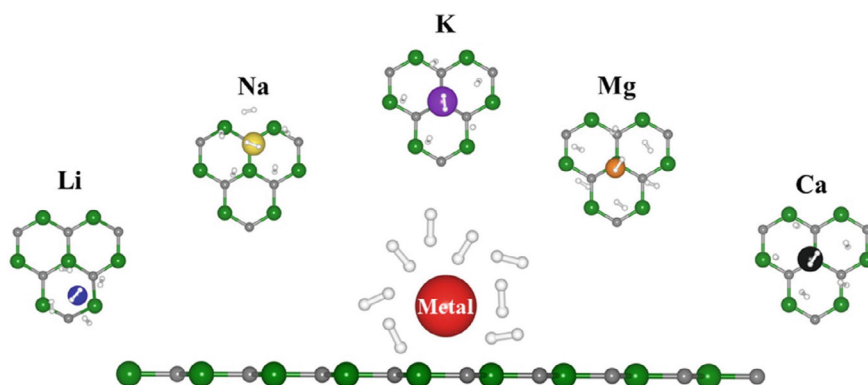


Fig. 19 The metal decoration as a strategy to improve the H_2 storage on the germanium carbide (GeC) monolayer. Reprinted with permission from [121]. Copyright 2024 Elsevier

Some traditional graphene-like semiconductors such as SiC, ZnO and GaN have been reported as excellent H₂ storage platforms. Analyzing the SiC monolayer, Arel-lano et al. [122] found a 1:10 proportion of metals (Li, Na, K and Ca) for hydrogen molecules, confirming its huge H₂. On the other hand, the same metal adatoms also, including Be and Mg, has improved the gravimetry density of ZnO monolayer, reaching 6.92 wt% for Li decoration [123], a superior value than reported for the pristine 2D ZnO of 4.70 wt% [124]. In the honeycomb GaN, the effects of Ni defect on the hydrogen storage properties were evaluated [125], and ab initio molecular dynamics (AIMDs) confirmed the stability of eight H₂ molecules above the Ni atom. In an experimental approach, Lei et al. [126] obtained by a sol–gel method 2–6 atomic layers of oxygen-doped hexagonal boron nitride (h-BN), corresponding to a great H₂ uptake of 5.6 wt% under 5 MPa at room temperature, and good storage cycling stability. A gravimetric density of 6.0 wt% of gravimetric density and a desorption temperature of 200 K was found for h-BN [127].

H₂ adsorption has also been explored in non-planar 2D materials, such as silicene and germanene. Under an external electric field, the Ca adatom can adsorb nine H₂ molecules on the silicene substrate, leading to 6.4 wt% of hydrogen capacity [128]. A similar capacity is found when the silicene nanosheet is decorated with Li and Na metals, 6.65 wt% [129]. Additionally, the K and Ca decoration results in the retention of eight and nine H₂ molecules per metal atom, respectively [130]. Employing an interesting approach, Sosa et al. [131] used both doping and metal decoration strategies to enhance the H₂ uptake on the germanene sheet and, therefore, achieved a gravimetric density of 5.32 wt% for an adsorption energy of -0.37 eV.

Also similar to graphene, borophene, a covalently bonded boron nanosheet, emerges as a promising candidate for energetic devices, with nanostructures experimentally obtained [132, 133]. At room temperature, the adsorption energy of H₂ on the Li-decorated borophene is around -0.35 eV, a typical strong physisorption, and maximum capacity is revealed at 9.22 wt% [134]. The H₂ adsorption mechanism on the Li-borophene can be accessed in Fig. 20. On the other hand, investigating the β_{12} phase, Liu et al. [135] appoint a decrease in the gravimetric density to 9.10 wt%. The decoration with N, Na, K and Ca was also reported in the mentioned boron-based structure to improve the H₂ uptake [136–138]. Other predicted structures such as B₄CN₃ [139], B₄C₃ [140], B₂N₂ [141], Si₂BN [142], MgN₄ [143], siligene [144, 145], polynitrides [146], among others [147–150], also emerge with appropriate hydrogen storage properties.

In general, the allotropic form of carbon is attractive to building new 2D materials with distinct properties and architectures than those reported for graphene, and a substantial increase in the surface area can be achieved. One of the most stable 2D carbon allotropes is graphdiyne (GDY), which was successfully synthesized in the last decade and is suitable for retaining hydrogen [151]. In view of increasing the interaction between and the pristine GDY substrate, Wang et al. [152] decorate the structure with Li/Na atoms using both DFT and Monte Carlo approaches, reaching 8.81/7.73% of gravimetric density. The Biphenylene carbon sheet (BPN) structure disposed of 4-, 6- and 8-membered rings, has also been the subject of many studies, in addition to its synthesis revealed in 2021 [153], and the hydrogen storage capabilities

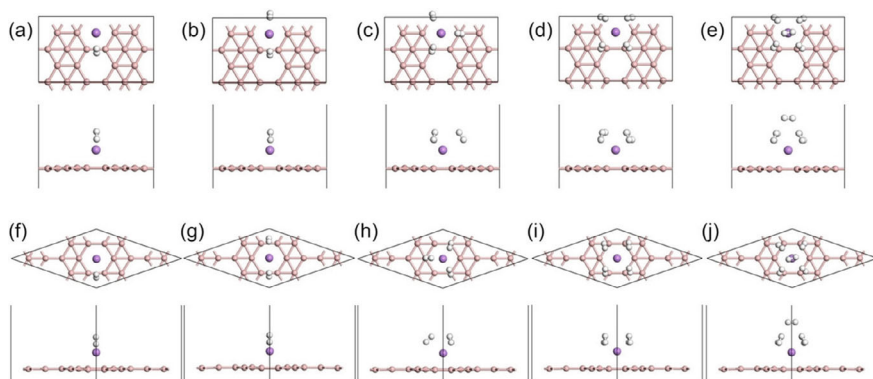


Fig. 20 H₂ adsorption on the **a–e** Li-β₁₂ borophene and **f–j** Li-χ³ borophene. Reproduced from [134] with permission. Copyright 2024 John Wiley & Sons

under substitutional doping or metal decoration being unveiled by theoretical calculations [154–156]. Graphenylene (GP), a similar carbon network to BPN with the difference in the largest ring (GP has 12 members instead of 8) [157], was obtained experimentally in 2017 [158], and investigated by first-principles as a hydrogen storage platform. Hussain et al. [159] indicate the metal decoration (Li, Na, K, Ca and Mg) as a good strategy to enhance the binding energy, leading to a moderate capacity (6.14 wt%) (see Fig. 21).

Inorganic counterparts of GP also can be used as hydrogen storage platforms, as verified in the graphenylene-like silicon carbide (IGP-SiC) [160] and boron phosphide (IGP-BP) [161] by a gravimetric density range from 6.78 up to 9.99 wt%. Many other promising carbon-based structures for H₂ storage with distinct geometry than graphene was unveiled by theoretical prediction, such as irida-graphene [162], Ψ-graphene [163], T-graphene [164], penta-graphene [165], PAI-graphene [166], 2D polyaramid [167]. Table 1 collects the hydrogen capacity and adsorption results of some monolayers discussed here, among other structures.

3 Summary and Outlook

This chapter highlights different strategies in green hydrogen production, mainly promising alternatives using 2D materials for hydrogen storage. Different strategies available were discussed based on theoretical and experimental reports, unveiling the key factors of hydrogen storage in distinct 2D structures such as graphene, TMDs, and Mxenes, as well as hexagonal lattices and recently predicted 2D carbon allotropes and derived inorganic compounds. Therefore, we can remark on the following details:

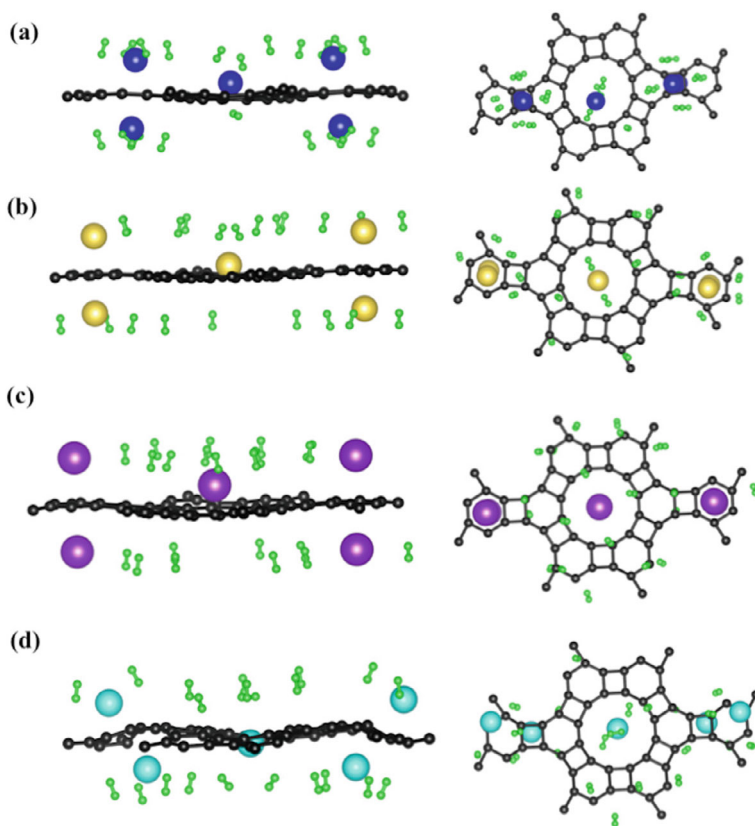


Fig. 21 Alkali and Alkaline earth metal decoration on the graphenylene (GP) monolayer. Reprinted with permission from [159]. Copyright 2024 American Chemical Society

- (1) The large surface area makes 2D materials ideal platforms to store H_2 , and many of them possess gravimetric densities exceeding the DOE hydrogen target (5.6 wt%).
- (2) The adsorption mechanism has a key role in guaranteeing high H_2 capacity, and a route between weak physisorption and chemisorption is desired in view of promoting great stability and facile desorption.
- (3) Metal decoration, doping, vacancies and other functionalizations are very useful in improving the interaction between hydrogen and the 2D substrate. In the Mxenes case, the use of functional groups and interlayer spacing engineering can enhance the gravimetric densities.
- (4) By theoretical frameworks, novel 2D materials with exciting architectures have been predicted to have huge H_2 capacity, which confirms the high demand for promising layered materials for hydrogen storage.

Table 1 Adsorption energies and H₂ storage capacity of some 2D materials

System	Average adsorption energy (eV)	H ₂ capacity (wt%)
2D SiC + Sc [168]	−0.27	6.32
2D ZnO [124]	−0.13	4.70
2D ZnO + Li [123]	−0.22	6.92
2D GaN + Ni [125]	−0.55 to −0.96	5.36
Silicene + Na/K/Mg/Ca [169]	−0.11 to −0.37	7.31–9.40
Germanene + K/Ca [130]	−0.09 to −0.15	12.52–13.97
Borophene + Li [134]	−0.35	9.22
Borophene + N [137]	−0.05 to −1.02	6.22
B-doped siligene + Ca [145]	−0.20	13.79
Si ₂ BN + Li/Na/Be/Ca [142]	−0.11 to −0.27	7.10–11.70
B ₂ N ₂ + Li [141]	−0.22 to −0.32	12.40
B ₄ C ₃ + Li [140]	−0.23 to −0.24	6.22
MgN ₄ + Li [143]	−0.15	6.86
Biphenylene + Li [154]	−0.13	7.40
Biphenylene + Ca [170]	−0.33	11.63
Graphenylene + Li/Na/Ka/C/Mg [159]	−0.30	6.14
SiC graphenylene + Li/Na [160]	−0.10 to −0.16	6.78–8.27
BP graphenylene + Li [161]	−0.15 to −0.20	6.99
Irida-graphene + Li [162]	−0.28	7.06
PAI-graphene + Li [166]	−0.25	15.70
N-doped penta-graphene + Li [165]	−0.10 to −0.40	7.88
Ψ-graphene + Zr [163]	−0.38	11.30
2D Polyaramid + Ca [167]	−0.29	6.9

Despite the abovementioned notes and the progress made in using 2D materials for hydrogen storage, several challenges remain. One major challenge is achieving the desired hydrogen storage capacity while maintaining good adsorption and desorption kinetics. Improving the stability of 2D materials under hydrogen storage conditions is another challenge that needs to be addressed. Additionally, the scalability and cost-effectiveness of 2D materials for large-scale hydrogen storage applications need to be evaluated.

Future research in this field should focus on developing novel 2D materials with enhanced hydrogen storage properties, optimizing the synthesis and processing methods to improve material performance, and exploring new strategies for enhancing hydrogen adsorption and desorption kinetics. By addressing these challenges, 2D materials have the potential to play a significant role in the development of efficient and sustainable hydrogen storage technologies.

Developing a hydrogen infrastructure, including production facilities, storage systems, and distribution networks, requires substantial investment. The existing infrastructure for fossil fuels is not readily adaptable to hydrogen, necessitating new pipelines, refueling stations, and storage solutions. Additionally, public acceptance and regulatory frameworks need to be developed to support the transition to a hydrogen economy.

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Biohydrogen Production in a Biorefinery Using Industrial Effluents: Challenges, Opportunities and Economic Analysis



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Abstract Hydrogen is one of the simplest sources of energy to produce, and one of the best alternatives to fossil fuels due to its high energy density, high combustion rate, low environmental impact, and high availability. Today, there are many routes of generation (coal, electrolysis, natural gas, heavy oils and naphtha), but very few have been promoted to the industrial level due to their competitive costs against gasoline or diesel. It has been estimated that the biohydrogen production using biomass from industrial wastes is the technology with the greatest potential for the generation of this energy vector, however, its conversion efficiency must be increased between 60 and 80% to be competitive at these levels. Numerous challenges have been recognized including that only 15% of the energy from the biomass source can be used for the biohydrogen yield, using a fermentative route. Another significant challenge is related to the capacity of microbial consortia, where there are no known natural biochemical pathways to achieve between 60 and 80% conversion efficiency, although genetic engineering could significantly contribute to achieving these values. From an economic standpoint, two strategies can be emphasized to improve the biohydrogen production using biomass, the use of free and low-cost substrates, and finding a market for a high portion of the substrates not used in conventional processes. Accordingly, technical and engineering challenges

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are herein revisited, as well as economic aspects that must be overcome to achieve a sustainable implementation of biohydrogen production.

Keywords Biohydrogen · Industrial effluent · Biomass · Biorefinery · Economic analysis

1 Introduction

The use of hydrogen dates back more than a century ago, both as a fuel in internal combustion engines and its conversion to generate electrical current in a fuel cell. Its consolidation in various energy applications was not achieved largely due to its storage problem, wide flammability range with little energy to start combustion, and the consolidation of fossil fuels for vehicle and industrial applications. Nevertheless, the serious environmental and economic problems emerged in the last decades due to the excessive consumption of petroleum products have once again turned attention towards hydrogen. In this context, batteries (mainly Li-ion) and bioenergy have been envisaged as transition energies towards a greener source of energy, with capacities and energy metrics similar to oil derivatives (e.g. gasoline, diesel, turbosine).

To date, hydrogen is one of the best options to replace fossil fuels due to its high combustion rate, green fuel, water as reagent, high energy density and calorific power [1]. In 2021, this fuel presented a demand of 94 million tons, representing 2.5% of global energy consumption, wherein the demands for new applications grew more than 60% compared to 2020 [2]. Certainly, the production of hydrogen needs to be from a sustainable source or some type of waste, to further increase its profitability. To this concern, biotechnological processes constitute a great opportunity since they demand low energy consumption and are profitable; however, they need to be optimized in terms of residence time and efficiency to maintain these advantages at the macroscopic level [3]. Apparently, the substrate selection (e.g. renewable resources, biomass) is one of the key factors in biohydrogen production because it directly impacts productivity. Substrates where biomass is involved are preferred due to their lower nutrient contents, favorable energy productions and higher organic loads, standing out industrial wastewaters due to their wide compositions [4]. Table 1 reviews different substrates obtained from industrial wastewaters used for biohydrogen production.

As observed in Table 1, the substrate used for biohydrogen production is particularly low-cost biodegradable matter. Industrial wastewaters can be extremely diverse in composition, requiring flexible microbial culture (biocatalyst), whereby its efficiency can be obstructed by methanogenic bacteria (i.e. fermentation). This can be inhibited by adequate selection of pretreatment (chemical, physical, biological) and microbial biocatalyst. On the other hand, several factors hamper the biohydrogen production including its consumption by the microbial consortia, and residual organics in the industrial discharges; while its improvement involves culture

Table 1 Substrate, bacteria and type of reactor used in the biohydrogen production

Substrate/ wastewater	Bacteria	Reactor	Yield	References
Citric acid	Facultative anaerobic bacteria	Upward-flow anaerobic sludge blanket	0.84 mol H ₂ /mol hexose	[5]
Chemical	Enriched mixed culture	Acidogenic bioreactor	13.29 mol/Kg CODR and 13.70 mol/Kg CODR (vitamins)	[6]
Cheese whey	Mesophilic fermentative bacteria	CSTR-type digester	2.9 L of H ₂ /L cheese whey	[7]
Rice mill	<i>Enterobacter aerogenes</i> and <i>Citrobacter ferundii</i>	Batch	1.74 mol H ₂ /mol reducing sugar	[8]
Beverage	<i>Clostridium</i>	CSTR	1.70 ± 0.03 mol H ₂ /mol hexose utilized (6 h)	[9]
Palm oil mill	<i>C. butyricum</i> LS2, <i>Clostridium</i> sp., <i>R. palustris</i>	CSTR	3.064 ml H ₂ /ml POME (dark-photofermentation)	[10]
Casava starch processing	<i>Brevibacillus</i> sp. <i>Caloranaerobacter</i> sp. and <i>Geobacillus</i> sp.	Up-flow anaerobic sludge blanket	465 ml H ₂ g/COD (fermentation and microbial electrolysis)	[11]
Sugar industry	–	Microbial electrolysis cell	–	[12]

enrichment, biocatalyst pretreatment, genetic engineering, bioaugmentation, two-stage fermentation, photoheterotrophic production, uses of microbial electrolysis cell (MEC) and microbial fuel cell (MFC). In this context, the following chapter revisits biohydrogen production from industrial wastewaters with particular focus on biocatalysts, reaction mechanisms, scale-up, challenges, and drawbacks of industrial processes, as well as economic aspects; with the intention to remark opportunities of biorefinery pathways for biohydrogen production.

2 Overview of Biohydrogen Production and Main Reaction Mechanisms.

The production of H_2 through biological methods (biohydrogen) can be carried out by using four main processes: (1) photo-fermentation (PF) with volatile fatty acids (VFA) as carbon source, (2) dark fermentation (DF) of organic waste, (3) biophotolysis (involving microalgae [13] recently reviewed in [14]), and (4) microbial electrolysis cells (MEC). Biohydrogen production has become a promising alternative to replace fossil fuel demands, thus, research has been focused greatly on all the biological processes that can produce it. Recently, within the concept of biorefinery one of the most revised alternatives as a source of organic matter has been industrial wastewater due to the two advantages that its utilization can derive: the treatment of the industrial effluent and the production of a “clean” potential fuel. The biological processes that have been studied to produce hydrogen are shown in Fig. 1. In this figure, the main aspects that have been subject of study regarding biohydrogen production, and derived discussion in the scientific community are also shown.

2.1 Photo-Fermentation (PF)

Organic compounds can be converted completely to hydrogen during PF, however, the requirements that the process needs to be successful can be a drawback to scale the process (Fig. 2). Photosynthetic purple non-sulfur bacteria, for instance, *Rhodospseudomonas palustris*, *Rhodospirillum rubrum*, *Rhodobacter sphaeroides* are prominent microorganisms that can produce hydrogen under appropriate conditions that involve the optimal conditions of light intensity and distribution in the bioreactor, pH and temperature control, and type of substrates. Under the optimal conditions, these microorganisms can produce hydrogen reaching an almost complete conversion of the substrates. Due to the constraints presented to achieve a successful PF process, particularly in regards of light distribution and intensity required by the bacteria, genetically modified strains of purple non sulfur bacteria have been studied. There are several strains of purple non sulfur bacteria that have been genetically modified to decrease the pigment content and therefore, improving light penetration and

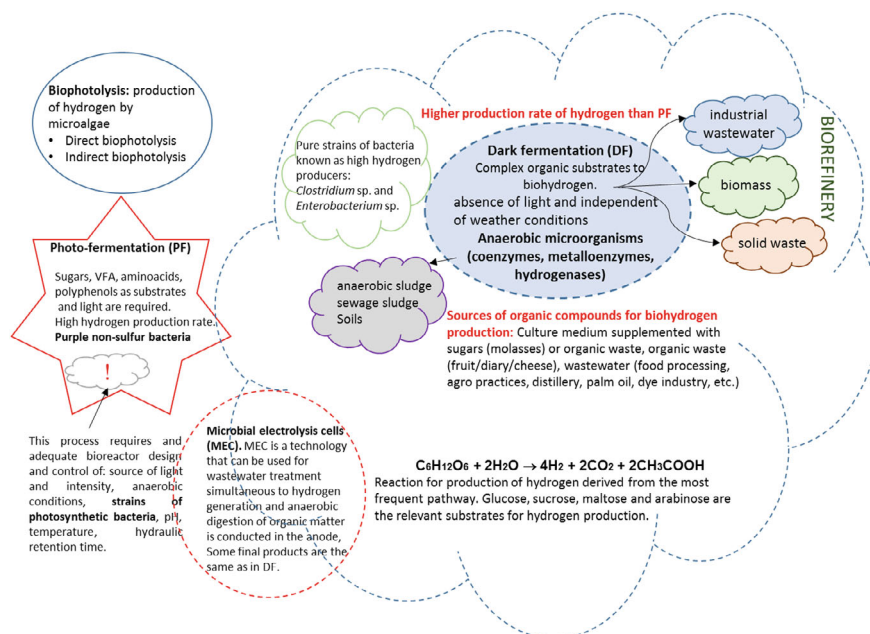


Fig. 1 Current technologies to produce biohydrogen

hydrogen production. Some of these microorganisms are *Rhodobacter capsulatus* SB1003, and *Rhodobacter sphaeroides* RV MTP4. Since nitrogenase and hydrogenase are the most relevant enzymes utilized in PF to produce hydrogen, genetic modifications to *Rhodobacter* sp. have been studied to increase the enzymatic activity, ammonium [15] and antibiotic tolerance [16], and ATP content [17] have also been modified using genetic engineering for purple non sulfur bacteria involved in the production of hydrogen.

To improve stability and reutilization of the bacteria in PF processes, immobilization of bacteria in gels and biofilms has also been used. In these cases, hydrogen production rate and yields have been maintained within an acceptable range, for example, the result obtained was 3.53 mol H₂/mol acetate and 1.3 mmol H₂/L h for *Rhodobacter capsulatus* YO₃ [18]. In view that the effluents from DF contain VFA that can sustain bacteria in the PF process, several studies have been conducted to link both processes and take advantage of the best characteristics of each one to improve the biohydrogen yield. This has been highlighted in one of the most recent and comprehensive reviews on PF [19]. Theoretically, in an integrated DF-PF sequential process, up to 12 mol of hydrogen can be produced per mol of glucose although this is difficult due to the several VFA produced from DF (Fig. 3). The mixture of VFA resulting from DF includes acetate, formate, propionate and butyrate, in addition to ammonium (depending on the mixture of organic matter that is fed to DF) [20, 21]. Ammonium inhibits the activity of the nitrogenase in bacteria in the

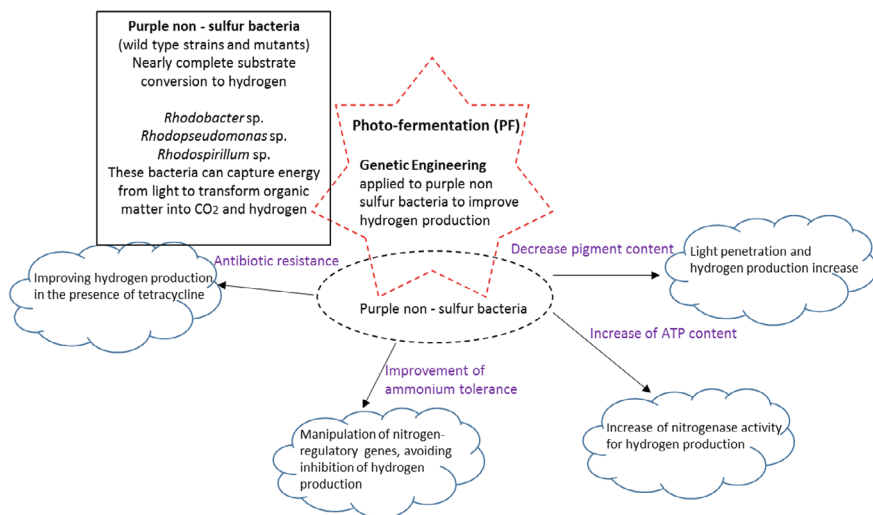


Fig. 2 Highlights on hydrogen production by photo—fermentation (PF)

PF, thus, diminishing the yield of hydrogen in the integrated DF-PF process [21]. Some of the integrated DF-PF processes that have used pure strain of bacteria, for example *Clostridium butyricum* for DF, and *Rhodopseudomonas palustris* for PF have reported hydrogen up to 8.3 mmol H₂/g COD. More recently, the improvement of hydrogen production in a DF-PF has been approached by utilizing a consortium composed of *Rhodopseudomonas palustris* (utilizes acetate) and *Syntrophomonas wolfei* (utilizes butyrate to produce acetate) taking advantage of a syntrophy between the two bacteria. Butyrate and nitrogen can be present in the effluent of DF, and the uptake of this stream in the PF stage can be carried out by *Rhodopseudomonas palustris* and *Syntrophomonas wolfei*.

When *Syntrophomonas wolfei* utilizes butyrate and generates acetate for *Rhodopseudomonas palustris* in a syntrophic metabolism, hydrogen production is facilitated. The regulation of nitrogen concentration during this study was also investigated to improve the PF process [22]. Single stage for DF-PF has also been investigated. In this type of coupling, the strain of bacteria that have been recently studied are *Rhodobacter sphaeroides*, *Clostridium acetobutylicum* and *Enterobacter aerogenes* MTCC2822 [23–25]. Hydrogen yields ranged from 164.4 ± 2.6 (supplemented with resin) to 197.15 ± 2.9 (supplemented with biochar) ml H₂/g VS with food waste as a feedstock [23]. For the same strains, using glucose as the source of carbon under batch conditions, up to 4.44 mol H₂/per mol glucose were obtained for the single stage integrated DF-PF [24]. On the other hand, *Enterobacter aerogenes* MTCC2822 and *Rhodobacter sphaeroides* O.U.001 in a single stage DF-PF generated up to 1580 ml H₂/L medium using 2 g/L of carbohydrate concentration (powdered potato waste) and 0.2 g/L FeCl₃, in which iron was supplemented to support the enzymatic activity of the bacteria [25]. Regarding PF, this process is still mostly reported at lab scale

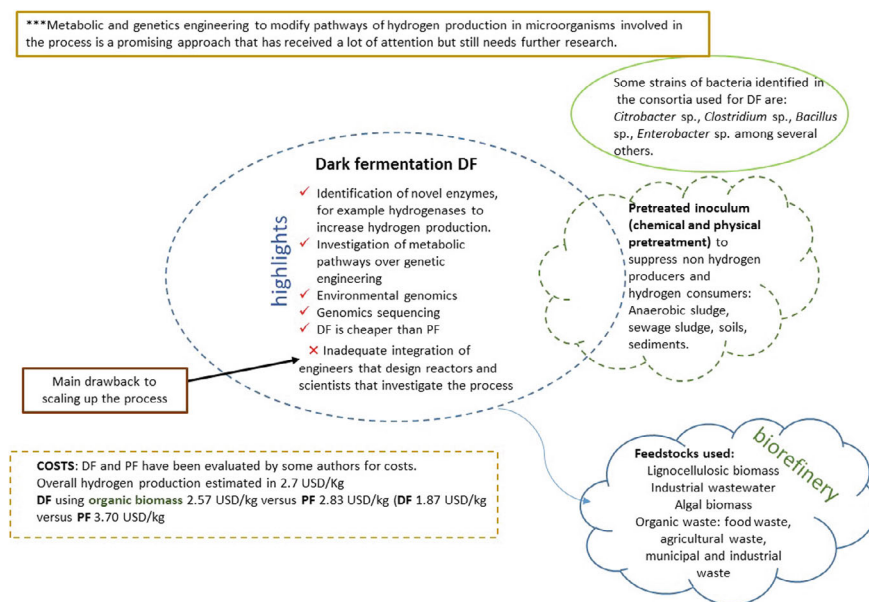


Fig. 3 Highlights of hydrogen production by dark fermentation (DF) [27, 30, 74]

and genetic engineering seems promising, and it has been utilized to enhance light transfer efficiency, control of enzymatic activity, tolerance to inhibitory compounds and ATP content [19].

2.2 Dark Fermentation (DF)

DF for production of hydrogen has been extensively studied and proved that higher hydrogen production rates are reached with this process compared to PF although the yields are lower (up to 4 mol Hydrogen/mol hexose). Thus, DF has been linked to PF to increase the conversion of organic compounds to hydrogen. This process has also been evaluated within the concept of biorefinery and several reports indicate that it is feasible taking lignocellulosic, algal, waste biomass and organics contained in wastewater effluents as feedstocks [26]. The efficiency of DF depends on several factors including the pretreatment of the inoculum, the source of sugars or organic compounds (feedstock), the content of sugars in the feedstock that can serve to produce hydrogen, and the consortium used in the process [27]. There are some microorganisms that play an important role in DF when in consortia are living in syntrophy with hydrogen consumers, thus, research has focused on the utilization of

highly hydrogen pure strains of bacteria to overcome the issue of hydrogen consumption, and the elimination of the pretreatment stage. On the other hand, the disadvantage of the utilization of pure strains of bacteria could be the risk of contamination of the inoculum, particularly at large scale. Some of the strains of bacteria that have been widely studied regarding hydrogen production are *Clostridium* sp. and *Enterobacterium* sp. Despite all efforts on scaling up the process, as a stand-alone technology DF. In addition to the integration of DF with PF, which was previously discussed, DF has been also integrated to the production of bioplastics to make a cost-effective process while obtaining an environmentally friendly material. These bioplastics, for example, the polyhydroxyalkanoates (PHA) and polyhydroxybutyrates (PHB) are mainly produced by pure strains of bacteria in a photoheterotrophic stage with an adequate source of carbon composed of volatile fatty acids (VFA), for example acetate and butyrate. Several recent investigations have shown that the microorganisms involved in this process are *Macelibacteroides fermentans*, *Rhodopseudomonas palustris*, *Clostridium propionicum* and *Acinetobacter* sp. [28]. Regarding increasing hydrogen and biopolymers productions, the inclusion of *Syntrophomonas wolfei* in the photoheterotrophic consortium has proved to be a good strategy [29]. In this work, the authors reported that hydrogen production was up to 520 ml H₂/g COD when *Syntrophomonas wolfei* was in the photoheterotrophic culture (composed also by *Clostridium pasteurianum* and *Rhodopseudomonas palustris*). Lactic, butyric, acetic and propionic acids were used as a source of carbon for DF while consuming over 50% of the acids corresponding to approximately 55% of COD removal [29]. Regarding the reaction mechanisms for hydrogen production, a comprehensive recent review [30] discusses the metabolic pathways that have been studied so far for hydrogen production and the role of the key enzymes. This includes the different types of hydrogenases such as [Fe]—hydrogenases, [Fe–Fe]—hydrogenases, [Ni–Fe]—hydrogenases are explained, and the microorganisms most utilized to study hydrogen production in which these enzymes are present are also mentioned along with the strains that have been genetically modified to improve the production of hydrogen. The knowledge on metabolic pathways of hydrogen production serves to identify the targets for genetic and metabolic engineering for those microorganisms that are subject of modification to enhance the hydrogen production. Fermentation of glucose (Fig. 4) is the most representative example for the reactions that occur in DF. As pointed out in [30], indeed DF can be related to methanogenic anaerobic digestion for which the intermediate products are formate, acetate, propionate and butyrate. Some highlights on DF are shown in Fig. 3.

2.3 Bio-Photolysis (Microalgae and Cyanobacteria to Produce Hydrogen)

Microalgae are of interest to produce biofuels, since the autotrophic metabolism and growth conditions can be utilized to generate hydrogen. Several strains of microalgae

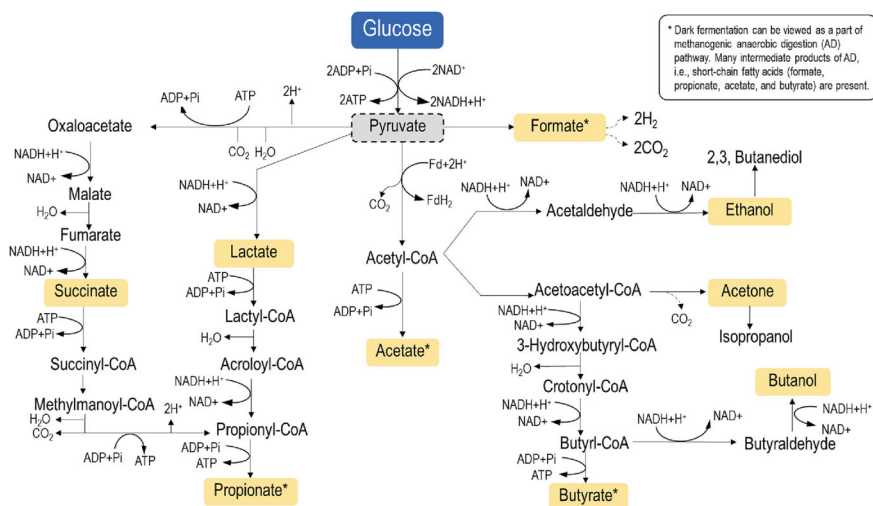


Fig. 4 Biochemical pathway to hydrogen production via dark fermentation [30] reproduced with permission of Elsevier

and cyanobacteria have been studied for hydrogen production, i.e., *Chlamydomonas* sp., *Scenedesmus* sp., *Chlorella* sp., *Tetraspora* sp., *Monoraphidium* sp., *Platymonas* sp., *Chlorococcum* sp. and the cyanobacteria *Synechocystis* sp., *Synechococcus* sp., *Anabaena* sp. and *Nostoc* sp. The production of hydrogen by these microorganisms largely depends on the hydrogenases and nitrogenases enzymes (Fig. 5) [14]. Hydrogenases are more effective regarding hydrogen production than nitrogenases. Along the same lines, the [Fe-Fe] hydrogenases are more effective than the [Ni-Fe] hydrogenases. The factors that highly influence the production of hydrogen by microalgae are oxygen concentration, since it inhibits the enzymatic activity, temperature, pH and an imbalance in nutrients C, N, P and S. It has been observed that under appropriate conditions, co-cultures of microalgae and bacteria can be effective in hydrogen production. Thus, co-cultures of *Chlamydomonas* sp. with *Pseudomonas stutzeri*, *Pseudomonas putida* and *Escherichia coli* have been studied for this purpose [31]. It was demonstrated that the bacteria inhibited the microalgae growth and improved hydrogen up to 46% in the presence of *Pseudomonas stutzeri*. This study demonstrated a microalgae-bacteria syntrophy in which the process depends on the capacity of the microalgae to uptake acetic acid, that is one of the final products of the metabolism of sugars used to maintain the culture. Generic engineering has been also studied for improving hydrogen production in microalgae, for example, *Chlorella* sp. [32], *Anabaena* sp., and *Chlamydomonas reinhardtii*. Genetic modifications have been focused on diminishing the inhibition of the enzymes caused by oxygen. Hydrogen yields obtained with microalgae and cyanobacteria are difficult to compare due to the different types of bioreactors and operational conditions used to culture the microalgae. However, some yields obtained in recent research can be mentioned as examples of the extent to which hydrogen is produced in these

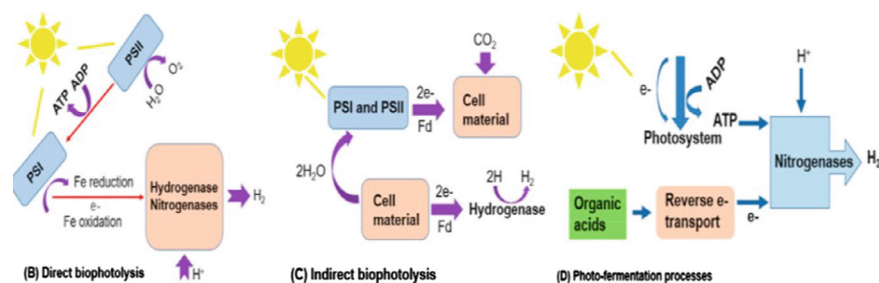


Fig. 5 Hydrogen production mechanisms in microalgae, adapted from reference [14]

systems. For example, *Spirulina platensis* hydrogen yield was of 1.92 ± 0.2 mmol H_2 /g COD removed [33]. *Anabaena* PCC7 120 (cyanobacteria) produced 14.9 mL/h at light of $456 \text{ mE/m}^2\text{s}$ with a light conversion efficiency of 0.042% [34], light conversion efficiency typically ranges 1–2% in this type of processes [35]. The production of hydrogen by microalgae and co-cultures microalgae—bacteria is still at early stages to be considered for scaling up. Research continues focusing on genetic engineering to improve the tolerance of oxygen of the enzymes and thus the hydrogen production in microalgae. The supplementation of Fe^0 (to be oxidized to Fe^{2+}) [36], and Ni nanoparticles has also been evaluated to enhance the hydrogenase activity [37]. As with the other processes to generate hydrogen biologically, the utilization of microalgae still needs further research to achieve an application at large scale. Some highlights of bio-photolysis are shown in Fig. 6, while Fig. 7 accounts for the difference between direct and indirect bio-photolysis from photo-fermentation. As observed in Fig. 6, direct photolysis consists of oxygenic photosynthesis to produce energy when microalgae convert solar energy into chemical energy. PSI and PSII reduce H_2O by transferring electrons to ferredoxin to produce hydrogen. In the indirect bio-photolysis, the production of hydrogen occurs in two steps. In the first step, microalgae growth occurs under appropriate conditions that involve an adequate amount of nutrients, light, pH, carbon source, temperature, etc. In the second stage, the production of hydrogen is carried out by microalgae under specific conditions such as for example in the case of *Chlamydomonas reinhardtii* supplemented with acetate (and sulfur deprived) can produce hydrogen for several days as long as the sulfur concentrations are favorable [14]. This indirect process depends on light conversion efficiency (desirable 10–15%) and is affected when microalgal biomass is accumulated.

2.4 Microbial Electrolysis Cells (MEC)

Microbial electrolysis cells (MEC) is a technology that has received great attention due to the simultaneous wastewater treatment and hydrogen generation that can be conducted in the process at relatively low cost [38]. In this process the organic matter

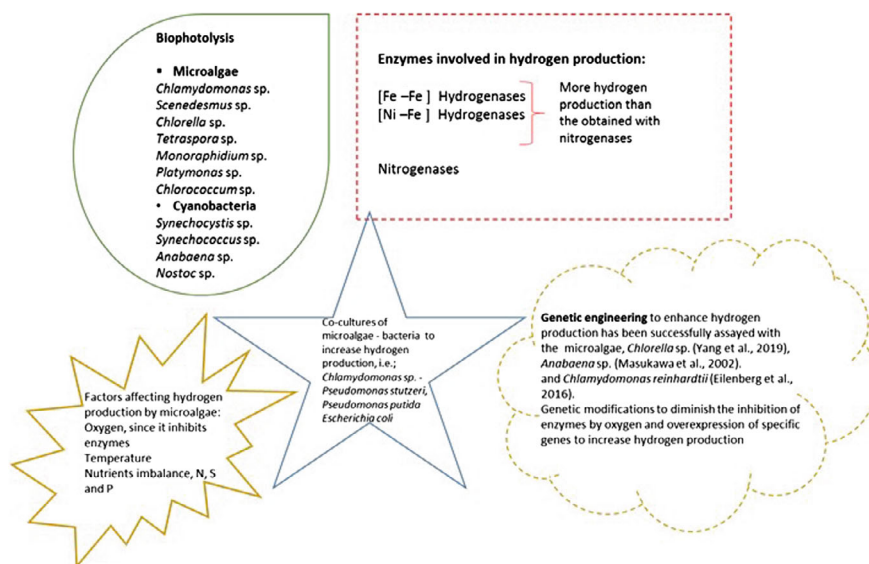


Fig. 6 Highlights of hydrogen production by biophotolysis [32]

is oxidized to carbon dioxide (CO_2) in the anodic chamber (on the anode a biofilm is formed) generating electrons and protons. The protons are transferred to the cathodic chamber through a membrane (proton exchange membrane) and hydrogen is formed on the cathode with applied voltage. Generally, low voltage is required to produce hydrogen. In view that this technology can be applied to wastewater treatment, it has been one of the main processes reviewed to bio-hydrogen generation from industrial wastewater streams [39]. The process can be coupled to dark fermentation when the wastewater that is loaded to the anaerobic digester (anaerobic bioreactor) produces acidic effluents rich in VFA, which could serve as a source of carbon to the MEC.

A research group that has conducted experiments in MEC coupled to DF reported that utilizing concentrations of 3000 mg/L of VFA in a MEC served to produce hydrogen. The applied potentials used were of 0.2, 0.5, 0.6, 0.8 and 1.0 V using anaerobic sludge as inoculum, the maximum hydrogen production rate was of 0.53 mmol/h and the cumulative hydrogen production was of 3.6 mmol [40]. For this process to succeed some key factors must be considered, such as the cathode design and material, the design of the MEC, the type of inoculum, the content of organic matter that is contained in the wastewater stream to be treated as well as an optimum membrane to exchange protons. Wastewater streams coming from the alimentary industry, i.e., cheese, fruit processing, sugar have been used to produce hydrogen via DF-MEC [41]. The results on bio-hydrogen production obtained from coupling these two processes (in batch mode) were up to 13 times higher than DF alone. The maximum yield for hydrogen production was reported to be 1608.6 ± 266.2 ml H_2 /g COD along with a $78.5 \pm 5.7\%$ COD removal. On the other hand, under continuous regime a system taking as a source of carbon cassava starch processing wastewater,

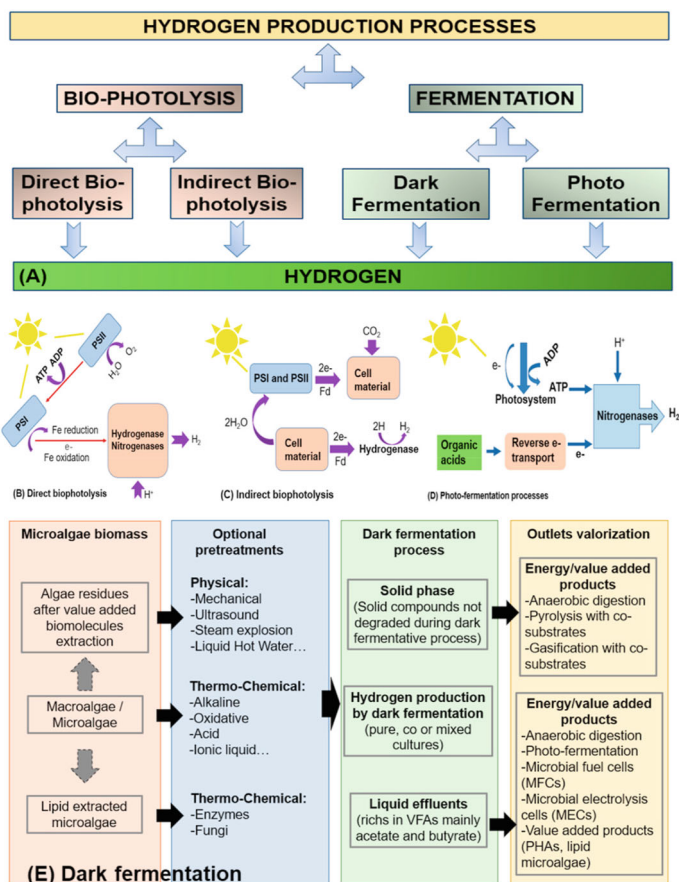


Fig. 7 Overview of different reaction mechanisms of hydrogen production processes, adapted from reference [14]

a DF-MEC system generated up to 182 ml H_2 /g COD at HRT 48 h and an energy recovery efficiency of 217% [11]. Some highlights on MEC technology for hydrogen production are shown in Fig. 8.

3 Scale-Up of Biohydrogen Processes

Figure 9 describes two different classifications for the type of hydrogen, such as gray and blue hydrogen, arising from methane, natural gas reforming, biomass and coal gasification. However, green hydrogen (i.e., water splitting) is produced by renewable resources, where this process has received considerable attention in recent years.

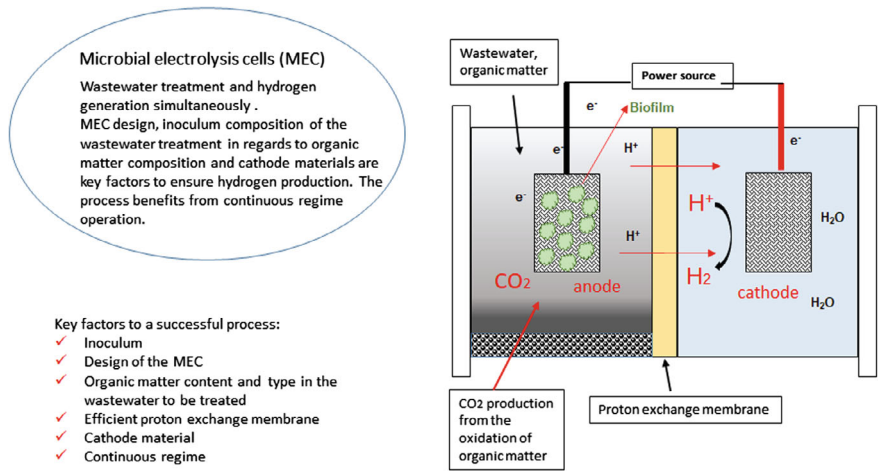
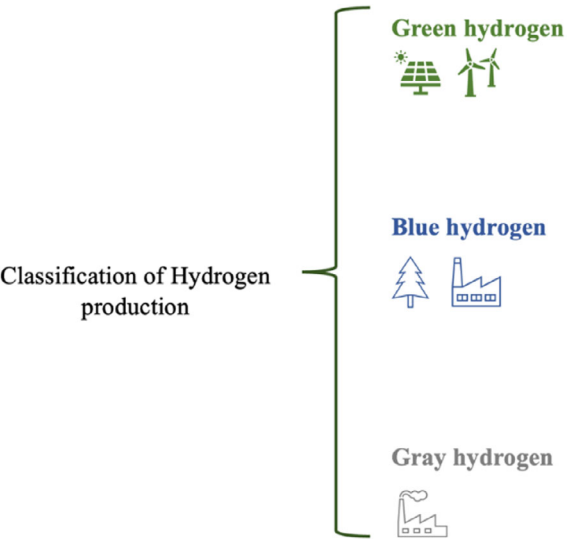


Fig. 8 Hydrogen production in microbial electrolysis cells

Here, the photo- and electrochemical electrolysis processes have a crucial role in producing green hydrogen [42].

In the same way, we can consider biohydrogen gas, which is produced by biological processes. Figure 10 shows the different pathways to produce biohydrogen gas [43]. In this context, we focused on scale-up of biohydrogen processes in dark fermentation, photo-fermentation, and bio-electrochemical processes.

Fig. 9 Different types of H₂ according to production



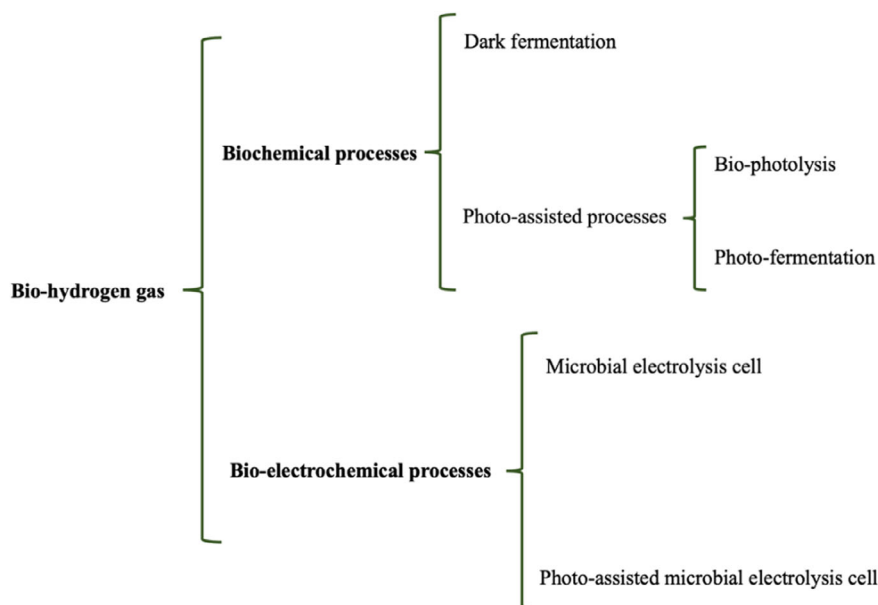


Fig. 10 Different processes to produce biohydrogen gas

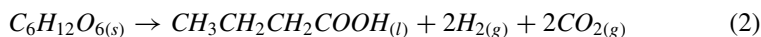
3.1 Dark Fermentation

This is a biological process involving the conversion of organic substrate into hydrogen gas in absence of light. Different aspects have been considered such as pH, temperature and hydrogen partial pressure. pH condition is the most important factor to be regulated in dark fermentation processes. Here, we know that this parameter affects the yields of hydrogen production and the microbial communities. In agriculture residues, the optimal hydrogen production takes place around neutral pH. Another process emerges like denitrification, Taotao Tang and co-workers found that optimal pH is 5 for biohydrogen production [44]. Table 2 depicts different sources to produce biohydrogen and the optimized pH.

Table 2 Experimental conditions used to yield H₂

Source	Reactor	pH range	pH optimum	References
Agriculture residues	Different reactors	4–9	5–8	[45]
Denitrification	Batch	4–11	5	[44]
Beverage wastewater	Internal recirculation batch	3–6 6.5 + 5.8*	5.5	[46] [47]
Food-processing industry	Batch	6–9	7.5	[48]
Swine wastewater	Batch	5–10	6	[49]

Temperature is another important parameter, which affects the scale-up in biohydrogen production. In dark fermentation, different end-products can be formed as shown in Eqs. (1 and 2), using glucose as substrate [50].



We can simulate the reactions 1 and 2 and considering ideal gas and enthalpy of reaction independent of the temperature. The Table 3 shows the thermodynamic properties at 298.15 K and 1 atm.

The overall equilibrium constant of reaction 1 and 2 only depends on hydrogen partial pressure, as shown Eq. 3. According to this equation, if both reactions occur in parallel, we need to add heat to the reactor.

$$\ln k = \ln \frac{k_{p,I}}{k_{p,II}} = \ln \left(\frac{p_{H_2}}{p^0} \right)^2 = -70.60 - \frac{16428}{T} \quad (3)$$

Le Chatelier's suggest that the curvature of the reaction's free energy profile (second derivate) is always positive (Eq. 4). This implies a minimum free energy at equilibrium. The extent of reaction (represented by the advancement of degree in Eq. 5) is influenced by the stoichiometric coefficients of the gaseous species.

$$G'' = \frac{\partial^2 G}{\partial \xi^2} \quad (4)$$

$$\left(\frac{\partial \xi}{\partial p} \right)_T = -\frac{\nu}{G''} \quad (5)$$

Since the overall stoichiometric coefficients (ν) for reactions 1 and 2 are 6 and 4, respectively. The total pressure needs to decrease to favor the reaction according to Eqs. 4 and 5. Therefore, thermodynamics of dark fermentation suggests these reactions proceed at higher temperature and lower pressure. In presence of microbial

Table 3 Thermodynamic properties for the species regarded in Eqs. 1 and 2

Species	$\Delta_f H_{298.15}^0 / \text{kJ mol}^{-1}$	$\Delta_f G_{298.15}^0 / \text{kJ mol}^{-1}$
$C_6H_{12}O_{6(s)}$	-1268	-910
$H_2O_{(l)}$	-285.83	-237.129
$CH_3COOH_{(l)}$	-484.5	-381.9
$CH_3(CH_2)_2COOH_{(l)}$	-533.92	-601.1
$H_{2(g)}$	0	0
$CO_{2(g)}$	-393.51	-394.36

consortia, Rami and co-workers found a relationship between temperature and pH [48]. In that case, the best temperature condition was 45 °C. Here, thermophilic and mesophilic bacteria play an important role. Hong Chen and co-workers found that thermophilic bacteria yield more biohydrogen gas than mesophilic bacteria, with an interval from 0.46 to 63.60 mL g⁻¹, in a volume of reactor of 0.6 L (Fig. 11a) [51]. In a review, Mei Guo et al. found that the optimal conditions to produce hydrogen was at 60 and 75 °C [45]. Table 4 summarizes some substrates to produce biohydrogen, where the optimum interval was in the range from 45 to 60 °C, considering the substrate and microorganism are important.

Due to the extremely poor solubility of hydrogen (1311 atm mol⁻¹ kg H₂O), the partial pressure is a key factor during biohydrogen production. Tinku Casper D’ Silva and coworkers reviewed that the optimal condition for the hydrogen partial pressure is close to atmospheric pressure [55]. In this case, the best way is to transfer hydrogen from one reactor to the collection tank.

In contrast, Martínez-Burgos and co-workers [56] utilized low-cost substrates like agro-industrial effluents, corn steep liquor, and cassava processing wastewater

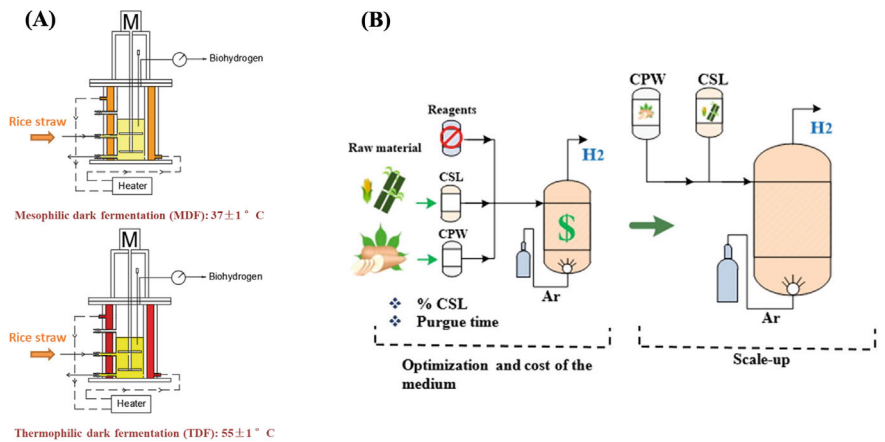


Fig. 11 a Schematic of batch reactor and b Optimization and scale up reactor experiment apparatus for dark fermentation process [51, 56]

Table 4 Optimized range of temperature used to produce biohydrogen

Source	Reactor	Range of temperature °C	Optimized temperature °C	References
Food-processing industry	Batch	35–45	45	[48]
Agriculture residues	CSTR	37–55	55	[51]
Waste wheat	Batch	35–55	45	[52]
Waste biomass	CSTR, Batch	35–75	55–75	[53, 54]

for biohydrogen production. Their microbial consortia achieved yields of 107 and 83.1 mL of biohydrogen/g COD removed. Additionally, they investigated scaling up biohydrogen production with the “Gal consortium”, which resulted in a greater than 40% increase, Fig. 11b. Notably, gas accumulation in the large reaction did not surpass 100 kPa, which is lower than the 300 kPa observed in the smaller reactor.

3.2 *Photo-Fermentation*

This process which has been gained attention for producing biohydrogen, using photosynthetic microorganisms. Here, microorganisms use the light energy to drive the oxidation of organic compounds and produce biohydrogen and CO₂ gas as end-products. As compared with DF, photo-fermentation presents the advantages due to atmospheric nitrogen fixation, and the broad use of the light spectrum. However, the low efficiency of light conversion and the energy costs for using a lamp infers a drawback [57]. There are some challenges to scale-up the photo-fermentation into the biohydrogen production. The photobioreactor design. The main advantage of the photobioreactor is production cost, comparatively low than other production method. Different types of photobioreactor have been developing, such as Tubular, Flat panel, Vertical Column, Airlift, Stirred tank/sharing flask photobioreactor [58]. Light intensity and distribution are a key factor, where a light absorption and scatter is promoted by algal cells [59]. Figure 12 depicts a tubular photo-bioreactor with a volume of 70 L. In addition, Changpeng Ren and coworkers found that the reactor batch yields more Hydrogen gas than that in continuous mode (i.e. 37.6 versus 20.7 mL H₂ L⁻¹ h⁻¹) [60]. They also concluded that the improving efficiency light is due to use an artificial light and sunlight source, at the same time.

3.3 *Bio-Electrochemical (BioEchem) Processes*

The use of microorganisms to catalyze electrochemical reactions is considered, assuming two different bio-electrochemical approaches used for biohydrogen production, including Microbial Electrolysis Cells (MECs) and Microbial Fuel Cells (MFCs) (Fig. 13) [61]. Raul Gautam and co-workers reviewed those systems [61]. Basically, MFCs generate electricity, while MEC require electricity to produce biohydrogen gas. There are some drawbacks such as MFC works best in aerobic conditions in contrast with MECs, in anaerobic conditions. This later condition capture more efficient energy, but they can also create methane, reducing the hydrogen output. Overall, MECs offers a more promising approach for generating valuable products and MFCs is better at removing larger organic compounds.

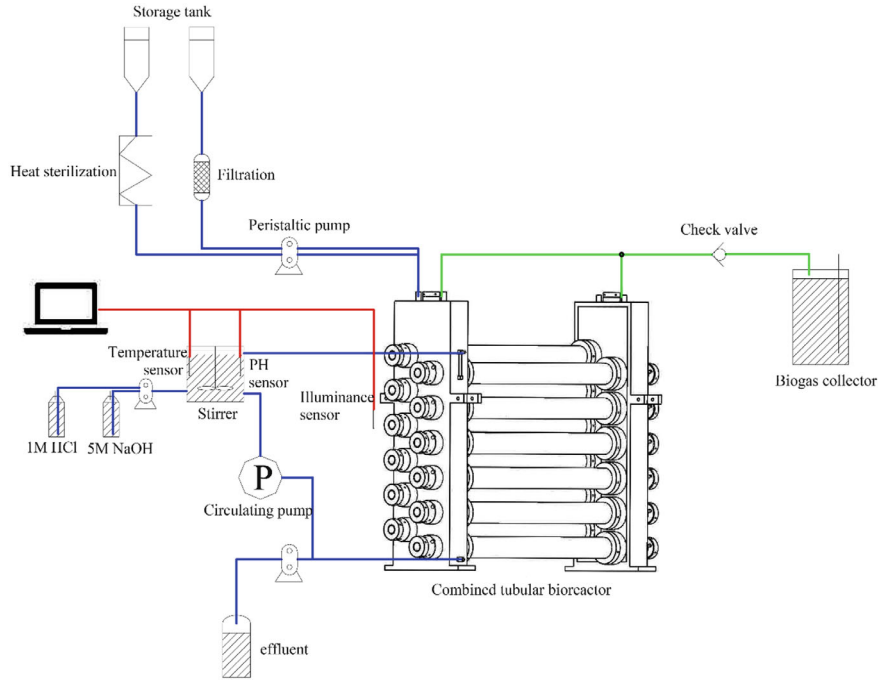


Fig. 12 Schematic diagram of continuous fermentation hydrogen production system in composite tubular reactor [60]

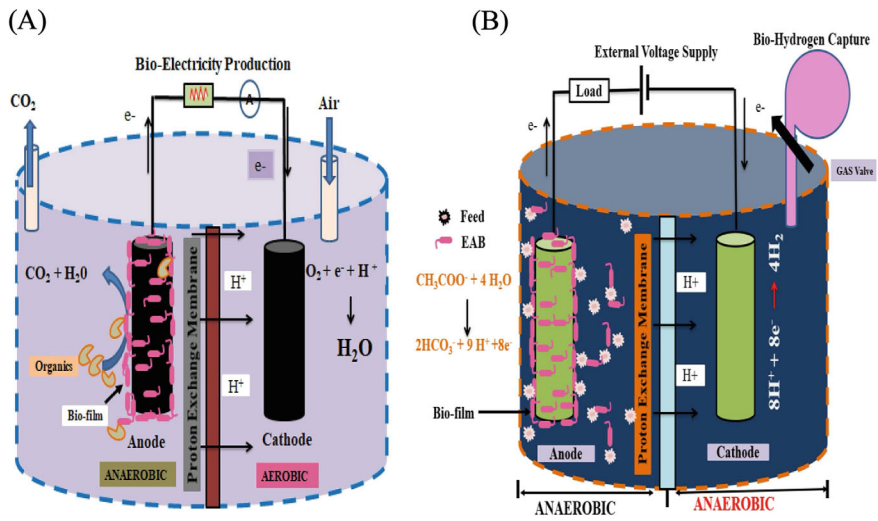


Fig. 13 Schematic representation of **a** microbial fuel cell (MFC) and **b** microbial electrolysis cell (MEC) [61]

4 Industrial Challenges and Drawbacks of Biohydrogen Production

The production of Biohydrogen promise to be an alternative as renewable and sustainable energy vector. However, there are some challenges and drawbacks that need to be addressed before its implementation. Here, we suggest some of them.

Low production rates. The challenges encountered during production are influenced by various factors such as the composition of substrate, method of substrate pretreatment, bio-reactor condition, physicochemical parameters [62].

Purification and storage. Hydrogen produced via different methods always requires purification/separation steps. Biohydrogen production such as DF, PhotoF and BioEchem yields a gaseous mixture. Therefore, Biohydrogen purification presents a challenge for industrial scale. On the other side, the storage is always a challenge and independently the production method. Hassan El Bari and coworkers has reviewed a lot of problems for purification and storage biohydrogen gas [63].

Culture enrichment. Nowadays, there is a difficulty to enrich the culture with specific microbes, which would allow to maximize the biohydrogen yield, and simultaneously suppress the growth of biohydrogen consuming organisms (i.e. methanogens, homoacetogens). To this concern, different strategies have been adopted: (a) adaptation strategies to favor the mixing and acclimatization of cultures to less biodegradable substrates [64], (b) dilution-extinction, where the original population of microbes is diluted to select microbes with a particular function [65], (c) enrichment of the strains in medium with a high content of nutrients [66].

Regarding the challenges that today continue at the industrial level for the production of biohydrogen, the most important continues to be the scaling of the processes at the macroscopic level. In this regard, studies have not been performed for an industrial scale on dark fermentation. And the studies of these bioprocesses have been very scarce at the pilot plant level. Studies have been carried out with a 100 m³ reactor from sugar cane distillery discharges using diverse co-cultures including *Rhodopseudomonas palustris* P2, and *Enterobacter aerogenes* E10 [67]. In this study, high biohydrogen yields of 2.76 mol H₂/mol glucose (0.53 kg/100 m³/h) were obtained. A different study generated 26.13 mol H₂/kg COD eliminated in the organic loading rate in a mixture of acetic acid and ethanol; and a maximal generation rate of 5.57 m³ H₂/m³ reactor/d in a continuous flow anaerobic reactor operating with 1.48 m³ (volume) during 200 days.

5 Economic Aspects of Biohydrogen Production

Analyzing the performance of biohydrogen is essential to determine its competitiveness with other energy sources, including hydrogen production by conventional methods such as pyrolysis. It is indisputable that biohydrogen has great potential to become a relevant alternative energy source in the coming decades, for which

some technical challenges must be solved. Using industrial wastewater as a substrate for producing dark fermentative biohydrogen effectively reduces production costs, being a promising solution to make biohydrogen production profitable. The main challenges in producing dark fermentative biohydrogen are its low yield and high production costs compared to other methods and fuels.

5.1 Temperature

This is a fundamental variable in the net energy production from biohydrogen based on dark fermentation. The processes to produce biohydrogen using room temperature (on average 25 °C or lower) generate a net output of positive energy since they usually do not require much energy to increase the operating temperature, which also considerably reduces the operational costs of bioreactors. It was reported that the optimal operating temperature of dark fermentation is, on average, 20°C [68]. However, depending on the type of reactor, the chemical activators added, and the substrate used, the temperatures associated with the operating conditions may vary. Likewise, several studies suggest that dark fermentation be complemented with other processes such as anaerobic digestion, bioelectrochemical systems, or photofermentation to ensure a positive energy gain and, with this, reduce production costs by facilitating the conversion efficiency of the substrate [39, 69, 70]; in addition, complementing this process can improve the elimination of COD, and with it, favor higher quality of the effluent.

5.2 Substrate

Biohydrogen productivity is strongly related to the substrate used for its production. Biomass substrates (rich in sugars and aminoacids) have higher productivity levels due to their high organic matter content, highlighting the wastewater from the sugar and brewing industries from the production of sweet candies, palm oil, and cotton processing (Table 5). The substrates provided by wastewater from these industries are biodegradable, abundantly available, and low cost [4, 10–12, 71]. Industrial wastewater can generate a net energy gain almost three times higher than what could be obtained from domestic wastewater, amounting to 51.84 kJ/g COD [72]. The processes and substrates used for dark fermentation must be improved to make biohydrogen production competitive. For this, it is necessary to evaluate the best microbial biocatalysts and chemical activators that increase productivity and efficiency in biohydrogen generation [39].

Table 5 Comparison of production costs depending on the substrate

Substrate	Production cost (USD/m ³)	H ₂ revenue (USD/yr)
Beverage wastewater	7,100	16,000
Agriculture waste	7,900	4,000
Solid biowaste	1.52	37,760
Sugar cane distillery effluent	35.40	37,070
Food waste	0.15	–

Source Hay et al. [73]; Shi et al. [74], Chang et al. [75], Li et al. [76]

5.3 Catalysts

Catalysts or photocatalyst can play a relevant role in making biohydrogen production more efficient [77, 78]. It has been reported that mixed crops as a biocatalyst minimize the costs of the biohydrogen production process, but they are used in the large-scale production. The reduction in costs due to the use of mixed cultures as biocatalysts are associated with the fact that they do not require medium sterilization, unlike the use of pure strains as biocatalysts (i.e., *Clostridium* spp and *Enterobacterium* spp) that require a selective use of substrates, in addition of aseptic conditions to avoid contamination problems; these requirements increase the total cost of the process. For biocatalytic pretreatment of the inoculum before dark fermentation process, although it increases biohydrogen production by inhibiting the growth of bacteria that consume hydrogen and enhance biohydrogen production from 6 to 16 times more, it rises production costs considerably, since high temperatures close to 100 °C are required for periods ranging from 1 up to 24 h [79–81]. Finally, metallic additives have also presented benefits in biohydrogen production by increasing its production, improving the formation of enzymes, increasing the transfer of electrons, and providing nutrients for the growth of microbes. Different metal additives include monomers (i.e., gold, silver, copper, iron, nickel, palladium), metal ions (i.e., Fe²⁺), and metal oxides. In particular, metal ions are considered effective additives to increase biohydrogen production [39, 82, 83]. These additives can increase biohydrogen production at a lower cost; for example, iron supplementation (257 mg Fe²⁺/L) in effluents from the palm oil industry increased biohydrogen production by 60% (O-Thong et al. 2008). In the case of the Ni-graphene nanocomposite, with a dose of 60 mg/L for an average effluent flow rate of 1000 m³/day and influent with mono-ethylene glycol (MEG) concentration of 6000 mgCOD/L, an increase in biohydrogen production of 104.86% [84].

5.4 Investment and Operating Costs

Biohydrogen production through dark fermentation is not the most cost-efficient method, with indirect biophotolysis and photofermentation being the most profitable, although they have fewer environmental co-benefits associated with the use of industrial wastewater. For example, while the cost of kg H₂ for indirect biophotolysis is 1.96 USD/kg H₂ with a total operating cost of 892.31 USD, for dark fermentation, it is 18.70 USD/kg H₂ with a total operating cost of 892.31 USD (Table 6).

More information on the investment and operating costs of treating industrial wastewater by dark fermentation is necessary for a complete economic analysis. As mentioned throughout the document, these costs can be reduced using cheaper renewable raw materials, such as wastewater, as substrate. The costs of the reactor and its condition influence the capital investments required; these will depend on the type of reactor, the wastewater that will be processed, and the volume that will be treated. The larger the reactor capacity, relevant economies of scale can occur that reduce investment and operating costs per m³. For example, Vatsala et al. [67] reported that a continuously stirred anaerobic bioreactor (CSTR) with a 100 m³ capacity had investment costs of 444 USD/m³ and operating costs of 135.46 USD/m³, while Han et al. [89] identified that a continuous mixed immobilized sludge reactor (CMISR) with a capacity of 10 m³ had investment costs of 47,820 USD/m³ and operating costs of 4,400 USD/m³. Therefore, increasing the treatment capacity of bioreactors can bring significant improvements in profitability for biohydrogen production. Economies of scale benefits explain why the cost per gallon of biohydrogen produced by the CMISR reactor with a capacity of 10 m³ amounts to 9.52 USD/gallon, while the cost per gallon of biohydrogen produced with the 100 m³ CSTR bioreactor is 0.9 USD/gallon [67, 89, 90]. Han et al. [89] found that production scales with reactors between 10 and 30 m³ are not profitable, identifying that hydrogen production plants with scales greater than 50 m³ could obtain positive benefits. Nevertheless, not only economies of scale are relevant to make biohydrogen production competitive, but

Table 6 Comparison of the efficiency of some biohydrogen production methods

Process	Direct biophotolysis	Indirect biophotolysis	Photofermentation	Dark fermentation
Glucose substrate	0.0	0.0	144.19	867.18
Hydrogen production	1.68 mmoles H ₂ /L day	8.64 mmoles H ₂ /L day	3.672 mmoles H ₂ /L day	2.904 mmoles H ₂ /L day
Total operating cost	64,063.38 USD	93.25 USD	176.10 USD	892.31 USD
Cost per kg H ₂	1,342.27 USD	1.96 USD	3.70 USD	18.70 USD
References	[85]	[86]	[87]	[88]

also the type of industrial effluent to be used as a substrate, the technology used, and the chemical activators added (Table 7).

More information on the production and investment costs for biohydrogen production with dark fermentation is needed to conduct a complete analysis to determine the best technologies. The cheapest sources to produce hydrogen are coal and natural gas, with 0.36–1.8 \$/kg and 2.48–3.17 \$/kg, respectively, while the highest costs are those of solar and wind energy. Therefore, hydrogen production costs are reduced when using waste materials or biomass as energy sources [92]. For now, biohydrogen production is still not profitable compared to other processes and other fuels; fossil fuels are the most competitive. However, it is essential to highlight that hydrogen production by pyrolysis is the energy with the highest cost-efficiency, showing the potential of biohydrogen production with industrial water. For now, the costs of producing hydrogen by the different methods available, except for pyrolysis, are higher than conventional methods for the production of fuels. For example, while the production of hydrogen by pyrolysis presents an energy efficiency of 0.0042 USD/GJ, this value is for gasoline 0.0064 USD/GJ, for biodiesel 0.0085 USD/GJ, and natural gas 0.0106 USD/GJ (Table 8). The production of biohydrogen by fermentation and the production of photobiological hydrogen is just as efficient as the production of natural gas (0.0106 USD/GJ). If current technologies and processes are improved, biohydrogen production can improve its profitability to favor its commercialization in the short term.

6 Conclusions

Although there are significant challenges for making biohydrogen production profitable in the coming years, this is the most viable alternative to replace fossil fuels, avoiding stronger effects in climate change, and uncertainties in fuel prices (improvement of energy security). Likewise, biohydrogen generation is CO₂ neutral, and free of any other greenhouse gas emission, whereby this technology is expected to play an important role to meet the mitigation commitments of numerous world's countries. Particularly, the exploitation of the industrial wastewater discharges to produce biohydrogen can underpin two future problems: the raw materials to produce energy, and the reuse/recyclability of polluted water which can contaminate other water resources. To this concern, further research and development are required to overcome low conversion and yield of biohydrogen production, by tackling the following factors: biocatalyst pretreatment, metabolic and genetic engineering, culture enrichment, and co-culture fermentation, particularly at the industrial level.

Table 7 Capital costs and production costs of dark fermentation

Type of industrial wastewater	USD/gallon of hydrogen	Reactor volume	Temperature	Reactor type	Investment cost (capital) (USD/m ³)	Production cost (USD/m ³)	References
Molasses residual water	9.52 USD per gallon	10 m ³	35 °C	Continuous mixed immobilized sludge reactor (CMISR)	47,820 USD/m ³	4,400 USD/m ³	[89]
Sugar beet molasses wastewater	–	15 m ³	70°C	Fermentor (THF—thermophilic fermentation, PHF—photofermentation)	34,790 USD/m ³	–	[91]
Wastewater from beverages	–	10 m ³	35 °C	Continuously stirred anaerobic bioreactor (CSTR)	161,550 USD/m ³	1,600 USD/m ³	[76]
Wastewater from the sugar industry	0.9 USD per gallon	100 m ³	28 °C	Continuously stirred anaerobic bioreactor (CSTR)	444 USD/m ³	135 USD/m ³	[67]

Source [67, 76, 89, 91]

Table 8 Cost of energy and raw materials involved during the H₂ production

Process	Raw material	Cost of energy USD/GJ
Pyrolysis for hydrogen production	Coal, biomass	0.0042
Photobiological H ₂ hydrogen	H ₂ O, organic acids	0.0106
Fermentative hydrogen	Molasses	0.0106
Hydrogen from water electrolysis	H ₂ O	0.0117
H ₂ from wind energy	Wind mill	0.0360
H ₂ from photovoltaic power	Solar energy	0.0445
H ₂ by biomass gasification	Biomass	0.0668
Other fuels	Raw material	Cost of energy USD/GJ
Gasoline	Crude petroleum	0.0064
Ethanol	Molasses	0.0334
Biodiesel	Jathropa seed	0.0085
Natural gas	Raw natural gas	0.0106

Source [86, 93, 94]

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In Silico Quantum Simulation on the MnTiO₃, FeTiO₃, and NiTiO₃ Materials in Ilmenite Structure as Candidates to Spintronic Devices



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Abstract The development of spintronic devices through the years made significant progress from quantum mechanics. Since improvement on the digital signal to quantic computation, the research on the spin control has promoted advances on the search by materials for such aim. Then, magnetic semiconductors oxides are candidates to build spintronic devices, mainly in antiferromagnetic state. The big promise attributed on the spintronic devices is the quantum processor because instead of using 1 and 0 to describe bytes, the introduction of the spin alpha and beta states creates more combination for the information processing. The evaluation of the MnTiO₃, FeTiO₃, and NiTiO₃ materials in Ilmenite structure investigated through DFT/B3LYP approach to discuss the magnetic ordering as candidate to make spintronic devices.

1 Introduction

1.1 Electronic Development

From the computational revolution started by the mathematician Alan Mathison Turing in 1912, the processing and cryptography of information rose exponentially, reaching a *status quo* in contemporary society. The implementation of the logic through automatic processing applied in an electromechanical machine allowed breaking enemy cryptography in the Second World War. Consequently, the increase of the velocity of the processing introduced high possibilities for many fields of knowledge, such as cryptography, statistics, mathematics, and biology.

The logic structure utilizes true and false answers for direct questions as tools to build logical paths. The extraordinary advance was transforming the abstract logical paths in electrical signals intermediate by mechanical response. The association

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between logic answers and electrical signals through true, there is electrical current and false, there is no electrical signal, indicates the possibility to test much tentative to reach an alphanumeric sequence. The mechanical procedure has the role of storing the “hit” or positive information. All this engineer is accessible from previous data analysis from the user, which manages the “direct questions” to be submitted to the machine. The insertion of a “question” or process was performed through electrical connectors or cables, making an electrical-logical system, and each time it reached an answer, the mechanical system locked the register storing the data. Consequently, the user reads the stored data set, analyzing the answer. Then, the knowledge about the question is fundamental to programming the logical-electrical-mechanical system, becoming the computational process a tool for answering questions. i.e., a big combination of results to find an answer showing that the computational structure works with tentative and error system, i.e., always the interpretative logic implemented from the electrical system verifies the tentative result, determining whether it is true or false. In this point, the number of possibilities for the combination among results is a key to estimating the calculation challenge. A strategy to accelerate the calculation is to reduce the search by a set of answers. For example, we need to know what number of dice with six sides satisfies the diversity between two even numbers. The data set can be all numbers of dice, 1–6, or only the even numbers because of the condition of the question. Each option leaves a computational cost connected to time for the resolution of the problem.

After a brief presentation on the first logical-electrical-mechanical machine, we emphasize that the crucial aspect of computation is the information found by human logic through advanced electrical systems representing true and false conditions. This advanced technology was called binary system, where true and false are associated with positive and negative electrical signals interpreted as 1 and 0 answers were intuitively called bits. The combination of these two possibilities allowed an advance technological never seen before: the creation of the electronic. The idea of producing a machine with electrical circuits to control the path of electrons and create a complex map of combinations between 0 and 1 answers promoted an innovative form to work and generate knowledge. The detection and organization of the electrons in a restricted physical box profoundly modified the processing of the information. A new era was created since the Turing machine, the integrated circuit, semiconductors, and nanoscience made faster, compact, and efficient computers through advanced memories, peripheral, and processor devices.

The union between logic and electrical circuits showed that the 0 (off) and 1 (on) developed the bit or the information in a computer device. However, 4 bits is the minor information or word that can be processed by a simpler processor. This word allows 2^4 or 16 combinations with expansion for 8 bits, and reached combinations are 256, etc. In another way, the high velocity of the electrical current generates combinations at the same velocity, causing a fast-testing process for a big set of answers. Therefore, all technological advances in computers were based on combinations and high velocity of the exchange of information into and among electronic devices. Still, the great agent is the electronic charge and wave behavior properties placed on this incredible subatomic particle. Today, the center of human civilization is the computer on high

or low scales. The control of the electron from nanotechnology determines our recent technological conditions with great solutions and big problems [1].

All aspects commented before are connected to the charge property of the electron. The electrical signals detected by electronic devices are charges displaced through metallic wires. The charge is controlled by the difference of potential measured by Volts, which is the electrical work into an electronic circuit to produce a flow of electrons. Such electronic flow should be withdrawn from the ground state in the materials to be detected, creating the electrical signal. The change of the physical state of the electron is an electronic transition. Here, the applied voltage is the driving force to transit or to excite the electrons from valence band to conduction band. The electrons with more energy create the flow of electrons allowing the electrical current. However, the electrons stay connected to metallic wires, while the wave behavior allows an electromagnetic field during the flow. Basically, the presence and absence of excited electrons are the fundament to digital processing into computers.

Semiconductor oxides won great importance for electronic devices through the diodes to control the electrical current and the Random Access Memory (RAM) to storage the information from ferroelectric polarization. The discovery of diodes was one of the most important parts of the evolution of computers. The type of chemical bonds and crystalline structure determine the properties of the semiconductors through the modulation of the band gap. The p-n junction domain until today the architecture of the electronic memories. The molecular structure of a p-n junction is a very particular union through the fit in between the valence and conduction bands of the n-type and p-type semiconductors making an electronic interface with features very interesting.

The organization of the energy bands around the bandgap is important to control the electrical flow through n-p junction. The position of the electronic bands builds a step on the chemical potentials or Fermi levels between the n-type and p-type semiconductors creating the p-n junction. In the semiconductor area, the scientist created the “engineering of band gap” concept to define the modulation of the band gap through cationic and anionic doping. Semiconductor oxides make n-p junction exactly on the interface between both oxides. If both semiconductor oxides have the same crystalline structure, for example cubic-cubic or tetragonal-tetragonal, is a homostructure creating a homojunction, where there are no mechanical tension crystalline structures. However, if the crystalline structures are different, such as cubic-tetragonal, the interface is a heterojunction from a heterostructure. Now, the interface is under a mechanical tension created by the fit among non-symmetric crystalline structures.

1.2 Spintronic Research

The electron has another property called spin because of the association with the intrinsic rotational movement. The consequence of spin is to create a magnetic moment generating a magnetic field around the particle. Such property is quantic with

discrete values of $+1/2$ and $-1/2$, which are many times denoted as alpha and beta or up narrow ($\uparrow$) or down narrow ($\downarrow$). Magnetism has been a fundamental feature of the nature knowledges since old times. Historical evidence reports first information on the magnetism at 4,000 b.C. in China and the development of the compass around 900 b.C. Curiosity on how magnetic interactions occur in the materials directed to scientific experiments with iron-based materials. One of these investigations studied by William Gilbert in 1600 concluded that the Earth is magnetic indicating the coverage of the magnetic field influencing the compass to point to North, but Gibert believed that magnetism and electricity were separated physic effects, then the magnetism would continue to be understand as isolated from electron. The connection between magnetism and electricity would be discovery by Hans Christian Oersted in 1819 through famous experience using a needle and a metallic wire, where the needle was oriented when the electrical current travelled by metallic wire [2]. Two aspects proved were that the magnetic effect occurs in closed electrical circuit and the interposition of materials between compass and metallic wire did not perturbate the magnetic effect.

Soon, in 1820, André-Marie Ampère noted that Oersted's experiment evidenced an orthogonal magnetic interaction between the compass and the conductor wire. Then, he created an experiment with an iron bar (magnetic) wrapped around a copper wire (non-magnetic) for electric current [3, 4]. This device was called a solenoid, which showed that both materials were transformed into magnets. If two solenoids were brought together, the same properties previously known attraction and repulsion magnetic as magnetic poles would be validated. One of the most advanced magnetic phenomena was the high-density magnetization by square inch introduced in HDD devices by IBM in 1997 through spin-valve. In such sense, the giant magnetic resistance (GMR) effect reported by Albert Fert and Peter Grünberg [5, 6] was the physical phenomenon most important in the decade 90's for information technology (IT). The high quantity of data storage in the new HDD devices increased in large scale the communications creating tools such as data analytics and machine learning. The GMR is a device made by a sandwich between ferromagnetic (FM) and non-magnetic (NM) materials distributed in FM-NM-FM layers forming a device. The quantum effects act in coherence through thin films of each material with the magnetic field determining the orientation. The amazing response occurs in electrical current, if the FM orientation is performed on the GMR device, the electrical is high; however, in AFM orientation there is the cut of the electrical current, a null current. In 2007, the scientists received the Nobel Prize for the development of the discovery.

Spintronics is one of the more challenges recent in computational development [7–9]. The primary idea is creating an electronic circuit able to control an electric flow oriented by spin, i.e., the electrical current should be oriented by alpha or beta spin moment. Today, the electrical current is chaotic as spin orientation because the signal is detected by charge. However, the inclusion of the spin orientation promotes a large advantage to information processing. In an electric circuit, the bites are 0 or 1, on or off. From the spintronics, bite 1 is duplicated in alpha 1 and beta 1 electrical currents; then, the total fundamental bites are 0, $+1$, and -1 , here the signal represents the spin orientation. The combination of the alpha and beta spins currents creates information

processing much more robustly introducing a new set of logical combinations. Now, we have new combinations among electrical current channels in independent form, for example, an electronic circuit can process information from two active electrical channels as $+1 +1$, $+1 -1$, $-1 +1$, and $-1 -1$, all this combines with 0 or null electrical current would be $0 +1$, $0 -1$, $+1 0$, and $-1 0$ totalizing 8 fundamental combinations instead of 4 combinations from modern computers.

Magnetic semiconductor oxides (MSO) are great candidates for spintronic devices. The magnetic ordering on such materials is accessible in ferromagnetic (FM) and antiferromagnetic (AFM) states. The advantage of the MSO is that the ground state as AFM ordering because the unpaired electrons on the magnetic cations are antiparallel aligned one each other in a spin net. It is interesting such condition because into to AFM MSO materials there are alpha and beta spin ordering generate divided electronic profiles in alpha and beta channels. In this point, the separation of the band structure in different spin moments should promote different electronic excitations through different bandgaps making possible a selection of one spin current. The Perovskite-type MSO has been investigated as candidates for spintronic technology mainly, the expanded structures as Ilmenites [10, 11], Lithium-Niobates [12], and Corundum [13].

2 Computational Methodology

The implementation of the DFT approach for open-shell calculations is consolidated as an option to represent the spin distribution of the magnetism in semiconductor oxides. There are two main types of functional the local as the Local Density Approximation (LDA) and Spin Local Density Approximation (LSDA) and, the non-local Generalized Gradient Analysis (GGA) functional, both are important tools to describe the electronic densities in singular chemical bonds as metal, semiconductor, and insulated materials [14, 15]. The GGA description can be classified as pure, hybrid, and meta functionals. To investigate the MTO, FTO, and NTO materials was implemented the Becke functional [16] empirically three-parametrized by Lee-Yan-Parr hybrid functional (B3LYP) [17] extensively mentioned as a good functional for the electronic properties in crystalline structures as bandgap [18–20], magnetism [10, 13, 21, 22], optical [23, 24], and electrical [25–28] features. The PBE0 is pure functional [29–31]. The HSE06 non-parametrized hybrid functional [32]. All simulations were performed on the CRYSTAL09 program [33] under the crystalline orbitals constructed from linear combination of atomic orbitals applying the periodic contour conditions through type-Gaussian basis-sets representing the electrons as 86-411(d31)G for the Ti [34], 86-411d41G for the Fe [35], 86-411d41G for the Mn, 86-411(41d)G for the Ni [36], and 8-411G [36] for the O atoms [37].

In polarized spin systems, the spin moment ordering can be ferromagnetic (FM) or antiferromagnetic (AFM) depending on the strength of the exchange–correlation (E_{xc}) energy. Low E_{xc} leaves for FM states, the spin moments are parallel causing the repulsion among magnetic sites decreasing the magnetic coupling (J). For AFM

state there is an increase on the E_{xc} promoting a high magnetic coupling among magnetic sites. In quantum simulations of solid-state materials with correlated highly electrons, the Ising model determines magnetic ground state in good agreement for semiconductors oxides in different crystalline structures [38–40]. To obtain good descriptions on the magnetic states, the relaxation of crystalline structures should be performed with spin moment ordering of the unpaired electrons well determined by the model allowing a spontaneous relaxation proceeding to describe the magnetic state of interest. Another point is the number of points to build the numerical mesh for the wave functions in the Brillouin zone through the Monkhorst–Pack approach [41] and Gilat net [42, 43]. The mesh choice was the $8 \times 8 \times 8$ implemented by SHIRINK option. Besides, the SPINATOM described the magnetic sites in FM and AFM states through alpha and beta ordering on the Mn, Ti, and Ni atoms; while the SPIN keyword connected to SPINLOCK option guarantees the confinable of the DFT calculations.

3 Results and Discussion

Ilmenite crystalline structure is featured in octahedra sites for 2^+ and 4^+ cations oriented in non-regular form. The unusual octahedra orientation is due to the deviation of Goldsmith rule for cation radius in Perovskite oxides. MTO, FTO, and NTO materials follow such distortion, in particular because of unpaired electrons on the 3d orbitals creating the $e_g^2-t_{2g}^3$, $e_g^2-t_{2g}^4$, and $e_g^2-t_{2g}^6$ electronic configurations for Mn^{2+} , Fe^{2+} , and Ni^{2+} , respectively. The 3d orbitals orientation induces the rotation of octahedral sites regarding regular octahedral sites found in non-magnetic ferroelectric Perovskite oxides, such as $SrTiO_3$ [44–46], $BaTiO_3$ [38, 47], $SrSnO_3$ [48], and $SrTi_{1-x}Sn_xO_3$ [49].

It is important to comment that the applied functionals can be classified as pure, PBE0, non-parametrized hybrid, HSE06, and parametrized hybrid, B3LYP. In DFT approach, the mathematical functions proposed for the calculation of the exchange–correlation energy is the main point on the description of the functionals. Then, for each mathematical formalism in each functional indicates singular exchange–correlation energies. The results for each PBE0, HSE06, and B3LYP functionals on the structural properties of the magnetic ground states found for the investigated materials is Table 1.

The predicted results from studied functionals indicated the decrease of the crystalline lattice parameters regarding the decrease of the ionic radius following the expected atomic behavior in chemistry of materials. The similarity among PBE0, HSE06, and B3LYP functionals on the lattice parameters is notable indicating that the ionic description from the functionals is close, i.e. the calculation on the attractive and repulsive forces strongly connected by the Hartree–Fock potential through the Coulomb integrals on the nuclear and electronic interactions are lowly affect by cationic exchange. The electronic representation of the 2^+ formal charge from each Mn, Fe, and Ni cations on the lattice parameters is in good level.

Table 1 Calculated lattice parameters for the MTO, FTO, and NTO in Ilmenite structure from PBE0, HSE06, and B3LYP functionals

Functional	Lattice param.	MTO	FTO	NTO
		10 ⁻¹⁰ m	10 ⁻¹⁰ m	10 ⁻¹⁰ m
PBE0	a = b	5.150	5.113	5.036
HSE06	a = b	5.150	5.114	5.037
B3LYP	a = b	5.187	5.149	5.076
PBE0	c	14.258	14.047	13.864
HSE06	c	14.258	13.889	13.708
B3LYP	c	14.414	13.893	13.704

In counterpart, the contribution of the exchange–correlation potentials implemented from the PBE0 and HSE06 were the same for the a and b lattice parameters, while the B3LYP showed different results. For c lattice parameters there was a progressive variation in according to e_g-t_{2g} configuration. The z orientation of this lattice parameter corresponds to the essential orientation on the e_g-t_{2g} degenerate electronic levels. Then, the rotation on the octahedra sites, [XO₆] X = Mn, Fe, Ni and [TiO₆] clusters, of the Ilmenite structure played a determinant role to represent the unpaired electrons on each magnetic cation.

The energetic difference between the energies of the ferromagnetic (FM) and antiferromagnetic (AFM) states determines the magnetic ground state (E_{mgs}) for MTO, FTO, and NTO materials.

$$E_{mgs} = E_{FM} - E_{AFM} \quad (1)$$

where, the positive value for Eq. (1) indicates that the AFM state is the magnetic ground state; meanwhile, a negative value suggests an FM state for the ground state [12]. An important point, the Ising Hamiltonian is the mandatory calculation for quantum simulations to calculate the magnetic ground state predicting the magnetic coupling constant since the Heisenberg Hamiltonian is not applied in silico approximations because it does not satisfy the eigenvalue condition from the mathematical formalism. The CRYSTAL17 program in this work employs a collinear method to simulate the magnetic ordering following the condition since there is a symmetric distribution of the unpaired electrons along the spatial coordinates on the unit cell. From the high spin ordering on the Ilmenite structure, Pauli's rule is true determining the removal of the spin coupling and, consequently, a null effect of the spin orbital coupling. The number of electrons in each material is the same allowing a comparison among total energies calculated by functionals. The total energies differences calculated for MTO, FTO, and NTO at same magnetic state for different functionals (ΔE_ρ) is defined in Eq. (2), where the total energies for each functional in each magnetic state is E_ρ(FM, AFM) for ρ = PBE0, HSE06, and B3LYP.

$$\Delta E_\rho = E_\rho(FM, AFM) - E_{\rho'}(FM, AFM) \quad (2)$$

Table 2 shows the calculated total energies, the E_{mgs} , and the magnetic ground stated for the MTO, FTO, and NTO. The calculated average ΔE_p between PBE0 and HSE06 for all materials is 0.11 eV (4.0 mhartree) for both FM and AFM states, since the PBE0 calculated higher total energies than the HSE06. However, the B3LYP calculated higher total energies than PBE0 and HSE06 with an average ΔE_p of 22.7 eV (0.83 hartree). Then, the exchange–correlation potentials for PBE0 and HSE06 calculate near total energies, while the description of the exchange–correlation potential through empirical parametrization into B3LYP calculates total energies much more than pure and non-parametrized functionals applied in this investigation. The calculated E_{mgs} indicate the magnetic ground states for the MTO as AFM, FTO as FM, and NTO as AFM. Such result can be considered robust because three different functionals predict equal magnetic ground states for each material. Furthermore, there are modulations regarding the quantitative interpretation of E_{mgs} . In MTO, the values alternated in 2 meV proposing that the representation of the unpaired electrons in $e_g^2-t_{2g}^3$ electronic configuration for the Mn atoms were analogously described for each functional. All electrons in the Mn^{2+} 3d orbital are unpaired causing a high electronic location because each electron occupies a single orbital in alpha spin. Changing of material, the Fe^{2+} cations present $e_g^2-t_{2g}^4$ electronic configuration in FTO determining one electronic pairing between two electrons into t_{2g} orbitals. The introduction of this effect changed the magnetic coupling among Fe atoms to become the magnetic state of the Ilmenite structure as FM. This is overall evidence of the break of the magnetic coupling among the $[\text{FeO}_6]$ sites. This single electron available for electronic pairing has a resonant feature because of the possibilities of pairing with any t_{2g} electrons. The E_{mgs} has an increase of energy at range of 3 meV among functionals with the B3LYP calculating the more value. In NTO, the Ni^{2+} cation shows a fill of the t_{2g} levels become $e_g^2-t_{2g}^6$ configuration. The complete occupancy of such levels promotes the absence of the contribution for the magnetic effect. Then, only the two electrons on the e_g levels in $[\text{NiO}_6]$ cluster have magnetic moment to create the magnetic effect. Again, there is an increase on the E_{mgs} regarding the inclusion of parametrization on the functional. Then, the electronic pairing on the t_{2g} levels has caused an increase between the FM and AFM magnetic states for the FTO and NTO materials according to parametrization method applied in the HSE06 and B3LYP regarding PBE0.

The calculated magnetic couplings (J) are in Table 2. The J calculated for the MTO presents an average of $5.2 \times 10^{-5} \text{ K}^{-1}$ calculated from closed values by functionals. The description of the $t_{2g}^3-e_g^2$ unpaired electrons present low influence on the exchange–correlation effects from the type of the functionals. However, the J for FTO indicates an increase of value according to description of the functional. The simple exchange–correlation term from PBE0 calculated the minor J, while the HSE06 and B3LYP increased the J in a rate of $0.2 \times 10^{-5} \text{ K}^{-1}$ suggesting that the parametrizations had influence on the electronic coupling energy from the $t_{2g}^4-e_g^2$. Probably, the correlation terms described in more detail on the HSE06 and B3LYP determined such difference among the J.

Table 2 Magnetic properties for the MTO, FTO, and NTO in Ilmenite structure calculated from PBE0, HSE06, and B3LYP functionals

ρ		MTO	FTO	NTO
PBE0	FM	-4451.7556044000	-4677.0872249360	-5166.2347037190
	AFM	-4451.7579196667	-4677.0867679290	-5166.2349932759
	E_{mgs} meV	+63.0 (<i>AFM</i>)	-12.4 (<i>FM</i>)	+7.9 (<i>AFM</i>)
	J (10^{-5} K^{-1})	5.1	-1.6	4.8
HSE06	FM	-4451.7513887420	-4677.0835515120	-5166.2306726100
	AFM	-4451.7537792952	-4677.0830229757	-5166.2309790169
	E_{mgs} meV	+65.0 (<i>AFM</i>)	-14.4 (<i>FM</i>)	+8.3 (<i>AFM</i>)
	J (10^{-5} K^{-1})	5.3	-1.8	5.1
B3LYP	FM	-4452.5598722470	-4677.914873484	-5167.0963017440
	AFM	-4452.5622556550	-4677.914304175	-5167.0966831531
	E_{mgs} meV	+64.9 (<i>AFM</i>)	-15.5 (<i>FM</i>)	+10.4 (<i>AFM</i>)
	J (10^{-5} K^{-1})	5.3	-2.0	6.5

Furthermore, the inversion of the J is due to the paired electron, which decreases the J because of the repulsion among $[\text{FeO}_6]$ magnetic clusters propagated by non-magnetic O atom. Similar effect occurred on the NTO, where there is the most value of J . The unpaired electrons on the $t_{2g}^6-e_g^2$ caused the increase on the magnetic coupling among $[\text{NiO}_6]$ clusters becoming the J near to the MTO. Again, the correlation energy from HSE06 and B3LYP increased the magnetic coupling values.

Table 3 shows the atomic magnetic moment (μ) for NTO, FTO, and MTO materials. The investigated functionals calculated closed m values for all atoms. The average μ for magnetic cations were 1.732 μ_B for Ni, 3.783 μ_B for Fe, and 4.764 μ_B for Mn. However, the more magnetic moment induced on the non-magnetic cations, O and Ti, was in the FTO, the only FM material. The atomic magnetic moment on the O atoms is less than Ti atoms in NTO and FTO. Nevertheless, in MTO, the O and Ti magnetic moments are very close. Such behavior indicates the role of the O atoms as magnetic connections between cations. In this study, the O atoms act despite the connections occurring between magnetic cations and non-magnetic Ti cations. In another hand, all functionals confirmed the unexpected atomic magnetic moments on the Ti atoms creating a new magnetic site on the $[\text{TiO}_6]$ clusters.

In Fig. 1 are the DOS projections by atom for MTO, FTO, and NTO with the Fermi's energy referenced at 0 eV. The PBE0, HSE06, and B3LYP functionals calculated wide semiconductors band gaps of 4.794, 3.984, and 3.962 eV for MTO, respectively. Clearly, the PBE0 overestimates the band gap, which is progressively decreased by the parametrizations included in the HSE06 and B3LYP. Such behavior is frequently reported in non-magnetic Perovskites. In AFM state, the description of the electronic structure is very near to non-magnetic systems. The correction introduced into hybrid functionals allows a more adjustment on the energetic levels on the Valence (VB) and Conduction (CB) Bands calculating values closer to experimental

Table 3 Magnetic moments on the magnetic sites of the transition metals of the MTO, FTO, and NTO materials in Ilmenite structured simulated by PBE0, HSE06, and B3LYP functionals

ρ	μ	MTO	FTO	NTO
PBE0	$\mu_{X(1)}$	4.777	3.792	1.745
	$\mu_{X(2)}$	-4.777	3.792	-1.745
	$\mu_{Ti(1)}$	-0.013	0.077	-0.019
	$\mu_{Ti(2)}$	0.013	0.077	0.019
	$\mu_{O(1)}$	0.010	0.044	0.007
	$\mu_{O(2)}$	-0.010	0.044	-0.007
HSE06	$\mu_{X(1)}$	4.771	3.789	1.740
	$\mu_{X(2)}$	-4.773	3.789	-1.740
	$\mu_{Ti(1)}$	-0.013	0.077	-0.019
	$\mu_{Ti(2)}$	0.013	0.077	0.019
	$\mu_{O(1)}$	0.011	0.045	0.008
	$\mu_{O(2)}$	-0.010	0.045	-0.008
B3LYP	$\mu_{X(1)}$	4.742	3.768	1.711
	$\mu_{X(2)}$	-4.742	3.768	-1.711
	$\mu_{Ti(1)}$	-0.011	0.080	-0.019
	$\mu_{Ti(2)}$	0.011	0.080	0.019
	$\mu_{O(1)}$	0.012	0.051	0.008
	$\mu_{O(2)}$	-0.012	0.051	-0.008

band gaps [15, 20, 50–54]. In this point, the FTO in FM state has two band gaps in alpha and beta channels calculated as wide semiconductor band gap of 4.576 eV in alpha channel and 2.422 eV on the beta channel from PBE0. Now, the HSE06 calculates 3.963 eV for alpha channel and 2.543 eV on the beta channel, while the B3LYP simulated 4.099 eV for alpha and 2.383 eV on the beta channel. The strongest unpaired electrons effect from the Fe atoms is evident. The decrease in the superexchange effect among magnetic sites leaving the FM state causes the separation of the electronic structure in alpha and beta channels because most electrons in a spin orientation, as alpha state. All functionals predicted a wide and an optical band gap for FTO. In AFM state for NTO, the PBE0 simulated the widest band gap of 5.034 eV, the HSE06 obtained 4.300 eV, and B3LYP calculated 4.366 eV. Such magnetic state shows the superexchange effect causing the magnetic coupling between Ni magnetic sites inducing the increase of the band gap through a high degenerescence of the electronic levels. Hence, in type-Ilmenite Perovskite structure, the magnetic cations on the A site possibility a band gap engineering through magnetic state controlled by the superexchange effect.

The MTO has the 3d Mn and 2p O states predominant on the top of the VB indicating an electron transference from Mn–O bonds to bottom of the CB dominated by 3d Ti orbitals. The contribution of the Ti–O bonds are low; then, the electronic

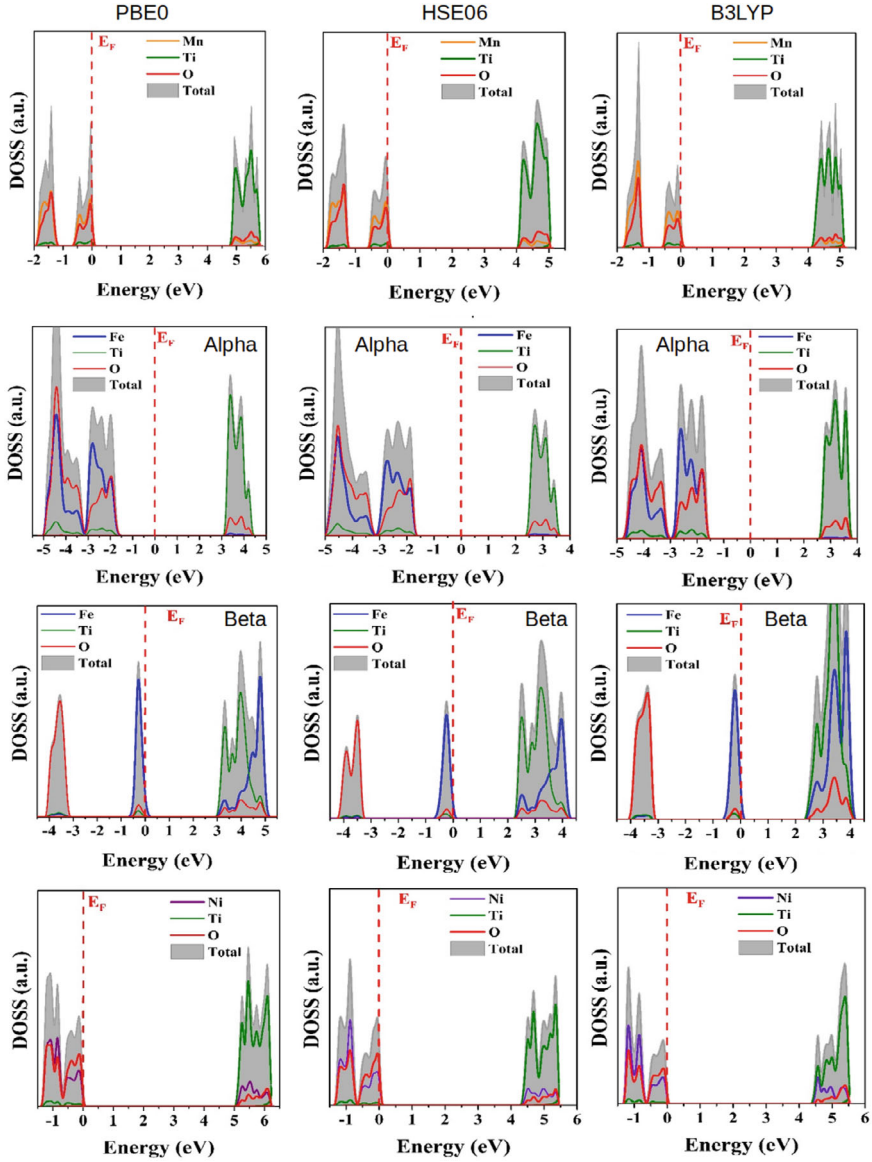


Fig. 1 Density of electronic states (DOS) of MTO, FTO, and NTO materials in Ilmenite structured simulated by PBE0, HSE06, and B3LYP functionals. MTO and NTO materials presents degenerated alpha and beta channels

location is more directed to 3d Ti orbital. In such aspect, there are 3d Ti orbitals on the top of VB confirming the calculated spin moment described in discuss of the Table 3. All functionals show a spontaneous electronic transference from 3d⁵ Mn atoms to 3d⁰ Ti atoms through a non-magnetic atom with features magnetic, the O atoms. The 0.5 eV value was the energetic difference calculated between $e_g^{2-}t_{2g}^{3-}$ levels. In semiconductor behavior on the alpha channel of the FTO, there is a strong decrease of energy states around 1.5 eV followed by a lowest energetic difference between $e_g^{2-}t_{2g}^{4-}$ levels. In beta channel, the e_g levels are in the board of the VB promoting a high contribution on the frontier states decreasing the bandgap. For NTO, the $t_{2g}^{6-}e_g^{2-}$ are closer in energy. The predominance of the electronic states on the top of VB are from 3d Ni orbitals and 2p O orbitals indicating the Ni–O bonds as source of electrons. In CB, the antibonding 3d Ti orbitals are predominantly on the antibonding 3d Ni orbitals. Therefore, an electronic excitation suggests that electrons can occupy both Ti and Ni states from the Ni–O bonds.

4 Conclusion

The PBE0 and HSE06 functionals described near a and b lattice parameters indicating that the electronic interactions calculated in such plane is close for pure and parametrized functionals; while the B3LYP functional overestimated the same lattice parameters suggesting that electronic interactions of type repulsions are more than attractive interactions. For the c lattice parameters there is a contraction according with the ionic radius and t_{2g} - e_g electronic configurations of the magnetic cations. Atomic magnetic moments showed a low spin moment on the Ti atoms suggesting a charge transfer for the empty 3d orbitals. The magnetic ground state for MTO and NTO were AFM with more magnetic coupling and FTO was FM with low magnetic coupling. Regarding optical properties calculated through band gap, the MTO and NTO have equivalent band gap for alpha and beta channels and the FTO has two band gaps, one wide band gap more than 4 eV and another band gap more 2 eV. In this direction, the NTO and MTO magnetic titanates are good candidates for spintronic devices, mainly because of the location of magnetic moment on Ti atoms and magnetic ground state.

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Theoretical and Experimental Insight for the Design and Characterization of Perovskite-Based Solar Cell



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Abstract The perovskite solar cells (PSCs) have excellent optoelectronic properties due to their tunable band gap, large carrier mobility, high absorption coefficient, and long diffusion lengths, making them great candidates for photovoltaic applications. Additionally, these devices have a simpler fabrication process and a lower cost compared to conventional silicon-based solar cells. Most PSCs consist of a perovskite material as the light absorber, sandwiched between a hole transport layer (HTL) and a electron transport layer (ETL). Recently, the highest efficiencies above 25% have been achieved for the organic–inorganic PSCs based on MAPbI₃ and FAPbI₃ as light-harvesting materials. In recent years, all-inorganic perovskite compounds like CsPbX₃ (X = I, Br, Cl) and new perovskite-based materials with

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low concentrations of Pb or Pb-free, such as $\text{CsGe}_n\text{Pb}_{1-n}\text{X}_3$ and $\text{CsSn}_n\text{Pb}_{1-n}\text{X}_3$ ($\text{X} = \text{I, Br, Cl}$; $0 \leq n \leq 1$), have attracted much attention from researchers to improve the long-term stability of the devices and to reduce their toxicity. In this work, we study all-inorganic PSCs based on different absorber material as perovskite layer. Specifically, we investigate and discuss the electronic properties of the perovskites CsBX_3 ($\text{B} = \text{Pb, Sn, Ge}$ and $\text{X} = \text{I, Br, Cl}$) materials, such as lattice parameter, formation energy and electronic band-gap, using a set of ab-initio calculations based on quantum mechanics within the framework of the Density Functional Theory (DFT). Furthermore, we carry out numerical simulations using SCAPS-1D software to study the performance of different architectures of PSCs. Several possible combinations of materials have been studied as HTL and ETL. The results obtained are helpful for improving the performance of PSCs and seeking an “eco-friendlier” alternative by using less Pb in them. Finally, results of the manufacturing and characterization of one of the materials studied as ETL (ZnO) are discussed.

1 Introduction

Energy is a strategic resource for the development of countries. The increasing demand of energy together with the associated environmental impact of fossil fuel combustion forces the use of new technologies [1–130] based on renewable energy to produce electricity, such as photovoltaic (PV) systems. For this reason, the study and development of solar cells has become a very important topic in recent years, in order to generate electric power directly from sunlight energy and to reduce the carbon dioxide (CO_2) emissions to the atmosphere. Nowadays, these devices are extensively used in both rural and urban areas. Also are used in specific applications, such as in the space environment, to feed the batteries of satellites orbiting the earth.

In recent decades research in the area of PV systems has given rise to a wide variety of solar cells that can be classified into three generations based on the materials and manufacturing methodologies employed. PV devices based on crystalline silicon (c-Si) are the first generation of solar cells, which have been widely manufactured and deployed. Presently, mono and poly-crystalline silicon wafers dominate the global solar cell market, constituting over 90% of its share. The performance of these devices closely aligns with the theoretical limits postulated by the Shockley–Queisser model, reaching approximately 31% efficiency under one sun illumination and approximately 41% under maximum concentration of suns [1].

The second generation of solar cells correspond to thin films based on materials such as cadmium telluride (CdTe), copper indium gallium selenium (CIGS), or amorphous silicon (a-Si). These devices are cheaper but less efficient than first-generation solar cells. Recently, Cd-free CIGS solar cells have reached the highest efficiency of 23.35% [2]. Their ability to be fabricated on flexible or non-flat substrates unlocks a multitude of novel applications.

Finally, the third generation encompasses diverse technologies including dye-sensitized solar cells (DSSCs), organic polymer-based solar cells, quantum dots solar cells (QDSCs), hot carrier solar cells, among others [3].

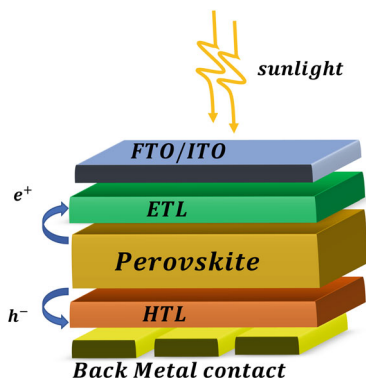
In recent years, perovskite solar cells (PSCs) have emerged as a new class of photovoltaic technology that promises to revolutionize the industry. These cells offer the promise of significantly lower costs and higher efficiencies compared to conventional crystalline silicon solar cells, positioning them as pioneers in the search for sustainable energy solutions.

Perovskite solar cells exhibit excellent optical and electronic properties such as tunable band gap, large absorption coefficient, high charge carrier mobility and long diffusion lengths [4–7]. These devices have had a significant increase in performance. The power conversion efficiencies (PCE) of PSCs have substantially increased from 3.8% in 2009 [8] to more than 25% today [9], confirming the upward trend of this promising technology.

The planar n-i-p structure of a PSC is shown in Fig. 1, which consists of a perovskite layer as a light-absorbing region sandwiched between a n-type electron transport layer (ETL) and a p-type hole transport layer (HTL). The light enters through the ETL. Both transport layers (ETL and HTL) link the perovskite layer with the electrodes. To ensure optimal carrier collection at the electrodes, ETL should block the flow of holes but allow electrons flow to travel towards the cathode. On the contrary, HTL should block the flow of electrons but allow the flow of holes towards the anode.

In general, perovskite layers have an ABX_3 structure, where A is an organic or inorganic cation, B is a metal cation, and X is a halide anion. Perovskite materials like methylammonium lead halide ($MAPbX_3$; X = I, Br, Cl) and formamidinium lead halide ($FAPbX_3$; X = I, Br, Cl) have garnered significant attention for PSCs in recent years [10–17]. These organic–inorganic hybrid PSCs have achieved efficiencies comparable to established market technologies such as thin-film solar cells based on cadmium telluride (CdTe), copper indium gallium selenide (CIGS), and even silicon solar cells. However, organic–inorganic hybrid PSCs face significant challenges related to long-term stability and toxicity. These materials are prone to

Fig. 1 Typical normal planar structure (n-i-p) of PSCs



degrading under the influence of common external stresses, with humidity as the worst problem, facing significant challenges for their development and commercialization [18–20]. Specially, MAPbI₃ has exhibited instability attributed to the high volatility of hydrophilic organic cations at elevated temperatures. On the other hand, FAPbI₃ perovskite has shown superior thermal stability and enhanced photoelectric properties [21], yet concerns persist regarding its phase stability. FAPbI₃ undergoes a structural phase transition from the black α -phase to the non-perovskite yellow δ -phase at room temperature [22, 23]. Recently, CsPbX₃ (X = I, Br, Cl) inorganic halide perovskites, created by replacing volatile organic MA⁺ and FA⁺ cations with cesium (Cs⁺), have garnered significant attention due to their inherent inorganic stability and impressive photovoltaic performance [24–28].

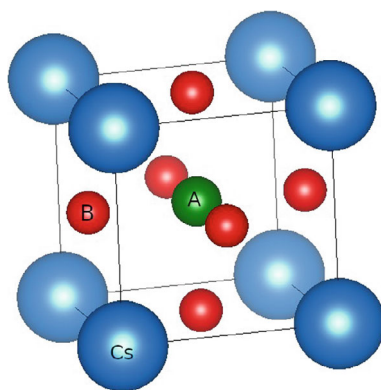
In particular, all-inorganic CsPbI₃-based PSCs have seen a significant efficiency improvement from 2.9% [29] to over 20% [30, 31] in a few years, positioning them as an ideal material for photovoltaic devices. However, PSCs based on CsPbX₃ (X = I, Br, Cl) are not yet industrialized for widespread adoption as clean and renewable energy sources due to their use of lead as a sunlight-absorbing material, which is harmful to the environment. Lead contamination can cause long-term ecological damage due to its persistence and high toxicity.

To address the well-known long-term stability and toxicity limitations of traditional lead-based perovskite solar cells, researchers are actively exploring alternative materials. One promising approach involves “All-Inorganic Lead-Free Perovskite Solar Cells” where lead (Pb) is substituted with environmentally friendly elements like germanium (Ge) and tin (Sn). This strategy aims to achieve a sustainable and high-performing solar cell technology.

In this study, we have investigated the lattice parameter, the formation energy and the electronic band-gap for the perovskites CsBX₃ (B = Pb, Sn, Ge and X = I, Br, Cl) materials, using the Density Functional Theory (DFT) method under the Vienna Ab-initio Simulation Package (VASP). In addition, a comparative theoretical analysis of all-inorganic CsBX₃ (B = Pb, Sn, Ge and X = I, Br)-based PSCs was conducted using modeling and simulation techniques. Three materials (Cu₂O, CuSCN, and CuI) were proposed as the hole-transporting layer (HTL), and two materials (ZnO and TiO₂) were used as the electron-transporting layer (ETL). These materials under consideration in this study are promising candidates for use in all-inorganic PSCs. The performance of the devices was evaluated for different thicknesses in the perovskite layer to optimize the design of the structure and to enhance the efficiency of these all-inorganic PSCs. Finally, results of the manufacturing and characterization of one of the materials studied as ETL (ZnO) are discussed.

The work is organized as follows: Sect. 2 outlines the used methodology. Section 3, presents and discusses the results. The paper ends with a summary of findings and conclusion.

Fig. 2 Crystal structure of α -crystal phase (space group $Pm\bar{3}m$) of the perovskites CsAB_3 ($A = \text{Pb, Sn, Ge}$; $B = \text{Cl, Br, I}$). Color legend: blue, green and red spheres represent Cs, A, and B atoms, respectively



2 Methodology

2.1 Electronic Band Structure Calculations

All calculations have been performed in the framework of the Density Functional Theory (DFT), where the self-consistent Kohn-Sham's equation was solved using the Pseudopotential and Plane Wave method implemented in the Vienna Ab-initio Simulation Package (VASP) [32–34]. The core-valence interaction was described using Projector Augmented Wave (PAW) pseudopotentials [35]. The cut-off energy for the wave-function was set to 500 eV for all calculation. The exchange-correlation (XC) part was described by the parametrization of Perdew-Burke-Ernzerhof [36] of the Generalized Gradient Approximation (GGA-PBE), in order to enhance the electronic band structure calculation and the band-gap of these materials, at the equilibrium we have used the meta-GGA SCAN XC functional [37]. The first Brillouin zone of the reciprocal space, was described by a dense mesh-grid of $12 \times 12 \times 12$ k-points, furthermore spin-orbit coupling effect was considered.

We have considered perovskites α -crystal phase (cubic) and space group $Pm\bar{3}m$ (221), like CsAB_3 where $A = \text{Pb, Sn, and Ge}$; and $B = \text{Cl, Br, I}$, as shown in the Fig. 2, with Wyckoff position $1a$, $1b$ and $3c$ for the atoms Cs, A and B, respectively.

2.2 Numerical Simulations of Solar Cells Using SCAPS-1D

Numerical design and modeling of solar cells are widely acknowledged as effective tools in photovoltaic research. These computational techniques offer a prototype-based approach to comprehending device operations and predicting their physical behavior. They consider numerous critical parameters that significantly influence solar cell performance, including carrier concentration, film thickness, band gap, electron affinity, defect density, work function, recombination velocity, shunt and

series resistances, as well as operating temperature. This approach minimizes costs, reduces tedious labor, and accelerates research timelines.

Several extensively developed and freely available simulation packages, such as PC-1D [38], SCAPS-1D [39], wxAMPS [40], Nexnano [41] and ATLAS [42], have played an important role in advancing the study of solar cells. Specifically, the Solar Cell Capacitance Simulator-one dimension (SCAPS-1D), developed by the University of Gent in Belgium, is one of the extensively used simulators to evaluate the PSCs performance [43–48].

In this software, the solar cell is represented as a layered structure characterized by parameters such as layer thickness, doping levels, and various physical properties of the materials involved. The SCAPS-1D software calculates essential performance metrics for the device, such as open circuit voltage (V_{OC}), short-circuit current density (J_{SC}), fill factor (FF), power conversion efficiency (PCE), maximum power point (P_M) and external quantum efficiency (EQE). This allows the characterization of the device behavior for different design and operating conditions. Additionally, it evaluates material characteristics such as band diagrams, carrier concentrations, electric fields, and currents by numerically solving a set of three coupled differential equations: Poisson's equation and continuity equations for holes and electrons subject to specific boundary conditions.

The current densities of holes and electrons are determined by Eqs. (1) and (2):

$$J_p = qp(x)\mu_p E(x) - qD_p \frac{\partial p}{\partial x} \quad (1)$$

$$J_n = qn(x)\mu_n E(x) + qD_n \frac{\partial n}{\partial x} \quad (2)$$

In this case, μ_p and μ_n are the hole and electron mobilities, respectively, $E(x)$ is the electric field and D_p and D_n are the diffusion coefficients of the holes and electrons, respectively.

Figure 3 shows the configuration of the all-inorganic PSC used in this work, which consists of a normal architecture (n-i-p) FTO/ETL/CsBX₃/HTL/Au, where B = Pb, Ge, Sn and X = I, Br. The standard AM1.5G spectrum has been used. Moreover, we employed Cu₂O, CuSCN, and CuI as hole transport layers (HTL), and ZnO and TiO₂ as electron transport layers (ETL), to determine the optimal FTO/ETL/CsBX₃/HTL/Au configuration yielding the highest efficiency.

Tables 1, 2, and 3 summarize the main parameters used in the simulations for the perovskite layers, and for the materials chosen as HTL and ETL, respectively. Here, E_G is the band gap energy, X is the electron affinity, ϵ_r is the relative permittivity, N_C and N_V are the effective conduction and valence band density of states, respectively, v_n and v_p are the thermal velocities of electrons and holes, respectively and N_T is the defect density. The band gap energy values presented in Table 1 are derived from our DFT-based calculations or experimental data found in the literature, as detailed in Table 5. The rest of the values in Tables 1, 2, and 3 are based on experimental

Fig. 3 Device architecture of regular (n-i-p) all-inorganic PSC used in this work

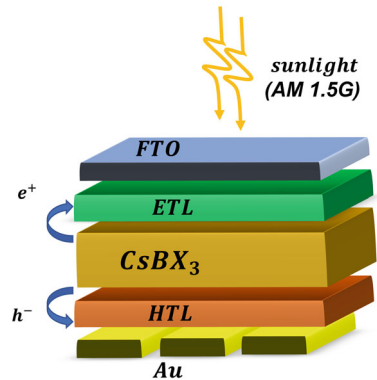


Table 1 Physical parameters of the perovskite materials used in the simulation [50–58]

Parameter	CsPbI ₃	CsPbBr ₃	CsGeI ₃	CsGeBr ₃	CsSnI ₃	CsSnBr ₃
Thickness (nm)	350	350	350	350	350	350
E_G (eV)	1.754	2.282	1.482	2.330	1.283	1.596
X (eV)	3.49	3.6	3.52	2.87	3.6	4.07
E_r	6	6.5	18	3.9	9.93	5.9
N_C (cm ⁻³)	1.61×10^{19}	4.94×10^{17}	1×10^{18}	1.9×10^{18}	1×10^{19}	1×10^{18}
N_V (cm ⁻³)	2.21×10^{18}	8.47×10^{18}	1×10^{19}	2.07×10^{18}	1×10^{18}	1×10^{18}
v_n (cm/s)	1×10^7	1×10^7	1×10^7	1×10^7	1×10^7	1×10^7
v_p (cm/s)	1×10^7	1×10^7	1×10^7	1×10^7	1×10^7	1×10^7
μ_n (cm ² V ⁻¹ s ⁻¹)	16	4500	20	20	4.37	0.1
μ_p (cm ² V ⁻¹ s ⁻¹)	16	4500	20	20	4.37	0.1
N_D (cm ⁻³)	–	–	–	–	–	–
N_A (cm ⁻³)	1×10^{15}	1×10^{15}	1×10^{15}	1×10^{15}	1×10^{15}	1×10^{15}
N_T (cm ⁻³)	1×10^{15}	1×10^{15}	1×10^{15}	1×10^{15}	1×10^{15}	1×10^{15}

and theoretical studies reported in the literature. The work functions of the front and back contacts are 4.4 eV (FTO) and 5.1 eV (Au), respectively [49].

3 Results

3.1 Electronic Band Structure Results

The total energy-volume curve for each compound was obtained fitting the data obtained from ab-initio calculations and using the third order Birch-Murnaghan [59, 60] equation of state. From the volume optimization we have obtained the optimized

Table 2 Physical parameters of HTL materials used in the simulation [44, 50]

Parameter	Cu ₂ O	CuSCN	CuI
Thickness (nm)	25	25	25
E_G (eV)	2.17	3.4	3.1
X (eV)	3.2	1.9	2.1
E_r	7.5	10	6.5
N_C (cm ⁻³)	2.5×10^{20}	1.7×10^{19}	2.8×10^{19}
N_V (cm ⁻³)	2.5×10^{20}	2.5×10^{21}	1×10^{19}
v_n (cm/s)	1×10^7	1×10^7	1×10^7
v_p (cm/s)	1×10^7	1×10^7	1×10^7
μ_n (cm ² V ⁻¹ s ⁻¹)	200	2×10^{-4}	100
μ_p (cm ² V ⁻¹ s ⁻¹)	80	2×10^{-1}	44
N_D (cm ⁻³)	–	–	–
N_A (cm ⁻³)	3×10^{18}	3×10^{18}	3×10^{18}
N_T (cm ⁻³)	1×10^{15}	1×10^{15}	1×10^{15}

Table 3 Physical parameters of ETL materials used in the simulation[44, 49, 52]

Parameter	ZnO	TiO ₂
Thickness (nm)	25	25
E_G (eV)	3.16	3.2
X (eV)	4	4
E_r	9	9
N_C (cm ⁻³)	4.5×10^{18}	1×10^{21}
N_V (cm ⁻³)	1×10^{18}	2×10^{20}
v_n (cm/s)	1×10^7	1×10^7
v_p (cm/s)	1×10^7	1×10^7
μ_n (cm ² V ⁻¹ s ⁻¹)	100	20
μ_p (cm ² V ⁻¹ s ⁻¹)	25	10
N_D (cm ⁻³)	3×10^{18}	3×10^{18}
N_A (cm ⁻³)	–	–
N_T (cm ⁻³)	1×10^{15}	1×10^{15}

lattice parameter for each compound that is presented in the Table 4 together with experimental values from the literature. In Fig. 4, we compared the DFT-based values of the equilibrium lattice parameter (a_0) with the experimental values, where a great linear correlation has been found.

In order to analyze the stability of the compounds, we have calculated the formation energy (Φ_f) for all CsAB₃ cases, using the equation

$$\Phi_f = E_{\text{CsAB}_3} - N_{\text{Cs}}E_{\text{Cs}} - N_{\text{A}}E_{\text{A}} - N_{\text{B}}E_{\text{B}} \quad (3)$$

Table 4 Equilibrium lattice parameter, a_0 (Å), from DFT for the perovskites CsAB_3 , A = Pb, Sn, Ge and B = Cl, Br, I, compared with experimental values from the literature

Compound	a_0 (Å)—DFT	a_0 (Å)—Experimental
CsPbCl ₃	5.7198	5.599 [62], 5.604 [63, 64], 5.605 [65–69]
CsPbBr ₃	5.9889	5.87 [70], 5.8733 [71], 5.8735 [72], 5.874 [63, 66, 68, 73–76], 5.8745 [77]
CsPbI ₃	6.3876	6.1769 [29], 6.23 [78], 6.2894 [79], 6.2950 [80]
CsGeCl ₃	5.3399	5.18 [81], 5.434 [63, 82], 5.444 [83], 5.446 [84], 5.478 [85]
CsGeBr ₃	5.6056	5.35 [81], 5.36 [86, 87], 5.635 [63, 82], 5.647 [84]
CsGeI ₃	5.9958	5.830 [63], 5.983 [82]
CsSnCl ₃	5.6204	5.504 [88], 5.56 [89], 5.5838 [90], 5.601 [63]
CsSnBr ₃	5.8853	5.795 [87, 91], 5.80 [89, 92, 93], 5.8043 [94], 5.808 [63]
CsSnI ₃	6.2739	6.18 [92], 6.2057 [95], 6.219 [96], 6.231 [63]

Table 5 Electronic band gap (E_g) from DFT, using GGA-PBE and metaGGA-SCAN for the XC functional, for the perovskites CsAB_3 , A = Pb, Sn, Ge and B = Cl, Br, I, compared with experimental values from the literature. All values in eV unit

Compound	GGA-PBE	SCAN	Experimental
CsPbCl ₃	2.257	2.855	2.96 [70], 3.00 [97, 98], 3.06 [99], 3.18 [100]
CsPbBr ₃	1.844	2.282	2.22 [71], 2.25 [76], 2.28 [101], 2.30 [98, 102], 2.32 [72], 2.33 [98]
			2.35 [103], 2.36 [70], 2.37 [104], 2.40 [103], 2.43 [100]
CsPbI ₃	1.572	1.754	1.73 [29, 105, 106], 1.76 [107], 1.77 [70], 1.88 [100]
CsGeCl ₃	1.198	1.353	1.34 [81], 2.03 [108], 2.10 [109], 3.43 [110], 3.67 [111]
CsGeBr ₃	0.908	1.012	0.89 [81], 0.9 [109], 1.91 [108], 2.32 [87, 111], 2.37 [110, 112]
CsGeI ₃	0.800	1.482	1.51 [113], 1.53 [111], 1.60 [114], 1.61 [112], 1.63 [115], 1.80 [108]
CsSnCl ₃	1.107	3.131	2.80 [93], 2.90 [116, 117], 2.98 [88]
CsSnBr ₃	0.782	1.596	1.36 [118], 1.55 [119], 1.75 [93], 1.78 [120], 1.92 [121]
CsSnI ₃	0.630	1.283	1.24 [122], 1.27 [119], 1.30 [123], 1.31 [122], 1.32 [124]

where, N_{Cs} , N_{A} , and N_{B} are the number of atoms Cs, A, and B, respectively, in the compound CsAB_3 , whereas, E_{CsAB_3} is the total energy of the compound CsAB_3 at the equilibrium volume, E_{Cs} , E_{A} and E_{B} the total energy per atom of the compound Cs, A, and B, respectively, in the corresponding equilibrium crystal structure. In Fig. 5, we represent the Φ_f for all compound CsAB_3 considered (A = Pb, Sn, Ge; B = Cl, Br, I), in all cases $\Phi_f < 0$ indicating that all compounds are thermodynamically stable. Moreover, the perovskites Pb-based (A = Pb, CsPbB_3) are the most stable, whereas the less stable are the Ge-based (A = Ge, CsGeB_3) perovskites.

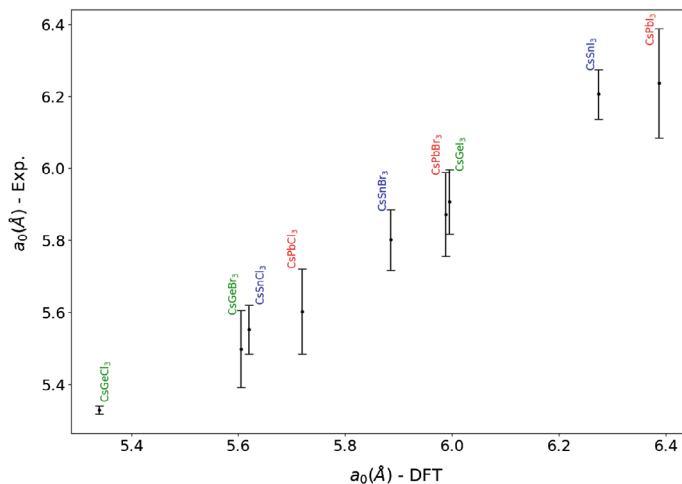


Fig. 4 DFT-based values of the equilibrium lattice parameter a_0 compared with experimental data from the literature for the perovskites CsAB_3 , $A = \text{Pb, Sn, Ge}$ and $B = \text{Cl, Br, I}$. Vertical bar represent the range of the experimental data reported

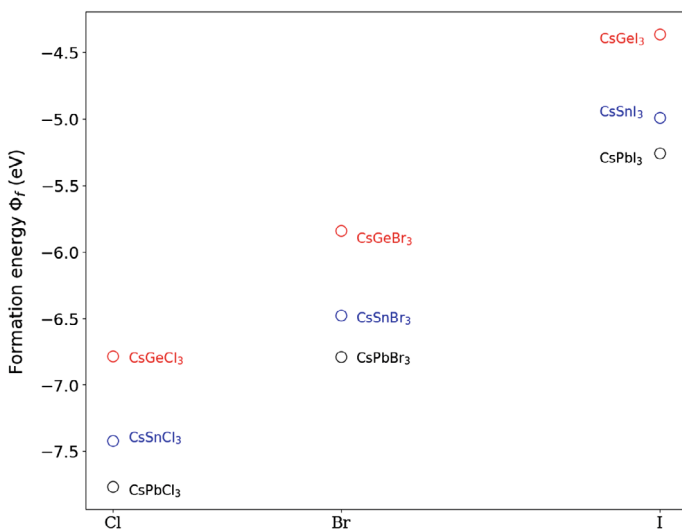


Fig. 5 Formation energy, Φ_f , for the perovskites CsAB_3 , $A = \text{Pb, Sn, Ge}$; $B = \text{Cl, Br, I}$

At the equilibrium lattice parameter a_0 , we have performed the electronic band structure calculation in order to determinate the corresponding electronic band-gap (E_g) for all perovskite-like compounds. It is very know that GGA underestimate the E_g description, in order to enhance the E_g , the exchange-correlation part was described using the metaGGA functional SCAN [37]. In Table 5 we report the value of

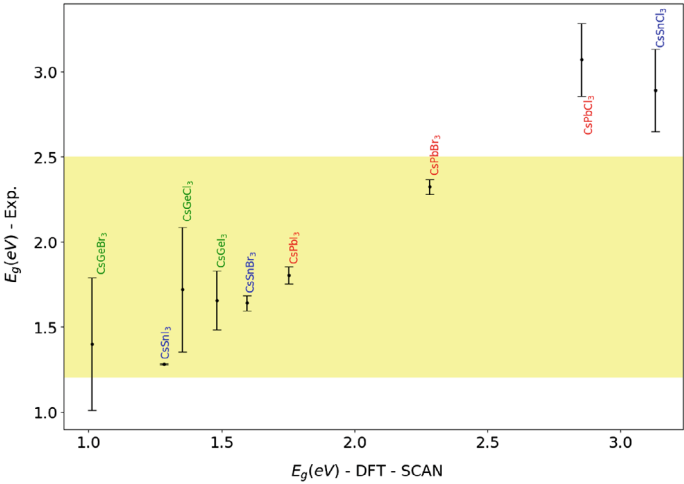


Fig. 6 Correlation between DFT-based value of E_g and the experimental data for the perovskite CsAB_3 , $A = \text{Pb, Sn, Ge}$ and $B = \text{Cl, Br, I}$. Yellow stripe represent the optimal values to be used in solar cell applications. Vertical bar represent the range of the experimental data reported

E_g using GGA and SCAN approximation for the exchange-correlation part, together with experimental values reported in the literature, whilst, in Fig. 6 represent the great correlation between the DFT-based value of E_g and the experimental data. The yellow band represent the optimal values of E_g required for the perovskite to be used in solar cell applications [61].

3.2 Analysis of SCAPS-1D Results

3.2.1 Effect of ETL and HTL on CsPbX_3 ($X = \text{I, Br}$)

In this section, the photovoltaic performance of the $\text{FTO/ETL/CsPbX}_3\text{/HTL/Au}$ structure (where $X = \text{I, Br}$) is evaluated by varying the ETL and HTL layers, according to the values in Tables 1, 2, and 3. Specifically, an analysis of the normal n-i-p all-inorganic PSCs was conducted using SCAPS-1D software (ver.3.3.11) for six different combinations of ETL/HTL for each CsPbI_3 and CsPbBr_3 absorber layers.

The results of the output parameters (V_{OC} , J_{SC} , FF and PCE) are summarized in Table 6. In this table, it is evident that among the ETL materials studied, ZnO achieved the highest PCE values. Both TiO_2 and ZnO exhibit similar properties such as band structure, electronic affinity, relative permittivity, and band gap energy. However, TiO_2 has a higher N_C and N_V , resulting in a greater intrinsic carrier concentration compared to ZnO. A higher intrinsic carrier concentration generally leads to increased recombination rates and reduced carrier collection efficiency under high electric field

Table 6 Output parameters of the PSC for the twelve different combinations FTO/ETL/CsPbX₃/HTL/Au, where X = I, Br

Structure	V_{OC} (V)	J_{SC} (mA/cm ²)	FF (%)	PCE (%)
ZnO/CsPbI ₃ /Cu ₂ O	1.2272	16.1323	77.77	15.40
ZnO/CsPbI ₃ /CuSCN	1.2234	16.0956	77.32	15.22
ZnO/CsPbI ₃ /CuI	1.2254	16.0980	77.58	15.30
TiO ₂ /CsPbI ₃ /Cu ₂ O	1.2295	16.1256	67.35	13.35
TiO ₂ /CsPbI ₃ /CuSCN	1.2260	16.0884	66.86	13.19
TiO ₂ /CsPbI ₃ /CuI	1.2278	16.0912	67.14	13.27
ZnO/CsPbBr ₃ /Cu ₂ O	1.6021	6.1867	90.87	9.01
ZnO/CsPbBr ₃ /CuSCN	1.7422	6.1860	73.61	7.93
ZnO/CsPbBr ₃ /CuI	1.7421	6.1871	75.98	8.19
TiO ₂ /CsPbBr ₃ /Cu ₂ O	1.6133	6.1918	85.36	8.53
TiO ₂ /CsPbBr ₃ /CuSCN	1.7459	6.1909	70.82	7.65
TiO ₂ /CsPbBr ₃ /CuI	1.7457	6.1920	73.19	7.91

conditions (near V_{OC} voltages). Consequently, TiO₂ exhibited lower FF and PCE values compared to ZnO. Based on these simulation results, ZnO emerges as the preferred ETL candidate for CsPbI₃ and CsPbBr₃ PSCs.

On the other hand, the selection of HTL also significantly impacts device performance. Table 6 reveals that, for a given ETL, CuSCN yielded the lowest PCE values, whereas Cu₂O achieved the highest. Therefore, Cu₂O can be considered as a good option for HTL. In the following simulations, the ZnO is used as the ETL and Cu₂O as the HTL.

3.2.2 Study of All-Inorganic Lead-Free Perovskite Solar Cells

In this section, four lead-free perovskite solar cells have been studied and compared with the Pb based halide perovskites solar cells. The photovoltaic performance of the FTO/ZnO/CsBX₃/Cu₂O/Au structure (where B = Pb, Ge, Sn and X = I, Br) is evaluated using SCAPS-1D software. Using the initial settings from Tables 1, 2, and 3, device performance characteristics are summarized in Table 7.

Figures 7 and 8 show the J-V characteristics (a) and EQE spectrum (b) for CsBI₃ and CsBBr₃ (with B = Pb, Ge, Sn) devices, respectively.

Figure 9 shows the variation in the electrical parameters (V_{OC} , J_{SC} , FF and PCE) when the perovskite thickness is increased from 250 to 1150 nm. The values displayed in this figure are normalized to those corresponding to the thickness of 350 nm, as shown in Table 7.

In Fig. 9a, we observe that as the CsPbI₃ thickness varies from 250 to 1150 nm, both V_{OC} and FF exhibit reductions of 4 and 18%, respectively. The decline in V_{OC} is attributed to its dependency on photogenerated current and dark saturation current.

Table 7 Performance parameters of the FTO/ZnO/CsBX₃/Cu₂O/Au devices (where B = Pb, Ge, Sn and X = I, Br)

Structure	V_{OC} (V)	J_{SC} (mA/cm ²)	FF (%)	PCE (%)
ZnO/CsPbI ₃ /Cu ₂ O	1.2272	16.1323	77.77	15.40
ZnO/CsGeI ₃ /Cu ₂ O	1.1700	20.2242	74.53	17.64
ZnO/CsSnI ₃ /Cu ₂ O	0.9723	25.0196	65.50	15.93
ZnO/CsPbBr ₃ /Cu ₂ O	1.6021	6.1867	90.87	9.01
ZnO/CsGeBr ₃ /Cu ₂ O	1.3060	5.8733	70.59	5.41
ZnO/CsSnBr ₃ /Cu ₂ O	1.1009	15.6299	47.63	8.20

Table 8 Optimized cell parameters with simulated PCE results of PSC

Structure	Optimized absorber thickness (nm)	PCE (%)
ZnO/CsPbI ₃ /Cu ₂ O	650	16.13
ZnO/CsGeI ₃ /Cu ₂ O	550	18.22
ZnO/CsSnI ₃ /Cu ₂ O	350	15.93
ZnO/CsPbBr ₃ /Cu ₂ O	950	11.95
ZnO/CsGeBr ₃ /Cu ₂ O	650	5.97
ZnO/CsSnBr ₃ /Cu ₂ O	250	9.29

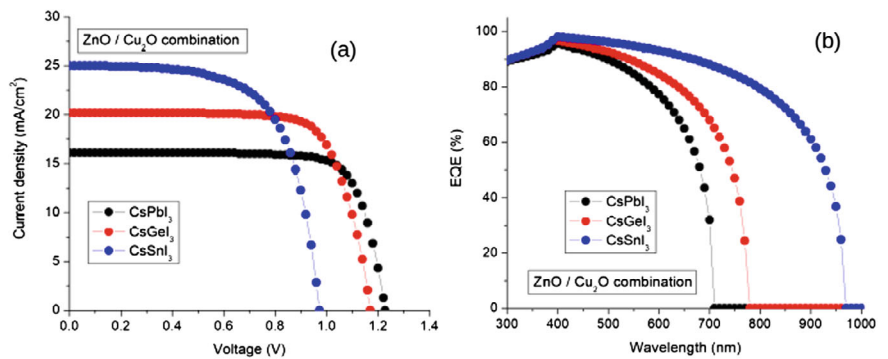


Fig. 7 **a** J-V characteristics and **b** EQE spectrum for CsBI₃ perovskite solar cells, where B = Pb, Ge, Sn

Increased dark saturation current encourages carrier recombination, thereby lowering V_{OC} as thickness increases. Additionally, the substantial decrease in FF with thicker active layers can be attributed to elevated series resistance. Conversely, J_{SC} experiences a notable increase of over 34% with thicker CsPbI₃ layers, owing to enhanced light absorption and higher concentrations of free carriers generated by photons and collected by the electrode. Regarding PCE, a significant 10% enhancement is evident with thickness increasing to 650 nm, after which it begins to decline for thicker films.

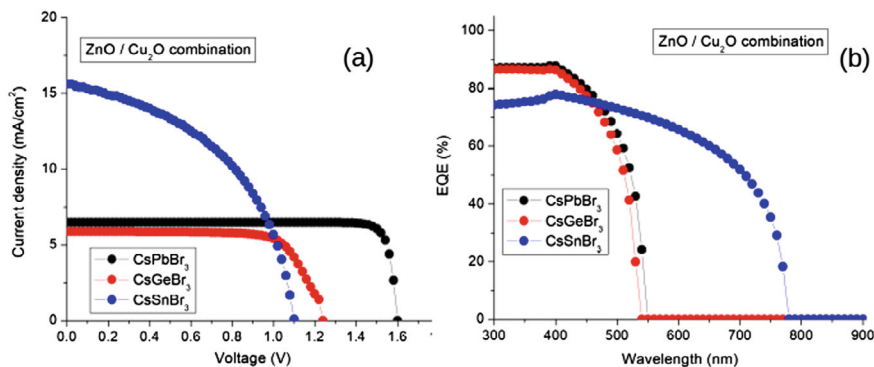


Fig. 8 **a** J-V characteristics and **b** EQE spectrum for CsBBr₃ perovskite solar cells, where B = Pb, Ge, Sn

Figure 9c and e depict the cases for lead-free CsBI₃ perovskite, with B = Ge, Sn, respectively. In these cases, as the perovskite thickness is increased from 250 to 1150 nm, V_{OC} decreases by 5% and FF by 17% for CsGeI₃, and by 3 and 23% for CsSnI₃, respectively. In contrast, the J_{SC} is strongly increased above 29 and 20% with increasing CsGeI₃ and CsSnI₃ thickness. Finally, the maximum PCE values were achieved at thicknesses of 550 nm for CsGeI₃ and 350 nm for CsSnI₃ devices, respectively.

On the other hand, in Fig. 9b, we observe that as the CsPbBr₃ thickness increases from 250 to 1150 nm, V_{OC} remains constant, while FF decreases by 6% and J_{SC} increases by 50%. Additionally, the PCE value improves by 44% for a CsPbBr₃ thickness of 950 nm. Figure 9d and f represent the cases of lead-free CsBBr₃ perovskites, where B = Ge and Sn, respectively. In these instances, as the perovskite thickness is increased from 250 to 1150 nm, V_{OC} remains unchanged, while FF decreases by 23 and 21% for CsGeBr₃, and CsSnBr₃, respectively. On the contrary, J_{SC} shows a significant increase of more than 50% but a reduction of 15% with increasing layer thickness of CsGeBr₃ and CsSnBr₃, respectively. Finally, the maximum PCE values were achieved at thicknesses of 650 nm for CsGeBr₃ and 250 nm for CsSnBr₃ devices.

The optimized parameters of absorber thickness and PCE are mentioned in Table 8.

3.3 Manufacturing and Characterization of ZnO

To initiate the fabrication process of the studied devices using SCAPS-1D software, in this section, we first conducted the fabrication and characterization of the zinc oxide (ZnO) material, which is a wide band-gap semiconductor, that finds extensive applications in sensors, optoelectronic devices, and photocatalysis due to its unique properties.

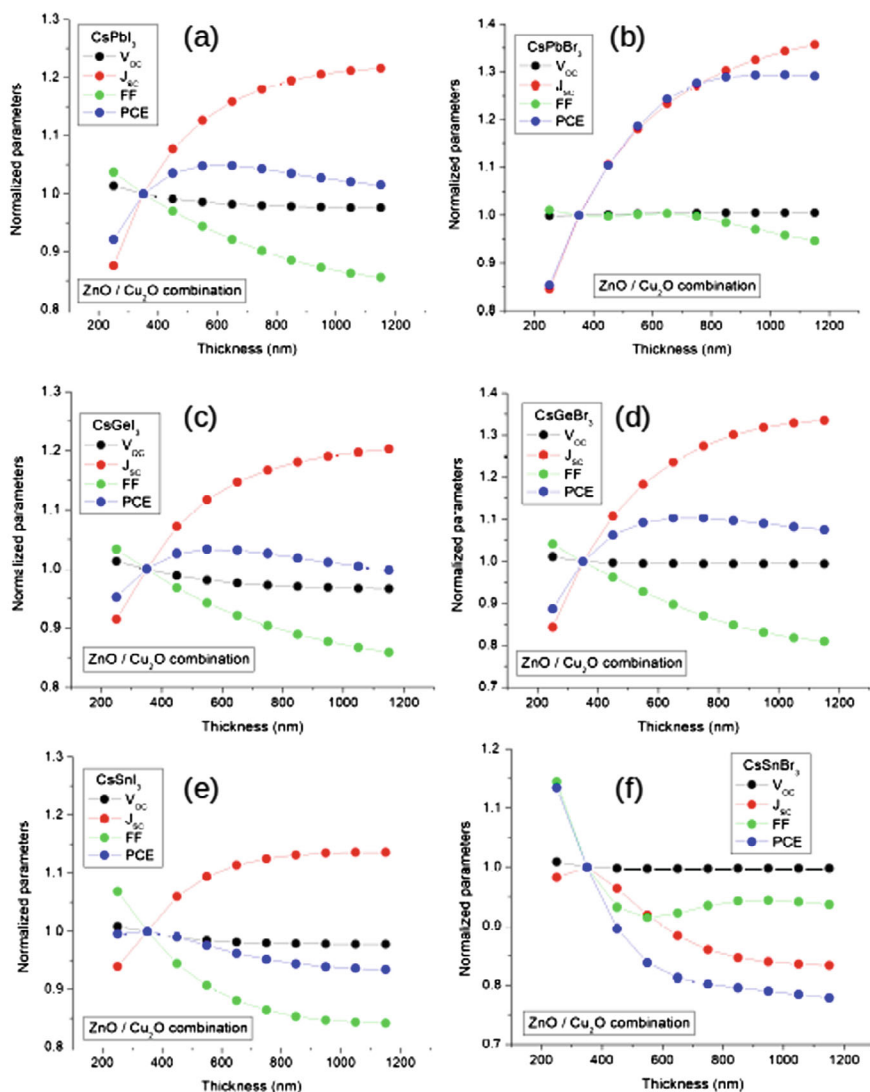
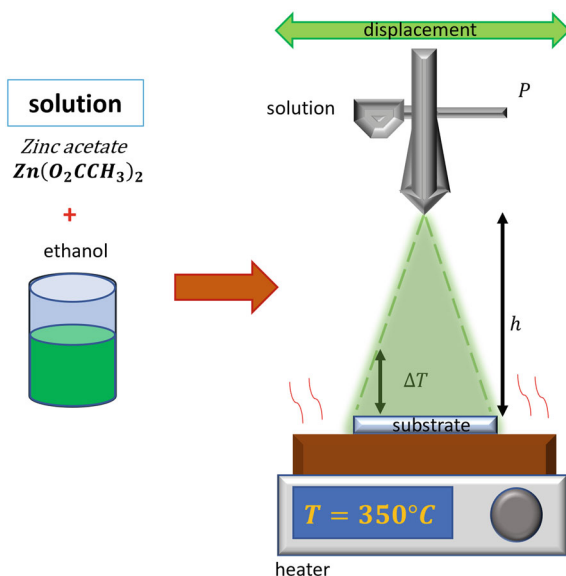


Fig. 9 Normalized electrical parameters as a function of the **a** CsPbI₃, **b** CsPbBr₃, **c** CsGeI₃, **d** CsGeBr₃, **e** CsSnI₃ and **f** CsSnBr₃ absorber thickness

Spray pyrolysis has emerged as a versatile technique for the synthesis of ZnO nanocoatings, offering precise control over film thickness and morphology. The spray pyrolysis process involves the thermal decomposition of precursor solutions sprayed onto a heated substrate. A typical experiment of ZnO nanolayer fabrication consist in spraying a zinc-containing precursor solution, such as zinc acetate or zinc nitrate, onto a substrate heated to temperatures ranging from 200 to 500 °C. The decomposition

Fig. 10 Experimental set-up of a spray pyrolysis system adequate to obtain nanolayers of ZnO



of the precursor under these conditions results in the formation of ZnO nanoparticles or thin films on the substrate surface. The case of using Zinc Acetate, a solution in absolute ethanol of a concentration 0.1M is prepared. The parameter such as substrate temperature, distance between sprayer and substrate (h , see Fig. 10) and the quantity of solution sprayed are all experimentally determined parameters.

ZnO is a good electrical conductor with a wide band-gap that do not interfere with the band-gap of perovskites. That is the reasons due to nanolayers of this material could be a good candidate to be used as electron transport layer in perovskite photovoltaic applications. However, there is a problem to overcome before this material be considered for the community.

The problem is the formation of OH^- coming from moisture once the ZnO is formed over the substrate. This cause many problem an several strategies where proposed to overcome the OH^- formation. Between the problems we can consider:

1. **Interface deterioration:** The presence of OH^- can deteriorate the interface between the ZnO layer and the perovskite, affecting the efficiency of solar energy conversion due to decreased efficient charge carrier transfer.
2. **Perovskite degradation:** OH^- can react with the light-sensitive perovskite and cause its degradation, thereby reducing the solar cell's lifetime and stability.
3. **Alteration of conductivity:** Formation of OH^- can alter the electrical conductivity properties of the ZnO layer, negatively impacting the overall performance of the solar cell.
4. **Impact on optical properties:** Hydroxide formation may also modify the optical properties of the ZnO layer, affecting its absorption capability and the solar cell's conversion efficiency.

To mitigate these issues, it is crucial to optimize the fabrication processes of ZnO nanolayers to minimize OH^- formation and maintain the desired interface properties in perovskite solar cells. This may involve adjustments in precursor composition, deposition conditions, and post-deposition treatment of ZnO layers.

For photovoltaic applications the nanolayers of ZnO should be of no more than 200 nm, this introduce a problem at the time of characterize them. For example, in Fig. 11 we show a X-ray diffraction (XRD) spectrum of ZnO, due to the X-ray photons used in this spectroscopy, have a deeper penetration in the material that the thick of the nanolayer, some information of the glass substrate is obtained. This is an issue since the final spectrum is a mixture between glass and ZnO signal (see Fig. 11).

Zinc oxide (ZnO) crystallizes in the wurtzite structure, which is a hexagonal crystal system with space group $P6_3mc$. The XRD spectrum of ZnO in the wurtzite phase is characterized by distinct peaks corresponding to specific crystallographic planes. Upon analysis the XRD pattern typically shows several prominent peaks at various 2θ angles, only one of them are shown in the experimental figure (see Fig. 11). The key peaks observed are typically located around 31.8° , 34.4° , 36.3° , 47.6° , 56.6° , 62.9° , and 67.9° . These angles correspond to the (100), (002), (101), (102), (110) crystallographic planes of the wurtzite structure, respectively. One of the most intense peak, often found at around 34.4° (002), indicates the preferred orientation of the ZnO

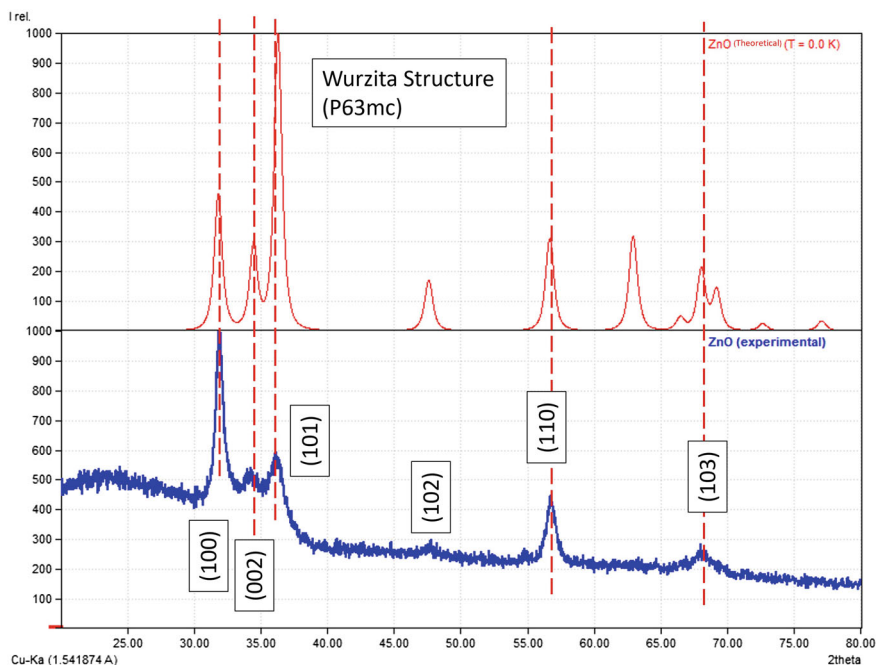


Fig. 11 XRD spectrum of nanolayers of ZnO. The thick of the layer analyzed is approximately 150 nm

crystals along the c-axis of the hexagonal lattice. This orientation is a characteristic feature of wurtzite ZnO and reflects the crystal's structural alignment. The XRD peaks are typically sharp and well-defined, indicating a high degree of crystallinity and structural integrity in the sample. Broadening or splitting of peaks may occur depending on factors such as grain size, strain, and defects within the crystal lattice.

In summary, the XRD spectrum of ZnO in the wurtzite phase provides valuable information about its crystal structure, orientation, and quality, making it a fundamental technique for characterizing semiconductor materials like ZnO in various applications such as optoelectronics, catalysis, and sensors.

3.3.1 Band-Gap Determination of ZnO

The band-gap of zinc oxide (ZnO) is a critical parameter that defines its optical and electronic properties. It is typically calculated using experimental data from techniques such as UV-Vis spectroscopy (see Fig. 12) or photoluminescence, as well as theoretical methods such as the Density Functional Theory.

The band-gap of ZnO refers to the energy difference between its valence band (highest occupied energy level) and its conduction band (lowest unoccupied energy level). This parameter plays a crucial role in determining ZnO's optical and electronic characteristics, including its transparency and conductivity. To calculate the band gap experimentally, UV-Vis spectroscopy is commonly employed. In this method, light of varying wavelengths is directed onto the ZnO sample, and the absorption spectrum is measured. The band gap energy (E_g) can be determined from the onset

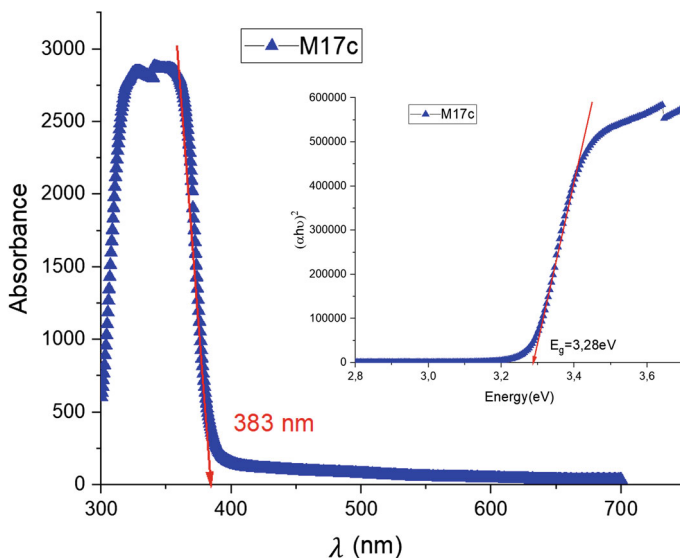


Fig. 12 UV-vis spectroscopy of the ZnO sample

of absorption, which corresponds to the wavelength (λ) at which the absorption coefficient sharply increases. The relationship between the band-gap energy and the corresponding wavelength is given by:

$$E_g = \frac{1240}{\lambda} \quad (4)$$

where E_g is in electron volts (eV) and λ is in nanometers (nm).

Alternatively, photoluminescence spectroscopy can also be used to estimate the band-gap indirectly. In this technique, the emission of light from the sample upon excitation is analyzed, and the energy of emitted photons provides information about the band-gap. The band-gap obtained was $E_g = 3.20$ eV that is in agreement with other experimental values reported, such as: 3.14 eV [125], 3.22 eV [126], 3.23 eV [127], 3.25 eV [128], 3.30 eV [129].

4 Conclusions

In this work, we have studied the electronic properties of the perovskites CsBX_3 ($B = \text{Pb, Sn, Ge}$ and $X = \text{I, Br, Cl}$) materials, in the framework of the Density Functional Density. Specifically, the equilibrium lattice parameter and electronic band-gap have been calculated and compared with experimental data from the literature. A great linear correlation has been found for the equilibrium lattice parameter for all perovskite-like compounds studied. Furthermore, the band-gap values based on DFT and experimental data for CsPbCl_3 and CsSnCl_3 perovskites are outside the range of optimal values to be used in solar cell applications. In addition, the formation energy for all CsBX_3 cases has been obtained in order to analyze the stability of the compounds. The results obtained show that although all the compounds studied are thermodynamically stable, the Pb-based perovskites ($B = \text{Pb, CsPbX}_3$) are the most stable, while the least stable are the Ge-based perovskites ($B = \text{Ge, CsGeX}_3$).

On the other hand, normal n-i-p all-inorganic PSCs based on CsBX_3 ($B = \text{Pb, Sn, Ge}$ and $X = \text{I, Br}$) perovskite were studied through SCAPS-1D simulations. Several potential inorganic materials for HTL and ETL were considered. The simulation results show that Cu_2O as HTL and ZnO as ETL have better performance than the other combinations considered in this study, with efficiencies of 15.40 and 9.01% for CsPbI_3 and CsPbBr_3 , respectively. In order to reduce the toxicity of the devices studied, four lead-free perovskite solar cells have been studied and compared with the Pb based halide perovskites solar cells. To optimize the design and to enhance the efficiency of the $\text{FTO/ZnO/CsBX}_3/\text{Cu}_2\text{O}/\text{Au}$ structure (where $B = \text{Pb, Ge, Sn}$ and $X = \text{I, Br}$), the optimal values of the thickness in the absorber layer were obtained. The $\text{FTO/ZnO/CsGeI}_3/\text{Cu}_2\text{O}/\text{Au}$ structure obtained the highest efficiency of 18.22%.

The results of this study highlight the significant potential of lead-free perovskites as effective adsorbent materials for high-efficiency perovskite solar cells (PSCs).

These insights enhance our understanding of lead-free PSCs and pave the way for the design and production of environmentally friendly, high-performance solar cells.

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Biohydrogen Generation from Agroindustrial Waste: Challenges and Opportunities



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Abstract Hydrogen emerges as a promising solution to address current environmental challenges and future energy demands in cities and industry. Biohydrogen, derived from agroindustrial wastes through microbial activity—particularly lignocellulosic materials—presents a viable avenue for valorizing such resources into green energy. However, despite its potential, real-world industrial exploration of this technology remains limited due to challenges such as relatively low yields and high process costs. This chapter explores the utilization of recalcitrant agroindustrial and urban biomasses for biohydrogen generation, identifies bottlenecks, discusses proposed strategies for enhancing cost-effectiveness, explores stimulation techniques for H₂-producing microbes, and considers future prospects in the field of microbial hydrogen.

Keywords Lignocellulosic materials · Green hydrogen · Resource recovery · Dark fermentation

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1 Introduction

Hydrogen is notable for its potential to produce energy without emitting greenhouse gases when burned or used in fuel cells [1]. There are several methods for producing hydrogen, each with unique processes and benefits, including Steam Methane Reforming [2], electrolysis [3], coal gasification [4], and biomass gasification [5].

In fuel cells, hydrogen combines with oxygen to produce electricity, water, and heat. Hydrogen can also power modified combustion engines and serve as an energy storage medium, storing excess renewable energy and reconverting it into electricity when needed.

Biohydrogen, produced from biological means, offers a renewable and sustainable alternative. It can be produced through various methods involving microorganisms or biomass [6]. Biophotolysis uses certain algae and cyanobacteria to split water into hydrogen and oxygen using sunlight [7]. Dark fermentation involves anaerobic bacteria breaking down organic matter, such as agricultural waste, in the absence of light to produce hydrogen, carbon dioxide, and organic acids [8]. Photofermentation uses photosynthetic bacteria to produce hydrogen from organic acids and light. Microbial Electrolysis Cells employ bacteria on electrodes to consume organic material and generate electricity, which is then used to produce hydrogen from water [9]. Enzymatic hydrogen production uses enzymes like hydrogenases and nitrogenases to catalyze the production of hydrogen from organic substrates [10].

Biohydrogen can be integrated into renewable energy systems, serving as a means to store and transport energy. It can also be used in industrial processes as a clean fuel and feedstock for chemical processes, similar to conventional hydrogen. Both hydrogen and biohydrogen can power fuel cells and modified combustion engines. The advantages of hydrogen and biohydrogen include their clean energy profile, producing only water as a byproduct when used in fuel cells, and their high energy density.

Great attention has been paid to byproducts and industrial waste valorization and greener technologies in the fossil fuel industry, such as carbon dioxide capture systems [11–14], which are increasingly important for climatic and environmental concerns. Biohydrogen is particularly beneficial due to its use of renewable resources and potential for waste management.

However, there are challenges to widespread adoption. Storing and transporting hydrogen is difficult due to its low density, necessitating high pressures or cryogenic temperatures. Production costs, especially for green hydrogen and biohydrogen, remain high compared to fossil fuels. Significant investment in infrastructure is needed for production, distribution, and refueling to support widespread use [15].

Biohydrogen production from lignocellulosic materials presents a promising pathway for sustainable and renewable hydrogen generation. Lignocellulosic biomass, which includes agricultural residues, forestry waste, and dedicated energy crops, is composed of cellulose, hemicellulose, and lignin. These components make it an abundant and low-cost feedstock for biohydrogen production [16].

Lignocellulosic biomass consists primarily of cellulose, a polysaccharide made up of glucose monomers; hemicellulose, a heterogeneous polysaccharide containing various sugar monomers like xylose, mannose, and arabinose; and lignin, a complex aromatic polymer that provides structural support and resistance to microbial attack. Before lignocellulosic biomass can be used for biohydrogen production, it requires pretreatment to break down the complex structure and release fermentable sugars. Pretreatment methods include physical pretreatment (mechanical milling or grinding to reduce particle size and increase surface area), chemical pretreatment (using acids, alkalis, or solvents to break down lignin and hemicellulose), physico-chemical pretreatment (such as steam explosion, ammonia fiber explosion (AFEX), or ionic liquids that combine physical and chemical treatments), and biological pretreatment utilizing microbes or enzymes to selectively degrade lignin and hemicellulose.

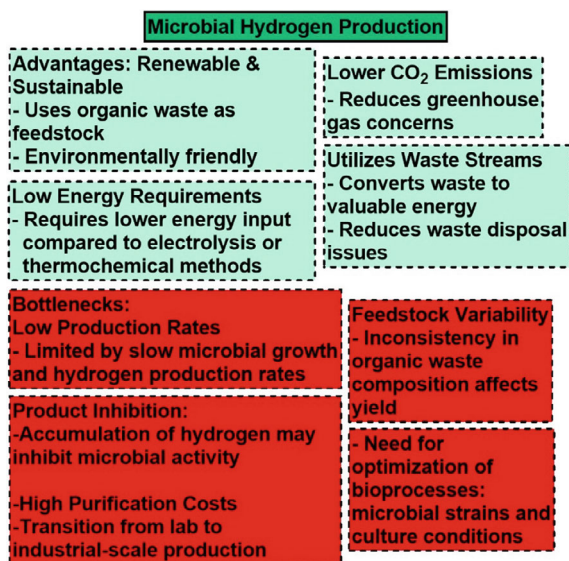
Once the biomass is pretreated and the fermentable sugars are released, several biological processes can be employed to produce hydrogen [17]. Dark fermentation involves anaerobic microorganisms breaking down sugars to produce hydrogen, carbon dioxide, and organic acids. This process does not require light and can be conducted using mixed or pure microbial cultures. The key steps in dark fermentation include hydrolysis, where enzymes break down complex carbohydrates into simpler sugars [18], acidogenesis, where microorganisms convert sugars into volatile fatty acids (VFAs), alcohols, hydrogen, and carbon dioxide, and hydrogen production by specific bacteria, such as *Clostridium* species, which produce hydrogen as a metabolic byproduct during the fermentation of VFAs. Dark fermentation can handle a wide variety of feedstocks and operates under relatively simple conditions, but hydrogen yields can be low due to the formation of byproducts like VFAs and alcohols, necessitating subsequent treatment of fermentation residues [19].

Photofermentation, another method of biohydrogen production, utilizes photosynthetic bacteria to convert organic acids produced during dark fermentation into hydrogen using light energy. This process is conducted by purple non-sulfur bacteria, such as *Rhodobacter* and *Rhodospirillum* species, which use organic acids as a carbon source and light as an energy source. Light-driven reactions in the photosynthetic apparatus produce hydrogen from organic acids. Photofermentation can achieve higher hydrogen yields by converting fermentation byproducts and integrates well with dark fermentation for a two-stage hydrogen production process. However, it requires light, which can complicate scaling up, and is sensitive to oxygen, necessitating careful control of process conditions [20].

Combining dark fermentation and photofermentation offers a synergistic approach to maximize hydrogen yields from lignocellulosic biomass. In this integrated system, biomass is initially fermented to produce hydrogen, organic acids, and other byproducts during the dark fermentation stage. The organic acids are then utilized by photosynthetic bacteria during the photofermentation stage to produce additional hydrogen. This integrated approach can enhance overall efficiency and hydrogen yield [20].

Biohydrogen production from lignocellulosic biomass offers numerous advantages, as well as technological bottlenecks (Fig. 1). Lignocellulosic materials are abundant and low-cost, making them an attractive feedstock. This process utilizes

Fig. 1 Advantages and bottlenecks of microbial hydrogen approach



renewable resources and reduces waste, contributing to environmental sustainability. Moreover, it can lower greenhouse gas emissions compared to fossil fuels. However, there are challenges to overcome. Efficiently breaking down lignocellulosic biomass requires energy-intensive and sometimes costly pretreatment processes. Developing robust microbial cultures capable of high hydrogen yields is essential. Additionally, efficiently integrating different production stages to maximize hydrogen yield remains a technical challenge [21].

This chapter delves into the utilization of recalcitrant agroindustrial biomasses for biohydrogen generation, focusing on lignocellulosic materials such as agricultural residues, forestry by-products, and certain industrial and urban wastes. These materials, composed of cellulose, hemicellulose, and lignin, are abundant and renewable, offering a sustainable feedstock for biohydrogen production. However, the inherent complexity and resistance to degradation of lignocellulosic biomass pose significant challenges. The chapter discusses the current methodologies employed to overcome these obstacles, such as pretreatment processes that break down the rigid structure of lignocellulosic materials to enhance microbial accessibility and enzymatic hydrolysis efficiency. Various pretreatment techniques, including physical, chemical, and biological methods, are evaluated for their effectiveness in improving biohydrogen yields. The future prospects of biohydrogen in the context of a broader green hydrogen economy are also considered, comparing its feasibility and scalability with other green hydrogen sources such as electrolysis using renewable energy. This comprehensive analysis aims to provide insights into how biohydrogen from lignocellulosic biomass can contribute to a sustainable and economically viable hydrogen economy.

2 Research from Biohydrogen from Lignocellulosci Materials

Most research on hydrogen and lignocellulosic biomass focuses on molecular biology techniques and synthetic biology. These fields have the potential to identify microorganism interactions and responses to external compounds, solving metabolic pathways in hydrogen production or even precisely designing microbial platforms to excel in hydrogen production [22]. Other approaches that have been thoroughly investigated in the past two years include various methods for treating lignocellulosic biomass and optimizing bioprocesses to enhance hydrogen production (Table 1).

One study observed hydrogen productivity in a bioreactor fed with wheat straw over 80 days. Initial hydrogen production peaked at 325 ± 43 mL/kg d but stabilized at 80 mL/kg d from days 36 to 52. Switching wheat straw cultivars on day 60 did not significantly impact productivity. Butyric acid was the main fermentation product, and *Lactobacillus* and *Clostridium* remained dominant. Canonical correspondence analysis linked hydrogen production with the presence of *Lactobacillus* and *Clostridium*. Agar diffusion tests showed minimal antagonism, suggesting a synergistic relationship where *Lactobacillus*'s lactic acid production supports *Clostridium*'s hydrogen generation. This cooperation underlies stable hydrogen production from lignocellulosic substrates [23].

Another study demonstrated converting bamboo into hydrogen through enzymatic treatment and microbial electrolysis. Laccase pretreatment removed 40.31% of lignin, enhancing biomass digestibility. Cellulase hydrolysis produced 99.54 ± 4 mg/dL of glucose, which was used in a microbial electrolysis cell with an iron oxide nanoparticle-coated graphene anode. Optimal conditions (0.8 V and 200 rpm mixing) yielded 224 ± 6 mL of hydrogen per gram of biomass. The iron oxide nanoparticle-coated anode improved hydrogen production by 1.14 times compared to an uncoated anode, suggesting that further improvements in the process could enhance hydrogen yields. These results demonstrate the importance of developing low-cost methods for functional enzyme production, such as using solid-state fermentation with various lignocellulosic materials [24, 25].

A study employed cassava stems as both a support material and a co-substrate in dark fermentation processes by *Clostridium* and *Hydrogenispora* strains. Adding cassava stems improved the concentration of fermentable carbohydrates in the medium, enhancing hydrogen production when used alongside cassava starch wastewater as a substrate due to optimal biomass retention and high carbohydrate conversion rates (>97%). The structural analysis revealed changes in cassava stem composition after hydrolysis and fermentation, indicating material degradation and increased surface area, facilitating the immobilization of hydrogen-producing microorganisms [26].

When utilizing glucose and lignocellulosic hydrolysate as substrates in a microbial electrolysis cell, optimal hydrogen yields were attained at lower organic loading rates (OLRs). Specifically, glucose at reduced OLRs yielded up to 83% hydrogen, while lignocellulosic hydrolysate reached yields as high as 91%. This study observed

Table 1 Microbial hydrogen production using lignocellulosic materials

Approach	Lignocellulosic material	H ₂ production	Source
<i>Clostridium</i> and <i>Lactobacillus</i> bacteria	Wheat straw	80 mL/kg d	[22]
Microbial electrolysis cell	Napier grass	0.71 L/L/D	[26]
Magnetite nanoparticle assisted fermentation, <i>Klebsiella</i> sp.	Rice straw hydrolysate	83.20 ± 2.19 mL/g substrate	[27]
Fe and Mo amendment, <i>Rhodobacter palustris</i>	Saccharified wheat straw	48.1 mL h ⁻¹ L ⁻¹	[41]
Nickel ferrite nanoparticles supplementation	Rice straw hydrolysate	5544.86 ± 37.03 mL/L	[29]
Lipopeptide Biosurfactants amendment, Firmicutes (54.74 to 86.38%)	Pretreated straw	4.68 mmol/2 g	[31]
<i>Rhodospseudomonas palustris</i>	Wheat straw hydrolysate	25.6 ± 0.3 mL h ⁻¹ L ⁻¹	[30]
<i>E. aerogenes</i>	Enzymatic hydrolyzed rice straw	0.085 mol H ₂ /mol glucose	[42]
Heat shocked biogas inoculum	Gelatin, cotton, wheat straw, and sour cabbage	450 mL/g VSS	[43]
Heat shocked biogas inoculum from a slaughterhouse treatment plant	Brewer's spent grain hydrolysate	4160 mL/L; 760 mL/L h; 5154.26 mL/g glucose cons	[44]
<i>C. flavigena</i> ; <i>K. aerogenes</i>	Hyacinth hydrolysate	69.2 mL/g-VS	[45]
Calcium oxide nanoparticles amendment; Cellulolytic <i>Rosellomorea marisflavi</i>	Raw wheat bran treated by <i>Rosellomorea marisflavi</i>	2.1 mol H ₂ /mol reducing sugar	[46]
<i>Acetivibrio thermocellus</i> DSM 1313 and <i>Thermoanaerobacterium thermosaccharolyticum</i> MJ1	Pre-treated sugarcane bagasse	259.57 mL/g substrate	[47]
<i>Enterobacteriaceae</i> , <i>Clostridium</i>	Arundo donax hydrolysates	0.30 ± 0.05 mol of H ₂ /mol glucose	[48]
Peanut shell as microbial carrier; Heat shocked biogas inoculum from a sewage treatment plant	Glucose: 5 g/L	1.7 L (6 h) 0.58 L/L/d	[49]
<i>Clostridiales</i> consortium	Sugarcane bagasse and food waste	1.71 mL H ₂ /mL-culture; 586.19 mL H ₂ /g-carbohydrate	[50]

(continued)

Table 1 (continued)

Approach	Lignocellulosic material	H ₂ production	Source
Consortium derived from agricultural digestate	Spent mushroom bed (wheat straw)	7.98 LH ₂ /(KgVS*d	[51]
Activated nanocarbon from rice straw; <i>Bacillus coagulans</i> and <i>Bacillus subtilis</i>	Enzymatic hydrolyzed rice straw	2745 mL/L (50 h)	[52]

volumetric hydrogen production rates ranging from 0.73 to 1.67 L/L/D, facilitated by an efficient reactor setup, optimal operating temperature, and minimal inhibitory compounds. Cathodic hydrogen recovery (CHR) fluctuated between 30 and 56%, influenced by background current and energy inefficiencies. Analysis of the microbial community identified dominant genera including *Cloacibacillus*, *Enterococcus*, and *Acetobacterium*. Notably, *Enterococcus* spp. played a pivotal role in achieving optimal hydrogen yield by serving as both electron donors and acceptors [27].

Adding magnetite nanoparticles significantly impacts hydrogen production using *Klebsiella* sp. WL1316, a bacterium renowned for its efficient hydrogen production from lignocellulosic hydrolysate. It enhances glucose and xylose consumption rates and alters key metabolites crucial for hydrogen production. Specifically, it elevates pyruvic acid concentration, diminishes formic acid concentration during the early-to-mid fermentation stages, and reduces succinic, lactic, acetic acids, and ethanol concentrations after 72 h. Furthermore, the addition boosts hydrogenase activity, reaching its peak at 24 h of fermentation. RT-PCR analysis reveals heightened expression levels of hydrogenase and formate-hydrogen lyase-related genes following the addition of nanoparticles [28].

Previous research has indicated that Ni/Al₂O₃ catalysts exhibit the highest H₂ yield and gasification efficiency. The introduction of Co to Ni/Al₂O₃ has been shown to enhance H₂ yield, with the optimal Co loading identified as 6%. Additionally, factors such as cellulose concentration, residence time, and feedstock type were investigated [29].

In another approach using green nanoparticles, increased enzyme activities, substrate conversion efficiency, cumulative hydrogen production, hydrogenase, and formate-hydrogen lyase activities were observed in biohydrogen production by *Klebsiella* sp. WL1316 when amended with 30 mg/L of gsNiFe₂O₄ [30].

Photofermentation of lignocellulosic material by *Rhodospseudomonas palustris* has proven to be an efficient method for bio-H₂ production. By optimizing key parameters using Response Surface Methodology (RSM), a high hydrogen production rate of $25.6 \pm 0.3 \text{ mL h}^{-1} \text{ L}^{-1}$ was achieved under specific conditions. The optimized conditions provided a stable microenvironment, enhancing enzyme activity and metabolic pathways, leading to high substrate degradation efficiency (89%) and H₂ yield ($1.34 \text{ mmol H}_2 \text{ mol}^{-1} \text{ carbon}$) [31].

Bacterial lipopeptide biosurfactants were utilized to enhance biohydrogen production from pretreated wheat straw biomass via anaerobic dark fermentation. These

compounds facilitated microbial degradation, reduced surface tension, and altered lignin characteristics. Microbial community analysis revealed Firmicutes as the predominant phylum in lipopeptide-assisted bioreactors, with genera such as *Ruminiclostridium* and *Bacillus* being promoted for their efficiency in lignocellulose degradation and hydrogen production [32].

Temperature influenced microbial structure and biochemical pathways for hydrogen production using raw wheat straw with native microbiota. Additionally, organic loading rate affected the abundance of hydrogen producers and productivity. *Lactobacillus*, *Clostridium*, and *Pichia* were key members at 37 °C, while *Clostridium*, *Thermoanaerobacterium*, *Pichia*, and *Clavispora* were prominent at 55 °C. The highest hydrogen productivity was achieved at an OLR of 15.0 g-TS/kg d at 37 °C and 10.0 g-TS/kg d at 55 °C [33].

Ultrasound combined with dilute alkali cooking resulted in the highest reducing sugar concentration (13.18 g/L) and a maximum biohydrogen yield of 133.6 mL/g total solids (TS), significantly higher than raw wheat straw (23.9 mL/g TS) [34].

A study investigated the potential of Napier grass as a substrate for biohydrogen production through dark fermentation, highlighting its cost-effectiveness and environmental friendliness. Delignification of Napier grass using sodium hydroxide improved glucose recovery, with a 2% NaOH solution found to be ideal. Optimal conditions for hydrogen production were mesophilic (35 ± 2 °C) and pH 5.5, resulting in the highest hydrogen yield of 763.34 mL [35]. However, the use of NaOH may not be the most environmentally friendly process, with alternative methods such as the use of greener obtained delignification enzymes suggested to reduce environmental impact [36].

A multi-stage process for producing biohydrogen from the organic fraction of municipal solid waste and pruning materials was developed. This process involved three stages: starchy dark fermentation, intermediary autogenous acidic pretreatment, and lignocellulosic dark fermentation. The optimized three-stage process yielded an overall of 5,427 mL hydrogen from 100 g of untreated substrate, which was 2.4-fold higher than one-stage dark fermentation [37].

Integration of dark and photo-fermentation processes led to high hydrogen production from waste biomass. Sequential fermentation outperformed co-culture, with the highest hydrogen yield achieved at an HRT of 3.5 days and pH 7.5. Effective cooperation between dark and photo bacteria enhanced hydrogen productivity [38].

Experiments using single sugars and sugar mixtures resembling lignocellulose hydrolysates found that pentose sugars, notably arabinose, exhibited the highest H₂ production rates, while glucose had the lowest HPR. Sugar mixtures had lower-than-expected HPR due to reduced arabinose consumption. Pentose sugars were preferred for H₂ production but were limited by the presence of glucose. The microbial community was dominated by *Clostridium* bacteria, with varying abundances depending on the substrate [39].

Various techniques, including autoclaving, acidification, alkalization, aeration, and fungal pre-treatment, have been employed to enhance hydrogen production via dark fermentation, utilizing food wastes as substrates [40].

An alternative approach, distinct from microbial hydrogen production, focuses on green hydrogen generation. It utilizes biomass-assisted solar systems to extract hydrogen by depolymerizing lignin from lignocellulosic biomass. This method not only yields valuable aromatic compounds but also facilitates electron extraction. Integration of this process with a perovskite-based photocathode enables bias-free photoelectrochemical hydrogen production, efficiently harnessing solar energy. Passivation of perovskite layers enhances stability, ensuring long-term performance. This integrated system achieves a record-high photocurrent density, demonstrating efficient solar-to-hydrogen conversion. Offering practical advantages over conventional methods, such as continuous chemical production from biomass and hydrogen generation, this approach eliminates the need for expensive catalysts and minimizes problematic oxygen generation [41].

Furthermore, improvements in hydrogen production by microorganisms are crucial. Optimization of hydrogen production from lignocellulosic biomass through photofermentation revealed that the addition of iron (Fe) and molybdenum (Mo) ions enhances hydrogen production by promoting nitrogenase activity [42].

3 Bottlenecks and Limitations of Hydrogen Production from Lignocellulosic Materials

Conventional hydrogen production processes, reliant on fossil fuels as substrates, are neither sustainable nor ecological. In contrast, biological processes offer a more attractive alternative, utilizing a wide variety of organic waste as substrates and enabling biohydrogen production using microorganisms under ambient conditions. Moreover, the debate over “food versus fuel” has spurred investigations into producing biohydrogen from abundantly available waste, such as lignocellulosic biomass substrates. These substrates have already been explored for materials development [53], serving as culture mediums for microbial growth and metabolite production [54–56]. However, current research focus on biofuel faces several challenges, including efficient pre-treatment of lignocellulosic biomass, the production process itself, and storage and transportation of biohydrogen [6, 57–60].

Lignocellulosic biomass and other organic materials, available year-round, hold promise for biohydrogen production. Their availability and acquisition as waste are crucial factors. Utilizing renewable plant biomass for biohydrogen production offers a viable alternative to fossil fuel-based methods [21]. Lignocellulosic biomass, comprising agricultural and vegetative waste, poses challenges due to its low biodegradability rate and high lignin content, hindering fermentation. While considered an ecological option, biohydrogen production from lignocellulosic biomass requires addressing certain pitfalls for industrial-scale application. The intricate structure of lignocellulosic biomass, primarily attributed to lignin, poses challenges in its utilization, tackled through various approaches such as pretreatment processes,

synthetic biology, and genetic engineering. The recalcitrant lignin must be removed before utilizing lignocellulosic biomass for biofuel fermentation [58, 61].

The integration of lignocellulosic biomass as a raw material in biorefinery approaches faces bottlenecks in terms of efficiency of pretreatment methods, cost implications, energy efficiency, and environmental sustainability. Biomass pretreatment stands as a crucial process to address the inherent recalcitrance in the structural components of lignocellulosic materials. Among available technologies, acid pretreatment is generally preferred due to its higher effectiveness in breaking down the structure of lignocellulosic materials and promoting the conversion of crystalline-amorphous cellulose. However, this method carries various deficiencies, such as the generation of inhibitory compounds and the discharge of highly acidic and potentially harmful effluents into the natural environment and human health. Researchers now advocate for biological methods for pretreatment and hydrolysis due to their feasibility, reliability, and commercial viability in large-scale production [57].

Currently, the majority of industrial hydrogen is derived from fossil fuels as an energy source, utilizing conventional methods like coal gasification, steam reforming of methane, and water electrolysis. However, these methods are characterized by their energy-intensiveness, high costs, and adverse environmental impacts. Dark fermentation offers a promising avenue for converting cellulosic and lignocellulosic materials, including waste, into hydrogen. This biological process presents the potential to merge waste management with energy production. Nonetheless, further development of existing technologies is necessary to achieve financial competitiveness and realize practical application and commercialization potential [62]. The primary bottleneck in hydrogen production through dark fermentation lies in the reduced hydrogen yield compared to the substrate consumed [63].

In the realm of biohydrogen production through photofermentation, various challenges have surfaced. Notably, photosynthetic bacteria encounter limitations in harnessing solar energy efficiently, resulting in a significantly low light-to-hydrogen conversion efficiency. These bacteria also necessitate specific environmental conditions and sterility for optimal growth and hydrogen generation. Moreover, the high activation energy demanded by nitrogenase enzymes requires a substantial energy input for photofermentation. Additionally, cellular shading effects impede light penetration into the photoreactor, diminishing light intensity and consequently reducing biohydrogen production performance [64]. While dark fermentation has shown improvements, the yield and production rate of H_2 in biological processes remain inferior to thermochemical methods, rendering them less competitive [64].

In a parallel vein to the biological fermentation of sugars obtained from carbohydrate hydrolysis (starch) or structural carbohydrates (cellulose, hemicellulose), thermochemical gasification of lignocellulosic biomass emerges as a more prevalent and promising avenue for biohydrogen production [59]. Gasification, thus, stands as a dependable method for biomass utilization to generate gas. Nevertheless, further studies are imperative to overcome technical barriers to this process, including tar formation and ash accumulation [65].

Following hydrogen production, storage and transportation pose significant challenges [60]. Given hydrogen's exceptionally low boiling point, storage becomes

intricate. Over time, stored hydrogen degrades container materials, complicating subsequent storage. With its extremely low density, hydrogen gas occupies considerable space—1 kg of hydrogen gas fills over 11 m³ at room temperature and atmospheric pressure—and its pure form is both flammable and explosive [60]. Therefore, increasing storage density is crucial for economically viable hydrogen storage. Several storage methods are currently under investigation [60], although hydrogen liquefaction and liquid hydrogen transportation necessitate expensive equipment [65]. Additionally, biomass conversion techniques (e.g., gasification and pyrolysis) come with inherent limitations and significant energy consumption [65].

In a study examining microbial hydrogen production, several bottlenecks were identified. Initially, there was a decline in productivity, with hydrogen productivity sharply dropping from 325 ± 43 mL/kg d to 125 mL/kg d after the first 4 days. Although the reactor reached a stable productivity of 80 mL/kg d between days 36 and 52, fluctuations occurred, resulting in peaks and drops in productivity. While changing the wheat straw cultivar caused a temporary productivity peak, it did not sustain and required recovery to steady-state performance. Inhibitory interactions among native microbial strains, particularly lactic acid bacteria (LAB), affected overall productivity, despite some positive synergisms like between *Lactobacillus* and *Clostridium*. The production of organic acids and bacteriocins by LAB could inhibit hydrogen production, as indicated by pH changes and antimicrobial activity. Additionally, the interactions between microbial species were complex and species-specific, complicating the achievement of consistent and optimal hydrogen production [23].

An analysis of hydrothermal pre-treated algae and lignocellulosic biomass for hydrogen production revealed that furfural substantially inhibits hydrogen production by consuming large amounts of NADH and inhibiting key enzymes. Although furfuryl alcohol is less inhibitory, it still negatively affects yield when present in high concentrations. Conversely, furoic acid enhances hydrogen production at low concentrations but extends fermentation time at higher levels. Efficient transformation and utilization of these compounds are crucial for optimizing hydrogen yield. High inhibitor concentrations disrupt microbial community structure, reduce hydrogen production efficiency, and alter metabolic pathways. Managing inhibitor concentrations to maintain microbial diversity and functionality is essential for effective hydrogen production [66].

Another study identified several obstacles hindering hydrogen production through microbial electrolysis cells. These hurdles included issues during reactor startup, fluctuations in anodic current densities, less-than-ideal hydrogen yield, low cathodic hydrogen recovery, energy inefficiencies, variations in microbial community composition, and their impact on electrochemical performance. During reactor startup, a delay in current density increase was noted, likely due to biofilm/sludge accumulation in electrode holders. Fluctuations in anodic current densities hinted at potential limitations in cathode performance, impacting overall electrochemical efficiency. While achieving relatively high hydrogen yields with glucose and lignocellulosic hydrolysate, discrepancies between substrate dosing and actual hydrogen production suggested inefficiencies in substrate utilization. Cathodic hydrogen recovery fell

short of theoretical maximums, possibly due to hydrogen scavengers and background currents, leading to energy losses (35–61%). The dominance of *Enterococcus* spp. in both anodic and cathodic communities influenced electrochemical performance, with synergistic fermentative bacteria also playing significant roles in electron transfer [27].

Although the hydrogen supply catalytic system using CaO and NiO/ γ -Al₂O₃ shows promise in increasing liquefaction yields of lignocellulose, several bottlenecks and limitations related to catalyst efficacy, biomass conversion rates, reaction mechanisms, efficiency of hydrogen donor solvents, and control over product distribution persist. Further research and optimization are needed to address these challenges for practical and scalable hydrogen production from lignocellulosic biomass [67].

4 New Trends to Overcome Bottlenecks of Biohydrogen from Lignocellulosic Materials

Despite significant technological advancements in biohydrogen generation from agricultural waste through biological approaches, there remain challenges in implementing these technologies on a commercial scale [68]. While the cost of raw materials can be reduced by utilizing lignocellulosic waste and municipal solid waste, a sustainable supply chain of raw materials with homogeneous quality is required for the continuous operation of large-scale hydrogen production facilities. Other inherent biomass parameters need consideration, including the type of raw material and the potential need for pretreatment steps. Ideally, biomass should be as inexpensive as possible, resulting in a sustainable yield after its utilization [62]. Establishing large-scale hydrogen generation plants helps reduce the high capital cost required for biochemical and thermochemical plants. The high market value of hydrogen can also attract investors, thus enabling faster dissemination of the technology [69].

Grouping biomass according to its chemical composition (carbohydrates, lignin, and others) can facilitate necessary pretreatment. When considering dark fermentation, pretreatment significantly impacts hydrogen production yield by hydrolyzing the raw material into low-weight molecules easily metabolized by microorganisms [62]. The use of microbial consortia greatly reduces pretreatment time and is advantageous due to improved hydrolytic efficiency, high adaptability, and increased substrate utilization rate [64]. However, pretreatment technologies are not entirely effective, as most result in process-inhibiting products that influence microbial metabolism and decrease hydrogen yield. Nonetheless, the application of biocatalyst immobilization can increase biohydrogen production and reduce bioprocess costs [68].

Saha et al. [64] assert that much research has been conducted on biological methods to develop and enhance hydrogen yield and production rate, but only at the laboratory scale. The authors advocate for further research to (1) make these methods

economically viable and highly efficient, (2) develop a common and effective saccharification technology for all raw materials, (3) isolate more microorganisms capable of saccharification and efficient production of biomass reducing sugars, (4) reduce the necessary treatment time for biomass saccharification, and (5) gain a detailed understanding of the genes and metabolic pathways involved in processes for robust biohydrogen production required for industrial applications.

Investing in rumen microorganisms presents a promising avenue in the biological process of hydrogen production from lignocellulosic waste. The rumen of ruminants harbors a plethora of microorganisms adept at cellulose degradation. Predominantly anaerobic and facultative anaerobic microorganisms thrive in the rumen environment, comprising bacteria, fungi, archaea, and protozoa [69]. Ganeshan et al. [59] isolated cellulose-degrading bacteria with high hydrogen concentration from the rumen of sheep and cattle, unveiling a novel strain of *Klebsiella pneumoniae* exhibiting cellulose degradation capability and robust hydrogen production. Its hydrogen production performance surpassed that of other fermentation strains commonly employed in biohydrogen production. The choice of reactor is also pivotal. Performance enhancements can be derived from innovative designs like catalytic membrane reactors, which augment gas conversion by modifying reaction equilibrium [65].

Both the public and private sectors must prioritize hydrogen storage development, as the progression of the hydrogen economy hinges upon adequate hydrogen storage solutions. The technical and economic feasibility associated with hydrogen presents significant challenges. The accessibility of green hydrogen production technology poses another obstacle. The cost of the integrated system responsible for green hydrogen production must be attainable for potential users. Various nations are formulating policies and enacting legislation to integrate hydrogen as a fuel in transportation and energy sectors, aiming for zero emissions of polluting gases within a decade. Consequently, increased research and development funding in this domain is imperative. Technological advancements in hydrogen production, storage, distribution, and transmission are crucial and necessitate governmental support. While hydrogen gas can be transported via pipelines and liquid hydrogen via ships, establishing a comprehensive network for transmission, storage, distribution, and delivery of hydrogen remains paramount to meet global demand, which is yet to be fulfilled [60].

Furthermore, electromagnetic, mechanical, and chemical methods such as ultrasound, pulsed electric field [70–72], magnetic fields, photostimulation [73–75], and hydrogenic imbalance stress [76] have been employed to enhance microbial bioprocesses. These methods influence cellular processes, metabolism, and mass transfer, leading to improved productivity and product yields in bacteria, fungi, and microalgae.

5 Future Prospects and Concluding Remarks

Hydrogen, particularly when derived from renewable biomass, presents a promising pathway for sustainable energy. Despite challenges such as storage and production costs, ongoing research endeavors to surmount these obstacles. The integration of biohydrogen into renewable systems not only bolsters energy security but also tackles waste management issues. As we refine processes and increase production scale, biohydrogen emerges as a substantial contender in the shift towards a greener hydrogen economy. The concluding remarks drawn from the discussions in the present chapters are as follows:

- (I) Various methods, including enzymatic treatment and microbial electrolysis, hold promise for efficient hydrogen generation. Integrating multiple stages and alternative approaches, such as biomass-assisted solar systems, present innovative pathways for green hydrogen production. Continued advancements in microbial performance and process optimization are crucial for realizing the full potential of hydrogen production from biomass.
- (II) Efficient pretreatment methods, addressing the recalcitrance of lignocellulosic biomass, and enhancing the overall production process are critical areas of focus. Biological methods for pretreatment and hydrolysis are gaining traction due to their feasibility and commercial viability. Despite progress, challenges persist in achieving competitive hydrogen yields and production rates, particularly in comparison to thermochemical methods.
- (III) Significant progress has been made in harnessing biohydrogen generation from agricultural waste through biological approaches, yet several challenges remain in scaling up these technologies for commercial viability. Addressing issues such as the sustainable supply chain of raw materials, effective pretreatment methods, and optimizing microbial processes is essential for the success of large-scale hydrogen production facilities.
- (IV) While dark fermentation and photofermentation show promise, further research is required to overcome efficiency limitations and achieve practical scalability. Thermochemical gasification of lignocellulosic biomass presents another viable option for biohydrogen production, albeit with its own technical barriers.
- (V) Innovative reactor designs and advancements in hydrogen storage solutions are also crucial components for the growth of the hydrogen economy. Collaboration between the public and private sectors, along with supportive policies and increased research funding, will be vital in overcoming technical and economic hurdles.

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Antiallergic, Antioxidant, Anti-Inflammatory and Immunostimulant Potential of Kefiran Postbiotic: Molecular Docking, Prediction of Pharmacokinetic Properties and Biological Activity



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Abstract Kefiran is an example of a postbiotic of kefir grains and is believed to be primarily responsible for the health-promoting effects. This study aimed to evaluate the antiallergic, antioxidant, anti-inflammatory and immunostimulant potential of the main component of kefir, in silico, through prediction of biological activity and molecular docking. The prediction of the biological activity of kefir showed significant results for antiallergic, anti-inflammatory, antioxidant and immunostimulant activities and the docking was carried out between this compound and the following biological targets; H1R1 iNOS, PLA2 and VEGF. It is concluded that kefir can be used properly in the development of drugs or functional foods, which results in a probable inhibitory action in the immune response and the treatment of children's food allergies.

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Keywords Food allergy · Kefiran · In silico

1 Introduction

Food allergy is an abnormal immunological reaction upon exposure to certain food proteins. It is broadly categorized into immunoglobulin E (IgE)-mediated, non-IgE-mediated, and mixed IgE/non-IgE-mediated [1, 2]. In turn, probiotics appear as potent functional agents in the human intestinal microbiota, as they produce immunological effects, resulting in increased resistance against pathogens. Thus, the use of probiotic bacterial cultures stimulates the multiplication of beneficial bacteria at the expense of the proliferation of potentially harmful bacteria, reinforcing the host's natural defense mechanisms [3].

Kefir is an example of a probiotic drink with beneficial therapeutic effects, Kefiran, an exopolysaccharide (EPS) present in kefir grains, is an example of a postbiotic and is believed to be primarily responsible for the health-promoting effects obtained with the use of kefir [4].

In silico methodologies have been increasingly used in studies involving the design of drug candidates and their mechanism of action. These methodologies allow the selection and optimization of molecules with greater development potential, reducing time and cost [5]. Thus, an in silico study of the main molecule present in milk kefir was carried out: kefiran, using molecular modeling strategies, in order to identify and characterize probable biological targets, mainly those related to antiallergic, anti-inflammatory, antioxidant and immunostimulant activities.

2 In Silico Study

2.1 Prediction of Biological Activity

The biological activity of kefiran was analyzed by online tools that predict profiles from the structural formula of a molecule, using the ligand-based drug design approach: the PASS Online (<http://www.akosgmbh.de/pass/index.html>) and Swiss Target Prediction (<http://swisstargetprediction.ch/>). PASS predictions possess high accuracy (70–80%) up to 2,000 biological activities for chemicals when compared to other organic drugs with similar structural and physicochemical features of the compounds submitted [6]. In this webserver it is possible to predict the biological effects of a compound based on its structural formula using MNA (multi-atomic neighbors) descriptors [7], suggesting that the biological activity depends on its chemical structure. Only predicted activities with Pa (Probability of being active) > Pi (probability of being inactive) for the selected compound were considered [8].

Swiss Target Prediction online tool (<http://swisstargetprediction.ch/>) was used to predict small molecules targets according to their 2D or 3D similarity with more than 370,000 known activities in well-defined experimental binding assays or more than 3000 proteins of different species [9].

2.2 Docking Molecular

The chemical structure of kefiran was drawn using the ChemSketch software. To obtain the most energetically stable chemical structure, this molecule was optimized using the semi-empirical method RM1 by HyperChem 6.02 software [10, 11].

The GOLD (Genetic Optimization for Ligand Docking) software (version 5.5, Cambridge Crystallographic Data Center) was used to perform docking simulations between the milk kefir molecule (kefiran) and the biological targets studied. This software uses the genetic algorithm for flexible ligand coupling experiments at protein binding sites [12].

The biological targets studied were obtained from the files deposited with the Protein Data Bank (PDB): Histamine H1 receptor (HRH1) complexed with doxepin (PDB ID: 3RZE, resolution: 3.10 Å) [13]; Nitric Oxide Synthase (iNOS), complexed with arginine (PDB: 1NSI, resolution: 2.55 Å) [14]; Phospholipase (PLA₂), complexed with AZD2716 (PDB: 5G3N, resolution: 1.80 Å) [15] and Vascular and endothelial growth factor (VEGF) complexed with imidazole [1,2-b]pyridazine derivative (PDB ID: 3VO3, resolution: 1.52 Å) [15]. To perform the molecular coupling, one adds hydrogen atoms and then removes water molecules. The molecules complexed with each of the proteins were used to calculate the fit coordinates, that is, the exact location of the site of protein activity. The coordinates for each biological target were: x: 16.66, y: 35.97, z: 22.31 for HRH1 (Radius: 20 Å), x: 9.79, y: 64.76, z: 15.90 for NOS (Radius: 10 Å); x: 7.48 y: 3.41 z: -0.16 for PLA₂ (Radius: 10 Å) and x: 26.91, y: -26.93, z: -14.59 for VEGF (Radius: 10 Å), extracted of the molecules originally complexed.

Subsequently, analyses of molecular interactions were performed using the Discovery Studio Visualizer software [16]. Validation of resulting docking poses has been carried out considering the Root Mean Square Deviation (RMSD) between the experimental and the conformation of the ligand that yielded the best pose after docking.

To identify the interactions between the compounds and the therapeutic targets, it was necessary to identify the amino acids making up the binding site of each of the proteins: HRH1: Tyr108, Phe435, Asp107, Thr112, Ile115, Trp158, Asn198, Phe424, Trp428, Phe432, Tyr431, Lys179 e Lys191 [13, 17]; iNOS: Tyr347, Pro350, Val352, Gly371, Trp372, Gln263, Arg266, Trp346, Tyr373, Glu377, Asp382 e Arg388 [14, 18]; PLA₂: Tyr52, Phe5, Ile9, Phe22, Ala102, Ala 103, Phe106, Gly29, His47, D48, Leu2, F5, His6, Ile9, A17, A18, and G22 His48, Asp49, Asp99 [15, 18]; VEGF: Leu840, Val848 Val867, Lys868 Ala866, Leu889, Val898 Val899, Val914 Val916,

Glu917, Glu885, Phe918, Leu1035 Ile1044, Cys919, Cys1045 Asp1046, Phe1047 [15, 19, 20].

3 Results and Discussion

3.1 *In Silico Study*

In our previously published study (Barros et al. 2022) [21], the results of the pharmacokinetic properties obtained by the Qikprop software and the toxicology of kefiran using the Derek software were presented. Kefiran demonstrated low absorption results, due to it being poorly soluble and not absorbable in the upper part of the GIT (mouth and stomach), as it is resistant to enzymatic hydrolysis. Therefore, this low oral solubility was already expected, as its action takes place in the colon where it is absorbed by specific bacteria present in the intestinal microbiota. Furthermore, kefiran also showed low values for logP (−8.048), MDCK (0.017 nm/s) and Caco-2 cells (0.072 nm/s), compared to reference values. These values confirm the hydrophilic and polar characteristics, which confers a viscosity characteristic that can hinder its passage through the plasma membrane. And, as excessively polar drugs cannot cross the blood–brain barrier, kefiran also presented satisfactory results, because as it is a molecule whose action does not need to occur in the CNS, with this result, it is less likely to cause side effects.

Toxicological parameters are also important to evaluate potential beneficial effects of a new therapeutic compound. Based on this, in the prediction of kefiran toxicity, using the Derek software, no structural alert state was detected for kefiran in the predictions of mutagenicity, carcinogenicity, hepatotoxicity, nephropathies in humans or other mammals, which suggests greater reliability and safety in the possible use of this molecule in new research [21].

3.2 *Prediction of Biological Activity*

The results of the prediction of the biological activity of Kefiran are shown in Table 1.

Table 1 Biological activities predicted by PASS for kefiran

Biological activity	Kefiran	
	Pa	Pi
Histamine release inhibitor	0.445	0.065
Antioxidant	0.918	0.003
Anti-inflammatory	0.759	0.009
Immunostimulant	0.909	0.003

PASS web server [22] provides data using chemical similarity calculations and it can show biological targets and various therapeutic effects for the test molecule. PASS uses prediction based on a set of molecules with known biological activities, measured experimentally, where a value of P_a (Probability of being active) and P_i (Probability of being inactive) are emitted for each studied molecule. So, the higher the value of P_a than P_i , the greater the probability of the studied molecule to present activities evaluated [23]. Analysis of the prediction of biological activity for kefiran allowed us to identify that this molecule presented excellent anti-allergenic (Histamine release inhibitor), anti-inflammatory, antioxidant and immunostimulating activities. For all of these biological activities, it was found that there was a greater likelihood of activity (P_a) than inactivity (P_i). For these last three, P_a values were close to 1.00, thus obtaining satisfactory results as shown in Table 1.

This immunostimulant activity found by PASS coincides with the result obtained by the Swiss Target Prediction, where the Vascular Endothelial Growth Factor (VEGF) was found as the most relevant target using this online tool. Thus indicating that the kefiran have a beneficial action in improving the immune response, which is directly related to the possible anti-allergenic activity.

These results served as a subsidy for the realization of docking with kefiran, seeking targets related to these identified activities in order to better understand the antiallergic, antioxidant, anti-inflammatory and immunostimulant activities of kefiran.

3.3 Molecular Docking

Figure 1 shows the 2D and 3D chemical structure of kefiran that was designed and optimized using the software: ChemSketch [10] and HyperChem 6.02 [11] respectively.

Molecular docking was used in this study to identify the mode of interaction and the degree of affinity of the Kefiran and the following biological targets: H1R1, iNOS, PLA₂ and VEGF.

Docking accuracy is indicated by the RMSD value, which indicates the accuracy of the poses calculated by the GOLD fitting algorithm compared with the experimentally poses of a compound evaluated to a biological target. RMSD values less than 2.0 Å indicates good precision according to Cole et al. [12]. In this study, the following RMSDs for the therapeutic targets HRH1, iNOS, PLA₂ and VEGF were 0.830, 1.841, 1.070 and 1.722, respectively, showing, therefore, that all values are within the recommended range, with HRH1 having the lowest value of RMSD.

Then, the solution that found the largest Goldscore for each therapeutic target was selected from the docking results of kefiran. This result classifies the orientations of the molecules in decreasing order of affinity with the corresponding binding site of the biological target for better understanding.

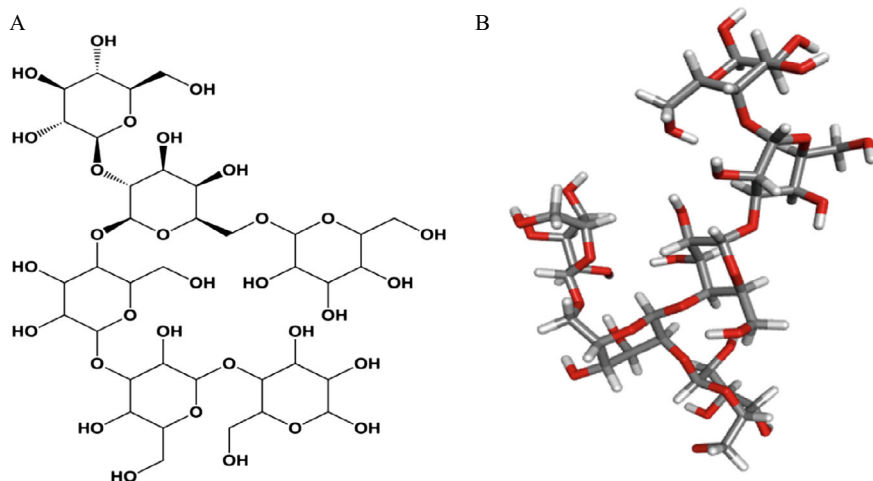


Fig. 1 2D **a** and 3D **b** structure of kefiran

The GoldScore values obtained on docking between kefiran and this biological targets HRH1, iNOS, PLA₂ and VEGF were 72.91, 88.1, 91.46 and 78.88, respectively. The greater the result, the greater the affinity of the molecule with the corresponding binding site of the biological target. This result ranks the orientations of molecules in descending order of affinity with the corresponding binding site of the biological target for better understanding The 2D images referring to the “docking” between the kefiran molecule and the ligands complexed to the four biological targets are shown below in Figs. 2 and 3, respectively. The discussion of these results was grouped by target studied.

3.4 Antiallergic Action

Histamine is a biogenic amine and an important mediator in physiological and pathophysiological conditions of allergy and inflammation, thus being the target of most drugs developed against allergies. There are 4 histamine receptors (H1, H2, H3 and H4 receptors) [24, 25]. In allergic reactions of type I hypersensitivity, the antigen comes in contact with Immunoglobulin E (IgE) which is present in mast cells and induces the release of histamine, which is responsible for the activation of the HRH1 receptor [26]. Thus, the H1 receptor is directly involved in the action of histamine, thus having greater pharmaceutical importance in search of research involving HRH1 antagonists, commonly known as antihistamines, these compounds inhibit the action of histamine on HRH1 to relieve the symptoms of allergic reactions, making this receptor one of the most validated targets for anti-allergenic action [13].

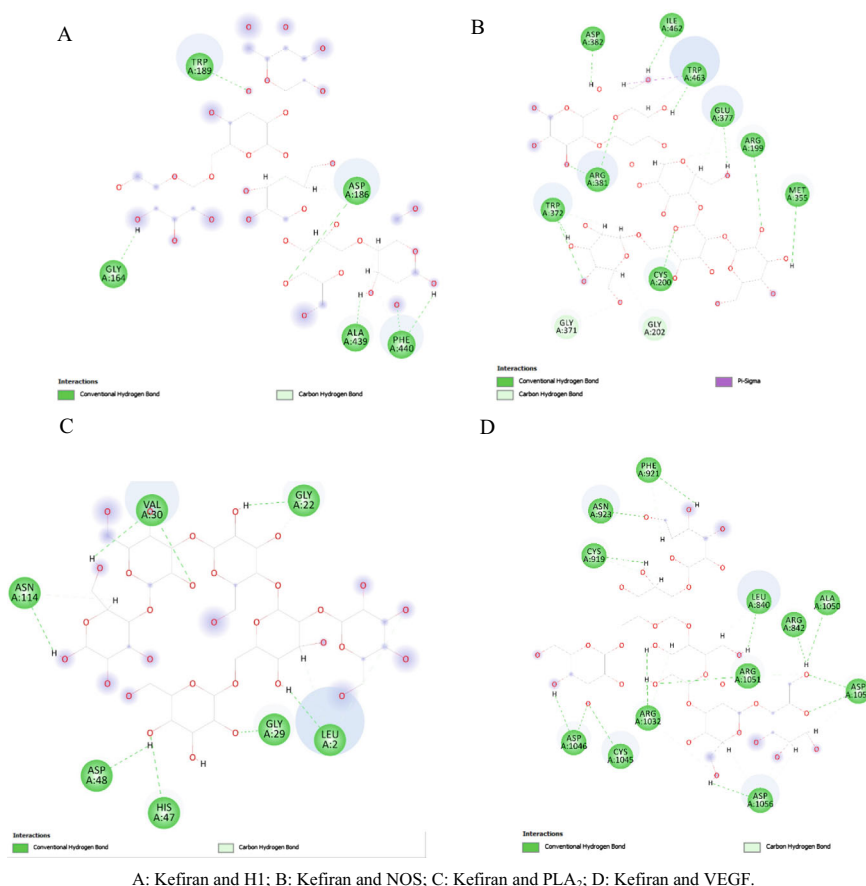
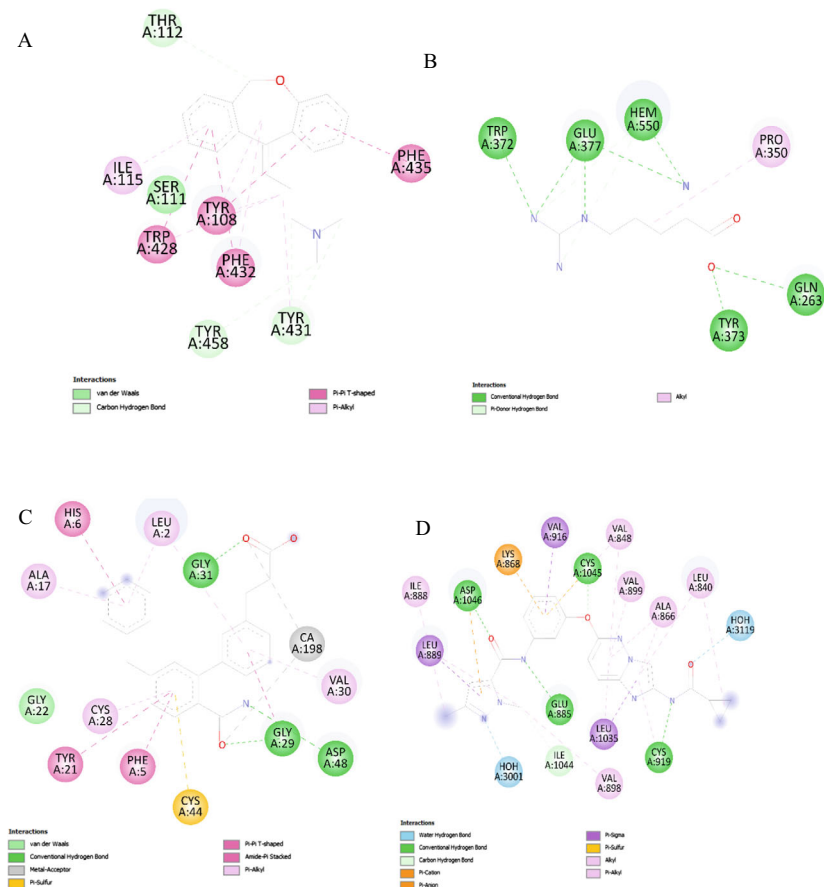


Fig. 2 2D images referring to the docking between the kefiran molecule and biological targets: HRH1, iNOS, PLA₂ and VEGF **a** Kefiran and H1; **b** Kefiran and NOS; **c** Kefiran and PLA₂; **d** Kefiran and VEGF

In the docking performed with the HRH1 target, the kefiran made 9 favorable interactions of hydrogen with the amino acid residues: Phe440, Ala439, Asp186, Gly164 and Trp189. In comparison with kefiran, doxepin (E), a complexed molecule, performed 15 interactions with the following active site amino acids: Asp107; Tyr108; Ser111; Trp428; Tyr431 and Phe432. The Goldscore result with doxepin (E) was 89.51. The GoldScore value obtained on docking between o kefiran and this biological target was 72,91, a value very close to the complexed Molecule and, a higher value than those found in the study by Thangapandian et al. [27], in which docking between H1R and inhibitors doxepin and 4-methyl diphenhydramine resulted on the in following GoldScore values: 49.604 and 52.912, respectively.

Hydrogen interactions are the most important non-covalent interactions for the target-binding group. These bonds are based on the interaction between positively



A: Doxepin complex (E) and HRH1; B: Arginine and iNOS complex; C: AZD2716 and PLA2 complex; D: Imidazole complex [1,2-b] derived from pyridazine and VEGF.

Fig. 3 2D images referring to the molecular docking between the complexed ligands to their respective biological targets: **a** Doxepin complex (E) and HRH1; **b** arginine and iNOS complex; **c** AZD2716 and PLA2 complex; **d** imidazole complex [1, 2] derived from pyridazine and VEGF

polarized (oxygen bound hydrogen) and negatively charged (oxygen) atoms. Conventional hydrogen bonding interactions may exist between a hydrogen bond donor atom and an acceptor atom such as, for example, N, O, P and S [28]. This type of interaction was the most observed in the molecular anchorages, possibly due to the molecules present in kefiran having in their molecular structure many polar points at its extremities, mainly because they are atoms of oxygen. This molecule did not interact with the amino acid residues of the H1R1 binding pouch, it is believed that due to the size of Kefiran, it interacts only on the surface of the histamine receptor, and may prevent this molecule from interacting with the amino acid residues located in the

binding site of the target studied, however, it is possible to promote a difficulty for the natural ligand to have access to the binding site.

3.5 *Antioxidant Action*

Nitric Oxide Synthases (NOSs) are a family of enzymes that produce nitric oxide (NO) from L-arginine. High levels of nitric oxide influence pathological processes, for example, inflammation and excessive iNOS expression can be found in many inflammatory processes, such as allergies [29]. In mammals, there are three isoforms of NOS: isozymes endothelial (eNOS) and neuronal (nNOS) that are constitutively expressed and inducible isozymes (iNOS), which is produced in response to cytokines or pathogens. NOS contains the heme group in its catalytic site, which is important for its enzymatic activity [30].

In comparison with the 1NS1 docking, between the complexed molecule and kefiran, it was noted that the Arginine Goldscore presented a lower value (58.66) than the result of kefiran (88.1) which demonstrates great affinity of kefiran for this target. Furthermore, while the complexed ligand had 9 interactions of hydrogen with amino acid residues, the kefiran had 19 favorable interactions with iNOS, with 18 hydrogen bonds and 1 hydrophobic, a quantity also higher than the complex molecule.

Furthermore, there were 8 interactions of hydrogen with the amino acid residues of the binding site of this target: Asp382 and Gly371; Trp372 and Glu377 with which substrates (L-arginine) form a hydrogen bond. This last residue (Glu377) is essential for the interaction because according to Li et al. [14], the inhibitor fits with the active site of this target by assuming that the ethanimidamide moiety mimics the guanidino group of arginine to hydrogen bond with Glu377, that the 5-amino group interacts with the heme propionate, that and the glycol moiety extends into the polar binding pocket where the carboxylate of L-Arg binds. Interactions with these amino acids are important for inhibitory activity. Thus, it may suggest that kefiran inhibits the action of iNOS, thus having an antioxidant action.

3.6 *Anti-Inflammatory Action*

Phospholipases A₂ (PLA₂) are enzymes that, due to their hydrolytic function, are responsible for the release of free fatty acids, arachidonic acid (AA), from the sn2 position of membrane phospholipids. AA is metabolized in different lipid mediators, such as bronchodilator and bronchoconstrictor, mainly through cyclooxygenases (COXs) and is a limiting factor for the production of eicosanoids in the airways, such as prostaglandins (PGs) and leukotrienes (LTs) [31, 32]. According to Nolin et al. [33], this enzyme is present in high concentrations in mucous secretions and is expressed in epithelial cells, macrophages and eosinophils in the airways of asthmatic patients.

Thus, the control of PLA₂ activities has been proposed for the treatment of inflammatory/allergic diseases.

For this biological target, kefiran had the highest GoldScore value: 91.46. This molecule made 9 favorable interactions, all the hydrogen bond types. 4 of these interactions are with the amino acid residues of the binding pouch: Leu2, Gly22 and His47. This last amino acid is particularly important for the interaction. According to Giordanetto et al. [34], the amino acid residue His47 is also a part of the coordination bond with the catalytic calcium ion, responsible for increased affinity and functional inhibition of the enzyme as a result of a bidentate calcium chelate and additional van der Waals contact with the enzyme. In addition to 1 interaction with the Asp48 amino acid residue that is close to the Asp49 active site residue.

About the interaction of PLA₂ with its co-crystallized ligand, 14 interactions were performed with 6 amino acid residues from the active site (Phe5; His6; Ala17; Gly22; Gly29 and His47), being 6 of hydrogen; 1 electrostatic and 7 hydrophobic. It is thus observed that kefiran obtained a GoldScore similar to the co-crystallized (107.14) ligand and maintained similar interactions to the same amino acid residues as well. In addition, kefiran made interactions of shorter distances (1.6 Å to 3.79 Å) when compared to the complexed molecule, which had interactions with distances of up to 5.28 Å. This demonstrates a high affinity of kefiran with the PLA₂ target.

It is also known that one of the common criteria among PLA₂ inhibitors is the interaction of hydrogen with the His47 residue. In this sense, the two molecules under study showed this type of interaction with this residue. However, kefiran manages to bind more strongly to this residue as it presents two hydrogen interactions with a distance of 2.03 Å and 2.74 Å, while the co-crystallized ligand presented only one with a distance of 2.77 Å. Therefore, kefiran has the potential to interrupt the bonds of other molecules in the active site of PLA₂, due to its key interaction with His47 and because of its large size, which can hinder the passage of other molecules to this binding pocket.

3.7 Immunostimulant Action

Vascular endothelial growth factor (VEGF) is a mediator of vascular permeability, which when at high levels can promote hypervascularity and edema, characteristics related to many chronic inflammatory diseases such as pulmonary disorders [35]. Furthermore, studies show that this biological target has a fundamental role in allergic diseases, especially urticaria, and has been considered as a potential blood biomarker for chronic spontaneous urticaria (CSU) diagnosis [36] vasodilator function is due to its ability to increase the production of nitric oxide, this vasodilator triggers some physiological responses such as mucus and atopic dermatitis, characteristics of the allergic process. In addition, as a pro-inflammatory mediator, it inhibits the recruitment and maturation of dendritic cells, thus leading to the differentiation of T cells in the TH2 profile, thus suggesting a role for VEGF in the pathophysiology of the disease [37].

Corroborating with these authors, Poza-Guedes et al. [38] in a prospective longitudinal study ($n = 20$), observed downregulation of VEGF after rapid desensitization by IgE in patients with cow's milk protein allergy, suggesting that it is a valuable method for obtaining temporary immunotolerance, as it was effective to prevent mast cell degranulation and protect against food allergy anaphylaxis.

The docking between kefiran and biological target VEGF revealed a total of 25 favorable hydrogen interactions, involving 13 different amino acids. 4 of these amino acids are part of the region of the active site of this target: residues Cys919, Leu840, Cys1045 and Asp1046.

The interaction between VEGF and the complexed imidazole [1,2-b]pyridazine derivative generated a GoldScore of 97.94 and 15 interactions with amino acid residues present in the active site (Val848; Ala866; Leu889; Val899; Val916; Cys919; Leu1035; Cys1045; Asp1046), of which they diversified into 1 hydrogen interaction; 1 electrostatic interaction and 13 hydrophobic interactions.

Despite the lower GoldScore value obtained for kefiran (78,88) about the complexed molecule, one has to take into account the complexity of the structure of this molecule. Another factor to be analyzed were the interactions made by the imidazole [1,2-b] pyridazine-derived molecule, mainly hydrophobic interactions, while kefiran had only hydrogen interactions, due to its molecular structure being rich in hydroxyls, being this type of interaction stronger in energy terms than hydrophobic interactions [39].

In addition, the kefiran interacts with the Asp1046 residue orients toward the ATP binding cleft. Despite not having the best Gold Score value, kefiran did not have the most interactions with this target, in addition to interacting with the largest number of amino acid residues on the active site, which suggests a greater affinity of kefiran with the analyzed target.

This demonstrates how kefiran has the potential to inhibit VEGF, even though it does not have superior results than the imidazole-derived molecule [1,2-b] pyridazine, as it interacts with key amino acid residues, mentioned above, to inhibit VEGF, leading to considering mainly the residue Asp1046, where the co-crystallized ligand obtained an ionic bond of 4.24 Å and the kefiran interacted with two interactions of strong and moderately strong hydrogen with distances of 2.14 Å and 2.85 Å [40]. Thus, it can satisfactorily prevent VEGF activity.

Thus, results found in this study point to kefiran as a pharmacologically important compound that can be used as alternative drug [41] or functional food in the treatment of food allergies. This is mainly due to the high affinity and stability of the receptor-ligand complex observed by evaluating the quality of the docking simulation and the corresponding RMSD and GoldScore values with the kefiran molecule.

4 Conclusion

Based on the *in silico* study, the Kefiran showed predictions of anti-allergic, antioxidant, anti-inflammatory and immunostimulant activities, with emphasis on the last action. These results served as a subsidy for choosing the biological targets used in docking. The results of this coupling indicated that the kefiran molecule can form hydrogen bonds with amino acid residues from the targets HRH1, iNOS, PLA2 and VEGF. Kefiran had good GoldScore values (>70) for all targets studied and had greater affinity with the biological target VEGF, corroborating the results obtained by predicting biological activity, suggesting that the molecule, in this study, can potentially modify its structure and promote the improvement of the immune response, thus preventing the differentiation of T cells in the TH2 profile, which results in a probable inhibitory action in the immune response and the treatment of food allergy.

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Computational Simulation of Molecular Docking and Adme/Tox Analysis of Triazole-Based Kojic Acid Analogues



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Abstract The enzyme Tyrosinase (TYR), a constituent of the biosynthetic steps of melanogenesis, is linked to various skin disorders in individuals. In particular, hyperpigmentation, melasma and skin cancer in the non-melanomic form and malignant melanoma stand out as some dyschromias caused by TYR kinetic changes. The vast majority of skin diseases are treated with substances containing depigmenting agents, which have demonstrated adverse effects on skin melanocytes, such as kojic acid (AK), which over the years has gained prominence in research into TYR inhibitors. As a result, TYR has become an important biological target for the development of new drugs. In this work, new TYR inhibitors were investigated, using AK analogues as a line of study, in this case, the 1,2,3-triazole analogues of AK, using the computational approaches of ADME/Tox, and molecular *docking*. ADME/Tox analyzes showed that AK triazole analogues exhibit drug-like properties, presenting no problematic warnings. Molecular docking calculations showed agreements with experimental data, with distances close to crystallography. All inhibitors presented distances close to 3.65 Å from the hydroxyl group of the AK part based on 1,2,3-triazole in relation to Cu²⁺, with conformations in line with the energy produced by AK in molecular redocking calculations, obtaining a $r^2 = 0.58$ in linear regression with ΔG versus MOLDOCK scoring. Theoretical Chemistry methods allowed validation for studies related to new TYR inhibitors.

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Keywords Tyrosinase · Triazoles · Molecular docking · ADME/Tox

1 Introduction

Melanin belongs to the class of multifunctional biomacromolecules, derived from quinones and phenols, and is found in fauna and flora (insects, fungi, plants, mammals) presenting a remarkable set of physicochemical characteristics [1]. Melanin is a protein responsible for the pigmentation of eyes, hair and skin color, therefore, its functions are to protect deoxyribonucleic acid (DNA) from the harmful action of ultraviolet radiation, thermoregulation, wound healing and elimination of free radicals, preventing interaction with other cellular elements [2–6].

Although it expresses beneficial actions, abnormalities in melanin synthesis are generally caused by the action of enzymes present in its biosynthetic pathway, which constitute one of the reasons why skin complications arise in individuals, including vitiligo, albinism, hyperpigmentation, lentigo, melasma, malignant melanoma and non-melanomic melanoma. In humans, melanins are closely linked to melanoma, a type of skin cancer (malignant and non-melanomic melanoma) with high mortality rates in its malignant form [7]. According to the World Health Organization (WHO), melanoma is in the category of skin disease that causes deaths on a global scale. Around 132 thousand cases of malignant melanoma are recorded annually, with an estimated 66 thousand deaths [8].

The National Cancer Institute (INCA) states that between 2023 and 2025 around 177 thousand cases of melanoma will be registered, therefore, taking into account that neoplastic cells are generally incurable, the stage at which carcinogenesis is found defines the strategy adopted to treatment, which is mainly done to alleviate symptoms and improve the individual's quality of life [9]. In Brazil, there is a 30% rate of malignant skin cancer, with non-melanoma cases being recurrent, which does not occasionally affect children and the black population, with a history of cases in the white population, redheads and blondes [9].

Melanoma in its non-melanomic form, when diagnosed early, is not lethal, but its treatment is painful as it affects areas of the body (neck, ears, etc.) that cause mutilation [9, 10]. Skin problems are treated with depigmenting substances, which often demonstrate toxicity to the skin, such as irritation of the patient's skin [11, 12]. Therefore, in recent years there has been an increasing need to develop new therapeutic substances that are not as aggressive for patients with this disease.

In this context, the melanogenesis process occurs through a specialized cellular pathway, represented by a series of enzymatic and chemical reactions, among which the enzyme tyrosinase (TYR) stands out [13]. The TYR enzyme appears as one of the essential sources for the steps of this synthesis, therefore, initially the amino acid L-tyrosine (L-Tyr) undergoes hydroxylation to L3,4-dihydroxyphenylalanine (L-DOPA) which in the presence of TYR is oxidized to DOPAquinone [3, 14]. Therefore, melanin is produced by the polymerization of DOPAquinones [15, 16].

In this sense, as it has great biological relevance, the TYR enzyme has been used as a target for studies with the aim of producing anti-skin cancer drugs [17–19]. Thus, kojic acid (AK) is an inhibitor that in its nature has the chelation of transition metal ions, as well as Cu^{2+} ions in the TYR structure [6]. However, AK showed low activity in tests, which resulted in studies using AK analogues [18, 20].

This work used molecular modeling methodologies such as ADME/Tox analysis and molecular docking to investigate TYR inhibitors, particularly the 1,2,3-triazole analogues of AK. Through modeling methodologies, it was possible to predict the binding mode between inhibitors and TYR, as well as allowing the recognition of the binding of these inhibitors, through the identification of the thermodynamic properties of these complexes. Aiming at the production of AK-based inhibitors with low toxicity, Theoretical Chemistry procedures have been widely used in the planning of new medicines, as they are tools used to maximize the effectiveness of medicines already known by the pharmaceutical industry [17, 19, 21, 22]

2 Material and Methods

2.1 Disorders in Melanogenesis: Melanoma

Melanogenesis occurs in all organisms and was first elucidated by Raper (1928) and recently by Cooksey et al. (1997) and Schallreuter et al. (2008), its starting point is the oxidation of the amino acid L-tyrosine, which, with a specialized reaction mechanism, has the TYR enzyme acting as a key point in this system [3]. In general, this biochemical process is responsible for the biosynthesis of melanin, a protein responsible for the pigmentation of organisms present in fauna and flora, which occurs in the basal layers of the skin, in melanocytes [13, 23]. Although in nature different functions are attributed (plants, animals, microorganisms, fungi), in humans the function of melanin is to protect the epidermis from radiation, absorbing ultraviolet rays and eliminating these free radicals, thus protecting DNA cells [5].

Biological, genetic and external factors are related to the melanogenesis pathway, the latter is closely linked to disorders of the catalytic pathway due to the overproduction of the TYR enzyme, which is one of the essential steps for the synthesis of melanin, in addition to producing it in an accelerated manner. The proteins TYRP1 and TYRP2 are biomarkers of dyschromia, with TYRP2 being related to melanoma. In this way, malignant melanoma appears in the melanocytes of the epidermis, manifesting itself from the pigmented region or healthy skin, in these cases respectively, the signs are perceived through the appearance of dark lesions, accompanied by itching and flaking and in case of pre-existence of dark spots on the skin, they increase in size presenting irregular shapes and color changes, when not diagnosed in time, treatment is carried out through surgery, however, when there is metastasis it is irreversible [9, 24].

In recent decades, cases of this malignant neoplasm have increased significantly, with the globalized world making it possible to quantify the percentages over the years. According to the latest update from the World Health Organization in 2020, the number of deaths from malignant melanoma has increased significantly, reaching 132 thousand cases annually, with a death toll of 66 thousand. The incidence of cases increases in countries where the culture of melanoma Tanning is increasing: Australia, New Zealand, North American countries and northern Europe [10]. In Brazil, according to the most recent data, it is assumed that by 2029 there will be 600 thousand cases of cancer, of which 170 thousand are non-melanoma skin cancer [9].

According to Carvalho (2004), many cases of skin cancer are linked to hereditary factors, as well as external factors. Similarly, cases of melanoma are caused by genetic and external factors, in terms of exposure to UV and UV-B radiation [10]. Statistically, cases of malignant tumors account for 3% of cases in organs and 30% of the frequency in the country, presenting a total of 1,978 deaths among men and women, in this sense, prolonged exposure to artificial tanning, as well as lifelong exposure to UV rays, being part of risk groups (white people, redheads, light-eyed people) and having a family history of skin cancer increase the risk of developing the disease [9]. Changes in certain daily sun exposure habits and adoption of the use of adequate protection (shade, clothing, sunscreen) directly contribute to the prevention of skin cancer [9].

Therefore, understanding the functioning mechanisms of the organism, in this case, the functioning of cellular communications and the action of enzymes in the biochemical course of melanin formation, is the importance of new studies to combat these pathologies. From this perspective, many scientific studies regarding the TYR enzyme have been carried out with the aim of developing new drugs to combat malignant melanoma [3, 23, 25–28].

2.2 A Multifunctional Glycoprotein: Tyrosinase Enzyme (TYR)

Tyrosinases (TYRs) or polyphenol oxidase are glycoproteins, that is, they are metalloenzymes found in animal and plant tissues (mammals, mushrooms and plants), they have two Cu^{2+} ions in their active site (Fig. 1) [29, 30]. The well-characterized TYRs originate from *Streptomyces glaucescens*, the fungi *Neurospora crassa* and *Agaricus Bisporus*, which catalyze the initial formation of melanin, as well as the TYR enzyme taken from the mushroom *A. Bisporus*, which is analogous to mammalian TYR, is widely used in assays related to melanogenesis. An interesting feature of TYRs from different sources is that their catalytic Cu^{2+} -binding site and amino acid residues are strictly conserved [3]. As it is a cupric-protein enzyme complex, TYR is synthesized in ribosomes and transferred through the endoplasmic reticulum to the Golgi apparatus [30].

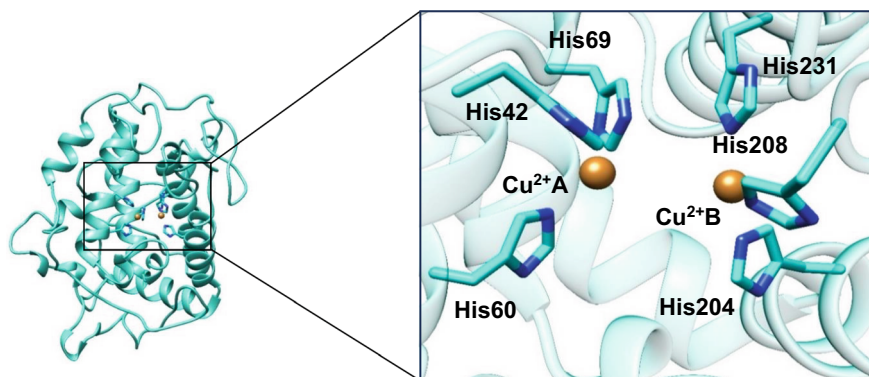


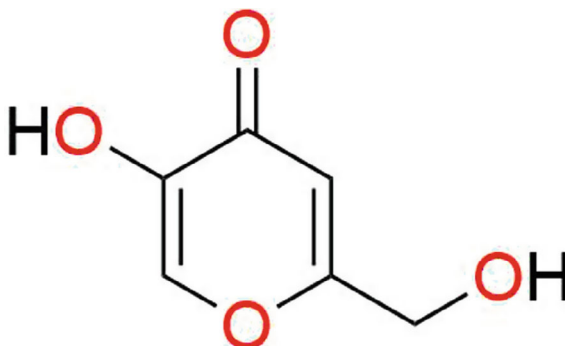
Fig. 1 Tertiary structure of TYRBm with Cu²⁺ ions (cofactors) coordinated by histidine residues present in the active site of the enzyme; Connection location

2.3 Chemical Compound of Natural Origin with Potential Activity Against Melanoma: Kojic Acid

AK is a metabolite found in fungal or bacterial species, which acts as a chelator of transition metal ions (Cu²⁺, Fe²⁺ and Zn²⁺) and is responsible for eliminating free radicals [2]. It is constantly used as a skin lightener and food additive [31]. AK is a hydroxy- γ -pyrone, which is a heterocyclic compound that has an unsaturated six-membered ring containing an oxygen atom and a ketone group, its IUPAC nomenclature is 5-hydroxy-2-(hydroxymethyl)-4-pyrone and molecular formula C₆H₆O₄, as well as its pK_a is 7.9 to 8.03 (Fig. 2) [32]. Acting in two steps, the first in the inhibition of tyrosinase, and the second, in the mixed form of difelonase, which demonstrates that the molecule has affinity with the active site of the enzyme [33]. Thus, it assumes antitumor and photoprotective functions [34, 35].

Despite demonstrating a good standard in experimental measurements, Ashooriha et al. (2020) clarify that the acid is a compound widely used among skin lightening

Fig. 2 2D structure of kojic acid (AK)



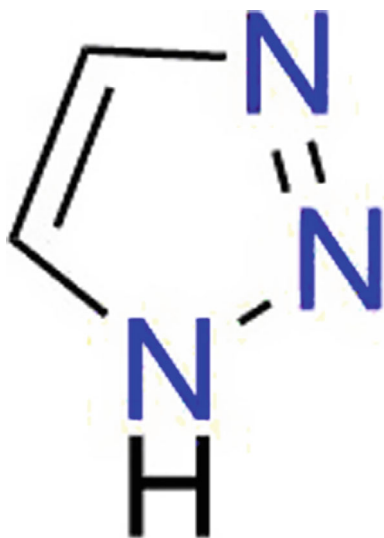
agents, but with low efficacy and stability in the formulation, in addition to insufficient penetration into the skin, there are also reports of highly sensitive effects, in which some in some cases, may cause irritant contact dermatitis. Therefore, due to the side effects caused by the high concentration levels of AK required to achieve results, studies are being carried out to produce potential AK-derived inhibitors with good biological results, mainly with potent antityrosinase activity, better efficacy and profile. safe and with few side effects [6, 36].

2.4 Click Reaction: 1,2,3-Triazole AK Analogues a Pharmacological Alternative

Triazoles (Fig. 3) are heteroaromatic substances, having six π electrons, species that do not present substituents on the nitrogen group, have three tautomeric structures, in addition to being composed of a five-member ring of two carbon atoms and three nitrogen atoms, are part of a class of substances generically called azoles [37]. 1,2,3-Triazole is of exclusively synthetic origin, being obtained by a 27-click reaction, which allows the stability and stereoselectivity of the molecule under biological conditions [26, 37].

As AK has the ability to chelate with Cu^{2+} ions, when it is in the active site of TYR, as it is structurally similar to the L-Tyr substrate, AK takes over the action of the amino acid in the melanogenic pathway and continues the melanogenic pathway. catalytic cycle, forming a chelate complex with the enzyme's cofactors. Ashooriha (2019) and his collaborators in the process of designing the compounds, preserved

Fig. 3 2D structure of Triazole proposed by ASHOORIHA and his collaborators



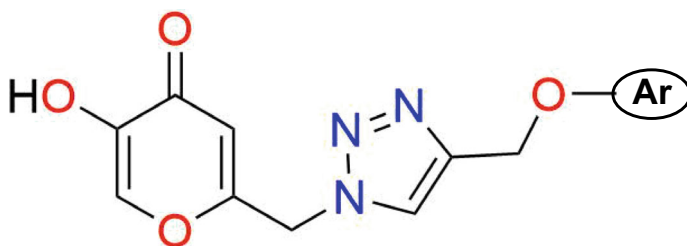


Fig. 4 2D structure of the compound proposed by ASHOORIHA and his collaborators

the structure of AK, replacing the hydroxyl group of AK with the 1,2,3-triazole structure (Fig. 4) [26].

2.5 Molecular Modeling in Drug Design

Transformations on a global scale have brought about the need to improve new forms of scientific investigation, ranging from virtual screening studies to genetic validation studies of medicines. Thus, research methods for medication planning have also undergone significant transformations [38]. Theoretical Chemistry participated directly in the technological revolution, with the formulation of means of data collection, with the creation of databases of proteins, inhibitors, providing the storage of information and allowing the monitoring of analyzes for the creation of new, more effective medicines, an example of this is the Protein Data Bank (PDB), responsible for collecting essential information on several existing chemical species [38, 39]. Molecular modeling develops from existing biological complexes, with the creation and manipulation of systems, creating approximate simulations of real systems, evaluating experimental and theoretical procedures. These tests constitute a set of mathematical and chemical algorithms, which are used to evaluate the results obtained in the constructed models. This allows us to understand the interactions of molecules that are biological targets [39].

2.6 Molecular Docking

Many commercial and free software are used for molecular docking simulations, such as AutoDock Vina [40], Molegro Virtual Docker (MVD) [41]. As it presented satisfactory results⁴⁷ in previous studies regarding TYR complexes, Molegro Virtual Docker (MVD) version 5.5 was used for the simulations carried out in this work.

What is the structure of tyrosinase taken from the database (5I38) [42] presents the AK inhibitor within its active site, the redocking calculation was performed as a starting point to analyze and define the behavior of the system within the

active site, identifying the structure's binding mode and its main molecular interactions. Redocking calculations for the crystallographic structure containing AK in its active site were performed using the MolDock Score function, a search algorithm implemented in MVD [41, 43].

Molecular docking simulations for the 1,2,3-triazoles analogous to AK (6A-6P) (Table 1) were performed based on AK redocking, in order to superimpose the crystallographic AK present in the active site of TYR, achieving a better comparison between molecules. Furthermore, for all anchorings carried out, the MVD automatically parameterized and treated the copper ions as spheres with a van der Waals radius, with their interactions being more spherical than electrostatic, preventing the ligands from occupying the spaces reserved for metallic ions present in the TYR.

2.7 *In Silico Modeling: ADME/Tox*

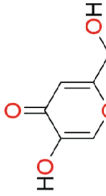
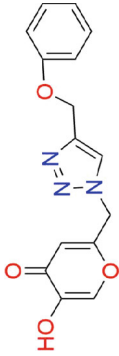
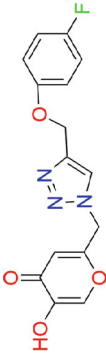
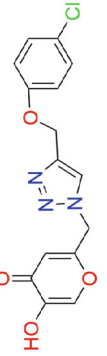
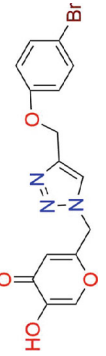
Using the free **SwissADME** web tool, which is integrated into the Swiss Drug Design workspace, among the various tools developed by the Molecular Modeling Group of the SIB Swiss Institute of Bioinformatics, it was possible to quickly but robustly determine the physical properties—chemistry, lipophilicity, pharmacokinetics, drug similarity and medicinal chemistry of AK and the inhibitors used in this study [44].

The properties found were: lipophilicity, size, polarity, solubility, flexibility and saturation, which are part of the Bioavailability Radar; Molecular and physico-chemical descriptors such as molecular weight; Solubility in water; Pharmacokinetic predictions, such as passive human gastrointestinal absorption and blood–brain barrier permeation; Qualitative assessment of similarity with the medicine; And finally, the medicinal chemistry of the compound evaluating the possibility of potentially problematic fragments [44, 45].

To complement the ADME/Tox study, toxicological predictions were carried out using the Lhasa company's **Derek** program. The program makes comparisons between the structures provided and the database of compounds or parts thereof that already have toxicological definitions established in analyzes by partner laboratories. The endpoints present toxicological filters for teratogenicity, carcinogenicity, genotoxicity, irritation, organic toxicity, neurotoxicity, reproductive toxicity, respiratory sensitization and skin sensitization. The warnings of teratogenicity, carcinogenicity, genotoxicity and skin sensitization were evaluated as priority due to the severity of toxicity and the intention to treat melanoma.

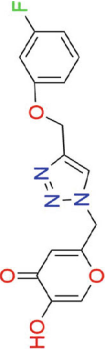
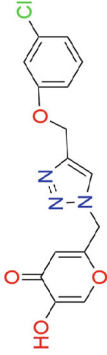
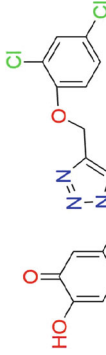
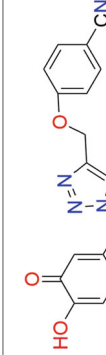
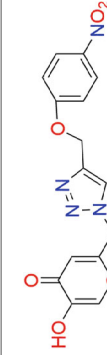
Additionally, the **Derek** program presents forecast alerts in ascending order of likely toxicities, such as: Impossible; Unlikely; Doubt; Mistaken; Plausible; Likely; Right. And for this study, predictions were evaluated that could be plausible, probable or certain as toxicity warnings for the compounds.

Table 1 Inhibitors studied in this work and their respective information

Ligands	IUPAC nomenclature	2D Structure	IC ₅₀ (μM)
AK	5-Hydroxy-2-(hydroxymethyl)-4H-pyran-4-one		9.28
6A	5-Hydroxy-2-(phenoxymethyl)-1H-(1,2,3-triazol-1-yl)methyl)-4H-pyran-4-one		1.33
6B	2-((4-(4-Fluorophenoxy)methyl)-1H-1,2,3-triazol-1-yl)methyl)-5-Hydroxy-4H-pyran-4-one		0.88
6C	2-((4-(4-chlorophenoxy)methyl)-1H-1,2,3-triazol-1-yl)methyl)-5-hydroxy-4H-pyran-4-one		0.69
6D	2-((4-(4-Bromophenoxy)methyl)-1H-1,2,3-triazol-1-yl)methyl)-5-hydroxy-4H-pyran-4-one		6.80

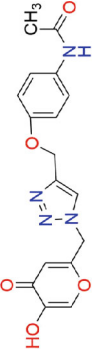
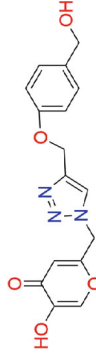
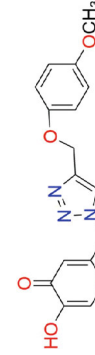
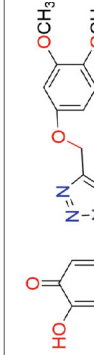
(continued)

Table 1 (continued)

Ligands	IUPAC nomenclature	2D Structure	IC ₅₀ (μM)
6E	2-((4-((3-Fluorophenoxy) methyl) -1H-1,2,3-triazol-1-yl) methyl) -5hydroxy-4H-pyran-4-one		1.07
6F	2-((4-((3-Chlorophenoxy) methyl)-1H-1,2,3-triazol-1-yl) methyl) -5hydroxy-4H-pyran-4-one		0.99
6G	2-((4-((2,4-Dichlorophenoxy) methyl) -1H-1,2,3-triazol-1-yl) methyl) -5-hydroxy-4H-pyran-4-one		1.12
6H	((1-((5-Hydroxy-4-oxo-4H-pyran-2-yl) methyl)-1H-1,2,3-triazol-4-yl) methoxy) benzonitrile		6.29
6I	5-Hydroxy-2-((4-((4-nitrophenoxy) methyl)-1H-1,2,3-triazol-1-yl) methyl) -4H-pyran-4-one		0.52

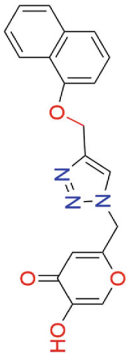
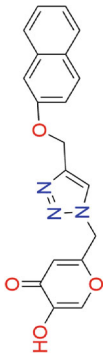
(continued)

Table 1 (continued)

Ligands	IUPAC nomenclature	2D Structure	IC ₅₀ (μM)
6J	N-(4-((1-(5-hydroxy-4-oxo-4H-pyran-2-yl)methyl)-1H-1,2,3-triazol-4-yl)methoxy)phenyl) acetamide		2.64
6K	5-hydroxy-2-((4-((4-(hydroxymethyl)phenoxy)methyl)-1H-1,2,3-triazol-1-yl)methyl)-4H-pyran-4-one		1.32
6L	5-hydroxy-2-((4-((4-methoxyphenoxy)methyl)-1H-1,2,3-triazol-1-yl)methyl)-4H-pyran-4-one		1.24
6M	5-hydroxy-2-((4-((3,4,5-trimethoxyphenoxy)methyl)-1H-1,2,3-triazol-1-yl)methyl)-4H-pyran-4-one		0.87
6N	5-Hydroxy-2-((4-((naphthalen-1-yl)oxy)methyl)-1H-1,2,3-triazol-1-yl)methyl)-4H-pyran-4-one		0.74

(continued)

Table 1 (continued)

Ligands	IUPAC nomenclature	2D Structure	IC ₅₀ (μM)
6O	5-Hydroxy-2-((4-((naphthalen-1-yloxy)methyl)-1H-1,2,3-triazol-1-yl)methyl)-4H-pyran-4-one		0.06
6P	5-hydroxy-2-((4-((naphthalen-2-yloxy)methyl)-1H-1,2,3-triazol-1-yl)methyl)-4H-pyran-4-one		0.30

Source Adapted from (Ashooriha et al. 2019)

3 Results and Discussions

3.1 Molecular Docking Calculations

Considering the technical procedures used to simulate molecular anchoring, the crystal structure used in this study was taken from the PDB, under code 5I38 [42]. To evaluate the simulation protocols, Kojic Acid (AK) underwent a *redocking* routine in the active site of the enzyme. According to molecular redocking procedures, the inhibitor binds to the active site of the TYR enzyme and develops interactions with Cu^{2+} ions, as well as the obtained conformations demonstrated agreement with experimental data, that is, values presented by mean square deviation (RMSD). Thus, the selection of the conformation started from the evaluation of the energy data generated by the MOLDOCK Score function, where the most negative value was chosen and the conformation of AK within the protein was aligned with the experimental AK.

The calculated values for AK were respectively -72.55 kcal/mol for MOLDOCK and 0.14 Å for RMSD, resulting in an average deviation close to zero, which demonstrates that the docking results are in agreement with previous studies [17, 18]. When evaluating the results of molecular redocking for the crystallographic inhibitor, it was possible to notice that the oxygen atom of the hydroxyl group of AK is positioned towards His56 with a distance of 2.36 Å, making a hydrogen bond with the nitrogen of the His56 ring. His56 (Fig. 5a). And the AK system is stabilized by π - π interactions in the active site of the enzyme with the His204 residue ring, the AK formed a distance with the Cu^{2+}B ion of 3.21 Å (Fig. 5b). The carbonyl group present in the inhibitor structure is oriented towards Cu^{2+}A at a distance of 4.80 Å.

In Fig. 6 it is possible to observe the TYR and AK cavity inside the hydrophobic surface of the enzyme, where the copper ions are arranged in relation to AK. It is evident that the cavity offers adequate space for AK to bind with greater dexterity, in

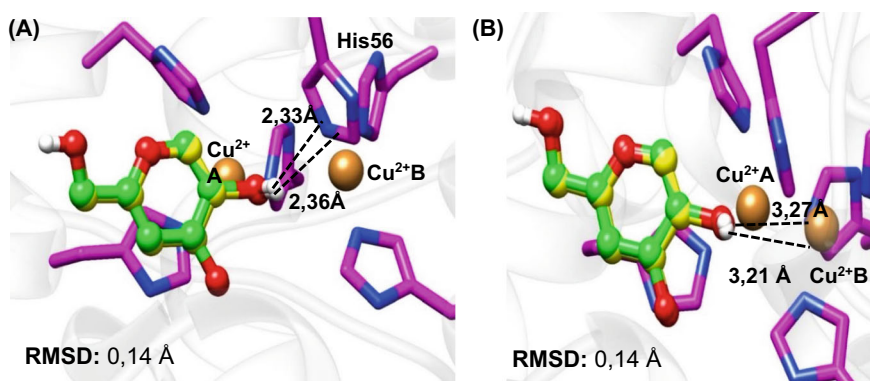


Fig. 5 a Overlay of crystallographic AK (green) and AK redocking (yellow), demonstrating the binding mode of AK in relation to His56; b distance between the crystallographic AK (green) and the redocking AK (yellow) in relation to the Cu^{2+}B ion

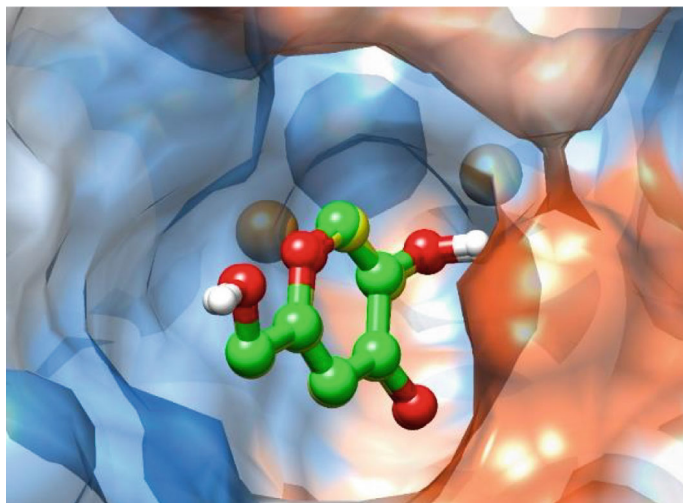


Fig. 6 TYR cavity like AK on the hydrophobic surface with the presence of cofactors. Histidine residues and hydrogens were hidden for better visualization

addition to its similarity to L-Tyr, which allows it to bind more easily to the protein [26].

As the molecular *redocking* results validated the binding mode of the crystallographic structure and showed cohesion with the experimental data [17, 18], the results for AK were adopted as parameters for carrying out molecular docking for AK analogues.

Molecular *docking* for the 1,2,3-triazole analogues of AK demonstrated that, like the reference structure, the inhibitors maintain the hydrogen bond of the –OH group present in the portion of AK with the nitrogen of His56 and produce a distance of the Cu^{2+} B Ion of about 3.30 Å. The selection of conformations was carried out based on the energy values that were calculated by the MOLDOCK Score function, and was based on the alignment that the AK analogues demonstrated with the experimental AK. For more accurate results on how the inhibitors behave in relation to TYR, MD simulations and LIE calculations were performed.

Table 2 presents the MOLDOCK values. It is possible to note that the energies vary according to the molecular structure of each inhibitor.

All analogous AK inhibitors showed interactions with amino acid residues (Arg205, Asn201, Gly196, Met180, Phe193 and Pro197) present in the TYR cavity. In relation to Cu^{2+} ions, it was observed that it did not interact with all inhibitors, making its energetic contribution only a few inhibitors exist. The TYR-1,2,3-triazole systems are stabilized by hydrogen bonds and hydrophobic interactions, with their main differences reflected in the values presented by the MOLDOCK score. Tables 3 and 4 show the main interactions that amino acid-TYR residues carry out in the 1,2,3-triazole-TYR system.

Table 2 MOLDOCK score values (kcal/mol) for the inhibitors and the main interactions of the inhibitors with the TYR cavity

Inhibitors	Moldock
6A	−127.9
6B	−135.9
6C	−123.7
6D	−124
6E	−121.7
6F	−124.8
6G	−136.4
6H	−139.06
6I	−129.9
6J	−135.60
6K	−134.2
6L	−125.93
6M	−120.5
6N	−130.17
6O	−130.15
6P	132.3

Table 3 Hydrogen bonds made by inhibitors during molecular anchoring calculations

Inhibitors	Amino acids	Hydrogen bonding
6A	Phe193 Arg205	F–H–C C–H–N
6B	His56 Arg205 Pro197 Gly196	N–H–O C–H–N HN ₂ F–H–C F–H–C
6C	His56 Pro197	N–H–O C ₆ H ₆ –N ₂ H ₂ –Cl–C
6D	His56 Arg205 Gly196	N–H–O C–H–N Br–H–N ₂ H ₂
6E	His56 Pro197 Phe193 Gly196	N–H–O HN ₂ –F–C H–F–C H–O–C
6F	His56 Gly196	N–H–O H–O–C
6G	His56 Gly196	N–H–O H–O–C
6H	His56 Arg205 Pro197	N–H–O C–H–N HN ₂ –H–O
6I	His56 Pro197	N–H–O C–H–NH ₂
6J	His56	N–H–O
6K	His56 Phe193 Pro197	N–H–O C=O–H–O N–H–C
6L	His56 Pro197 Phe193 Gly196	N–H–O H–O–C C=O–H–O N–H
6M	His56 Pro197 Phe193 Gly196	N–H–O C–H–N O–H–C H–O–C
6N	His56 Arg205 Pro197	N–H–O C–H–O C–H–O
6O	His56 Met180 Gly196	N–H–O O–H–C H–N–H–O
6P	His56 Phe193 Pro197	N–H–O H–O–C N ₂ H–O–C

Source Data extracted from the research

Table 4 Interactions and distances presented by the inhibitors in relation to the amino acid residues present in the TYR cavity

Inhibitors	Residue	Distances (Å)	Interactions
6A	Gly196 Phe193 Pro197 Arg205	4,7 4,3 3,9 3,3 2,0	Hydrophobic
6B	Gly196 Arg205 Pro197	4,4 3,2 3,3	Hydrophobic Peptide bond with Pro197
6C	Pro197 Phe193	3,4 2,3 3,6 4,5	Carbon-halogen bond with Pro197 Hydrophobic
6D	Phe193 Gly196	3,3 4,8 5,5	Hydrophobic with Phe193; hydrophilic with Gly196
6E	Gly196 Pro197 Phe193	5,7 4,0 4,3	Carbon-halogen bond between Pro197 and Fluorine Hydrophobic (Phe193) e hidrofílicas (Gly196)
6F	Gly196 Phe193 Pro197	6,2 4,1 4,2	Carbon-halogen bond between Phe193 and Cl
6G	Phe193 Pro197	3,6 e 4,0	Carbon-halogen bond and the carbon ring Phe193 and Br
6H	Cu ²⁺ A Arg205 Pro197 Phe193	4,9 3,3 2,6 3,5 e 4,4	Peptide bond with the Phe193 ring; H bond with Arg205; hydrophobic with Pro197
6I	Pro197	2,1 2,5	Hydrophobic
6J	Cu ²⁺ A	3,0 4,8	Electrostatics between Cu–O
6K	Phe193 Pro197 Arg205	3,3 2,0 3,0	Hydrophobic (Phe193) and Hydrophilic (Pro197); H bond with the carbonyl of Phe193
6L	Cu ²⁺ A Pro197 Phe193 Gly196	4,8 2,8 1,7 2,5 4,7	Electrostatics between Cu–O Hydrophobic Hydrophilic
6M	Cu ²⁺ A Pro197 Phe193 Gly196	5,0 3,3 1,7 2,9 4,5	Hydrophilic (Pro197) and Hydrophobic
6N	Gly196 Pro197 Arg205	6,3 2,7 3,6	Hydrophilic Hydrophobic Polarity with Arg205
6O	Met180 Pro197 Phe193	3,8 e 4,1 1,9 2,7	Hydrophobic
6P	Phe193 Pro197 Gly196	2,3 2,9 6,9	Hydrophobic

Source Data extracted from the research

The inhibitor 6A interacts with four amino acid residues of a hydrophobic nature, with Phe193 being responsible for the dipole–dipole interaction with the triazole ring. What differentiates 6A from 6B (Fig. 7) is that the 6A inhibitor has Fluorine in its structure, which makes the region of the molecule more electronegative, enabling interactions with hydrophobic amino acids (Gly196 and Pro197) in this region of the inhibitor. Arg205 involves the AK ring region and the triazole ring of 6B, 6C and 6D (Fig. 7), a region close to the TYR cofactors. In relation to the 6C inhibitor, it performs intermolecular interactions with Phe193, this interaction influences the energy value (−135.9) and IC₅₀ (0.69) that the structure presents.

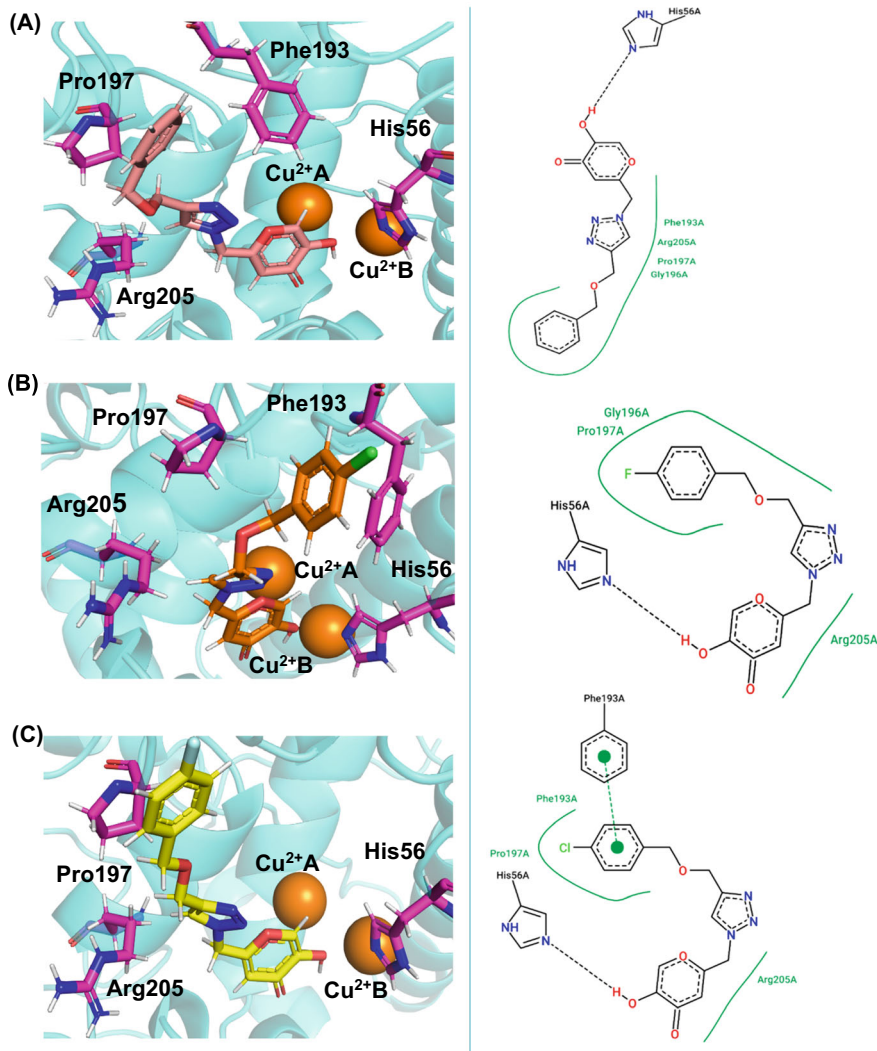


Fig. 7 Inhibitors 6A **a**, 6B **b**, 6C **c**, 6D **d** and hydrogen bonds and hydrophobic interactions with amino acid residues within the active site of TYR

The MOLDOCK values (Table 2) are related to the presence of halogens in the molecular structure of the inhibitors (6B, 6C, 6D, 6E, 6F and 6G), as they present variations in inhibition, as represented in the IC_{50} (0.88–0.69–6.80–1.07–0.99–1.12) in terms of energy, which is linked to electronegativity and the position (ortho, meta and para) that each atom (Cl, F, Br) occupies in the inhibitor structure (Fig. 8).

Systems 6J, 6L, 6M (Fig. 9) have distances of 3.0 Å from Cu^{2+} A. Acetamide (6J), despite presenting greater electronegativity compared to the other inhibitors

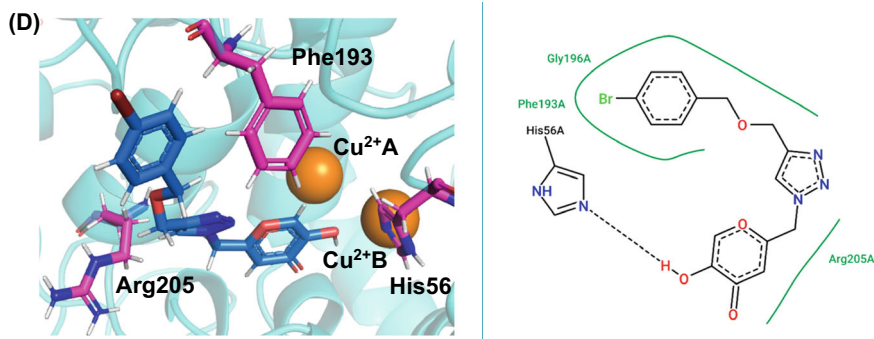


Fig. 7 (continued)

(6L and 6M), and interacting only with the Cu²⁺ A ion and His56, resulted in a higher MOLDOCK score compared to the other inhibitors, interactions with the amino acid residues that the 6L and 6M inhibitors carry out are factors that influence the different energies.

Hydroxymethyl showed that the 6 K inhibitor expresses an effective inhibitory activity and that the hydroxyl group interferes with affinity and this interaction is reflected in the energy calculated by MVD in relation to the 6N and 6P inhibitors. 3,5-dimethyl (6N) presented a good MOLDOCK score, interactions with Arg205, Pro197 and Gly196 are related to the energetic value that the inhibitor presents, just as the IC₅₀ shows that it has good inhibitory activity (Fig. 10).

The 6P inhibitor expresses a second more potent activity (IC₅₀), Gly196, Pro197 and Phe193 surround the β -naphthyl ring on a hydrophobic surface, which is reflected in the energy calculated by MVD.

The calculated value for MOLDOCK of 6H (−139.06 kcal/mol) and the IC₅₀ value (6.29) show that the cyano group influences the low affinity of the inhibitor in relation to TYR, as well as the methoxy group linked to the 6I showed that CH₃O interferes with the high affinity in the TYR-6I system, the calculated energy (−129.9 kcal/mol) and IC₅₀ (0.52 kcal/mol) validate the activity in the system. 6O is the inhibitor that demonstrates the greatest activity (IC₅₀) and in biological assays it showed much higher activity than AK [26], the direct interaction between the Phe193 ring and the α -naphthyl ring is responsible for the good energy calculated for the MolDock score (Fig. 11).

The molecular docking results demonstrated that the ligand 6O has higher affinity (IC₅₀), while 6H has low affinity (IC₅₀) for the enzyme, both presented coherent conformations within the TYR active site, in accordance with crystallographic AK (Fig. 12).

The MOLDOCK score presented satisfactory results in relation to the prediction of the best conformations and prediction of the bonds between the TYR targets, showing that the interactions related to the 1,2,3-triazole analogues of AK were close to the crystal inhibitor, demonstrating that the results were consistent with previous

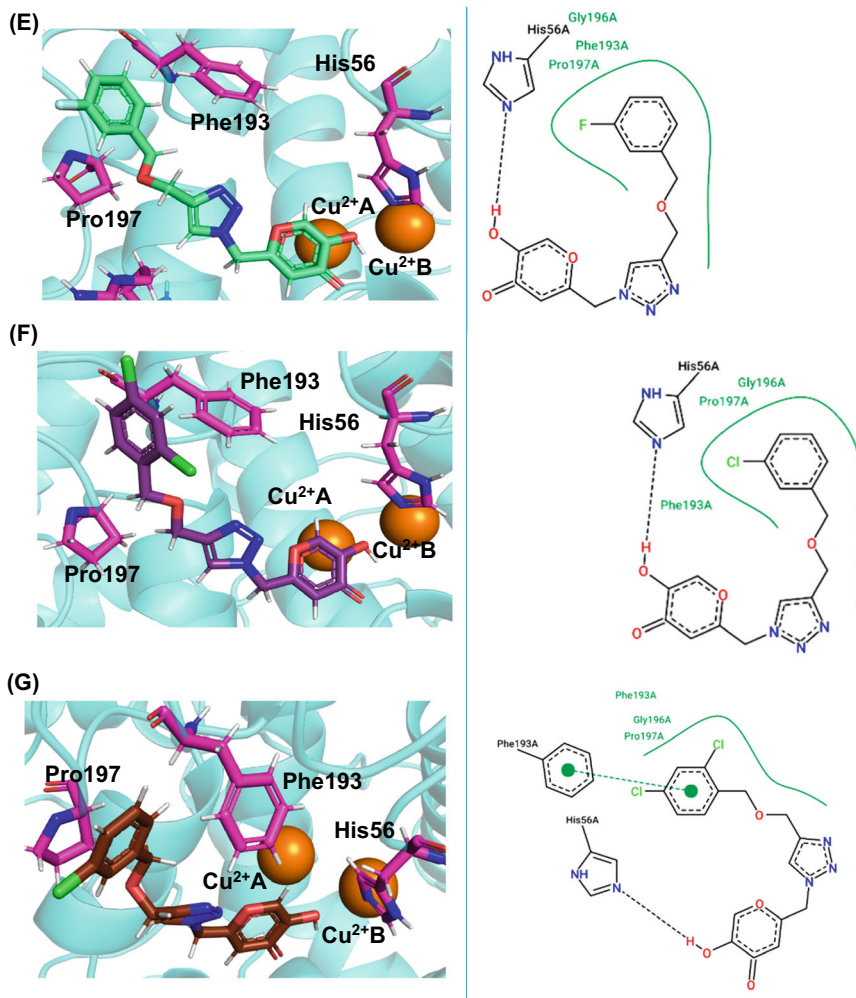


Fig. 8 Inhibitors 6E **e**, 6F **f**, 6G **g** and hydrogen bonds and hydrophobic interactions with amino acid residues within the TYR active site

studies [17, 18]. When performing an analysis of $\Delta\text{G}_{\text{EXP}}$ versus MOLDOCK score, a slight positive correlation was obtained with $r^2 = 0.58$, which means that there is a relationship between the energies obtained during the molecular docking calculations and the experimental data (Fig. 13).

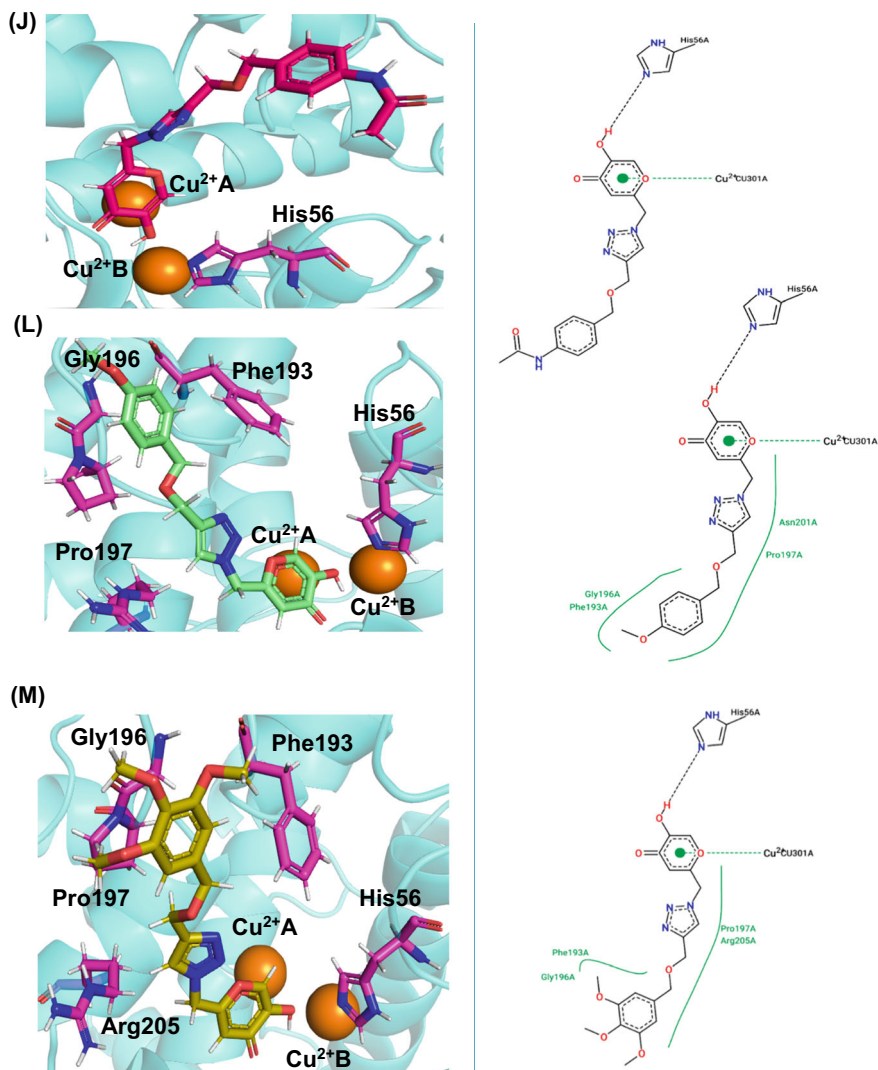


Fig. 9 Inhibitors 6J **j**, 6L **l** and 6M **m** and their interactions within the TYR active site

3.2 ADME/Tox Analysis

According to the properties resulting from the free web tool, **SwissADME**, it is worth highlighting the physicochemical characteristics of AK in which the compound shows good similarity with medicines, since most of the coordinates of the inhibitor's bioavailability radar graph are entirely within of the ideal range (Fig. 14), except for establishment. With lipophilicity values of -0.64 (the range of $\log Po/w$ (XLOGP3)

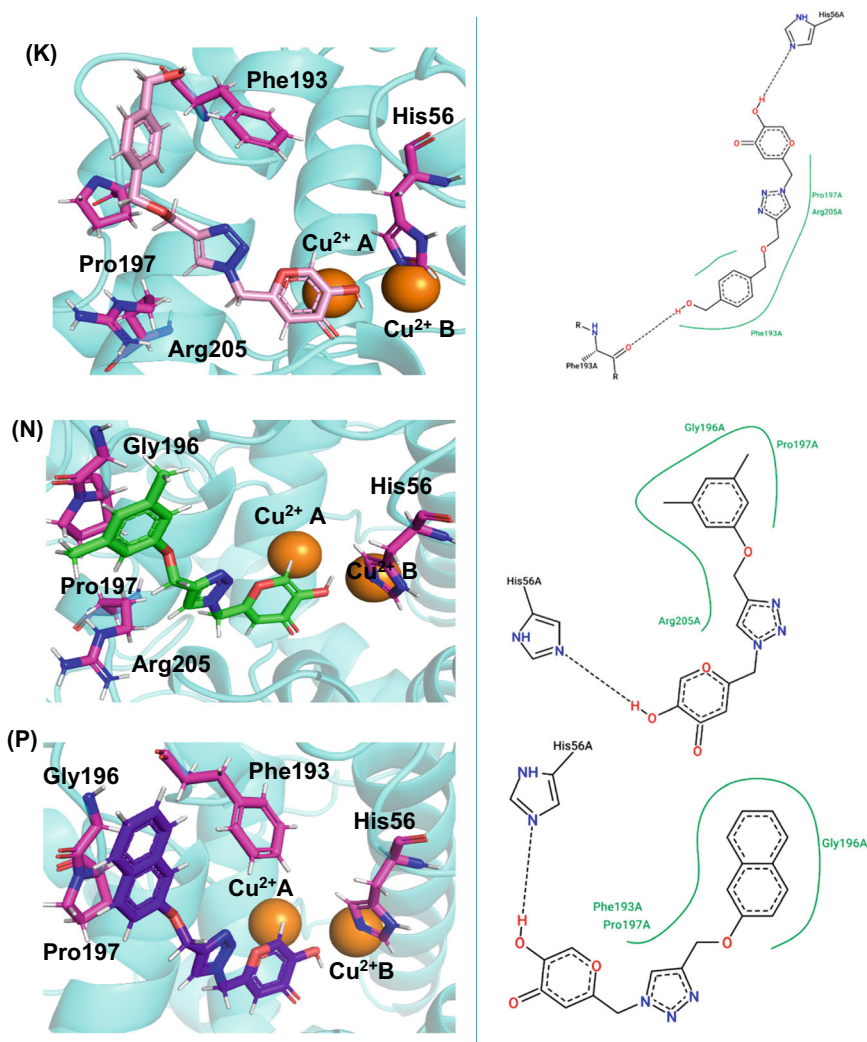


Fig. 10 Inhibitors 6 **K**, 6 **N** and 6 **P** and its main interactions within the TYR active site

is -0.7 and $+5$) which represents the pharmacokinetic capacity of the compound within the expected range. However, the size or molecular weight of 142.11 g/mol (range between 150 and 500 g/mol) was below the desired minimum, the polarity of 70.67 Å (TPSA between 20 and 130 Å) was within the range, solubility equal to -0.37 (log S not exceeding 6) being considered very soluble in water, flexibility with only 1 rotational bond (not having more than 9 bonds) and saturation with difference in properties shows that the compound is very unsaturated [44].

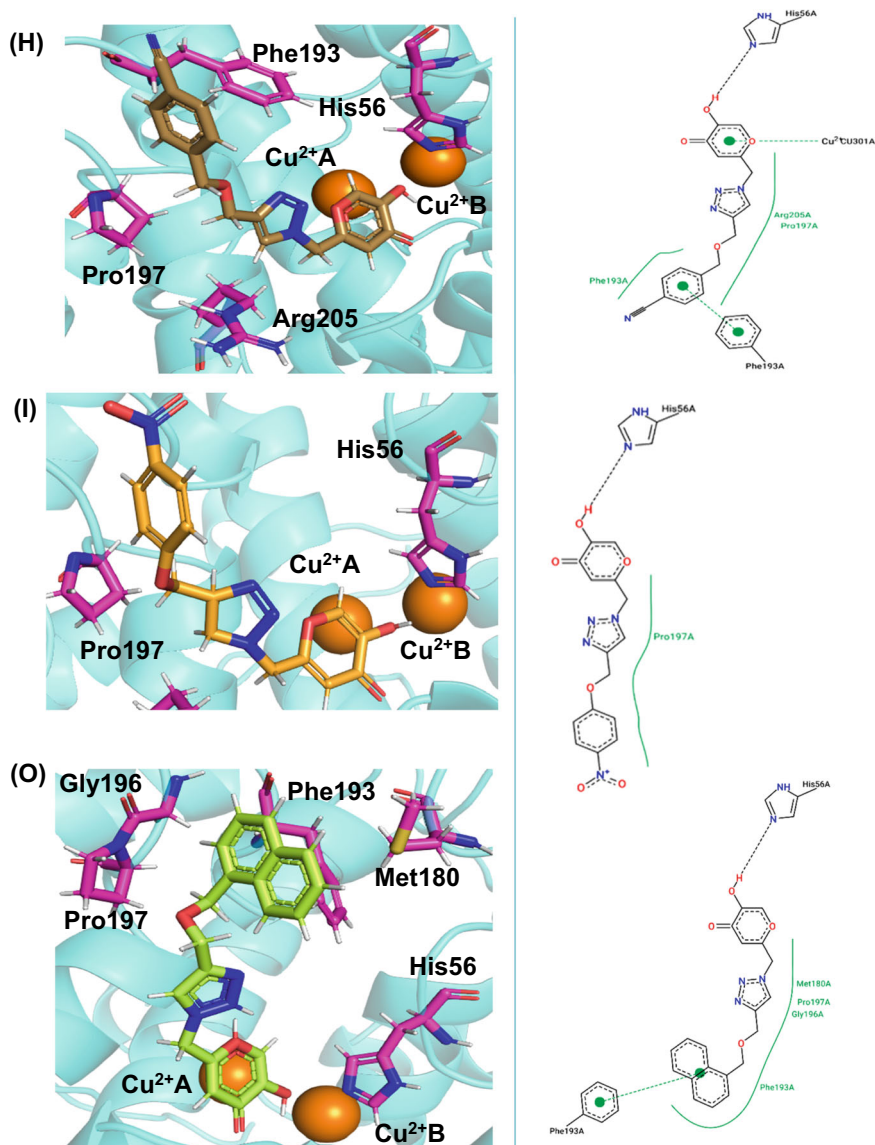


Fig. 11 Main interactions of inhibitors 6H **h**, 6I **i** and 6O **o** in the active site of TYR

The pharmacokinetic predictions described by the program demonstrate that AK has a high level of human gastrointestinal absorption (HIA), which does not necessarily influence the essential characteristics to be considered for the use of the acid in skin treatment. Therefore, the skin permeability result ($\log K_p$) of -7.62 cm/s indicates a low permeability index of the inhibitor, as the more negative the K_p value,

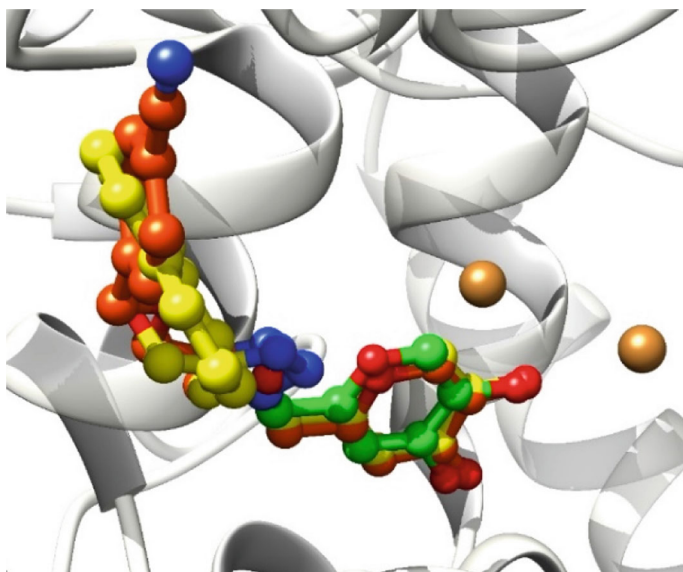


Fig. 12 3D overlay of AK (green), 6H (orange) and 6O (yellow) in the TYR active site by molecular anchoring calculation. Cu^{2+} ions are described as brown spheres. H and other residues were hidden for better visualization

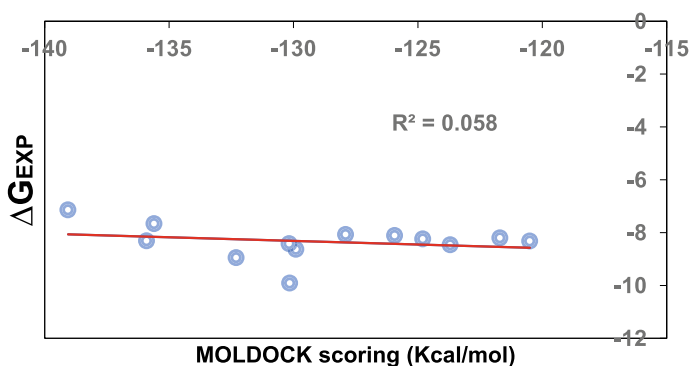
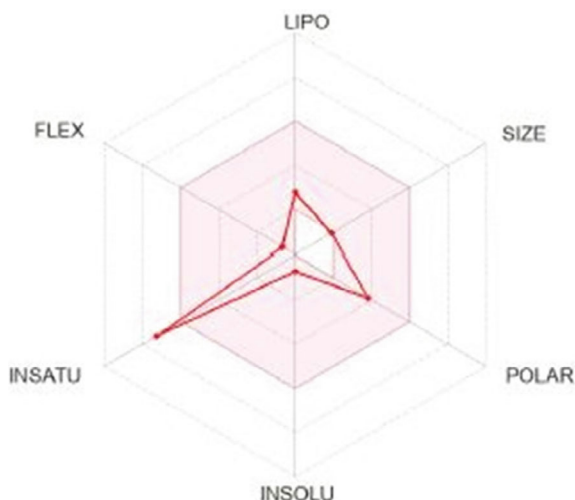


Fig. 13 Regression plot between MOLDOCK score (kcal/mol) and ΔG_{EXP} for TYR systems

the less permeable the compound is called. In addition to not showing permeation of the blood–brain barrier (BBB), but which may be favorable for evaluating the flow between biological membranes [44].

AK demonstrates the lowest probability of causing unwanted toxic or adverse effects due to the low clearance and accumulation of the drug or its metabolites, in addition to not having shown inhibitory activity to the permeability glycoprotein (P-gp) substrate, which is important for protecting the nervous system center

Fig. 14 AK bioavailability radar graph



of xenobiotics, as well as being expressed in tumor cells and leading to super-resistant cancers, in addition to not inhibiting the family of main isoenzymes, namely: CYP1A2, CYP2C19, CYP2C9, CYP2D6, CYP3A4, which are part of the first phase of metabolism. Finally, the medicinal chemistry alert PAINS (Pan Assay Interference Structures) assessed that there is no possibility of the compound having potentially problematic fragments [44].

Furthermore, the properties resulting from the **SwissADME** tool indicate that the physicochemical characteristics of all inhibitors show good similarity with drugs, as the coordinates of the bioavailability radar graph are favorable to the ideal range, especially the 6M and 6N inhibitors, the others showed unsaturation. For lipophilicity (the range of $\log P_{o/w}$ (XLOGP3) is -0.7 and $+5$) all inhibitors showed this pharmacokinetic capacity within the expected range. The size or molecular weight (range between 150 and 500 g/mol) are very favorable, as they all have a value in the range between 299.28 g/mol and 389.36 g/mol [44].

Regarding polarity (TPSA between 20 and 130 Å), all are within the desired range, with the majority having similar values and some with few variations. Most showed moderate solubility ($\log S$ not greater than 6), with the exception of the inhibitors 6O and 6P which were considered highly soluble in water. Thus, water-soluble compounds favor the development of medicines and influence their absorption. And they all have good flexibility [44].

The results of skin permeability ($\log K_p$) for AK analogues showed that only the 6M inhibitor showed high skin permeability with a value of -8.13 cm/s, while the other inhibitors were below the value presented by AK, in addition to not showing permeation of the blood–brain barrier (BBB) and high gastrointestinal absorption (HIA). Regarding the inhibition of P-gp, the inhibitors 6I, 6J, 6L, 6M and 6N showed inhibitory capacity against this important glycoprotein, which can leave the

body susceptible to the development of tumor cells, while the other analogues do not inhibit it.

Although 6O and 6P are the two most active, they inhibited all the main isoenzymes, while the other analogues showed inhibitory affinity with small differences in inhibition between CYPs, that is, some were inhibited and others were not. However, all molecules inhibited CYP1A2, CYP2D6 and CYP3A4, which are responsible for 50% of drug metabolism reactions. Although these warnings indicate possible toxic effects, in the future, through synthesis studies, these effects could be minimized and the molecules tested in more biological assays.

Complementing the ADME/Tox analyses, computational studies were carried out with the **Derek** program from the company Lhasa. As a result, AK analogues did not present warnings within the pre-established standards (plausible, probable or certain) for chromosomal damage, nephrotoxicity and most for carcinogenicity, with the exception of the 6I analogue which presented a plausible warning. Regarding the prediction of hepatotoxicity, the analogues 6H, 6I and 6J presented plausible warnings.

However, for the prediction of skin sensitivity, analogues 6A, 6D, 6E, 6K, 6N and 6O presented plausible warnings that can be considered as causing skin toxicity. Analogues 6B, 6C, 6F, 6H, 6I and 6L were predicted to be strong sensitizers, but with the note that they should be verified and 6G, 6J, 6M and 6P were found to be moderate sensitizers and also with the note that they should be checked for confirmation.

In summary, inhibitors indicate a good medicinal characteristic and are not likely to cause problems as medicines, since *PAINS* showed that there is no warning for problematic fragments. However, for **Derek**'s toxicity predictions, analogues 6A, 6D, 6E, 6H, 6I, 6J, 6K, 6N, and 6O could be ruled out as compounds for the treatment of melanoma due to toxicity predictions of skin sensitization and hepatotoxicity. The others could be used, but studies would be needed, such as synthesis, to inhibit the chains that generated the alerts and improve the compounds for use in the treatment of melanoma.

4 Conclusion

TYR is main path for melanogenesis, via disequilibrium of the enzymatic mechanism and is responsible for cutaneous pathologies. [13, 17, 32]. Therefore, this study aimed to investigate TYR inhibitors based on 1,2,3-triazole analogues of AK synthesized via click reaction [26] based on ADME/Tox analysis and molecular docking calculations.

The Theoretical Chemistry methodologies used showed satisfactory results for the TYR-1,2,3-triazole systems. And ADME/Tox analyzes of AK demonstrated that it has good similarity with medications, but low efficacy, instability and low permeability in the skin. However, AK does not act on the inhibitory activity of the permeability glycoprotein substrate or on the five main isoenzymes, reducing the likelihood of causing toxic or unwanted effects, as well as the absence of potentially problematic fragments.

Likewise, AK analogues showed good similarities with medicines, showed high bioavailability, demonstrating that they were within physicochemical and pharmacokinetic standards. The 1,2,3-triazoles showed low permeability in the skin, with the exception of the inhibitor 6M, in addition to demonstrating high values for HIA and only 6I, 6J, 6L, 6M and 6N can cause risks when inhibiting P-gp. CYPs did not undergo inhibitory activity from all AK analogues, but 6O inhibited all five CYPs, which showed that it can cause toxic effects.

However, for the toxicological predictions resulting from the **Derek** program, they demonstrated that the analogues require further studies to propose improvements in their structures, as they all present some toxicity warning. Among all AK analogues, those that presented the best toxicological results were 6B, 6C, 6F, 6G, 6L, 6M and 6P, among them it is worth highlighting 6M with high permeability in the skin and 6P as one of the most active, important properties for the development of future treatments.

Molecular docking was validated as it was consistent with experimental data, where the MOLDOCK scoring function generated good conformations and binding coordinates, demonstrating the main interactions of the inhibitors with the residues present in the TYR catalytic site [18, 42], thus, when carrying out an analysis between the Δ GEXP and MOLDOCK scores, an $r^2 = 0.58$ was obtained, which indicates that the anchoring calculations represented a correlation of 58% between the experimental data and the energies obtained in the calculations.

In this way, the results obtained in this study intend to make a significant contribution to other research on TYR enzyme inhibitors, encouraging new work such as the regulation of dyschromias related to the melanin synthesis pathway.

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Moonlighting Proteins: Importance and Current Scenarios



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Abstract This chapter investigates the current scenario of moonlighting proteins, elucidating some molecular mechanisms and structural adaptations, besides highlighting their profound impact on pathogenesis for different organisms and situations. The related databases and main forms of prediction are explored, using some characteristics observed in described moonlighting proteins and applying them to bioinformatics tools. Lastly, the potential applications of moonlighting proteins for therapeutic purposes are discussed. In conclusion, moonlighting proteins prediction, as well as their validation as targeting molecules and practice application remain as great challenges for future research and therapeutic innovation.

1 Introduction

Proteins are structures that permeate all aspects of life and play essential roles in virtually all biological systems. As essential biomolecules in the cells and organisms, they are intimately involved in pathological processes such as genetic diseases [1],

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cancer [2] and viral infections [3], making them important targets for diagnosis and therapy. A class of protein has attracted attention in recent decades: the moonlighting proteins.

Moonlighting proteins represent a fascinating category of biomolecules with multiple and diverse functions. They are initially identified by canonical functions, by which they play specific roles in cellular processes, however, over time, they revealed additional functions, not directly related to their primary activity [4]. Such proteins have been described in several organisms, either in prokaryotes [5, 6] as in eukaryotes such as animals [7], plants [8] and yeasts [9].

The term *moonlighting protein* was coined in 1999 by Constance Jeffery [5] for a protein that has multiple functions in different locations, therefore excludes those proteins that have a single function in different locations or use different substrates. More recently, this term has been defined for a protein that performs independent and sometimes unrelated functions and these functions are not attributed to different protein's domains [4]. The characteristic of a moonlighting protein having independent functions implies that the inactivation of one of the functions cannot affect the other [10]. The functions performed by the protein can be catalytic or non-catalytic in nature and even a combination of them [11].

The first descriptions of moonlighting proteins were for those proteins that have a well-defined function in a biochemical pathway [12, 13], in protein folding [14] or in essential physiological processes [15]. In fact, this first described group was remarkably diverse and the shift between activities could vary depending on the protein's subcellular location [16], conformational state [17], product concentration [18], interactions with genetic materials [19] or with the membrane [20], and environmental factors [21].

According to Singh and Bhalla [22], the best candidates for moonlighting functions are those proteins that are constitutively expressed and belong to metabolic pathways such as the glycolytic pathway. These enzymes have been reported as having additional functions in apoptosis, as a regulator of cell mobility, as transcription factors, among other functions [23].

These emerging functions of moonlighting proteins offer an additional layer of complexity to the understanding of both cell biology and disease mechanisms [1]. The functional versatility of these proteins means that they can influence a wide range of biological processes, often in unexpected ways. This possible multifunctionality can contribute to the resilience and adaptability of cells in response to different environmental and physiological conditions, as well as to pathological situations. Several moonlighting proteins have also been reported to be involved in virulence of both bacterial and fungal pathogens. Of the 700 moonlighting proteins classified in 2018, a quarter of them are related to pathogen virulence [24]. One of the largest groups of identified moonlighting proteins are chaperones, enzymes responsible for folding proteins inside the cell and involved in a different role outside [25–27]. Secretion of these enzymes were not related to known mechanisms responsible for targeting proteins outside the cell and, at the same time, the mechanism by which this occurs is not well-understood [28]. There are reports mentioning that, once released abroad, they mediate cell signaling and adhesion [29] and can even act as toxins [30].

Given their involvement in diverse cellular and pathological processes, moonlighting proteins are emerging as important biological targets for biomedical research. The better understanding of their functions can open new avenues for the development of innovative and efficient therapies. For example, in diseases where the canonical function of a moonlighting protein is well known, and there are emerging functions that contribute to pathology, therapeutic approaches can be developed to modulate the additional functions. Furthermore, identification of new moonlighting functions may reveal new biomarkers for disease diagnosis and prognosis, as well as new targets for therapeutic intervention.

Moonlighting proteins exemplify the complexity and elegance of cell biology. Their canonical and multiple emerging functions are probably essential for cellular homeostasis and response to various stimuli, highlighting them as crucial targets in the search for new treatments for a variety of diseases. Continued studies about these proteins are important to reveal the intricate mechanisms of life and ways to utilize these processes to improve human health or other biological processes.

2 Functional Features of the Moonlighting Proteins

Since the discovery of secondary functions for some proteins were reported [4], a substantial number of moonlighting proteins in a variety of organisms were documented [5–9]. This led researchers to investigate the molecular mechanisms that permeate these events, which seems to be related to specific factors, such as subcellular localizations, post-translational modifications (PTM), protein–protein interactions (PPIs), and structural plasticity.

The subcellular localization has shown to be an important factor that affects the moonlighting function of a protein. The expression or even the translocation of a protein to a different compartment within the cell can bring a new role to the protein. Some cytosolic proteins for example, especially involved on a metabolic pathway and/or as part of ribosomal subunits, after phosphorylation are translocated to the nucleus and have a role in gene transcription serving as an expression regulator [30]. One phenomenon described in several bacteria is the presence of proteins, known to be intracellularly located, on the bacterial surface, having a moonlighting activity. The alteration of the subcellular localization of such proteins plays a role in bacterial infection, being involved in cellular invasion, biofilm formation or virulence [31]. Proteins like the DnaK, a highly conserved chaperone, was described in the cell surface in *Mycoplasma hyorhinis* [32], showing a higher expression in more pathogenic strains or under infectious conditions.

Post-translational modifications (PTMs), as cited above, can also influence the moonlighting function of proteins, and since they are easily reversible, they tend to be easily adaptable within the cell [10]. In humans, Signal Transducers and Activators of Transcription 3 (STAT3) are moonlighting proteins [33]. When phosphorylated, they are translocated to the nucleus where they bind to a few gene promoters to regulate their expression. But it has been shown that cellular STAT3 is also present

on the mitochondria, where its absence interferes with the activity of complexes I and II in the electron transport chain and diminishes the mitochondrial oxidative phosphorylation [34].

Protein–protein interactions (PPIs) can alter the role of one protein and its partners (being as one single partner or gathering in polypeptide chains), leading to a distinct function of both. An example of this interaction is the thioredoxin in *E. coli* that reduces ribonucleoside diphosphates to deoxyribonucleoside phosphates, but also can be recruited by T7 phage to work as a subunit of a heterodimeric DNA polymerase, whose coupling is necessary for the enzyme function [35].

Another example is the glycolytic metalloenzyme enolase in *Staphylococcus aureus*, that catalyzes the interconversion of 2-phosphoglycerate and phosphoenolpyruvate, which has been shown to have a role in pathogenicity by binding with proteins of the extracellular matrix components, such as laminin and collagen II, acting like an adhesin [36, 37]. However, an alternative function of an alpha-enolase was detected in *Lactobacillus plantarum*, a constituent of the healthy human microbiota, playing a role in binding to fibronectins. This function contributes to the adhesion of the bacteria to intestinal tissues, to favor the colonization process of this commensal bacteria [38].

Multiple binding sites within a protein can contribute to its multifunctionality in biological systems. These sites can bind different substrates and cofactors [20]. Histones have been described to have an activity unrelated with their catalytic event in *Xenopus laevis*. Besides its role in DNA packaging, they were also described as having a copper binding site right at the histone tetramer adjoining point, which confers to the protein a copper reductase activity [39]. The Proline utilization A (PutA) protein is a membrane-associated protein that catalyzes the catabolism of proline but is also a transcriptional repressor of other protein utilization genes, and the interchange of roles is regulated by the concentration of proline [40]. Lee and co-authors [41] have solved the structure of PutA and identified the proline catalytic site, inferring the location of DNA-binding surface [6].

The discovery of moonlighting proteins raised questions on how one protein, which does not suffer any changes on polypeptide sequence, can perform more than one function without losing its first function [31]. Analysis of protein structures showed that proteins with moonlighting roles can suffer changes in conformations depending on the role they are performing. The DegP (HrtA) heat shock protein present in *Escherichia coli* is a cytoplasmic protein with protease and chaperone activities [42]. The analysis of the crystal structure shows a hexamer with a central cavity, with subunits arranged as dimers and trimers. On the hexamer there is a large cavity where lies the active site of the protease activity, but it has a loop that covers this space and when the protease is activated, it moves away [43]. These transient modifications can happen to allow multiple roles in one protein.

All the types described above can contribute in combination to provide the moonlighting function to a protein. One example that can summarize the molecular changes and interactions that allow a protein to have other functions besides the one originally described is the enzyme glyceraldehyde-3-phosphate dehydrogenase (GAPDH). This enzyme has a significant role in glycolysis, where it catalyzes the conversion of

glyceraldehyde-3-phosphate (G3P) to 1,3-bisphosphoglyceric acid (BPGA) in the presence of NAD^+ and inorganic phosphate, mediating the formation of NADH and ATP [44]. The widespread presence of this enzyme through the organisms allowed us to uncover its moonlighting roles and describe how these changes happen in different organisms and different cell types [45, 46].

The cytoplasmic GAPDH, when undergoing oxidative stress goes through a reversible PTM on a cysteine residue that is redox-sensitive, suffering a S-thiolation in its sulfhydryl groups, which allows the enzyme to influence the cellular signaling pathways related to this stress [47, 48]. In monocytes, GAPDH facilitates a shift in the metabolism from anaerobic respiration to the pentose phosphate pathway [49]. Accumulated mitochondrial GAPDH interacts with Voltage-Dependent Anion Channel 1 (VDAC1) protein and induces a pro-apoptotic mitochondrial membrane permeabilization [50]. This accumulation also causes the collapse of the inner membrane potential, as well as matrix swelling, resulting in permeabilization of the mitochondrial inner membrane and release of pro-apoptotic proteins [50]. The nuclear GAPDH has many roles described such as regulation of cell death [51], DNA repair [52], gene expression [53], cell cycle control [54] and even telomeric DNA protection [55, 56]. In mammals, in some *Staphylococcus* species and in *Lactobacillus plantarum*, GAPDH was found on the cell surface, acting as receptor for the iron carrier transferrin on the first two types of organisms [57] and in adherence to human colonic mucin in the latter organism [38].

These several mechanisms reveal the intricate and dynamic nature of the molecular and cellular processes that influence the moonlighting protein's function. The knowledge of such functional features can drive the development of novel therapeutic strategies when these aspects were involved in pathogenesis. Elucidation of the roles of the moonlighting proteins can help to gain deeper insights into cellular regulation and adaptability, leading to advancements in medical research and even biotechnology approaches.

3 Moonlighting Proteins in Silico Analysis and Prediction Computational Tools

Even after 30 years since the first citations about proteins with additional functions, there are still few databases and tools directly related to moonlighting proteins, despite the growing number of studies detecting known proteins in atypical cellular regions, with uncertain functions in these locations. The most common way to characterize proteins as having multifunctional functions is looking for indirect information through computational tools and comparing them with already characterized proteins in other organisms.

The main physical and biochemical characteristics evaluated using these tools are sequence similarity, amino acid composition, PTM, PPI [58–61], three-dimensional structure and folding type, sequence motifs, signal peptide, molecular weight, protein

length, isoelectric points (pI), aliphatic index, hydropathy, and transmembrane domain [62].

According to an extensive comparative analysis with these mentioned characteristics for proteins described as moonlighting, it has been reported that most of the intracellular proteins that are detected on the cellular surface do not exhibit evident distinct characteristics in one or another environment. Despite the active site of the studied proteins have remained conserved, the other exposed areas contained structural modifications. Such modifications seem to favor the stability of the protein to raspy environments (such as the cell surface) and contribute to the development of new binding and interaction sites, which may be related to the emergence of a new function. Another common ability of moonlighting proteins detected on the cell surface is to bind to plasminogen, which is related to a replacement of some superficial or C-terminal residues by lysine, whose intrinsic positive charge at neutral pH could contribute to the new function. Curiously, none of the moonlighting proteins analyzed have been found to possess a known signal peptide for its targeting to the cell membrane or a sequence motif for attachment to the cell surface [62].

Bioinformatics tools are essential for the detection of surface regions of proteins involved in secondary functions and for clarification of an alternative routing to different cell locations [26, 63]. *In-silico* analyses include sequence-based functional prediction and the analysis of conformational and interactional characteristics acquired by a protein throughout its evolutionary history. A compilation of available databases for moonlighting proteins and bioinformatics tools that facilitate its characterization and prediction are described in Table 1. Two other databases for moonlighting proteins are mentioned in literature, MultitaskProtDB [63–65] and PlantMP [66], but the links were not active at the time of the access attempts.

In general, bioinformatics tools facilitate the establishment of a relationship among the sequence, structure, and possible function of the protein, having as its main basis the similarity and homology with other described proteins. However, since moonlighting functions are often not conserved among protein homologues [65–67] it is crucial to improve the prediction of multifunctional proteins by creating tools that correlate protein's function to each cellular location, integrating information on physical and biochemical characteristics with the general environmental context of protein detection [98].

An example of application of the bioinformatics tools for the indirect analysis of a possible moonlighting protein was carried out with the enzyme malate dehydrogenase (MDH), a known enzyme of the Krebs Cycle. This enzyme was found to be more abundant in the cellular surface of cells interacting with the citrus host than in non-infective cells from the bacterium *Xanthomonas citri* subsp. *citri* (Xcc), the causative agent of citrus canker [27]. A partial characterization of Xcc's MDH was performed [99], using the MDH amino acid sequence of this bacteria, available in NCBI Website (National Center for Biotechnology Information—ORF XAC1006). Some molecular characteristics of MDH, such as molecular weight, cellular location, primary function, identity with other proteins, phosphorylation sites, and presence of signal peptide were investigated.

Table 1 Moonlighting proteins databases and related computer-based tools

Platforms names and access links	Description	Articles
MoonProt http://moonlightingproteins.org	Database of human, plants, archaea, protozoa, and bacteria proteins with multiple functions (over 500 annotated proteins). Includes information about structure, disordered regions, predicted transmembrane helices, and relationship of the protein to some disease. Also allows BLAST search and provides a list of links to published articles about moonlighting proteins	[65–67]
MoonDB http://moondb.hb.univ-amu.fr/index.php/	Database of Extreme Multifunctional (EMF) and moonlighting proteins. Contains 351 protein entries of human, mouse, fly, worm, and yeast	[61, 68]
MPFit https://kiharalab.org/MPfrediction/	Makes moonlighting protein prediction with missing feature imputation. Provides instructions for reading source codes of predicted moonlighting proteins, by MPFit model, from yeast, <i>Caenorhabditis elegans</i> , human, and MPs taken from Uniprot and MoonProt Databases	[69]
IdentPMP http://identpmp.aie.lab.net	This is a web application for plant moonlighting protein prediction	[70]
DextMP http://kiharalab.org/DextMP	Predicting tool for moonlighting proteins based on its textual features, extracted from scientific literature and the UniProt database. Just like MPFit, this tool also provides instructions for reading source codes of predicted moonlighting proteins, by DextMP model, from yeast, <i>Xenopus laevis</i> , human and moonlighting proteins taken from Uniprot and MoonProt Databases	[71]
Conserved Domain Database (CDD) https://www.ncbi.nlm.nih.gov/Structure/cdd/wrpsb.cgi	Protein annotation resource that consists of a collection of well-annotated multiple sequence alignment models for ancient domains and full-length proteins	[72–74]
AlphaFold Protein Structure Database https://alphafold.ebi.ac.uk	Prediction tool for protein's 3D structure	[75]
DeepTMHMM https://dtu.bio.lib.com/DeepTMHMM	Enables transmembrane topology prediction and classification	[76]
PSORTb 3.0 https://www.psort.org/psortb/	Predicts protein subcellular localization from bacteria and archaea	[77]
PSIPRED Workbench http://bioinf.cs.ucl.ac.uk/psipred/	Secondary structure prediction tool, which includes disorder regions and transmembrane helix packing; contact analysis; fold recognition; structure modeling; prediction of domains and function; prediction of protein-metal ion contacts; protein–protein hotspot residues; and membrane protein orientation	[78, 79]

(continued)

Table 1 (continued)

Platforms names and access links	Description	Articles
APID http://apid.dep.usal.es:8080/APID/init.action#tabr2	Agile Protein Interactomes Data Server provides a comprehensive collection of protein interactomes for more than 400 organisms based in the integration of known experimentally validated protein–protein physical interactions (PPIs)	[80–83]
CATH-Gene3D https://www.cathdb.info	Platform that considers the information in CATH (a classification of protein structures) to predict the locations of structural domains on protein sequences available in public databases. Allows the inclusion of additional annotations such as functional information and active site residues	[84, 85]
I-TASSER https://zhanggroup.org/I-TASSER/ I-TASSER-MTD https://zhanggroup.org/I-TASSER-MTD/	Is another common tool for protein structure prediction and structure-based function annotation. The MTD platform performs the same structural and functional prediction, but with proteins that have multiple domains	[86–88]
SwissModel https://swissmodel.expasy.org	Facilitate protein structure homology-modeling	[89–93]
ScanProsite https://prosite.expasy.org/scanprosite/	Scans protein of interest against motifs collection and patterns	[94, 95]
SignalP https://services.healthtech.dtu.dk/services/SignalP-6.0/	Server that predicts the presence of signal peptides and the location of their cleavage sites in proteins from Archaea, Gram-positive Bacteria, Gram-negative Bacteria and Eukarya	[96, 97]

The main objective was to search for information that could correlate this enzyme with the bacterial infectious process when it is surface-located, or that could give some clue about the movement mechanism of MDH to the surface and the possible acquisition of a new function. The InterproScan tool [100] confirmed the MDH enzyme as a member of the L-lactate/malate dehydrogenase and Malate dehydrogenase, type 2, families with a NAD (P)-binding domain. Using Psort tool [77, 101, 102], the predicted cell location of MDH was the cytoplasm. PHobius [103, 104] and SignalP [96, 97] tools indicated that there is no any signal peptide in the MDH sequence [99].

Without a signal peptide, there is no direct transport association of the enzyme to the cell surface via a classical transport pathway. With this partial information, obtained by the mentioned tools, there is still no evidence about the displacement of Xcc's MDH by known mechanisms. It was also not possible to infer whether the function performed by MDH on the cell surface (when interacting with the host) is the same as that prescribed for MDH located in the bacterial cytoplasm (reversible malate dehydrogenation) or whether it is a new function acquired by PTM or interaction with other proteins. This could perhaps be inferred through a PMT analysis tool, such as PSIPRED and APID (Table 1).

A quick present search for MDH in the MoonProt database detected information about the human enzyme (ID 513) which, in addition to the primary enzymatic function in the mitochondrial matrix related to malate metabolism, also has a secondary function described as binding to mRNA. However, there are no further details on the literature about studies of this secondary function, nor on the location of the functional site or the cell conditions in which this additional function is required. In the MoonDB database, no record was found about the multifunctionality of the MDH enzyme, which raises an alert about the emerging need to deposit and update information about moonlighting proteins in databases.

Text mining methods are also used to identify moonlighting proteins, using several information sources, and analyzing database entries, literature, and big data omics [60]. Aiming to forecast moonlighting proteins of varied proteomic conditions, a computational framework was created using Gene Ontology (GO) annotations of known moonlighting proteins (that were clustered into multiple groups), gene expression, PPIs, phylogenetic profile, genetic interaction networks, structural characteristics, intrinsic disordered regions, and ligand binding sites [105].

One of the findings of the study was that moonlighting proteins physically interact with a great number of distinct functional classes of proteins and its partners tend to share a functional similarity of its primary or secondary function. For example, an analyzed mouse PMS2 moonlighting protein (P54279), a mismatch repair endonuclease, had only one interacting protein (which is also a moonlighting protein), called Mlh1 (Q9JK91), another DNA mismatch repair protein, and both proteins are involved in hypermutation of somatic cells. Therefore, these two moonlighting proteins that interact with each other perform the same primary and secondary functions. An important observation was that moonlighting proteins tend to interact with other moonlighting proteins. Thus, identification of novel moonlighting proteins may be more effective through analysis of PPIs of moonlighting proteins. They have concluded that this framework method enables identification of novel moonlighting proteins from current functional annotations in public databases and that potential moonlighting proteins without sufficient functional annotations can be identified by analyzing available omics-scale data [105].

A recent method used a resource called SVMProt-188D to extract information on the protein composition and some physicochemical properties and analyzed these properties by linear discriminant analysis (LDA—used to reduce the dimensionality of the feature set to separate positive and negative samples). Then, a support vector machine (SVM) was used to classify the moonlighting proteins. This model reached a great accuracy and was applicable to plant moonlighting proteins. Although, the depth of the machine learning model was considered shallow [106].

The last moonlighting proteins identifying method, named MPKNN, is a K-nearest neighbors (KNN) based method that finds the *k* nearest neighbors of the query sample, from the class of training samples. The adjacency values between the query sample and all classes are compared and the query sample is assigned to the class to which it has the lowest adjacency value. The weighting of this method is calculated

from features such as the composition of certain amino acids, extended pseudo-amino acids and FEGS features (Feature Extraction based on Graphical and Statistical features), which are physicochemical characteristics of the proteins extracted by this specific model, based on graphical representation and statistical characteristics. It has demonstrated high efficiency and accuracy, according to its creators [107].

4 Moonlighting Proteins' Role in Diseases and Pathogenicity

Proteins with multiple functions can contribute to the interaction of distinct biological processes, which can influence many aspects in the overall diversity of living systems. As many diseases and pathologies are related to moonlighting features, it's important to identify and understand protein multifunctionality in organisms [108].

A large-scale identification of moonlighting proteins established that 30% of the human interactome consists of multifunctional proteins, 3% of which is composed of predicted moonlighting proteins. Those that adhere to the strict definition of a moonlighting protein are mostly involved in more than one disease, whereby the diseases in which they are related are phenotypically different according to their gene annotations [109].

The effect of mutations induced by disease state on innate moonlighting protein functions can lead to complex phenotypes [109]. The shift of function of a moonlighting protein can be called "neomorphic", where a mutation or conformational change in the protein structure can lead to a second function that does not play the same role in healthy and disease conditions. Furthermore, neomorphic moonlighting functions were identified because of their connection to disease. For example, in Alzheimer's and Parkinson's disease, these kinds of functions can lead to conformational change in protein structure responsible for a new type of PPI, resulting in the formation of protein aggregates in disease condition. Thus, distinct functions can lead to cancers or severe neurological damages, although the mechanisms involved have not been fully elucidated [110].

As stated by Sriram et al. [7], the prediction of phenotype from genotype can be directly affected by single-gene disorders, especially in metabolic disorders scenarios. If the metabolic enzyme responsible for the disorder has moonlighting activities, the additional functions could endorse to the complexity of such disorders and muddle the clinical phenotype. For example, hemolytic anemia is an autosomal recessive genetic disorder caused by scarcity of the enzyme Phosphoglucose Isomerase (PGI), that can also manifest neurological defects due to the enzyme's moonlighting role as neuroleukin, a neurotrophic factor. A severe mutation can lead to the misfolding of the protein configuration and the consequent loss of both enzymatic activities, while a less severe mutation affects only the PGI activity and leads to a phenotype that is not related to neurological deficiency [7, 111]. This example is schematically illustrated in Fig. 1.

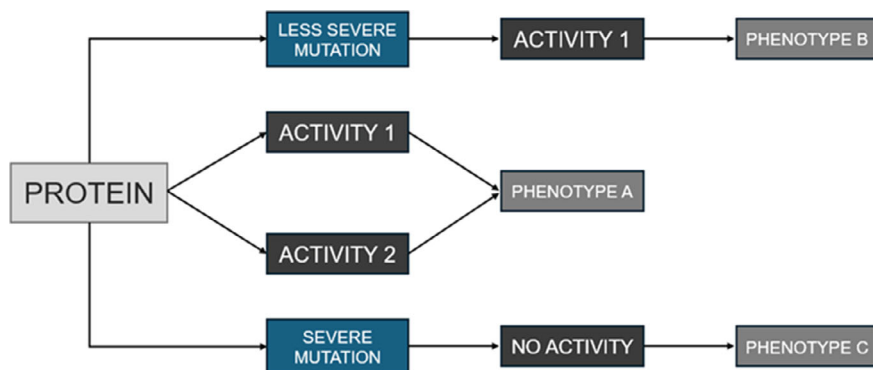


Fig. 1 Scheme of how mutations can disturb the clinical phenotype when the metabolic enzyme related to the disorder has moonlighting activities

By analyzing moonlighting and canonical functions of bacterial proteins in existing databases for moonlighting proteins, it has been reported that proteins involved in key biological functions and metabolic pathways, such as glycolysis and Krebs cycle, commonly present moonlighting functions, some of which are listed in Table 2. Although many of these enzymes are not always associated with pathogen virulence, 25% of the total examined proteins exhibited virulence-related behavior as the corresponding moonlighting activity [24, 112]. Pathogenic microorganisms often comprise an intricate set of moonlighting proteins related to virulence, which exhibit high structural conservation with those of the host. These facts may be directly related to protection against the host immune system. Besides, as it is expected that different epitopes of a virulence-related protein might be shared between the host and the pathogen, the pathogen is able to hide from the host's immune system protective action [112].

The significant increase in bacterial infections concomitant with the increase in drug-resistant pathogens is currently a worrying health problem, in addition to the increasing occurrence of multidrug-resistant organisms like *Mycobacterium tuberculosis*, *Staphylococcus aureus* and *Escherichia coli* [63]. The ability of pathogenic bacteria to colonize and cause disease is related to their production of virulence factors, which are often specific proteins with a very selective range of actions such as adhesins, invasins, evasins and toxins. Moreover, in most bacterial pathogens, these virulence factors are generally moonlighting proteins, mostly including highly conserved proteins such as enzymes of glycolytic pathway, tricarboxylic acid cycle and a range of other metabolic enzymes as proteases, transporters, and molecular chaperones [25, 26].

In several species of bacteria like *Streptococcus* and *Mycoplasma*, the glycolytic enzyme enolase has been identified on the cell surface playing a key role in binding important proteins related to infection [30]. Furthermore, most pathogenic bacteria are able to first interact and colonize host cells and tissues using adhesins present on the cell surface. These bacterial adherence factors were first identified as intracellular

Table 2 Canonical and moonlighting functions of microorganism proteins

Protein	Canonical function	Moonlighting function	Organism	References
Glucose-6-Phosphate isomerase	Glycolysis	Adhesion protein, laminin, collagen	<i>Lactobacillus crispatus</i>	[113]
		Virulence	<i>Xanthomonas oryzae</i>	[114]
GAPDH	Oxidative phosphorylation	Binds with mucin and plasminogen	<i>Bacillus anthracis</i>	[115]
		Transferrin	<i>Lactobacillus acidophilus</i>	[116]
			<i>Staphylococcus aureus</i>	[117]
DnaK	Chaperonin	Binds to plasminogen	<i>Bifidobacterium</i>	[118]
			<i>Mycobacterium tuberculosis</i>	[119]
Superoxide dismutase	Antioxidant	Adhesin	<i>Mycobacterium avium</i>	[120]
Ef-Tu (elongation factor Tu)	Protein synthesis	Interact with cells and mucins	<i>Lactobacillus johnsonii</i>	[121]
		Binds with fibronectin, factor H, and plasminogen	<i>Mycoplasma pneumoniae</i>	[122]
Aldolase	Oxidative phosphorylation in glycolysis	Pathogen-host interaction	<i>Candida albicans</i>	[123]
			<i>Streptococcus pneumoniae</i>	[123]
Enolase	Carbohydrate degradation and glycolysis	Binds to plasminogen	<i>Leishmania mexicana</i>	[124]
			<i>Streptococcus pneumoniae</i>	[37]
6-Phosphogluconate dehydrogenase	Pentose phosphate pathway	Adhesin	<i>Streptococcus pneumonia</i>	[125]

chaperones or other intracellular enzymes engaged in central metabolic pathways. In most of the cases, it is unknown how they are secreted and become attached to the cell surface due to the lack of signal sequences for secretion or known sequence motifs for binding [126]. The glycolytic pathway enzyme GAPDH was the first intracellular protein detected on the surface of pathogenic streptococci and whose

surface moonlighting activity was correlated with the colonization, internalization, and the subsequent proliferation of group A streptococci [126, 127].

With the emergence of diverse antibiotic-resistant strains, *Staphylococcus aureus* has become a great threat to public health due to its relevance in hospital-acquired infections [128]. *S. aureus* infections can be classified into superficial, toxin-mediated, and invasive infections, in which are employed a large number of moonlighting proteins that contribute to the pathological invasion and the following virulent cycle [129].

In humans, *S. aureus* is responsible for many biofilm-associated infections, characterized by a decrease in metabolic activity of the biofilm-embedded cells and a surrounding Extracellular Matrix (ECM) with protective nature, what makes it difficult to treat [130]. As stated by Graf et al., *S. aureus* secretes virulence factors and ribosomal proteins that exhibit a moonlighting function that contribute to biofilm integrity and stabilization. This effect is mediated by an acidic ECM environment with the release of fermentation products like lactate and acetate. The acidic environment promotes positive charges on alkaline proteins and benefits the interaction of proteins with negatively charged cell surfaces and other metabolites [130].

Mycobacterium tuberculosis, the causative agent of tuberculosis, is a highly evolved pathogen that has gathered over the time several adaptive skills that provided mechanisms to trick the host immune system. Although there is absence of typical virulent factors, the bacterium owns many elements that modulate host physiological processes for its survival and consequential disease progression [131], many of which are moonlighting proteins acting as key players in virulence. It has been shown that the moonlighting activity of the protein PP2, one of the proline-glutamic acid and proline-proline-glutamic acid (PE/PPE) family, can cause sufficient damage to lead to the establishment of the infection and disease progression by itself [132].

The potential of the moonlighting functions of molecular chaperones in mycobacteria pathology is also reported in literature. In its canonical function, molecular chaperones are intracellular proteins involved in protein folding control. On the other hand, when secreted as signaling molecules, can both control the composition and actions occurring at the cell wall, as well as playing a key role in the entry of the bacterium into the macrophage during infection [133].

Another essential enzyme produced by *M. tuberculosis* that has important roles in bacterial growth and survival is acetylornithine aminotransferase (ArgD) originally from arginine biosynthetic pathway. Among many of its essential activities, ArgD's moonlighting function is utilized to modulate host innate immunity, increasing production of immune effector molecules and pro-inflammatory cytokines when exposed to macrophages cells. Moreover, *in-silico* analysis predicted it to be a highly antigenic protein, inducing pro-inflammatory response, apoptosis and stimulating macrophage activity, properties that make this protein a significant subunit vaccine applicant. Additionally, this protein has essential roles for *M. tuberculosis* survival, which could be an important chokepoint to drug target against tuberculosis [133].

From certain points of view, as discussed in Gupta et al. (2019) [134], as drug is specifically designed to inhibit the function of a protein and it is often difficult to

predict the impact it will have on its moonlighting activity, the moonlighting feature turns out to be a limiting factor in new drug discovery approaches. Nevertheless, others suggest that targeting moonlighting proteins can benefit the development of innovative molecules for therapeutic applications, which can become an important tool that brings hope in an era of increasing drug resistant bacteria [63].

5 Moonlighting Proteins in Phytopathogens as Highlighted by Subproteomic Approaches

Looking for moonlighting proteins involved in diseases caused by phytopathogenic organisms in the MoonProt database, only one protein, Phosphoglycerate Kinase (PGK), was deposited. This protein was detected in a filamentous phytoplasma bacterium, *Spiroplasma citri* [135], that causes a disease known as “Stubborn”, which affects several species of plants. Its transmission depends on insect vectors with polyphagous habits like different species of leafhoppers, within which the bacteria multiply [136]. By confocal microscopy of the salivary gland of an infected insect vector, it was observed that the bacteria co-localized with actin filaments present in the insect’s membrane. The PGK protein bound to actin was isolated by affinity chromatography and then identified by LC–MS/MS. It was concluded that bacterial PGK has no effect on its binding to the cell surfaces of the vector, but the internalization of bacteria in eukaryotic cells is reduced with the presence of PGK in a dose-dependent manner (higher concentrations decreased internalization), which may have some effect on the transmission process of *S. citri* [135].

A proteomic analysis of the cell surface of Xcc identified several of the moonlighting proteins mentioned above and related them to the phytopathogenicity of this bacterium. Bacterial cells were cultivated *in vivo* (plant leaf infection) and *in vitro* (nutrient rich medium) and the differential surface proteins were analyzed. Among the proteins differentially detected in response to host infection, proteins DnaK, 60 kDa-chaperonine and MDH had the highest relative abundance on *in vivo* condition and did not exhibit a signal peptide in their structure. Moreover, MDH was found to be a signature protein for the infectious condition. It has been suggested that these proteins may be involved in the interaction of bacteria with the host and that they may undergo PTMs, which would justify their transport to the surface and, eventually, the acquisition of a new function as moonlighting proteins [27].

Another proteomic analysis with Xcc compared the enriched periplasm fraction of the bacterial cells that were cultivated in pathogenicity-inducing medium (XAM-M) and in non-pathogenicity-inducing medium (NB) [137]. Several differentially expressed proteins that were upregulated in XAM-M were related to the virulence of Xcc, among them were the moonlighting proteins GAPDH, that was related with the induction of known pathogenicity factors—*hrp* genes, and Superoxide Dismutase (SOD), described as potentially involved in Xcc defense against Reactive Oxygen Species (ROS), produced under stress conditions.

Based on the results of differential proteomic analysis of enriched periplasmic fractions of Xcc and *X. fuscans* subsp. *aurantifolii* XauB (causes a milder citrus canker), some proteins stood out as being present only in Xcc, the more virulent bacterium [138]. Among the proteins identified as expressing only in Xcc are some moonlighting proteins, such as enolase and elongation factor Tu (EF-Tu).

The role of enolase in the periplasm fraction of Xcc has not been explored in depth and its involvement with the pathogenicity of Xcc is not well-understood. Regarding the Elongation factor Tu (EF-Tu), it has been characterized as a PAMP molecule (Pathogen-Associated Molecular Pattern), which can induce plant immune responses and be involved in Xcc biofilm formation and also be associated with the Type III Secretion System (T3SS) in this genus [138].

Some Xcc proteins, detected in the aforementioned proteomic works when the bacteria were cultured under infectious conditions, are represented in Fig. 2. All proteins represented in Fig. 2 are described as moonlighting proteins in other organisms and are present in the database MoonProt, but their secondary function in Xcc has not yet been determined. In other organisms some mechanisms of their functions have been reported to be related to host adhesion, binding to plant receptors, or favoring bacterial colonization.

Through the NCBI PubMed platform, the query “phytopathogen moonlighting proteins” recovers only two recent articles. In one of them, two distinct roles of GAPDH (or Gap) of *Pseudomonas syringae* pv. *tomato* (Pto) DC3000 were reported. Gap1 protein has demonstrated a prevailing glycolytic activity but depending on the physiological state of the bacteria it was highly involved in motility, biosurfactant production and biofilm formation. NADPH-dependent Gap2 presented a more canonical gluconeogenic activity and appeared to have an essential role for bacterial growth within the plant. The two forms of the protein were identified in the extracellular medium of Pto grown in different defined media conditions, highlighting their additional roles related to the viability of the infection [139]. The other article is not specifically about moonlighting proteins from a plant pathogen organism, but rather about the discovery of a new antimicrobial activity detected for a protein from *Streptomyces lividans* (TK24) against *Clavibacter michiganensis*, a phytopathogen that causes canker in tomatoes [140].

This lack of articles highlights the need for new studies involving a better characterization of moonlighting proteins in phytopathogens and the subsequent deposit of the information obtained in highly visible databases.

6 Moonlighting Proteins as Therapeutic Targets

Proteins as key molecular factors of biological processes can be targeting biomolecules when selected as critical indicators or biomarkers, being crucial for drug discovery. This emphasizes the importance of studies to search for such targets and outline key definitions and analytical criteria for biomarker selection. Since

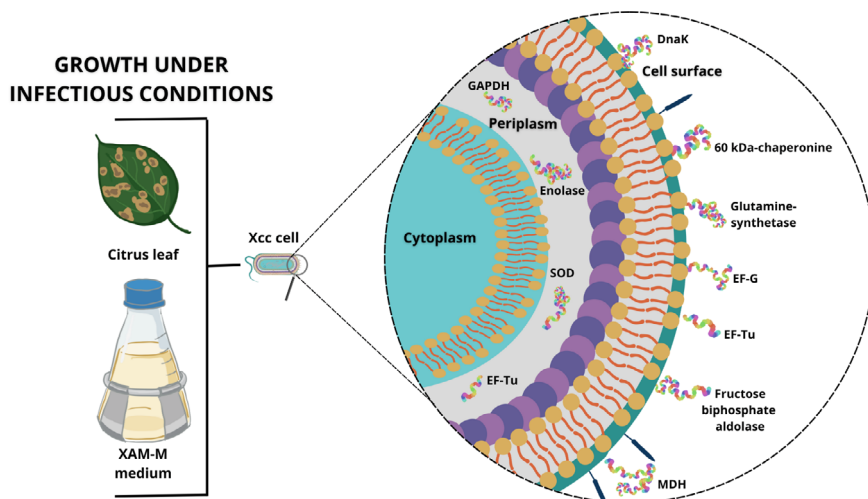


Fig. 2 Putative moonlighting proteins related to the phytopathogenicity of *Xanthomonas citri* (Xcc), a Gram-negative bacterium that infects citrus, detected by differential proteomic analyses of the cellular surface and periplasm fractions. Cells were cultured under infectious conditions, i.e., in vivo (plant leaf) or in vitro under pathogenicity-inducing condition (medium—XAM-M), relatively to the non-infectious condition (NB rich medium, not indicated in figure). The proteins represented on the cell surface of Xcc (DnaK, 60KDa-chaperonine, MDH, Elongation factor G—Ef-G, Glutamine-synthetase, Fructose biphosphate aldolase and EF-Tu) were detected in cells taken from the exudate of leaves, seven days after inoculation, when disease symptoms were visible [27]. The proteins that are in the periplasm (GAPDH, SOD, Enolase and EF-Tu) were found after culturing the bacteria in XAM-M medium [137, 138]. According to the MoonProt tool (<http://moonlightingproteins.org>), all these proteins has been reported to be moonlighting proteins in other organisms

moonlighting proteins can be involved in human disease processes, strategies for therapeutic approaches based on these proteins could be considered.

According to Franco-Serrano and coworkers [141], from databases research including the MP databases (MoonProt and MoonDB) [65, 68, 142], DrugBank [143] and human gene mutation [144], more than 78% of the proteins known at the present as moonlighting proteins are involved in human diseases, with 48% targeted by existing drugs [141], while over 25% contribute to the virulence of pathogenic microorganisms [145]. Given that 48% of known moonlighting proteins are targets of existing medications, it is possible that moonlighting proteins can help to explore drug designs for human diseases, detect toxicity and efficacy at an early stage, and identify patient populations that respond to such treatments [146].

Protein-driven biological processes, particularly those involved in socially relevant human diseases such as cancer, neuropathies, viral and bacterial diseases, require the developing of valuable therapeutic strategies [147]. To be a drug target, a protein must have “druggable” properties, including folds that favor interactions with small drug-like molecules, and presence of binding sites [148, 149]. Understanding the functional features related to these interactions is crucial for evaluating and selecting

effective therapeutic strategies in the drug development process [150, 151]. Additionally, key steps in developing strategies to explore therapeutic targets include the use of unbiased biomarkers and clinical trial endpoints (e.g. survival rates, improved quality of life, symptom relief, and tumor regression) throughout the drug discovery and development process [152, 153].

Targeting of moonlighting proteins or other molecular target is a systematic process that includes four important steps: Discovery, Development, Practice (e.g. clinical) Validation, and Qualification [146]. This process, which resembles a continuum, begins with biomarker discovery, encompassing initial identification and proof-of-concept studies. The early stages of research are commonly referred to as the discovery phase [154].

As highlighted on preceding sections, moonlighting proteins are found in a broad spectrum of organisms, ranging from prokaryotes to eukaryotes and in both unicellular and multicellular organisms. Yadav and coworkers [63] have discussed how certain bacterial moonlighting proteins contributes for survival and virulence by participating in metabolic processes, related to host–pathogen interaction and evasion of the immune system [63]. This versatility underlines their potential as targets for new antibacterial treatments and gives hope in the fight against the increasing antibiotic resistance of the human pathogens.

Major studies across the biological activity of moonlighting proteins clarify their role in mediating transitions between biochemical pathways in response to cell dynamics and have shown some understanding of biological function changes under pathological contexts [155], such as disease-promoting immune system activities [156]. A case in point is GAPDH, which was related to cancer progression and immune system regulation by modulating the functions of macrophages through immunosuppressive enzyme upregulation, reduction of TNF- α release, and suppression of neutrophil infiltration. Additionally, GAPDH and indoleamine 2, 3-dioxygenase 1 (IDO1) have been shown to be promising drug targets among the pivotal proteins in the cancer immunology, though still in the early stages [156]. Notably, these alterations may correlate with the activation of numerous moonlighting proteins, based on molecular interactions, significantly influencing disease evolution [145].

PPIs are crucial for central nervous system activity. Allosteric receptor-receptor interactions within heteroreceptor complexes allow a single G Protein-Coupled Receptors (GPCR) protomer to switch its coupling to different G proteins, affecting gene expression, synaptic plasticity, and immune responses. For example, D1- and D2-dopamine heteroreceptor complex switches its G protein coupling upon coactivation, influencing calcium levels and synaptic plasticity. These interactions highlight the protein's versatility and its significant role in physiological regulation and neurological disorders [157]. This means that moonlighting proteins have distinct functions due to the dynamics of interactions at different times in non-pathological or pathological processes, in different organism's cells and subcellular compartments (e.g. nucleus), when secreted or associated with membranes [158].

Despite the added complexity in interpreting their molecular mechanisms, understanding of the moonlighting proteins is essential due to their significant impact

on cellular activities and broader implications in medicine, agriculture, and other human endeavors [158]. Omics sciences are fundamental for unraveling the mechanisms of proteins and can be applied for the study of moonlighting proteins functions and their diverse roles. The key to future drug discovery lies in our ability to leverage new technologies to integrate data from sequenced genomes, functional genomics, protein profiling, metabolomics, and bioinformatics, enabling a comprehensive systems-based analysis to deepen our understanding of health and disease complexities [159]. This holistic approach is vital for developing effective therapeutic strategies and innovative applications in various fields.

7 Future Perspectives and Challenges

Moonlighting proteins are an intriguing and rather diverse group of biomolecules that have posed questions to the classical view on protein roles. Due to their versatility, they can be frequently involved in several unrelated actions either in normal or pathologic biological processes. In this section, we discuss the prospects and issues that remain to be resolved in moonlighting protein studies, specifically in relation to their therapeutic implications.

7.1 *Therapeutic Applications and Existing Issues*

The concept of moonlighting proteins has gained a lot of interest in the development of therapeutic targets since most of them are involved in various cellular processes and disease situations. As already mentioned, GAPDH, for example, is not only involved in well-known metabolic processes but is also involved in pathogenesis and immune evasion in different pathogens. Due to its multifunctional nature, this enzyme may be considered a candidate for drug targeting [160, 161].

However, when considering therapeutic intervention, several issues arise when moonlighting proteins are considered possible target candidates to drugs. One major concern is the risk of side effects arising from the blocking of their primary activities, which could be essential to the cell. As also mentioned, the glycolytic enzyme enolase is involved in cell energetic metabolism but also in bacterial adhesion to the host, depending on its location, during infections by *Staphylococcus aureus* [161] and fungi such as *Candida albicans* [162]. Targeting enolase to prevent infection might lead to adverse effects on the general metabolism where enolase plays an important primary role.

Furthermore, the processes by which proteins can transit between these functions are not well understood. The poor understanding about the distinction between different states of the protein complicates the possibility of developing drugs that can modulate one function without interfering with the other one. The regulation

of function switching through PTMs, changes in subcellular localization, and PPIs further complicates this challenge.

7.2 Advances in Bioinformatics and Experimental Techniques

To overcome these challenges, there is a need to adopt necessary improvement in bioinformatics and experimental methods to infer a putative protein function. The computational tools like moonlighting database MoonProt can offer the idea of moonlighting functions based on protein sequence, structure, and interaction [66]. These tools can predict new functions and can point out potential drug targets.

Other approaches suitable to identification and characterization of moonlighting proteins include tandem mass spectrometry as well as NMR spectroscopy. These methods can detect proteins localized in odd compartments or having other than expected levels of expression, thus awakening for a possible multifunctional feature. Further, moonlighting functions can be characterized by means of the 3D structures of proteins through X-ray crystallography and NMR where researchers can make guesses regarding the binding domains and possible functional sites to moonlight functions [160]. However, due to the recent advances in detecting interactions with other proteins, tools that predict PPIs are those that most contribute to the characterization of moonlighting proteins.

7.3 Case Studies and Potential Applications

Several examples were mentioned in this manuscript explaining the potential therapeutic value and the challenges with the moonlighting proteins. Additionally, the aldolase A (ALDOA), a muscle-specific isoform, has been described to be up regulated in cancer and have a significant role in cancer advancement. Inhibiting ALDOA directly affects the formation of the actin cytoskeleton and also kills tumor cells. This approach has been promising in preclinical model suggesting that the inhibition of moonlighting functions can be used as a cancer therapy [163–165].

Likewise, ribosomal proteins, referred to as RPs, have been identified to perform roles apart from their main function of being components of the ribosome, which includes DNA repair and apoptosis. These proteins can have an effect either negative or positive on the formation and development of a tumor if they can interact with non-ribosomal components. Awareness of these interactions can be key to creating new targets for cancer treatments that take into account the RP moonlighting function [166].

7.4 *Future Directions and Scientific Gaps*

There are still several scientific gaps that need to be filled to understand the potential of moonlighting proteins. One of them is to determine the circumstances under which these proteins can adapt to multiple roles. The mechanisms by which these switches are turned on and off within the cell context are critical to develop selective inhibitors that selectively target the pathologic activities of these proteins.

Another promising area to be explored is the study of moonlighting proteins in less-probed organisms and diseases. For instance, several advances were made in defining the functions of moonlighting proteins in bacteria and fungi, but little is known about those protein roles in parasitic infections and plant pathogens. Extending knowledge of such proteins to these situations or organisms potentially holds the key for a better understanding of protein promiscuity. The integration of omics data including genomics, proteomics, and metabolomics with use of high-end bioinformatics tools can help in the understanding of proteins role and cross talk.

8 Conclusion

Future investigations of moonlighting proteins can open opportunities to a greater comprehension of cellular mechanisms and possibilities of treatment approaches. Characterization of the multifunctionality of these proteins could make possible to attack their selective targeting and understand the mechanisms of function switching. We expect that what we see today only in dull gray light will become a bright spotlight on moonlighting proteins, enabling consecutive steps of discovery through computational simulations and experimentation.

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Use of Phytotherapeutics in the Amazon with a Focus on the Metropolitan Region of Belém: Traditional Applications and Computational Studies of Bioactive Compounds



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Abstract The use of phytotherapeutics in the metropolitan region of Belém is deeply influenced by the rich Amazonian biodiversity and the vast traditional knowledge about medicinal plants. This context offers a wide range of natural and accessible therapeutic alternatives, benefiting public health and promoting environmental and economic sustainability. The appreciation of phytotherapeutics not only respects cultural traditions but also encourages responsible agricultural practices and the conservation of local biodiversity. The integration of phytotherapeutics into the public health system can bring numerous benefits, including the provision of more economical treatments with fewer side effects. To optimize the discovery and development of new bioactive compounds, computational strategies have proven to be indispensable. Tools such as molecular modeling, molecular dynamics, and molecular docking allow the virtual screening of thousands of compounds, accelerating the process of identifying potent therapeutic substances. Furthermore, cheminformatics and bioinformatics play crucial roles in organizing and analyzing large volumes of chemical and biological data, revealing important patterns and relationships that may

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not be detected by traditional methods. These approaches are particularly valuable in the Amazon, where the systematic exploitation of native plants still has enormous untapped potential. Therefore, the purpose of the study is to carry out a literature review, seeking to evaluate and include scientific data, combining the traditional use of phytotherapeutics with advanced computational strategies offers a promising path for innovation in medicine, promoting health and well-being while preserving the biodiversity and cultural knowledge of the Amazon.

Keywords Phytotherapeutics · Computational studies · Bioactive compounds · Amazon region

1 Introduction

The traditional practice of using phytotherapeutics, which consists of products derived from medicinal plants, is widespread in several global cultures as a way to treat and prevent various health conditions. The growing adoption of phytotherapeutics reflects the search for natural and alternative approaches to traditional medicine, also reflected in the increase in the supply of herbal products in the global market for dietary supplements and medicines [1].

Many people seek natural alternatives to treat health problems and promote well-being, but the perception that phytotherapeutics are automatically safe due to their natural origin is a misconception [1], since these products are derived from different parts of plants, such as roots, leaves, and seeds, and their efficacy and safety need to be proven through pharmacological and toxicological studies [2]. Therefore, several factors need to be considered when evaluating the use of phytotherapeutics, and this includes the risks associated with poisoning by these substances [1].

Phytotherapeutics have biological properties that include stimulating immune system reactions, such as immunomodulatory actions, resulting in a reinforcement of the body's immune response. These immunomodulatory actions, which can strengthen the host's defense mechanisms, are known as immunostimulants. They activate innate and acquired immunity, through the activation of specific cells and mediators present in phytotherapeutics [3].

Another important consideration is that, even if the biological activity of a plant is proven, this may not manifest itself in a prepared phytotherapeutic due to the influence of several factors. Among these factors, planting, cultivation, harvesting, drying of plant species, medication preparation methodology, and storage conditions stand out, which can affect the quality and effectiveness of the phytopharmaceutical [4].

In Brazil, the search for and interest in phytotherapy is justified by the culture, tradition of use, and the country's broad biodiversity [5]. Approximately 82% of Brazilians use preparations and products based on medicinal plants [2]. This data is in line with estimates from the World Health Organization (WHO), which suggests that a large part of the world population, especially in developing countries, uses

traditional practices, such as herbal medicine, and suggests new studies in these countries on this topic [6].

From 2013 to 2015, there was a 161% increase in the search for treatments with medicinal plants and phytotherapeutics through the Brazilian Unified Health System (SUS) [7]. This is largely due to the creation of public policies and legislation that encourage the use of phytotherapeutics by the population, among which we can mention the National Policy on Integrative and Complementary Practices (PNPIC) and the National Program of Medicinal Plants and Phytotherapeutics [2, 8].

The Amazon region is highlighted in terms of Medicinal Plants and Phytotherapeutics. In recent decades, scientific interest in local biodiversity resources has been growing [9]. This interest in Amazonian biodiversity is also related to the possibility of studies that can assist in the development of phytotherapeutics, given its wide diversity in terms of flora. The State of Pará, located in the Brazilian Amazon region, has the Metropolitan Region of Belém (RMB) which has been made up since 2023, of the municipalities Belém (the capital), Ananindeua, Marituba, Benevides, Santa Bárbara do Pará, Santa Izabel do Pará, Castanhal, and Barcarena, these municipalities have great study potential for the development of new phytotherapeutics.

The development of medicines is often directly or indirectly related to natural sources that contain bioactive compounds, capable of providing treatment for various diseases, playing a crucial role in the discovery of new medicines. It is estimated that one-third of the most sold medicines currently come from natural products [10, 11]. Amid research into medicines based on natural products, computational approaches are fundamental. Continuous advances in bioinformatics tools and molecular network methods have made it possible to identify new bioactive compounds in an increasingly efficient manner [10].

Currently, the Global Social Molecular Network of Natural Products (GNPS) has been used in research to create bioactive molecular networks. These networks provide structures and predict potential chemical compounds, both known and unknown, with high bioactivity, enabling efficient drug discovery and generating hypotheses for new bioactive compounds. The current process for discovering natural bioactives requires bioassay-guided fractionation, which involves numerous steps ranging from metabolite extraction, chromatographic fractionation, screening of each fraction, isolation and identification of compounds, to verification of bioactivity. Due to the wide variety of bioinformatics tools, pre-processing and post-processing data vary in each publication, although GNPS is a common step in creating these networks [10].

Bioactive compounds from natural sources represent a promising basis for the development of new and more effective analogs, using strategies that aim to optimize structural characteristics, employing modern medicinal chemistry approaches, including molecular modeling and combinatorial chemistry. Although a variety of homologs can be obtained from natural sources, only a small fraction of available plants have been investigated for biological activity screening in the search for new therapeutic agents. Currently, with modern advances in the search for new drugs derived from natural products, recently isolated compounds that exhibit considerable bioactivity are subjected to Structure–Activity Relationship (SAR) analyses and molecular modeling processes. These steps aim to design and develop analogs

that present greater potency, lower toxicity, and more favorable pharmacokinetic profiles [12].

Other research explores the application of computational studies using advanced tools and databases to identify potential bioactive molecules in plants for the treatment of cancer, for example. These investigations employ diverse strategies such as ADMET analyses (absorption, distribution, metabolism, excretion, and toxicity), Molecular Docking, homology modeling, Molecular Dynamics, protein–protein interactions, and topological characteristics of plant bioactive proteins in comparison with human protein [13].

The purpose of the study is to carry out a Literature Review, which aims to evaluate and include scientific data currently available on the use of phytotherapeutics in the Amazon, especially in the Metropolitan Region of Belém (RMB), addressing their positive and negative impacts, and scientific validation of its effectiveness, in addition to addressing computational strategies that are being used in the discovery of bioactive compounds present in phytotherapeutics. To conduct this study, a systematic approach was employed with pre-established criteria, including language, types of study such as literature reviews, observational, clinical, and in silico trials, formulation of the research question, presentation of main results, and relevant final considerations for research.

2 Material and Methods

2.1 Identification of the Literature Review Research Question

In this study, the central research question is defined as: “What are the most used phytotherapeutics in the Metropolitan Region of Belém (RMB) of Pará, which is part of the Brazilian Amazon Region and computational studies in the identification of bioactive compounds in phytotherapeutics?”. Where, the authors seek to discuss the use of phytotherapeutics in the RMB, regarding their effects, effectiveness, and the computational strategies that are being used to identify bioactive compounds present in phytotherapeutics on an international scale.

2.2 Selection of Studies and Extraction of Relevant Data

The research covers academic databases: Scielo, Capes, and Google Scholar, where it used the keywords: “Phytotherapeutics”, “Phytotherapics in the Amazon”, “Belém Metropolitan Region”, “Computational Studies”, “Bioactive Compounds”. The studies were selected according to pre-established criteria, which were: studies in Portuguese, English, and Spanish; complete works published in journals, book chapters, or academic events in recent decades; kind of study; and according to relevance

to the research question. The selection of articles was carried out in two stages: by tracking the titles and abstracts of each selected article and, subsequently, by completely evaluating the selected articles.

2.3 Results Presentation

Both the qualitative and quantitative characteristics of the data collected were examined, whenever relevant. The findings were presented, emphasizing the essential discoveries and scientific progress associated with the use of phytotherapeutics in RBM and the computational study strategies that can help in the identification of bioactive compounds in these herbal medicines. Based on these results, a critical analysis was carried out, taking into account the restrictions of the reviewed studies, potential biases, and gaps in the literature, as well as future perspectives for the area.

3 The Culture of Use of Phytotherapy

3.1 Phytotherapeutics: A General and Global Overview

Traditional medicine encompasses the study of traditional medicines practiced by different ethnic groups, dating back to the beginnings of human history and historically based on resources such as medicinal plants [14]. Since ancient times, natural herbs have been extensively used in the treatment and prevention of various diseases. The evaluation of the merits and demerits in this field has led to the creation of new phytotherapeutics that provide health benefits [15].

Some studies discuss that throughout history, herbs, generally defined as any form of plant or plant product, as well as plant extracts, constituted the basis of the first medicines used in traditional medicine systems of different cultures and civilizations, as plants and herbs have always been a common source of medicines, whether in the form of traditional extracts or pure active compounds [14]. Thus, the practice of herbal medicine dating back centuries shows that people have sought natural remedies to treat common illnesses, such as colds, allergies, stomach aches, and toothaches. This trend of resorting to herbal treatments continues to grow steadily [16].

The first records of drugs of natural origin date back to Mesopotamia, around 2,600 BC, describing the use of around 1000 compounds derived from plants. One of the most detailed records of the use of natural extracts in therapy is found in the Ebers papyrus of the Egyptians, dating from around 1500 BC, which documents more than 700 natural medicines, mainly of plant origin. The Chinese Materia Medica record (BCE 1100), dating from around 1100 BC, describes 52 natural medicinal preparations, while the Indian Ayurvedic record, from around 1000 BC, describes

more than 800 natural substances and medicinal extracts [17, 18]. Hippocrates also applied phytotherapy, that is, healing with herbs, in his treatments [18].

In China, the use of traditional herbal medicines accounts for approximately 30–50% of total medicine consumption. In countries such as Ghana, Mali, Nigeria, and Zambia, phytotherapeutics make up to 60% of primary home care. It is estimated that in regions such as Europe, North America, and other developed countries, more than 50% of the population has used herbal approaches at least once in their lives [16, 19].

Other studies report that in San Francisco, London, and South Africa, approximately 75% of patients affected by HIV/AIDS resort to herbal formulations, while in Canada and Germany, between 70 and 90% of the population have already used medications phytotherapeutics at some point in their lives, and in the United States, an estimated 158 million adults use herbal products, and this trend is constantly growing. Globally, the phytotherapeutics market currently exceeds 60 billion dollars annually and continues to grow [16, 19].

3.2 Phytotherapeutics: Definitions and Characterization

The World Health Organization (WHO) definition of herbal medicine is a finished and labeled medicine that contains an active ingredient, which comes from the aerial or underground parts of a plant, other plant material, or combinations of these elements, such definition has already been addressed in some articles in the literature [16, 20].

Studies indicate that approximately 80% of the world's population turns to some type of plant in search of relief from symptoms or pain, due to their easy accessibility, low cost, and the perception that they are harmless by a large part of the population [21]. The growing interest in phytotherapeutics reflects the search for natural and alternative approaches to conventional medicine [1, 14]. However, it is essential to understand that, although derived from natural sources, phytotherapeutics are not without risks. Phytotherapeutics poisoning is a legitimate health concern, as inappropriate use of these substances can result in adverse side effects, toxicity, and unwanted drug interactions [1].

The active chemical components present in herbal medicines, which are responsible for the clinical efficacy of these medicines, are defined as the pharmacodynamic material basis of phytotherapeutics [22]. It is crucial to elucidate this pharmacodynamic material basis to evaluate the therapeutic efficacy of herbal medicines. Due to their unique effects resulting from the interaction of multiple components with multiple targets, herbal medicines have received increasing attention [23].

A wide range of chemically diverse secondary metabolites have been purified from plant bioactives and optimized to exert biological effects; however, there is still much to be explored regarding its clinical use. Despite this, recent scientific evidence, technological advances, and research directions indicate that compounds of natural origin will continue to be crucial sources of new medicines [18, 24].

From this perspective, recent research reports have driven the renewed popularity of traditional herbs and plant-derived medicines among researchers, even with advances in combinatorial chemical synthesis and the production of modern synthetic drugs [14]. Computational methods, for example, such as pharmacophore-based virtual screening and molecular docking, are consolidated tools that assist in the selection of plant materials with high potential for biological activity during drug discovery [23].

Another motivation for the interest in medicinal plant products is the fact that, in their struggles to combat disease amidst scarcity of socioeconomic resources, rural communities in developing countries have found traditional herbal therapies and plant derivatives a valuable source of resources [14].

3.3 Phytotherapeutics in Brazil

Medicinal and phytotherapeutic plants emerge as excellent alternatives to address issues related to sustainable objectives and goals to be achieved by 2030, especially regarding access and use of safe and quality medicines [25]. Considering that the phytotherapeutics industry is expanding globally and may represent an opportunity to strengthen the Brazilian pharmaceutical sector, given the abundance of biodiversity combined with the knowledge accumulated about the biological activities of plants by science and technology institutions, as well as by civil society [26].

The Alma-Ata Declaration [27] played a significant role in health advances, consisting of 10 points that emphasize primary health care. This document was crucial for the recognition of primary health care, thus strengthening the implementation of its guidelines in various health contexts around the world. In Brazil, legislation provided support for its implementation, with contributions from debates held at the 8th National Health Conference (CNS). The Organic Law of the Unified Health System (SUS) of 1990 came to implement these initiatives through its objectives and guidelines, being continuously implemented to achieve satisfactory results [28].

In the almost 30 years of the SUS's existence, one of the health policies implemented was the creation, in 2006, of the National Policy on Integrative and Complementary Practices (PNPIC). This policy, in constant evolution, aims to include alternative and complementary practices in health, with its last update dated 2018. This policy is fundamental to strengthening traditional knowledge and promoting the development of alternative treatments within the SUS [29].

The implementation of policies related to traditional medicine is a phenomenon observed throughout Brazil and has produced several results, whether positive or not, especially within the scope of primary health care in the Brazilian public health system. In Brazil, the National Health Surveillance Agency (ANVISA) assumed responsibility for monitoring the origin, effectiveness, and safety of the use of this class of medicines [30]. However, its implementation represents a challenge for management and multidisciplinary teams [31].

In 2020, 220 pharmaceutical laboratories operating in Brazil producing medicines were identified, of which 75.86% are aimed at treating anxiety and depression [25]. Some of the most commonly used classes of phytotherapeutics also include psycholeptics (sedatives, anxiolytics, and antidepressants), antivaricose and anti-hemorrhoidal drugs, digestive and hepatic aids, antispasmodics, tonics, laxatives, decongestants, anti-flu and cerebral circulation agents [30].

Phytotherapeutics, used for prevention, cure, symptom relief, or diagnosis, has been recognized by the World Health Organization since 1978 [27]. They have always been considered a valuable resource and widely sought after by human beings, who have found sources in nature to improve their quality of life. The advancement of knowledge about phytotherapeutics and their molecules boosted the development of traditional medicine, as new molecules were created based on substances from plants [30].

From studies carried out, it was possible to verify that there are cases of hypersensitivity observed as one of the most common side effects among users of herbal medicines, generally caused by the irrational use of medicinal plants, its effects can vary from temporary dermatitis to anaphylactic shock, in this way Thus, in favor of user safety, the Ministry of Health classified 2000 Phytotherapeutics Products as Herbal Medicines, establishing specific legislation. In the period from 2004 to 2008, the National Toxic-Pharmacological Information System (SINITOX) recorded 8,501 cases of plant poisoning in Brazil [30].

3.4 National Perspectives and Environmental Questions

As defined by the Ministry of Health, phytotherapy is “therapeutics characterized by the use of medicinal plants in their different pharmaceutical forms, without the use of isolated active substances, even if of plant origin” [32]. Despite this, many people mistakenly refer to herbal medicine as “alternative therapy” or “poor man’s medicine”. This therapy represents one of the oldest and richest methods used in natural medicine [30].

Without a doubt, the strengthening of alternative practices in the SUS enables closer interaction between health professionals and communities through this knowledge [28]. These health policies are relevant within the context of the Amazon, considering its vast biodiversity, and diversity of people, populations, and communities that share particularities in terms of beliefs, customs, and values transmitted between generations [33].

The importance of this policy is undeniable, including as a means to promote affirmation of people and populations, recognizing and respecting their knowledge, and allowing their choices in their therapeutic itinerary to be considered under their way of life [28]. The growing interest in Amazonian biodiversity is driven not only by its natural richness but also by the promise of advances in the field of phytotherapy. The vast variety of Amazonian flora offers fertile ground for research aimed at developing new phytotherapeutics, taking advantage of the medicinal benefits of its plants.

In the State of Pará, located in the Amazon region of Brazil, the Metropolitan Region of Belém (RMB) emerges as a prominent point for such studies. With its unique biodiversity and access to natural resources, Belém and its surroundings offer a conducive environment for the discovery and development of new herbal treatments. Furthermore, collaboration between research institutions, universities, and local communities can further enhance these efforts, promoting not only scientific advances but also the socioeconomic development of the region.

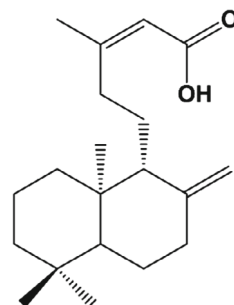
Research indicates that the population of Belém has access to a wide variety of medicinal and ritual plants, which are used both in home medicine and in religious rites. However, it has been observed that herbal sellers' therapeutic knowledge of the uses and doses of plants is limited, which may pose a problem regarding the effectiveness and risks associated with this practice [34].

4 Some Phytotherapy Medicines Used by the Amazon Population of the Rmb

Among the wide variety of natural compounds used by the population of the Amazon Region, especially in the Metropolitan Region of Belém, some phytotherapeutics commonly found in commerce and widely used by local residents stand out. Initially, it is important to discuss the species of “copaifeiras” or “copaifbas”, as the Amazonian species of *Copaifera* are known, mainly due to their emollient, anti-inflammatory, bactericidal, and anti-melasma pharmacological effects, in addition to their wide use in cosmetics [35].

Many researchers have already been carried out on the chemical constituents of copaíba (*Copaifera* spp.), including pharmacological studies of oleoresins, which are widely recognized for their anti-inflammatory, antitumor, and homeopathic antibiotic properties. Oleoresins are predominantly composed of sesquiterpenes and diterpenes, with copalic acid (Fig. 1) acting as the main biomarker [35].

Fig. 1 Chemical structure of diterpene *Copalic Acid*



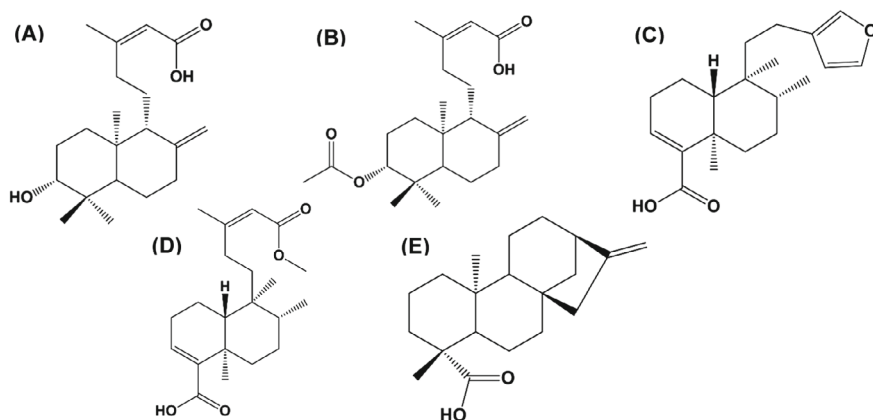


Fig. 2 Chemical structures of copaiba diterpenic acids (*Copaifera* spp.)

Studies have also identified other chemical compounds such as diterpenic acids, responsible for contributing to anti-inflammatory effects and other biological activities, including 3-hydroxy copalic acid (**A**); 3-acetoxy-cephalic acid (**B**); hardwickiic acid (**C**); kolavic acid 15-methyl ester (**D**); kaurenoic acid (**E**) (Fig. 2) [35].

Also worth highlighting is the broad spectrum of applications of oil from andiroba seeds (*Carapa guianensis* Aubl.), a plant native to the Amazon. In addition to its use in the wood industry and its ecological value, this oil has several medicinal and cosmetic benefits. Among the main chemical compounds present in andiroba oil are palmitic, oleic, linoleic acid, and limonoids. The latter are highly oxygenated tetranortriterpenoid compounds known for their wide range of biological activities. Although limonoids are predominantly found in plants belonging to the Meliaceae family, it is interesting to note that compounds of this type are found in only four other families: Rutaceae, Cneoraceae, Simaroubaceae, and Ptaeroxylaceae [36].

The biological properties of andiroba oil (there are also reports on the flower) are widely recognized, encompassing a series of therapeutic benefits. To date, highly effective analgesic, antibacterial, antifungal, and antimalarial activities have been identified, as well as anti-inflammatory, anticancer, antitumor, and anti-allergic properties. The oil has also been shown to be effective as a leprostatic, anti-ulcer, antipyretic, and anti-irritant. Additionally, it has been found useful in treating wounds, bruises, rheumatism, and ear infections. Its insecticidal and repellent properties are well known, along with its ability to inhibit the growth of pathogens. Studies also indicate its safety in acute and subacute doses [37].

In Fig. 3, some constituents of andiroba are represented with different therapeutic properties such as antileishmanial [36], insecticidal, anti-allergic, and anti-malarial, among others [37]. Are they: 17-Glycolyldeoxygedunin (A); Andirobin (B); Gedunin (C); Deoxygedunin (D); Deacetyldihydrogedunin (E); 7-Deacetoxy-7-hydroxygedunin (F); Hydroxy_acetoxy_diacetoxygedunin (G).

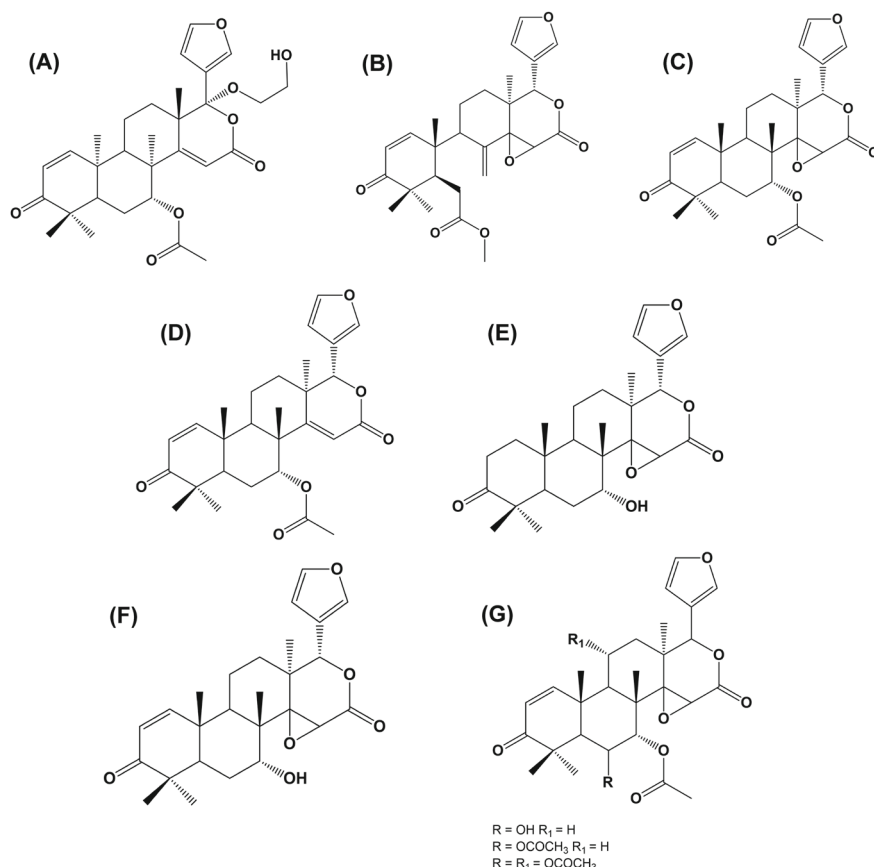


Fig. 3 Limonoids present in andiroba (*Carapa guianensis* Aubl.)

The Belém municipal health network has a Municipal Medicines List (REMUME), which presents nine phytotherapeutics for use in care at health units. These medicines are also part of the National Medicines List (RENAME), which has twelve phytotherapeutics and is updated every two years. RENAME selects and standardizes medicines indicated for the treatment of diseases or health problems in the population at all levels of care and lines of care within the scope of the SUS. The phytotherapeutics on the RENAME list are **Artichoke** (*Cynara Scolymus* L.); **Aloe vera** (*Aloe Vera* (L) Burm f.); **Mint** (*Menta x Piperita* L.); **Aroeira** (*Schinus Terebinthifolia* Raddi); **Holy Cascade** (*Rhammus Purshiana* D.C); **Espinheira Santa** (*Maytenus Illicifolia* Mart. ex Reissek); **Guaco** (*Mikania Glomerata* Sprengel); **Soy Isoflavone** (*Glycine Max* (L) Merr); **Cat Nail** (*Uncaria tomentosa* (Willd. ex Roem. & Schult)). In RENAME three more are added: **Devil's Claw** (*Harpagophytum Procumbens* DC ex Meissn); **Plantago** (*Plantago Ovata* Forssk) and **Willow** (*Salix Alba* L.) [38].

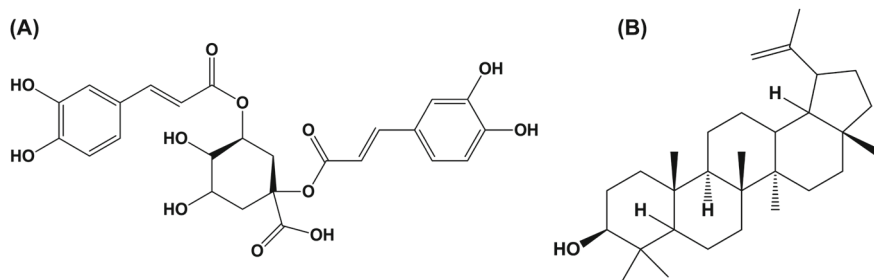


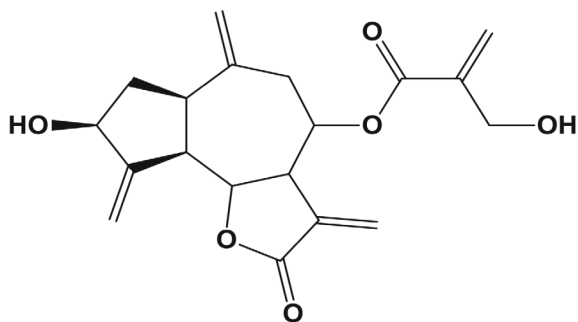
Fig. 4 Chemical structures of artichoke compounds

Some of REMUNE's phytotherapeutics were selected to present their pharmaceutical properties and images of important chemical constituents. Starting with the artichoke (*Cynara Scolymus* L.), its main chemical constituents in crude methanolic extract of the plant's leaves revealed the presence of phenolic acids, flavonoids, and sesquiterpenes. Cynarin (Fig. 4a) is reported as the active principle of artichoke, demonstrating hypolipidemic, antioxidant, choleric, cholagogue, and hepatoprotective activities, among others. Just as the triterpene lupeol (Fig. 4b) demonstrated activity in the proliferation of keratinocytes in the skin, nephroprotective activities in certain pathologies could cause damage to the renal tubule. Lupeol also has antitumor, antioxidant, and immune response mediator-releasing activities [39].

Another compound identified is cynaropicrin (Fig. 5), which can be found in several species of the Compositae family, which are rich in sesquiterpene lactones. Although these substances may be responsible for some toxic actions, the isolated extract of cynaropicrin showed inhibitory properties of tumor necrosis, which is involved in anti-inflammatory processes, comparable to the drug prednisolone (a potent anti-inflammatory glucocorticoid). Several authors also attribute other anti-tumor, antifungal, and antimicrobial pharmaceutical properties to cynaropicrin. However, in some cases, it can cause allergic dermatitis, cytotoxic effects, and inhibition of platelet secretion [39].

Other studies reveal several biological properties of the mastic tree (*Shinus terebinthifolia* Raddi), mainly in extracts made from the plant's bark and essential

Fig. 5 Chemical structure of the artichoke Cinaropicrin



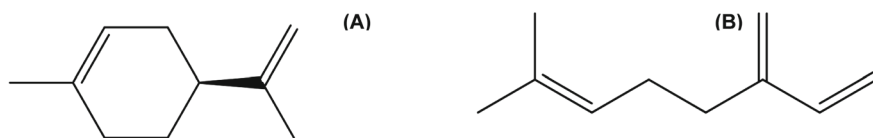


Fig. 6 Chemical structures of Aroeira

oil. The most abundant chemical compounds in these extracts are limonene and myrcene, represented respectively by A and B in Fig. 6. Other major compounds are α -pineno; β -pineno; germacreno-B; δ -3-careno; and β -elemeno [40].

Aroeira has anti-allergic, anti-inflammatory, and antimicrobial properties, the latter being possibly related to the presence of terpenes, tannins, and flavonoids. Flavonoids are also associated with the plant's anti-inflammatory and anxiolytic potential. Furthermore, the mastic tree has insecticidal activity against vectors of African malaria. Its leaves are used in folk medicine against fever, flu, headaches, body aches, sinusitis, rheumatism, and gastric discomfort. The plant can also be a flavoring, stimulant, diuretic, and carminative [40].

Guaco (*Mikania glomerata* Sprengel) became official in the first Brazilian Pharmacopoeia due to its notable therapeutic properties, including bronchodilator, anti-allergic, and anti-asthmatic. In addition to these, guaco has other therapeutic properties, such as antihemorrhagic, antivenom, antiviral, and antimicrobial actions. Phytochemical studies revealed that the *Mikania* genus contains important compounds such as lactones, coumarin, diterpenes, and sesquiterpenes [41].

Among the compounds presented in Fig. 7, 2H-1-Benzopyran-2-one (B) stands out, as one of the main constituents of guaco. This compound contributes significantly to several pharmacological properties of the plant, highlighting its therapeutic importance [41].

Studies conducted by Santana et al. (2014) attribute anxiolytic properties to *Mikania*, evidenced by its ability to reduce glutamate and aspartate concentrations in the hippocampus of mice. This reduction may contribute directly and/or indirectly to the anxiolytic effect of the plant [41].

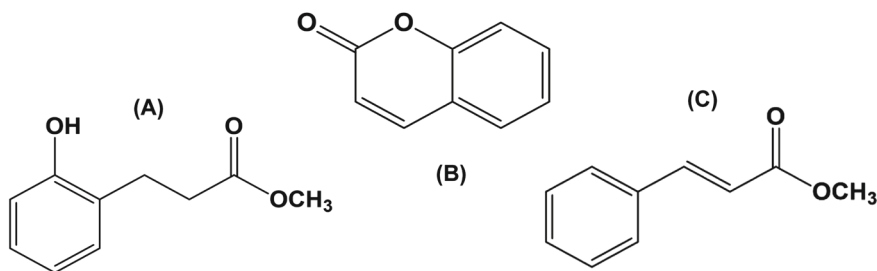


Fig. 7 Guaco chemical structures (2-hydroxyphenyl) methyl propionate (a); 2H-1-Benzopyran-2-one (b); Methyl cinnamate (c)

It is also important to highlight the remarkable properties of a cat's claw (*Uncaria tomentosa* (Willd. ex Roem. & Schult)), a plant from the Rubiaceae family. Originally from the Amazon rainforest and other areas of South and Central America, this climbing plant is rich in several phytoconstituent compounds, including glycosides, fatty acids, steroids, proanthocyanidins, and triterpenes. [42].

Phytotherapeutics derived from a cat's claw is available on the market in various forms, such as extracts, capsules, teas, decoctions, and tinctures. Its pharmaceutical properties have traditionally been employed in the treatment of a wide range of conditions, including fever, urinary tract and viral infections, asthma, abscesses, and wounds. Furthermore, Cat's Claw is recognized for its anti-inflammatory, antimicrobial, and antioxidant properties and its ability to strengthen the immune system [42].

Several studies also report the plant's effectiveness in treating cancer, gastric ulcers, inflammation, arthritis, and infections. Various chemical constituents in *Uncaria Tomentosa* extracts are reported to associate them with a wide range of biological activities. Among them, the chemical compounds are represented in Fig. 8, Uncarine F (A); Speciophylline (B); and Mitraphylline (C). They are responsible for anticancer, antineoplastic, anti-inflammatory, and antioxidant pharmacological activities [42].

Officially, the municipality of Belém implemented and implemented the Integrative and Complementary Health Practices program—PICS, especially phytotherapy, through ordinance no. 1151/2021-GABS/SESMA/PMB, late in 2021. Other Brazilian municipalities were already in the process of institutionalizing these practices since 2006. The Municipal Health Unit (UMS) in the Jurunas neighborhood was

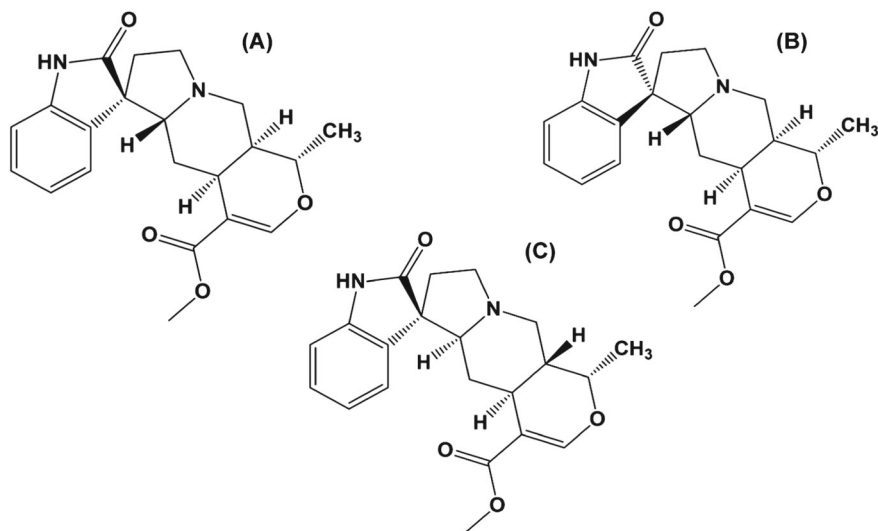


Fig. 8 Chemical constituents of cat's claw that have pharmacological activities

a pioneer in phytotherapy services in Belém. Later, this service was also implemented at the UMS of Carananduba, in the municipality of Mosqueiro [38].

Integrative and Complementary Practices (PICs) are of immense importance, especially in primary care with the use of phytotherapeutics, as they provide comprehensive care and promote the humanization of health care. Several studies highlight the numerous benefits of PICs, which not only improve physical and mental well-being but can also represent a more economical and sustainable treatment for different regions, thanks to the rich national biodiversity [38].

5 Computational Strategies in the Identification of Bioactive Compounds of Phytotherapeutics

According to Han et al. (2020), technological and health science advances have demonstrated significant evidence of the limitations associated with single-target drug discovery. As a result, the multi-drug, multi-target approach is emerging as the next step in drug discovery. Thus, the pharmacological efficacy of phytotherapeutics arises from the simultaneous binding of multiple chemicals to multiple sites.

Herbal medicines are gaining increasing prominence due to their distinct effects, multifaceted composition, and multiple targets. Unlike the approach of contemporary Western medicine, the use of these medications is largely grounded in practices and theories rooted in ancient philosophies. In the era of evidence-based medicine, scientific evaluation of herbal medicines, in line with modern approaches to medical practice, plays a crucial role in their development [23].

However, herbal medicines have a great diversity of chemical components and complex interactions with the human body; the information available on their pharmacokinetics, pharmacodynamics, efficacy, and safety is limited. Therefore, this gap in scientific evidence regarding therapeutic efficacy has become a concern for health authorities and the general public. Therefore, efforts are being made to understand the therapeutic efficacy of phytotherapeutics, as the combination of chromatographic separation techniques and spectroscopic methods is important to identify the natural products [23].

In the nineteenth and twentieth centuries, fractionation by bioassay was essential in the discovery of the active ingredients of phytotherapeutics. During this period, the important discovery of artemisinin, a bioactive component extracted from Qinghaosu, recognized for its antimalarial action, stands out. The antimalarial drug was subsequently developed, which resulted in a significant reduction in mortality associated with malaria [23].

With the development process of modern biotechnology, high-throughput screening has been fundamental in the investigations of bioactive herbal compounds. One of the prominent techniques was the integration of the cell membrane with

liquid chromatography, enabling the dynamic separation of compounds. Furthermore, this approach allowed observing the interactions of the target with the receptors, expanding the understanding of the mechanisms of action of phytotherapeutics [23].

Furthermore, computational methods, especially pharmacophore-based virtual screening and molecular docking, play a significant role in the discovery of new drugs and the selection of natural product compounds with high biological activity. These computational approaches allow for predicting the positioning of a ligand within the protein binding pocket and estimating the strength of the interaction through a coupling score, which makes it possible to elucidate the mechanisms of action of the investigated compounds [23].

Another promising alternative is network pharmacology, an approach that integrates network biology and polypharmacology in drug design. This technique has stood out as a powerful means to analyze the pharmacological mechanisms of herbal medicines systemically and at a molecular level. Network pharmacology was applied to investigate the molecular mechanisms of Pudilan (PDL), a formula composed of four herbs, with therapeutic potential against coronavirus 2, SARS-CoV-2. Integrated bioinformatics and network pharmacology analysis revealed that PDL can act to inhibit the cytokine storm by regulating and targeting several cytokines and chemokines. Furthermore, PDL has demonstrated the ability to prevent the virus from entering host cells and modulate the immune system, which may contribute to reducing the duration of COVID-19 [23].

Other chemical compounds of natural origin were analyzed due to their potential activity against SARS-CoV-2, being an alternative therapy or complement in combating the disease. One of the compounds was ivermectin, which has antiparasitic properties and may have antiviral activity, but its effectiveness is the subject of debate and investigation. Other studies suggest quercetin, a flavonoid from fruits, vegetables, and herbs, which has antiviral and anti-inflammatory properties. Another compound is curcumin, which is extracted from the turmeric root and has anti-inflammatory, antioxidant, and antiviral activities [43].

The polyphenol present in green tea (Epigallocatechin-3-gallate) has also been studied to inhibit the entry and replication of the COVID-19 virus in host cells. Furthermore, berberine has shown potential antiviral and anti-inflammatory activity and can be found in the roots of *Coptis* plants and the bark of trees such as *Phellodendron*. Finally, another natural compound is allicin, which can be found in garlic and has been studied for its antiviral and immunomodulatory properties [43, 44].

6 Conclusion

The use of phytotherapeutics in the Amazon Region, mainly in the Metropolitan Region of Belém, is a striking example of how traditional medicine can coexist and complement modern medicine. The richness of the Amazon flora, with its unique medicinal plants, represents an invaluable treasure for public health and

science. Indigenous and riverside communities, over the centuries, have developed vast knowledge about the therapeutic use of these plants, knowledge that has been increasingly recognized and valued by scientific research. Furthermore, over the years, other medicinal plants from different parts of the world have been incorporated into the local region.

The integration of phytotherapeutics into the public health system can provide several benefits. Firstly, it offers an accessible and often more economical alternative to synthetic medicines, which is especially relevant in areas with lower purchasing power. Furthermore, phytotherapeutics may have fewer side effects, making them a safer option for many patients.

The development and regulation of phytotherapeutics in the region also encourages environmental preservation. The sustainable demand for medicinal plants can promote responsible agricultural practices and the conservation of natural habitats, contributing to the preservation of Amazonian biodiversity. Sustainable cultivation and natural resource management programs not only protect the environment but also create jobs and strengthen the local economy.

It is essential, however, that the use of phytotherapeutics is accompanied by rigorous scientific research to ensure their effectiveness and safety. Research institutions and universities have a fundamental role in investigating the medicinal properties of plants used in the Amazon, as well as in training health professionals capable of integrating this knowledge into clinical practice.

Therefore, computational strategies have become indispensable tools in the identification of bioactive compounds in phytotherapeutics, playing a crucial role in the advancement of pharmacology and the bioprospecting of new medicines. These strategies offer an efficient and precise approach to analyzing the vast chemical diversity present in medicinal plants, optimizing the process of discovering and developing new drugs.

One of the main advantages of computational strategies is the ability to perform virtual screening of thousands of compounds simultaneously. Techniques such as molecular modeling, molecular dynamics, and molecular docking make it possible to predict the interaction between bioactive compounds and specific biological targets, such as proteins or enzymes. This not only speeds up the identification of substances with therapeutic potential but also significantly reduces the costs and time associated with experimental research.

Furthermore, chemoinformatics, which involves the use of computational tools to store, organize, and analyze chemical data, allows the creation of databases of bioactive compounds and their pharmacological properties, such as the Molecular Network (GNPS). Analyzing this data through statistical and machine learning methods can reveal patterns and relationships that would not be easily identifiable through traditional methods, facilitating the discovery of new drug candidates.

In the Amazon region, where biodiversity is immense and still little explored, computational strategies can be especially valuable. They allow for a more systematic and comprehensive exploration of the medicinal properties of native plants, enhancing the discovery of new compounds that can be developed into innovative medicines. Furthermore, these tools can help value traditional knowledge, integrating

it with modern scientific approaches to validate and optimize the use of medicinal plants.

In conclusion, computational strategies are essential for the identification and development of bioactive herbal compounds, offering an efficient and innovative approach to pharmacological research. They complement and enhance traditional and experimental knowledge, contributing to the discovery of new medicines and the sustainable valorization of natural resources. Valuing the use of phytotherapeutics in the metropolitan region of Belém is not only a matter of health but also of sustainable development and cultural and environmental preservation. By integrating traditional knowledge with modern science, it is possible to create a more holistic and inclusive health system that respects and takes advantage of the vast natural and cultural resources of the Amazon.

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Drug-Like Properties and Therapeutical Potential of *Calendula officinalis* L. Active Ingredients



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Abstract *Calendula officinalis* L. is a medicinal herb from the Asteraceae family with a long history of usage in culinary and folk medicine. The plant originates from southern Europe and the Eastern Mediterranean area and contains a wide variety of bioactive compounds, including flavonoids, triterpenoids, glycosides, saponins, carotenoids, volatile oils, amino acids, carbohydrates, steroids, sterols, and quinines. These phytoconstituents have broad applicability in the food and cosmetic industries, as they produce a matrix of multifaceted biological effects with therapeutic applications. The active ingredients of *Calendula officinalis* L. are generally associated with anti-cancer, antimicrobial, antiparasitic, antioxidant, anti-inflammatory, hypoglycemic, hypolipidemic, and wound healing properties in experimental and clinical trials. They emerge as an interesting source of drug-like chemical entities for the conception and design of medicines to treat especially inflammatory, antineoplastic, and infectious diseases. This chapter aims to describe the drug-like and some pharmacokinetic properties of the main active ingredients from *Calendula officinalis* L. as well as their therapeutic properties.

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1 Introduction

Medicinal plants have a long history of use in the treatment of diseases [1, 2]. Plants provide a diverse array of natural products with various therapeutic properties, which are continually researched for the development of new drugs. For centuries, traditional medicines have relied on these natural products to treat numerous diseases. Today, many pharmaceutical drugs are derived from these natural sources. Natural products are composed of numerous bioactive compounds, which exhibit biological activity against a range of disease-causing agents [3, 4].

Calendula officinalis L., commonly known as marigold, is an annual herbaceous plant with notable medicinal properties. The genus *Calendula* derives from the Latin word “*Calendae*,” meaning “first day of each month,” and includes about 25 species such as *C. officinalis*, *C. sellata*, *C. arvensis*, and *C. suffruticose*. Belonging to the Asteraceae family, *C. officinalis* is also referred to as wonder, verrucaria, and golden daisy [5, 6].

Calendula officinalis is extensively cultivated in China and India and is prevalent in Mediterranean regions and Central Europe. Historically, it was grown by Greeks, Arabs, Egyptians, and Hindus, with its flowers used medicinally since the twelfth century. The plant thrives in sunny locations across North America and Europe, blooming from May to October [7, 8].

The plant grows to a height of 30–60 cm. Its leaves are slightly toothed, lanceolate, and alternate, with glandular hairs on both sides. The lower leaves are spatulate, while the upper leaves are oblong to lanceolate. The stem is solid, hairy, and angular, ending in a composite flower head measuring 5–7 cm. This flower head consists of tapered lanceolate sepals covered in glandular hairs and yellow-orange tubular florets. The peripheral flowers are ligulate and fertile, whereas the central flowers are tubular and sterile. The plant produces achene-like fruits that are curved and narrowly oblong. Flowering is most intense from late spring to late summer, with flowers closing at dusk and opening at sunrise. The powder form of *C. officinalis* is yellowish-brown with a slightly bitter taste and characteristic odor, containing corolla fragments, pollen grains, stigma fragments, sclerenchymatous cells, and ovary wall fragments with brown pigment [5, 6].

Calendula officinalis is a rustic species, preferring clay soils but not demanding specific soil types. It germinates best at temperatures between 18 and 24 °C but can withstand higher temperatures later. The plant prefers temperate climates and can resist drought and frost, growing from sea level up to 1.000 m. Historically, it was used as an anti-inflammatory and antiseptic agent during the First World War and the American Civil War, with its medicinal use becoming more widespread from the thirteenth century, especially for wound healing [6, 9].

Calendula exhibits various pharmacological activities, including anti-inflammatory, antioxidant, wound healing acceleration, neuroprotective, hepatoprotective, immunostimulant, antimutagenic, molluscicidal, genotoxic, larvicidal, antiparasitic, nephroprotective, hypoglycemic, gastroprotective, antibacterial, antifungal, antiviral, and insecticidal effects. Ethnotherapeutically, it is known for

its excitant, emmenagogue, antispasmodic, antiscorbutic, ophthalmic, antiseptic, sudorific, bile duct stimulant, antiemetic, analgesic, skin-toning, vasodilator, emollient, astringent, antiallergic, resolutive, diuretic, and diaphoretic properties [6, 10].

2 Therapeutical Profile of *Calendula officinalis* L.

The butanoic fraction of *C. officinalis* has a significant free radical scavenging and antioxidant activity. *C. officinalis* flowers are helpful in reducing inflammation and speeding up wound healing. As an antiseptic agent, it is used in the treatment of skin conditions, from eczema to skin ulcerations and burns. In addition, *C. officinalis* is used for the treatment of inflammation and ulcers of the stomach [11]. The anti-inflammatory and healing activities of *C. officinalis* are related to a great diversity of active ingredients, mainly triterpene alcohols, triterpene oligoglycosides, monoesters, triterpenoids and flavonoids. In addition, the use of the hydroglycolic extract of the plant has been associated with a high rate of healing of diabetic foot ulcers [12, 13].

The species also provides chemical constituents with antitumor potential. Plant extract showed a potent in vitro inhibition of tumor cell proliferation when tested on a wide variety of human and murine tumor cell lines. The mechanisms of inhibition are related to cell cycle arrest in G0/G1 phase and caspase-3 induced apoptosis. In vivo experiments in nude mice showed that *C. officinalis* extract inhibited tumor growth of Ando-2 melanoma cells and prolonged the survival day of the animals [14].

Calendula officinalis shows as well potential for restoring fertility in women with polycystic ovary syndrome (PCOS) and ovulation failure and can be used alone or in combination with other medications. In mice with PCOS and non-ovulation, the plant extract improved oxidative stress, restored folliculogenesis, and increased ovulation. In addition, the extract reduced lipid infiltration in liver tissue, modulated oxidative stress, and influenced the lipid profile. In treatment of periodontal disease, a mixture of extracts of *C. officinalis* and *H. perforatum* decreased inflammation, alveolar bone loss, and tissue oxidative stress, with an impact on the cytokine IL-1 β , which is associated with periodontal attachment loss [15–17].

In Brazil, calendula is widely used in folk medicine, in the form of flower tea, as an expectorant, spasmodic and to treat nervous anemias. It is also used to fight inflammation of the viscera, wounds, cancerous wounds, sore throat, allergy and jaundice; The petals macerated in alcohol swabs are used for open wounds [6].

Phytochemical studies with calendula flowers and receptacles record a broad spectrum of chemical compounds, especially flavonoids, polysaccharides, carotenoids, triterpenes, triterpene saponins, tannins, coumarins and phenolic acids, as well as esters of hydrocarbons, fatty acids, sesquiterpene glycoside, sterols, polyacetylenes

and a volatile oil that contains a large number of alcohols and sesquiterpene hydrocarbons. The anti-inflammatory properties of this species are due to its carotenoids and flavonoids [5, 6, 10].

The flavonoids have high anti-inflammatory properties and prevent the release of histamine and harmful enzymes that cause swelling and tenderness and improve pain and redness [11]. In terms of anti-inflammatory activity, the herbal formulation effectively inhibited the production of pro-inflammatory mediators, including prostaglandin E2 (PGE2), tumor necrosis factor-alpha (TNF-alpha) and interleukin 6 (IL-6). *C. officinalis* extract inhibits pro-inflammatory cytokines, such as IL-6, TNF-alpha, and interferon-gamma (IFN-gamma), cyclooxygenase-2 (COX-2), and subsequent prostaglandin synthesis. Previous research has demonstrated the ability of marigold flower extracts to inhibit nitric oxide production through the modulation of inducible nitric oxide synthase (iNOS) activity. At low concentrations, it has been proven to have antioxidant properties, which act by inhibiting reactive oxygen species (ROS) and reactive nitrogen species (RNS) [18].

Early research on *Calendula officinalis* has shown that extracts of this plant species have an inhibitory effect on acetyl and butyrylcholinesterase. In this context, the methanolic extract of *C. officinalis* flowers showed the more pronounced activity. It is noticeable also that *C. officinalis* extracts exhibit a broad spectrum of biological activity in the central and peripheral nervous system. In particular, the protective effect of extracts against monosodium glutamate (MSG)-induced neurotoxic oxidative stress and excitotoxic brain damage has already been demonstrated [19].

3 Bioactivity of *Calendula officinalis* Molecules

Calendula officinalis contains large list of active metabolites such as flavonoids, coumarins, terpenes, quinones, saponins, and carotenoids, which have inherent pharmacological properties that justify the therapeutic use of the plant (Table 1).

3.1 Antioxidant Activity

Key compounds with antioxidant properties include carotenoids like lutein, neoxanthin, and their derivatives, α -carotene, β -carotene, and γ -carotene; flavonoids such as isorhamnetin, quercetin, rutin and their derivatives; α -tocopherol, a quinone; and esculetin, a coumarin [10, 20, 21]. These compounds scavenge free radicals, reducing oxidative stress and potentially preventing degenerative changes caused by lipid peroxidation [18, 21].

Table 1 Bioactivity of *Calendula officinalis* L. most important bioactive ingredients

Chemical class	Molecules	Bioactivity
Carotenoids	Lutein	Antioxidant, anti-proliferative, neuroprotective, nephroprotective
	Neoxantin (and derivatives)	
	Alpha-carotene	
	Gamma-carotene	
	Beta-carotene	Antioxidant, anti-proliferative, neuroprotective, nephroprotective activities, potential anti-metastatic
Coumarins	Esculetin	Antioxidant, anti-inflammatory
	Scopoletin (and derivatives)	Antimicrobial, anti-inflammatory, antioxidant, neuroprotective, immunomodulatory, anti-proliferative, analgesic, hypotensive
Flavonoids	Isorhametin (and derivatives)	Antioxidant, neuroprotective
	Quercetin (and derivatives)	
	Rutin (and derivatives)	Antioxidant, neuroprotective antiplatelet, antiviral, antihypertensive
Quinones	Phylloquinone (vitamine K)	Coagulative
	α -tocopherol (vitamine E)	Antioxidant, anti-inflammatory
Triterpen glycosides	Calendulaglicosyde A (and derivatives)	Antiedemateous, anti-inflammatory
	Calendulaglicosyde B (and derivatives)	Antiedemateous, anti-inflammatory, anti-ulcerous
	Calendulaglicosyde C (and derivatives)	Antiedemateous, anti-inflammatory
	Calendulaglicosyde F (and derivatives)	Antiedemateous, anti-inflammatory, cytotoxic (Calendulaglycoside F6'-O-nbutyl ester)
	Calendulaglycoside G6'-O-methyl ester	Cytotoxic
Triterpenic saponins	Faradiol	Antiedemateous, anti-inflammatory, anti-proliferative
	Taraxasterol (and derivatives)	Antiedemateous, anti-proliferative
Triterpenoids	Oleanic acid (and derivatives)	Antibacterial, anti-helminthic, anti-ulcerous

3.2 Antiproliferative/Cytotoxic Activity

Carotenoids such as beta-carotene and lutein exhibit antiproliferative effects against certain tumor cells by regulating pro-inflammatory cytokines. Additionally, triterpenes like Calendulaglycoside F6'-O-nbutyl ester and Calendulaglycoside G6'-O-methyl ester show antitumor activity, inhibiting tumor cell proliferation and inducing apoptosis, results that characterize such molecules as promising agents against leukemia, colon cancer, and melanoma [21].

Helianol, a triterpene isolated from the flowers of this species, can be considered a potent antitumor agent. The PF2 molecule, isolated from this same plant, has been indicated as a probable stimulant of human lymphocytes, helping in the treatment of cancer [6]. Despite this, its use in the management of human cancer is generally limited to the treatment of radiochemotherapy-induced side effects [22]. However, its efficacy for the prevention and treatment of radiodermatitis is still classified as “probable” by the Oncology Nursing Society, a category in which more empirical studies and synthesis of evidence on its efficacy are needed [23].

3.3 Neuroprotective and Nephroprotective Activity

Due to their antioxidant properties, carotenoids like lutein, neoxanthin, alpha, beta, and gamma-carotene, as well as esculetin and flavonoids such as isorhamnetin, quercetin, and rutin, have shown potential in experimental studies for nephroprotection and in preventing neurodegenerative diseases caused by oxidative stress [21, 24]. Chlorogenic acid, 3,4-dicaffeoylquinic acid, rutin, isorhamnetin 3-O-glucoside and calendulose E obtained from *C. officinalis* flower extracts seems also to bind to the pocket region of the Hsp 90-alpha protein and then concurrently activate the PI3K/Akt signaling pathway and inhibit the ERK signaling pathway. As multifunctional agents these molecules are promising candidates for improving the symptoms associated with Parkinson's disease [25].

3.4 Anti-inflammatory/Anti-edematous Activity

Triterpenoid glycosides and saponins are key anti-inflammatory metabolites in this species, with compounds like taraxasterol, faradiol, and their derivatives being highly active. These substances reduce the expression of interferon-gamma and cyclooxygenase-2, which is also inhibited by alpha-tocopherol and esculetin, thereby modulating the inflammatory response. Carotenoids such as lycopene reduce inflammatory cytokines and prostaglandins. Edema inhibition is linked to the anti-inflammatory activities of bioactive ingredients such as alpha-tocopherol, calendulaglycosides A, B, C, F, faradiol, and taraxasterol, enhancing macrophage proteolytic

activity [5, 6, 10, 21, 24, 26]. The esters faradiol and taraxasterol obtained from *Calendula* flowers showed superior anti-edematous effect to indomethacin when inducing edema in the ear of rats [6].

3.5 Antimicrobial Activity

Esculetin, rutin, and oleanolic acid exhibit antibacterial activity by inhibiting proliferation and inducing lysis of Gram-positive bacteria. These substances also demonstrate antiparasitic activity by inhibiting helminth larval states and coenzyme Q synthesis by the protozoan *Plasmodium*, and show insecticidal and acaricidal effects [6, 21]. The high concentration of saponins is commonly related to *C. officinalis* antiparasitic activity [27]. *Calendula* extract also inhibits the reverse transcriptase of HIV and has shown potential in vitro to suppress the replication of influenza viruses APR-8, influenza A2, and herpes simplex, indicating also an antiviral potential [21].

3.6 Immunomodulatory Activity

In addition to its antimicrobial, anti-inflammatory, antioxidant, and antiproliferative activities, esculetin also has immunomodulatory effects by enhancing macrophage phagocytic activity and reducing splenocyte proliferation and adaptive immune system activation [21, 26].

3.7 Anti-ulcer Activity

Calendulaglycoside B and oleanolic acid have anti-ulcerative effects against venous, cold-induced, and gastrointestinal ulcers, with effectiveness comparable to ranitidine, a commonly used drug to treat ulcers, erosive esophagitis and gastroesophageal reflux disease that reduces the secretion of gastric acid by reversible binding to histamine H2 receptors [28]. This activity is due to the gastroprotective effects of increased mucus production and decreased acid secretion stimulated by these metabolites [21].

3.8 Coagulant/Antihypertensive/Antiplatelet Activity

Esculetin and rutin are known for their antihypertensive properties by inducing vasodilation, thereby lowering blood pressure. Additionally, rutin is part of a group of flavonoids that inhibit platelet activation and aggregation by binding to platelet receptors [24, 26, 29]. Phylloquinone, also known as vitamin K, is a cofactor for

gamma-carboxylase, which modulates and activates clotting factors II, VII, IX, and X, making its primary activity coagulant, useful in controlling hemorrhages and coagulation disorders [30].

4 Drug-Like and Pharmacokinetic Properties

The clinical efficacy and safety of a drug is determined by its molecular properties and targets in humans [31]. Calendula molecules have a variety of drug targets and pharmacological properties characterized by its broad range of bioactive compounds and drug candidates, such as flavonoids, terpenoids, coumarins, quinones, and volatile oils that contribute to its diverse therapeutic potential.

The role of pharmacokinetics in drug discovery is to support the optimization of ADME (absorption, distribution, metabolism and excretion) properties of drug candidates to attain molecules with clinical feasibility that achieve a concentration–time profile in the body adequate for manifesting the desired efficacy and safety profile [32]. Furthermore, some drug candidates due to inadequate pharmacokinetics properties can induce drug–drug interactions that limit their use in clinical practice by producing unpredictable pharmacological effects and lead to adverse events that have potential to cause irreversible damage to the organism [33].

ADME screening has become increasingly important in identifying and optimizing drug candidates. ADME prediction using *in silico* models has become essential and is ideally used to evaluate a large number of molecular structures according to very diverse parameters in order to steer the selection of which chemicals to synthesize, test and promote, with the final goal to identify those with the best chance to become an effective medicine for the patients [34–36].

It is evident that Calendula is a plant species with high therapeutic value, due to its countless applications as a medicine in both traditional and popular medicine. Thus, the chemical compounds extracted from this plant species emerge as drug candidates and can naturally also inspire the development of new therapeutic chemical entities, especially for inflammatory, infectious, metabolic, endocrine and carcinogenic diseases [21]. Then, it becomes essential to understand the drug-like properties of its major medicinal ingredients in order to fully recognize molecules with a favorable profile for rational drug development. This task could be addressed by employing *in silico* prediction of drug-like and pharmacokinetic properties (Table 2).

In general, the bioactive molecules from Calendula most likely to have oral bioavailability are found among coumarins, flavonoids, and volatile oils. For example, quercetin, a flavonoid present in foods like fruits, seeds, vegetables, bracken fern, coffee, and tea, has been extensively studied for its antidiabetic, antibacterial, anti-inflammatory, anti-Alzheimer's, anti-arthritis, antioxidant, anticancer, cardiovascular, and wound healing properties [37]. Therefore, Calendula is a promising source of lead compounds with broad bioactivity and chemical features suitable for the development of orally administered drugs targeting inflammatory, infectious, neurological, neoplastic, and endocrine diseases.

Table 2 Drug-likeness and pharmacokinetic profile of *Calendula officinalis* L. medicinal ingredients predicted with SwissADME web tool [36]

Molecule	Source	Chemical group	Lipinski filter approval	GI absorption	CYP inhibition	BBB permeant
Lupeol	Flower	Terpinoids	Yes	Low	No	No
psi-Taraxasterol			Yes	Low	No	No
Erythrodiol			Yes	Low	No	No
Calendulaglycoside A			No	Low	No	No
Calendulaglycoside B			No	Low	No	No
Cornulacic acid acetate			Yes	Low	CYP2C9	No
Isoquercetin	Flower	Flavonoids	No	Low	No	No
Rutin			No	Low	No	No
Calendoflavoside			No	Low	No	No
Quercetin			Yes	High	CYP1A2 CYP2D6 CYP3A4	No
Isorhamnetin			Yes	High	CYP1A2 CYP2D6 CYP3A4	No
Isorhamnetin-3-O-beta-D glycoside			No	Low	No	No
Narcissin	Flower	Coumarins	No	Low	No	No
Esculetin			Yes	High	CYP1A2	No
Scopoletin			Yes	High	CYP1A2	Yes
Umbelliferone			Yes	High	CYP1A2	Yes
Cubenol	Flower	Volatile oils	Yes	High	CYP2C19 CYP2C9	Yes
Alpha-cadinol			Yes	High	CYP2C19	Yes
Oplopanone			Yes	High	No	Yes
Sabinene			Yes	Low	No	Yes
Limonene			Yes	Low	CYP2C9	Yes
Alpha-pinene			Yes	Low	CYP2C9	Yes
p-cymene			Yes	Low	CYP2D6	Yes
Nonanal			Yes	High	No	Yes
Carvacrol			Yes	High	CYP1A2	Yes
Geraniol			Yes	High	No	Yes
Nerolidol			Yes	High	CYP1A2 CYP2C9	Yes
t-muurolol			Yes	High	No	Yes
Palustrol			Yes	High	CYP2C19 CYP2C9	Yes

(continued)

Table 2 (continued)

Molecule	Source	Chemical group	Lipinski filter approval	GI absorption	CYP inhibition	BBB permeant
Phylloquinone	Leaves	Quinones	Yes	Low	CYP1A2	No
Plastoquinone			No	Low	No	No
Alpha-tocopherol			Yes	Low	No	No
Ubiquinone			No	Low	No	No
Calenduloside B	Roots	Terpenoids	No	Low	No	No
Calenduloside C			No	Low	No	No

Inhibition of cytochrome P450 (CYP450) enzymes is the most common mechanism leading to drug-drug interactions. Cytochrome P450 represents a family of isozymes responsible for biotransformation of drugs via oxidation. The enzymes are heme-containing membrane proteins, which are located in the smooth endoplasmic reticulum mainly in the liver. Numerous factors affect drug biotransformation, including cytochrome P450 inhibition that in general results in decreased metabolism and increased serum levels of the unmetabolized chemical entity, leading to a greater potential for toxicity. On the other hand, for drugs whose pharmacological activity requires biotransformation from pro-drug form, inhibition can lead to decreased efficacy [38].

Each type of interaction causes an intrinsic consequence to human body and involves a distinct clinical management strategy. In clinical practice, the need for the use of multiple drugs is common, as patients often present with numerous chronic diseases. This is why a comprehensive understanding of mechanisms of CYP450-mediated metabolism inhibition is needed to prevent or mitigate eventual harmful interactions [39].

Considering the CYP450 inhibition profile suggested for the bioactive molecules of *Calendula* (Table 2), it is evident that there is an inherent inhibitory potential in compounds across various chemical classes (terpenoids, flavonoids, coumarins, volatile oils, and quinones). This is particularly notable for the isoforms CYP1A2, CYP2D6, CYP2C9, CYP2C19, and CYP3A4, all of which are related to drug metabolism in the human body. This indicates that *Calendula* may have the capacity to inhibit the metabolism of other drugs. This is extremely important because *Calendula* is often used against clinical conditions requiring drug combinations, such as chronic inflammatory diseases. Consequently, one potential outcome is the inhibition of the metabolism of other drugs or prodrugs, which can lead to increased toxicity or reduced efficacy, respectively. Both scenarios are problematic within any pharmacotherapeutic protocol.

Given that *Calendula* has been investigated for its therapeutic potential against neurodegenerative diseases, the ability of its molecules to cross the blood–brain barrier (BBB) and reach adequate concentrations in the central nervous system is a

highly desirable characteristic. The molecules most likely to cross the blood–brain barrier belong to the coumarins and volatile oils groups (Table 2).

One of the most vital components synthesized by aromatic plants are essential oils that can be extracted from various parts of the plants, such as flower, bark, leaf, root, or peel. The volatile oils from diverse species have been proposed as prevention and treatment strategies for anti-ageing and neurodegenerative disorders, including Alzheimer’s disease, Parkinson’s disease, Huntington’s disease and amyotrophic lateral sclerosis [40]. Thereby, the volatile oils from *Calendula* seems to be an interesting source of potential neuroprotective agents for drug design purposes.

5 Conclusion

Calendula officinalis L. emerges as a valuable source of bioactive drug-like molecules with the potential to inspire the development of medicines capable of treating various public health conditions, such as inflammatory, rheumatic, endocrine, metabolic, carcinogenic, and infectious diseases. These diseases have standard drug therapies that are usually associated with limitations, particularly related to the induction of harmful side effects. Many active ingredients from *Calendula* exhibit chemical features aligned with the concept of drug-likeness, especially flavonoids, coumarins, and volatile oils, making them promising candidates for screening bioavailable drug candidates. This can help expand the therapeutic options available for globally relevant clinical conditions. However, some of these molecules may have the capacity to inhibit the metabolism of other drugs and generate drug–drug interactions capable of increasing toxicity or reducing the efficacy of traditional drugs. Volatile oils and coumarins are the most likely to cross the blood–brain barrier and demonstrate bioactivity in the central nervous system.

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Herbal Medicines: From History to Current Research—A Comprehensive Survey



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Abstract The use of natural resources, particularly plants, for prophylactic and curative purposes dates back to ancient times and has been employed by various indigenous peoples. To survive, plants produce a variety of substances known as secondary metabolites, which are known for their distinct biological properties useful in treating diverse pathologies. Integrating empirical, traditional, and scientific knowledge has led to an increased use of natural resources as therapeutic solutions. Consequently, regulating the production processes of herbal medicines is vital to ensure their safety and efficacy. With this context, the primary aim of this chapter is to explore herbal medicines and related aspects, offering a historical perspective alongside the advances and processes involved in their extraction and production.

Keywords Natural resources · Secondary metabolites · Herbal medicines

1 Introduction

Throughout the history of humanity, plants have been used in people's daily lives, from clothing to decoration and energy resources [1]. Over time, people also discovered their medicinal properties and their usage depended on the species available regionally.

Some of these known plants were selected and cultivated, improving knowledge about them and their healing or poisonous properties. There are writings in Asia and

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Europe that prove these transfers of information through documents. In contrast in Africa and America, this knowledge transfer took place through word of mouth for generations [2].

Phytotherapy, the study of the medicinal properties of plants, has been a cornerstone of traditional medicine. Additionally, with the advent of pharmacological research, some herbal medicines have transitioned into modern treatments. For instance, artemisinin, extracted from *Artemisia annua*, a plant long used in Chinese medicine to treat fevers, is now a key component of modern antimalarial drugs. Herbal medicine and plants also encompass fungal products, bee products, minerals, shells, and even specific parts of animals [3, 4].

Herbal medicines are a prevalent aspect of healthcare worldwide. In highly developed nations, they often serve as a complementary therapy [2]. Conversely, in developing countries, an estimated 80% of individuals rely on herbal medicine as their primary healthcare [5]. This global prevalence underscores the importance of understanding the use, challenges, and regulation of herbal medicines.

Despite their widespread use, there remains a gap in scientific evidence regarding the safety and efficacy of many herbal medicines used today and clear standards concerning their purity and dosage [6]. This lack of detailed information becomes even more problematic as the global interest in these medicines as a therapeutic option grows. The increasing demand highlights the need for high-quality, effective, safe products. Complicating matters, the multi-component nature of herbal medicines can lead to complex interactions with the human body and metabolism, which raises concerns about their potential toxicity [7].

This chapter will delve into the history of herbal medicines, spanning from antiquity to the present day. We will explore the main active ingredients used, the processes of extraction and production, the challenges and advances in herbal medicine research, and the regulatory agencies' roles in product regulation and quality control. This comprehensive overview will equip readers with a thorough understanding of the subject matter.

2 The History of Herbal Medicines

Since the dawn of humanity, plants have served dual roles in human civilization: as both food and medicine. While the exact timeline of when plants were first incorporated into human routines is unclear, the discovery of medicinal plant residues dating back approximately 65,000 years ago in Neanderthal archaeological sites in the Shanidar caves of northern Iraq provides some of the earliest evidence of their use [8].

In these sites, archaeologists identified the presence of pollen from plants of the species yarrow (*Achillea millefolium* L., *Asteraceae*), chamomile (*Matricaria recutita* L., *Asteraceae*), centaury (*Centaurium erythraea* Rafn., *Gentianaceae*), mallow (*Malvaceae*) and ephedra (*Ephedraceae*) [9]. This discovery was corroborated when

they found traces of yarrow and chamomile in Neanderthal teeth at an archaeological site in Spain dating back 50,000 years [10].

Remains of a fruiting body of the birch polyp (*Piptoporus betulinus*) were found in a 5300-year-old Iceman, a natural mummy named Ötzi, which was recovered from a thawed glacier in the Thiolese Alps [8]. Nowadays, we know that a fungus found with it contains bioactive compounds that have anti-inflammatory and laxative effects [11].

In addition to analyzing humanoid species, a Sumerian clay slab from Nagpur composed of 12 recipes for preparing medicines, in which more than 250 plants are used [12] was also found. Other references to plants with medicinal uses can be found in the Indian sacred books Vedas and also the Chinese book written by Emperor Shen Nung around 2500 B.C. [13].

In antiquity, Dioscorides, known as the “father of pharmacognosy” was the most prominent writer on plant drugs. As a doctor and soldier in the Roman army, on his travels, he studied medicinal plants, and around 65–77 A.D., he wrote a document called *Materia Medica* of Dioscorides, which served as the basis for phytotherapy practiced until 1600 A.D. [12]. This document contains more than 600 herbal medicines and is considered one of the most influential herbal medicine books in history and a precursor to all modern pharmacopeias [8]. In addition to the description of the plants, there is information about where they occur and how they are cultivated.

A contemporary of Dioscorides, called Pliny the Elder, also contributed to the use of plants as natural medicines. In his book “*Historia Naturalis*” there are around 1000 medicinal plants described. Both works, by Dioscorides and Pliny, incorporate all the knowledge of the time in which they lived [14].

In Europe, medicinal knowledge about plants did not change significantly until the seventeenth/eighteenth centuries [15]. The studies of herbal medicinal plants were restricted to monks as the monasteries had gardens in which medicinal herbs were grown and where herbal cures were produced [16].

In the mid-seventeenth century, the physician and botanist Nicholas Culpeper wrote the book “*The English Physician*”, in which he related herbal medicines to the signs of the zodiac, causing the peak of popularity in herbal treatments [16]. In this document, the doctor identified several herbs that have therapeutic effects, such as wintergreen, which contains the analgesic salicin and was also described by Hippocrates around 400 B.C. [17].

The discovery of the American continent had a significant impact on herbal medicine. Both tobacco and chocolate were seen as medicines in Europe. According to Nicholas Monardes, a Spanish doctor, tobacco could cure 36 health problems [18]. Another herbal medicine from Peru also reached Europe in the 1630s: the ‘Jesuit bark’, which contained quinine, an alkaloid that was used to prevent and treat malaria [19, 20].

On the Asian continent, mainly in traditional Chinese medicine, the Chinese Manual of *Materia Medica* was compiled during the Han Dynasty, which describes about 252 herbal medicines [21]. Later, during the Ming Dynasty, another compilation was also written and is considered one of the most significant *Materia Medica*

ever written: the *Compendium of Materia Medica*. This compilation was organized by the doctor Li Shizhen and contains around 12 thousand herbs and their prescriptions [8].

In addition to China, there is evidence of the use of herbal medicines in India, such as *Charak Samhita*, which is a pharmacopeia used by traditional Indian medicine that shows the action of more than 100 thousand plants and their derivatives [22]. In America, herbal medicine gained popularity only in the nineteenth century through the actions of Samuel Thomson, an American herbalist and botanist who developed “Thomsonian Medicine”, an alternative system of medicine [23].

At the beginning of the nineteenth century, a turning point occurred in the knowledge and use of medicinal plants due to the discovery, proof, and isolation of poppy alkaloids, quinine, and other plants. This marked the beginning of pharmaceutical research, which, fueled by modern chemical techniques, led to the discovery of active plant substances like tannins, vitamins, and oils [12].

With the advancement of phytotherapy, in the mid-nineteenth century, there was the emergence of apothecaries in which medicines were prepared based on plants by extraction and concentration. Production of medicines such as morphine, quinine and strychnine were common in these places. It was in these small apothecary environments that pharmaceutical companies had their origins [8].

At the end of the nineteenth century, there was a risk of eliminating the use of medicinal plants as therapy. Several authors wrote that medicines obtained from them were deficient due to changes during the drying process. By the early twentieth century, methods to stabilize dried plants were proposed, and significant efforts were made to optimize their cultivation and manufacturing conditions [24].

3 Classification of Medicinal Plants

Medicinal herbs can be classified in various ways, such as by their usage, decorative value, constituents, life cycle, and botanical taxonomy [25]. Given their multifaceted benefits, many plants fit into multiple categories. One approach to classifying medicinal plants is by their therapeutic properties:

- **Adaptogenic plants:** help the body adapt to physical, mental and environmental stress, promoting balance and homeostasis.
- **Analgesic/anesthetic plants:** provide pain relief through a direct analgesic effect or by reducing inflammation.
- **Anti-inflammatory plants:** help reduce inflammation in the body and may be useful in treating conditions such as arthritis and skin inflammation.
- **Antimicrobial plants:** have properties to combat microorganisms such as bacteria, fungi, and viruses.
- **Hypotensive plants:** help reduce blood pressure, promoting cardiovascular health.

In the following paragraphs, we will explore each of these categories in more detail.

3.1 Adaptogenic Plants

Adaptogens are pharmacologically active compounds often derived from plant extracts that can increase the body's stability against physical loads without increasing oxygen consumption. These compounds improve the body's ability to adapt to stress, normalize metabolic functions, and improve mental and physical performance [26–28].

Adaptogens are categorized into plant-based and synthetic types. Plant-derived adaptogens have been used for centuries [29]. In the 1980s, scientists Breckham and Dardimov identified several key plants with adaptogenic effects, including *Panax ginseng*, *Eleutherococcus senticosus*, *Rhaponticum carthamoides*, *Rhodiola rosea*, and *Schisandra chinensis* [30].

Adaptogenic plants have a very rich phytochemical composition and present compounds such as triterpenoid saponins (in *Panax ginseng*—ginsenosides; in *Eleutherococcus senticosus*—eleutherosides); phytosterols and ecdysone (in *Rhaponticum carthamoides*); lignans (in *Schisandra chinensis*); alkaloids; flavonoids, vitamins, and more [31, 32].

Some studies highlight these substances' action on the hypothalamic–pituitary–adrenal axis [33] and various stress mediators [34]. Due to their beneficial impact, they have potential applications in treating mental illnesses, behavioral disorders, and stress-induced diseases such as anxiety, cardiovascular diseases, and diabetes [31, 32, 35–39]. However, the mechanisms behind these compounds are complex and not yet fully understood, continuing to be an area of active research.

3.2 Analgesic/Anesthetic Plants

The term “analgesic” refers to the relief of pain without a loss of consciousness, while “anesthetic” denotes the temporary elimination or blocking of sensitivity, including pain. Several plants have been used for these purposes, with the usage of herbal medicines for such applications dating back to 40–90 A.D. [40].

The anesthetic agents currently used are derived from or associated with plants. As an example, we have cocaine, which was isolated from coca (*Erythroxylum coca*, *Erythroxylaceae*), thymol which is found in thyme (*Thymus vulgaris*, *Lamiaceae*) and eugenol which is present in cloves (*Syzygium Aromaticum*, *Myrtaceae*). These last two are similar, both in structure and mechanism, to intravenous phenolic anesthetics.

In 2001, Almeida, Navarro and Barbosa-Filho [40] listed 166 plants belonging to 79 families with analgesic activity that act on the central nervous system. The

mechanism of action of these herbs can be categorized as general or local anesthesia [41].

Some plants that have local analgesic effects in addition to cocaine are cloves (*Syzygium aromaticum*), which are traditionally used for toothache relief [40], plants of the species *Spilanthes* [42], and plants of the *Cinchona* species, which have quinine as their constituents.

Conversely, there are plants known for their systemic effects: *Passiflora incarnata*, used worldwide to treat anxiety, neuralgia, and insomnia [43]; *Valerian officinalis*, historically utilized for brain disorders since the late sixteenth century [44]; ginseng (genus *Panax*), which enhances the human body's ability to handle fatigue and physical stress [40]; and St. John's wort (*Hypericum perforatum*), utilized for centuries due to its anti-inflammatory, sedative, analgesic, diuretic, antimalarial, and antidepressant properties [45], and is also regulated as a dietary supplement [46].

3.3 Anti-inflammatory Plants

Inflammation is a survival mechanism [47] involving complex tissue changes aimed at eliminating the cause of cellular injury [48, 49]. This process may manifest as local redness, swelling, pain, heat, and loss of function may occur [50].

In recent decades, several articles and research have been published on the anti-inflammatory activities of plants [50–53]. Some species listed with compounds that have anti-inflammatory action are *Myracrodruon urundeuva* Allemão, *Schinus terebinthifolius* Raddi, *Spondias mombin* L., *Spondias purpurea* L., *Spondias tuberosa* Arruda, *Euphorbiaceae* *Acalypha hispida* Burm. f., *Acalypha indica* L., *Achyranthes aspera* L., *Alternanthera brasiliana* (L.) Kuntze (family *Amaranthaceae*), among others [54–61].

These species have been identified to contain several secondary metabolites such as phenols, triterpenes, flavonoids, saponins, tannins, and cinnamic acid, which contribute to their anti-inflammatory action [54–56]. Although extensive research has been conducted, the specific plant parts and compounds responsible for these anti-inflammatory effects remain not fully elucidated.

3.4 Antimicrobial Plants

Due to the increasing incidence of multi-resistant microorganisms, several medicinal plants can become an outlet for antimicrobial, antifungal, and antiviral products thanks to their secondary metabolites [62].

Plants such as *Cinnamomum cassia*, *Rumex nervosa*, *Ruta graveolens*, *Thymus serpyllum* [63], *Allium sativum*, and *Punica granatum* [64] have antimicrobial activities while other species such as *Punica granatum* [65–69], *Chamaecyparis obtusa*,

Chrysanthemum boreale, *Cryptomeria japonica* [70] have antifungal and antiviral activity.

The phytochemical constituents present in these medicinal plants are flavonoids, alkaloids [71], and phenolic compounds [72, 73]. Several studies show the biological action of these compounds against bacteria, fungi, and viruses, suggesting potential pathways for developing new biologically active compounds to combat these microorganisms [62].

3.5 Hypotensive Plants

Hypertension is a chronic medical condition characterized by elevated arterial pressure [74]. Sustained high blood pressure can lead to strokes, heart attacks, heart failure, arterial aneurysms, and kidney failure [75], consequently decreasing life expectancy. However, modifications in diet, lifestyle, and medication can substantially improve blood pressure control and reduce the risk of these severe health issues.

In recent decades there has been a search for plants that can be used as anti-hypertensives, such as *Allium sativum* [76], *Artocarpus altilis* [77], *Blond psyllium* [78], *Carum copticum* [79], among others. Phytochemical studies are still needed to understand which metabolites are involved in the action of these species against hypertension.

4 Main Phytochemical Constituents and Their Biological Properties

To defend against external aggressions, abiotic factors, pathogens, and pests, plants produce substances known as secondary metabolites, which help the species to survive and reproduce in the environment [80]. Researchers continuously study and characterize these substances and their derivatives to uncover potential properties useful for pharmaceutical, cosmetic, and food industries.

Given the vast diversity of secondary metabolites, which vary by species and environmental conditions, these compounds can be categorized based on their chemical structures and biosynthetic pathways [80, 81]. The subsequent sections will discuss some of the classes and their intrinsic characteristics.

Alkaloids

Alkaloids are substances primarily synthesized by plants but also found in animals, fungi, and bacteria [82]. Their main chemical characteristic is the presence of at least one nitrogen atom incorporated into the cyclic structure, which is why they are called

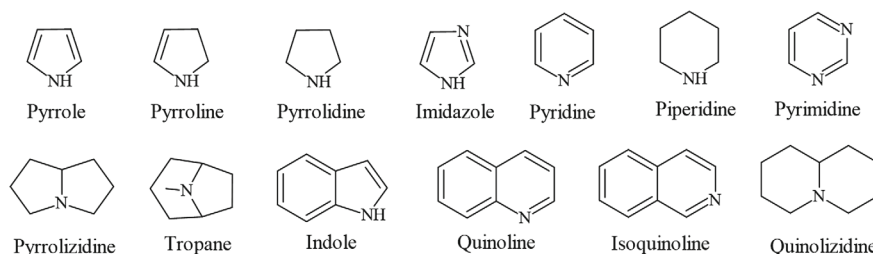


Fig. 1 Characteristic chemical groups of alkaloids

alkaloids due to their alkaline or basic nature [83]. These substances are predominantly produced by larger plants in families such as Leguminosae, Menispermaceae, Ranunculaceae, Loganiaceae, and Papavaraceae [84, 85].

In terms of biosynthesis, they originate from a variety of precursors including amino acids like ornithine, lysine, phenylalanyl, tyrosine, tryptophan, and histidine, as well as polyterpene, polyketide units, and purines [83]. They can be broadly classified based on their origins and chemical structures. Origin-wise, they can be classified as true alkaloids (originating from amino acids), proto-alkaloids (containing nitrogen from an amino acid), and pseudo-alkaloids (not derived from amino acids) [86]. Chemically, alkaloids can be classified into pyrrolidine, pyridine, quinoline, isoquinoline, quinazoline, imidazole, and tropane [87] (Fig. 1).

These substances possess various biological properties and hold significant industrial interest, particularly because many alkaloids can easily penetrate the blood–brain barrier, acting at the level of the central nervous system [86].

Polyphenols

Polyphenols, widely distributed in nature, are chemical compounds known for containing phenolic groups in their structures [88]. These substances are produced by plants for various protective reasons: they confer resistance against pathogens and predators, offer protection against environmental factors like rain and ultraviolet radiation, increase the astringency of foods to deter consumption, and play crucial roles in plant reproduction and pigment formation [89, 90].

Polyphenols are classified into several groups based on their structural characteristics, origin, and biological function. Notable groups include flavonoids, phenolic acids, and anthocyanidins, among others [88, 89]. Biosynthetically, they originate from two primary pathways: the shikimic acid and the acetate pathways. Typically, the A ring in their structures is derived from a resorcinol or a phloroglucinol molecule (acetate pathway), while the B ring originates from the shikimate pathway (Fig. 2) [89]. The complexity of their chemical structures varies greatly, ranging from simple phenolic acids to complex tannins, and they can exist in free forms or be conjugated with sugars, acids, or other biomolecules [91].

The interest in these substances by the scientific community is therefore vast since they have the potential to be used for the treatment of various diseases and also in

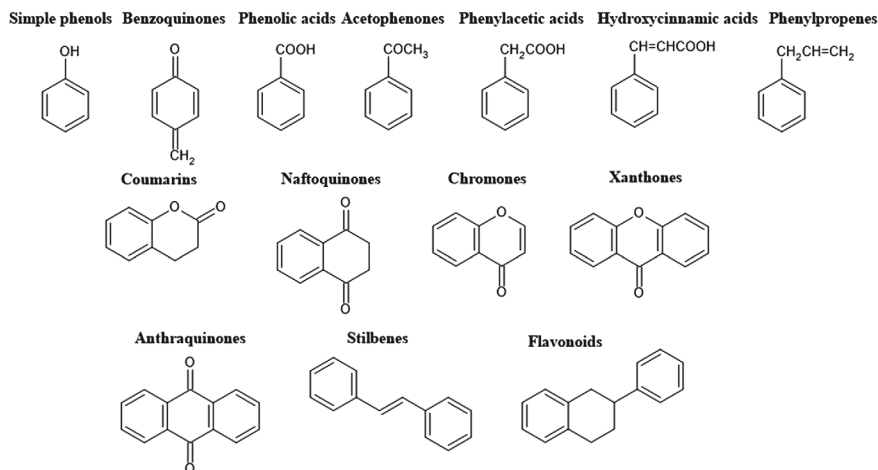


Fig. 2 Classes belonging to the polyphenol group

the prevention of chronic diseases, such as cardiovascular and neurodegenerative diseases, diabetes mellitus, cancer, and osteoporosis [89, 90, 92]. These properties are mainly attributed to their antioxidant action, as they are free radical scavengers and metal chelators, as they transfer electrons to free radicals and activate antioxidant enzymes [90, 91].

Simple phenolic acids

Phenolic acids, non-flavonoid compounds, are classified into benzoic or cinnamic acid derivatives (Fig. 3) [88, 93]. Compounds derived from benzoic acid are characterized by the presence of carboxylic groups, such as gallic acid and protocatechuic acid [90, 93]. Cinnamic acid derivatives are characterized by two carbon skeletons ($C_6H_5CHCHCOOH$) with at least one hydrogen in the molecule replaced by a hydroxyl, such as coumaric, caffeic, and ferulic acid [90, 91, 93].

These compounds are attributed with a range of potential therapeutic activities, including antidepressant, antihypertensive, anti-inflammatory, neuroprotective, antihyperglycemic, anticancer, and anti-diarrheal effects [91].

Fig. 3 Precursors of simple phenolic acids

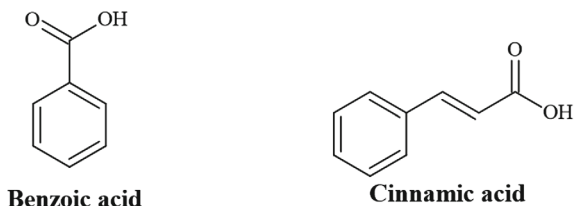
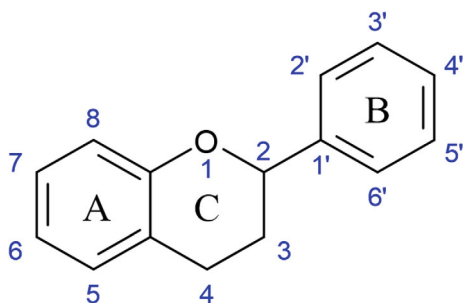


Fig. 4 Basic skeleton of flavonoids



Flavonoids

Flavonoids represent the most prevalent group of polyphenols. Their fundamental structure consists of a diphenyl propane containing 15 carbon atoms in the main nucleus (C6–C3–C6). It also comprises two phenolic rings, A and B, connected by a heterocyclic pyrene ring that contains an oxygen atom [88, 90, 94] (Fig. 4).

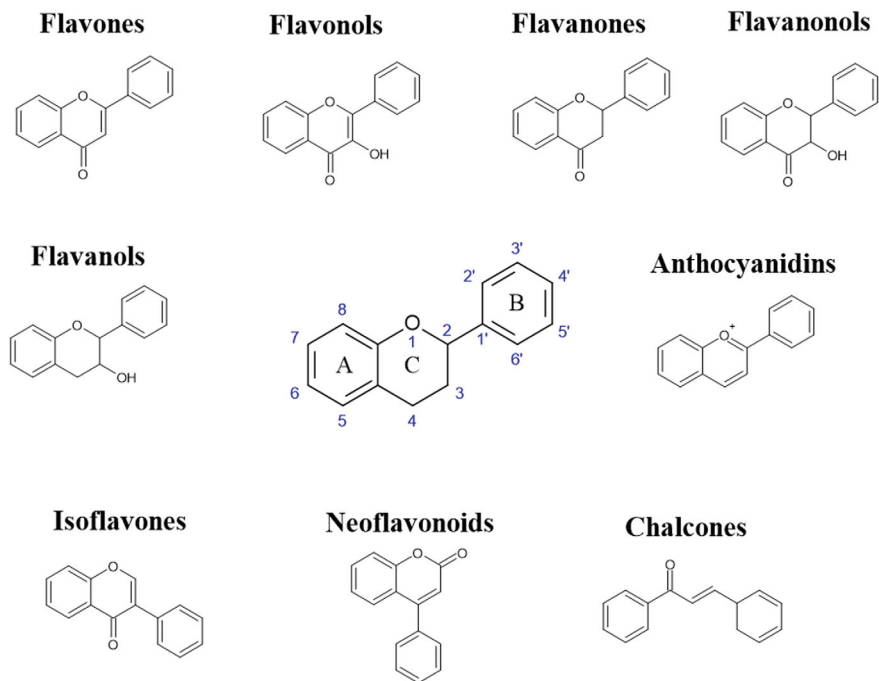
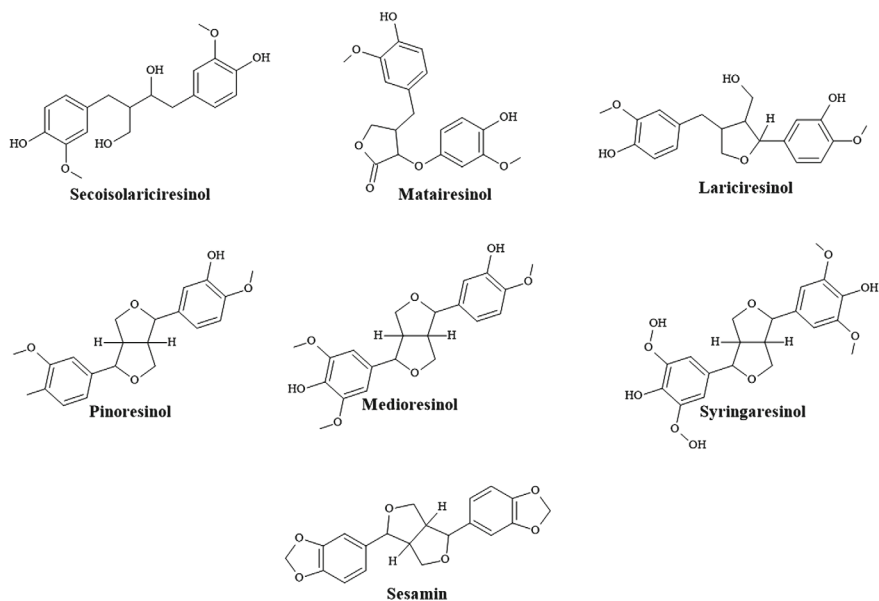
Depending on the position of the bond between rings B and C, flavonoids can be classified into isoflavones (bonded at carbon 3 of ring C), neoflavonoids (bonded at carbon 4 of ring C), and when bonded at position 2, they can be classified into flavones, flavonols, flavanones, flavanols, flavanols or catechins, anthocyanins and chalcones [87] (Fig. 5).

Although the basic structures of flavonoids are aglycones, many of these compounds exist as glycosides [88]. In terms of their biological properties, flavonoids can have antitumor, antimicrobial, antiviral, antiangiogenic, antioxidant, antimalarial, and neuroprotective effects, among other therapeutic purposes [95].

Lignans

Lignans are secondary metabolites produced through the oxidative dimerization of two phenylpropanoid units (C6–C3) [96–99]. Various plant species produce lignans in diverse parts, including roots, rhizomes, bark, stems, leaves, flowers, fruits, and seeds [99]. Because they have structures similar to estrogen and can bind to estrogen receptors in the human body, they are called phytoestrogens [100, 101]. Lignans can be divided into lignans proper when the two C6–C3 units are linked by a 3β-(8–8') bond; neolignans when the units are not linked by 3β bonds; and finally, hybrid lignans when the C3–C6 units are present but linked in different structures such as flavolignans [102]. Lignans can be divided into seven main types: secoisolariciresinol, pinoresinol, matairesinol, kedioresinol, sesamine, siringaresinol and lariciresinol (Fig. 6) [101].

These compounds are attributed to various biological properties, including anti-tumor activity, antiviral activity, inhibition of the activity of some enzymes, reduction of oxidative stress, analgesic, anti-inflammatory, relaxing, hepatoprotective, hypertensive, hypolipidemic, and phytoestrogenic activities. The latter having effects on breast and prostate cancer, cardiovascular diseases, osteoporosis and menopausal symptoms [97, 99, 102].

**Fig. 5** Classes belonging to the flavonoid group**Fig. 6** The chemical structure of lignans

Terpenes

Terpenes are a class of hydrocarbons made up of isoprene units (C_5H_8) [103, 104], which are found in a variety of plants, fungi, microorganisms, and insects. They can also be presented in the form of oxygenated derivatives, i.e., they have carbonyl, hydroxyl, ketone, and aldehyde groups, which are then called terpenoids [105, 106]. Terpenes can be divided into hemiterpenes (C_5), monoterpenes (C_{10}), sesquiterpenes (C_{15}), diterpenes (C_{20}), sesterterpenes (C_{25}), triterpenes (C_{30}), tetraterpenes (C_{40}) and polyterpenes ($>C_{45}$), depending on the number of forming units (Fig. 7). They are soluble in organic solvents (lipophilic characteristics) and have an asymmetric carbon with optical rotations [106].

Terpenes are produced by higher plants or spermatophytes from pyrophosphate and its isomer dimethylallyl pyrophosphate via plastidic methylerythritol phosphate and the cytosolic mevalonate pathway [103, 107].

The physicochemical characteristics of compounds within this class vary depending on their molecular weight. Low molecular weight molecules, such as monoterpenes and sesquiterpenes, are typically volatile oily liquids with boiling points that increase relative to their molecular weight and degree of unsaturation. In contrast, compounds with higher molecular weights, such as diterpenes, triterpenes, and polyterpenes, are generally found as solid crystals [106].

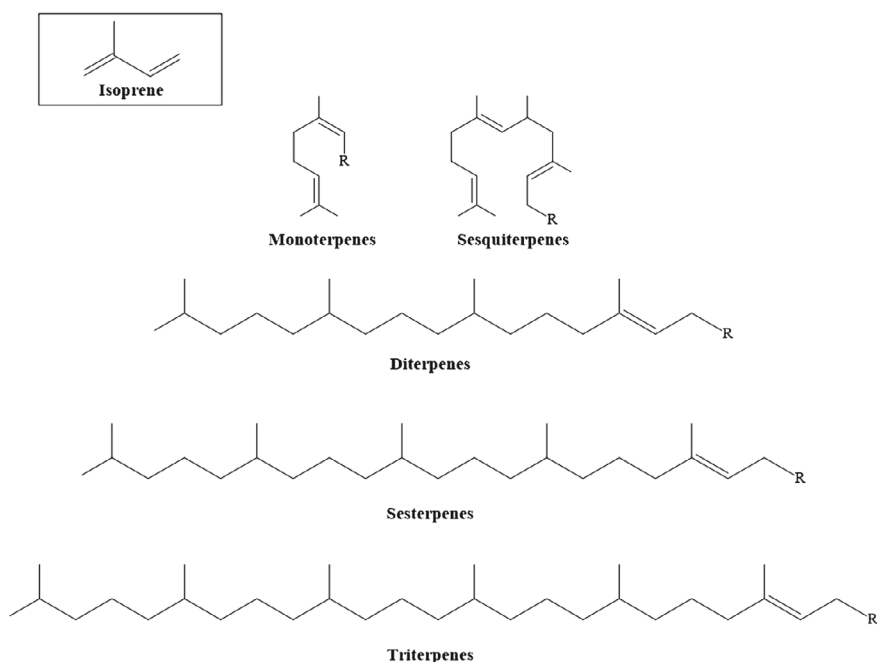


Fig. 7 Classification of terpenes according to the number of isoprene-forming units

In the pharmaceutical industry, terpenes are widely used as anti-cancer, antimicrobial, antifungal, antiviral, antihyperglycemic, analgesic, anti-inflammatory, and anti-parasitic agents [105–107].

5 Extraction and Production Process

To identify and extract the chemical components of a plant, a series of steps must be followed, including collecting plant material, sample preparation, extraction, and subsequent characterization [108]. During this process, careful handling and proper preparation of the material are critical to enhance extraction efficiency and achieve higher yields of higher quality and purity [109].

Extraction methods differ from one another, each with its own specificities and particularities. Depending on the purpose, some are more commonly used than others [108]. Nowadays, new approaches and methods that are safer and more environmentally friendly are being increasingly valued [110].

The range of extraction techniques spans from traditional to more contemporary approaches. Conventional methods, such as immersion, maceration, percolation, hydrodistillation, and Soxhlet extraction, are relatively straightforward to execute but tend to be more time-intensive. These methods can also lead to the degradation of sensitive components and typically require substantial volumes of solvents [108, 110, 111].

It is important to note that several factors can influence and affect extraction processes, the most common being matrix properties, the solvent used, temperature, and pressure of the medium [108]. This is why the choice of method is essential for the effectiveness of extraction processes.

The use of solvents for the extraction of plant components is commonly used for both qualitative and quantitative analysis for different purposes, including the development of herbal medicines [111]. The choice of solvent is crucial to the success of the extraction. Depending on the characteristics of the target substance, such as polarity, molecular affinity between solvent and solute, and mass transfer, there is a need for a co-solvent. In addition, other aspects are essential and are increasingly taken into account when choosing a method, such as environmental safety and low potential toxicity [108].

Soxhlet extraction, first proposed by Ritter Von Soxhlet in 1879 [108], employs the reflux technique, primarily for lipid extraction [110]. Maceration involves soaking plant material in an extracting solvent at room temperature to obtain the extract. Percolation, on the other hand, consists of the controlled passage of the extracting solvent through the plant material by the percolator, effectively exhausting the material to collect the extract [110]. Decoction is the process of exposing the material to boiling water. Hydrodistillation, which involves hydrofusion, hydrolysis and decomposition by heat [108], consists of boiling the material in water to obtain the bioactive compounds [110].

Non-conventional extraction methods are considered to be more environmentally friendly, as they typically require fewer synthetic chemicals and organic solvents. These methods also tend to be faster and yield more efficient results than traditional techniques. Non-traditional methods include ultrasound, pulsed electric field, enzymatic digestion, extrusion, microwave heating, supercritical fluids, and accelerated solvents [108, 110].

Ultrasound extraction utilizes the energy emitted by ultrasound baths and probes, which induces cavitation—the formation, growth, and collapse of bubbles under specific pressure and temperature conditions [110]. The collapse of these bubbles generates high pressure (5000 K at 2 atm) and temperature, potentially damaging plant cell structures and releasing active components [110, 112]. This method offers several advantages, including reduced extraction time, lower energy and solvent usage, yielding a more uniform extract, and allowing for greater process control [108].

Pulsed electric field extraction involves disrupting cell membrane structures using an electric field. This process allows for the separation of molecules based on their electric potential. This method is widely used for the extraction of intracellular components from plant tissue and for thermolabile components since an increase in temperature can damage the cell membrane [108].

Enzymatic digestion is a method that consists of adding specific enzymes to extract the material, break down the cell wall, and hydrolyze components such as polysaccharides and lipids [108, 110].

Microwave-assisted extraction utilizes microwave energy from 300 MHz to 300 GHz to extract soluble products from a fluid [108, 110]. This method operates on the principle of dielectric heating, where the energy released during particle collisions, through the emission of electromagnetic energy, causes the cell walls of plant components to release active compounds [110, 112]. It is known for its efficiency, yielding good results using less solvent [110].

The supercritical state occurs when a substance is subjected to pressure and temperature beyond its critical point (defined as the temperature and pressure at which there is no distinction between the liquid and gaseous phases), i.e., supercritical fluid cannot be classified as a liquid or gas, as it has low viscosity as a gas and high density as a liquid [110]. As a result, supercritical fluid cannot be liquefied by varying the temperature and pressure and can be applied as a compound extractor in a short time and with optimum yield [108], as it can penetrate plant materials and solubilize active components. Carbon dioxide (CO₂) is generally used as the ideal fluid since its critical temperature (31 °C) is close to ambient and allows it to operate at moderate pressures [108]; it is easily obtained, low-cost, inert, and non-flammable [109, 110]. However, it is most commonly used to assess non-polar components due to the low polarity of carbon dioxide and its low yield compared to other methods [108–110].

Associated with these methods, the need to use more environmentally friendly solvents is gaining momentum. Deep eutectic solvents are a combination of two or more natural, inexpensive, and biodegradable components capable of self-associating through hydrogen interactions to form eutectic mixtures with a much lower melting

point than the individual compounds. These solvents are advantageous because they are inexpensive to obtain and easy to prepare, with positive physicochemical characteristics when considering an extraction process, such as low pressure, temperature, stability, non-volatile and flammable, as well as being biodegradable. These solvents can be used in ultrasound-, microwave- and heat-assisted extraction, for example [113].

6 Scientific Evidence and Research

Herbal medicine, a cornerstone of traditional therapy across various cultures, has seen a dynamic shift with the integration of modern scientific methodologies. Initially, the validation and use of herbal medicines were heavily reliant on anecdotal evidence and traditional use. However, over the past few decades, the focus has expanded to a rigorous scientific approach aimed at understanding and proving herbal treatments' pharmacological basis and clinical efficacy. Contemporary research into herbal medicines now encompasses a broad array of studies, including *in vitro*, *in vivo*, and clinical trials, where researchers are committed to verifying traditional uses and uncovering novel therapeutic potentials for complex diseases such as cancer, Alzheimer's, and chronic inflammatory conditions.

This expansion in research scope and depth has been greatly facilitated by the advent of advanced analytical techniques and methodologies, including high-resolution techniques, functional assays, and phenotypic screenings, which enable detailed phytochemical profiling of medicinal plants to identify active compounds and decode their mechanisms of action. Additionally, computational-aided drug discovery, systems biology, and network-based polypharmacology have been increasingly employed to unravel how secondary metabolites from plants may act synergistically in therapeutic applications and shed some light on the scientific basis behind the efficacy of traditional knowledge of herb mixtures.

The integration of scientific research with traditional knowledge is pivotal for several reasons. Firstly, it provides a foundational understanding that enhances the research hypothesis, often pointing scientists toward specific plant species or extracts for similar indications. For example, the anti-malarial properties of *Artemisia annua* were utilized in traditional Chinese medicine long before the discovery of artemisinin, which has transformed malaria treatment globally. Second, this integration respects and preserves indigenous wisdom while subjecting it to scientific scrutiny, which not only helps in substantiating traditional claims but also in ensuring the safety of these treatments when used within modern therapeutic contexts.

This robust foundation of combining traditional wisdom with modern science sets the stage for the forthcoming sections that delve deeper into landmark clinical research, elaborate on the intricacies of pharmacokinetics and pharmacodynamics (PK/PD), and explore the sophisticated realm of computational methods. These sections aim to further elucidate the evolving landscape of herbal medicine research,

demonstrating its historical significance and potential to address contemporary health challenges through innovative scientific approaches.

6.1 Landmark Clinical Trials and Systematic Reviews

Landmark clinical trials and systematic reviews have significantly advanced our understanding of the efficacy and safety of herbal medicines. For instance, saffron (*C. sativus*), traditionally used in Persian medicine to enhance mood, has shown antidepressant, anxiolytic, and hypnotic effects in preclinical animal models using ethanolic extracts of saffron and its constituents safranal and crocin [114]. A study by Schmidt et al. [115] suggested that crocin's mode of action is via reuptake inhibition of dopamine and norepinephrine, while safranal is via serotonin reuptake inhibition. Finally, studies predominantly conducted in Iran have demonstrated saffron's effectiveness in treating depressive symptoms, showing similar efficacy to conventional antidepressants [116]. However, the limitations of this study, including small sample sizes and short durations, suggest a need for further research to fully establish saffron's therapeutic benefits.

Turmeric (*Curcuma longa*) and its primary active constituent, curcumin [117, 118], have demonstrated significant potential in treating a range of conditions, from depression to inflammation and autoimmune disorders [119–124]. Indeed, in a recent meta-analysis conducted by Ng et al. [125], six clinical trials lasting between 4 and 8 weeks each and involving 377 patients with depression were reviewed. Their analysis revealed that turmeric significantly reduced depressive symptoms and provided anti-anxiety benefits, outperforming a placebo.

St. John's Wort (SJW), scientifically known as *H. perforatum*, is widely recognized for its antidepressant properties and is frequently used in Europe as a first-line treatment for mild to moderate depression [126–129]. The efficacy of SJW is attributed to its complex pharmacological action, particularly its ability to nonselectively inhibit the reuptake of key neurotransmitters such as serotonin, dopamine, and norepinephrine.

This broad-spectrum action is attributed to compounds such as hyperforin and hypericin, along with various flavonoids that are thought to play critical roles in these mechanisms, suggesting a synergistic effect when the whole herb is consumed [130–133]. Several randomized controlled trials (RCTs) and systematic reviews support its comparable effectiveness to conventional antidepressants, although some studies suggested no substantial superiority over a placebo, which led the herbal medicine to be commonly recommended only for mild to moderate depression [126–129].

On top of its efficacy, SJW also presents a safety profile [134]. For instance, Schulz [135] concluded after reviewing 16 postmarketing surveillance studies that SJW is tenfold safer than synthetic antidepressants, displaying adverse effects ranging only from 0.1 to 2.4%. However, to the best of our knowledge, a direct pharmacodynamic interaction has not yet been established.

Ashwagandha (*Withania somnifera*) has garnered increasing attention recently for its adaptogenic properties, particularly in alleviating stress and anxiety [136, 137]. Such effects have been demonstrated in an animal study where ashwagandha showed adaptogenic properties during a stress-inducing procedure. This research noted a reduction in stress-related parameters, including cortisol levels, mental depression, and sexual dysfunction [138]. A systematic review by Pratte et al. [139] analyzed five human RCTs with 400 participants and highlighted ashwagandha's effectiveness in reducing stress and anxiety symptoms compared to control conditions. More recently, a double-blind RCT in 52 people with chronic stress by Choudhary et al. [140] reinforced these findings, showcasing the effectiveness of 600 mg of ashwagandha extract daily compared to a placebo over eight weeks.

Likewise, roseroot (*Rhodiola rosea*), another herbal medicine with adaptogenic properties, has shown promising antidepressant activity through its ability to modulate monoamine oxidase A [141] and cortisol levels [142, 143], which are crucial in stress response and mood regulation. In a 12-week double-blind RCT comparing *Rhodiola Rosea*, sertraline, and placebo for major depressive disorder, sertraline showed a slight advantage in effectiveness, although roseroot demonstrated significantly milder side effects, suggesting it as a more tolerable alternative for managing depression [144].

Finally, another example is Kava (*P. methysticum*), a South Pacific medicinal plant that has been traditionally used as a relaxant and as a treatment for anxiety [145]. Its mode of action is believed to be through the interaction of kava's active constituents, the kavalactones, with various neurological pathways. This includes modulation of GABA voltage-gated sodium ion channels, enhanced ligand binding at GABA-a receptor subtypes, and the inhibition of excitatory neurotransmitter release through the blockade of calcium ion channels [146–148].

Numerous RCTs have demonstrated kava's anxiolytic properties, with consistent findings indicating significant activity against placebo in various anxiety disorders [149–151]. Additionally, comparative trials have shown that kava's efficacy and safety are comparable to standard treatments like buspirone [152]. Despite its therapeutic benefits, kava's safety profile has been under scrutiny, particularly regarding hepatotoxicity concerns after spontaneous reports of liver toxicity [145]). However, clinical trials have generally shown kava to be well-tolerated, with adverse events being mild and reversible, similar to those reported for placebos [149, 153].

6.2 Pharmacokinetics and Pharmacodynamics of Herbal Medicines

As we delve deeper into understanding how herbal medicines exert their therapeutic effects, mechanistic studies become indispensable. A common misconception is that herbal medicines are safe and lack adverse effects because they are natural. Yet,

when these supplements are used concurrently with prescription drugs, such as anti-cancer agents, anticoagulants, or antidepressants, the potential for serious herb-drug interactions increases, which are often not recognized until they manifest clinically, sometimes with severe consequences [154]. For example, the coadministration of SJW with the immunosuppressant cyclosporine has led to organ rejection in transplant patients due to the induction of metabolic enzymes that reduce the drug's efficacy [155].

Furthermore, the complexity of understanding the modes of action of herbal medicines is exacerbated by the diverse array of secondary metabolites they contain and the ongoing uncertainty regarding which specific compounds are responsible for their therapeutic effects [156–159].

Therefore, understanding such molecules' pharmacokinetics and pharmacodynamics is crucial for elucidating direct herb-drug interactions in the body and revealing the broader biochemical pathways impacted by these interactions. The former consists of interactions that alter the absorption, distribution, metabolism, or elimination of a drug—the well-known ADME mechanism, which usually involves the inhibition or induction of hepatic and intestinal enzymes, specifically those in the cytochrome P450 (CYP) family, or drug transporters such as P-glycoprotein (P-gp) [160, 161]. For instance, SJW is known to induce CYP3A4 and P-gp, leading to decreased bioavailability and effectiveness of various drugs, including chemotherapeutic agents like irinotecan and targeted therapies like imatinib [162].

Conversely, pharmacodynamic interactions happen when an herbal compound and a pharmaceutical have additive, synergistic, antagonistic, or adverse effects. These interactions can sometimes be predicted from known pharmacological properties but remain challenging to manage clinically [163, 164]. For instance, certain herbal compounds may enhance the effects of anticoagulants like warfarin, increasing the risk of bleeding by targeting common pathways such as vitamin K epoxide reductase [165].

For a full list of pharmacokinetics and pharmacodynamic interactions, see [154, 156, 161, 166–168].

Building on the understanding of pharmacokinetics and pharmacodynamics in herbal medicine, nanoformulation technology has also emerged as a pivotal advancement to further enhance the bioavailability and therapeutic efficacy of natural products, which usually present poor solubility, stability, and bioavailability [169]. For instance, despite their potent therapeutic properties, curcumin and silymarin face challenges in clinical efficacy due to rapid metabolism and poor absorption [169, 170]. Thus, by incorporating these compounds into nanoformulations like liposomes, micelles, and solid lipid nanoparticles, researchers have successfully improved their solubility, controlled release, and target-specific delivery [171–175].

Finally, another alternative for tackling the intricate puzzle of herb-drug interactions within herbal medicines involves employing *in silico* methods. These computational tools enhance our capacity to preempt adverse interactions through systematic analyses of interaction dynamics and metabolic pathways, which are now possible due to the continuous expansion of molecular data in public databases. This wealth of information stems from foundational basic research, preclinical and clinical

studies, and detailed pharmacokinetic and pharmacodynamic investigations that have progressively elucidated how herbal compounds interact within the body. Thus, anecdotally, as happens in many herbs' modes of action, this research integration produces a synergy that enables and potentializes more effective therapies in healthcare.

6.3 *Computational-Aided Drug Discovery*

The challenge in natural product research has traditionally been identifying and characterizing secondary metabolites and active components, many of which remain unknown or only partially studied. As such, computational methods have proven essential in deconvoluting this complex process, particularly during the early phases of drug discovery, including hit discovery, hit-to-lead, and lead optimization. These techniques prioritize plant materials for testing, thereby conserving valuable time and resources, enhancing the identification of novel active components, and enabling drug repositioning. Approaches like molecular modeling, docking simulations, and cheminformatics are pivotal in predicting binding modes, understanding molecular mechanisms of action, and conducting target fishing.

For instance, Su et al. [176] employed virtual screened novel rho kinase inhibitors from natural sources, including herbal compounds. They narrowed the initial set of molecules from over 100,000 natural constituents to 25,000, and eventually, they confirmed the biological activity for five out of six compounds selected for experimental validation. Similarly, Amparo et al. [177] and Zhang et al. [178] utilized QSAR, molecular docking, and ADMET analyses to investigate the antiviral properties of compounds derived from Brazilian and Chinese medicinal herbs against COVID-19, respectively. Their coordinated efforts identified promising molecules, including, for instance, 19 triterpenes and phenolic compounds, with potential anti-SARS-CoV-2 activity.

Additionally, system biology and polypharmacology networks, in particular, have emerged as powerful approaches to integrate the complex interactions of herb's secondary metabolites within biological systems into drug discovery. These approaches leverage multiple layers of biological data, from genetic and proteomic profiles to metabolic pathways, cellular behavior, and broader physiological responses, shifting the paradigm from a 'one-target, one-drug' to a 'network target, multi-component' model much like the holistic thinking of the Traditional Chinese medicine (TCM) [179–184]. For instance, Cai et al. [185] developed a system pharmacology-based framework to identify TCM-derived natural products for cancer immunotherapy. They found that 32 out of 49 most promising compounds predicted to potentially modulate the tumor microenvironment are validated by public experimental data from clinical studies, in vitro and in vivo assays. Interestingly, Wang et al. [186], using a similar approach, have predicted synergistic targets and pharmacological profile of four herbs in cardiovascular diseases.

However, it is noteworthy that the success of these computational strategies hinges on the availability of accurate and comprehensive data. Errors in initial data can

lead to misguided conclusions, highlighting the importance of high-quality, reliable databases and experimental validation.

Not surprisingly, in recent years, a plethora of curated databases designed explicitly for medicinal herbs and natural products have been proposed, encompassing a wide range of critical information, including biological activity and chemical composition data such as detailed profiles of active compounds, pharmacological effects, and potential therapeutic uses of herbs. Among them are TCM database@Taiwan [187], KNApSACk-3D [188], TCMSP [189], NuBBEDB [190], NPASS [191], and COCONUT [192], just to name a few.

Other comprehensive and generalist databases widely employed in drug discovery include BindingDB [193], ChEMBL [194], DrugBank [195], InterPro [196], KEGG [197], Pfam [198], RCSB PDB [199], STRING [200], UniProt [201], and ZINC [202].

In summary, computational methods hold immense potential in streamlining the discovery and characterization of therapeutically active compounds in herbal medicine. Nonetheless, it is worth emphasizing once again that their success deeply relies on extensive and accurate data availability. Therefore, the ongoing integration with experimental methodologies is crucial to ensure the reliability of computational predictions and continually enrich public databases, especially given that many herbs and secondary metabolites' potential is still underexplored, paving the way for uncovering novel therapeutic opportunities.

7 Regulations and Quality Control

7.1 *Local and International Regulations*

Herbal medicines are defined by the World Health Organization (WHO) as “the aerial or underground parts of the plant or other plant material containing an active ingredient as a labeled finished drug” [203].

Since herbal medicines are complex products derived from medicinal plants, they require stringent regulation to ensure safe usage. To this end, numerous standards, regulations, and official documents, such as pharmacopeias, have been established, detailing criteria for the quality, safety, and efficacy of these products [204].

The World Health Organization (WHO) has issued The International Pharmacopoeia to standardize specifications and enhance the scope and harmonization of regulations worldwide. Additionally, national and regional pharmacopeias, such as those from Brazil, Europe, America, and China, further tailor these specifications to reflect each country or region's unique geographical and economic characteristics [205, 206].

Brazil

In Brazil, due to its great biodiversity, the use of medicinal plants is a widely accepted and valued practice. For this reason, public policies have come into force to regulate the use of herbal medicines safely and effectively [207].

In 1986, the 8th National Health Conference proposed the implementation of alternative practices in health services, including phytotherapy, in order to facilitate and democratize access to various therapies, according to the preferences of each user. As a result of these discussions, herbal medicine was introduced as a health service by means of *Comissão Interministerial de Planejamento e Coordenação* (CIPLAN) resolution nº 08/1988 [208].

At the 10th National Health Conference in 1991, it was recommended to incorporate integrative and complementary practices, such as herbal medicine, into the *Sistema Único de Saúde* (SUS). Since then, this practice has been discussed at various conferences and scientific events to regulate these practices, with a focusing on the safe and effective use of medicinal plants and herbal medicines [208].

A milestone regarding medicinal herbs in the country was the creation of the *Política Nacional de Práticas Integrativas e Complementares* (PNPIC), published by the Ministry of Health through Decree No. 971/2006 [209], which considers phytotherapy as a therapeutic resource arising from the use of medicinal plants, in their different pharmaceutical forms, and proposes the incorporation of medicinal plants and phytotherapy into health services. In the same year (2006), the Ministry of Health published the National Policy for Medicinal Plants and Herbal Medicines by decree no. 5813/2006 [208], whose primary purpose was to implement the rational and safe use of herbal medicines to ensure their quality and safety.

Once this practice was incorporated into health systems, care regarding these resources became a source of discussion and regulation. The purpose of Collegiate Board Resolution (RDC) No. 18 of 2013 and its updates [210] is to establish Good Practices for the Processing and Storage of Medicinal Plants, as well as the preparation and dispensing of magisterial and officinal products made from plants and herbal medicines in the so-called “*farmácias vivas*” in the SUS.

RDC No. 26/2014, issued by the *Agência Nacional de Vigilância Sanitária* (ANVISA), outlines the process for registering herbal medicines and notifying about traditional herbal products, establishing specific rules and requirements regarding deadlines and updates for registrations to ensure proper regulatory compliance [211]. According to this resolution, applicants must meet several requirements before registering. In case the constituents of the herbal medicine are not included in the Brazilian Common Denomination (BCD), applicants must first petition the Brazilian Pharmacopeia to have these constituents listed. Subsequently, the formal registration process can be initiated, which involves submitting a comprehensive set of documents and reports. These include technical documentation on the plant raw material, packaging, manufacturing location, stability studies, production and quality control reports, and safety and efficacy reports where necessary. Additionally, a description of the pharmacovigilance system and a quality control report for one batch of herbal medicine from each supplier are required.

The first version of the Brazilian Pharmacopeia was established in 1926 by Decree n°, 17.509. It specifies the minimum quality requirements for inputs, medicines, and health products [212]. It is currently in its 6th edition, approved by RDC 609 of 2022 [213].

Brazilian legislation, specifically RDC No. 511/2021 [214], facilitates the adoption of international standards in cases where the Brazilian Pharmacopoeia lacks official monographs for raw materials, pharmaceutical forms, correlates, and general methods. In such instances, an official monograph from the most recent edition of international compendia, including German, American, Argentine, British, European, French, International (WHO), Japanese, Mexican, and Portuguese Pharmacopoeia can be adopted.

United States of America (USA)

In the USA, herbal medicines are generally regulated under the category of “dietary supplements” by the US Food and Drug Administration (FDA), regulated by the Dietary Supplement Health and Education Act of 1994 [215–217]. Dietary supplements include vitamins, minerals, amino acids, enzymes, metabolites, extracts, concentrates, and herbs. The FDA is a federal government public health agency whose function is to regulate the efficacy and safety of human drugs, biological products, and foodstuffs. In the US, herbal medicine is defined as “a plant or plant part used for its aroma, flavor or therapeutic properties and which is sold in the form of powders, capsules, teas, extracts, and fresh or dried plants” [218].

The existing regulations are relatively flexible, as these products are not required to undergo pre-market safety and efficacy assessments [215, 217]. To monitor safety post-marketing, a surveillance system is in place: the FDA Adverse Event Reporting System (FAERS). This system maintains a comprehensive database that collects reports of adverse reactions associated with using various drugs and biological products approved by the FDA [219].

European Union (EU)

In the European Union, the regulation of herbal medicines began with the need to simplify the registration process for these drugs. According to Directive 2001/83/EC, there is a Community code on medicinal products for human use, which establishes that applications for authorization to place medicinal products on the market, according to their categories, must be accompanied by a dossier containing a range of information on the product, the results of physicochemical, biological, microbiological, pharmacological and toxicological tests, as well as clinical trials, which prove the quality, safety and efficacy, in cases where there are no validated scientific references proving the efficacy and safety of the medicinal products [220].

Since many medicines, including those used traditionally, lack detailed information on clinical usage, safety, and efficacy, regulations have been proposed to simplify the registration process for certain conventionally used medicines. This approach aims to increase community access to these medicines, which are presumed safe due to their historical use [220]. Herbal medicines are defined as any medicine

that contains one or more herbal substances exclusively as its active ingredient or a combination of several herbal substances [206, 221].

Directive 2004/24/EC, enacted on March 31, 2004, by the European Parliament and the European Council, amends the 2001 directive and establishes the regulatory framework for herbal and traditional medicines in the EU [204, 222, 223]. This directive targets medicines with a well-established tradition of medicinal use, which are supported by scientifically recognized data on their efficacy and safety [204, 221].

Directive 2004/24/EC also addresses critical regulatory aspects of herbal medicine use by defining traditional herbal medicines, simplifying the registration procedure, and detailing provisions concerning monographs and the Community list of herbal substances and preparations [222]. Additionally, it established the Committee for Herbal Medicinal Products (HMPC) as part of the European Medicines Agency (EMA) under Council Regulation EEC No. 2309/93. The HMPC's responsibilities include the registration and authorization of herbal medicines, the development of Community monographs, and the compilation of a list of herbal substances, preparations, and combinations that are deemed safe for therapeutic use due to their traditional applications, according to Decision 2008/911/EC [204, 221].

China

The use of plants for therapeutic purposes in China is a well-established and ancient practice. This traditional knowledge, deeply valued and passed down through generations, remains a cornerstone of Chinese culture [224, 225].

In China, herbal medicines are distinctly classified as functional foods or medicines, each adhering to specific regulatory standards that include approval processes, production management, labeling, advertising, and specifications. Functional foods are regulated by the Provisional Law of the People's Republic of China on Food Hygiene and the Food License Bureau. Herbal medicines, on the other hand, fall under the jurisdiction of the Traditional Chinese Medicines (TCMs) Division and Ethno-Medicines, managed by the Drug Registration Department. This department outlines specific pre-marketing requirements to ensure these products meet standards set by the Drug Administration Law of the People's Republic of China, enacted in 2001, which aims to guarantee the quality and safety of medicines for human use [225, 226].

Herbal medicine products in China are regulated by the State Food and Drug Administration (SFDA), the country's authoritative regulatory body. The current regulations, established in 2007 (Order No. 28), provide practical guidelines for registering medicines [224, 226]. It is important to distinguish between traditional Chinese medicines, which refer to medicinal substances and their preparations used according to traditional Chinese medical theory, and natural drugs, whose usage follows modern medicine guidelines. However, regardless of this distinction, both must follow the standards set by the country's regulations [226].

The latest version of the Chinese Pharmacopoeia, the 11th edition, was published in 2020. Approved by the National Medical Products Administration (NMPA) and the National Health Commission of the People's Republic of China, it provides

regulations for the development, production, and use of medicines across China [227].

7.2 *Importance of Quality Control*

In order to maintain the purity and quality of the components of herbal medicines (the product in its final form), quality control is carried out at all stages of production, from collection to marketing, with the aim of correctly identifying the plant species and verifying the presence of possible contaminants in the material analyzed, always in accordance with the established limits [205, 207].

The species identification stage includes sensory, botanical (microscopic and macroscopic), and chromatographic analysis, with characterization of the components. Contamination can be of a chemical nature (e.g., heavy metals and pesticides), microbiological (e.g., mycotoxins), or through foreign bodies, many of which can pose safety risks, hence the need for quality control of processes [205, 207, 228].

A critical point to consider is that depending on the conditions in which a species is exposed, the amount of metabolites it produces can vary, whether due to abiotic factors or not. Exposure to solar radiation, temperature, soil conditions, humidity, salinization, pathogens, chemical products such as fertilizers or pesticides, growing conditions in general, and geographical location can influence species' quality and chemical composition [228].

Given these issues, the development of uniform procedures and the implementation of validated tests and protocols are crucial to ensuring the efficacy and safety of these products, minimizing potential risks related to contamination and adulteration of the materials [229, 230].

Generally speaking, three stages are crucial to the quality of a herbal product: cultivation, manufacture, and marketing or circulation. Implementing Good Agricultural and Collection Practices (GACP) and applying Good Manufacturing Practices (GMP), both issued by the World Health Organization (WHO), are paramount for the quality of these products. GMP consists of guidelines and standards that ensure quality and safety in processing materials and production. This includes identifying raw materials, managing facilities and equipment, adhering to standard operating procedures, maintaining records and documentation, and conducting stability tests. Furthermore, Good Supply Practices (GSP) are essential during the marketing stage. GSP involves applying the standards of each region, tailored according to each area's specific needs [228].

All these measures are essential and indispensable to ensure that consumers are not exposed to harmful products or those that cause unintended therapeutic effects. This situation highlights the need for improved safety in the professional prescription of these medicines, which stems from insufficient analytical criteria. Therefore, there is an urgent need for robust quality control grounded in validated standards and protocols for herbal medicines [229, 230].

8 Conclusion

The use of medicinal plants for treating various pathologies has been a continuous practice since ancient times thanks to their numerous properties, many of which are attributed to secondary metabolites—substances that can trigger multiple biological pathways through molecular interactions. Consequently, developing techniques to obtain and characterize these natural products is crucial and highly valued. This includes both traditional methods and modern, environmentally conscious techniques that consider the safe exploitation of resources. Regulatory agencies must, therefore, enforce standards and quality control in producing herbal medicines, keeping therapeutic safety and environmental concerns in mind.

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The Role of Artificial Intelligence in the Discovery of New Drugs



Mariana Pegrucci Barcelos and Carlos Henrique Tomich de Paula da Silva

Abstract The drug discovery and development process is a long and complex journey that involves several steps, from identification of biological targets to clinical approval and commercialization. This process traditionally takes more than a decade and requires billion-dollar in investments. Artificial intelligence (AI) is revolutionizing the process by accelerating multiple phases by analyzing vast amounts of biological and chemical data to identify patterns that would be impossible to detect manually. Machine learning tools, for example, can predict the interaction between compounds and biological targets, optimize the properties of candidate drugs and suggest new molecular structures based on large databases of chemical compounds. But the use of AI in drug development raises significant ethical and regulatory questions. Challenges include the transparency of AI algorithms, which are often seen as “black boxes,” and the need for rigorous validation to ensure the safety and effectiveness of medicines developed with the help of AI. Furthermore, there is concern about the privacy and security of the data used and generated during the development process. Thus, the future of AI in the pharmaceutical industry is promising, with the potential to radically transform the way medicines are discovered and developed, but it is crucial to address the ethical and regulatory challenges to ensure that these technologies are used in a responsible and beneficial way for the pharmaceutical industry and society.

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1 Introduction to Drug Discovery

Nature is a rich source of chemical substances, of plant and animal origin, with pharmacologically active action [1]. Extracts from plants and shrubs constituted the first herbal medicines used by humanity and came from human beings' search for food and survival. The oldest record of therapeutic treatment is in the Ebers papyrus, which describes a preparation using various herbs, shrubs, leaves, minerals and animal excrement [2] to treat illnesses of the ancient Egyptians around 1500 B.C.

Years later, apothecaries continue with the great challenge of isolating and characterizing the active ingredients present in medicinal plants. In 1805, history changed when a 23-year-old apothecary, Friedrich Willhelm Sertüner, managed to isolate meconic acid from raw opium [3]. This meconic acid would later be renamed "morphine" in honor of the Greek god Morpheus [4] because one of its effects is to make living beings enter a state of deep sleep. After obtaining morphine, other plant alkaloids were also isolated from opium, such as codeine and papaverine.

Modern drug discovery involves a few steps, but can be divided into two major stages: the discovery stage and the development stage. The discovery stage comprises the selection of the biological target, the discovery of compounds that act against this target [5, 6], optimization studies of results to increase properties such as affinity, selectivity, efficacy, among others and also the beginning of *in vitro/in vivo* tests. After these data and their analysis, the development stage takes place in which more *in vivo* studies and clinical trials are used and proceeds to approval by the country's regulatory agency and then to the commercial market.

As can be seen in Fig. 1, the entire development process for a medicine has several stages, resulting in high time and costs throughout the process. Nowadays, there are libraries of compounds developed synthetically or extracted from plants and animals, and this fact, combined with cloning techniques and large-scale expression of proteins, has revolutionized the discovery of new medicines. The technique of high-throughput screening (HTS) of compound libraries against biological targets has become widely used.

The main function of HTS is to select which compounds are capable of acting on a given biological target and how selective these compounds are for that target. This selectivity test is called cross-screening and is important because it shows the probability of toxicity of this compound once it progresses to clinical testing [9]. Toxicity would occur if the compound manages to bind to targets other than the intended one.

The chance of a candidate emerging from these initial tests is negligible. Thus, studies such as structure–activity relationship (SAR) are used to improve pharmacokinetic and toxicological properties of the main compound [10–12].

Another method used to screen compounds is high-throughput virtual screening (HTVS). This method is nothing more than the computational analogue of HTS. In it, the screening is carried out using computer-generated models in which they try to

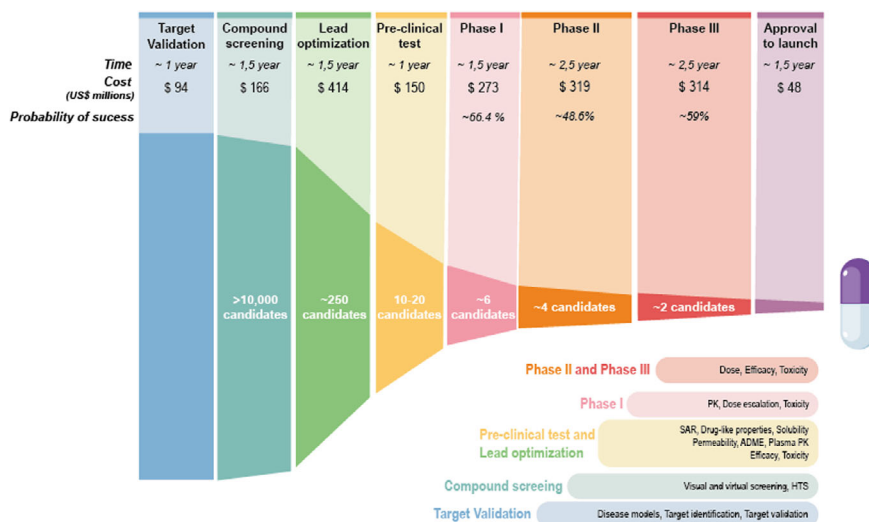


Fig. 1 Process of discovering and developing a medicine in relation to its time and cost. Source adapted from Sun et al. (2022) [7] and Paul et al. (2010) [8]

find the best fit between the ligand and the biological target through techniques such as molecular docking and also molecular dynamics.

The advancement of computational chemistry has profoundly transformed the drug discovery process, evolving from basic simulations of molecular interactions to the use of complex and precise models. Initially, molecular dynamics and molecular docking techniques allowed the prediction of how different compounds could interact with specific biological targets, accelerating the screening of drug candidates. The arrival of artificial intelligence (AI) marked an inflection point, enabling the analysis of large volumes of biological and chemical data with unprecedented speed and accuracy. Today, AI algorithms are capable of identifying complex patterns in data, predicting pharmacological and toxicological properties of new compounds, and even suggesting new molecular structures, revolutionizing pharmaceutical innovation and significantly reducing the time and costs involved in developing new medicines.

2 The Artificial Intelligence in Drug Design

In the current context of developing new methodologies for introducing medicines onto the market, the computer-assisted design of new molecules has become a potential opportunity [13–16] and the use of artificial intelligence is present to revolutionize this environment [17, 18]. The main question is whether these approaches would be able to improve the process and help design new drug candidates [19, 20].

Artificial intelligence is a term used to designate when a machine presents cognitive behaviors such as learning and problem solving similar to human beings [21]. This technology has become increasingly present in our daily lives, such as in image and speech recognition, natural language processing, among others [21] and in pharmaceutical research it would be no different [22].

As previously stated, the process of discovering and developing a drug is complex and time-consuming, depending on intensive techniques such as HTS experiments and also trial and error [23], in addition to often producing results with low precision [24]. Over the last two decades, HTS has presented hit rates ranging between 0 and 0.1%, with these values depending on aspects such as the nature of the target, type and readout of assay, and the quality of the screening compound sets [25, 26]. Another factor that impacts the process is the limited availability of compounds to be used as tests and the difficulty in predicting how their behavior would be in the organism [27].

Artificial intelligence offers the ability to accelerate and improve the entire process through accurate and more efficient analysis of large amounts of data [28]. AI-based algorithms such as learning, evolutionary, or rule-based methods can help explore this large amount of data in different ways [27, 29, 30]. For example, methods such as machine learning (ML) and deep learning (DL) can be employed along the way to predict the efficacy of drug compounds with high accuracy [31] and predict the toxicity of drug candidates [24].

Deep learning, in addition to its use in toxicity, can also be used in the design of new compounds with specific properties and activities. Recently, this technique was used by American and Canadian researchers together with a set of data composed of compounds with known medicinal properties to propose new therapeutic molecular structures [29] with solubility and activity characteristics.

Thus, AI is capable of transforming each stage of the drug development process. Using advanced machine learning algorithms, AI can analyze large volumes of biological and chemical data to identify complex patterns and relationships that are invisible to traditional methods. Furthermore, AI accelerates the screening of candidate compounds, optimizing their properties and significantly reducing the time and costs associated with drug development. Although there are still challenges to be overcome, such as the interpretation of AI models, the application of artificial intelligence in drug design promises to transform the pharmaceutical industry, offering more effective and personalized treatments for various diseases.

3 The Data Revolution in Drug Design

For long-term success, one of the challenges in AI drug design is obtaining appropriate data [32]. The construction of predictive models requires obtaining appropriate data sets for the generation of new chemical entities (NCE) [33, 34] and these data must be evaluated in relation to their scopes and limitations so that they are capable of producing useful results [35, 36].

One of the most recent examples has been the construction of models to predict the toxicity of candidate drug molecules [37]. This data is important because being able to predict the toxicity of a compound can reduce the cost and time required for animal testing and impact the drug discovery process [38, 39]. However, many models are based on results from *in vitro* tests [40] and, in some cases, there is no clearly established relationship between the *in vitro* results and the toxicological response obtained in *in vivo* tests [41].

It is not only toxicological models that address these questions involving the relationships between results obtained in *in vitro* experiments and results obtained in *in vivo* experiments. This type of issue is very present in animal models in which there may be a limited relationship with outcomes observed in patients during drug discovery [42].

In vitro assays also face problems involving the relevance of the data as it is possible that biochemical assays with proteins may reflect differently in relation to the protein found in cells in their native form. Generally, in this case, predictive models are built based on HTS experiments. The use of these data has advantages due to the size of the set, but results may be affected due to decisions that were made to increase assay throughput [43, 44]. Thus, understanding the context of the experiment in which the data set was generated is important to use useful data.

An important factor in building predictive models is understanding that technical errors and biological variability associated with the underlying data are possible, as is accidental misreporting of data. In order to make use of the data, it is also necessary to understand the accuracy and precision of the measurements [45, 46].

Incorrectly reported data can distort the results of a predictive model and its curation with identification of potential errors is one of the areas that AI can also be relevant for drug discovery. At this stage, AI used for fraud detection can be used to detect patterns that are outside those commonly found [32].

Drug discovery is an optimization problem. To obtain a medicine, a molecule that is active against a specific biological target, that has a required pharmacokinetic profile and does not produce adverse results *in vivo* is desired. As a result, several sets of data related to assays are generated, however, much data is lost as only those with the highest yields are sent to *in vitro* or cellular assays.

One of the problems with experimental compound data sets is that they are often unbalanced as they have a small number of active compounds and a large number of inactive compounds. This imbalance is more latent when searching for this data in the scientific literature, as there are fewer reports of negative results.

In recent years, several databases have emerged that contain thousands of biological results, such as ChEMBL [47] and PubChem [48], which can be used for machine learning models to predict biological activities or physicochemical properties for similar molecules.

The data revolution in drug design represents a profound and accelerating transformation in the way new medicines are discovered and developed. The integration of big data, artificial intelligence and machine learning enables the analysis of vast volumes of biological and clinical information, providing valuable insights that were previously inaccessible. These advances enable the optimization of chemical compounds

and improve the selection of candidates for clinical trials, significantly reducing the time and cost of drug development. While challenges like data quality persist, the data revolution is paving the way for a new era by transforming the pharmaceutical industry and improving global health.

4 Artificial Intelligence Methods Used in Drug Design

The arrival of artificial intelligence in drug design brought highly reliable computational methods, allowing the correction of complex chemical problems in the drug identification process [49]. Scientists around the world are developing new methods and algorithms to obtain suitable molecules within a short period of time and also with a cost–benefit ratio compatible with expectations. As a result, the introduction of artificial intelligence, machine learning, deep learning and computational chemistry itself had significant impacts on the process [50].

One of the advances in the area is the emergence of the concept of applicability domains [51–53], after the understanding of algorithms and their statistical interpretation, which allowed computational scientists to increase the transparency of the capabilities of the tools used in the search by computational drugs. This fact led to a steady increase in successful applications using machine learning for the discovery and design of new medicines in the 2000s [54].

With computational advances, the inclusion of GPUs and the increasing amount of RAM in computers, training larger neural networks has become viable. Artificial neural networks (ANNs) were developed based on the human brain and its connections [55], being highly used in molecular modeling and drug design.

In the context of pharmaceutical sciences, ANNs can solve complex problems associated with statistical models used in HTVS, in quantitative structure–activity relationships (QSAR) and pharmacokinetic and pharmacodynamic studies [55]. Its application tends to provide high reliability in results [56], since it analyzes extensive banks of molecules.

In order to recognize a pattern in data, Machine learning (ML) makes use of several methods and algorithms. ML can be classified into several types, with Deep Learning being one of its subfields.

The Deep Learning method is related to artificial networks that interconnect computing elements [57] and is a well-established mathematical model as it can detect the patterns available in the data provided and apply methods to predict future data. In the Kaggle challenge, a team from Toronto demonstrated a complex deep learning AI method [58] by winning a SAR challenge set by Merck. At this time, deep neural network becomes a useful tool for computational molecular design from biological target conception to prediction of pharmacokinetic and toxicological properties.

4.1 Protein Structure

Machine Learning can be used in protein structure to analyze the characteristics of the binding site of a receptor (biological target) [59, 60] and also to identify the biological target. An example worth mentioning was carried out by Costa et al. [61], in which a computational model was developed capable of predicting which genes encode proteins that could be putative molecular targets for drugs.

After identification, the next step would be to predict the 3D structure of the biological target. Recently, deep learning technologies have been able to perform feature extraction and predict secondary structure [62] and main chain twist angle [63]. This method associated with one-dimensional and two-dimensional CNN was able to predict residue contacts [64–66].

The emergence of Alpha Fold also somewhat changed the scenario for predicting 3D structures. Alpha Fold is a software platform developed by DeepMind that aims to help advance our understanding of biological targets [67]. It is composed of an algorithm that, together with AI, uses data from protein sequences to propose their three-dimensional structure and, it is expected, that this will accelerate the development of new medicines and help advance personalized medicine.

One of the most studied interactions in recent times refers to protein–protein interactions, which are important for biological processes and are involved in several diseases [68, 69]. Thus, the prediction of these interactions becomes an important target for the development of new drugs that are different from traditional ones that are focused on coupling to G proteins, kinases, receptors and ion channels [70, 71]. Information regarding these interactions is limited [72] and computational methods are used such as eFindSite [73], a web server that makes predictions of these protein–protein interaction interfaces using support vector machine models. (SVM) and naive Bayesian classification (NBC). Other examples of software that can be used to predict the protein–protein interaction interface through docking methods are ZDOCK [74] and SymDockv [75], as long as the structures have already been resolved [76].

With the advancement of technology, deep learning methods can be applied in cases of protein–protein interactions as they would be able to predict these interfaces. To search for sites for positioning the substance, fragment coupling and direct coupling can be used as done by Bai et al. (2016) [77]. For this purpose, a tool called iFitDock was developed in which hot spots for fitting fragments are searched and these locations are scored using a function based on an evolutionary conservative level.

4.2 Hit Discovery

Hit Discovery is one of the initial stages of the drug discovery and development process that consists of the discovery of compounds (hit) and their selection. One of the most used techniques currently in this selection process is virtual screening,

which refers to the application of algorithms and software that scans libraries of compounds to find bioactive molecules [59].

In general, the process of computer-aided drug design (CADD) and compound selection can be classified into two aspects: structure-based (SBDD) and ligand-based (LBDD) [78]. In both methods, algorithms, scoring functions and force fields are used to classify and evaluate compounds in molecular systems.

When the 3D structure of the protein is available, molecular docking is widely used [79] in SBDD. Although there are successful virtual screening applications with this technique [80], the method still has some limitations such as the flexibility of the protein [79], the fact that the scoring functions cannot accurately predict the binding affinities [81] and residence time ignorance [82]. These limitations mean that the rate of false positive substance results from virtual screening based on molecular docking is high [79, 83].

Machine learning techniques are being employed to improve molecular docking scoring functions [84] such as random forest (RF) [66] and SVM [85, 86]. One of the methods used is the convolutional neural network (CNN) and is being used to predict protein–ligand binding affinities [87].

On the other hand, when the three-dimensional structure of the protein is not available, ligand-based virtual screening studies can be used. In this technique, the molecular descriptors of the compounds are mapped [59] and machine learning methods such as SVM have been used [88–90] with good results [91].

Deep learning methods are also being applied [92, 93] in cases of sparse distribution of active compounds in databases. For this use, a new library of molecules similar to the training molecules was generated through a long- and short-term memory network model based on the similarity between natural language and the simplified molecular input line input specification (SMILES) [93].

Another methodology to be used in hit discovery is drug repositioning as these compounds can have multiple targets [94] and these targets can have multiple effects. In this case, due to the diversity of interactions, the use of networks can be useful [95] as similar medications have similar effects or targets [96].

An example to be cited is DTINet, in which, through the dimensionality reduction approach and the network diffusion algorithm, a heterogeneous network was developed that integrates information from multiple networks and was used to predict indications of substances and a new target [97]. When applying this network in the search for compounds capable of inhibiting cyclooxygenase, the results found were the compounds alendronate, chlorpropamide and termisan, which were experimentally verified and provided results for the prevention of inflammation.

4.3 *Hit to Lead Optimization*

QSAR are mathematical methods used to study the quantitative mapping between the structures, physicochemical properties and biological activity of compounds and are widely used in the optimization process of these substances [98]. The first time the

mathematical method was used was in 1964 with Hansch et al. [99], in which linear regression and physicochemical descriptors were used to describe a 2D structure–activity relationship. That same year, the Free-Wilson method was developed with the relationship between chemical structure and bioactivity [100].

With the emergence of machine learning techniques, other mathematical models were built [101–103] using SVM and RF (random forest). Deep learning methods were also employed using multitask deep neural network (DNN) [58, 104, 105] and they presented good results. The use of this methodology will play important roles in future compound optimization studies.

In the new design part, deep learning methods are also being used to generate new structures with desired properties [106]. Molecules from the ChEMBL database were used to train the recurrent neural network (RNN) and associated with deep reinforcement learning methods were used to generate molecules with predicted biological activity [107]. The use of autoencoders has also been used [29].

In the synthetic evaluation part, a model based on neural networks was developed to find out whether the synthesis of that compound was accessible. The model was trained with a database of chemical reactions and more than 22 million pairs (reagent, product) were used for this training [108], with its result presented by a learned function (SCScore) that describes the complexity of the synthesis process. This model helps chemists in the synthesis processes by helping to eliminate very complex molecules for the development of a medicine.

AstraZeneca is one of the pharmaceutical companies that is betting on the use of artificial intelligence to synthesize compounds. In December 2019, it entered into a partnership with DeepMatter with the aim of increasing the efficiency of synthesis through the use of digital technologies licensed for artificial intelligence [109].

4.4 Evaluation of ADMETox Properties

The risk of failure in drug discovery can be reduced once the physicochemical, pharmacokinetic and toxicological properties of the compound being used in studies are known. Several deep learning-based approaches have been developed to predict these intrinsic properties of molecules [110].

Regarding physicochemical properties, models have been developed to predict aqueous solubility and Caco-2 permeability, for example. In the case of aqueous solubility, the first model used convolutional neural network (CNN) associated with an artificial neural network (ANN) to extract information with good predictive performance while the second surpassed it, as it used information in greater atomic level [111]. For Caco-2 permeability, Wang et al. (2016) [112] collected 1272 compounds that presented this data and used Boosting, multiple linear regression (MLR), SVM and partial least-squares (PLS) to build prediction models with 30 descriptors.

Pharmacokinetic parameters such as bioavailability are extremely important when it comes to drug development. Bioavailability reflects the degree of absorption of a compound and helping to predict this property can help optimize this factor. Multiple

linear regression (MLR) models have already been used in this property based on a data set of 1014 molecules and showed good predictive performance [113].

Another important piece of information to consider when developing a medicine is its distribution within the body. Partial least-squares (PLS) and Random Forest (RF) models were used to predict the steady-state distribution based on 1096 molecules, but the results were not satisfactory [114]. Something that may have generated this result is that there are many unknown factors that can affect the distribution of a compound within an organism and it would not be possible to predict just based on its chemical structure.

The path a medicine takes after being administered into the body is to be metabolized. In this metabolization process, the medication may lose function or even produce toxic metabolites. Being able to predict metabolization helps medicinal chemists optimize the molecule to ensure metabolic stability of the compound. Machine learning methods are employed such as neural networks that can provide information of small molecules being metabolized by cytochromes P450s [115, 116].

The last stage of a medicine in the body is its excretion, that is, its elimination from the body. The principal component analysis (PCA) method was used to predict the primary clearance mechanism and the model presented results with a predictive accuracy of 84%. The partial least-squares (PLS) model for total human clearance was also used and performed well compared to animal scaling methods [117].

Toxicity is another parameter that must be taken into account, as it causes the elimination of approximately 1/3 of the compounds that are in the process of being evaluated to become a medicine [118]. Thus, as with other parameters, its prediction becomes useful as it helps in optimizing and reducing failures during drug development.

Generally, toxicity profiles are predicted through rule-based knowledge and structural alerts, and this type of prediction can cause false positives. Therefore, deep learning models present a revolution in relation to this data in addition to showing good performance in this type of prediction.

Acute oral toxicity prediction models using convolutional neural networks showed better results than those based on SVM [119]. Another model developed was multitask DNN, which also showed good results for toxicity prediction [120].

4.5 Clinical Trials

Clinical trials are important to ensure the safety and efficacy of new medicines, but some factors such as underwriting, side effects and contradictory data contribute to their failure. They are divided into 4 phases named phase I, phase II, phase III and phase IV.

In phase I, the medicine or therapy is tested on a small number of patients between 20 and 80 people and the aim is to observe the safety of the tested compound and its negative effects. In phases II and III, the patient sample increases to between 100 and 300 for phase II and between 1000 and 3000 for phase III and, at this point, the

efficacy and safety of the compound are evaluated in addition to monitoring adverse effects. Finally, in phase IV the effect on the general population is monitored to determine the best forms of application and this phase only begins after approval by the Regulatory Agency [109].

The success of a clinical study depends on several factors such as the selection of participants, as good identification allows the potential effectiveness of the trial to be increased. Artificial intelligence can be used at this stage by analyzing data and identifying which patients would respond well to a particular clinical study [109].

Another important role in which artificial intelligence can be used is in the analysis of data from clinical studies [121, 122] and also in the quality of this data, since it is submitted by patients [123]. This technology can also be used in biosimulations to measure the human reaction to a certain compound before it is tested in a living human [124].

In contrast, integrating artificial intelligence into clinical studies is still a challenge. Organizing the database for artificial intelligence training is a critical part of the process. For a patient to be selected, several factors are taken into account, such as their history and comorbidities, and there is a risk that the intelligence models will not analyze this data correctly and may affect the results of the study [109]. Furthermore, the presence of imperfect databases can compromise the entire procedure, since patients of European and Caucasian descent are overrepresented in medical studies and genetic databases [125, 126].

Therefore, artificial intelligence has the capacity to revolutionize clinical trials, but some factors may restrict its use and scalability.

5 Ethical and Regulatory Challenges in Using Artificial Intelligence in Drug Discovery

One of the most discussed topics in recent times refers to the regulation of the use of artificial intelligence in all fields and healthcare would be no different. The use of technology has ethical implications [127, 128], as a computer would make decisions that affect people's health and well-being.

Some recurring concerns about the use of artificial intelligence in the pharmaceutical industry are the question of which drugs to develop, which clinical studies to carry out and how to market the drugs. If we look at the databases present in clinical studies, for example, we realize that the majority of them refer to data from people of Caucasian and European descent, which could cause unequal access to medical treatment as the development would be aimed at individuals with these characteristics [125, 126].

There are also questions being raised regarding privacy and data security as there is a risk of personal information being used inappropriately, since artificial intelligence to be trained depends on a large amount of data. On the other hand, within the

pharmaceutical industry, there is concern about automation, which could lead to job losses [129–130].

Therefore, the use of artificial intelligence within the discovery and development of medicines requires special care and the adoption of different approaches that are capable of guaranteeing the use of diverse and representative data, in addition to regularly carrying out audits on the systems and implementing a security protocol data. With these attitudes taken, artificial intelligence can be used ethically and responsibly.

6 The Future of Drug Discovery with Artificial Intelligence

The future of drug discovery is being radically transformed by artificial intelligence (AI). This revolution promises to accelerate the development of new treatments, improve precision in compound creation and significantly reduce associated costs.

As explored throughout the chapter, traditionally, discovering new drugs is a long and expensive process, which can take more than a decade and cost billions of dollars. AI has the potential to shorten this time by analyzing large volumes of biological and chemical data quickly. Machine learning algorithms can identify patterns in complex data, enabling the discovery of new therapeutic targets and candidate compounds with greater efficiency than conventional methods.

Another important capability of the technology is that it is able to predict how different compounds will interact with biological targets, which is essential for evaluating the efficacy and safety of potential medicines. Advanced predictive models can simulate virtual clinical trials, identifying potential toxicity issues or adverse effects before compounds reach human testing. This not only speeds up the process, but also increases the safety of the medicines developed.

Despite the promises, the application of AI in drug discovery also faces significant challenges. Data quality is a critical factor—AI algorithms are only as good as the data that feeds them. Additionally, there are ethical concerns surrounding the use of AI including the privacy of patient data and the potential for biases in algorithms that could affect equity in access to new treatments.

Therefore, the integration of artificial intelligence into drug discovery represents an exciting frontier in modern medicine. With the potential to transform every step of the development process, from identifying new compounds to personalizing treatments, AI promises to make the creation of new medicines faster, safer and more cost-effective. However, to fully realize this potential, it will be crucial to address the technical and ethical challenges that accompany this revolutionary technology.

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Laccase: A Catalyst for Change in Eco-Friendly Industrial Processes



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Abstract The oxidoreductase enzyme laccase demonstrates versatile catalytic abilities under mild conditions, making it a well-investigated catalyst for a range of industrial applications. Its adaptability extends from pulp and paper to textile, bioremediation, and bioenergy processes. Due to changes in consumer patterns, this chapter will present bioprocesses, genetic engineering, and other approaches to boost enzyme yields. Future perspectives, along with challenges to reduce costs, will also be discussed to facilitate its wider adoption in biotechnology. Overcoming these challenges can enhance the position of laccase as a pivotal player in sustainable biotechnological processes, paving the way for eco-friendly solutions and the generation of advanced products across various industries.

Keywords Oxidoreductase enzyme · Sustainable biotechnology · Green chemistry · Bioprocesses

1 Introduction

Laccase (benzenediol:oxygen oxidoreductase; EC 1.10.3.2) is a blue-copper oxidoreductase enzyme renowned for its ability to catalyze the oxidation of a broad range of substrates, including phenolic compounds, while simultaneously reducing molecular oxygen to water [1]. The fungal variant of laccase contains two disulfide bonds and four copper atoms, which are organized into three distinct centers: T1, T2, and T3. The T1 center is mononuclear and features an absorption band around 610 nm, imparting the enzyme with its characteristic blue hue. The T2/T3 centers form a

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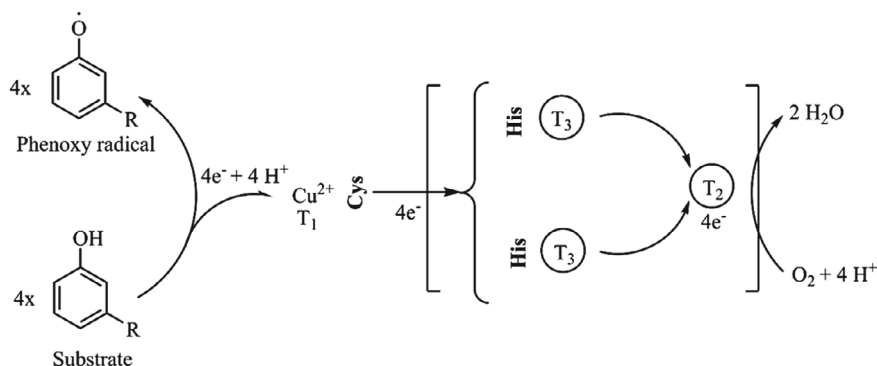


Fig. 1 Simplified mechanism of substrate oxidation in laccases

trinuclear cluster. Oxidation of the substrate takes place at the T1 center, facilitated by a tripeptide sequence His-Cys-His. The electrons extracted during this process are then transferred to the T2/T3 site, where molecular oxygen is reduced to water (Fig. 1) [2]. This intricate structure endows laccase with remarkable catalytic efficiency and versatility, making it highly valuable in various industrial, environmental, and biomedical applications.

Historically, this enzyme, classified as a copper-containing polyphenol oxidase, was first discovered in the exudates of the Japanese lacquer tree, *Rhus verniciflua*, in 1883 by Yoshida [3]. Later, it was also identified as a fungal enzyme by Bertrand and Laborde in 1896. Laccase is present in a diverse array of organisms, including plants, fungi, insects, bacteria, lichens, and sponges. Its ability to catalyze the oxidation of a wide range of phenolic and non-phenolic compounds, as well as environmental pollutants, makes it valuable for various industrial and environmental applications [4].

The use of laccases in industrial processes is widely recognized as an environmentally sustainable choice, perfectly aligning with the concept of a circular economy based on the principles of “reduce, reuse, recycle”. A recent report on the global market for industrial enzyme applications projected significant growth over the next five years, with an estimated market value of \$8.7 billion by 2026, reflecting a compound annual growth rate (CAGR) of 6.3% from 2021 to 2026 [5]. Beyond these impressive figures, it is important to highlight the innovative uses of laccases across various industrial sectors. For instance, they play a crucial role in the production of biofuels, the treatment of industrial effluents, and the manufacture of pharmaceuticals and food products. This versatility makes laccases a promising target in the global enzyme market. In 2020, the size of this global market was estimated at about \$3 billion, with projections to reach \$4 billion by the end of 2027, maintaining a CAGR of 4.3% between 2021 and 2027 [6].

Developing and studying enzymatic catalysts is essential for sustainability and environmental protection. They reduce emissions, enable cleaner production, and enhance energy efficiency, supporting renewable energy and green chemistry while

reducing reliance on fossil fuels. These catalysts lower costs, drive innovation, and are crucial in pollution control, carbon capture, and the hydrogen economy, aiding in mitigating climate change and advancing science and the economy [7, 8]. Consequently, laccases not only offer a sustainable choice but also provide significant benefits across various industries. In this chapter, we will delve into laccases' roles in different industrial sectors, their mechanisms of action, and the challenges and opportunities related to their production and application.

2 Laccase Applications in Industry and Products, Services

Laccase, a versatile enzyme widely distributed in nature, continues to attract significant interest due to its remarkable effectiveness in various industrial, environmental, and biomedical applications. Its variable molecular weight and dual regulation, combining constitutive and inducible characteristics, enable its use across diverse fields (Fig. 2).

From paper processing to the decolorization of water and wine [9], and even in the production of anticancer drugs [10], laccase has proven essential. In the industrial sector, it is indispensable for textile finishing and biofuel production [11], contributing to biochemical pulp bleaching [12] and the delignification of plant products [13]. Additionally, its ability to degrade xenobiotic compounds and pollutants [14] makes it crucial for depolymerization and humification processes [15], playing a key role in sustainable waste management and environmental preservation [16].

In medicine, laccase applications are even more fascinating, ranging from the preparation of antiviral and antioxidant agents to the synthesis of anticancer drugs and hormonal derivatives [17]. Its use as a marker in immunoassays, and its applications

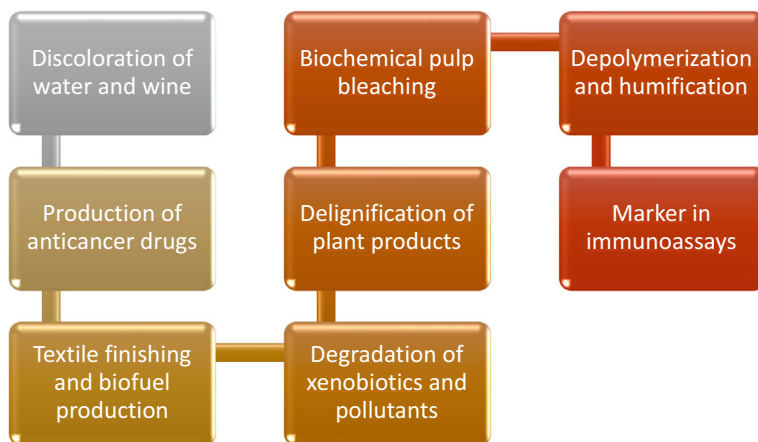


Fig. 2 Versatility of laccase applications

in nanotechnology [18] and energy transformation systems, highlight its potential in innovative biomedical and pharmaceutical fields, promising significant advances in disease treatment and the development of more effective therapies [19]. Thus, laccase not only addresses industrial and environmental challenges but also offers exciting opportunities for advancements in medicine and materials science.

3 Bioprocesses for Laccase Production

Laccase is found across a wide spectrum of organisms and plays pivotal roles in diverse biological and ecological settings. Found in bacteria, fungi, plants, and insects, laccases are predominantly extracellular enzymes, although they can sometimes be intracellular [20]. Both Gram-positive and Gram-negative bacteria are known to produce laccase, inhabiting various terrestrial and aquatic environments [21]. These enzymes are particularly abundant in different bacterial phyla, encompassing α - and γ -proteobacteria, *Firmicutes*, *Cyanobacteria*, *Aquificae*, *Deinococcus-Thermus*, and even *Archaea* [2].

Fungal laccases, primarily produced by *Basidiomycota* and *Ascomycota* species, participate in various biological processes such as defense, virulence, pigmentation, and sporulation. Their ability to degrade lignocellulose is vital for the carbon cycle in the biosphere [22]. Along with other catalysts like xylanolytic and glucanolytic and cellulolytic enzymes, they are essential for decomposing lignocellulose polymers [23–25].

In plants, laccases are commonly found in gymnosperms and angiosperms, but they have also been detected in green algae like *Tetracystis aeria*. These enzymes are present in various higher plant species, contributing to critical biological processes [2].

In the animal kingdom, insect laccases, predominantly found in the orders *Hymenoptera*, *Diptera*, *Lepidoptera*, and *Coleoptera*, play diverse roles in physiological and defense processes, facilitating ecological adaptation and interaction with their environment [21].

Several types of bioreactors and operation strategies, ranging from laboratory to pilot scales, have been studied to support microorganisms [26–32] and enhance the production of their metabolites. These include submerged, tray, expanded bed, and rotating drum bioreactors, each presenting unique advantages and challenges concerning efficiency and control of cultivation conditions. Despite advancements, significant improvements are still warranted. To advance ligninolytic enzyme production, novel bioreactor models that consider efficiency and the importance of aeration in macromycete fungi cultivation are indispensable. Bioreactors equipped with more effective aeration systems can create optimal conditions for fungal growth and metabolite production, markedly boosting ligninolytic enzyme production such as laccase [33]. Furthermore, attention must be paid to the specific effects of laccase-producing microorganisms on the modification of different kinds of lignocellulosic biomass, as has already been reported with other bioprocesses [34].

The production of laccases in bioreactors using solid-state fermentation (SSF) is less documented in the scientific literature compared to other enzymes, with the submerged fermentation process being more commonly reported for their production [20]. Traditionally, laccase has been produced by white-rot fungi using batch submerged fermentation, enabling controlled conditions and heightened enzyme production [35]. However, large-scale industrial application of laccase has been hindered by low production levels, natural enzyme expression, and high costs [36]. Achieving high production, quality, and enzyme performance is crucial for industrial laccase applications, yet current methods have not attained widespread adoption. Hence, ongoing research endeavors focus on exploring strategies to enhance laccase production [20].

Currently, the commercial production of laccases is predominantly carried out using fungi, particularly species belonging to the genera *Trametes* and *Pleurotus*. These organisms are cultivated in large industrial fermenters, where the necessary nutrients are provided for their growth and laccase production [37]. The processes are then compared with commercial prices to determine their economic viability. This approach not only identifies the most effective production methods but also establishes a framework for potential bioproducts derived from microbial sources, with laccases being a prominent example. Among the fungal species studied for laccase production, *Trametes versicolor*, *Trametes hirsuta*, and *Pleurotus pulmonarius* stand out as important sources of knowledge and established commercial production. Additionally, more recently studied species such as *Pseudolagarobasidium acaciicola*, *Funalia trogii*, and *Marasmiellus palmivorus* show promise for future industrial applications [20].

4 Trends in Laccase Research

Laccase plays diverse roles in industry, leading to ongoing evolution in research concerning its application and production. Apart from its conventional uses in biotechnology and the food sector, laccase finds application in medical diagnosis biosensors, biofuel cells, bioremediation of herbicides, pesticides, and toxic soil chemicals, as well as in the production of biopolymers and high-value-added chemicals [20, 38, 39].

Addressing challenges related to the stability of this enzyme and its gradual loss of activity over time, researchers have implemented immobilization procedures [2, 20, 40–42]. Additionally, investigations into protein engineering methods aim to enhance laccase properties such as stability, activity, and specificity, potentially involving modifications to the enzyme's structure to optimize its function under varying conditions [43, 44].

Moreover, there is a growing focus on developing more sustainable approaches to laccase production. Metabolic engineering is being utilized to enhance laccase production in host organisms like bacteria and yeast. In addition to traditional fungi such as *Trametes versicolor* and *Pleurotus ostreatus*, researchers are exploring new

frontiers, including bacteria and fungi isolated from extreme environments. This diversification broadens the scope for industrial laccase production, offering potential benefits in terms of enzyme stability and performance [4, 45–47].

The production of laccase using *Penicillium brevicompactum* CCDCA 11,400 through solid-state fermentation with sugarcane bagasse and atrazine resulted in significant laccase production and degradation of sugarcane bagasse, particularly at lower temperatures (19 °C) and acidic pH (pH 4) [48]. Studies like this address issues such as the traditional reliance on expensive substrates, now replaced by exploring renewable and alternative sources of carbon and nitrogen, such as agro-industrial residues, to make laccase production more accessible and sustainable [49]. Other examples are reported in Table 1.

Another innovative study presented the characterization of a new yellow laccase (Lac_CB10) from *Chitinophaga* sp. CB10, which demonstrates remarkable stability and functionality under extreme conditions. The enzyme exhibited optimal activity at an alkaline pH (10.5) and high temperatures (80–90 °C), maintaining more than 50% activity for 5 h at these temperatures [50]. Lac_CB10 is capable of significantly degrading lignin, achieving 52.27% degradation in 32 h. Similarly, the production and characterization of a new CotA-laccase enzyme from *Bacillus altitudinis* SL7, which was cloned and overexpressed in *E. coli* BL21 (DE3) to increase its concentration and potential for lignin degradation, showed thermostability and reduced lignin by more than 31% at 55 °C and pH 5.0 [51]. This demonstrates the potential of these enzymes for industrial applications, such as biorefinery processes and bioremediation of effluents from pulp and paper mills.

Overall, these trends are helping to overcome the bottlenecks associated with laccase production, making it more accessible, sustainable, and versatile. Research is exploring more advanced applications of laccase in biotechnology, including the enzymatic synthesis of high-value chemical compounds and the conversion of lignocellulosic biomass into useful products. This can facilitate its adoption in various environmental applications, from wastewater treatment to the generation of commercially valuable chemicals.

4.1 Importance of Discovering and Isolating New Microbial Strains with Enhanced Laccase Production and Biochemical Properties

The Protein Database presently hosts 1,571 structures associated with the keyword “laccase”. These structures are accessible at the Research Laboratory for Structural Bioinformatics within the Protein Database (RCSB PDB) [20]. The identification and isolation of novel microbial strains with heightened laccase production and biochemical attributes carry significant implications.

Uncovering new microbial strains capable of laccase production aids in the exploration and comprehension of microbial diversity across diverse ecosystems.

Table.1. Laccase production under different conditions

Microorganism	Enzyme (s)	Growth medium	Cultivation conditions	Optimal enzyme conditions	References
<i>Dictyopanus pusillus</i> LMB4	Laccase	Palm empty fruit	Temp. 28 °C and 60% humidity in the dark for 30 days	Temp. 45 °C, pH 3 to 5 and 26 days	[52]
<i>Alkalophile</i> strain	Laccase, Lignin peroxidase (LiP) and manganese peroxidase (MnP)	Wheat bran	Temp. 45 °C, 90% humidity	LiP and laccase: temp. 55 °C and pH 7.0 and for MnP: temp. 60 °C and pH 8.6	[53]
<i>Aspergillus niger</i>	Laccase	Sugarcane bagasse, reed grass, rice bran, wheat bran, wheat straw, corn cob, guava leaves and peanut shells	Temp. 35 °C, 30% humidity and 5 days	Temp. 45 °C and pH 5.5	[54]
<i>Bhargavaea bejingsis</i>	Laccase	Century Pulp and Paper Mill, Lalkuan, Uttarakhand, India	Temps. (25 °C, 30 °C, 35 °C, 45 °C, 50 °C, 55 °C), pH (3–11), incubation time (24–120 h)	Temp. 37 °C, pH 7.0 and 72 h	[55]
<i>Megasporoporia</i> sp.	Laccase	Olive and potato leaves	Temp. 35 °C and 14 days	Temp. 25 °C, pH 7 and a reaction time of 60 min	[56]
<i>Stenotrophomonas maltophilia</i> strain E1	Laccase	Sawdust, coconut husk, rice, wheat and barley bran	Temp. 35 °C for 36 h	Temp. 35 °C and pH 5.0	[57]
<i>Hexagonia hirta</i> MSF2	Laccase	Coconut pith waste	Temp. 28 °C, pH 4.6 and 75% moisture content	Temp. 40 °C and pH 4.0	[58]
<i>Fomes fomentarius</i>	Laccase	Corn husk	Temp. 30 °C, 10 days and 50% humidity	Temp 35–40 °C and pH 7.0	[59]

Colonies isolated from distinct environments may possess biochemical traits tailored to specific conditions such as temperature, pH, substrate presence, and other environmental factors [12, 55, 60].

TOKER et al. [60] isolated strains of six marine fungi (*Alternaria* sp. D11, *Alternaria* sp. D21, *Cadophora* sp. D43, *Cadophora luteo-olivacea* D51, *Phoma* sp. K2BR, *Phoma* sp. K21) from a lagoon in Turkey and demonstrated their promise as a source of laccase, showing significant potential for biotechnological applications in the decolorization of industrial dyes, with 98% efficiency for methyl orange and 93% for acid green. JIANG et al. (2023) [61] investigated the production of a new laccase from *Vibrio* sp. (L01) and found that it maintained significant enzymatic activity at lower temperatures (10–35 °C) and across a wide pH range (4.5–10.0).

Other studies have identified bacterial laccases with unique characteristics compared to fungal laccases. For instance, the laccase from *T. thermophilus* demonstrated high thermophilicity, with an optimal reaction temperature of 92 °C and thermal tolerance exceeding 14 h at 80 °C [39]. Debnath & Saha [62] explored production strategies for laccases from mesophilic and extremophilic microorganisms under various conditions, concluding that laccases from different sources possess unique properties despite similar structural and catalytic characteristics.

5 Methods to Enhance Laccase Production

There are various techniques for enhancing laccase production in microorganisms that produce it. One such method involves optimizing culture media by adjusting their composition to create optimal conditions for growth and enzyme production. Additionally, precise control of parameters like pH, temperature, and aeration in the culture medium can stimulate laccase growth and production [54]. This approach is already employed to optimize the production of various enzymes, including laccase, across different microorganisms, even those not typically known for producing this enzyme. It underscores the flexibility in adjusting conditions to enhance enzymatic activity and utility. Examples of such practices are summarized in Table 1. Another strategy is enzyme induction, where specific inducers are used to boost laccase production [56, 63, 64]. Conversely, technologies such as LED, laser, ultrasound, pulsed electric field, magnetic electric fields, and alkaline stress have been investigated to increase microbial enzyme production. Exposure to these external stimuli can trigger changes in microbial cells, activating metabolic processes linked to enzymes and other metabolite production [65–72]. These strategies hold promise for enhancing enzymatic synthesis, thereby improving efficiency and economic viability for various industrial and environmental applications. However, it's crucial to carefully optimize and assess these techniques for their impact on microbial growth, cell viability, and the quality of the produced enzyme.

LED or laser irradiation has been explored as a means to stimulate microbial metabolism and enzyme production [73]. These techniques can enhance laccase production by influencing cellular processes such as gene expression and protein

synthesis. The specific wavelengths and intensities of light can be optimized to promote the growth and activity of laccase-producing microbes. For example, certain wavelengths may activate photosensitive molecules or trigger photobiological responses, leading to increased metabolite production [74, 75]. Such responses are intrinsic to each microorganism, as different species may have distinct molecular mechanisms of detection and response to light. Effects observed include changes in mycelium growth and reduced laccase activity when green light was irradiated on cultures of *Pleurotus eryngii*, while a 4.4-fold enhanced activity was achieved when employed on *Pleurotus pulmonarius* [76]. Furthermore, only modest laccase production by *Cerrena unicolor* was reported when irradiated with different LED colors [77]. Additionally, the effects of red light (630 nm) as a means to improve the catalytic activity of commercially available laccase from *Trametes versicolor* were reported [78].

Ultrasonic mechanical waves can disarrange cellular structures and enhance mass transfer, thereby improving nutrient uptake and waste removal in microbial cultures. This can create favorable conditions for enzyme production or secretion. Ultrasound can also induce physical or chemical stress on microbial cells, triggering a cellular response that may include increased enzyme synthesis as a survival mechanism [79]. Ultrasound irradiation through a liquid medium generates reactive free radical species, promoting local turbulence/liquid microcirculation and enhancing mass transport process rates. Combined laccase and ultrasound processes lead to the reduction of the consumption of chemicals, energy, and water, presenting promising systems for reducing the environmental impact and operational costs of conventional industrial processes [80]. A study devised an effective intermittent ultrasonic method to enhance laccase production from *Trametes versicolor* mycelia cultures. The optimized approach involved subjecting 2-day-old cultures to two 5-min ultrasonic sessions, spaced 12 h apart, at a constant power and frequency (120 W, 40 kHz). After 5 days, a 1.8-fold increase in laccase was detected due to increased pellet porosity, aiding nutrient and metabolite transfer, and stimulated laccase gene expression (lcc), leading to a significant rise in both extracellular and intracellular laccase activity [81]. *Trametes versicolor* immobilized in alginate beads was stimulated by ultrasonic treatment during 36-h-old bead cultures with 7-min ultrasonic treatments twice with a 12-h interval (120 W, 40 kHz), resulting in an increase higher than 2.1-fold. The authors also interpret this result due to the improved mass transfer of nutrients and product between the liquid medium and the gel matrix [82]. Other works reported improved laccase bleaching efficiency when irradiated with low-intensity ultrasound (7 W) for 30 min, suggesting improved diffusion of the enzyme from the liquid phase to the solid material present in the reaction medium [83], while improved laccase activity against 2,4-dichlorophenol (2,4-DCP) in soil was reported due to ultrasound treatment [84].

Finally, magnetic fields can influence cellular processes by exerting forces on charged particles within cells. By applying magnetic fields to microbial cultures, it may be possible to modulate metabolic pathways involved in laccase production. Ultrasound applied for 10 min per day allowed for enhanced laccase production by *Lentinula edodes* under submerged fermentation [85].

Random mutagenesis techniques, such as chemical mutagens or radiation, can induce genetic variations in laccase-producing microbes. Mutants with improved laccase production capabilities can be selected through screening or directed evolution approaches, leading to strains with enhanced enzyme secretion. To date, laser radiation at a power density of 0.24 mW/mm² for 40 min has been employed to generate mutant strains of *Phellinus igniarius* producing higher laccase enzyme (+36.8%) [86]. Other work reported the use of light and magnetic fields for mutagenesis of *Phellinus* spp., generating mutants with higher growth and laccase production [87].

6 Overview of Synthetic Biology, Genetic Engineering for Laccase Production

Despite extensive studies on molecular techniques for monitoring microbial and metabolic studies [88], laccase production is primarily driven by synthetic biology and genetic engineering, both of which play vital roles in this process. In synthetic biology, biological systems are deliberately designed and constructed to execute specific functions, such as enhancing enzyme production. This involves designing synthetic metabolic pathways that favor the synthesis of proteins and bioactive compounds, ultimately yielding final products with desirable functional and structural characteristics, including enzymes [89, 90].

Synthetic biology harnesses genetic and computational tools, such as accessible genome sequencing, gene synthesis, genome editing, and directed evolution technologies, to optimize microbiological metabolic engineering. This optimization aims at enhancing the production of both native and foreign biomolecules, such as proteins and metabolites [90]. Conversely, genetic engineering manipulates genetic material to alter organism traits [91].

Advanced genome editing techniques enable precise modifications to be made to the DNA of laccase-producing microorganisms, thereby contributing to increased efficiency and secretion capacity [91]. Additionally, heterologous expression involves introducing laccase genes from one species into a different host, like transferring fungal genes into bacteria or yeast, to enhance enzyme production in more scalable fermentation systems [92, 93].

Furthermore, host optimization involves maximizing enzyme production and secretion by modifying metabolic pathways, protein secretion systems, and selecting laccase-producing strains with desired traits [89–91]. These integrated approaches have the potential to improve the efficiency and economic viability of laccase production, benefiting a wide array of industrial and environmental applications.

Laccases in microorganisms are encoded by genes (such as *pcoA*, *copA*, and *lac-AB*) that are metabolically associated with the utilization of plant substances and secondary metabolites exhibiting antimicrobial [94], antitumor, antibiotic, and antioxidant activities [19].

A laccase was cloned from cDNA of *Fusarium oxysporum* HUIB02 and expressed in *P. pastoris*, yielding an extracellular glycosylated protein with a production of 21,966 U/L, aimed at dye removal [95].

An approach that does not follow a rational design is error-prone PCR, which enabled the creation of a recombinant laccase with four mutation sites, resulting in a 22% improvement in expression compared to the control recombinant enzyme [96].

A laccase from *Agrocybe pediades* was synthesized de novo and expressed in *Saccharomyces cerevisiae* using an improved signal peptide and enzyme-directed evolution, resulting in improved activity at neutral pH and high tolerance to inhibitors [97].

A constructed laccase plasmid from *Bacillus licheniformis* utilized modified target genes via error-prone PCR for site-directed mutation to achieve higher enzyme activity (426.13 U/mg), resulting in enhanced thermal stability, a higher optimal temperature (70 °C), and improved dye decolorization ability for acid violet dye [98].

Another example of heterologous expression involves the cloning and expression of the Laccase gene from *Bacillus licheniformis* NS2324 in *E. coli* using pUC18 as the cloning vector and pet15b as the expression vector for Methylene blue degradation [99].

A study aimed to increase laccase productivity from white-rot fungi for commercial use by inserting the lacA gene from *Trametes* sp. AH28-2 into *Trichoderma reesei*, utilizing the strong promoter Pcbh1 from the cbh1 gene. Initially, lacA integration did not yield detectable laccase expression but triggered intense ER stress. To address this, the expression cassette was modified to include Tcbh1, resulting in lacA:: Δ cbh1, which replaced cbh1. This alleviated secretion pressure, leading to successful lacA integration and transcription with minimal UPR and ERAD activation. Consequently, laccase activity peaked at 168.3 U/L in the fermentation broth, indicating the efficacy of reducing secretion pressure by deleting endogenous protein-encoding genes for enhanced heterologous protein secretion in these industrial fungi [92].

Furthermore, small laccase genes from *Streptomyces coelicolor*, *Streptomyces viridosporus*, and *Amycolatopsis* 75iv2, with their native signal sequences, were expressed in Gram-positive *Bacillus subtilis* and *Streptomyces lividans* host organisms. ScLac, SvLac, and Amlac expressed in *S. lividans* reached 1950 ± 99 U/L, 812 ± 57 U/L, and 12 ± 1 U/L, respectively [100].

Another approach involves the use of genetically modified insects like *Drosophila melanogaster* to overexpress laccase from the fungus *Trametes trogii*, reducing the inputs required for microbial bioprocesses and enzyme production, as dried powders from the fly can be used to obtain laccase extracts [1].

7 Concluding Remarks

This chapter highlights the vital role of laccases in traditional applications such as textile processing and bioremediation, as well as in the production of value-added products and innovative bioremediation approaches. Due to regulatory pressures and carbon-neutral targets set by various nations, a significant increase in the demand for enzymes like laccase is anticipated in the coming years. Research into laccase production is varied and should be integrated across several well-funded areas: basic bioprocessing, where new raw materials and strains are investigated; genetic engineering to enhance production or alter enzyme functionality; and cutting-edge biotechnology, where synthetic biology could generate non-natural replicating structures to maximize laccase enzyme yields. Additionally, a less-explored top-down approach involves designing and optimizing microbial platforms for improved production through mechanical and electromagnetic stimulation. Advances in the exploration of extremophiles could also lead to the discovery of new types of laccases, which could either be replicated in more efficient producers or used as models to improve existing enzymes. Biotechnology, bolstered by advancements in computational power, is poised to contribute to the fourth industrial revolution in the coming years, improving the accessibility, cost-effectiveness, and efficiency of microbial laccase enzymes.

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Photostimulation Applied to the Production of Biopolymers (Exopolysaccharides)



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Abstract The wide range of applications of microbial biopolymers and the extensive world market, have been directing efforts to obtain techniques to optimize their biosynthesis. The photobiomodulation of bioprocesses is a promising tool to consider, has been gaining a prominent place in health, food, environmental and industrial areas. Low power irradiation, emitting in certain parameters (wavelength, energy density, power, irradiated surface, among others), can present biostimulatory results. Stimulating specific targets that encompass terminal enzymes of the respiratory chain, such as cytochrome c oxidase, increasing the electron flow rate and consequently increasing the quantity of available cellular ATP. Radiation emission may also activate intracellular metalloproteins that actively participate in biosynthetic pathways. As a consequence of cell proliferation, chemical signalling mechanisms develop through self-inducers that alter gene expression and regulate metabolic activities, activating cooperative behaviour between micro-organisms. Photostimulation provides a vast field of opportunities, of great economic benefit, establishing sustainable practices, often contributing to the circular economy system, increased productivity and yield from bioprocesses.

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1 Introduction

Biologically degradable polymers, produced by living organisms without causing any negative effect on the environment are defined as biopolymers. They are macromolecules formed by the covalent union of small molecular units called mesomers, which were obtained through a process of polymerization from smaller molecules called monomers [1].

The first biopolymers to be used were those derived from marine and terrestrial plants, later appearing a new generation, in the 1950s, the polysaccharides of microbial origin [2]. The great diversity of chemical structures capable of being elaborated by microorganisms makes it possible to obtain water-soluble polymers with different characteristics and consequently properties. Being among these, the exopolysaccharides (EPS), polysaccharides secreted by microorganisms of special physicochemical properties (viscosity, stability, ability to form gels) and with remarkable applications in food and petroleum industries (xanthan gum) [3], cosmetics (FucoPol and FucoGel) [4, 5], pharmaceutical and biomedical (dextran, cellulose, hyaluronic acid and levan) [6–8]. Due to environmental, socioeconomic factors and their biocompatibility and degradability characteristics, researches have been developed aiming to replace synthetic polymers of petrochemical origin, by biopolymers. One of the barriers to the consolidation of biopolymers in the contemporary market is due to the costs of production, extraction and purification compared to synthetic polymers [9]. For this reason, research has been carried out to make the process economically viable, investigating residual substrates or industrial by-products as a source of carbon/nitrogen, optimising its biosynthesis conditions and developing methods capable of influencing the productive metabolic pathways, such as photobiomodulation [10].

Photostimulation has proved to be a useful tool in biotechnology due to its ability to optimise biological processes, interacting with the physical–chemical sciences, inducing biochemical and metabolic pathways capable of interfering with cell proliferation and protein activation, improving phenotypes without altering genetics, cell proliferation and protein activation, improving the phenotype without altering the genotype. A promising and ecologically sustainable method, the emission of laser and LED irradiation can improve the adaptability of different microorganisms, participate in the production of cellular ATP, as well as increase the production of various biocompounds [11, 12].

2 Microbial Exopolysaccharides

A wide range of microorganisms have the ability to synthesise intracellular polymeric, structural and EPS substances, mainly driven by environmental signals [13].

EPS are classified as homopolysaccharides when they are composed of repeated units of a single monosaccharide, the most common being dextran, curdlan and cellulose. Heteropolysaccharides are branched and consist of repeated units of more than one monosaccharide, mainly glucose, fructose and galactose, with various differentiating carbohydrate groups [14].

EPS has been gaining wide interest due to its great diversity of structural and functional properties and diversity of application areas, food, health, environment, agriculture and oil extraction.

Biofilm-secreting microbes are encapsulated by a matrix of extracellular polysaccharides that participate in their protection against antimicrobial agents, such as antibiotics, UV radiation, heavy metals, etc. Furthermore, these polysaccharides play an active role in the colonization and formation of complex and dynamic biofilms that comprise a well-organized community with numerous specialized configurations [15]. Increased cell density favours the communication of chemical signals with cells (quorum sensing) that respond to social interactions in biofilms, probably adding another level of complexity [16].

Bioprocess engineering has been trying to understand the complexity of biofilm formation in order to be able to intervene in a way that enhances the production of microbial exopolysaccharides.

3 Optimisation of Production

Scientific research in the field of bioprocess engineering has explored the ability to develop technologies that can help improve microbial metabolism and growth in order to obtain an increase in biomass and/or desired metabolites or bioactives. Studies have provided advances in the field of cell biostimulation, through substrates and imposed conditions that alter the production strategy [17] or through physical processes [18], electrical, magnetic, ultrasound [19, 20] and light emission [10]. EPS production depends on factors such as medium composition (carbon/nitrogen source), culture growth conditions (pH, temperature, oxygen concentration, time, agitation speed) and type of strain used. By promoting an alkaline imbalance, researchers were able to induce the production of exopolysaccharides by *Enterobacter* and *Xanthomonas* bacteria. Such a response may be related to a homeostasis mechanism, as EPS possess acidic groups, allowing a reduction of medium pH [17]. *X. campestris* under alkaline stress conditions (pH 9.5) showed an increase in EPS production of 16.7% [17]. The carbon/nitrogen ratio or the limitation of these nutrient sources are preponderant factors that directly influence the differential production of EPS by the same strain

[21]. For example, nutritional imbalances, such as a high carbon/nitrogen source ratio, are highlighted as inducing the production of exopolysaccharides [22, 23].

EPS biosynthesis is directly related to the different phases of microbial growth, with time becoming a determining factor. According to a study by Sugumaran et al. (2023) on the production of pullulan in batches, using coconut kernel as a substrate, there was a decrease after 144 h, suggesting that *Aureobasidium pullulans* may consume the EPS produced itself [24]. On the other hand, studies employing sucrose and crude glycerin as carbon sources have reported higher EPS production after reaching the stationary phase of bacterial growth. This suggests a trigger of the metabolic machinery towards EPS production itself, instead of biomass production [25, 26].

Physiological stress through changes in pH may be an essential condition for an induced increase in EPS production [27]. The use of detergents in the culture medium can increase oxygen transfer and improve the production of xanthan gum and its rheological characteristics, such as viscosity [28].

New strategies have emerged to stimulate metabolic pathways and obtain bioactives, such as the application of electric current in fermentative processes, called electro-fermentation. This technology is characterized by the supply or removal of electrons from cells, through the application of low energies, interfering with the permeability of the cell membrane, which can increase the absorption of nutrients and excretion of bioactive. Enzymatic activity can also be influenced, especially in the presence of metallic cofactors. On the other hand, the induction of microbial stress can develop biological responses that culminate in greater production of the final product [29, 30]. Through electrical stimulation in pilot-scale reactors for *Saccharomyces cerevisiae* growth, improved ethanol production was achieved at a frequency of 50 Hz and an intensity of 990 μA [31]. Other work investigating *Hanseniaspora* sp. demonstrated a shortening of the stationary phase by 10 h and a higher microbial growth rate when the culture was subjected to pulsed electric fields [32]. Using a similar approach but with electromagnetic fields, increased substrate consumption, ethanol production (33%), and proton efflux were reported, culminating in reduced batch duration [33].

The ultrasound technique, through mechanical, thermal and chemical stimuli, has been used as a method of enhancing microbial growth and metabolic activity, and can be a valuable method in microbial bioprocesses. Ultrasound can increase EPS production by increasing the permeability of cell membranes, improving mass transfer and promoting interactions between enzymes and their substrates [34]. Employing this approach has allowed improved yield of GABA by *Lactobacillus brevis* through enhanced metabolic activity [35]. When *S. cerevisiae* culture was exposed to 28 kHz ultrasound, improved biomass and a significant 30.8% increase in ethanol production were achieved. The reported physiological changes related to better metabolic activity include higher intracellular Ca^{2+} concentration, changes in cell surface, and improved enzymatic activities related to sugar metabolism [36]. Similar stimulation of *S. cerevisiae* has been reported using ultrasound at 132 W, 37 kHz, and a 50% duty cycle for ethanol production [37].

4 Photostimulation

Low-power laser or LED irradiation, emitting at certain parameters (wavelength, power, energy density and irradiated surface), indicate biostimulatory results [11, 12, 38–40].

Laser radiation is characterised by a greater concentration of energy (fluence) in a small spectral range. Whereas the energy density in LED emission is distributed over a wider electromagnetic band, making it possible to interact with a larger group of specific photoreceptors. Despite the different energy distribution of laser emission compared to LED radiation, there is a wide window of light absorption in biological cells, allowing for a biomodulatory action [41].

The physiological state and specificity of each cell presents a certain survival threshold, which can be modulated according to the energy offered. The molecular effect of low intensity energy absorption in biochemical structures of the cell, results in the search for maintenance or normalization of its functions, biomodulation. For the biomodulation process to occur, it is important that the target cells are not in homeostatic conditions. When irradiating a living organism, we introduce energy into a biological system, which uses and transforms this energy for its own consumption. The effectiveness of irradiation is strongly related to the appropriate fluence, applied regularly and gradually, so that low or high fluences may produce no effects or generate damage, respectively. The fluence often defines the frontier of obtaining a stimulatory or inhibitory process [42].

One of the functions of cell regulation happens through photoreceptors such as flavoproteins, porphyrins, cytochromes, tyrosines, asparagines, among others, capable of absorbing a photon of a specific wavelength, causing a transformation in the functional and metabolic activity of the cell. The specific photoreceptors interfere in cellular metabolism without the need for light energy, but if a specific wavelength radiation falls upon them, they are capable of absorbing the photons, assuming an electronically excited state, causing alterations in cellular metabolism. These processes are defined as biostimulation [43, 44].

Light basically acts as a “trigger”, triggering cellular responses, primary reactions, which propagate through consecutive secondary reactions in the cytoplasm and nucleus of cells, setting off a cascade process [43].

On the other hand, according to Karu (1987) [45], the cellular response to photostimulation is not associated with specific properties of laser light, such as coherence. This property is lost in the interaction of the light with the biological material, and is not an essential condition for the photostimulation process. Thus, it is considered that there are similar biostimulatory effects generated at a cellular level, by the laser and LED, when compared under the same wavelengths, intensity and time of irradiation [46]. Studies developed by NASA (National Aero Space Agency) in the United States have shown strong evidence of the efficiency of LEDs in processes of cellular photostimulation similar to that of low-power lasers [47].

Photobiostimulation has shown a series of positive results at the cellular level, such as an increase in the proliferation rate [40], an increase in the rate of RNA and DNA synthesis [45], protein activation and ATP synthesis [48].

5 Photostimulation in the Production of Microbial Exopolysaccharides

When biosynthesis of biopolymers involves aerobic or facultative microorganisms, one of the targets of irradiation in the red spectrum are terminal enzymes of the respiratory chain such as cytochrome c oxidase, increasing the rate of electron flow and consequent increase in the amount of cellular ATP available (Fig. 1). Photoexcitation induces allosteric changes in cytochrome c oxidase and/or in flavin components of the chain, such as NADH-dehydrogenase, and may cause other redox changes and modulate biochemical reactions through a chain of transduction and amplification of the light signal towards biological macroeffects, such as increased cell proliferation via increased DNA synthesis [42, 43].

Membrane sodium channel enzymes increase the ratio between sodium and hydrogen, with an increase in sodium, potassium and intracellular ATPase, causing a decrease in the concentration of hydrogen ions. This induces changes in the concentration of intracellular calcium and cAMP, which are signals for the synthesis of DNA and RNA, by activating the synthesis of nucleic acids and proteins (Fig. 1) [49].

Radiation emission can also activate intracellular metalloproteins that actively participate in the biosynthetic pathway of exopolysaccharide production (Fig. 1).

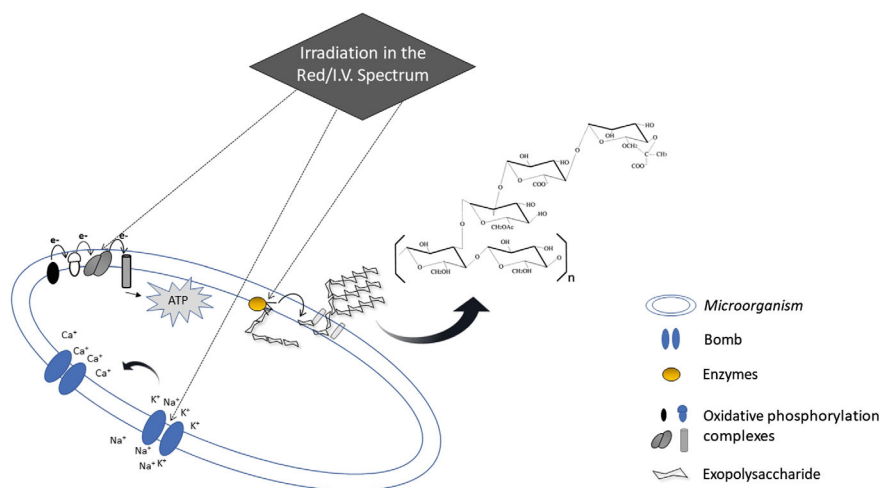


Fig. 1 Main photostimulation pathways in the production of exopolysaccharides

The absorption of photons by the prosthetic groups can increase the anabolic capacity or help in the maintenance of the three-dimensional conformation of enzymes making them more resistant to the denaturation promoted by the metabolic action [50, 51]. Thus, irradiations performed during the process of expolysaccharide production can promote the excitation of prosthetic groups, mediators of oxireduction and activation of metalloproteins that participate in the catalytic action of the polymerization, excretion and formation of pentasaccharide polysaccharides (Fig. 1). Photon absorption by prosthetic groups and/or aromatic peptide residues can increase anabolic capacity or help maintain the three-dimensional conformation of enzymes, making them more effective [51]. On the other hand, effects produced by light on the activation of transcriptional and translational processes have also been described [52, 53].

The use of light emission in a pre-production phase (microbial activation) is a factor to be taken into account, because it promotes the activation of metabolic pathways in cascade and consequently the increase of microbial proliferation for the optimization of the biosynthesis process of biopolymers [10]. In conditions of high cell density, chemical signalling mechanisms through autoinducers that alter gene expression and regulate metabolic activities, activating a cooperative behaviour among microorganisms, i.e. irradiation may also infer in the quorum-sensing process influencing the production process [54, 55].

6 Conclusion

Photostimulation has been shown to be a useful tool in biotechnological methods, due to its ability to optimise biological processes, as in cell proliferation and protein activation, improving the phenotype without altering the genotype. A promising and ecologically sustainable method, the emission of Laser and LED irradiation, can improve the adaptability of different microorganisms, participate in the production of cellular ATP, as well as increase the production of various biocompounds.

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Sustainable Horizons: Navigating Challenges in Butanol Production from Lignocellulosic By-Products



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Abstract This chapter delves into unlocking the latent potential of lignocellulosic by-products, exploring the innovative production of butanol. These by-products are abundantly generated across various industries, making them a promising low-cost feedstock for the production of this sustainable biofuel. Despite the vast potential, the current process faces challenges, notably the relative cost of biomass pretreatment through enzymes and physico-chemical methods, inhibition effects of the solvent, and the intrinsic nature of wild strains. Potential approaches to overcome these bottlenecks, involving technological advancements and process optimization for widespread adoption, will also be explored. These approaches include synthetic biology, nanotechnology, enzymology, and boosting techniques for butanol microbial enhancement, such as ultrasound and electrofermentation processes. Additionally, the chapter will address the future perspectives of this promising bioenergy avenue, emphasizing its contribution to environmental concerns by repurposing biomass waste and fostering a cleaner, greener future.

Keywords ABE fermentation · Biobutanol · Sustainability · *Clostridium* · Agro-industrial coproducts

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1 Introduction Overview of Butanol

The production of butanol began in the early twentieth century through the fermentation of acetone-butanol-ethanol (ABE) by bacteria of the *Clostridium* genus. This process became an industrial landmark until the 1960s, especially for acetone and butanol production. However, the rise of the petrochemical industry led to a sharp decline in butanol production via ABE fermentation [1]. Currently, the search for sustainable alternatives to fossil fuels has rekindled interest in biobutanol derived from renewable sources [2].

Butanol stands out as a promising biofuel for the automotive industry due to its higher viscosity, low vaporization enthalpy, and high carbon content, which translate into lower volatility, better efficiency, and greater fuel economy compared to other alcohols [3]. Its excellent combustibility and favorable thermochemical properties make it a viable substitute for gasoline. Additionally, butanol is a valuable chemical platform used in the chemical, pharmaceutical, and food industries, playing a crucial role as a solvent in the preparation of active pharmaceutical ingredients [4].

The global butanol market was valued at USD 8.5 billion in 2023 and is projected to reach USD 4.41 billion by 2030, with an annual growth rate of 5.0% between 2023 and 2030, according to Vantage Market Research [5]. The Asia-Pacific region led the market in 2023 due to high demand in the automotive and pharmaceutical sectors, driven by economic growth and urbanization. In North America, significant growth is expected due to butanol adoption in the automotive industry and advancements in manufacturing and construction, increasing its use in paints and adhesives [6].

Grand View Research highlights the dynamism of the butanol market, driven by major companies such as The Dow Chemical Company, BASF SE, OXEA GmbH, Sasol Ltd, Grupa Azoty, Mitsubishi Chemical Corporation, and KH Neochem Co Ltd. These companies lead the market with innovations and advanced technologies, significantly contributing to meeting the growing worldwide demand for butanol [7].

Currently, industrial butanol production mainly utilizes the ABE fermentation method, based on the fermentation of starchy substrates such as corn and sugarcane [2]. However, this production method faces significant challenges affecting its economic viability. The high cost of substrates, the complexity of processing steps, and product inhibition due to solvent toxicity are major obstacles. Additionally, ABE fermentation generates unwanted by-products and requires complex nutrients, increasing costs and hindering microbial growth [4].

Challenges include low butanol concentration, yield, and productivity, along with significant purification costs to obtain a commercially viable product. These factors limit the economic competitiveness of butanol compared to other biofuels and petrochemical products [4]. The three main factors determining competitiveness are substrate cost, butanol yield and productivity, and separation and purification costs. Without significant technological advancements or cost reductions, biobutanol production via ABE fermentation will struggle to become economically viable in the biofuels and industrial chemicals market [3].

Attention has been given to bio-based solutions for treating and valorizing urban and industrial wastes and by-products, enabling resource recovery and sustainability. Wild or engineered microorganisms can treat a wide range of solid and liquid wastes or by-products to yield value-added biomass [8–10] or metabolites such as polysaccharides [11, 12], solvents [13, 14], and enzymes [15–17].

Lignocellulosic biomass is a promising feedstock for biofuels due to its abundance, low cost, and minimal competition with food [18]. Cellulose and hemicellulose can be hydrolyzed into fermentable sugars for butanol production [2]. Using this biomass maximizes waste utilization, reduces competition with food, and decreases greenhouse gas emissions [19]. Advanced pretreatment and fermentation technologies can further improve conversion efficiency, making lignocellulosic biomass crucial for the economic and ecological viability of butanol production [20].

Carbon capture using bio-based technologies, such as microalgae [21], cyanobacteria [22], and anoxygenic phototrophic bacteria [23], aims to mitigate climate change and promote sustainability. Recent studies suggest that butanol can participate in carbon capture through solvent-based absorption, effectively absorbing CO₂ from flue gases and contributing to carbon capture efforts [24–26]. Additionally, specific butanol producers can be used for carbon capture [27].

Butanol production has evolved significantly since its origin in ABE fermentation, facing challenges and adapting to market demands and technological innovations. This chapter has addressed the historical trajectory of butanol, its beneficial properties as a biofuel, and its diverse industrial applications. Despite economic and technical obstacles, the use of lignocellulosic biomass and technological advancements present a promising path for sustainable biobutanol viability. Thus, butanol remains a relevant and potentially revolutionary option in the global biofuels and industrial chemicals landscape.

2 Lignocellulosic Materials for Butanol Production, Pretreatments and Employed Microorganisms

Butanol has drawn significant interest from the scientific community due to its potential as a fuel, either as an additive or a gasoline substitute [28]. It offers several advantages over ethanol: higher calorific value, low solubility in water, less corrosiveness, and similar properties to gasoline, enabling transportation through existing ethanol pipelines. Additionally, butanol has various applications, including use as a solvent in industries, cosmetics, cleaning products, pharmaceuticals, and as a chemical intermediate [28]. These characteristics make butanol an attractive and promising product, especially since it can be produced from renewable sources.

Lignocellulosic materials derived from agro-industrial waste are a focus of research due to their abundance, low cost, and potential for producing biofuels like ethanol and butanol [29–31]. The increasing cost of fossil fuels further enhances

the appeal of butanol as an alternative fuel [32, 33]. Various studies have documented biobutanol production from agricultural residues such as sugarcane bagasse, eucalyptus, rice straw, corn stover, and cassava bagasse [34–37].

Biobutanol can be produced from non-food sources like lignocellulosic biomass, which consists of cellulose, hemicellulose, and lignin. Cellulose, made up of D-glucose units, is crystalline and resistant to enzymatic hydrolysis. Hemicelluloses are amorphous polymers that strengthen the cell wall, and lignin, a hydrophobic polymer, connects cellulose and hemicellulose. Pretreatment is necessary to make fermentable sugars available for fermentation due to the complex structure of lignocellulose [38].

The process of butanol production from lignocellulosic materials involves ABE (acetone, butanol, ethanol) fermentation, typically performed by *Clostridium* bacteria under anaerobic conditions [39]. These microorganisms can metabolize various carbon sources, potentially enhancing fermentative efficiency. However, lignin remains a barrier, reducing enzyme availability for hydrolyzing polysaccharides [39].

To overcome this, pretreatments are applied before enzymatic hydrolysis to reduce recalcitrance. Techniques aim to remove lignin, disintegrate crystalline cellulose, and convert cellulose and hemicellulose into fermentable sugars [40]. Effective pretreatments should promote polysaccharide conversion to monosaccharides, prevent cellulose loss, inhibit microbial inhibitors, and offer a favorable cost–benefit ratio for industrial implementation [41].

Pretreatments are classified into physical, chemical, physicochemical, and biological categories. The pretreatment step is crucial for neutralizing the negative impact of hemicellulose and lignin on cellulose enzymatic hydrolysis [42]. Choosing a pretreatment method involves considering the lignocellulosic material characteristics and the advantages and disadvantages of each approach, including economic and environmental factors.

ABE fermentation by *Clostridium* bacteria faces challenges like low solvent tolerance and slow microbial growth, which restrict performance [43]. *Clostridium acetobutylicum* is well-studied for its metabolism related to butanol production, with other solventogenic species like *C. beijerinckii* and *C. saccharobutylicum* also recognized for their solvent production capabilities, especially under specific fermentation conditions [44, 45].

3 Bottlenecks in Butanol Production

Biotechnological production of butanol faces several challenges, including substrate choice, microorganism selection, product toxicity, and medium composition [46]. The use of lignocellulosic materials offers advantages due to their low cost and widespread availability, making them preferred for large-scale production. However, inconsistencies in biomass availability and composition pose significant challenges for the biobutanol industry, affecting the overall process cost [47].

Lignocellulosic materials, composed primarily of lignin, cellulose, and hemicelluloses, are derived from agricultural residues rich in cellulose and hemicelluloses

but low in lignin, making them highly suitable for butanol production [48]. However, the pretreatment of these materials to reduce their recalcitrance involves high temperatures, pressures, and extensive chemical use, leading to high capital costs and energy consumption. Methods such as steam explosion, supercritical CO₂ explosion, and wet air oxidation are energy-intensive and can degrade hexose and pentose sugars, rendering them less effective for biomass with high lignin content [49].

Chemical pretreatment methods, including acid/alkaline hydrolysis, ozonolysis, and organosolv pretreatment, require substantial amounts of chemicals like concentrated acids and oxidizing agents, necessitating additional neutralization steps to prevent interference in hydrolysis and fermentation. These chemicals are toxic, corrosive, and hazardous, demanding corrosion-resistant reactors and complicating the fermentation process due to the presence of toxic substances [46, 49, 50].

4 Advancements and Proposals to Overcome Butanol Production Limitations

Figure 1 shows some approaches under investigation to overcome biobutanol production bottlenecks. Advancements in metabolic engineering and synthetic biology tools have significantly improved strain development for butanol production. These improvements focus on enhancing tolerance to toxic compounds, optimizing substrate utilization, and increasing butanol selectivity and productivity from lignocellulosic biomass residues [51]. The constant availability of biomass throughout the year is essential for the lignocellulosic biobutanol industry. This is crucial not only for meeting daily production demands but also for addressing storage issues. Implementing new agricultural methods, crop rotation, and replacing or combining different types of lignocellulosic residues can ensure an uninterrupted biomass supply year-round, supporting continuous biobutanol production without impacting current agricultural practices [52].

Biomass pretreatment is another critical challenge as it directly affects the efficiency of converting these residues into fermentable sugars. Progress in developing more efficient enzymes and optimizing pretreatment processes has shown potential to enhance this efficiency and reduce operational costs [53]. Utilizing low-cost raw materials for microbial fermentation and lignocellulolytic enzyme secretion continues to be a well-explored theme in recent literature [54–57]. Concurrently, the resistance of microorganisms to end products, such as butanol and other toxic byproducts, remains a significant bottleneck. Research focuses on developing genetically modified microbial strains with higher tolerance to these compounds, which could increase productivity and make production more economically viable [58].

Investigations into new setups and operations of packed bed reactors, which are common in anaerobic bioprocesses, show potential for bacterial butanol production [59–61]. Reducing operational costs is vital to making butanol competitive in the biofuel market. Strategies to optimize the supply chain and enhance the logistics

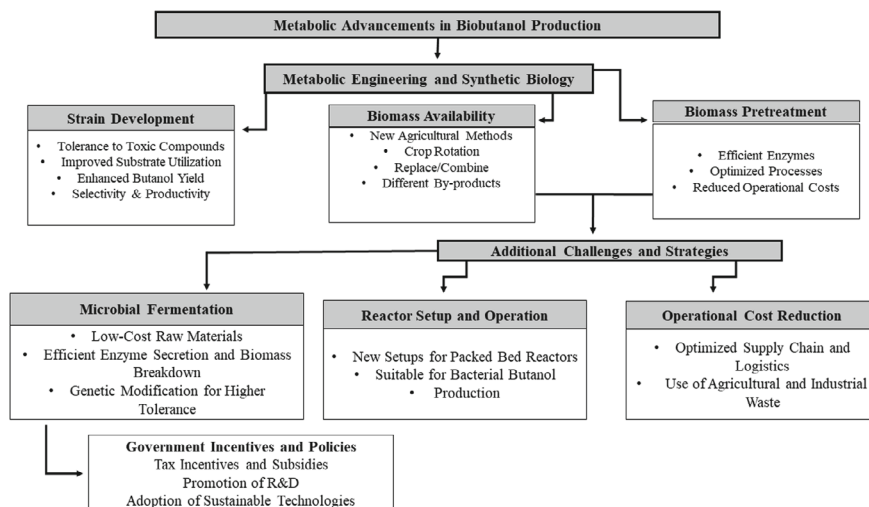


Fig. 1 Advancements and proposals to overcome butanol production limitations

of waste transportation and storage are essential. Utilizing more efficient production processes and harnessing agricultural and industrial waste can significantly contribute to cost reduction [52]. Additionally, government incentives and policies to promote research and development are fundamental to overcoming these obstacles. Creating tax incentives and subsidies can accelerate the adoption of cleaner and more sustainable technologies, driving the development of the biofuel industry.

5 Stimulation Techniques for Butanol Production

Several studies have examined the potential of electromagnetic and mechanical radiation in bioprocesses, such as phototransformation of compounds and assisting in the breakdown of lignocellulosic materials. However, few have explored their use in modulating the metabolic activity of microorganisms to improve bioproduct yields [62–65]. Photostimulation techniques have been used to alter the physiology of bacteria, promoting the secretion of industrial metabolites and lignocellulosic material degradation [66–70]. Hydrogenic imbalance stress has been shown to induce the production of compounds triggering homeostasis [71].

Pulsed electric fields and currents have been suggested to alter cell membrane potential and pores, change protein activity, boost mass transfer processes, and assist in product secretion [72–74]. Fine-tuning ultrasound modulation is essential, as it can either increase or decrease butanol production depending on the targeted approach [75–77]. Changing extracellular redox potential through cathodes during fermentation has been shown to enhance butanol production without affecting biomass

yield [77, 78]. Electro-fermentation processes promote electron transfer, increasing intracellular NADH levels crucial for butanol synthesis [77, 78].

Cathodic electro-fermentation with *Clostridium beijerinckii* NCIMB 8052 resulted in excellent butanol production [78]. Gene expression analysis highlighted significant up or downregulation of ABE pathway genes under electro-fermentation, influencing butanol production [79–81]. Supplementing fermentation medium with elements like gadolinium or rare earth elements, activated carbon adsorption, and cell immobilization has shown to enhance biomass production and butanol synthesis [81, 82].

Thermoactivated *Clostridium saccharoperbutylacetonicum* with ZnO nanoparticles and a Hybrid System Assembly of CdS nanoparticles attached to *C. beijerinckii* cells increased butanol production [81, 83]. CuS/Cu₂S nanoparticles improved the intracellular NADH/NAD⁺ ratio and enzyme activity in *Clostridium beijerinckii* [84]. Nanoparticles have also been employed as catalysts to process lignocellulosic biomass before fermentation [85]. Table 1 shows stimulation techniques employed in butanol production.

Table 1 Stimulation techniques employed in butanol production

Microrganism	Induction technique	Production	References
<i>C. acetobutylicum</i>	Ultrasound 35 kHz and 10% duty cycle (test)	0.233 g/g (Ultrasound) 0.149 g/g (Control)	[75]
<i>C. saccharoperbutylacetonicum</i> N1-4	Electro-fermentation: cathode potential at – 600 mV versus Ag/AgCl	5.8 g/L	[78]
<i>Clostridium beijerinckii</i>	Electro-fermentation: cathode potential at – 0.60 V (vs Ag/AgCl)	10.85 g/L	[79]
<i>C. beijerinckii</i> NCIMB 8052	electro-fermentation: (–0.60 V cathode potential)	7.17 ± 0.55 g/L	[80]
<i>C. acetobutylicum</i> ATCC 824	Supplement of 1 × 10 ^{–5} M gadolinium (Gd(III))	15.26 ± 0.43 g/L	[81]
<i>Clostridium beijerinckii</i> NCIMB 8052	Butyric acid and 0.20 g/L Cu ²⁺	Butanol/acetone yield: 3.90 ± 0.21	[83]
<i>Clostridium saccharoperbutylacetonicum</i>	ZnO nanoparticles	11.58 g/L	[86]
<i>Clostridium beijerinckii</i> NCIMB 8052	CdS nanoparticles	10.42 ± 0.15 g/L	[84]
<i>Clostridium beijerinckii</i>	CuS/Cu ₂ S nanoparticles	14.6 g/L	[85]

6 Recent Works Using Genetic Modification and Syntetic Biology Approachs

Recent advancements in molecular biology and synthetic biology have revolutionized various aspects of bioprocess monitoring, microbial control, and genetic manipulation [87–89]. These tools have been crucial in the pursuit of sustainable fuel sources in the twenty-first century, particularly in butanol research. To achieve this, cutting-edge biotechnology tools have been employed to either insert butanol genes into non-naturally ABE fermenting microorganisms or design new microbial platforms. While different rationales exist for obtaining butanol from ABE fermenting systems, integrating these studies is essential for comprehensive evaluation and comparison.

For instance, researchers have introduced the n-butanol biosynthesis pathway into *Rhodospseudomonas palustris* TIE-1, yielding high n-butanol quantities [90]. Similarly, engineering phototrophic microorganisms like *Synechocystis* sp. PCC 6803 via heterologous biosynthetic pathways has shown promise, achieving significant butanol production through photosynthesis [91].

Furthermore, mutation approaches, such as carbon ion beam irradiation on *Clostridium acetobutylicum*, have enhanced butanol production substantially while maintaining cell membrane integrity [92]. Additionally, genetic manipulations, like integrating the *Clostridioides difficile* gene *tcdR* under lactose control, have shown potential in enhancing butanol productivity [93].

Transcriptional analysis has identified key genes in the Leloir pathway, leading to the development of strains with improved galactose utilization for butanol production, thus opening avenues for sustainable biorefinery routes [94]. Optimization of co-utilization systems has further enhanced glucose and galactose utilization, overcoming carbon catabolite repression.

Moreover, optimizing co-utilization systems of Fe^{3+} and butyrate has increased ABE mixture yield and productivity in *Clostridium* strains [95]. Heterologous introduction of genes has shown significant improvements in acetone, butanol, and ethanol production. Synthetic inducible promoters have been developed for fine-tuning gene expression in *Clostridium tyrobutyricum*, facilitating efficient regulation of key enzymes and acetate assimilation [96].

In addition, understanding the genetic mechanisms behind butanol production, such as gene disruptions of sporulation-related genes in *Clostridium saccharoperbutylacetonicum*, has offered insights into enhancing butanol yield [97].

Mitigating hydrogen formation during ABE fermentation is crucial, and various strategies including optimizing conditions, employing engineered strains, selecting suitable substrates, and implementing specialized reactor designs have been proposed to steer metabolic pathways towards desired products while minimizing hydrogen production [98].

7 Concluding Remarks

Advancements in biomass utilization and technological innovations have paved promising pathways forward for butanol, from its origins in ABE fermentation to its current status as a sustainable biofuel and industrial chemical. Despite facing challenges, ongoing research into lignocellulosic biomass utilization, pretreatment, and fermentation optimization are vital for realizing its promise as a renewable energy source. Recent breakthroughs in biotechnological approaches, such as electromagnetic and mechanical radiation, genetic modification, and synthetic biology, offer exciting prospects for enhancing butanol production efficiency. Techniques like ultrasound exposure and cathodic electro-fermentation show promise in improving yields and sustainability. Butanol's versatility and superior properties make it a compelling alternative to ethanol in the quest for greener energy solutions. Overall, with continued research and innovation, butanol remains a relevant player in shaping a more sustainable future for both energy and chemical industries, driving its evolution as a key component of the green economy.

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The Role of Composting in Sustainable Agriculture and Waste Management



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Abstract The projected increase in the human population in certain emerging areas poses a serious risk to sustaining food supply, managing waste, and using fertilizers. Composting urban and industrial organic wastes, along with lignocellulosic biomass, represents a sustainable approach to recycling and nutrient cycling, thereby generating biofertilizers with microbial-aided properties. This chapter will examine the latest global trends in composting such materials and explore the obstacles hindering broader adoption of this approach. Major materials for composting and innovative research on enhancing the fermentation process will highlight the potential of this technology to improve sustainability, carbon capture, and integration with the bioenergy, food, and waste treatment industries.

Keywords Waste management · Resource recovery · Nutrient cycling · Bio-based fertilizers

1 Introduction

The increasing urbanization and population growth, along with changes in consumption patterns, have led to a considerable rise in daily waste production [1–3]. Annually, about 2.1 billion tons of solid waste are generated worldwide, of which 33% are improperly managed. Studies indicate that this production could increase to 3.4 billion tons by 2050 [1]. For the same aforementioned reasons, there have been changes in the exploitation of natural resources and an intensification of agricultural

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production to meet population demands. However, according to the Food and Agriculture Organization, about 14% of the food produced (valued at around 400 billion dollars per year) is lost during harvest and before reaching the final consumer. The United Nations Environment Programme estimates that an additional 17% is wasted by retailers and consumers, mainly by households [4, 5].

Currently, waste management and environmental sustainability are significant issues discussed by society. The 2030 Agenda is committed to reducing the negative impacts of urban activities that are dangerous to human health and the environment [6]. One of the pillars of this development is the reduction and recycling of waste, including organic matter and residual nutrients [7].

Population growth results in more intense waste production, including a substantial amount of organic waste from food, gardens, and other biodegradable materials [8]. Conventional, generally low-cost methods like landfills and open burning are often employed to manage this waste. However, these approaches pose environmental threats due to issues such as leachate production, greenhouse gas emissions, loss of valuable resources like land, water, and energy, and the spread of diseases [7, 9–11].

To address the environmental issues caused by the rampant production of waste, the basic principles of the circular economy must be pursued, with composting being a well-established and elegant approach. Composting is a natural and sustainable process for treating organic waste, converting biodegradable materials such as food scraps, grass clippings, and leaves into a valuable resource known as biofertilizer [8, 12]. The composting process involves the controlled biological decomposition of organic waste by microorganisms such as bacteria, fungi, and worms, under appropriate conditions of moisture, temperature, and aeration [2, 12]. The resulting biofertilizer is a natural and balanced source of nutrients that helps improve soil structure, increase its water retention capacity, and promote beneficial microbial activity. Additionally, it reduces the amount of improperly disposed waste [6, 9].

The market capitalization of the composting industry has seen significant growth in recent years, driven by the search for sustainable solutions for waste management. The global biofertilizer market size was valued at \$1.5 billion in 2022 and is expected to grow at a compound annual growth rate (CAGR) of 12.8% from 2023 to 2030 (<https://www.grandviewresearch.com/industry-analysis/biofertilizers-industry>).

Forecasts for the future of composting are promising, with projections indicating a continuous increase in demand for composting-related services and products. Additionally, government policies, stricter environmental regulations, and, above all, concerns about food security are expected to further drive the growth of the composting market, encouraging the adoption of sustainable waste management practices on a global scale to improve the quality of soil available for agriculture.

In this context, this chapter will discuss the main issues related to composting and its broader adoption, the treatment and generation of biofertilizer, and future perspectives for this process, bringing an overview of the effective implementation of circular economy policies and the development of innovative technologies based on practices that benefit both farmers and the environment, resulting in healthier soils, more productive crops, and agricultural systems more resilient to climate change [10].

With the emergence of new technologies and innovative approaches, it is expected that composting will become even more efficient and accessible, opening opportunities for expansion in various sectors such as agriculture, landscaping, horticulture, and the food industry.

2 Primary Applications by Type of Waste for Composting Worldwide

Composting is the most versatile and promising method for managing biodegradable solid waste, and it is profitable for various types of vegetative, animal, and synthetic waste, from residential bins to large corporations [13].

Food waste represents a significant portion of urban solid waste in many countries. Every year, an incredible amount of food is lost to waste. One-third of the food produced for human consumption worldwide is lost or wasted [14]. Composting food waste is an effective solution to reduce waste and produce high-quality organic fertilizer. They have enormous potential to improve biofertilizer production because they are present in biodegradable forms and can effectively accelerate microbial metabolic activity. Restaurants, supermarkets, school cafeterias, and households are some of the places where food waste composting can be successfully implemented [9, 14].

Lignocellulosic material waste mainly consists of residues from agro-industries and the production and processing of wood, such as sawmills, carpentries, and paper and pulp industries [15]. Grass and wood clippings, leaves, straw, and vegetable pulp are excellent raw materials for composting. The main characteristic of these residues is their composition. They consist mainly of 35–50% cellulose, 25–30% lignin, and 25–30% hemicellulose. These materials provide a rich source of carbon and nitrogen and contribute to the structure and aeration of the compost. Composting of lignocellulosic materials is widely used in agriculture, landscaping, and gardening to improve soil quality and promote healthy plant growth [2, 16].

Composting of industrial waste encompasses a wide range of materials generated before, during, or after production processes, or as by-products of various industries, such as poultry processing, slaughterhouses, sugar industry, fish processing, sludge from wastewater treatment plants, textile industry, among others. Slaughterhouse residues, such as hair, skin, feathers, horns, and hooves, contain large amounts of organic matter, animal fat, and protein, for example, and are promising in the composting process [2].

Waste is produced during handling, cutting, heat treatment, transportation, packaging, or due to discoloration, microbial invasion, and biochemical reactions. Companies from different sectors, such as agribusiness, food, textile, and pharmaceutical, are adopting composting practices to reduce waste disposal costs, comply with environmental regulations, and promote sustainability in their operations [2].

Composting can be seamlessly integrated with the valorization of biogas and hydrogen leftovers that are unsuitable as feedstock or substrate in the feed industry. Lignocellulosic materials, such as plant biomass composed of cellulose, hemicellulose, and lignin, serve as abundant and renewable resources for biogas production [17] and biohydrogen [18]. The resulting residual waste, rich in lignin, poses digestion challenges and isn't fit for animal feed; however, it can undergo composting and become a valuable soil conditioner [17].

Biomass derived from microalgae wastewater treatment plants can be utilized as feedstock for biodiesel, bioethanol, and biogas production. However, the residual waste, abundant in algal biomass, is unsuitable for animal feed due to its high mineral content and potential toxicity [19]. Composting presents a viable option for converting this waste into fertilizer [19].

Municipal solid waste, encompassing household waste, restaurant waste, and other organic waste, can serve as a substrate for biogas production [20]. The residual waste from this process can undergo composting and be repurposed as fertilizer [20].

Organic waste from farms and restaurants can be directed to covered pools, where its decomposition yields biogas. The remaining biodigestate is utilized as fertilizer [21]. Further composting can enhance its biological activity and beneficial effects on crops. This integrated approach not only boosts resource efficiency and reduces waste but also fosters sustainability within the bioenergy sector, concurrently generating economic value through compost sales.

Finally, carbon capture, which is a process that involves capturing carbon dioxide (CO₂) emissions from large point sources such as power generation or industrial facilities [22], is assisted by composting through reducing methane emissions, sequestering carbon in soil, boosting plant growth, decreasing fertilizer use, supporting soil microbes, recycling carbon, and enhancing soil storage [23–25]. It prevents methane production by decomposing organic waste aerobically, adds stable organic matter to soil for long-term carbon storage, and improves soil health to promote plant growth that captures carbon dioxide. By lowering reliance on chemical fertilizers, composting reduces greenhouse gas emissions associated with their production and use. It also enriches soil with microbes that convert organic matter into stable carbon compounds, creating a sustainable carbon loop by converting waste into compost. Regular compost application further improves soil's capacity to store carbon.

3 The Role of Composting in Sustainable Agriculture

Composting is an exceptionally effective method for treating agricultural waste, offering substantial benefits for soil improvement compared to other techniques. This process not only enhances soil structure but also accelerates geochemical processes that increase the availability of essential nutrients for crops, thereby boosting soil fertility [26].

Composting transforms food scraps and plant materials into nutrient-rich compost, providing farmers with a renewable source of organic fertilizer. This compost

improves soil structure, increases water and nutrient retention capacity, and promotes microbial biodiversity, enhancing the overall health of the agricultural ecosystem [7].

In urban farming areas, composting can also serve as a vital source of carbon and nitrogen, especially where leaf and garden waste are routinely removed, creating nutrient hotspots [27]. Additionally, composting significantly reduces the amount of organic waste sent to landfills, mitigating the environmental impacts of anaerobic decomposition [28]. By diverting this waste from improper disposal, composting helps reduce methane production, a potent greenhouse gas, and decreases the need for unsustainable, chemically produced fertilizers [29, 30].

The reduction in the carbon–nitrogen (C/N) ratio within the compost matrix is a key factor in controlling greenhouse gas emissions. Incorporating carbon-rich additives like wood chips, mushroom waste, rice bran, biochar, and minerals into the composting system is essential for regulating nitrogen loss and minimizing greenhouse gas emissions [31].

In sustainable agriculture, composting also enhances food security and resilience in agricultural systems. The compost produced provides essential nutrients for plants, reducing reliance on synthetic fertilizers that can harm the environment and human health. This organic fertilizer improves soil quality, fostering the growth of healthy food and improving food quality [26]. Furthermore, by enhancing soil quality and its water retention capacity, composting helps farming operations become more resilient to climate change and extreme weather events.

Multiple studies have delved into various facets of composting and its potential to tackle environmental issues. The utilization of composted sewage sludge has emerged as a promising solution for restoring degraded sites [32]. Microbiology and molecular biology tools can be utilized to address the microbial profile throughout the bioprocesses like the composting or to assess its viability in the presence of toxic compounds [33–36]. Concurrently, synthetic biology and computational methodologies have been explored to break down cellulose [33, 34, 37], finding applications in composting and underscoring the significance of Artificial Intelligence and Machine Learning in data analysis [37]. Moreover, community-based models of urban composting have stressed citizen engagement and sustainability as feasible strategies for managing organic waste in urban settings.

Additional research has focused on assessing the effectiveness of microbial inoculants [38], biochar [39], and food waste compounds [40] in sustainable drainage systems. Furthermore, studies have evaluated the impacts of composting chicken litter [41] and biochar [42] on reducing gas emissions and remediating contaminated soils. The comparative effectiveness of wheat straw biochar and chicken waste compost in remediating cadmium-contaminated soil has also been examined, contributing to enhanced understanding and implementation of sustainable waste management practices [43].

Furthermore, the efficacy of microbial agents derived from tomato straw composting in promoting compost maturation and enhancing compost quality was investigated, offering valuable insights for the composting process [40]. These endeavors collectively advance our comprehension and application of sustainable waste management practices.

4 Major Bottlenecks and Limitations of Composting for Wider Adoption

Stimulation techniques, such as electromagnetic and electric current stimulation, have proven to be effective tools in enhancing the growth and performance of microalgae. Microalgae are a valuable source of biofuel, food, and pharmaceutical products, and as such, finding ways to increase their yield and productivity is critical. Electromagnetic and electric current stimulation techniques can help by increasing the efficiency of photosynthesis, promoting cell division, and enhancing nutrient uptake. Additionally, these techniques are non-invasive and have minimal environmental impact, making them an attractive option for sustainable microalgae cultivation. Below we will show some examples of this approach.

Composting encounters numerous challenges that undermine its efficiency and positive impact. One major hurdle is the escalating volume of food waste, which decomposes rapidly, thereby escalating the workload for waste management companies in terms of collection and sorting, thus raising concerns about domestic hygiene [44]. Additionally, composting facilities often struggle to find end users for the produced compost, leading to financial difficulties in covering operational costs. Certain composting methods, such as the Windrow method, operate outdoors and are susceptible to climatic variations, making it difficult to control the quality of the final product and generating unpleasant odors that may provoke objections from local communities [45].

The quality of the final compost significantly depends on its maturity, an aspect that is challenging to standardize due to the complexity of the materials and composting environments. Assessing compost maturity is crucial before its use as organic fertilizer; however, the techniques for determining it are diverse and challenging to standardize [26]. Immature compost can release phytotoxic compounds that harm plants, underscoring the importance of its maturity and stability [46].

Optimizing composting parameters, such as transformation time, thermal phases, C/N ratio, and moisture, is essential for improving the process. Transformation time plays a crucial role, as periods that are either too short or too long can compromise decomposition efficiency and the quality of the resulting compost [47].

Despite the clear benefits, composting still faces significant challenges that limit its widespread adoption. The lack of adequate infrastructure for the collection, transportation, and processing of organic waste is a major obstacle, especially in urban areas. This often results in the disposal of these materials in landfills, causing environmental problems and hindering the sustainable development of cities [28].

Furthermore, raising awareness and educating the public about the benefits of composting are essential for its acceptance and adoption. Regulatory issues and concerns about compost quality also pose challenges, as inconsistent regulations and variable standards make its commercialization and large-scale use in sustainable agriculture difficult [48].

Composting faces a series of challenges that can hinder the process and impact its effectiveness. According to Ayilara et al. [45], these challenges include pathogen

detection, low nutrient levels, extended composting duration, prolonged mineralization, and odor production. Furthermore, the presence of toxic substances and the need to find sustainable solutions are highlighted as technical challenges by Policastro's research [49].

Determining the maturity and stability of the compost is another area presenting challenges in composting research, as pointed out by Sayara [50]. These challenges extend to the effects on agricultural land fertility and suppressive effects. Additionally, the variability of materials used in composting, the lack of common standards, and potential environmental risks are additional difficulties identified by Dandeniya et al. [51].

The viability of techniques for detecting compost maturity is also questioned, as evidenced in a recent study [46]. Detection based on odor production may be inadequate, especially considering that carbon-rich waste may not emit odors during the composting process.

Beyond these challenges, integrating agricultural waste into a circular economy model presents significant obstacles. Limitations related to technology, business model, supply chain, analytical tools, and stakeholder engagement have direct or indirect impacts on this process. Anaerobic digestion is the most commonly used technique on a commercial scale, but its effectiveness may be limited by restrictions on the use of specific raw materials, requiring the incorporation of other waste valorization techniques [52].

5 Trends in Research Involving Composting

Composting is a metabolic process involving the humification, stabilization, and maturation of organic matter through microbial action and fermentation at high temperatures, under appropriate conditions. This method has proven effective in treating organic solid waste generated in human routine and everyday life being widely adopted. The biological transformation during composting is a complex process, where organic matter serves as a substrate, while various factors and the collaboration of multiple functional microorganisms are involved. Microorganisms play a crucial role in the decomposition of organic matter during the composting process. Both the metabolic capacity and the structure of microbial communities are key factors influencing the effectiveness of this process [53].

Various microorganisms can decompose lignocellulose, converting it into carbon dioxide and water under aerobic conditions, and into carbon dioxide, methane, and water under anaerobic conditions. Most lignocellulolytic microorganisms are bacteria or fungi, but anaerobic microorganisms that degrade lignocellulose, such as protozoa, have also been identified in the rumen [54].

Aerobic bacteria, actinomycetes, fungi, and others oxidize small fermentation substrate molecules to support biological growth. Factors like aeration, temperature, humidity, pH, C/N ratio, and particle size also aid in the growth and reproduction of microorganisms [55–58]. They decompose macromolecular organic materials and

generate additional bacteria and fungi to enhance the fermentation process [26, 59]. Attention has been paid to the thematic of optimizing fermentation processes for producing cellulolytic enzymes from lignocellulosic byproducts [60, 61]. Solid-state fermentation of such products can be utilized for efficient fungal enzyme production [62, 63], as well as enzymatic saccharification of lignocellulosic residues using enzymes from fungi cultivated on such wastes [64]. Additionally, there is a focus on the production of high-value metabolites from lignocellulosic biomass [65].

Moreover, there is growing research on waste mixing techniques to optimize composting and produce compost with specific characteristics, such as higher nutrient content or water retention capacity. This includes studying different proportions of organic waste and adding mineral-origin materials to improve compost quality. For example, Carvalho et al. [66] used *Aspergillus oryzae* with a mix of sugarcane bagasse, rice husks, and cocoa shell waste to produce enzymes focused on saccharification.

Another expanding area of research is the development of innovative technologies to facilitate and accelerate the composting process [67]. This includes using biostimulants and selected microorganisms to promote decomposition and create high-quality compost in a shorter time. Both fungi and bacteria have gained increasing attention for their ability to produce a wide variety of enzymes [68, 69]. These lignocellulolytic microorganisms play a crucial role in the biosphere by recycling lignocellulose, the most abundant organic substance on earth [16].

Additionally, there is growing interest in exploring the environmental, economic, and social implications of composting on both local and global scales. This includes studies on its potential to mitigate climate change, promote food security, and create green jobs in rural and urban communities. Coordinating urban waste management and food production can enhance food security while simultaneously addressing climate change, although current waste and food systems are often isolated politically, financially, and logistically [27]. These research trends are driving innovation and expanding the potential of composting as a fundamental tool for sustainable agriculture and waste management.

Recent studies have investigated various strategies to optimize composting and increase its effectiveness in producing valuable humic compounds. The addition of composite microorganisms accelerated the degradation of proteinoids and lignocellulose, promoting the humification of cow manure bagasse co-compost [70]. The incorporation of enzymatic inducers during straw composting resulted in a significant increase in humic substance formation [69]. Zhuo et al. [71] employed a composite reagent to enhance humification, increasing the production of humic acids. Additionally, adding exogenous microbial agents to a compound pig carcass waste, may result a substantial increase in nitrogen contents and plant germination index [72]. Finally, Ahmed et al. [73] evaluated the effectiveness of bacterial inoculants combined with tricalcium phosphate in nitrogen retention and enzymatic activity during kitchen waste composting. These studies highlight the importance of composting strategies to promote efficient degradation of organic waste and the production of humic compounds beneficial for soil.

Continuing, an innovative strategy involves the inoculation of *Bacillus safensis* YM1 in food waste compounds, aiming to alleviate the inhibitory effects of salt and oil. This salt-tolerant and oil-degrading strain demonstrated efficacy in improving the compost maturity index, significantly reducing oil content and Cl^- concentration compared to uninoculated compounds [74]. Another study investigated Fenton ultrasound treatment during corn stalk composting, analyzing its influence on polyphenol and humic acid production. Using a ^{13}C -glucose tracer, researchers examined the potential for microbial assimilation of polysaccharides to generate polyphenols, highlighting the role of Fenton ultrasound in this process [75].

Furthermore, evaluating gas emissions and enzymatic dynamics in sheep manure composts amended with peach kernel biochar revealed valuable insights into the interaction between these materials during composting [76]. Another innovative approach employed bioelectrochemical technology to promote efficient conversion of lignocellulose into humus during composting. This study investigated the effect and mechanism of electrification on carbon conversion across different composting stages, providing unique insights into the application of bioelectrochemical methods in this context [77]. Despite its potential, electric field-assisted composting still faces significant challenges, such as the emission of nitrogenous gases and limitations in electrical conductivity [78]. These challenges underscore the ongoing necessity for research and development to surmount obstacles and enhance the effectiveness and sustainability of composting across diverse applications. This research demonstrates the potential of composting as a sustainable approach for organic waste management and reducing environmental impacts, offering important insights for future composting practices and the development of policies related to biodegradable plastics [79].

6 Boost and Stimulation Techniques for Enhancing Composting Processes

The application of ultrasound stimulation on yeasts has been shown to significantly enhance their growth, mass transfer, sugar metabolism, enzyme activity, and bioproducts production, as demonstrated by various studies [16, 75, 77–79]. Ultrasound promotes the transportation of small molecules and the exchange of materials between intracellular and extracellular environments, which accelerates material and energy metabolism, resulting in a faster propagation rate [75, 80].

Microorganisms can thrive in diverse harsh conditions and compounds through adaptation mechanisms [34, 58, 80]. Several stimulation techniques could be used to boost the composting process. Light radiation is known not only to be effective in the photodegradation process of green wastes but also in altering the metabolism of microorganisms involved in such processes.

The influence of radiation with different spectral wavelengths (using light-transmitting films to filter sunlight) on composting efficiency has been investigated. A no-UV-A treatment produced the best-quality compost product in only

34 days, increasing porosity, humus coefficient, optimal particle size, germination index, available phosphorus, available potassium, and nitrate nitrogen, while diminishing residual organic matter and ammonium nitrogen. Additionally, a change in the enzymatic profile was evidenced, showing increased dehydrogenase, cellulase, and laccase activity [81].

LED and LASER radiation have also been effectively used to boost a thermocellulolytic Bacterial Consortium responsible for the degradation of cellulose in coconut shells [82, 83], with enhanced protein synthesis, enzymatic profile, and activity of microorganisms [84–86]. Similarly, green light (515–530 nm) was used to improve lignocellulose-decay enzymes by *Pleurotus* spp., which has significant applications in the treatment of green wastes [87].

Electric fields affect microorganism structures such as membranes and proteins, altering their metabolism [88–90]. This method has also been used to improve lignocellulose degradation and humic acid formation in the co-composting of food waste and agricultural waste. Pulsed electric fields leveraged the intermediate products of lignocellulose degradation to provide a stable core structure for humic acid formation, promoting the highest humification degree due to enhanced ROS formation. Microbial community analysis indicated that the enriched dominant phylum Firmicutes in EF was responsible for crude fiber degradation. Metagenomics analysis further suggested that the upregulation of carbohydrate-active enzymes with the signal peptide and reactive oxygen species formation triggered by microorganisms contributed to the crude fiber decomposition [91].

Other work used a similar approach of electric field-assisted aerobic composting to increase the composting temperature to 69.3 °C and improve the germination index by 15%. This process took at least 40% less time to reach standard compost maturity. The fluorescent spectroscopy results showed that Pin-EAC could increase humic acid and fulvic acid by 33 and 37%, respectively, also increasing the diversity of thermophilic bacteria during composting [92]. This approach has attracted significant attention, with another recent article revealing that electric fields can act as an activator of extracellular enzymes, enriching carbon-converting functional microorganisms by 1.5 to 2.8-fold, ultimately promoting a 23.0 to 32.9% improvement in lignocellulose conversion to humus [77].

An alternating electric field promoted uniform water distribution and further raised the temperature to achieve hyperthermophilic composting at 90 °C, maintaining temperatures above 80 °C for 4 days. The physicochemical properties and maturity index showed that the alternating electric field improved the biodegradation and humification of organic matter due to the generation of metabolic heat. The alternating electric field enriched thermophilic bacteria [93].

7 Concluding Remarks

Composting stands as a pivotal solution to tackle waste issues and advance sustainability efforts. Its significance lies in its ability to alleviate environmental risks, yield valuable biofertilizer, and foster a circular economy ethos. With an increasing demand and ongoing innovation, composting holds the promise of steering us towards a greener horizon, characterized by healthier soil and enhanced agricultural productivity. It serves as a versatile and lucrative means of handling biodegradable waste, encompassing food scraps, lignocellulosic materials, and industrial residuals. Not only does it offer environmental advantages, but it also yields premium organic fertilizer and seamlessly integrates with biogas and bioenergy initiatives. From individual households to large-scale industries, composting emerges as a beacon of hope for sustainable waste management and the creation of economic value. Nevertheless, while composting presents a beacon of promise for waste management and agricultural enhancement, it encounters obstacles such as the volume of food waste, the identification of compost users, and susceptibility to weather conditions. Standardizing maturity assessments, refining composting parameters, enhancing infrastructure, and fostering awareness are pivotal for its success. Confronting technical challenges and incorporating composting into circular economy frameworks are equally crucial strides. Recent studies underscore its potential to generate valuable humic compounds, curtail environmental footprints, and champion circular practices. Innovations like bio electrochemical technology offer promising avenues for bolstering composting efficiency. The exploration of electromagnetic radiation and electric fields has demonstrated potential in enhancing both the efficiency and quality of composting processes. The adaptability of microorganisms to diverse conditions underscores their pivotal role in optimizing waste decomposition. These discoveries suggest that embracing innovative approaches can usher in more sustainable composting practices, carrying the promise of enriching waste management strategies and fostering resource recovery. Embracing composting paves the way toward a more sustainable future by optimizing resource utilization and addressing environmental challenges head-on.

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Exploring the Landscape of Fermentation: Products, Processes, and Outlook



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Abstract Fermentation processes generate a wide range of valuable products, including beverages, ethanol, enzymes, modified proteins, and pharmaceuticals. Advances in genetic engineering have enabled the production of drugs and high-value molecules through microorganisms, while synthetic biology has created non-natural microbial platforms functioning as biofactories. Despite significant progress in the past decade, exponential growth in knowledge and technology is expected to drive the 4th industrial revolution, vastly expanding biomolecule production possibilities. However, challenges in fermentation processes persist, presenting opportunities for new process development, business ventures, and value creation worldwide. This chapter explores the current status of submerged and solid-state fermentation bioprocesses, their approaches, and challenges, and provides a forecast for the biotechnology industry's future in this field.

Keywords Bioprocesses · Submerged fermentation · Solid-state fermentation · Metabolites · Bioproducts

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1 Historical Overview

The bioprocess sector of industrial biotechnology has applied huge efforts to develop scalable technologies for the improvement of microbial metabolism and growth to achieve higher biomass and desired metabolites. Considerable progress has been obtained in the field of biostimulation of microorganisms, either through the supply of substrates that alter the productive profile [1] or through the application of stimuli such as mechanical force [2], laser [3, 4] or LED [5].

In ancient civilizations like Mesopotamia, Egypt, and China, fermentation was essential for food preservation and religious rituals. Egyptians brewed beer as a vital part of religious ceremonies and daily life [1]. In China, fermentation was used to make soy sauce, vinegar, and rice wine [2]. These early societies passed down fermentation knowledge through generations, setting the stage for future advancements.

The Greeks and Romans refined fermentation techniques, especially in wine-making. They controlled environmental factors such as temperature and vessel cleanliness to produce high-quality wines. The spread of winemaking knowledge throughout the Roman Empire led to vineyards proliferating across Europe [3, 4].

During the Middle Ages, monasteries advanced fermentation practices in brewing and cheese-making. Monks documented their brewing techniques and experimented with ingredients, contributing to the diversity of fermented products. Types of cheese like Trappist and Maroilles were created in monasteries [5]. The introduction of distillation expanded the range of alcoholic beverages, leading to the creation of spirits such as brandy and whisky [6].

The Renaissance period saw significant contributions from scholars like Louis Pasteur, whose work on microbiology and fermentation laid the foundation for modern understanding of the process [7]. Pasteur's discoveries advanced food safety and industrial fermentation processes [8].

By the Industrial Revolution, fermentation was established worldwide, with technology driving mass production of fermented foods and beverages [9]. Innovations such as refrigeration and pasteurization improved food preservation [10], while industrial-scale breweries and distilleries transformed alcohol production. Mechanization and scientific advancements led to more controlled and predictable fermentation outcomes [1]. Advancements in transportation and refrigeration facilitated the global distribution of fermented products, expanding the reach of beer and cheese [11–13].

The twentieth century saw industrial-scale fermentation processes for producing antibiotics, vitamins, and other pharmaceuticals [14]. The discovery of penicillin by Alexander Fleming in 1928 marked a milestone, ushering in the antibiotic era and saving countless lives [15]. Fermentation also became central to biofuel production as concerns over fossil fuel dependence grew [16]. Ethanol fermentation gained prominence as a renewable alternative to gasoline, with large-scale facilities converting agricultural feedstocks into ethanol [17, 18].

The latter half of the twentieth century saw the industrialization of fermentation in the food industry, with mass production of products like yogurt, sauerkraut, and soy sauce [19]. Advances in biotechnology led to the development of genetically modified organisms optimized for fermentation, enhancing productivity and product consistency [20]. Fermented foods gained popularity for their flavor, nutritional benefits, and perceived health benefits from probiotics [21].

The twenty-first century brought further innovation in fermentation through bioprocessing and synthetic biology [22]. Novel fermentation techniques and bioreactor designs optimized production, minimized waste, and ensured food safety [22–24]. Computational modeling and simulation accelerated bioprocess development, reducing costs [25, 26].

Fermentation has emerged as a platform for sustainable food production and waste valorization. Bioconversion technologies generate value-added products such as biogas, biofertilizers, and animal feed supplements from organic waste [27]. These biorefinery approaches contribute to the circular economy by minimizing waste and maximizing resource utilization.

Fermentation remains vital in artisanal and craft food production, as consumers seek unique and authentic products with distinct flavors [28]. Small-scale fermenters and homebrewers experiment with fermentation, leading to a renaissance in artisanal cheese-making, bread-baking, and vegetable preservation [13, 29].

Recently, there has been growing interest in fermentation for sustainable protein production, with companies exploring microbial fermentation to produce alternative proteins like fungal mycoprotein and microbial biomass [30, 31]. The US FDA considers mycoproteins as GRAS since 2001 [32]. These proteins offer an environmentally friendly and efficient alternative to conventional animal agriculture, addressing concerns related to land use, water consumption, and greenhouse gas emissions [31].

Looking ahead, fermentation is poised to address global challenges like food security, environmental sustainability, and public health [9, 31]. Continued advancements in fermentation science and technology, along with a growing awareness of the benefits of fermented foods and bioproducts, will drive further innovation and adoption of fermentation-based solutions in the years to come [33].

In conclusion, fermentation processes have been integral to human civilization, providing a diverse array of products from everyday staples to life-saving pharmaceuticals [34]. Industrialization of fermentation has led to remarkable advancements in biotechnology, allowing us to harness microorganisms for valuable compounds on an unprecedented scale. Genetic engineering has unlocked new possibilities for fermentation, enabling the creation of designer microbes for specific tasks.

This chapter will explore the current landscape of submerged and solid-state fermentation products, examining the processes, industry challenges, and future prospects for biotechnological innovations. From traditional fermented foods to cutting-edge biopharmaceuticals, the scope of fermentation applications is vast and expanding. Understanding fermentation complexities and staying abreast of technological developments will harness the full potential of microbial fermentation to

address societal needs and propel the biotechnological industry forward into a more sustainable and prosperous future.

2 Economics of Fermentation

Solid-state fermentation (SSF) is considered an effective method for converting agricultural residues and by-products into value-added products such as bioactive compounds, bioplastics, enzymes, and biofuels [35]. SSF processes often utilize cheap agro-industrial by-products as culture media, and the low water volume in SSF significantly impacts process economics by reducing bioreactor size, downstream processing, and sterilization costs [36, 37]. This makes SSF particularly attractive for biomass valorization and cost-effective production of biosurfactants.

Martínez et al. [38] conducted a simplified techno-economic analysis for producing sophorolipids via SSF using sustainable resources. They developed a model to process 750 tons of winterization oil cake annually, comparing three support materials: wheat straw, rice husk, and coconut fiber. The unit production costs were 5.1, 5.7, and 6.9 USD/kg of sophorolipids for WS, RH, and CF, respectively. Production capacity ranged from 325 to 414 tons per year, with financial parameters including a CAPEX of 6.7 million USD, OPEX of 1.9 million USD/year, and a payback period of 3.2 years. The primary operational cost was sophorolipid recovery from the solid matrix, while fermentation equipment shaped capital costs. Implementing the sophorolipid production system at the winterization oil cake production site using SSF supports a circular bioeconomy by valorizing organic solid wastes and by-products [37].

In another study, Taiwo et al. [39] evaluated the economics of cellulase production via SSF using batch and semi-batch processes. They found that the semi-batch approach had a payback period of less than 2 years, an internal rate of return of 39.14%, and a net present value of 142.089 million USD, making it more economically viable than the batch process.

SSF has seen a resurgence in recent years, particularly in the production of cellulosic ethanol, enzymes, and antimicrobial compounds [40–43]. Major companies involved include Novozymes, DuPont, DSM, Monsanto, Genentech, Nestec, Xyleco, and MS Technologies [38]. The focus of SSF research has shifted from Japan, Bangladesh, the USA, Egypt, and the UK before the 1970s to the USA, China, India, Brazil, and Spain in recent years, likely due to the availability of cheap, high-quality substrates in developing countries and the need to valorize them through bioprocesses. China, South Korea, India, and the USA are the major patent-filing countries, with China holding 89% of the patent documents [44].

Submerged fermentation (SmF) involves microorganism growth in a liquid broth medium enriched with necessary nutrients, occurring in closed reactors with high oxygen concentration [45, 46]. Although SmF dominates current industrial enzyme production, studies indicate that SSF can produce higher enzyme yields at the laboratory scale [47]. Despite challenges like foam formation, which can be mitigated with

antifoam agents, SmF offers higher productivity and yield, reduced capital, maintenance, and labor costs, and lower contamination risk. It is less sensitive to medium composition changes, allowing a broader range of substrates. SmF is mainly operated as a batch system, including the shake flask technique for optimizing fermentation conditions [48, 49].

A prominent challenge in SmF is decreased enzymatic activities in fungal enzyme production due to shear stress on mycelial cells, large mycelium clumps, and carbon catabolite repression [50]. While bacteria, yeasts, and filamentous fungi are used to generate a wide range of industrial enzymes mainly through SmF, the costs remain high due to complex production processes targeting highly pure enzymes and utilizing expensive simple sugars [47].

3 Fermentation Applications

3.1 *Fermentation for Various Purposes*

Fermentation is often more accessible, economical, and environmentally friendly for obtaining specific organic molecules compared to conventional chemical synthesis methods. This has led to increased research in producing, optimizing, and processing pharmaceuticals, biological products, biofuels, biofertilizers, and biodegradable polymers through fermentation [1]. Driven by the need to create a decarbonized economy, the market for bio-based products obtained through fermentation processes has been steadily increasing in recent years [51, 52].

Industrial fermentation employs pure or mixed microbial cultures, where the type of microbial catalysts and raw materials used influence the metabolic pathways and the type of products generated, including modifications in the substrate and by-products [53, 54].

An innovative field of biotechnology known as electrofermentation has emerged, utilizing electrochemical means to guide and control fermentation. Initially developed to improve the transformation, production, and preservation of food, electrofermentation now represents a broader biotechnological platform where organic materials are transformed by microorganisms into various chemical products [55]. These products include organic acids, amino acids, solvents [56], vitamins [57], polysaccharides [58–60], secondary metabolites [61], enzymes [62], biofuels [63], and biopolymers. These valuable products have applications in various industrial sectors, including chemical, food, pharmaceutical, medical, textile, leather processing, and plastic manufacturing [53, 59].

Future fermentation industries will likely operate continuously, with automation reducing human involvement and preventing errors. Low-cost materials like plastic, cement, or ceramics may replace steel. Advancements in packed bed reactors may

offer interesting approaches for both submerged (biofilm formation) [64] and solid-state fermentation (aeration) [65], as well as for stressing and stimulation strategies, enabling various kinds of bioprocesses for better metabolite production [66–68]. Additionally, artificial intelligence will control fermentation, providing more efficient, sterile processes with reduced energy, water, and labor requirements [20].

3.2 Fermentation for Food-Related Purposes

Fermentation remains widely practiced for its ability to extend shelf life and enhance the organoleptic properties of foods. Fermented foods are created through controlled microbial growth and enzymatic conversion of food components [20]. With growing global demand and increased awareness of food safety, standardizing fermentation processes has become necessary, leading to industrial control of production procedures. Fermented foods contain functional microorganisms that provide probiotic, antimicrobial, and antioxidant benefits, among other health advantages [69, 70]. Fermenting agricultural and food processing waste can also yield healthy, value-added foods and sustainable food sources [71].

Fermentation can occur spontaneously through naturally present microorganisms in raw foods, or through the addition of starter cultures to ensure standardized products and controlled industrial protocols [1]. Revolutionary tools in the industry include genome editing and CRISPR-based microbiome manipulation, improving process control and final product properties, thus enhancing sensory aspects, quality, shelf life, nutritional content, and safety [1]. Metagenomic approaches identify functional microorganisms that increase nutrient bioavailability, sensory quality, and biopreservative effects, and improve food safety and human gut health [72].

The interest in precision fermentation technology is driven by the growing market for sustainable alternative food ingredients. Genetically modified organisms are used to improve yields and efficiency in the alternative meat and dairy sectors, as well as to produce dyes and flavors for the food industry [73]. In developing countries, small-scale fermentation has been enhanced through practical knowledge. For instance, the consortium method has improved small-scale soy sauce fermentation in Thailand by facilitating collaboration between industry and scientists [9]. The European Union and Indian Government fund projects like NAMASTE-EU and NAMASTE-India to develop technologies for converting agricultural waste into healthy food ingredients [71].

Scientific research suggests that diets including fermented foods can reduce the risk of non-communicable diseases like cardiovascular diseases, type 2 diabetes, cancer, and allergies. Interest in fermentation technology has also grown for adding value to food waste by generating valuable by-products [9]. Microalgae are being developed as protein sources due to their nutritional composition, functional diversity, and metabolic flexibility [74]. In 2021, \$1.69 billion was invested in fermentation companies, with costs of precision fermentation proteins expected to be competitive by 2025 [73].

From 2017 to 2022, market and dietary habit changes drove the success of fermentation ingredients for human and animal nutrition. The global market for food and beverages, agriculture, and consumer products were valued at \$0.43 billion, \$0.39 billion, and \$0.35 billion, respectively. The food and beverage market is expected to grow at a compound annual growth rate (CAGR) of 51.3%, reaching \$5.7 billion by 2026 [20].

Fermentation has been crucial in creating plant-based meat and dairy alternatives. The market for fermented plant-based products was valued at \$329.29 million in 2021 and is expected to reach \$422.26 million by 2026, with a CAGR of 5.0% [75]. This growth, driven by health benefits like immune system support and improved gut health, was accelerated by the COVID-19 outbreak [76]. Europe leads the market for plant-based fermented products, followed by Asia-Pacific, North America, Latin America, the Middle East, and Africa. In 2020, Europe generated \$96.99 million in revenue from these foods, projected to reach \$129.47 million by 2026 [75]. The global market for plant-based yogurt was estimated at \$2.02 billion in 2020, with Europe dominating about 50% of the market. The fermented “cheese” market was valued at \$2.70 million in 2019 and is expected to reach \$4.58 billion by 2025, with a CAGR of 8.91% [77].

The increasing demand for fermented products has highlighted the need for new regulations emphasizing safety, transparency, and quality. Current regulatory frameworks are often influenced by geographical, political, and cultural factors, and are primarily restricted to certain types of fermented foods, such as dairy products [78]. To maintain consumer trust, there is an urgent need for clearer, consistent, and harmonized regulations to guide safe production, storage, transport, and distribution. Evidence-based legislation could promote growth and innovation in the fermented foods sector, fostering consumer and investor confidence and catalyzing global economic progress.

3.3 Fermentation for Enzyme Production

Enzymes are proteins or biocatalysts that determine the rates of biological reactions and control complex metabolic pathways and other cellular processes. In the presence of enzymes, reactions occur much more rapidly than in their absence. These biological molecules are highly specific in recognizing the substances they need to transform [79]. Over time, enzymes have spontaneously evolved, making some available as low-cost raw materials.

Microbial cells are extensively used in fermentation processes to produce various industrial products. Microbial enzyme production is widespread and rapidly growing due to the potential and advantages microorganisms offer. Microorganisms are preferred for industrial enzymes due to their rapid growth, limited space requirements, and broad range of catalytic activities [80]. Additionally, microbial enzymes are more stable compared to those from plants and animals. Microorganisms can be grown in large quantities in a short period and genetically manipulated to generate

new enzymes with desirable properties, including extremophilic enzymes that can withstand harsh industrial conditions such as high temperatures, low water activity, alkaline pH, or organic solvents [20]. Their continuous availability, rapid cell growth, and ease of scalability from bench to industrial scale enable cost-effective industrial enzyme production. In contrast, plant and animal enzymes face limitations due to the seasonality of raw materials and ethical issues, respectively [1].

Commercially, industrial enzymes are produced using both submerged and solid-state fermentation techniques. Submerged fermentation has drawbacks such as high costs and energy requirements for downstream processing. Conversely, solid-state fermentation offers advantages like ease of product extraction, low capital investment, low energy consumption, minimal water usage, low waste output, and higher productivity of certain enzymes, leading to lower overall production costs. This highlights the importance of solid-state fermentation for future industrial enzyme production [1].

Optimizing fermentation conditions is essential for microbial enzyme production. Critical factors influencing fermentation include medium composition, pH, temperature, agitation, and aeration. The medium should contain all necessary nutrients for microbial growth and enzyme production, including carbon and nitrogen sources, minerals, and growth factors. The pH affects the microorganism's growth and the enzyme's activity and stability. Temperature influences microbial growth, enzyme activity, and stability [81, 82]. Agitation and aeration are essential for providing oxygen to the microorganism and distributing nutrients evenly throughout the medium [1]. Reports have demonstrated the effectiveness of using light and electric currents to improve microbial activity [63, 83].

Recent advances in enzyme production focus on developing new and improved enzymes for various industrial applications. One approach involves exploring new microbial sources, including extremophiles, which produce enzymes with unique properties. Extremophiles thrive in extreme environments and their enzymes, known as extremozymes, are more stable and active under harsh industrial conditions. Another approach involves genetic engineering of microorganisms to enhance enzyme production. This includes introducing genes encoding desired enzymes into microorganisms, modifying regulatory pathways, and engineering enzyme properties through protein engineering. Advances in fermentation technology, such as bioreactors, immobilization techniques, and process optimization, have also contributed to increased enzyme yields and reduced production costs [53, 78].

3.4 Fermentation for Pharmaceuticals Development

In recent decades, fermentation has become essential in the pharmaceutical industry for producing various drugs and high-value molecules [84]. Major companies have harnessed the efficiency of fermentation processes to manufacture pharmaceuticals at scale. Pfizer, for example, is known for using fermentation in the development and production of vaccines, antibiotics, and other biopharmaceuticals [85]. Their

expertise was crucial in responding to the COVID-19 pandemic by scaling up vaccine production using fermentation-based methods [86].

Similarly, Novartis, a multinational pharmaceutical company, has heavily invested in fermentation technology for producing biopharmaceuticals like monoclonal antibodies, recombinant proteins, and gene therapies [87, 88]. Through partnerships with biotechnology firms, Novartis has accelerated drug development and commercialization using fermentation platforms, delivering life-changing therapies worldwide.

Biogen specializes in neurological diseases and is known for its pioneering work in fermentation-based biologics production. Their expertise in cell culture and fermentation processes has led to the efficient production of complex protein therapeutics such as monoclonal antibodies and enzyme replacement therapies. Biogen's commitment to innovation in fermentation technology has made it a key player in the biopharmaceutical industry.

Merck & Co., Inc., also known as Merck Sharp & Dohme (MSD) outside the US and Canada, has significantly contributed to pharmaceutical fermentation. Their biopharmaceutical division focuses on developing and producing innovative therapies using fermentation platforms. Merck's extensive fermentation infrastructure has facilitated the production of vaccines, antibodies, and therapeutic proteins [89, 90] for various conditions, from infectious diseases to cancer and autoimmune disorders.

Pharmaceuticals produced using fermentation technology include insulin, antibiotics, biologics, vaccines, enzymes, hormones, and growth factors. Insulin, now produced using genetically modified strains of *Escherichia coli* or *Saccharomyces cerevisiae*, was traditionally extracted from animal pancreases. Antibiotics like penicillin, erythromycin, and tetracycline are produced through fermentation processes involving microorganisms such as *Penicillium chrysogenum*, *Streptomyces* spp., and *Saccharopolyspora* spp. [91–95].

Biologics, including monoclonal antibodies and therapeutic proteins, are manufactured using mammalian cell culture systems like Chinese hamster ovary (CHO) cells, which produce complex proteins with human-like glycosylation patterns [96]. Vaccines for influenza, hepatitis B, and HPV are produced using cell culture systems [97], while bacterial vaccines like those for diphtheria, tetanus, and pertussis are produced through bacterial fermentation processes [98, 99].

Enzymes produced through fermentation have applications ranging from digestive aids to clot-busting drugs like tissue plasminogen activator (tPA), produced using recombinant DNA technology in mammalian cell cultures or yeast [100, 101]. Enzyme replacement therapies for genetic disorders are also produced through fermentation [102]. Hormones and growth factors, such as human growth hormone (hGH) are produced using recombinant DNA technology in cell cultures or microorganisms [103, 104].

The development of biosimilars, biologic products similar to existing approved biologics, has also become significant. Produced using fermentation processes, biosimilars provide more affordable treatment options while maintaining efficacy and safety, expanding patient access to essential biologic therapies [105].

Despite advancements in fermentation technology, the development of new pharmaceuticals faces challenges. Producing biologics requiring post-translational modifications [106] like glycosylation can be complex and costly. Regulatory hurdles from agencies like the FDA and EMA add time and cost to the development process. Fermentation processes must address product purity, consistency, and yield, which can be influenced by various factors, requiring careful process optimization and quality control.

Innovative approaches are being explored to overcome these challenges. Synthetic biology enables the design of custom genetic circuits and metabolic pathways within microbial hosts for more efficient pharmaceutical production [107]. Novel fermentation platforms and bioreactor systems improve process efficiency and scalability, with advanced bioreactor designs offering better control over fermentation conditions [108–110].

Sustainable production methods using renewable sources like lignocellulosic biomass and waste streams are being investigated to reduce reliance on traditional substrates and minimize environmental impact. High-throughput screening and omics-based approaches, combined with computational modeling, are driving innovation in fermentation-derived pharmaceutical development [111, 112].

Ongoing advancements in genetic engineering, bioprocess optimization, and fermentation scale-up continue to expand the pharmaceutical industry's portfolio of fermentation-derived therapies, improving patient outcomes and advancing global healthcare.

3.5 Fermentation for Bioenergy Production

Fermentation processes are widely utilized for producing bioenergy compounds as renewable alternatives to fossil fuels. One prominent bioenergy compound is ethanol, which is used as a gasoline additive or as a standalone fuel in ethanol-blended gasoline [113]. Ethanol is typically produced by microorganisms such as *Saccharomyces cerevisiae* or *Zymomonas mobilis*, which convert sugars from biomass sources like corn, sugarcane, or cellulosic feedstocks into ethanol [59, 114]. This process also produces carbon dioxide as a byproduct, making ethanol a sustainable fuel option.

Butanol, another bioenergy compound, offers superior fuel properties compared to ethanol, such as higher energy density and lower vapor pressure. It is produced via acetone-butanol-ethanol (ABE) fermentation, where *Clostridium* spp. convert sugars into acetone, butanol, and ethanol [115, 116]. Butanol has applications as a biofuel additive, solvent, and chemical feedstock, with sustainable and cost-effective production compared to petrochemical processes [117].

Methane, the primary component of natural gas [118], can be produced through anaerobic digestion of organic materials like agricultural residues, wastewater sludge, microalgae biomass [119, 120], and organic waste. This process involves microbial degradation of organic matter by anaerobic microorganisms, resulting in methane and carbon dioxide production. Methane can be used for electricity generation, heating,

and transportation, offering a sustainable alternative to fossil fuels. Biogas plants with anaerobic digesters facilitate large-scale methane production from organic waste streams, aiding waste management and renewable energy efforts [121].

Hydrogen gas (H_2) is a promising bioenergy compound produced through microbial fermentation by bacteria [122], microalgae [123], and cyanobacteria [124]. This process involves the metabolism of organic substrates by hydrogen-producing microorganisms like *Clostridium* spp. or *Enterobacter* spp. under anaerobic conditions. Hydrogen gas serves as a clean energy carrier with applications in fuel cells and chemical synthesis [125, 126]. Biohydrogen production offers environmental benefits over conventional methods, utilizing renewable biomass feedstocks to mitigate greenhouse gas emissions [127–129].

Short-chain fatty acids like acetic, propionic, and butyric acids can also be produced through microbial fermentation of organic substrates. These acids have various industrial applications, including as chemical precursors, food additives, and platform chemicals for bio-based polymers and biofuels. Microorganisms like *Clostridium* spp. and *Acetobacter* spp. are commonly used for their production. Optimizing fermentation conditions and strain selection improves production yields and cost-effectiveness [130].

Microbial fermentation can produce bio-based chemicals and specialty compounds for industries such as pharmaceuticals, cosmetics [131] and agriculture. By leveraging fermentation technology and the metabolic capabilities of microorganisms [132], researchers are developing sustainable bioproducts, contributing to a bio-based economy [133, 134].

The selection of culture medium and carbon sources is crucial for bioenergy production efficiency. In South America, sugarcane and corn are primary carbon sources for ethanol production, particularly in Brazil, a global leader in bioethanol production [132, 135, 136]. In North America, corn is the predominant feedstock for ethanol, with cellulosic biomass like corn stover being explored for future use [137–141].

In Europe, bioenergy production varies regionally, with biogas production from agricultural residues and biodiesel production from oilseed crops being common. In Asia, countries like China and India utilize agricultural residues and waste materials for biogas and ethanol production. Oceania focuses on using agricultural residues, forestry biomass, and organic waste for biofuel and biogas production. Regional approaches to bioenergy development highlight the diversity based on feedstock availability, economic factors, and policy incentives.

Recent research focuses on developing new biofuels with superior properties. Butanol and isobutanol offer higher energy density and compatibility with existing infrastructure, produced through microbial fermentation [115, 116]. Bio jet fuel, produced from biomass feedstocks through thermochemical or biochemical processes, is emerging as a renewable alternative to conventional aviation fuels [142]. Advanced biofuels like cellulosic ethanol [143] provide sustainable alternatives to traditional biofuels, offering significant environmental benefits.

Fermentation for biohydrogen production is promising for renewable energy generation. However, challenges such as low yield, complex lignocellulosic biomass

utilization, and microbial strain optimization need addressing for widespread industrial implementation. Advances in synthetic biology, metabolic engineering, and systems biology are crucial for improving biohydrogen production processes and maximizing their potential.

4 Genetic Engineering and Synthetic Biology Trends for Fermentation

The invention of PCR by Kary Mullis in 1983 revolutionized molecular biology and had a profound impact on synthetic biology. PCR allows for the rapid and precise amplification of DNA for various purposes. It supports gene cloning, manipulation, and genome editing technologies like CRISPR-Cas9, crucial for creating new traits in organisms. Additionally, PCR plays a vital role in metabolic engineering, synthetic genomics, and DNA sequencing, broadening genetic research and diagnostics [144–146].

Genetic engineering and synthetic biology have revolutionized fermentation processes, offering unprecedented control over microbial metabolism and the ability to design custom microorganisms. Trends in this field include the development of advanced genome editing tools like CRISPR-Cas9 for precise manipulation of microbial genomes and the integration of omics technologies with computational modeling for systems-level understanding and optimization of microbial metabolism. Synthetic biology platforms and automation technologies facilitate the design, construction, and characterization of custom genetic circuits and microbial pathways for fermentation [147–149].

Expansion of genetic code technologies and non-natural amino acid incorporation systems is opening new frontiers in synthetic biology and fermentation, allowing the production of biomolecules with tailored properties. These technologies have been applied in various industrial biotechnology fields, enhancing processes such as xylose fermentation, starch saccharification, ethanol production, and antibiotic production [111, 150, 151].

Furthermore, genetic manipulation has led to improvements in various fermentation processes, such as increased riboflavin and L-malic acid production, antibiotic-free production of enzymes, and enhanced production of bioactive compounds. Examples include the production of mammalian hemoglobins, lactic acid, succinic acid, lycopene, B12 vitamin, erythritol, and 3-hydroxypropionic acid [152–154].

These advancements highlight the power of genetic engineering and synthetic biology in improving fermentative bioprocesses, contributing to the ongoing 4th industrial revolution. They enable the development of sustainable and efficient biomanufacturing processes for various industrial and biomedical applications [155].

5 Concluding Remarks

Fermentation, spanning from ancient civilizations to modern biotechnology, profoundly impacts human culture, industry, and sustainability. It offers a promising solution to global challenges and advances biotechnological frontiers. Solid-state fermentation, a cost-effective and sustainable method, valorizes agricultural residues, enabling the production of valuable compounds. With its accessibility, cost-effectiveness, and environmental friendliness, fermentation is poised to shape various industries, crucial for transitioning to a decarbonized economy. The growing interest necessitates cohesive regulatory frameworks ensuring safety, transparency, and quality, vital for industry growth and innovation, bolstering consumer confidence and economic advancement. Enzyme evolution and fermentation technology advancements revolutionize industrial enzyme production, pharmaceuticals, and healthcare, promising enhanced efficacy, accessibility, and sustainability. The synergy between genetic engineering and fermentation drives sustainable biomanufacturing, fostering innovation in bioenergy and bioproducts for a greener future.

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A Narrative Assessment of Free Webservers for Pharmacokinetic and Toxicological Predictions Using In Silico Tools



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Abstract Pharmacokinetic and toxicological data are critical to the rational design of efficient medications as well as the identification of lead compounds in pharmaceutical research. However, due to physicochemical and structural properties that hinder the absorption, distribution, metabolism, and elimination processes, as well as occasionally impose toxicity, evaluating substances in vitro and in vivo with the goal of achieving therapeutic success is seldom successful. Preventing a product from reaching the market. Computational methods, in this respect, can furnish rapid and dependable data for multiple compounds, with the objective of selecting and prioritizing lead compounds and streamlining the drug discovery process to produce molecules that are both safe and effective. Specifically, this work intends to examine four open-access online servers (ADMETlab, PreADMET, SwissADME, pkCSM and admetSAR) that perform pharmacokinetic and toxicological predictions. It will discuss the features, benefits, and drawbacks of these servers as well as instances in which researchers in medicinal chemistry have used them to accomplish their goals. Thus, considering the aforementioned, the ease with which various qualities can be predicted in an optimal and cost-free manner makes the usage of such technologies beneficial for both interested companies and the academic community.

Keywords Drug discovery · Pharmacokinetics · Medicinal chemistry · In silico

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1 Introduction

In clinical practice, the pharmacokinetic and toxicological behavior of candidate substances for new drugs is a fundamental parameter for dose selection, dosing schedules, dosage adjustments, and adequacy, as well as for therapeutic monitoring with the same substance. The aims to achieve and maintain the drug's plasma concentrations within the parameters of each therapeutic window [1–3].

Moreover, knowledge of the pharmacokinetic behavior of substances of therapeutic interest is useful in the pharmaceutical industry for planning and developing formulations that maximize bioavailability, increase metabolic stability through the synthesis of pro-drugs, delay the release and absorption profile, prolong the pharmacological action, or even allow for the site-specific release of drugs, as in the case of enteral release tablets and capsules [4, 5].

In the process of finding and studying new drugs, *in silico* pharmacokinetic and toxicological predictions represent feasible substitutes for screening a large number of molecules without requiring significant amounts of test substance samples used in animal models. These biological matrices must be subjected to instrumental analysis involving a large volume of solvents and energy required for operation of chromatographs assays [6–9]. This approach reduces the time, costs, and inherent ethical risks associated with using animal models in biomedical research.

In this regard, the work attempted to conduct a narrative evaluation of the literature pertaining to free *in silico* tools employed in investigations for screening compounds with therapeutic potential. The goal to choose drug candidates for *in vitro* validation testing in an assertive manner, in accordance with the application of the R's (Replacement, Reduction, and Refinement) in pharmaceutical research [10].

2 Material and Methods

The foundation of this study was the creation of a narrative literature review. Andrade [11] asserts that literature reviews are crucial tools for organizing and disseminating scientific knowledge, as they provide a comprehensive list of references that are easy to consult and apply. They also serve as a summary of the latest research on the subject under study.

The narrative review [12, 13] stands out in this context as an effective strategy for mapping, formulating, and structuring knowledge. It is particularly useful because it allows the mapping of knowledge production on a specific topic without strict criteria for systematizing searches, selection, and data extraction.

Descriptors such as “webserver”, “ADMETlab”, “PreADMET”, “SwissADME”, “pkCSM”, “admetSAR”, “pharmacokinetics”, and “toxicity”, together with the Boolean operators “and” and “or” were used to search the PubMed, Google Scholar, and Embase databases. The selection criteria for studies included relevance to the

research question and the accessibility of the data presented in the methodology and findings sections.

Consequently, the results and the discussion will focus on the presenting of the main free in silico tools for pharmacokinetic and toxicity predictions, considering their parameters, usability, functionalities, and the outcomes of studies available in the reviewed literature of the following tools: ADMETlab, PreADMET, SwissADME, pkCSM and admetSAR.

3 Results and Discussion

Structural and physicochemical data are essential for predicting pharmacokinetic and toxicity profiles, such as Lipinski's Rule of Five, which is a primary analysis parameter for prioritizing and selecting lead compounds in pharmaceutical research focused on substances with a satisfactory oral absorption profiles. However, this classic chemometric parameter depends on the availability of experimental data and the conditions under which these experiments were carried out [14].

Additionally, this traditional approach overlooks active transport mechanisms in the absorption process, and does not allow inferences about other pharmacokinetic processes, such as distribution between body compartments and interference in xenobiotics metabolism mediated by CYP enzymes. These aspects, however, can be assessed through computational methods [15–19].

In this context, the development of absorption distribution metabolism, excretion and toxicity (ADME/Tox) modeling strategies, such as ADMETlab, PreADMET, SwissADME, pkCSM and admetSAR, represents significant advancements in chemometrics and chemoinformatics. This field has progressed from traditional methods to sophisticated machine learning techniques and other computational methods, making a substantial contribution to scientific research in hit-to-lead and lead optimization stages in drug discovery [20].

It is worth noting that methodologies such as artificial intelligence (AI) and machine learning (ML) can be integrated with various molecular modeling methods (including those mentioned above), facilitating the detection of patterns that predict favorable outcomes and significantly accelerating the process of identifying potential drug candidates [21, 22]. This allows researchers to focus on candidates with higher druglikeness, low toxicity, and favorable pharmacokinetic properties for designing new pharmaceutical products [23].

Computational approaches do not eliminate the need for in vitro and in vivo pharmacokinetic studies; however, they have the potential to reduce costs and execution time in the early stages of drug discovery. They expedite the conversion of initial results and leads into qualified development candidates, especially in high-throughput approaches, where a significant number of molecules are analyzed [24].

Thus, the aim of this work was to comprehensively address the advantages, limitations, and successful examples in medicinal chemistry research that demonstrate

the importance of in silico predictive methods applied in pharmaceutical research, as well as the ongoing need for research and improvements in tools developed for this purpose.

4 ADMETlab

ADMETlab is a free platform available at <https://admet.scbdd.com/home/index/> designed to thoroughly evaluate the absorption, distribution, metabolism, excretion, and toxicity (ADME/Tox) properties of chemical substances. In addition to integrating an extensive ADME/Tox database, the platform provide an easy-to-use interface for researchers and professionals working in drug discovery and development [25].

Developed and launched online in 2018 is maintained by the Computational Biology and Drug Design Group at the Xiangya School of Pharmaceutical Sciences, Central South University. The Web server allows pharmacokinetic and toxicological predictions of molecules in SMILES format, or designed in JMSE format.

Among the benefits offered by the ADMETlab platform are the rapid and efficient evaluation of multiple compounds in virtual drug screening, reduced research costs by minimizing the costly experiments, and greater potential for success in the selecting of drug candidates [26].

However, like other ADME/Tox prediction tools, ADMETlab has limitations regarding data availability, modeling techniques, and system understanding. Nevertheless, it enhances drug discovery efforts by linking structural modifications to changes in response [27].

In this context, Boopathi et al. [28] demonstrated the utility of ADMETlab, along with other in silico tools, in screening of substances with inhibitory activity against the SARS-CoV-2 virus's adhesion to host cells. This inhibition occurs via the interaction of the S protein with the angiotensinogen-converting enzyme and the transmembrane protease TMPRSS2, which proteolytically cleaves the S protein into the S1 and S2 subunits, promoting membrane fusion and virus entry into the host cell.

Boopathi et al. [28] investigated natural substances from the *Panax giseng* plant with the potential to inhibit ACE-2 and TMPRSS2 gene expression using in silico analysis, followed by validation by in vitro tests on A549, Caco-2, and MCF7 models. These analyses revealed reduced expression of the selected targets and low cytotoxicity, suggesting that the ginsenoside compound CK has therapeutic potential against COVID-19.

In another study focused on the antiparasitic activity of the organotellurium compound RF07 against Leishmania, Nunes et al. [29], conducted predictive pharmacokinetic and toxicity analyses in PreADMET, showing good oral absorption and satisfactory permeability in Caco-2 cells, inhibition of CYP2C19 and CYP2C9 enzymes, a high distribution profile with possible central nervous system activity, and positive results in the AMES test, though out of range in a rodent carcinogenicity model. In in vivo tests on dogs with visceral leishmaniasis, RF07 was shown to

clinically reverse the infection's manifestations, and various mechanisms of anti-inflammatory activity were proposed, with no histopathological findings of liver or kidney toxicity.

This study highlights RF07's potential as a treatment for visceral leishmaniasis and also demonstrates how the physicochemical characteristics and ADME/Tox predictions from PreADMET can aid in planning in vivo studies.

The parameters that can be analyzed by ADMETlab are presented in Table 1.

According to Dulsat et al. [30], ADMETlab is one of the platforms offering the broadest coverage for academic and small business purposes in medicinal chemistry. The study highlights that ADMETlab demonstrated higher accuracy and precision in comparative analyses with other free online ADME/Tox prediction tools.

Table 1 Parameters analyzed by ADMETlab and supported formats

ADMETlab (https://admet.scbdd.com/home/index/)	
Input format	Analysis parameters
SMILES JMSE	<i>Physico-chemical properties</i>
	LogS (Solubility)
	LogD 7.4 (Distribution Coefficient D)
	LogP (Distribution Coefficient P)
	<i>Absorption</i>
	Caco-2 Permeability
	Pgp-inhibitor
	Pgp-substrate
	HIA (Human Intestinal Absorption)
	F (20% Bioavailability)
	F (30% Bioavailability)
	<i>Distribution</i>
	PPB (Plasma Protein Binding)
	VD (Volume Distribution)
	BBB (Blood–Brain Barrier)
	<i>Metabolism</i>
	CYP 1A2 inhibitor
	CYP 1A2 substrate
	CYP 3A4 inhibitor
	CYP 3A4 substrate
	CYP 2C9 inhibitor
	CYP 2C9 substrate
	CYP 2C19 inhibitor
	CYP 2C19 substrate
	CYP 2D6 inhibitor
	CYP 2D6 substrate
	<i>Excretion</i>
	CL (Clearance)
	T 1/2 (Half Life)

(continued)

Table 1 (continued)

ADMETlab (https://admet.scbdd.com/home/index/)	
Input format	Analysis parameters
	<i>Toxicity</i>
	hERG (hERG Blockers)
	H-HT (Human Hepatotoxicity)
	AMES (Ames Mutagenicity)
	SkinSen (Skin sensitization)
	LD50 (LD50 of acute toxicity)
	DILI (Drug Induced Liver Injury)
	FDAMDD (Maximum Recommended Daily Dose)

5 PreADMET

According to its website, PreADMET was launched online in 2002 and is available at <https://preadmet.webservice.bmdrc.org>. The platform's core developers include Professor Sung Kwang Lee and researchers Young-Mook Kang, Gue Su Chang, In Hye Lee, Se Ho Park, Jaeseong Park, Kil Yean Sung, Hyun Ji Kim, Jea Eun Chung, and is managed by team leader Sungbo Hwang and researcher Hyun Kil Shin.

PreADMET (Prediction of Absorption, Distribution, Metabolism, Excretion, and Toxicity) enables the *in silico* evaluation of compounds in early development stages. It has a wide range of applications, from selecting of drug candidates to optimizing leads compounds by prioritizing ligands with the best pharmacokinetic profiles. Additionally, it supports the planning of structural modifications to optimize physicochemical properties that impact ADME/Tox processes, as well as the interaction with pharmacological targets [25].

This platform can reduce the costs and time involved in drug development by identifying compounds with a low probability of success in laboratory tests, helping to identify potential safety and efficacy issues before clinical development, and thus increasing the probability of drug success (Table 2).

Validating PreADMET tools is crucial for ensuring their reliability. Several studies combine *in silico* predictions with *in vivo* experimental data to assess the performance of these tools. However, the accuracy of the PreADMET tool has some limiting factors, including the quality of input data, user training, and the methodology used. Notably, this digital platform accepts input files in MOL format.

An example of the tool's application can be seen in a study by Ali Zahedi et al. [31], where a set of coumarin-dihydropyridines 6a-m were synthesized as potential new inhibitors of cholinesterase isoforms and human carbonic anhydrases hCA I and II for the treatment of Alzheimer's disease. PreADMET, along with other computational tools, revealed promising results for compound 6f (1-allyl-5-oxo-1,5-dihydro-2H-chromeno[4,3-b]pyridine-2,3-dimethyl dicarboxylate) against acetylcholinesterase (AChE). Compound 6i (1-(4-fluorobenzyl)-5-oxo-1,5-dihydro-2H-chromeno[4,3-b]pyridine-2,3-diethyl dicarboxylate) demonstrated satisfactory inhibitory activity against butyrylcholinesterase (BChE) and hCA I and II.

Table 2 Parameters analyzed by PreADMET and supported formats

PreADMET (https://preadmet.webservice.bmdrc.org)	
Input format	Analysis parameters
MOL	<i>Absorption</i>
	HIA (Human intestinal absorption)
	Caco-2 cell permeability
	MDCK cell permeability
	Skin permeability
	<i>Distribution</i>
	PPB (Plasma Protein Binding)
	BBB (Blood–Brain Barrier)
	<i>Toxicity</i>
	Amy test
	Rodent carcinogenicity assay

These findings demonstrate the importance of computational methods in the research, development and design of new drugs.

6 SwissADME

The Swiss Bioinformatics Institute has launched an impactful initiative called Swiss-DrugDesign, which is making significant contributions to drug development research through its technological tools. This project offers a range of publicly accessible tools for computer-aided drug design. While some of these web-based tools, such as SwissTargetPrediction and SwissSimilarity, were specifically developed for virtual screening, others, like SwissADME, SwissDock, SwissParam, and SwissBioisostere, are also useful in related fields such as pharmacology, toxicology and medicinal chemistry [32].

In particular, the SwissADME web tool provides free access to a variety of fast and reliable predictive models for physicochemical properties, pharmacokinetics, drug likeness and medicinal chemistry parameters, presenting results in in-house models such as iLOGP, Bioavailability Radar and BOILED-Egg. Additionally, the tool features a user-friendly interface that is accessible on the website without the need for login (<http://www.swissadme.ch>), enabling easy and efficient data input and interpretation [33].

However, SwissADME is somewhat limited in the range of analysis parameters available compared to other in silico pharmacokinetic and toxicological prediction software.

Nevertheless, when used alongside other in silico tools, it can yield reliable data on physicochemical properties as well as pharmacological and toxicological attributes pertinent to drug development. This capability was

demonstrated by Asgher and Iqtadar [34], who performed *in silico* analyses of tryptamine derivatives on SwissADME and admetSAR. Their research identified compounds SR41 (O=C(c1cccc1)NCCc1c[nH]c2c1cccc2) and SR43 (Cc1ccc(cc1)C(=O)NCCc1c[nH]c2c1cccc2. Cl) as candidates for central nervous system activity, as these compounds were the only ones out of six tested to show blood–brain barrier permeability in computational studies.

Similarly, Mvondo et al. [35] evaluated the physicochemical properties of quinine derivatives and benzoyl quinine derivatives using SwissADME and pkCSM with the aim of discovering treatments of parasitic infection by *Plasmodium* spp., where resistance to antiparasitic drugs has become a serious public health concern. Their study highlighted nine substances with potential for further *in vitro* pre-clinical research and possibly clinical trials to alicate their therapeutic effectiveness for malaria treatment.

The functionalities available in SwissADME are summarized in Table 3, along with the input files format compatible with the web server. Notably, SwissADME can analyze multiple ligands simultaneously and within a short processing time.

Table 3 Parameters that can be analyzed in SwissADME

SwissADME (http://www.swissadme.ch/index.php)	
Input format	Analysis parameters
SMILES	<i>Physico-chemical properties</i>
	Formula
	Molecular weight
	Num. heavy atoms
	Num. arom. heavy atoms
	Fraction Csp3
	Num. rotatable bonds
	Num. H-bond acceptors
	Num. H-bond donors
	Molar Refractivity
	TPSA
	<i>Lipophilicity</i>
	Log $P_{o/w}$ (iLOGP)
	Log $P_{o/w}$ (XLOGP3)
	Log $P_{o/w}$ (WLOGP)
	Log $P_{o/w}$ (MLOGP)
	Log $P_{o/w}$ (SILICOS-IT)
	Consensus Log $P_{o/w}$
	<i>Water Solubility</i>
	Log S (ESOL) / Solubility Class
	Log S (Ali) / Solubility Class
	Log S (SILICOS-IT) / Solubility Class

(continued)

Table 3 (continued)

SwissADME (http://www.swissadme.ch/index.php)	
Input format	Analysis parameters
	<i>Pharmacokinetics</i>
	GI absorption
	BBB permeant
	P-gp substrate
	CYP1A2 inhibitor
	CYP2C19 inhibitor
	CYP2C9 inhibitor
	CYP2D6 inhibitor
	CYP3A4 inhibitor
	Log K_p (skin permeation)
	<i>Druglikeness</i>
	Lipinski
	Ghose
	Veber
	Egan
	Muegge
	Bioavailability Score
	<i>Medicinal chemistry</i>
	PAINS
	Brenk
	Leadlikeness
	Synthetic accessibility

7 PkCSM

The pkCSM web server (<https://biosig.lab.uq.edu.au/pkcsml/>) is a methodological tool for predicting pharmacokinetic and toxicological outcomes. It utilizes graph-based signatures to represent the chemistry and topology of small molecules, employing 14 quantitative regression models that provide numerical outputs, and 16 predictive classification models that yield categorical results. These models are designed to predict a broad range of ADME/Tox properties for novel and diverse molecules [36].

pkCSM accepts structural files in SMILES format, and the web server states that once a molecule is uploaded by the user, no molecular information is retained in the system [36]. The parameters available for analysis in pkCSM are listed in Table 4.

Dulsat et al. [30] analyzed the results of 24 tyrosine kinase inhibitors across 18 free web serves, comparing these results with experimental data available from the FDA. The authors reported that pkCSM offers one of the widest ranges of pharmacokinetic data, covering absorption, distribution, metabolism and elimination parameters, though it has a lower predictive accuracy compared to SwissADME, which excels in predicting physicochemical and absorption properties.

In a successful application of pkCSM, Yeni and Rachmania [36] investigated the pharmacokinetic parameters of compounds from the plant *Hemigraphis alternata*,

Table 4 Parameters that can be analyzed in pkCSM

pkCSM (https://biosig.lab.uq.edu.au/pkcsim/)	
Input format	Analysis parameters
SMILES	<i>Absorption</i>
	Caco-2 Permeability
	Water solubility
	Intestinal absorption Human
	P-glycoprotein substrate
	P-glycoprotein I inhibitor
	P-glycoprotein II inhibitor
	<i>Distribution</i>
	VD (Volume Distribution)
	Fraction unbound human
	CNS permeability
	BBB permeability
	<i>Metabolism</i>
	CYP 1A2 inhibitor
	CYP 3A4 inhibitor
	CYP 3A4 substrate
	CYP 2C9 inhibitor
	CYP 2C19 inhibitor
	CYP 2D6 inhibitor
	CYP 2D6 substrate
	<i>Excretion</i>
	CL (Clearance)
	Renal OCT2 substrate
	<i>Toxicity</i>
	Rat LD50
	AMES (Ames Mutagenicity)
	Minnow toxicity
	<i>T. Pyriformis</i> toxicity
	Maximum tolerated dose
	Hepatotoxicity
	Skin sensirisation
	Oral rat chronic toxicity
	hERG I and II innibitors

known for its anti-inflammatory activity observed in experimental studies. Using predictive analysis on the pkCSM web server, they identified 5 compounds out of 22 with satisfactory pharmacokinetic profiles suitable for advancing to in vitro research and subsequent pharmaceutical development for clinical studies.

The screened compounds by Yeni and Rachmania [36] included 8a-methyl-3,4,4a,5,6,7-hexahydro-2*H*-naphthalene-1,8-dione, (E)-3,7,11,15-tetramethylhexadec-2-en-1-ol, 2-methylenecholestan-3-ol, 5-(hydroxymethyl) furan-2-carbaldehyde and 2,3-dihydro-2,5-dimethyl-5*H*-1,4-dioxepin.

In another study, Silva and Padilha [37] conducted virtual screening to discover new drug candidates for leukemia treatment. They found that molecules derived from

orthoquinones and dihydroartemisinin exhibit high antitumor activity; however, pkCSM indicated these compounds lack satisfactory pharmacokinetics for clinical application. The authors propose enhancing the pharmacokinetic parameters through rational drug design, whereby subsequent modifications could improve the ligand molecule's specificity and pharmacological properties for a target.

Additionally, the utility of pkCSM in pharmaceutical research is illustrated in a study by Uygun et al. [38] investigating the antiproliferative activity of 11 pyrazolo[1,5-a]pyrazin-4(5H)-one derivatives. The researchers conducted structure-activity relationship studies for pharmacokinetic prediction, considering parameters such as absorption, volume of distribution, CYP inhibition, bioavailability, clearance, OCT2 and hERG-I-II channels inhibition, as well as skin sensitization. They assessed synthesis protocols and the impact of structural attributes on the pharmacokinetic profiles of these substances. Of the 11 tested compounds, 5c, 5e, 5f and 15e showed promising antiproliferative activity at a concentration of 160 μM , making them suitable for more advanced in vivo studies.

8 AdmetSAR

AdmetSAR was created to predict ADMET/Tox chemical characteristics as a comprehensive, free tool for environmental risk assessment and new drug discovery. Since its launch in 2012, it has been widely used by the chemical and pharmaceutical industries [39]. This online platform, accessible at <https://lmmd.ecust.edu.cn/admetSar2/>, is free to use and features a continuously updated database searchable by text and structure. It collects, organizes, and maintains ADME/Tox data available in published literature, carefully selecting proteins, species or other biological subjects linked to ADME/Tox information from numerous academic sources. The database offers a user-friendly interface for searching specific chemical profiles using common names, CAS registry numbers, or structural similarities. It includes five highly predictive quantitative regression models and twenty-two qualitative classification models to assess ecological and mammalian ADME/Tox characteristics for new compounds [40].

The most recent update in 2024, admetSAR3.0 includes interface improvements organized into modules, to option to input molecules in JSME format, and enhanced neural network graphs with multi-task integration for more robust ADME/Tox predictions. It also provides a suite analyses for human, environmental and, cosmetics-related toxicities [41].

Table 5 details the pharmacokinetic and toxicological properties that can be analyzed in admetSAR.

As an example of the applicability of the admetSAR web server, the study conducted by Rocha et al. [42] demonstrates the tool's usefulness in the initial screening of molecules with high druglikeness, favorable pharmacokinetics, and low toxicity.

Table 5 Parameters that can be analyzed in admetSAR

AdmetSAR (http://lmmd.ecust.edu.cn/admetSAR2/)	
Input format	Analysis parameters
SMILES JSME CASRN Common name of the substance	<i>Absorption</i>
	Human intestinal absorption (HIA)
	Human oral bioavailability (HOB)
	Caco-2 permeability
	<i>Distribution</i>
	Plasma protein binding (PPB)
	P-glycoprotein substrate Inhibitor
	Blood–brain barrier penetration (BBB)
	Blood-placenta barrier (BPB)
	<i>Metabolism</i>
	CYP 1A2 inhibitor
	CYP 3A4 inhibitor
	CYP 3A4 substrate
	CYP 2C9 inhibitor
	CYP 2C9 substrate
	CYP 2C19 inhibitor
	CYP 2D6 inhibitor
	CYP 2D6 substrate
	Pharmacokinetics transporters: BRCPi, BSEPi, OCT1i, OCT2i, MATE1i, OATP1b1i, OATP1b3i, OATP2b1i
	<i>Excretion</i>
	Half-life time
	Renal clearance
	<i>Toxicity</i>
	Organ toxicity
	Drug-induced liver injury
	hERG I and II channels Inhibition
	Acute toxicity
	Eye damage and eye corrosion
	Genomic toxicity
	Ames
	Carcinogenesis
	Micronucleus assay
	Eco-toxicity
	<i>Tetrahymena piriformis</i> toxicity
	Phytoplankton toxicity
	Honey bee toxicity
	Avian toxicity
	Biodegradation

In this context, Rocha et al. [42] investigated natural compounds found in the species *Amburana cearensis*, including ayapin, protocatechuic and vanillic phenolic acids, 6- coumaryl-protocatechuate, phenolic heterosides amburoside A and B, the flavonoids 4'-methoxy-physetin, isokaempferide, kaempferol, quercetin, afrormosin, β -sitosterol glycosides, the biflavonoids amburanin A and B, 6-hydroxycoumarin, o-coumaric acid and its analogs, (Z)-o-coumaric acid glycoside and (E)-o-coumaric acid, as well as the esterified analog of amburoside. The goal was identify compounds with the best kinetic toxicity profiles and therapeutic potential for treating infections, inflammation, and antioxidant activity. They used the admetSAR tool to assess whether these substances acted as inhibitors or substrates for CYPs 1A2, 2C19, 2C9, 2D6, and 3A4, phase II metabolism by the enzyme UDP-glucuronosyltransferase (UGT), and excretion through inhibition of the organic cation transporter (OCT2).

Out of the 20 substances analyzed, no activity as a substrate was detected for the 2C9 and 2D6 isoforms, while 12 substances showed potential as CYP3A4 substrates. The compounds protocatechuic acid, vanillic acid, amburoside A and B, stigmasterol glucoside, (Z)-o-coumaric acid glucoside, and (E)-o-coumaric acid glucoside did not inhibit any of the evaluated CYPs. None of the substances inhibited the OCT2 transporter, although nearly all of them showed metabolism by conjugation, with the except ayapin, 4'-methoxy-fisetin, and kaempferol.

This computationally gathered information is critical for predicting potential drug-drug interactions during clinical practice at the compound screening phase. It supports rational drug development strategies from the initial discovery stages [43].

Additional studies highlight the effective use of admetSAR in medicinal chemistry research, particularly in discovering new drugs. For instance, the investigations by Abdul-Hammed et al. [44] and Oso et al. [45] aimed to identify natural compounds with therapeutic potential against SARS-CoV-2 infection. Abdul-Hammed et al. [44] focused on the main protease Mpro, while Oso et al. [45] considered both Mpro and angiotensin-converting enzyme type II as molecular targets.

As a result, Abdul-Hammed et al. [44] identified the compounds lupeol, lupenone, hesperetin, apigenin and castasterone as probable SARS-CoV-2 inhibitors based on ADME/Tox and molecular docking data. Oso et al. [45] identified resveratrol, xylopic acid, ellagic acid, kaempferol, and quercetin as potential therapeutic showing lower acute oral toxicity than chloroquine in the ADME/Tox analyses. This suggests evidence that the tested natural products are comparatively safer and could be repurposed as antiviral agents to prevent SARS-CoV-2 entry and replication [46, 47].

9 Conclusion

Predictive analyses of absorption, distribution, metabolism, excretion, and toxicity (ADME/Tox) are made possible by computational modeling, reducing costs associated with late-stage failures and improves decision-making in the research and development of new drugs. Furthermore, the advancement of pharmaceutical research

depends on the synergy between computational and experimental methodologies, in which academics, researchers and pharmaceutical companies to confidently explore drug development with precision, using sophisticated techniques and extensive data sets. This synergy drives innovative approaches in the search for safer and more effective drugs.

The research presents highlights that advancements in computational chemistry not only reduce time and costs in drug discovery but also improve robustness in the precision and accuracy of predictive data. Methodological integration with artificial intelligence and machine learning tools provides a powerful means of analyzing thousands of compounds from various chemical libraries and databases, facilitating the discovery of promising drug candidates.

In conclusion, this work offers readers with academic, commercial, and industrial interests a comprehensive overview of the use of these platforms. It cautions that selecting the appropriate platform should be carefully aligned with research objectives, emphasizing the importance of staying updated on computational technique developments to keep up with technological advances in medicinal chemistry.

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